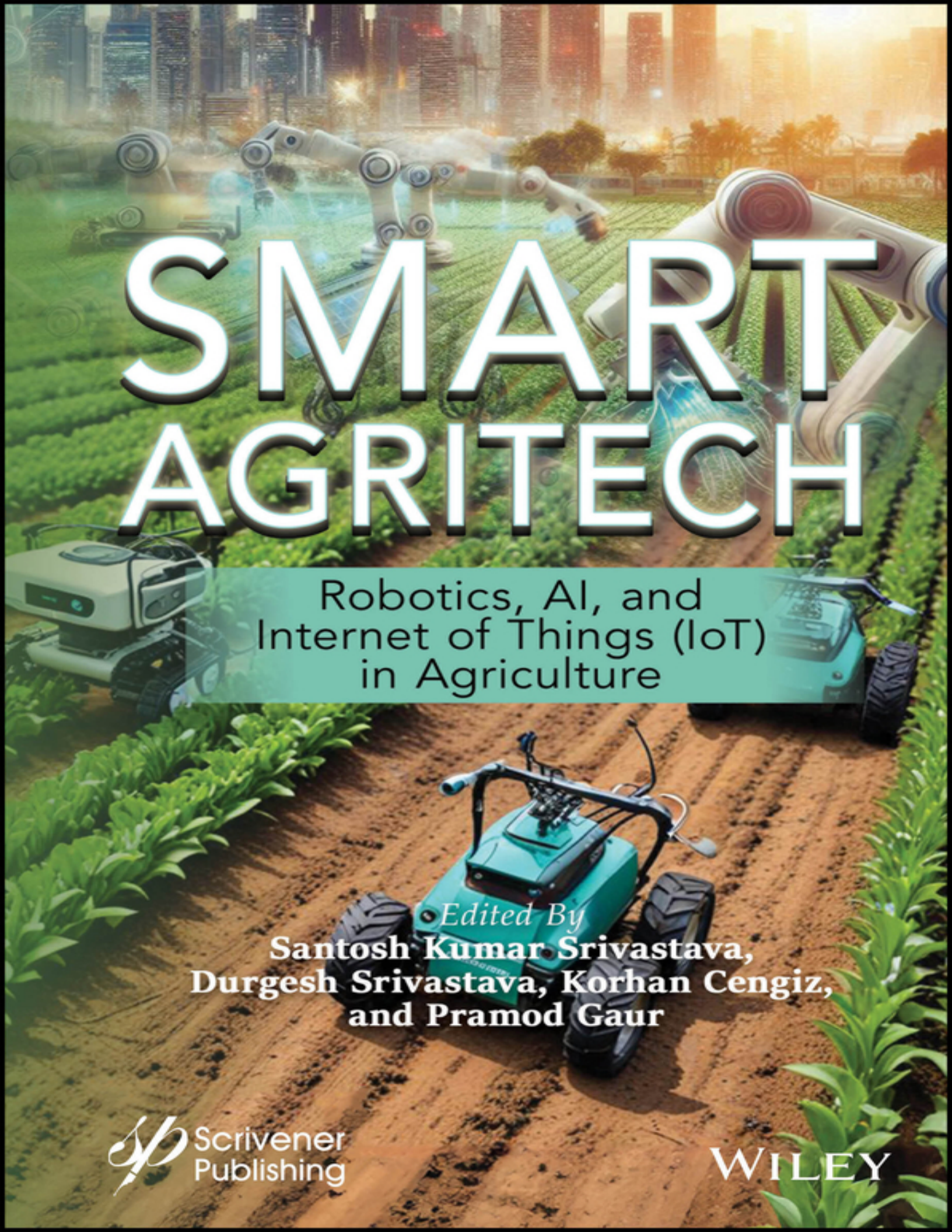
SMART AGRITECH
Robotics, AI, and Internet of Things (IoT) in Agriculture
Edited By
Santosh Kumar Srivastava, Durgesh Srivastava, Korhan Cengiz, and Pramod Gaur
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Table of Contents

1. Cover
2. Table of Contents
3. Title Page
4. Copyright Page
5. Preface
6. 1 Introduction to Smart Farming: Definition, Importance and Trends
    1. 1.1 Introduction
    2. 1.2 Smart Farming
    3. 1.3 Internet of Things
    4. 1.4 Technologies Used in Smart Farming
    5. 1.5 Importance of Smart Farming
    6. 1.6 Role of IoT in Advanced Farming Practices
    7. 1.7 Trends of Smart Farming
    8. References
7. 2 Overview of Robotics in Agriculture: Types and Applications
    1. 2.1 Introduction
    2. 2.2 Background
    3. 2.3 Types of Robotics in Agriculture
    4. 2.4 Applications of Robotics in Agriculture
    5. 2.5 Advantages of Robotics in Agriculture
    6. 2.6 Limitations of Robotics in Agriculture

7. 2.7 Future of Robotics in Agriculture
8. 2.8 Case Studies and Examples
9. 2.9 Implications and Recommendations
10. 2.10 Conclusion
11. References

8. 3 Digital Farming: The New Era of Agriculture-Opportunities and Challenges
    1. 3.1 Introduction
    2. 3.2 Tools and Methods Used in Digital Farming
    3. 3.3 Pandemic Effects on Traditional Farming
    4. 3.4 New Scope at Digital Farming
    5. 3.5 Challenges or Difficulties for Digital Farming
    6. 3.6 Future Scope and Benefits
    7. References
9. 4 Challenges and Barriers to Smart Farming Adaptation: A Technical, Economic, and Social Perspective
    1. 4.1 Introduction
    2. 4.2 Technical Challenges in Smart Farming Adaptation
    3. 4.3 Economic Barriers to Smart Farming Implementation
    4. 4.4 Social Obstacles to Smart Farming Adoption
    5. 4.5 Environmental Considerations in Smart Farming
    6. 4.6 Future Prospects for Smart Farming
    7. 4.7 Conclusion and Recommendations
    8. References

10. 5 Sustainable Development in Agriculture: Soil Management
    1. 5.1 Introduction
    2. 5.2 Reviewing the Need for Global Food Production to be Upgraded
    3. 5.3 Soil Quality and Its Impact
    4. 5.4 Emerging Technologies
    5. 5.5 Introduction and Theory of IoT
    6. 5.6 Several Sensors and How They are Used in Agriculture
    7. 5.7 Centralized Agriculture System
    8. 5.8 Conclusion
    9. References
11. 6 Concepts of Robotics, AI, and Internet of Things (IoT) in Agriculture
    1. 6.1 Introduction
    2. 6.2 General Challenges Faced in the Agriculture Industry
    3. 6.3 Role of Robotics, AI, and IoT in Agriculture
    4. 6.4 Benefits of Robotic and AI in Improving Agriculture
    5. 6.5 Collecting Data and Performing Analyses With the Help of the Internet of Things
    6. 6.6 Relationships of IoT and AI Strengthen the Agriculture Sector
    7. 6.7 Benefits of IoT in Agriculture
    8. 6.8 The Role of AI in Tomorrow's Farming
    9. 6.9 Robots in Agriculture—Perceptions and Pros, Cons

10. 6.10 Complications Associated With Robots Used in Agriculture
11. 6.11 Conclusion
12. References

12. 7 Data Analytics in Agriculture: Predictive Models and Real-Time Decision-Making
    1. 7.1 Introduction
    2. 7.2 Data Collection and Management in Agriculture
    3. 7.3 Challenges in Collecting and Managing Agricultural Data
    4. 7.4 Strategies for Effective Data Collection and Management
    5. 7.5 Predictive Models in Agriculture
    6. 7.6 Applications of Predictive Models in Agriculture
    7. 7.7 Real-Time Decision-Making in Agriculture
    8. 7.8 Integration of Predictive Models and Real-Time Decision Making in Agriculture
    9. 7.9 Future of Data Analytics in Agriculture
    10. 7.10 Concluding Remarks
    11. 7.11 Future Directions for Research and Practice
    12. References
13. 8 Examining the Role of IoT and AI in Revolutionizing Agriculture: A Smart Farming Approach
    1. 8.1 Introduction

2. 8.2 IoT in Agriculture
3. 8.3 Major Advantages of Using IoT in Agriculture
4. 8.4 Major Disadvantages of Using IoT in Agriculture
5. 8.5 AI in Agriculture
6. 8.6 Benefits of Drones in Agriculture
7. 8.7 Main Part of AI in Agriculture
8. 8.8 Effects of AI in Agriculture
9. 8.9 Combination of AI and IoT
10. 8.10 Automatic Irrigation
11. 8.11 Crop Health Monitoring
12. 8.12 Supply Yields to the Demanded Areas
13. 8.13 Conclusion
14. References

14. 9 Smart AgriTech: Sensors & Networks
    1. 9.1 Introduction
    2. 9.2 IoT in Agriculture
    3. 9.3 Utilization of Sensor-Based IoT Devices in the Agricultural Sector
    4. 9.4 Networks for IoT in Agriculture
    5. 9.5 Communication Protocols Used in Agriculture
    6. 9.6 Case Studies and Examples of IoT Applications in Agriculture
    7. 9.7 The Future of IoT in Agriculture
    8. 9.8 Conclusion

9. References
15. 10 Internet of Things in Agriculture: Sensor, Network and Communication Protocol
    1. 10.1 Introduction
    2. 10.2 IoT in Agriculture
    3. 10.3 Physical Design of IoT Architecture
    4. 10.4 IoT Network Levels and Stages
    5. 10.5 Characteristics of IoT
    6. 10.6 IoT Enabling Technology
    7. 10.7 IoT Challenges
    8. 10.8 IoT Communication Protocol
    9. 10.9 Sensors for IoT-Based Smart Agriculture
    10. 10.10 Sensor Interfacing and Control
    11. 10.11 Conclusion
    12. References
16. 11 An Irrigation System: Design, Implementation, and Benefits
    1. 11.1 Introduction
    2. 11.2 Smart Irrigation Composition
    3. 11.3 Advantages of Intelligent Irrigation Systems
    4. 11.4 Irrigation Methods
    5. 11.5 Review on Smart Irrigation Systems
    6. 11.6 Smart-Based Irrigation Techniques for Urban and Rural Areas

7. 11.7 Barriers of Smart Irrigation Systems
8. 11.8 Conclusion
9. References

17. 12 Implementation of Smart Irrigation Systems
    1. 12.1 Introduction
    2. 12.2 Literature Survey
    3. 12.3 Irrigation System Design
    4. 12.4 Irrigation System Implementation Benefits
    5. 12.5 Irrigation System Benefits
    6. 12.6 Results and Discussion
    7. 12.7 Conclusion
    8. References
18. 13 Smart Plant-Based Wastewater Treatment for Agricultural Irrigation Systems
    1. 13.1 Introduction
    2. 13.2 Water Quality Parameters for Agricultural Irrigation
    3. 13.3 Wetland Method for Wastewater Treatment
    4. 13.4 Importance of Artificial Intelligence in Wastewater Treatment
    5. 13.5 Economic Feasibility of Smart Plant-Based Treatment for Agricultural Irrigation
    6. 13.6 Challenges Associated with Artificial Intelligence Application in Wastewater Treatment
    7. 13.7 Conclusion

8. References
19. 14 Drones in Agriculture: Mapping, Monitoring, and Decision-Making
    1. 14.1 Introduction
    2. 14.2 Opportunities
    3. 14.3 Drone Mapping for Transforming Agriculture
    4. 14.4 Significance of Drone Monitoring in Precision Agriculture
    5. 14.5 Leveraging Drone for Real-Time Decision-Making in Agriculture
    6. 14.6 Common Usages of Agricultural Drones
    7. 14.7 Challenges
    8. 14.8 Summary and Scope
    9. 14.9 Conclusion
    10. References
20. 15 Investigation of Banana Plant Disease Detection Using Transfer Learning
    1. 15.1 Introduction
    2. 15.2 Literature Survey
    3. 15.3 Transfer Learning
    4. 15.4 Methodology
    5. 15.5 Result and Discussion
    6. 15.6 Conclusion
    7. References

21. 16 Exploring Dataset for Apple Leaf Disease Detection: A Focus on Horticulture
    1. 16.1 Introduction
    2. 16.2 Literature Review
    3. 16.3 Apple Leaf Disease Detection Mechanism
    4. 16.4 Significance of Dataset
    5. 16.5 Discussion
    6. 16.6 Conclusion
    7. References
22. 17 Optimizing Crop Yield Prediction Using Machine Learning Algorithms
    1. 17.1 Introduction
    2. 17.2 Related Work
    3. 17.3 Methodology
    4. 17.4 Machine Learning Algorithms for Crop Yield Prediction
    5. 17.5 Experimental Results and Analysis
    6. 17.6 Optimization Techniques
    7. 17.7 Challenges and Future Directions
    8. 17.8 Conclusion
    9. References
23. 18 Analyzing Smart Farming Technologies: A Study on Indian Farmers' Adoption Trends
    1. 18.1 Introduction

2. 18.2 Research Methodology
3. 18.3 Objectives
4. 18.4 Literature Review
5. 18.5 Data Analysis
6. 18.6 Key Findings
7. 18.7 Discussions and Recommendations
8. 18.8 Conclusion
9. References

24. 19 Leveraging Linear Regression Model to Address Food Insecurity in the United States: A Smart Agritech Approach
    1. 19.1 Introduction
    2. 19.2 Literature Review
    3. 19.3 Data Sources
    4. 19.4 Preprocessing
    5. 19.5 Data Dictionary
    6. 19.6 Exploratory Data Analysis
    7. 19.7 Feature Engineering
    8. 19.8 Predictions
    9. 19.9 Conclusion
    10. References
25. About the Editors
26. Index
27. Also of Interest
28. End User License Agreement

List of Tables

1. Chapter 3
    1. Table 3.1 Remote sensor data studies for crop yield analysis [10].
    2. Table 3.2 Assessment of vegetable prices during the pandemic between...
    3. Table 3.3 Precision crop growing sensor models [1].
    4. Table 3.4 Smart farming monitoring system services [1].
2. Chapter 7
    1. Table 7.1 Challenges for collecting data and strategies.
3. Chapter 11
    1. Table 11.1 Barriers of smart irrigation systems.
4. Chapter 12
    1. Table 12.1 Comparison of IoT transmission technologies.
    2. Table 12.2 Comparison of irrigation technologies.
    3. Table 12.3 Descriptive statistics analysis results.
5. Chapter 13
    1. Table 13.1 Water quality guidelines for agricultural crop irrigation [5].
6. Chapter 15
    1. Table 15.1 Specification of hyper parameter.
    2. Table 15.2 Collection of captured banana leaf images data.
    3. Table 15.3 Performance comparison analysis for the pre-trained network models.

4. Table 15.4 Computational time for various models.
7. Chapter 16
    1. Table 16.1 Exploration of existing apple leaf dataset.
8. Chapter 17
    1. Table 17.1 Summary of analysis.
9. Chapter 18
    1. Table 18.1 Analysis of variance between the variables.
    2. Table 18.2 Tests of between-subjects effects.
10. Chapter 19
    1. Table 19.1 List of datasets and their sources.
    2. Table 19.2 Data dictionary used in the analysis.
    3. Table 19.3 Average FI rate in different communities.
    4. Table 19.4 VIF factor of selected features.
    5. Table 19.5 Performance table for all the models.
    6. Table 19.6 Highest coefficients alongside their corresponding features.
    7. Table 19.7 Performance data.
    8. Table 19.8 Comparison table for FI rate and predicted FI rate.

List of Illustrations

1. Chapter 1
    1. Figure 1.1 Agricultural decision support system framework.

2. Figure 1.2 Obstacles that impede the adoption of smart agriculture technology.
3. Figure 1.3 A depiction of IoT applications for smart farming.
4. Figure 1.4 Implementing innovative technologies at various stages.
5. Figure 1.5 A smart farming network based on the Internet of Things (IoT).

2. Chapter 2
    1. Figure 2.1 Ancient farming [6].
    2. Figure 2.2 Tractor [8].
    3. Figure 2.3 Reaper [9].
    4. Figure 2.4 Drones [9].
    5. Figure 2.5 Autonomous tractors [11].
    6. Figure 2.6 Drones [11].
    7. Figure 2.7 Robot arms [14].
    8. Figure 2.8 Harvesting robots [15].
    9. Figure 2.9 Milking robots [16].
    10. Figure 2.10 Weeding robots [18].
    11. Figure 2.11 Vision robotics [19].
    12. Figure 2.12 Blue river technology [20].
    13. Figure 2.13 Harvest automation [21].
    14. Figure 2.14 Strawberry harvesting robot [22].
    15. Figure 2.15 Farm Wise robots [23].

16. Figure 2.16 Naïo weeding and harvesting robot [24].
3. Chapter 3
    1. Figure 3.1 (a) Manual process of sowing seeds. (b) Automated...
    2. Figure 3.2 (a) Soybean yield field [2]. (b) Stages of production...
    3. Figure 3.3 Pandemic’s effects on the vegetable supply chain and agricultural...
    4. Figure 3.4 Architecture of precision farming [1].
4. Chapter 4
    1. Figure 4.1 Different economic barriers in the implementation of smart farming.
    2. Figure 4.2 Percentage cost division in the implementation of smart farming.
5. Chapter 5
    1. Figure 5.1 Role of AI in modern agriculture and farming [30].
    2. Figure 5.2 Layered architecture/ecosystem of agro-IoT [32].
    3. Figure 5.3 Different sensors.
6. Chapter 7
    1. Figure 7.1 Applications of predictive models in agriculture.
    2. Figure 7.2 Benefits of integrating predictive models and real-time...
    3. Figure 7.3 Challenges of integration.

7. Chapter 8
    1. Figure 8.1 AI and IOT in agriculture [21].
    2. Figure 8.2 Arduino board parts for sensor connections [2].
    3. Figure 8.3 Column graph for decreased water usage in agriculture...
    4. Figure 8.4 Bar graph for increase in agriculture using IOT sensors.
    5. Figure 8.5 IOT sensors used for smart farming.
    6. Figure 8.6 Crop diagnosis cycle using AI.
    7. Figure 8.7 Task done by drone.
    8. Figure 8.8 Pie chart of AI in agriculture.
    9. Figure 8.9 Process of AI with IOT in an agricultural field.
    10. Figure 8.10 Connection between a physical farm and a virtual farm.
    11. Figure 8.11 IOT in agriculture market size, 2021 to 2030 (USD billion).
8. Chapter 9
    1. Figure 9.1 IoT-based climate condition monitoring for agri-farm...
    2. Figure 9.2 IoT-based soil pattern monitoring system...
    3. Figure 9.3 IoT-enabled devices for irrigation monitoring (source: Bing).
    4. Figure 9.4 IoT-enabled drones for smart agriculture (source: Kim *et al.* [60]).

5. Figure 9.5 IoT in greenhouse monitoring (source: Farooq *et al.* [14]).
6. Figure 9.6 IoT-enabled devices for livestock monitoring...
7. Figure 9.7 Application of IoT in different fields of agriculture...
8. Figure 9.8 Mobile apps used for smart agriculture (source: Farooq *et al.* [14]).
9. Figure 9.9 Diagram of an MQTT system in agriculture.

9. Chapter 10
    1. Figure 10.1 Application of Internet of Things in the modern world.
    2. Figure 10.2 Physical design architecture of Internet of Things.
    3. Figure 10.3 Network-level and stages of IoT-based system.
    4. Figure 10.4 Embedded system computing devices.
    5. Figure 10.5 IoT communication protocol architecture. (a) MQTT, (b) CoAP...
    6. Figure 10.6 Various types of sensors for IoT agriculture application.
    7. Figure 10.7 Pin description of Arduino UNO Revision 3.
    8. Figure 10.8 Interfacing of soil moisture sensor using Arduino UNO board.
    9. Figure 10.9 Interfacing of PIR sensor for motion detection using...

10. Figure 10.10 Interfacing of light dependent resistor as automatic light control...
11. Figure 10.11 Interfacing of ultrasonic sensor for object detection using ...
12. Figure 10.12 Interfacing of gas sensor using Arduino Uno board.
13. Figure 10.13 Interfacing of TMP36 sensor for temperature measurement...

10. Chapter 11
    1. Figure 11.1 Smart irrigation modules.
    2. Figure 11.2 Types of sensors and controllers.
    3. Figure 11.3 Soil moisture sensor ...
    4. Figure 11.4 (a) Evaporation sensors...
    5. Figure 11.5 Rain sensors ...
    6. Figure 11.6 Wind speed sensors...
    7. Figure 11.7 Climate-based controller...
    8. Figure 11.8 Benefits of smart irrigation systems.
    9. Figure 11.9 Irrigation methods.
    10. Figure 11.10 Drip irrigation smart system...
    11. Figure 11.11 Sprinkler irrigation system [9].
    12. Figure 11.12 Tank irrigation...
    13. Figure 11.13 Tube well irrigation...
    14. Figure 11.14 Canal irrigation...

15. Figure 11.15 Block diagram of a solar panel pumping module.

11. Chapter 12
    1. Figure 12.1 Irrigation system layers and components.
    2. Figure 12.2 Flow chart of implementing the step of the auto-watering model.
    3. Figure 12.3 Block diagram of the proposed irrigation system.
    4. Figure 12.4 Soil moisture level in automatic and manual irrigation methods.
12. Chapter 14
    1. Figure 14.1 NDVI range.
    2. Figure 14.2 Plant health through light sensors.
    3. Figure 14.3 Rice field drone mapping for NDVI determination.
    4. Figure 14.4 Crop health monitoring.
    5. Figure 14.5 Real-time livestock monitoring by drone.
    6. Figure 14.6 Thermal cameras arm drone for cattle scouting.
    7. Figure 14.7 Field surveying.
    8. Figure 14.8 Seed planting.
    9. Figure 14.9 Crop spraying.
    10. Figure 14.10 Land management.
    11. Figure 14.11 Biomass estimation.

13. Chapter 15
    1. Figure 15.1 Sample input images from real-time data set for four...
    2. Figure 15.2 Deep-learning-based overall framework for plant disease detection.
    3. Figure 15.3 Data augmentation of banana leaf image.
    4. Figure 15.4 Recognition accuracy and loss for VGG19.
    5. Figure 15.5 Recognition accuracy and loss for Inception Net.
    6. Figure 15.6 Recognition accuracy and loss for Mobile Net.
    7. Figure 15.7 Recognition accuracy and loss for ResNet152.
    8. Figure 15.8 Confusion matrix of models: (a) ResNet152 model...
14. Chapter 16
    1. Figure 16.1 Various types of apple leaf disease.
    2. Figure 16.2 Thresholding segmentation.
    3. Figure 16.3 Edge-based segmentation.
    4. Figure 16.4 Process of support vector machine.
    5. Figure 16.5 Process of random forest machine learning.
    6. Figure 16.6 Working of a convolutional neural network.
    7. Figure 16.7 Comparison between different datasets with size and author.
15. Chapter 17
    1. Figure 17.1 Model evaluation.

2. Figure 17.2 Relation between fertilizer usage and crop yield.
3. Figure 17.3 Crop yield distribution.
4. Figure 17.4 Comparison between different models.
5. Figure 17.5 Hyperparameter tuning.

16. Chapter 18
    1. Figure 18.1 Means plots for (a) external funding ...

17. Chapter 19
    1. Figure 19.1 Business dataset before cleaning.
    2. Figure 19.2 Business dataset after cleaning.
    3. Figure 19.3 Descriptive statistics of the distribution of each feature.
    4. Figure 19.4 Boxplot graph for features with significant outliers.
    5. Figure 19.5 Heat map for inspecting multicollinearity.
    6. Figure 19.6 Line plot graph of FI rate, unemployment, and houseless rates...
    7. Figure 19.7 Line plot graph of grocery, restaurants, and wholesalers, by year.
    8. Figure 19.8 Bar chart of rent and meal cost, by year.
    9. Figure 19.9 Line chart of rent and meal cost, by year.
    10. Figure 19.10 Line chart of FI rate in different communities, by year.

11. Figure 19.11 Line chart of unemployment rate in different communities, by year.
12. Figure 19.12 Heat map for food insecurity rates in five counties.
13. Figure 19.13 Choropleth map for US unsheltered rate in 2010.
14. Figure 19.14 Choropleth map for US unsheltered rates in 2019.
15. Figure 19.15 Choropleth map for unemployment rate by county in 2019.
16. Figure 19.16 Choropleth map for unemployment rate by county in 2020.
17. Figure 19.17 Choropleth map for percent people of color by county...
18. Figure 19.18 Choropleth map for food insecurity rate for the year 2009.
19. Figure 19.19 Choropleth map for food insecurity rate for the year 2018.
20. Figure 19.20 Histogram for food insecurity distribution, by restaurant count.
21. Figure 19.21 Food insecurity distribution, by one-bedroom rent prices.
22. Figure 19.22 Histogram for FI distribution, by unemployment rate.

23. Figure 19.23 Histogram for FI distribution, by houselessness rate.
24. Figure 19.24 Histogram for FI distribution, by race.
25. Figure 19.25 Scatter plot for FI rate vs. unemployment rate.
26. Figure 19.26 Scatter plot for FI rate vs. rent.
27. Figure 19.27 Scatter plot for FI rate vs. black population.
28. Figure 19.28 Scatter plot for FI rate vs. log rent.
29. Figure 19.29 Scatter plot for FI rate vs. white population.
30. Figure 19.30 Boxplot for the features with the largest outliers.
31. Figure 19.31 Boxplot of features after the reduction.
32. Figure 19.32 Model predicting food insecurity rates of the year 2018.
33. Figure 19.33 Model predicting food insecurity rates of the year 2020.
34. Figure 19.34 Feeding America's projected FI rate for the year 2020.

Smart Agritech

Robotics, AI, and Internet of Things (IoT) in Agriculture

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Preface

Agriculture has always been a vital sector of the global economy, providing food and raw materials for industries and households. However, with the growing population, changing climate conditions, and limited resources, the agriculture sector is facing numerous challenges. To address these challenges, farmers and agricultural companies are turning to advanced technologies such as robotics, artificial intelligence (AI), and the Internet of things (IoT).

This book entitled "*Smart Agritech: Robotics, AI, and Internet of Things (IoT) in Agriculture*" aims to provide a comprehensive overview of the latest technological advances in agriculture, focusing on the use of these three cutting-edge technologies. The book will explore the potential benefits of these technologies in improving agricultural efficiency, productivity, and sustainability.

Furthermore, the book will be written in a clear and concise manner that is accessible to a wide range of readers, from farmers and agribusiness professionals to researchers and students. It will be an ideal resource for those who wish to understand the potential of smart agriculture techniques and how they can be implemented to improve farm productivity,

sustainability, and profitability. Finally, this book will provide a unique perspective by including case studies from developing countries, demonstrating how smart agriculture techniques can be adapted and applied in a variety of contexts. This sets it apart from many other books that focus primarily on developed countries and their agricultural systems.

Chapter 1

This chapter discusses how advanced technologies such as sensors, drones, and IoT are utilized in Smart farming to boost agricultural efficiency, productivity, and sustainability while addressing global food demand and environmental issues. It facilitates data collection and analysis to make informed decisions, optimizing crop growth and resource management by monitoring factors like soil moisture, temperature, and pests. Precision agriculture, employing data-driven farming techniques, and the use of autonomous robots and drones for tasks are highlighted. The IoT plays a vital role in monitoring variables, aiding waste reduction and productivity enhancement. Smart farming's benefits include increased yield, cost reduction, and improved sustainability, making it a crucial solution for future food demands amidst population growth.

Chapter 2

This chapter explores the diverse range of robotic technologies revolutionizing agricultural operations. From autonomous drones to robotic harvesters, it examines the various types and their applications across different stages of farming. By providing insights into the integration of robotics, this chapter aims to showcase the potential benefits for farmers, including increased efficiency, reduced labor costs, and sustainable practices. Ultimately, this overview serves as a gateway to understanding how robotics is reshaping the landscape of agriculture worldwide.

Chapter 3

This chapter explores how digital innovations are reshaping traditional farming practices, presenting both opportunities and challenges. By embracing digital tools such as IoT, AI, and data analytics, farmers can enhance productivity, efficiency, and sustainability. However, this digital revolution also poses challenges such as data security, access, and the digital divide in rural areas. Through insightful analysis and case studies, this chapter navigates the complexities of digital farming, offering a comprehensive understanding of its potential and hurdles.

Chapter 4

This chapter explores the hurdles faced by farmers as they navigate technical complexities, economic constraints, and social factors in adopting smart farming practices. Through an in-depth examination, we aim to shed light on the intricate challenges inhibiting the widespread adoption of these transformative technologies in agriculture. By analyzing the interplay between technology, economics, and society, this chapter offers valuable insights into overcoming barriers and fostering the successful integration of smart farming solutions. Our exploration underscores the importance of understanding and addressing the diverse challenges encountered on the path toward sustainable and efficient agricultural practices.

Chapter 5

This chapter explores innovative strategies and technologies aimed at preserving soil health while maximizing productivity. By addressing soil degradation and nutrient depletion, we aim to foster resilient farming practices in the face of environmental challenges. Through a comprehensive examination of soil management techniques, we seek to empower farmers with the knowledge and tools needed to cultivate healthy soils and secure food production for future

generations. Join us on a journey to unlock the potential of sustainable soil management in shaping the future of agriculture.

Chapter 6

This chapter explores the transformative potential of robotics, AI, and IoT in agriculture. We delve into how these cutting-edge technologies are revolutionizing traditional farming practices. From autonomous robots to AI-driven decision-making, uncover the ways in which agriculture is evolving. The integration of IoT devices offers real-time monitoring and data analysis for optimized crop management. Join us on a journey to discover the conceivable advancements reshaping the landscape of modern agriculture.

Chapter 7

This chapter delves into the transformative potential of data analytics within agriculture, focusing on predictive models and real-time decision-making. It explores how harnessing data can revolutionize farming practices, enhancing efficiency and sustainability. By employing advanced analytics techniques, farmers can make informed decisions based on real-time insights, optimizing crop yield and resource allocation. This

chapter discusses the integration of data-driven approaches into traditional farming methods, highlighting the benefits of predictive modeling for anticipating crop performance and mitigating risks. Through case studies and examples, it illustrates the practical applications of data analytics in modern agriculture, paving the way for a more resilient and productive food system.

Chapter 8

This chapter explores the transformative impact of IoT and AI on modern agriculture, focusing on smart farming approaches. By examining the integration of advanced technologies like sensors, drones, and AI, it explores how they enhance efficiency, productivity, and sustainability in farming practices. Through data-driven insights and informed decision-making, smart farming optimizes crop growth and resource utilization while addressing global food demand and environmental concerns. This exploration underscores the pivotal role of precision agriculture and autonomous systems in revolutionizing traditional farming methods. Ultimately, this chapter offers a comprehensive analysis of how IoT and AI are reshaping the agricultural landscape, paving the way for a more sustainable and productive future.

Chapter 9

This chapter provides a comprehensive examination of how IoT is transforming agriculture, revolutionizing farming practices through the interconnection of physical devices *via* the internet. It delves into the applications of IoT in agriculture, showcasing its advantages such as heightened efficiency, reduced waste, and increased crop yields, while also addressing challenges like infrastructure costs and rural connectivity limitations. Emphasizing the pivotal role of sensor-based IoT, the chapter illustrates how sensors collect crucial data on crops, soil, weather, and livestock, thereby optimizing agricultural techniques. It also highlights the significance of selecting appropriate network and communication protocols tailored to diverse farm locations. Through real-world case studies spanning countries like India, Thailand, China, Malaysia, the Philippines, and the United States, the chapter vividly demonstrates IoT's transformative influence on farming practices worldwide.

Chapter 10

This chapter delves into the transformative role of the Internet of things (IoT) in modern agriculture, focusing on sensors, networks, and communication protocols. It explores how IoT

technologies revolutionize farming practices, offering insights into enhanced productivity, resource optimization, and sustainability. Through an examination of sensor deployment, network infrastructure, and communication protocols, we uncover the intricate web of interconnected systems driving agricultural innovation. This chapter provides a comprehensive overview of the IoT's integration into agriculture, offering practical insights for farmers, researchers, and industry professionals. By understanding the dynamics of IoT in agriculture, readers gain valuable perspectives on the future of farming and its pivotal role in addressing global food security challenges.

Chapter 11

This chapter offers comprehensive insights into the design and implementation of irrigation systems, highlighting their significance in agricultural settings. Readers will explore various irrigation techniques, their practical applications, and the benefits they bring to crop cultivation. From traditional methods to cutting-edge technologies, this chapter provides a holistic view of irrigation systems' evolution and their role in enhancing agricultural productivity. Whether you are a seasoned farmer or a curious enthusiast, this chapter aims to

broaden your understanding of irrigation systems and their invaluable contributions to sustainable agriculture.

Chapter 12

This chapter introduces a real-time monitoring and auto-watering system employing IoT technology and predictive mathematical models to optimize crop growth by regulating water usage efficiently. The system, which monitors soil moisture levels and automatically irrigates crops, conserves water and promotes interoperability among various sensing data streams for comprehensive agricultural analytics. With water consumption reduced by over 75% and enhanced irrigation efficiency, the system proves invaluable, especially in remote and desert regions facing water and labor shortages. It not only addresses water scarcity but also fosters sustainable farming practices and economic growth in the agricultural sector through improved production and lowered maintenance costs. This proposed system stands as a sustainable and effective solution, ensuring a brighter future for smart agriculture.

Chapter 13

This chapter explains the innovative approach of utilizing plant-based wastewater treatment systems for agricultural irrigation. By focusing on smart technologies and sustainable practices, this explores how these systems offer a viable solution to water scarcity and pollution in agricultural settings. Through a combination of scientific research and practical applications, we aim to provide insights into the potential benefits and challenges of implementing such systems. From enhancing water quality to promoting eco-friendly agricultural practices, this chapter offers a comprehensive exploration of smart plant-based wastewater treatment methods. Ultimately, the goal is to contribute to the advancement of sustainable agricultural practices through the integration of innovative wastewater treatment solutions.

Chapter 14

This paper introduces the transformative role of drones in modern farming practices. The chapter delves into how drones revolutionize agriculture through precise mapping, real-time monitoring, and data-driven decision-making. It highlights the increasing significance of drone technology in enhancing crop management, resource utilization, and overall farm efficiency.

Through this exploration, readers gain insights into the evolving landscape of agricultural innovation and the potential of drones to address pressing challenges in food production and sustainability. This preface sets the stage for an in-depth examination of the multifaceted applications and benefits of drones in shaping the future of agriculture.

Chapter 15

This chapter explains an innovative approach: leveraging transfer learning to identify banana plant diseases. By harnessing the power of deep learning models, we aim to revolutionize disease detection in one of the world's most vital crops. Through meticulous investigation and experimentation, this chapter elucidates the potential of transfer learning in bolstering agricultural resilience. This chapter is dedicated to the investigation of banana plant disease detection through the lens of transfer learning.

Chapter 16

This chapter explains the meticulous examination of a dataset tailored for the identification of apple leaf diseases, illuminating its potential horticultural significance. Through the scrutiny of images depicting various afflictions like apple scab,

cedar apple rust, and powdery mildew, this research endeavors to unlock insights vital for detecting and managing diseased foliage. Employing an array of computer vision techniques and algorithms, the dataset undergoes thorough evaluation to unveil pertinent characteristics and patterns crucial for leaf damage detection. The anticipated outcome of this endeavor is the development of an automated system poised to offer farmers and agronomists a reliable tool for real-time disease surveillance and management in orchards. Ultimately, the fruits of this research endeavor aspire to bolster the productivity and sustainability of the apple industry, fostering benefits for producers and consumers alike.

Chapter 17

This chapter delves into the realm of enhancing the accuracy and efficiency of machine learning algorithms for predicting crop yields. Through meticulous research, we explore the effectiveness of various methods including regression models, ensemble techniques, deep learning models, and support vector machines. Employing rigorous metrics such as MAE, RMSE, R2, precision, and recall, we meticulously evaluate these models against a meticulously curated dataset of agricultural parameters. Leveraging the “crop yield prediction dataset” alongside techniques like feature engineering and selection, we

refine our models' predictive capabilities. Our findings underscore the superiority of ensemble methods, deep learning models, and other advanced algorithms over traditional approaches, shedding light on critical factors influencing crop yield. The study emphasizes the significance of comprehensible models, laying the groundwork for future advancements in crop yield prediction and ultimately empowering farmers, agricultural planners, and policymakers to make informed decisions and foster sustainable agricultural practices.

Chapter 18

This chapter delves into the intricate world of smart farming technologies, examining their adoption trends among Indian farmers. As technology continues to reshape agricultural landscapes, understanding the dynamics of adoption becomes imperative. Through meticulous analysis and research, we aim to shed light on the factors influencing Indian farmers' embrace of these innovations. From IoT sensors to AI-driven solutions, each technology holds the promise of revolutionizing traditional farming practices. Our exploration aims to provide valuable insights into the challenges and opportunities surrounding the integration of smart farming technologies in India.

Chapter 19

In the realm of tackling societal challenges, few issues are as pressing as food insecurity. This chapter delves into a pioneering application of the linear regression model, merging it with cutting-edge agritech solutions to confront this issue head on. By leveraging data-driven insights and technological innovations, we embark on a journey to transform the landscape of food access in the United States. This interdisciplinary approach bridges the gap between traditional agriculture and modern data science, offering a promising avenue for sustainable change. Join us as we explore the potential of smart agritech in addressing one of humanity's most fundamental needs.

1
Introduction to Smart Farming: Definition, Importance and Trends

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Abstract

Smart farming is a cutting-edge agricultural approach that integrates advanced technologies to enhance efficiency, productivity, and sustainability. This method addresses the rising global demand for food due to population growth while aiming to conserve natural resources and mitigate climate change impacts. Utilizing technologies such as sensors, drones, robots, precision agriculture, and the Internet of Things (IoT),

smart farming collects and analyses data to make informed decisions and optimize agricultural processes. By monitoring factors like soil moisture, temperature, nutrients, and pests, smart farming empowers farmers to manage crop growth more effectively. This technology-driven approach reduces input costs, boosts yields, and minimizes environmental impacts, presenting a sustainable alternative to traditional farming practices. With the global population projected to reach 9.7 billion by 2050, a 70% increase in food production will be necessary. Traditional methods, which often lead to soil degradation, water pollution, and biodiversity loss, are insufficient to meet this demand. Smart farming offers a viable solution by enabling higher food production with fewer resources. Precision agriculture is a significant aspect of smart farming, involving the collection of data on soil, crops, and weather through sensors, drones, and satellite imagery. This data is used to create detailed field maps, allowing for tailored farming techniques for specific zones. Another critical component is the use of autonomous robots and drones, which can automate labour-intensive tasks such as planting, harvesting, and monitoring crops. These technologies operate around the clock, providing real-time data and enabling rapid responses to changing conditions. The IoT plays a crucial role in smart farming by using sensors and devices to gather data,

which is then analyzed centrally. This application includes monitoring soil moisture and temperature, managing irrigation systems, and optimizing equipment use. IoT integration helps farmers refine their practices and reduce waste, contributing to greater sustainability.

***Keywords*:** Smart farming, sustainability, climate change, sensors, drones, Internet of Things (IoT), data analysis, crop management

1.1 Introduction

Smart farming is an innovative method of agriculture that utilizes the latest technologies to enhance the effectiveness and eco-friendliness of farming techniques. Traditional farming has always been a labor-intensive and time-consuming process, with farmers relying on experience and intuition to make decisions about planting, irrigation, fertilization, and pest control. However, the advent of new technologies has enabled farmers to gather and analyze vast amounts of data to make more informed decisions about their farming practices.

Smart agriculture encompasses the utilization of cutting-edge technologies such as the Internet of Things (IoT), machine learning, artificial intelligence (AI), and big data analytics to

supervise and control agricultural activities. The term “Internet of Things” (IoT) denotes the networking of devices, sensors, and machines, which collect and exchange information through the internet. In agriculture, the IoT can be utilized to observe environmental variables like soil moisture, temperature, humidity, and other factors, offering farmers real-time insights into crop conditions. Utilizing big data analytics allows for the examination of vast volumes of data amassed through the IoT, empowering farmers to discern trends and patterns in areas such as crop development, weather fluctuations, and market dynamics. By implementing AI and machine learning algorithms, it is possible to predict crop yields, detect pests and diseases, and improve irrigation and fertilization techniques.

Smart farming has the potential to transform agriculture, offering numerous benefits to farmers, consumers, and the environment. Smart farming can enhance agricultural productivity and profitability by optimizing farming techniques, resulting in increased yields and reduced costs, while simultaneously reducing waste and enhancing crop quality. For consumers, this can mean a more stable and affordable food supply, as well as healthier and more sustainable food choices. Smart farming can have positive environmental impacts, including decreased usage of pesticides

and fertilizers, water conservation, and mitigation of greenhouse gas emissions.

One of the key aspects of smart farming is precision agriculture, which involves using data-driven insights to optimize farming practices. Precision agriculture enables farmers to make informed decisions about when, where, and how to plant crops, as well as how much water and fertilizer to use. By adopting this method, it is possible to decrease waste and enhance yields, all while minimizing the environmental impact.

Vertical farming, an essential component of smart farming, entails cultivating crops in vertically stacked layers under controlled environments using artificial lighting. Vertical farming is a technique that can optimize the utilization of scarce space and resources, minimize water consumption, and reduce reliance on pesticides and herbicides. This approach can be particularly beneficial in urban areas, where space is at a premium and there is a growing demand for locally grown produce.

Smart farming is not without its challenges, however. Small- and medium-sized farmers may encounter a significant obstacle in adopting new technologies due to the high costs involved in their implementation. Furthermore, an imperative

exists for increased education and training to facilitate farmers in comprehending the proficient utilization of these novel technologies. Furthermore, there are apprehensions about the security and confidentiality of data, along with the possibility of technology exacerbating current economic and social disparities.

Despite encountering numerous obstacles, smart farming has the capability to revolutionize agriculture by offering a more sustainable, lucrative, and effective approach to farming. As technology continues to advance and become more accessible, we can expect to see more farmers adopting smart farming practices in the years to come.

1.2 Smart Farming

In the past, agriculture was primarily focused on producing food on cultivated lands to ensure human survival and breeding animals. This era, referred to as traditional agriculture 1.0, heavily relied on manpower and animals. Basic farming tools, including sickles and shovels, were utilized to carry out farming activities. The majority of work was done through manual labor, resulting in low levels of productivity, as illustrated in Figure 1.1.

The advent of steam engines in the 19th century introduced novel machinery to the agricultural sector, which signified the commencement of the second agricultural era. During this era, farmers extensively employed agricultural equipment and chemicals, leading to significant enhancements in the efficiency and productivity of farms. Despite this, the overuse of chemicals had negative consequences, including chemical contamination, ecological deterioration, depletion of natural resources, and increased energy consumption.

The 20th century marked the advent of Agricultural era 3.0, driven by the swift progress in computation and electronics. This period marked the advent of programmed agricultural machinery, robotic techniques, and other innovative technologies that revolutionized agricultural processes, resulting in a significant improvement in efficiency. Policies were implemented to tackle the problems that arose during Agricultural era 2.0, emphasizing measures such as workload distribution, precise irrigation, decreased chemical usage, effective pest control technologies and location-specific nutrient application.

Agriculture 4.0, the prevailing phase of agriculture, is distinguished by the seamless amalgamation of state-of-the-art technologies, including the Internet of Things (IoT), big data

analytics, cloud computing and remote sensing artificial intelligence. Utilizing inexpensive network and sensor platforms, the adoption of these cutting-edge technologies has led to noteworthy improvements in agricultural techniques, which have optimized production efficiency and reduced water and energy consumption, all while minimizing environmental impact [1]. Within intelligent agriculture, big data furnishes instant insights into agricultural scenarios, empowering farmers to make well-informed choices. Simultaneously, real-time programming rooted in artificial intelligence is ingrained within IoT devices, aiding farmers in arriving at suitable decisions [2].

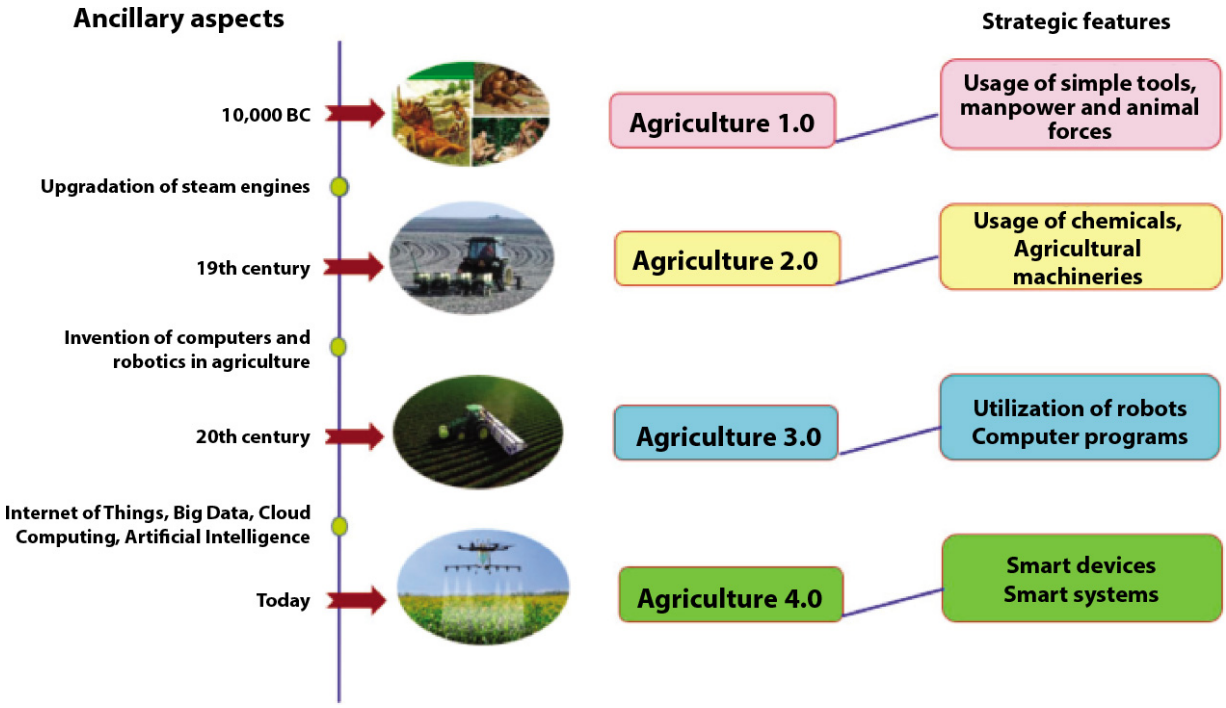

Figure 1.1 Agricultural decision support system framework.

Advanced technology is utilized in smart farming to facilitate precision agriculture, enabling farmers to monitor their crops remotely. The automation of sensors and machinery, through this method, has transformed agricultural processes, elevating efficiency in areas such as harvesting and crop yields, resulting in a more productive farming workforce [3]. The fusion of technology has instigated a technological revolution in agriculture, replacing conventional farming methods with automated devices. The arrival of the Internet of Things has revolutionized traditional techniques and substantially transformed contemporary farming practices [4].

Throughout the early phases of the crop's growth cycle, sensors diligently observe the crops to identify any unfavorable conditions with greater accuracy. Intelligent tools are employed throughout the entirety of the farming process, spanning from planting and cultivating crops to harvesting, storing, and transporting them. The efficient and profitable operation of the entire process is attributed to the precise use of various sensors that enable accurate monitoring capabilities. Moreover, online-available sensors that quickly gather data can offer location-specific and crop-specific insights for further evaluation in precision agriculture.

The agricultural industry has undergone a revolutionary transformation through smart agriculture technology, as it offers effective solutions to a myriad of crop production challenges. By employing sophisticated technologies like the Internet of Things (IoT), sensors, GPS, robotics, and data analytics, smart agriculture allows farmers to closely monitor their crops and make informed decisions. This approach optimizes spatial management practices, reduces excessive use of fertilizers and pesticides, and ultimately enhances crop production. An example of smart farming is the use of artificial neural network (ANN) models to manage irrigation water through a system called smart irrigation water management (SIWM). SIWM delivers up-to-the-minute data on irrigation

effectiveness of water supply and demand, water productivity metrics to facilitate irrigation scheduling and decision-making. Climate-smart agriculture (CSA) is gaining popularity in developing countries due to its potential to enhance food security, bolster farming system resilience, and reduce greenhouse gas emissions. With the integration of Internet of Things (IoT) technology, smart agriculture enables real-time monitoring and data collection for various agricultural activities, such as irrigation, plant protection, fertilization, and disease forecasting. This combination holds promise for sustainable and efficient agricultural practices in the face of climate challenges. The employment of this technology offers numerous advantages, such as precise assessments of crops and soil, remote monitoring capabilities for farmers, effective utilization of natural resources like water, improved agricultural and livestock production, and real-time data collection on crops. Smart agriculture is an advanced form of precision agriculture that employs sophisticated technologies to gather diverse data on farming practices and allow remote management. It also incorporates efficient real-time maintenance solutions into its operations.

1.3 Internet of Things

Smart farming has emerged as a result of the introduction of a new technology through the Internet of Things (IoT), which enables remote device connectivity [5]. This innovation has gradually impacted numerous industries, such as healthcare, trade, energy, communications, and agriculture, by enhancing their proficiency and performance transversely diverse sectors.

Information on the current impact and potential future applications of IoT technology in farming is available, but as technology continues to advance, its role is expected to become increasingly important. This entails employing communication infrastructure, sensors, and intelligent devices, acquiring data, utilizing cloud-based intelligent information, leveraging mobile devices, and automating agricultural processes. All of these elements work together to facilitate decision-making based on data-driven insights (as illustrated in Figure 1.2).

By utilizing IoT technology, farmers can remotely monitor plants and animals, as well as retrieve information from mobile devices. Through the application of sensors and instruments, farmers can assess weather conditions and predict production levels. The Internet of Things (IoT) assumes a vital function in water management, overseeing and controlling water flow,

gauging crop water needs, and refining water utilization [6]. By means of cloud connectivity and sensors, the condition and supply of water can be monitored from a distance, aligned with the demands of the soil and plants. While it is impossible for farmers to manually monitor every plant for nutrient deficiencies, pests, and diseases, IoT technology has allowed farmers to achieve a new level of precision in modern agriculture [7].

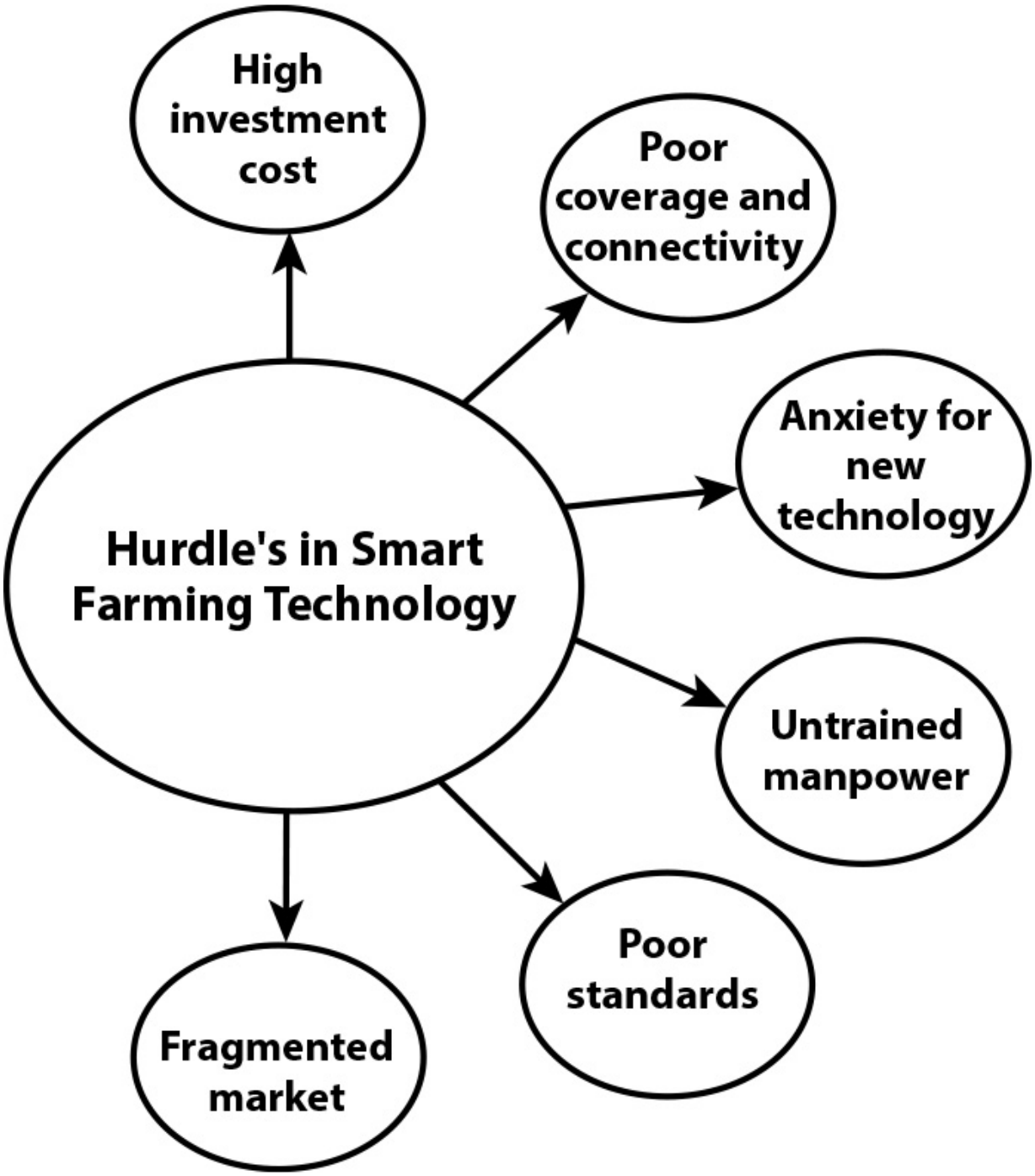

Figure 1.2 Obstacles that impede the adoption of smart agriculture technology.

The agricultural industry has been transformed by the widespread adoption of IoT technologies, which have greatly impacted the communication infrastructure. The agricultural

sector has benefited from the use of smart devices, remote data collection, and the integration of sensors and vehicles with mobile devices and the internet. In addition, a range of capabilities, including cloud-based intelligent analysis, mechanization of various farming operations, interface development and decision-making have been integrated. These advancements have enabled resource optimization, climate impact control, and increased crop yields, making a significant impact on the industry.

Researchers have proposed a range of methods, systems, and tools to monitor and transmit crop data during different growth stages, tailored to specific crop and field characteristics. Manufacturers provide an assortment of communication devices, including robots, sensors, drones, and heavy machinery, designed for data collection and sharing purposes. In order to guarantee food and environmental safety, food and agriculture organizations, in conjunction with governmental bodies, have established policies and guidelines to govern the application of these technologies [8].

1.3.1 Fundamentals of IoT Applications in Smart Farming

An affordable and user-friendly tracking platform has been developed to offer comprehensive information on conventional

farming practices, methods, tools, crop diseases and pests, etc. This information is sourced from various channels to support sustainable agriculture. Interactive agriculture enables users to conveniently access the data *via* multiple devices, including computers and mobile phones [9].

- Robust models: The agricultural industry is distinct in its range of unique attributes, which include complexity, diversity, uncertainties and spatio-temporal variability surrounding the most effective harvesting methods and facilities.
- Scalability: Achieving scalability of results is vital, particularly considering the vast differences in farm sizes. As such, planning and testing must be incrementally increased while minimizing costs.
- Affordability: Ensuring affordability is crucial for the success of farming endeavours, and pricing must be reasonable to provide significant support. Employing standardized platforms, facilities, products and tools can aid in achieving equitable pricing.
- Sustainability: Ensuring sustainability is a crucial concern given the intense economic pressures and fierce competition present global.

1.4 Technologies Used in Smart Farming

1.4.1 Global Positioning System (GPS)

Accordingly, GPS technology is capable of precisely recording latitude, longitude, and elevation data. By using signals transmitted from global positioning system satellites, GPS receivers can determine their position in real-time and continuously track their position while in motion. This information provides farmers with the ability to identify the exact location of field data, including pest occurrences, soil types, weeds, and other obstacles. With this system, farmers can attain accurate application of inputs, such as fertilizer, pesticide, seed, herbicide, and water, in precise areas of their fields, thereby enhancing resource utilization and efficiency [10].

1.4.2 Sensor Technologies

To determine soil texture and structure, nutrient levels, vegetation, temperature, humidity, air, vapor, and other related factors, various techniques such as photoelectricity, electromagnetics, conductivity, and ultrasound are employed. Remote sensing data facilitates the distinguishing of diverse crop varieties, categorization of pests and weeds, recognition of

zones with soil and plant stress, and surveillance of drought conditions [11].

Taking into account elements like soil moisture, precipitation, nutrient concentrations, light availability, humidity, and leaf color is essential to guarantee the vitality and well-being of plants. A pragmatic method for monitoring plant health is to optimize temperature and low-intensity lighting while preserving energy and water *via* micro-irrigation. Specific sensors are employed to identify particular parameters, and when any of these parameters surpass a predetermined threshold, the sensors detect the variation and transmit the data to the microcontroller. The microcontroller then executes appropriate actions until the constraint returns to its optimal level [12].

Sensors are widely used in agriculture to collect information on various factors such as humidity, moisture, temperature, pressure, airflow, light, soil quality, location, and CO_2 levels. Wireless sensors are widely utilized to gather data on crop conditions and other pertinent details due to their essential characteristics such as memory, coverage, portability, reliability, durability, and computational efficiency, which make them well suited for agricultural purposes. These sensors can either operate independently or integrate with sophisticated

agricultural equipment and heavy machinery, depending on specific application requirements.

1.4.3 Variable-Rate of Technology (VRT) and Grid Soil Sampling

In agriculture, variable-rate technologies (VRT) utilize GIS-generated maps to forecast input delivery rates, ensuring that inputs are distributed at varying levels, at the right times, and in the correct locations. The systematic method of gathering soil samples and generating a map for each soil parameter is achieved through soil sampling in a grid pattern. A variable-rate applicator uses these maps as a foundation and uploads them for utilization in variable rate technology (VRT). With the aid of a computer and GPS receiver, the amount of fertilizer product or delivery is regulated and controlled according to the characteristics of the map [13].

Incorporating innovative technologies such as grid soil sampling and variable rate technology can enhance soil fertility management and evaluate nutrient distribution and yields. The process of grid sampling entails dividing a field into smaller cells or areas using overlaid grid lines and gathering composite samples from each area at the intersections. Interpolation methods are used to estimate soil-test values for unsampled

locations based on grid sampling. This helps to account for field-specific variability of nutrients such as phosphorus and potassium, allowing for customized fertilizer and manure applications that enhance precision agriculture and nutrient management practices [14].

1.4.4 Geographic Information System (GIS)

The GIS is a system with hardware and software that compiles, stores, retrieves, and analyzes location and attribute data, generates maps and statistics to facilitate spatial analyses, and identifies the relationship between various factors that influence crop production in a given field. Beyond its functions in data storage and visualization, GIS is employed to assess various management approaches by integrating and modifying layers of data, contributing to informed decision-making [15].

1.4.5 Crop Management

By utilizing satellite imagery, farmers can obtain data on soil condition and crop performance changes that are attributable to topography in a field. This empowers them to carefully oversee factors that impact production, such as seeds, fertilizers, and pesticides, resulting in higher yields and efficiency.

Satellite imagery enables farmers to obtain real-time information on crop performance and soil conditions at a regional scale. Leveraging the reflective characteristics of vegetation, especially in red and near-infrared spectra, vegetation indices offer a means to predict the connection between crop biomass and yield experiments. The normalized difference vegetation index (NDVI) stands as the most widely employed metric to assess crop health and productivity, given its close association with leaf area index (LAI) and photosynthetic activity. Remote-sensing-derived indicators are interpreted and compared to previous or normal seasons in crop monitoring methods. Early crop yield valuation can be accomplished by examining the connection between vegetation indices and biomass before harvest. Automated field management comprises a range of functions, from basic tasks like crop production (yields), monitoring profit and loss, and forecasting weather to more complex responsibilities such as field mapping and soil nutrient monitoring [16].

1.4.6 Soil and Plant Sensors

New sensors have been designed with distinctive characteristics that distinguish them from existing sensors, as precision agriculture depends heavily on sensor technology to deliver data on soil properties, water status, and fertility [17].

Precision agriculture entails an agricultural methodology that leverages information technology to enhance both crop yield and quality, all the while minimizing resources like water, fertilizers, and pesticides. A vital element of precision agriculture lies in the real-time tracking of physical and chemical indicators within soil and plants, empowering farmers with informed insights to make strategic choices in crop management. By utilizing insights derived from data pertaining to soil attributes, fertility, and water conditions, farmers can fine-tune crop growth conditions, alleviate both biotic and abiotic stresses, and ultimately enhance yields. New sensors with enhanced features have been developed to improve monitoring effectiveness. These sensors, including optoelectronic, acoustic, nanostructured biosensors and impedance, can be used to detect pests. Furthermore, sensors for soil apparent electrical conductivity (ECa) can continually gather field surface data by penetrating deviations in soil texture and brininess. By leveraging insights obtained from data on soil properties, water status and fertility farmers can optimize crop growth conditions, mitigate biotic, increase yields, and abiotic stresses.

1.4.7 Rate Controllers

Rate controllers are used in a variety of applications, including agriculture, construction, and mining. In agriculture, they are used to apply fertilizers, pesticides, and herbicides at the correct rate. In construction, they are used to apply concrete, asphalt, and other materials at the correct rate. In mining, they are used to load and unload trucks and other vehicles at the correct rate.

1.4.8 Precision Irrigation in Pressurized Systems

Machines with motion control, sensor technologies, wireless communication, and GPS-based controllers have been developed for irrigation systems to monitor soil and climate conditions, as well as irrigation parameters such as flow and pressure, in order to improve water usage efficiency for crops. Although these advancements have the potential to provide significant benefits, further progress is needed before they can be widely adopted in the commercial sector [18].

1.4.9 Yield Monitor

Yield monitors, comprised of sensors, data storage components, computers, and user interfaces, gauge and record yield data. Sensors ensure a consistent measurement of grain mass or

volume flow. GPS receivers generate yield maps by utilizing data from specific locations. Mass flow sensors employ microwave energy beams to quantify the energy reflected upon striking the grain flow.

The yield monitor is a device that measures the amount of grain that is harvested from a field. It does this by using sensors to measure the mass or volume of the grain flow. The yield monitor is then connected to a mobile app that displays the data in real time. The app can be used to track the progress of the harvest, identify areas of low yield, and make adjustments to the harvester settings. The data from the yield monitor is also automatically uploaded to a web-based platform. This platform can be used to store the data, generate reports, and share the data with other stakeholders. Farmers can use this application to generate accurate yield maps, which can be shared with agronomists, and to export other farming data for analysis. In horticulture, monitoring fruit growth is crucial to assess crop development. To track fruit conditions, estimate maturation, identify the ideal harvesting time, and determine the appropriate market, color images are employed. Additionally, satellite images present a promising solution for monitoring crop yield in real-time over large areas.

The system for estimating crop yield includes components of both hardware and software. A yield estimator software program, along with a Bluetooth terminal, is utilized by a mobile application to calculate the yield of crops using mathematical algorithms. The use of spectral signatures obtained from satellites has been demonstrated to accurately predict crop yields, with the estimates being as reliable as actual yields. Machine learning techniques and the assimilation of satellite-derived data into crop models have been successful in predicting maize yields in different environments.

1.4.10 Software

The software has various functions, including mapping, interfacing with display controllers, processing and analyzing data, and interpreting information. It is frequently utilized to create maps that show soil possessions and nutrient position, yield information, and flexible rate application plans for inputs. Additionally, the software can overlay different types of maps with innovative geostatistical capabilities [19].

1.5 Importance of Smart Farming

Incorporating contemporary sensor and IoT technologies in agriculture can improve all aspects of conventional farming

practices. By incorporating wireless sensors and IoT technologies, smart farming can provide solutions to various challenges faced by traditional agriculture, such as monitoring land suitability, pest control, irrigation, drought, and optimizing yield. The hierarchical structure of smart agriculture applications, facilities, and devices is illustrated in Figure 1.3. Incorporating advanced technologies at multiple stages within these applications can enhance productivity and transform the agriculture industry.

1.5.1 Soil Mapping and Plant Monitoring

Soil analysis is critical in assessing the nutrient levels of a field by incorporating GPS location and site-specific data. Soil fertility is influenced by several factors, including topography, soil type and texture, cropping patterns, fertilizer use, and irrigation. Mapping the soil is crucial for assessing crop suitability, identifying the appropriate planting depth, and comprehending the soil's physical, biological, and chemical characteristics to enhance resource management. Farmers possess an array of sensors and instruments at their disposal to oversee soil attributes like water-holding capacity, texture, and absorption speed. This empowers them to uphold soil quality and avert degradation arising from erosional, alkaline, acidic, salinization, and pollution-related factors. Furthermore,

drought represents a significant concern, exerting an impact on crop yield and productivity. Soil moisture information can be garnered, and agricultural drought in remote regions can be evaluated *via* remote sensing techniques. Satellite data can be harnessed to produce soil moisture maps, facilitating the computation of the soil water deficit index (SWDI) and allowing the formulation of predictive models based on soil physical properties [20]. Soil characteristics and nutrition play a vital role in determining rice yield and quality. A mobile application based on IoT technology can provide real-time information to farmers for better crop management. The setup includes IoT-connected T-Beam microcontrollers that are linked to temperature sensors and electrical conductivity (EC) sensors [21].

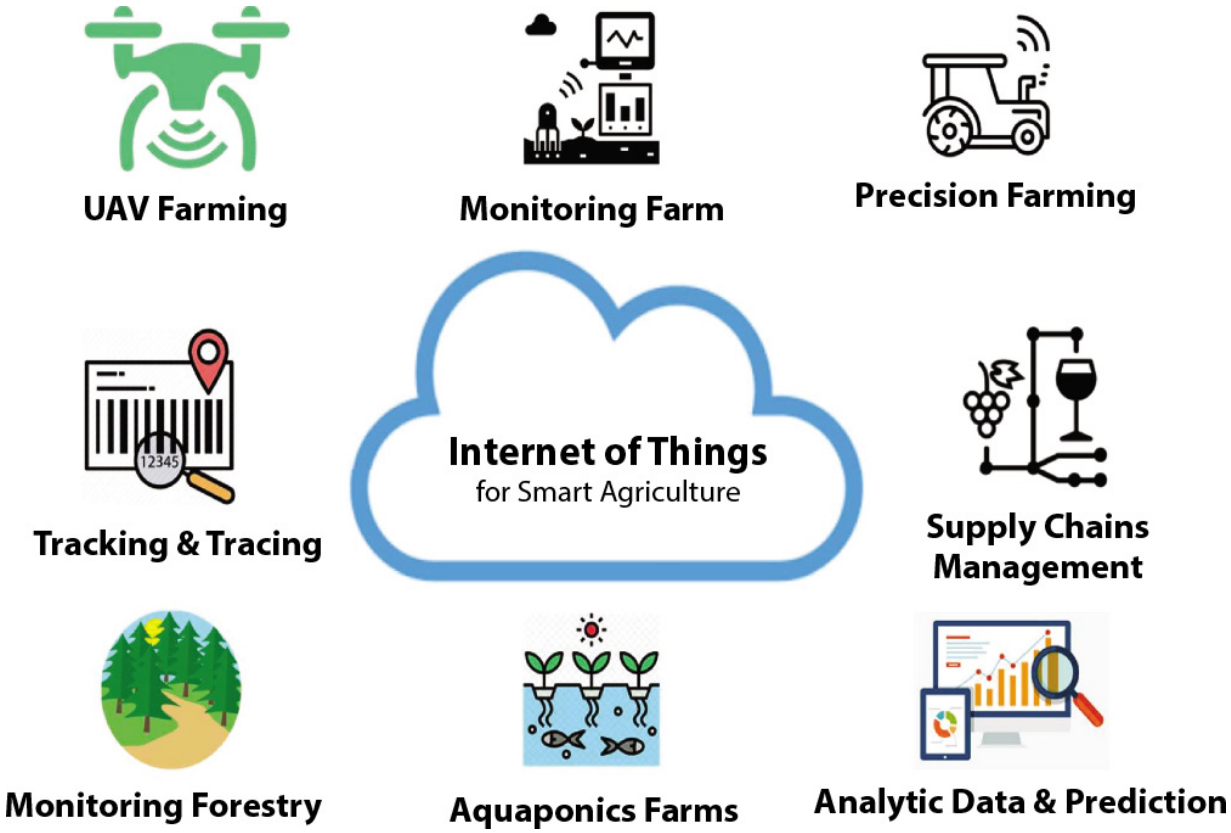

Figure 1.3 A depiction of IoT applications for smart farming.

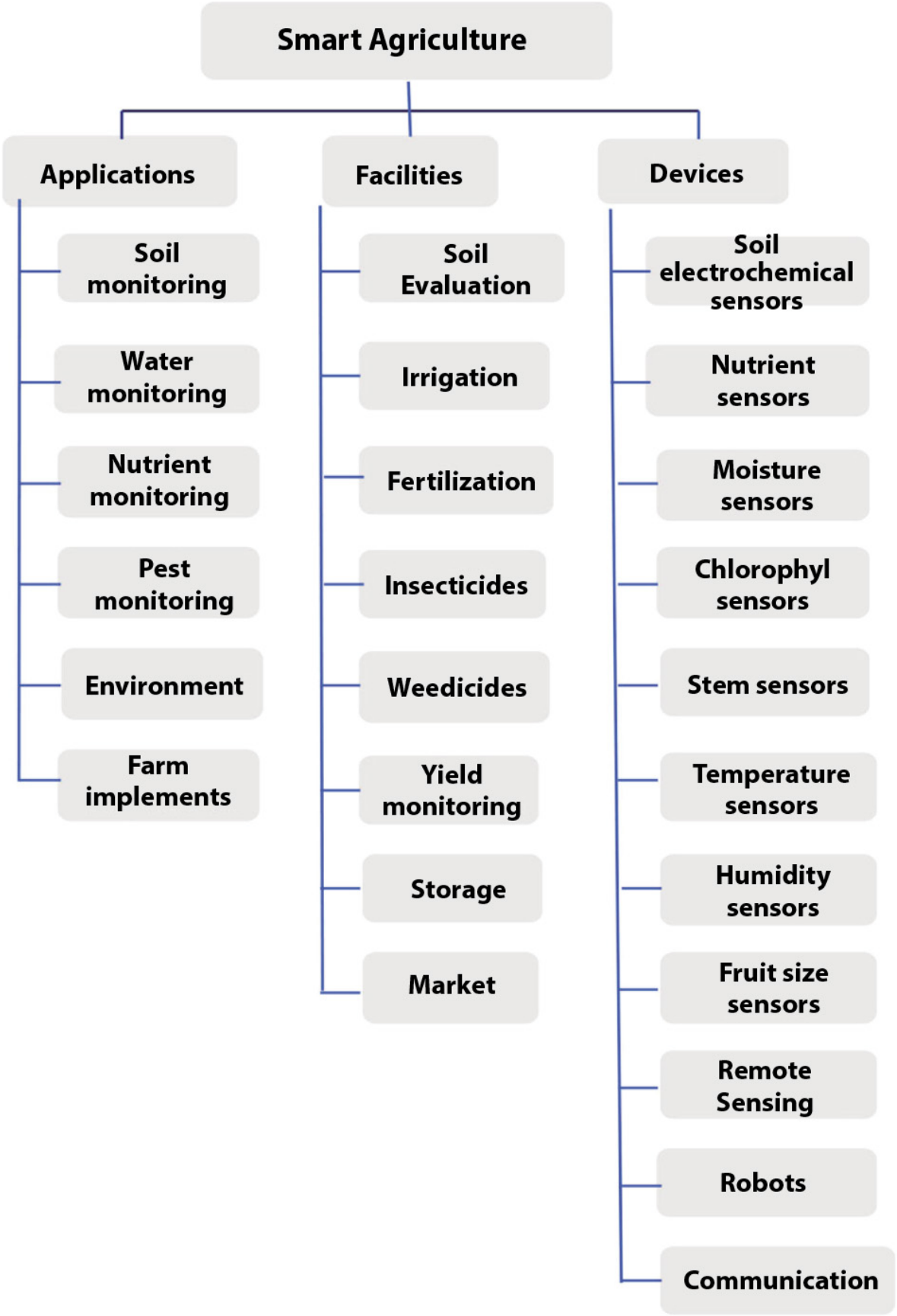
Smart Agriculture
Applications
Soil monitoring
Water monitoring
Nutrient monitoring
Pest monitoring
Environment
Farm implements
Facilities
Soil Evaluation
Irrigation
Fertilization
Insecticides
Weedicides
Yield monitoring
Storage
Market
Devices
Soil electrochemical sensors
Nutrient sensors
Moisture sensors
Chlorophyl sensors
Stem sensors
Temperature sensors
Humidity sensors
Fruit size sensors
Remote Sensing
Robots
Communication

Figure 1.4 Implementing innovative technologies at various stages.

Farmers can now collect information about their environment and soil using IoT ADCON-based stations and mobile devices like tablets and smartphones to improve grape productivity and crop quality during the growing season. The transmission system for data enables the analysis of interactions between soil, plants, and the atmosphere, which is essential for maximizing agricultural productivity. Comparisons were made between data collected by sensors that measured carbon dioxide, soil moisture, temperature, and light in a greenhouse, and the levels of CO_2 during day and night, as well as the opening and closing of greenhouse doors and windows, which were determined based on the soil moisture levels [22].

1.5.2 Irrigation

The UN Convention to Combat Desertification (UNCCD) predicts that, by 2030, desertification will affect around 168 countries, and nearly 50% of the world's population will inhabit regions facing severe water scarcity. Providing water to areas with limited water resources while meeting the growing demand for agriculture and other activities has become a critical concern. Water-efficient and controlled irrigation systems, such as drip and sprinkler irrigation, help conserve water resources [23].

Determining the water requirements of crops involves considering several factors, including soil type, rain, irrigation technique, crop variety, and soil moisture preservation. The use of wireless sensors in soil and air moisture control systems can enhance water usage efficiency and promote crop health. Currently, IoT methods like crop water stress index (CWSI)-based water management are anticipated to substantially boost crop yields. CWSI values are determined by evaluating crop canopy at different development phases and air temperatures, employing climate data, satellite, and sensors imagery to estimate water demand. To enhance the efficiency of water usage, one can predict the irrigation index values for each field by considering factors such as slope or soil variability.

1.5.3 Site-Specific Nutrient Management

Fertilizers, whether natural or synthetic, are substances that enhance plant growth and soil fertility. However, their improper use can lead to adverse effects on soil health, plant growth, and the environment. Smart agriculture's site-specific soil nutrient fertilization can estimate the needed amount of nutrients with precision, minimizing adverse effects on the soil and environment. Several factors affect site-specific soil nutrient measurements, including soil and crop types, exchange capacity, yield targets, fertilizer type, and weather conditions

[24]. IoT-based fertilization techniques are used to approximate the spatial distribution of nutrients. The satellite images utilize Normalized Difference Vegetation Index (NDVI) to monitor crop nutrient status, vegetation vigor, plant density, and soil nutrient levels [25, 26]. In IoT-based smart fertilization, cutting-edge technologies such as GPS, geo-mapping, variable rate technology (VRT), and autonomous vehicles play a pivotal role. These advancements facilitate precise and efficient fertilizer application by utilizing soil nutrient measurements specific to each location. To enhance fertilization productivity, effective practices like fertigation and chemigation are employed, utilizing water-soluble fertilizers in soil amendments and pesticides, respectively [27].

1.5.4 Crop Pest and Disease Management

As reported by the Food and Agriculture Organization (FAO), the global utilization of pesticides and other agrochemicals results in an annual decline in crop yields ranging from 20% to 40%, with adverse implications for both human and animal well-being in addition to environmental concerns [28]. The use of IoT-based devices, including drones, robots, and wireless sensors, can facilitate real-time monitoring and modeling of pests and diseases, resulting in more efficient pest control approaches. Data on plant health and pest incidence

throughout the crop period are collected using remote sensing imagery and field sensors in the IoT-based process, which involves detection and image processing. Automated traps equipped with IoT technology can be used to capture, count, and classify various types of insects. After collection, the data is uploaded to the cloud for analysis. The implementation of an IoT management system empowers agricultural robots, outfitted with multispectral image sensors and precision spraying nozzles, to enhance the precision in detecting and addressing pest challenges within the realm of agriculture [29].

1.5.5 Yield Monitoring and Forecasting

Efficient pollination is essential in monitoring crop yield, as it influences the produce quality, moisture content, and yield, especially when faced with changing environmental conditions [30]. Crop forecasting helps farmers predict yield before harvest, allowing them to plan for the future, make informed decisions, and perform further yield quality analyses. Precise monitoring of crop growth at different stages, including the measurement of parameters like fruit size and color, is essential in identifying the optimal time for harvesting. Precisely predicting the harvesting time maximizes crop quality and production, while also enabling effective market management strategies. Knowing the precise harvest time of crops is crucial

for farmers to ensure profitability. A visual representation in Figure 1.5 demonstrates the idea of a farm area network that facilitates live tracking of the complete farm.

The yield monitor of the harvester is coupled with a mobile application that transmits real-time data on the harvest to the manufacturer's web-based platform [31]. For estimating and monitoring crop production over large areas, satellite imagery is utilized. In the context of fruit crops, RGB satellite images are especially advantageous for monitoring the condition of the fruit, such as size and color, which are crucial for determining the fruit's maturity, deciding when to harvest, and identifying market prospects. Furthermore, optical sensors are utilized for monitoring fruit shrinkage when exposed to drying conditions.

1.5.6 Enhanced Production Rates

Agricultural production can be improved with the help of a smart agriculture system. By utilizing this system, it is possible to increase output while minimizing input—for instance, monitoring soil moisture, plant health, and weather conditions with a smartphone can aid in deciding the optimal timing for applying fertilizers, water, and pest control products. This approach can lead to a reduction in the required amount of chemical inputs while enhancing crop yields.

1.5.7 Water Conservation

The utilization of weather forecasts and soil moisture sensors restricts water consumption to the required time and place. Despite familiarity with the land, natural phenomena like weather can be unpredictable, leading to challenges such as insufficient rainfall during critical periods.

Ensuring that water is used effectively in agriculture requires accurate information about weather patterns and soil moisture levels. However, relying solely on weather forecasts can be risky. It is important to have a backup plan in case the forecast is inaccurate or rainfall is insufficient. This is where rain gauges come in handy. Through the measurement of rainfall in a particular area, farmers can make informed decisions regarding the timing and amount of water to use for their crops.

Rain gauges function by collecting raindrops on a particular surface and measuring their weight. Soil moisture sensors can be combined with rain gauges to gauge the moisture level of the soil. When integrated with an IoT-based system, these sensors enable farmers to gather and access information on rainfall and soil moisture from any location.

1.5.8 Real-Time Data and Insights

By utilizing the Internet of Things (IoT), farmers can obtain real-time information on production levels, soil moisture, sunlight intensity, and other pertinent data, allowing them to make informed decisions regarding resource allocation and operations. The IoT provides visualizations that enable farmers to quickly and intelligently evaluate their operations. Compared to traditional methods that required longer periods to gather and analyze data, this represents a substantial improvement.

1.5.9 Reduction in Operation Costs

An excessive application of fungicides can lead to wastage. Implementing an automated system that monitors fungicide concentration in the soil at regular intervals during the growing season can aid growers in optimizing their fungicide usage. The use of automation technology can help reduce resource consumption, minimize human error, and lower overall costs.

1.5.10 High-Quality Production

Produce consistency and high quality is the ultimate aim of farming operations. A study carried out by the US Department of Agriculture reveals that efficient management practices can increase farmers' yields by up to 20%. As an example, farmers

can overcome difficulties in producing high-quality crops, such as low-protein wheat, by utilizing specific fertilizer combinations that have been proven to increase protein content.

1.5.11 Accurate Farm and Yield Evaluation

Accurate measurement of nutrient levels in the field is essential for successful fertility management. Production rate charts provide farmers with a means to monitor production per unit area and assess the yield potential of each field. By adopting effective field management techniques, farmers can improve the field's productivity, resulting in higher profits.

1.5.12 Improved Livestock Production

The use of sensors and machines enables farmers to detect reproduction and health events in their animals at an earlier stage than previously possible. Additionally, geofencing location tracking can enhance livestock monitoring and management. Furthermore, the emotional states of animals can be determined with the use of behavioral biosensors, enabling farmers to identify potential issues. Biometric sensors can detect diseases like foot-and-mouth disease, avian flu, and swine influenza while also monitoring the animals' health

status. Biosensors utilizing biometric data can identify signs of stress, illness, or injury in animals. The potential applications of these biosensors in healthcare are manifold, enabling the diagnosis and treatment of various health issues in both humans and animals—for example, farmers may equip livestock with blood pressure cuffs to monitor and address elevated blood pressure levels.

1.5.13 Reduced Environmental Footprint Impact

Efforts to reduce water usage and increase production efficiency per unit of land are essential for environmental conservation and have a positive impact. In comparison to industrial activities, agricultural and forestry operations have a more significant impact on climate change. Improved agricultural land utilization and enhanced land, labor, and energy management practices are effective means of achieving increased production efficiency per unit of land. By adopting more efficient equipment, irrigation systems, and rainwater harvesting methods, farmers can reduce water consumption. Sustainable agricultural practices, forest management, and land use in agriculture, forestry, and rural areas have the potential to significantly contribute to climate change adaptation and mitigate emissions from agricultural practices, deforestation, and forest degradation. These practices also address pressing

environmental concerns such as soil erosion, biodiversity depletion, deforestation, desertification, and pollution. Food and water security are critical drivers of climate change, and it is vital to implement measures to address these issues [32].

1.5.14 Remote Monitoring for Easy Management

Farmers now have the ability to make important decisions about their farming operations from anywhere in the world. They can do this while on the move, watching their favorite TV shows, or even while exercising. They no longer have to wait to reach their office or attend a meeting to make these conclusions. By utilizing remote monitoring, farmers can make informed decisions and take prompt action in the field. This signifies the future of agriculture, where a straightforward web-based user interface or mobile app can monitor up to 100 locations globally. Farmers can receive warnings when glitches occur, enabling them to respond quickly and yield necessary actions. Managing the entire business becomes easy from a single location, no matter where they are located.

1.5.15 Expensive Asset Monitoring

Effective management of farming equipment is crucial for maintaining production rates, labor efficiency, and predicting

equipment failures. The Internet of Things (IoT) is a revolutionary technology capable of revolutionizing numerous industries by allowing for real-time data collection, process automation, and delivering valuable insights using low-cost sensors and IoT platforms. The agricultural and farming sector can greatly benefit from this technology. By implementing IoT solutions, farmers can efficiently monitor and maintain their equipment, which can lead to increased production rates and reduced equipment downtime.

1.6 Role of IoT in Advanced Farming Practices

Research has indicated that the use of sensor and IoT-based technologies can increase crop yields more effectively than conventional agricultural techniques. These advanced technologies are especially effective in controlled environments, and they are instrumental in improving the eminence and amount of produce.

1.6.1 Greenhouse Farming and Protected Cultivation

While the practice of cultivating plants in controlled environments has historical roots, its prominence surged during the 19th and 20th centuries, particularly in areas characterized by adverse weather conditions. Indoor crop

cultivation is less susceptible to external factors, and the usage of sensors and communication tools allows for year-round growth in any location. Achieving successful crop production in a controlled environment necessitates several factors, including suitable shed structures, wind control materials, aeration systems, precise monitoring parameters, and decision support systems. Monitoring environmental parameters with accuracy is a significant challenge in greenhouse farming, and it necessitates multiple measuring points to maintain and regulate the local environment. IoT-based greenhouses utilize sensors to measure and monitor internal parameters, including humidity, pressure, light, and temperature [33]. By implementing smart greenhouses, farmers can automate farming operations and shield plants from harsh weather and pest conditions without manual intervention. By using sensors to regulate factors such as light, temperature, and humidity, hibiscus plants can be grown during the night under optimal conditions. A study found that water usage can be reduced by 70–80%, and the IoT facilitates direct communication between farmers and consumers, enabling more efficient and profitable farming practices. Furthermore, the efficiency of rose plants cultivated in greenhouses has been enhanced by IoT-based automated systems that continuously supervise and regulate multiple factors, including humidity, UV light intensity, EC and

pH values, CO_2 levels, nutrient levels in water, temperature, and pesticide usage. Sensors are used to provide more precise diagnosis and finding [34].

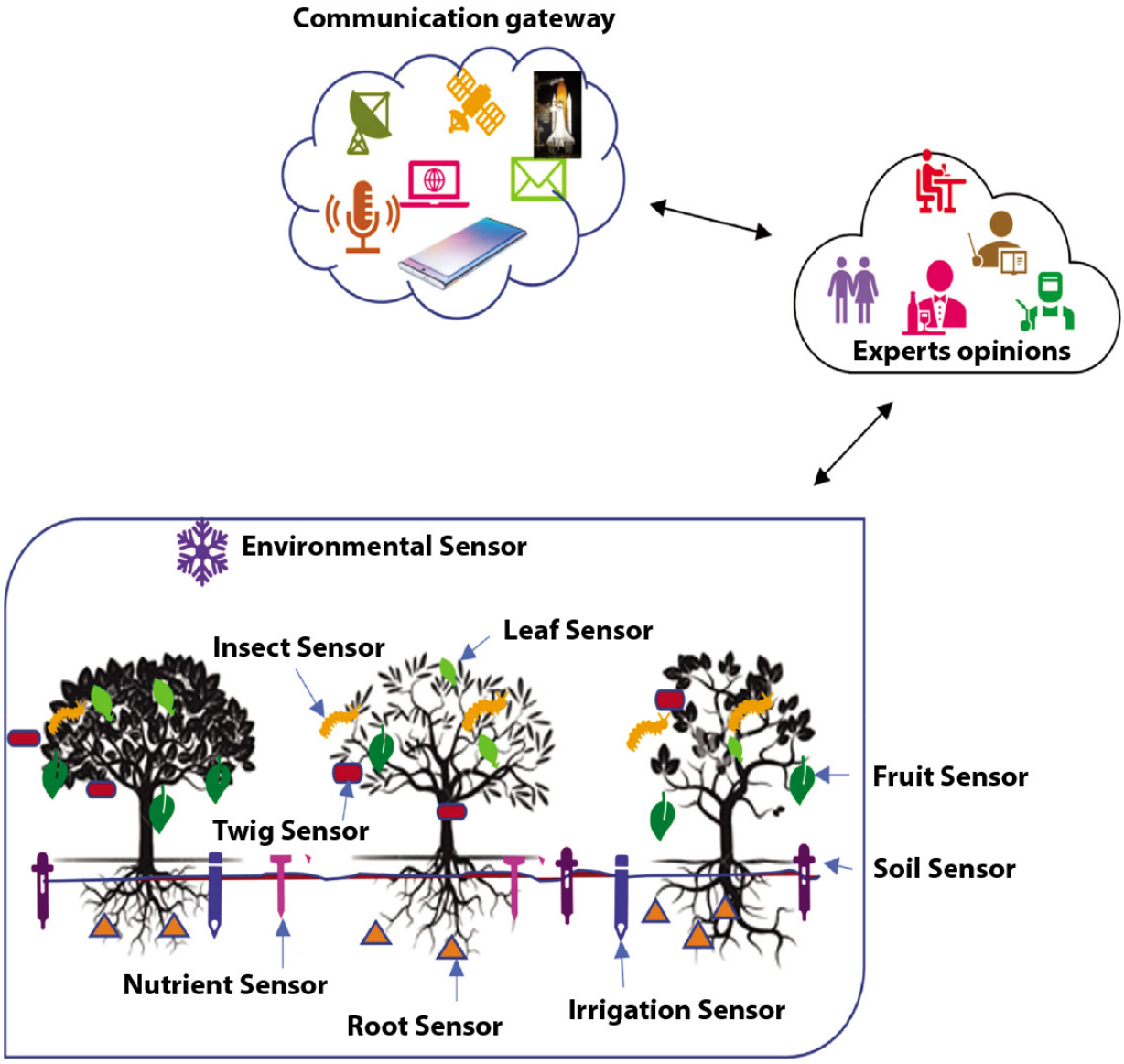

Figure 1.5 A smart farming network based on the Internet of Things (IoT).

1.6.2 Hydroponics

Hydroponics, a soilless form of hydroculture, has transformed

greenhouse farming. Through hydroponic irrigation systems, nutrients are dissolved in water and evenly delivered to plant roots. Advanced sensor technologies enhancing this process are now available, which can identify various parameters and perform data analysis at regular intervals [35]. For ensuring plant growth, it is crucial to precisely measure and monitor the nutrient content of the solution, taking into account the plant's requirements. A model based on wireless sensors [36] has been developed to monitor the levels of nutrient concentration and water in soilless cultivation in real time.

IoT-enabled smart hydroponics systems comprise three crucial elements: input data, output data, and a cloud server. Smart hydroponics systems with IoT integration allow remote monitoring of lettuce cultivation through the internet using cloud servers, input data, and output data. The real-time analysis of various parameters, such as pH levels, humidity, water quality, room temperature, and nutrient-rich water-based solutions can be conducted from any location. Integrated with Raspberry Pi, sensors gather information concerning plant growth variables like pH level, temperature, humidity, and water levels within the hydroponic reservoir. This data pertains to the deep flow technique hydroponic system, where plant roots are immersed in deep water layers, ensuring a continuous circulation of the nutrient solution. Real-time monitoring and

automated data processing ensure that water circulation is properly maintained.

1.6.3 Vertical Farming

Conventional farming practices have been shown to degrade soil quality more rapidly than natural processes can replenish it, which can result in soil erosion, reduced arable land, and increased stress on water resources. Vertical farming, on the other hand, offers a sustainable solution that minimizes resource consumption and increases production through precise environmental control, allowing crops to be grown in stacked layers. Compared to traditional farming practices, vertical farming is more effective in terms of yield and water usage. Monitoring and controlling carbon dioxide levels in this method is critical, and non-dispersive infrared (NDIR) CO_2 sensors are used for this purpose.

1.6.4 Phenotyping

Phenotyping is a cutting-edge approach in crop engineering that integrates plant genomics, agronomy, and ecophysiology to study the quantitative behavior of crops, such as resistance to pathogens and the weight of grains. According to [37], plant phenotyping is highly effective in analyzing measurable

characteristics such as growth, stress resistance, yield, and quality, which are not adequately addressed by genetic and molecular tools that have revolutionized crop breeding. Sensors and image-based phenotyping techniques are being used to screen and understand the mode of action of bio-stimulants. IoT-based phenotyping is developed to monitor crop and trait measurements, offering opportunities for digital agriculture and crop breeding, as mentioned in [38]. Trait investigation algorithms and modeling are used to govern the associations among genotypes, phenotypes, and growing conditions.

1.7 Trends of Smart Farming

This article will delve into the contemporary developments within smart agriculture. Smart farming entails the fusion of cutting-edge technologies like AI, IoT, and automation with conventional agricultural practices, aiming to enhance efficiency, minimize wastage, and boost productivity. Over the past few years, it has garnered worldwide acclaim and is projected to further burgeon in the years ahead.

1.7.1 Precision Agriculture

Agricultural fields can be analyzed and managed with greater accuracy and efficiency by utilizing technology in a practice

known as precision agriculture. The technology includes GPS systems, drones, sensors, and big data analytics. It enables farmers to monitor soil moisture levels, nutrient content, and crop growth patterns in real time, which helps them enhance the usage of resources and increase crop yields. Precision agriculture enables farmers to optimize the use of resources by targeting specific areas of their fields that require more attention while minimizing the use of resources in less demanding areas. This leads to a more efficient use of resources and improved crop yields.

1.7.2 Automation

Automation is another significant trend in smart farming. The use of automated systems in agriculture has been increasing rapidly, with the introduction of self-driving tractors, robotic harvesters, and drones. Automated systems help farmers to reduce labor costs, improve efficiency, and increase productivity—for example, automated irrigation systems can be programmed to water crops only when needed, based on real-time weather conditions and soil moisture levels. This reduces the amount of water wasted, saves time and money, and improves crop yields.

1.7.3 Internet of Things (IoT)

In the realm of intelligent agriculture, the Internet of Things (IoT) assumes a pivotal function by interconnecting sensors and devices, enabling farmers to actively track real-time environmental conditions and the well-being of their crops. Crop growth can be influenced by several factors, including soil moisture, temperature, and humidity. Sensors and devices can be used to measure these factors. Using AI and machine learning algorithms, the information collected from these sensors can be analyzed to forecast crop yields and optimize farming methods in the domain of smart farming. Additionally, IoT technology can automate various farming tasks such as planting, watering, and harvesting.

1.7.4 Big Data Analytics

Big data analytics entails the process of scrutinizing vast datasets to discern patterns and extract insights for informed decision-making. In the context of intelligent agriculture, big data analytics is harnessed to analyze information collected from diverse sensors and devices, with the aim of fine-tuning agricultural techniques, enhancing crop productivity, and minimizing inefficiencies. Farmers can utilize big data analytics to evaluate crop performance by analyzing sensor data, which

can assist in making informed decisions on crop selection and resource allocation.

1.7.5 Vertical Farming

Vertical farming is a farming method that involves cultivating crops in a controlled environment using artificial lighting, with plants grown in stacked layers. This approach enables farmers to grow crops throughout the year, irrespective of weather conditions. It also reduces the amount of land needed for farming, making it an ideal solution for urban areas with limited space. IoT sensors and automation systems are essential in vertical farming to monitor and control the growing environment.

1.7.6 Aquaponics

Aquaponics is a farming technique that involves cultivating fish and plants together in a mutually beneficial relationship. The plants in an aquaponic system play a dual role by utilizing the fish waste as a nutrient source and purifying the water for the fish. By using aquaponics, farmers can reduce waste and increase efficiency, as this method requires less water and fertilizer than traditional farming. In addition, the implementation of IoT technology in aquaponics can enable

farmers to monitor various factors such as water quality and nutrient levels, which have an impact on the health of both plants and fish.

In conclusion, smart farming is rapidly evolving, driven by advancements in technology and a growing need to increase efficiency and reduce waste in traditional agricultural practices: precision agriculture, automation, IoT, and big data analytics.

References

1. Ferrandez-Pastor, F.J.; Garcia-Chamizo, J.M.; Nieto-Hidalgo, M.; Mora-Pascual, J.; MoraMartinez, J. Developing ubiquitous sensor network platform using Internet of Things: Application in precision agriculture. *Sensors,* 2016, 16, 1141.
2. Liakos, K.G.; Busato, P.; Moshou, D.; Pearson, S.; Bochtis, D. Machine learning in agriculture: A review. *Sensors,* 2018, 18, 2674.
3. O'Grady, M.J.; O'Hare, G.M.P. Modelling the smart farm. *Inf. Process. Agric.,* 2017, 4, 179–187.
4. Quy, V.K.; Hau, N.V.; Anh, D.V.; Quy, N.M.; Ban, N.T.; Lanza, S.; Randazzo, G.; Muzirafuti, A. IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges. *Appl. Sci.,* 2022, 12, 3396.

5. Patil, K.A.; Kale, N.R. A model for smart agriculture using IoT. In *Proceedings of the 2016 International Conference on Global Trends in Signal Processing, Information Computing and Communication*, Jalgaon, India, 22–24 December 2016; IEEE: Jalgaon, India, 2016; pp. 543–545.
6. Yong, W.; Shuaishuai, L.; Li, L.; Minzan, L.; Ming, L.; Arvanitis, K.G.; Grorgieva, C.; Sigrimis, N. Smart Sensors from Ground to Cloud and Web Intelligence. *IFAC Pap. OnLine,* 2018, 51, 31–38.
7. Mittal, A.; Singh, A. Microcontroller based pest management system. In *Proceedings of the Second International Conference on Systems (ICONS'07)*, Martinique, France, 22–28 April 2007; IEEE: Martinique, France, 2007; p. 43.
8. King, T.; Cole, M.; Farber, J.M.; Eisenbrand, G.; Zabaras, D.; Fox, E.M.; Hill, J.P. Food safety for food security: Relationship between global megatrends and developments in food safety. *Trends Food Sci. Technol.,* 2017, 68, 160–175.
9. Chandhini, K. A Literature Study on Agricultural Production System Using IoT as Inclusive Technology. *Int. J. Innov. Technol. Res.,* 2016, *4*, 2727–2731.
10. Lang, L. GPS + GIS + remote sensing: An overview. *Earth Obs. Mag*., 1992, 1, 23–26.
11. Tripodi, P.; Massa, D.; Venezia, A.; Cardi, T.; Sensing Technologies for Precision Phenotyping in Vegetable Crops:

Current Status and Future Challenges. *Agronomy*, 2018, 8, 57.
12. Navulur, S.; Sastry, A.S.C.S.; Giri Prasad, M.N. Agricultural Management through Wireless Sensors and Internet of Things. *Int. J. Electr. Comput. Eng.,* 2017, 7, 3492–3499.
13. Berntsen, J.; Thomsen, A.; Schelde, K.; Hansen, O.M.; Knudsen, L.; Broge, N.; Hougaard, H.; Horfarter, R. Algorithms for sensor-based redistribution of nitrogen fertilizer in winter wheat. *Precis. Agric.,* 2006, 7, 65–83.
14. Mallarino, A.P.; Wittry, D.J. Use of DGPS, yield monitors, soil testing and variable rate technology to improve phosphorus and potassium management. In *Proceedings of the Integrated Crop Management Conference*; Iowa State University Extension and Outreach: Ames, IA, USA, 1997; pp. 267–275.
15. Ojo, O.I.; Ilunga, M.F. Geospatial Analysis for Irrigated Land Assessment Modeling and Mapping. In *Multi-Purposeful Application of Geospatial Data*; Rustamov, R.B., Ed.; IntechOpen: London, UK, 2018; pp. 65–84.
16. Chowdhury, M.E.H.; Khandakar, A.; Ahmed, S.; Al-Khuzaei, F.; Hamdalla, J.; Haque, F.; Reaz, M.B.I.; Shafei, A.A.; Emadi, N.A. Design, Construction and Testing of IoT Based Automated Indoor Vertical Hydroponics Farming Test-Bed in Qatar. *Sensors,* 2020, 20, 5637.
17. Adamchuk, V.I.; Hummel, J.W.; Morgan, M.T.; Upadhyaya, S.K. On-the-go soil sensors for precision agriculture. *Comput.*

Electron. Agric., 2004, 44, 71–91.

18. Hassan, A.; Aitazaz, A.F.; Farhat, A.; Bishnu, A.; Travis, E. Precision Irrigation Strategies for Sustainable Water Budgeting of Potato Crop in Prince Edward Island. *Sustainability*, 2020, 12, 2419.
19. Ferrández-Pastor, F.J.; García-Chamizo, J.M.; Nieto-Hidalgo, M.; Mora-Martínez, J. Precision Agriculture Design Method Using a Distributed Computing Architecture on Internet of Things Context. *Sensors,* 2018, 18, 1731.
20. Martínez-Fernández, J.; González-Zamora, A.; Sánchez, N.; Gumuzzio, A.; Herrero-Jiménez, C.M. Satellite soil moisture for agricultural drought monitoring: Assessment of the SMOS derived Soil Water Deficit Index. *Remote Sens.,* 2016, 177, 277–286.
21. Pallavi, S.; Mallapur, J.D.; Bendigeri, K.Y. Remote sensing and controlling of greenhouse agriculture parameters based on IoT. In *Proceedings of the International Conference on Big Data, IoT and Data Science (BID)*, Pune, India, 20–22 December 2017; pp. 44–48.
22. Ioana, M.; George, S.; Cristina, M.B.; Ana-Maria, D.; Marius, A.D. IoT Solution for Plant Monitoring in Smart Agriculture. In *Proceedings of the IEEE 25th International Symposium for Design and Technology in Electronic Packaging*, Cluj-Napoca, Romania, 23–26 October 2019; pp. 194–197.

23. Yuan, G.; Luo, Y.; Sun, X.; Tang, D. Evaluation of a crop water stress index fordetecting water stress in winter wheat in the North China Plain. *Agric. Water Manage.,* 2004, 64, 29–40.
24. Lavanya, G.; Rani, C.; Ganeshkumar, P. An automated low cost IoT based Fertilizer Intimation System for smart agriculture. *Sustain. Comput. Inform. Syst.,* 2020, 28, 100300.
25. Pinheiro Lisboa, I.; Melo Damian, J.; Roberto Cherubin, M.; Silva Barros, P.P.; Ricardo Fiorio, P.; Cerri, C.C.; Eduardo Pellegrino Cerri, C. Prediction of Sugarcane Yield Based on NDVI and Concentration of Leaf Tissue Nutrients in Fields Managed with Straw Removal. *Agronomy,* 2018, 8, 196.
26. Bruno, B.; Benjamin, D.; Davide, C.; Andrea, P.; Francesco, M.; Luigi, S. Environmental and Economic benefits of variable rate nitrogen fertilization in a nitrate vulnerable zone. *Sci. Total Environ.,* 2016, *545–546*, 227–235.
27. Raut, R.; Varma, H.; Mulla, C.; Pawar, V.R. Soil Monitoring, Fertigation, and Irrigation System Using IoT for Agricultural Application. In *Intelligent Communication and Computational Technologies*; Springer: Singapore, 2017; pp. 67–73.
28. Newlands, N.K. Model-Based Forecasting of Agricultural Crop Disease Risk at the Regional Scale, Integrating Airborne Inoculum, Environmental, and Satellite-Based Monitoring Data. *Front. Environ. Sci.,* 2018, 6, 63.

29. Marinelli, M.C.; Scavuzzo, C.M.; Giobellina, B.L.; Scavuzzo, C.M. Geoscience and Remote Sensing on Horticulture as Support for Management and Planning. *J. Agron. Res.,* 2019, 2, 43–54.
30. Wietzke, A.; Westphal, C.; Gras, P.; Kraft, M.; Pfohl, K.; Karlovsky, P.; Pawelzik, E.; Tscharntke, T.; Smit, I. Insect pollination as a key factor for strawberry physiology and marketable fruit quality. *Agric. Ecosyst. Environ.,* 2018, 258, 197–204.
31. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
32. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
33. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini Robotic Face Recognition System,” 2023 *World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.

34. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
35. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
36. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics,* 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
37. Torbick, N.; Chowdhury, D.; Salas, W.; Qi, J. Monitoring Rice Agriculture across Myanmar Using Time Series Sentinel-1 Assisted by Landsat-8 and PALSAR-2. *Remote Sens.,* 2017, 9, 119.
38. Theopoulos, A.; Boursianis, A.; Koukounaras, A.; Samaras, T. Prototype wireless sensor network for real-time measurements in hydroponics cultivation. In *Proceedings of the 7th International Conference on Modern Circuits and Systems Technologies (MOCAST)*, Thessaloniki, Greece, 7–9 May 2018.

Note

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2
Overview of Robotics in Agriculture: Types and Applications

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Abstract

Robotics in agriculture is an emerging field that aims to revolutionize farming practices by introducing automation, precision, and efficiency. With a growing global population, the demand for food is increasing, and agriculture, along with the development of rural infrastructure, must become more productive to meet this growing demand. Robotics can address this challenge by improving crop management, monitoring, and harvesting, leading to higher yields and reduced labor costs. One of the main applications of robotics in agriculture is precision farming, which involves the use of sensors, GPS, and machine learning to optimize crop management. Precision farming can improve yield and quality by monitoring soil and crop health, identifying pests and diseases, and providing targeted irrigation and fertilization. This approach reduces waste and environmental impact as well as labor and resource costs. Robots can also automate certain tasks in agriculture, such as planting, weeding, and harvesting. Autonomous vehicles, equipped with advanced technology, can efficiently traverse agricultural fields, execute tasks, and gather data with minimal human intervention. These robotic systems have the capability to operate seamlessly day and night, regardless of weather conditions and even in hazardous environments, thereby mitigating the risk of injuries to farmers. Moreover, the

integration of robotics in agriculture, coupled with the development of rural infrastructure, holds the potential to enhance crop quality and yield by minimizing errors and ensuring consistent performance. Beyond physical tasks, another realm where robotics can make a significant contribution to agriculture is in the realm of data analytics. By collecting and analyzing vast amounts of data, robots can help farmers make better decisions about crop management, pest control, and resource allocation. This can lead to more efficient and sustainable farming practices as well as reduced waste and improved profitability. In conclusion, robotics in agriculture has enormous potential to transform the way food is produced, leading to increased productivity, efficiency, and sustainability. While the adoption of robotics in agriculture is still in its early stages, ongoing research and development are likely to lead to significant advancements in the coming years.

***Keywords*:** Robotics, agriculture, sensors, rural infrastructure, sustainable farming, crop management

2.1 Introduction

Robotics in agriculture refers to the use of autonomous machines, such as drones, robots, and other mobile devices, to perform various tasks related to crop production and

management. The use of robotics in agriculture has gained significant attention in recent years due to its potential to enhance efficiency, reduce labor costs, and improve productivity. These machines are equipped with sensors, GPS, cameras, and artificial intelligence that allow them to operate without human intervention [1]. The primary applications of robotics in agriculture include precision agriculture, crop management, livestock management, and irrigation management. Overall, the integration of robotics in agriculture offers numerous benefits, including increased efficiency, reduced labor costs, and improved crop and livestock quality. As technology continues to advance, it is likely that we will see even more innovative uses of robotics in agriculture in the future.

Robotics in agriculture is an important topic due to its potential to transform the agricultural industry by increasing efficiency, productivity, and sustainability. The global population is expected to reach 9.7 billion by 2050, which will result in an increased demand for food production [2]. Agriculture will have to meet this demand while minimizing environmental impact, reducing waste, and improving the livelihoods of farmers. Robotic technologies offer a promising solution to these challenges. They can perform various tasks more quickly and accurately than humans and can operate autonomously

without the need for human intervention. This leads to improved efficiency and productivity as well as reduced labor costs. In addition, robots can collect and analyze data on soil health, crop growth, and climate conditions, allowing for precision agriculture and more sustainable farming practices. The integration of robotics in agriculture also has significant potential to improve the safety and working conditions of farmers. By automating dangerous or physically demanding tasks, such as pesticide application or heavy lifting, farmers can avoid injuries and health hazards associated with manual labor. Furthermore, robotics in agriculture can help address the challenge of attracting and retaining a skilled labor force in rural areas. As younger generations are drawn to urban areas, farming communities are struggling to find and retain workers. Robotics can help alleviate this labor shortage by replacing some of the manual labor that is necessary for agriculture [3]. Overall, robotics in agriculture is an important topic because it has the potential to improve the efficiency, productivity, sustainability, safety, and profitability of agriculture while addressing key challenges facing the industry. The purpose of this chapter on robotics in agriculture is to provide an overview of the current state of the field, including the types of robotics used in agriculture, their applications, advantages and limitations, and potential future developments. The chapter

aims to explore the benefits and challenges of integrating robotics in agriculture, including their potential to enhance productivity, efficiency, and sustainability as well as their impact on farmers and rural communities.

The chapter will also examine case studies and examples of successful applications of robotics in agriculture as well as the challenges and limitations that need to be addressed in order to fully realize the potential of this technology. In addition, the chapter will provide recommendations for policy makers and industry leaders on how to promote the adoption of robotics in agriculture and how to address the challenges and concerns that may arise. Overall, the purpose of this chapter is to contribute to the understanding of the potential of robotics in agriculture and to provide a foundation for further research and discussion on this important topic.

2.2 Background

2.2.1 Historical Perspective of Agriculture and Technology

Agriculture has been a critical part of human society since the beginning of civilization, with evidence of organized farming dating back over 10,000 years. Throughout history, advances in

technology have played a significant role in the development and transformation of agriculture.

Figure 2.1 Ancient farming [6].

One of the earliest technological innovations in agriculture was the development of the plow, which allowed farmers to more efficiently till the soil and plant crops. Farmers would use traditional wooden plows, often pulled by animals like oxen or buffaloes, to till the soil and create furrows as illustrated in Figure 2.1. This invention was followed by the introduction of irrigation systems, which allowed farmers to cultivate crops in arid regions. In the Middle Ages, crop rotation and the use of manure as fertilizer became common practices, further improving agricultural productivity.

The Industrial Revolution in the 18th and 19th centuries brought about significant changes in agriculture, with the introduction of machinery such as the tractor and the reaper as illustrated in Figure 2.2 and Figure 2.3. These machines greatly increased the efficiency of farming, allowing farmers to cultivate larger areas of land with fewer laborers [4]. The 20th century saw the development of new technologies such as hybrid crops, chemical fertilizers, and pesticides, which further improved agricultural productivity.

Figure 2.2 Tractor [8].

Figure 2.3 Reaper [9].

In the latter half of the 20th century, the development of computer technology and precision agriculture began to transform agriculture once again. With the use of satellite imagery, GPS, and other sensors, farmers were able to collect and analyze data on soil health, crop growth, and weather patterns, allowing for more precise and sustainable farming practices [4].

Today, the integration of robotics in agriculture represents the latest technological advancement in the field. These machines are capable of performing a variety of tasks autonomously, such as planting, harvesting, and monitoring crop health, thereby further enhancing efficiency and productivity [5, 6]. Overall, the historical perspective of agriculture and technology highlights the critical role that technology has played in the development and transformation of agriculture and underscores the potential of robotics to further enhance the field.

2.2.2 Overview of the Current State of Agriculture and Technology

The current state of agriculture and technology is rapidly evolving, with new advances in technology offering solutions to the challenges facing the agricultural industry. One of the most significant technological advances in recent years has been the development of precision agriculture, which involves the use of sensors, GPS, and other technologies to collect and analyze data on crop health, soil health, weather patterns, and other factors [7]. Precision agriculture enables farmers to make more informed decisions about planting, fertilizing, and harvesting crops, which leads to improved efficiency, productivity, and sustainability [8, 9]. In addition, the use of precision agriculture technologies helps reduce waste and minimize the

environmental impact of farming practices. Another area of technological advancement in agriculture is the development of robotics. Robots are being developed and deployed to perform a wide range of tasks, from planting and harvesting crops to monitoring crop health and applying pesticides [10]. By automating labor-intensive tasks, robots help reduce labor costs and improve the safety of workers.

Figure 2.4 Drones [9].

Other areas of technological advancement in agriculture include the use of drones as illustrated in Figure 2.4 for crop monitoring and management, the development of genetically modified crops for increased yield and resilience, and the use of block chain technology to improve transparency and

traceability in the food supply chain. Overall, the current state of agriculture and technology is characterized by a rapidly evolving landscape of technological advancements, offering farmers new tools and solutions to improve efficiency, productivity, and sustainability in agriculture.

2.2.3 Benefits and Challenges of Robotics in Agriculture

Robotics technology has the potential to revolutionize agriculture by improving efficiency, increasing productivity, and reducing costs. Here are some benefits and challenges of robotics in agriculture:

Benefits:

- Precision farming: Robotic systems can be programmed to perform precise and repeatable tasks such as planting, fertilizing, and harvesting. This can reduce waste, improve yield, and increase profitability.
- Labor shortage: With a growing shortage of farm labor, robots can help fill the gap. They can work around the clock, without breaks or vacations, and reduce the need for human labor.
- Safety: Robotic systems can perform dangerous tasks such as spraying pesticides, without exposing human operators to

harmful chemicals.
- Data collection: Robots can collect data on soil conditions, crop growth, and other environmental factors, which can be used to optimize crop production and reduce waste.
- Sustainability: Robotic systems can be designed to reduce the use of water, fertilizer, and pesticides, making agriculture more sustainable [31, 34].

Challenges:

- High cost: Robotics technology can be expensive to develop and implement, which can be a barrier for many farmers.
- Limited flexibility: Robots are designed for specific tasks and may not be able to adapt to changing conditions or unexpected events.
- Maintenance: Robotic systems require regular maintenance and repair, which can be time-consuming and costly.
- Technical expertise: Farmers may require specialized technical expertise to operate and maintain robotic systems, which can be a barrier for many.
- Ethical concerns: There may be ethical concerns around the use of robots in agriculture, such as the displacement of human labor and the potential impact on animal welfare.

2.3 Types of Robotics in Agriculture

Robotics technology has the potential to revolutionize agriculture by improving efficiency, increasing productivity, and reducing costs. Here are some types of robotics in agriculture:

- Autonomous tractors: Autonomous tractors are self-driving machines that can navigate through fields, plow, plant, and harvest crops without human intervention as illustrated in Figure 2.5. These tractors are equipped with sensors, GPS, and cameras that help them navigate and operate. Autonomous tractors can significantly reduce labor costs and increase productivity [11]. With pinpoint precision, they navigate fields, minimizing travel time and maximizing efficiency. They enhance precision agriculture by spotting changes in the landscape, weeds, and crops using computer vision. By minimizing the need for human labor and allowing for constant operation, even in difficult conditions, autonomous tractors increase productivity. Protecting humans and cattle, adjusting to different landscapes, and overcoming regulatory and financial hurdles are all obstacles that must be surmounted. Despite these challenges, they may change the way modern farming is done by increasing

productivity, decreasing resource waste, and lessening environmental impact.

- Drones: Drones are unmanned aerial vehicles that can monitor crop health, detect crop diseases, and estimate yield. Drones are quickly becoming a game-changing tool in the agricultural sector, with several potential uses that boost output and longevity. Drones may take comprehensive aerial images of crops using high-resolution cameras, multispectral sensors, and LiDAR technology, enabling early disease diagnosis, identifying nutrient shortage, and predicting yield as illustrated in Figure 2.6. They help farmers make informed choices, leading to more efficient use of resources and less waste. Drones may also be used for the precise application of fertilizers and pesticides, with the added benefit of lowering both chemical use and pollution levels. These flying machines can swiftly and cheaply cover enormous areas of land, significantly reducing the need for human labor. Problems arise when trying to comply with intricate legal frameworks, maintain secure airspace, and make sense of an overwhelming amount of data. However, unmanned aerial vehicles (UAVs) have the potential to radically alter the agricultural sector by facilitating precision farming, expanding monitoring capacities, and bolstering efforts toward more environmentally friendly methods.

Figure 2.5 Autonomous tractors [11].

Figure 2.6 Drones [11].

- Robot arms: Robot arms have found valuable applications in modern agriculture, transforming various aspects of the industry. These versatile machines are equipped with specialized tools and sensors to perform tasks such as planting, pruning, and harvesting with high precision [14]. Robot arms enhance labor efficiency by reducing the need

for manual work and addressing labor shortages. They can adapt to different crop varieties and sizes, increasing flexibility in operations. By using advanced computer vision systems, robot arms can identify ripe fruits and vegetables, ensuring optimal harvest quality as illustrated in Figure 2.7. Challenges include programming for complex tasks, ensuring gentle handling of delicate crops, and navigating challenging terrains. Despite obstacles, robot arms have the potential to revolutionize agriculture by improving yield, reducing resource wastage, and promoting sustainable practices.

Figure 2.7 Robot arms [14].

- Harvesting robots: Robotic harvesters are a game-changing

innovation that is changing the way crops are gathered, leading to greater productivity in the agricultural sector. Robots with the right sensors, grippers, and arms can pick and sort crops like fruits and vegetables without human intervention [15]. Using AI and computer vision, they can tell ripe from unripe fruit, which improves production quality and decreases food waste as illustrated in Figure 2.8. By reducing the need for as much human labor during peak harvesting times, harvesting robots help alleviate labor shortages. Because of their adaptability, they can keep running no matter the terrain or the weather. However, there are still obstacles to overcome in the creation of robots that can handle delicate objects, adjust to different types of crops, and fit into preexisting agricultural infrastructure. Despite several drawbacks, harvesting robots have the potential to greatly improve agricultural efficiency and output by reducing post-harvest waste.

- Milking robots: Milking robots are revolutionizing the dairy sector by automating cow milking. These innovative devices recognize and connect to cows' udders using sensors, cameras, and robots, assuring efficient and gentle milking as illustrated in Figure 2.9. Milking robots have several advantages, including greater milking frequency, decreased labor needs, and enhanced cow welfare. The robots collect

information on milk output, quality, and cow health, allowing farmers to make better management decisions [16]. While early setup expenses and upkeep are obstacles, the technology ultimately improves operating efficiency and minimizes manual labor. Milking robots work around the timetables of individual cows, minimizing stress and boosting herd well-being [32, 33]. Their combination with data analytics and cloud technologies improves herd management and optimizes milk output. Despite obstacles, milking robots demonstrate how automation improves animal care and agricultural sustainability.

Figure 2.8 Harvesting robots [15].

- Weeding robots: Robotic weeders are a game-changer for farmers since they effectively eliminate weeds in a sustainable way. These robots employ AI, sensors, and precise tools to detect and eradicate weeds without human intervention [17]. Damage to crops is reduced by the use of computer vision to identify crop varieties from weeds as illustrated in Figure 2.10. Robotic weeders encourage sustainable agriculture by reducing the use of harmful chemical pesticides [18]. They work nonstop to eliminate

weeds as effectively as possible. Managing a wide variety of weed species and adjusting to different crop patterns are two of the difficulties that farmers face. However, weeding robots can cut down on labor expenses and the need for all that backbreaking hand weeding. Because they can operate with minimum oversight, their incorporation into agricultural management systems increases productivity. These robots help farmers become more productive and environmentally conscious by assisting with long-term weed management.

Figure 2.9 Milking robots [16].

Figure 2.10 Weeding robots [18].

2.4 Applications of Robotics in Agriculture

- Precision agriculture: Precision agriculture is the cornerstone of robots' role in today's agricultural revolution. Precision planting, fertilizing, and spraying are all made possible by robots outfitted with cutting-edge sensors, GPS, and AI technology. Soil conditions, crop health, and resource allocation are all evaluated through meticulous data collection and analysis. This method reduces waste and protects the environment while increasing productivity. In precision agriculture, robots may drive themselves around fields and make decisions about what to do next based on the data they collect as they go. Integrating robots with current farm equipment and managing complicated data streams are two examples of the difficulties that must be overcome. While they are not without their problems, precision agricultural robots help farmers reduce their reliance on harmful chemicals and provide better growing environments. Their round-the-clock availability marks a revolutionary change in farming practices, raising output and ensuring uniformity in crop management.
- Crop monitoring: Crop monitoring is a significant application of robotics in contemporary agriculture, where robots

outfitted with sensors and cameras capture real-time data on crop growth, health, and environmental conditions. These machines provide precise information on factors such as moisture levels, nutrient content, and the presence of disease. By analyzing this data, producers can make informed decisions regarding irrigation, fertilization, and insect control, thereby optimizing crop yield and resource use. Crop monitoring machines can efficiently survey expansive areas, identifying problems that might be overlooked during manual inspections. Data management and integration with farm management systems are obstacles. However, these robots considerably improve the precision and efficacy of crop management, leading to more sustainable agricultural practices and higher crop yields.

- Crop management: Crop management is an essential use of robotics in contemporary agriculture, whereby robots equipped with sensors and artificial intelligence are employed to enhance and optimize several facets of farming. These automated systems observe and evaluate data pertaining to the condition of soil, the health of crops, and the patterns of development, with the purpose of providing insights for making informed choices about irrigation, fertilization, and pest management. By implementing actions with precision, the reduction of resource waste and

environmental effect is achieved. Crop management robots have the ability to independently traverse agricultural fields, therefore offering up-to-the-minute observations that facilitate prompt decision-making and subsequent action. One of the primary obstacles in this context is the integration of heterogeneous data sources as well as the imperative of establishing effective communication channels with pre-existing agricultural systems. However, these robots have the potential to revolutionize agricultural practices by increasing output, lowering expenses, and supporting sustainable farming through effective resource allocation and reduced reliance on chemicals.

- Harvesting: Harvesting is one of the most important ways that robots are used in modern agriculture. Robots with advanced sensors, computer vision, and precise tools can gather crops like fruits and veggies on their own. These robots can work around the clock and can tell when fruits and vegetables are ready to be picked. This makes sure that crops are picked quickly and efficiently. They cut down on the amount of work needed during busy times and improve crop quality by reducing damage. Adapting to different kinds of crops and taking care of fragile crops are two challenges. However, harvesting robots have a lot of benefits, including greater output, less waste after gathering, and better use of

labor. Their use with data analytics improves farm management as a whole and leads to more safe farming practices.

- Livestock management: Livestock management is an important use of robotics in modern agriculture, utilizing robots equipped with sensors and artificial intelligence to monitor and care for animals. These robots monitor important characteristics such as health, behavior, and eating patterns, offering real-time information for illness identification and nutrition optimization. They may distribute food, provide medication, and even help with activities such as milking. Livestock management robots minimize labor needs while improving animal care by automating mundane chores. Designing robots that are appropriate for various animal sizes and behaviors is one of the challenges. Despite the challenges, these robots improve operational efficiency, contribute to healthier livestock, and enable data-driven decision-making for more sustainable and productive agricultural practices.
- Environmental monitoring: Environmental monitoring is an important use of robotics in contemporary agriculture, including the use of robots outfitted with sensors and data gathering equipment to check elements such as soil quality, air quality, and water levels. These robots deliver real-time

data to farmers, allowing them to make more educated choices regarding irrigation, fertilizer application, and general land management. They can roam fields independently, efficiently covering enormous regions and detecting environmental changes that affect crop development. Data analysis and integration with current agricultural systems are among the challenges. Nonetheless, environmental monitoring robots play an important role in supporting sustainable agricultural practices by reducing resource waste and maintaining optimum crop growing conditions, resulting in higher yields and a lower environmental impact.

2.5 Advantages of Robotics in Agriculture

Robotics technology has the potential to revolutionize agriculture by improving efficiency, increasing productivity, and reducing costs. Here are some of the advantages of using robotics in agriculture:

- Increased efficiency: Robots can work around the clock without the need for rest, breaks, or vacations. They can perform precise and repetitive tasks with greater accuracy and consistency than humans, leading to increased efficiency and productivity.

- Improved accuracy: Robots can be programmed to perform specific tasks with high precision and accuracy, such as planting seeds, applying fertilizer, or harvesting crops. This can lead to better crop yields, improved quality, and reduced waste.
- Reduced labor costs: As labor costs continue to rise, robots can help reduce the reliance on human labor, leading to reduced labor costs.
- Better safety: Agricultural robots can perform dangerous tasks, such as spraying pesticides, without exposing human operators to harmful chemicals, improving safety in the workplace.
- Optimized resource utilization: Agricultural robots can help optimize the use of resources such as water, fertilizer, and pesticides, resulting in more sustainable farming practices.
- Increased data collection: Agricultural robots can collect vast amounts of data on soil conditions, crop growth, and environmental factors. This data can be used to make more informed decisions on crop management, leading to increased efficiency and productivity.
- Adaptability: Robots can be adapted to various agricultural applications, from precision farming to crop monitoring, leading to increased flexibility in the industry.

2.6 Limitations of Robotics in Agriculture

While there are many potential benefits of using robotics in agriculture, there are also several limitations that must be addressed. Here are some of the main limitations of using robotics in agriculture:

- High costs: Developing and implementing robotics technology can be expensive and may not be feasible for small-scale farmers with limited resources.
- Technical complexity: Robotics technology requires specialized technical expertise to operate and maintain, which may be a barrier for many farmers who are not familiar with the technology.
- Limited adaptability: Agricultural robots are often designed to perform specific tasks and may not be able to adapt to changing conditions or unexpected events.
- Environmental factors: Agricultural robots may face challenges operating in certain environmental conditions, such as muddy or wet fields, steep slopes, or uneven terrain.
- Ethical concerns: The use of robots in agriculture may raise ethical concerns, such as the displacement of human labor and the impact on animal welfare.
- Limited sensory capabilities: Agricultural robots may lack the sensory capabilities of humans, such as touch, smell, and

taste, which may limit their effectiveness in some tasks.
- Dependence on technology: The use of robotics in agriculture may lead to a dependence on technology, which could be problematic in the event of technical failures or malfunctions.

2.7 Future of Robotics in Agriculture

The future of robotics in agriculture is promising, as the technology continues to evolve and improve. Here are some trends and developments that are likely to shape the future of robotics in agriculture:

- Revolution in precision agriculture: Precision agriculture is one of the most revolutionary elements of agricultural robots in the future. This shift will be led by robotics, which will allow for precise and data-driven interventions at every stage of the agricultural process. By examining real-time data from the field, robots equipped with cutting-edge sensors, cameras, and AI algorithms will optimize planting, fertilization, pest management, and watering. This degree of accuracy will lead to less waste of resources, more effectiveness, and better agricultural yields.
- Spread of autonomous machinery: The use of autonomous machines will proliferate in the agricultural environment of

the future. Drones, harvesters, and even tractors will work continuously and independently, doing activities that formerly needed human involvement. By removing downtime caused by human labor limits and allowing 24/7 operations, especially during crucial planting and harvesting seasons, this autonomy will considerably increase output.

- Customization that adapts: The next generation of agricultural robots is expected to demonstrate enhanced levels of flexibility and customization. The agricultural machinery will be specifically engineered to accommodate a wide range of crop varieties, varying in size and adaptable to different field conditions. The inherent flexibility of robots will enhance their versatility and applicability in various agricultural operations, including both extensive monoculture farming and more niche-specialized crop cultivation.
- Data-driven decision-making and AI: Robots will be able to make intelligent judgments based on in-the-moment data analysis thanks to the combination of artificial intelligence and machine learning. Robots will be capable of pattern recognition, illness detection, and agricultural practice optimization. This will result in better-informed and more accurate decisions, which will help to allocate resources more effectively and improve crop health.

- Collaboration using swarm robotics: The utilization of swarm robotics, a collaborative approach using several robots, is expected to have a significant impact on the future of agriculture. The use of robotic swarms enables the execution of tasks that surpass the capabilities of a solitary robot due to their magnitude or complexity. The individuals will collaborate in a collective manner, effectively coordinating their efforts in order to attain predetermined objectives. The use of this collaborative strategy is expected to enhance efficiency and flexibility, hence enabling the simultaneous handling of diverse jobs.
- Data integration that is seamless: The future of agricultural robots will be characterized by their ability to effectively integrate data [26, 27]. The synthesis and analysis of data obtained from diverse sources such as robotics, sensors, satellite photos, and weather forecasts will be utilized to generate practical insights for the purpose of farm management. The use of this integrated strategy is expected to improve the process of decision-making, leading to increased effectiveness in the allocation of resources and improved overall management [28, 29].
- Encouragement of sustainability: In the future, agricultural robots' role in fostering sustainability will be one of their most important contributions. Robots will be essential in

minimizing pesticide use, cutting waste, and optimizing resource allocation as the agriculture sector confronts pressure to lessen its environmental impact. Robotics will make it possible to apply inputs more precisely and strategically, improving crop health, lowering environmental impact, and increasing sustainability.

2.8 Case Studies and Examples

There are several examples of robotics being used in agriculture today. Here are some case studies and examples:

- Vision Robotics: Vision Robotics is a California-based company that has developed a range of robots for use in agriculture. The integration of powerful camera systems and AI algorithms in vision robots has the potential to greatly transform current agriculture as illustrated in Figure 2.11. These autonomous machines use computer vision technology to process and analyze visual data obtained from the agricultural environment. This capability allows them to do various activities such as monitoring crops, detecting diseases, and assessing crop yields. With the use of cameras, LiDAR technology, and other sensors, these devices have the capability to accurately recognize and classify crops, weeds, and soil conditions. Vision robotics enhances resource

allocation by the facilitation of focused tasks, such as the exact application of insecticides and herbicides. One of the primary difficulties is the process of adjusting to a wide range of field circumstances, including varied environmental factors. Additionally, the presence of illumination fluctuations poses a significant challenge. Furthermore, the integration of data from numerous sources presents a complex task that must be addressed. Nevertheless, these robotic systems provide farmers with the ability to make informed choices based on data, resulting in improved crop management, enhanced operational effectiveness, and a decreased ecological footprint. Vision robots play a crucial role in advancing precision agriculture by improving accuracy and minimizing human errors, hence facilitating the transition toward more sustainable and productive agricultural practices.

- Blue River Technology: Blue River Technology is a company that has developed a precision spraying robot called the “See & Spray.” Blue River Technology, which has recently become a subsidiary of John Deere, is an innovative leader in the field of agricultural technology. Blue River, a highly regarded company, has created the See & Spray technology, a cutting-edge solution that utilizes computer vision and machine learning to accurately identify and selectively apply

herbicides only to weeds as illustrated in Figure 2.12. This innovative approach effectively minimizes the amount of chemicals used, hence fostering sustainable practices. This technological advancement allows the instantaneous recognition of cultivated plants and unwanted vegetation, hence improving accuracy and reducing potential harm. The technologies developed by Blue River aim to tackle prevalent issues in agriculture, including herbicide resistance and labor shortages, therefore revolutionizing farmers' strategies for weed control. The amalgamation of Blue River's technological advancements with John Deere's domain knowledge exemplifies the prospective synergy between robots and agriculture, facilitating enhanced efficiency and ecologically sustainable practices in farming [20].

Figure 2.11 Vision robotics [19].

- Harvest Automation: Harvest Automation has developed a range of robots for use in greenhouse operations [12]. Their products include a mobile platform that can move pots and trays of plants as well as a sorting and packaging robot as illustrated in Figure 2.13 [21]. Their solutions focus on automating tasks like plant and crop handling, transplanting, and container movement. By utilizing mobile robots equipped with sensors and advanced algorithms [13], Harvest Automation [30] streamlines the movement of plants within greenhouses and nurseries, reducing the need for manual labor and improving operational efficiency. These robots navigate autonomously and work collaboratively to transport plants while minimizing damage. The technology addresses challenges such as labor shortages, seasonal demands, and the need for careful plant handling. Harvest Automation's innovations showcase the potential of robotics in enhancing productivity, reducing costs, and revolutionizing plant management practices in modern agriculture.

Figure 2.12 Blue river technology [20].

- Agrobot: Agrobot is a company that has developed a strawberry harvesting robot. The robot uses sensors to identify ripe strawberries and then uses robotic arms to gently pick the fruit as illustrated in Figure 2.14 [22]. Agrobot robots combine computer vision, sensors, and gentle grippers to identify ripe berries and harvest them without damaging the plants. This approach improves efficiency, reduces labor costs, and minimizes waste. Agrobot's solution addresses challenges posed by labor shortages and the fragility of berry crops. The company's dedication to research and development has led to breakthroughs in berry harvesting technology, showcasing the potential of robotics to revolutionize specialized agricultural processes. Agrobot's contributions reflect the ongoing evolution of modern

farming practices, emphasizing sustainable practices and increased productivity through automation.

Figure 2.13 Harvest automation [21].

Figure 2.14 Strawberry harvesting robot [22].

- FarmWise: FarmWise is a company that has developed a range of robots for use in vegetable farming. Their products include a weeding robot that uses machine learning algorithms to identify and remove weeds as illustrated in Figure 2.15 [23]. The primary objective of the organization is to enhance the productivity and environmental viability of agriculture *via* the automation of labor-intensive activities. FarmWise has successfully engineered robotic systems that are equipped with advanced camera technology and machine learning algorithms. These sophisticated robots possess the capability to accurately detect and subsequently eliminate unwanted weeds from agricultural areas, all without resorting to the use of chemical pesticides. The use of this technology facilitates the adoption of sustainable weed control practices, hence mitigating the environmental repercussions associated with agricultural activities. FarmWise's robots effectively optimize resource allocation and improve crop output by accurately targeting weeds. The company's strategy effectively tackles several issues encountered in modern agriculture, including the emergence of herbicide resistance, scarcity of labor, and the growing need for sustainable agricultural practices. FarmWise's dedication to state-of-the-art technology exemplifies the potential of robots in revolutionizing conventional

agricultural methods toward a future characterized by enhanced productivity and sustainability.

Figure 2.15 Farm Wise robots [23].

- Naïo Technologies: Naïo Technologies is a French company that has developed a range of robots for use in vineyards and vegetable farms. Their products include a weeding robot and a harvesting robot as illustrated in Figure 2.16 [24]. The robots they have developed exhibit innovation in their design, enabling them to do activities such as weeding and hoeing with a high degree of accuracy and little adverse effects on the environment. These robotic systems use sophisticated sensory technology, computerized visual

perception, and artificial intelligence algorithms to discern between cultivated crops and undesirable vegetation, therefore enabling precise and selective eradication of the latter without causing any detrimental effects on the land [25]. The technology developed by Naïo aims to facilitate weed management without the use of chemical substances, therefore minimizing the reliance on herbicides and fostering the use of organic agricultural methods. The adaptability of robots to diverse crops and field circumstances enables them to effectively tackle difficulties such as labor shortages and the need for sustainable agriculture practices. Naïo Technologies serves as a prime illustration of the transformative capacity of robots in the realm of conventional agriculture, as it effectively enhances resource allocation, mitigates ecological impact, and fosters the development of more productive and environmentally sustainable farming methodologies [35].

Figure 2.16 Naïo weeding and harvesting robot [24].

These are just a few examples of the ways in which robotics technology is being used in agriculture today. As technology continues to evolve and improve, it is likely that we will see even more innovative applications in the years to come.

2.9 Implications and Recommendations

The implications and recommendations of using robotics in agriculture are many and varied. Here are some key implications and recommendations to consider:

- Increased productivity: One of the main benefits of using robotics in agriculture is increased productivity. Agricultural robots can perform tasks more quickly and efficiently than humans, leading to higher crop yields and reduced labor costs.
  Recommendation: Farmers should consider investing in robotics technology as a way to increase productivity and profitability.
- Environmental impact: Robotics technology has the potential to reduce the environmental impact of agriculture by reducing the use of pesticides and fertilizers and promoting more sustainable farming practices.
  Recommendation: Farmers should prioritize the adoption of sustainable farming practices and invest in robotics technology that supports these practices.
- Human displacement: The use of robotics in agriculture may displace human labor, particularly in areas where labor is already in short supply.
  Recommendation: Governments and industry stakeholders should work to address the social and economic implications of using robotics in agriculture, including retraining programs for displaced workers and support for affected communities.

- Cost-effectiveness: The cost of robotics technology can be a barrier to adoption for many farmers, particularly smallscale farmers with limited resources.
  Recommendation: Governments and industry stakeholders should work to make robotics technology more accessible and affordable to farmers through subsidies, grants, and other support programs.
- Technical expertise: The operation and maintenance of robotics technology require specialized technical expertise, which may be a barrier for many farmers who are not familiar with the technology.
  Recommendation: Governments and industry stakeholders should invest in education and training programs to help farmers learn how to operate and maintain robotics technology.

2.10 Conclusion

In conclusion, robotics in agriculture represents an exciting and rapidly growing field that offers a range of benefits and opportunities for farmers and the agricultural industry as a whole. The use of robotics technology in agriculture can help improve efficiency, productivity, sustainability, and safety while reducing labor costs and minimizing the environmental impact of farming practices. However, the adoption and use of robotics

in agriculture also present a range of challenges, including high initial costs, technical complexity, dependence on technology, ethical concerns, and regulatory challenges. To fully realize the benefits of robotics in agriculture, it is essential to address these challenges and provide support and resources to farmers and other stakeholders to promote the adoption and responsible use of robotics technology. Overall, the continued development and adoption of robotics in agriculture are likely to play an increasingly important role in meeting the growing global demand for food while promoting more sustainable and efficient farming practices. As technology continues to evolve and become more accessible, it is likely to become a critical tool for farmers in meeting the challenges and opportunities of the 21st-century agricultural landscape.

References

1. Zhang, Z.B.; Li, P.; Zhao, S.L.; Lv, Z.M.; Du, F.; An, Y.J. An Adaptive Vision Navigation Algorithm in Agricultural IoT System for Smart Agricultural Robots. *CMC Comput. Mater. Continua*, 66, 1043–1056, 2021 [CrossRef]
2. Wang, Z.; Xun, Y.; Wang, Y.; Yang, Q. Review of smart robots for fruit and vegetable picking in agriculture. *Int. J. Agric. Biol. Eng.,* 2022, 15, 33–54.

3. The Latest State of Food Security and Nutrition Report Shows the World Is Moving Backwards in Efforts to Eliminate Hunger and Malnutrition. Available online: https://www.who.int/news/item/06-07-2022-un-report–global-hunger-numbers-rose-to-as-many-as-828-million-in-2021/ (accessed on 28 October 2022).
4. Kumar, K.N.; Pillai, A.V.; Narayanan, M.K.B. Smart agriculture using IoT. *Mater. Today Proc.,* 2021. [CrossRef]
5. Zhou, H.; Wang, X.; Au, W.; *et al*. Intelligent robots for fruit harvesting: recent developments and future challenges. *Precision Agric*., 23, 1856–1907 (2022). https://doi.org/10.1007/s11119-022-09913-3
6. Sustainable Development Indicator Group. https://www.hq.nasa.gov/iwgsdi/Agricultural_Land.html, 1996 (accessed 07.03.2020).
7. Science and Technology Options Assessment (Precision Agriculture). http://www.europarl.europa.eu/thinktank/infographics/precisionagriculture/public/, 2020 (accessed 07.03.2020).
8. Geng, A.; Hu, X.; Liu, J.; Mei, Z.; Zhang, Z.; Yu, W. Development and Testing of Automatic Row Alignment System for Corn Harvesters. *Appl. Sci.,* 2022, 12, 6221.
9. Automated Agriculture: Robots and the future of Farming. https://www.theautomationengineer.com/marketssectors/aut

omated-agriculture-robots-future-farming/, 2017 (accessed 07.03.2020).

10. Meivel S, Dinakaran K, Gandhiraj N and Srinivasan M 2016 Remote sensing for UREA Spraying Agricultural (UAV) system. *3rd International Conference on Advanced Computing and Communication Systems (ICACCS),* vol 1 (IEEE) pp 1–6, 2016.
11. Meshram, A.T.; Vanalkar, A.V.; Kalambe, K.B.; Badar, A.M. Pesticide spraying robot for precision agriculture: A categorical literature review and future trends. *J. Field Robot.,* 2022, 39, 153–171. [CrossRef]
12. Gat G, Gan-Mor S and Degani A. Stable and robust vehicle steering control using an overhead guide in greenhouse tasks. *Comput. Electron. Agric.,* 121 234–44, 2016.
13. Vakilian K A and Massah J A farmer-assistant robot for nitrogen fertilizing management of greenhouse crops. *Comput. Electron. Agric.,* 139 153–63, 2017.
14. Van Henten E J, Hemming J, Van Tuijl B A J, Kornet J G, Meuleman J, Bontsema J and Van Os E A An autonomous robot for harvesting cucumbers in greenhouses. *Auton. Rob.,* 13 241–58, 2002.
15. Bhat, A.G. Arecanut tree-climbing and pesticide spraying robot using servo controlled nozzle. In *Proceedings of the*

2019 Global Conference for Advancement in Technology (GCAT), Bangalore, India, 18–20 October 2019; pp. 1–4.

16. Li H, Wang H, Shang Z, Li Q and Xiao W. Application of RFID in agricultural seed quality tracking system. *8th World Congress on Intelligent Control and Automation* (IEEE), pp 3073–7, 2010.
17. Mohanraj I, Ashokumar K and Naren J Field monitoring and automation using IOT in agriculture domain. *Procedia Comput. Sci.,* 93 931–9, 2016.
18. Prathibha S R, Hongal A and Jyothi M P, IOT Based monitoring system in smart agriculture. *International Conference on Recent Advances in Electronics and Communication Technology (ICRAECT),* (IEEE) pp 81–4, 2017.
19. Baranwal T and Pateriya P K Development of IoT based smart security and monitoring devices for agriculture. *6th International Conference-Cloud System and Big Data Engineering (Confluence),* (IEEE) pp 597–602, 2016.
20. Suma N, Samson S R, Saranya S, Shanmugapriya G and Subhashri R IOT based smart agriculture monitoring system. *Int. J. Recent Innov. Trends Comput. Commun.,* 5 177–81, 2017.
21. Cambra C, Sendra S, Lloret J and Garcia L An IoT service-oriented system for agriculture monitoring. *2017 IEEE International Conference on Communications (ICC),* (IEEE) pp 1–6, 2017.

22. Gondchawar N and Kawitkar R S IoT based smart agriculture. *Int. J. Adv. Res. Comput.Commun. Eng.,* 5 838–42, 2016.
23. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
24. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
25. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini Robotic Face Recognition System,” *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.
26. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
27. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.”

Design and Development of Efficient Energy Systems (2022): 245-263.

28. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics,* 2023, 13, 3485.
29. M. Vega-Heredia, R. E. Mohan, Y. W. Tan, J. S. Aisyah, A. Vengedesh, S. Ghanta and S. Vinu, Design and modelling of a modular window cleaning robot, *Autom. Constr.*, 103, 268-278, 2019.
30. Tangarife H I and Díaz A E Robotic applications in the automation of agricultural production under greenhouse: A review. *2017 IEEE 3rd Colombian Conference on Automatic Control (CCAC),* (IEEE) pp 1–6, 2017.
31. Ruiz-Canales A and Ferrández-Villena M New proposals in the automation and remote control of water management in agriculture: Agromotic systems. *Agric. Water Manage.,* 1–3, 2015.
32. Singh S, Chana I and Buyya R Agri-Info: cloud based autonomic system for delivering agriculture as a service arXiv preprint arXiv:1511.08986, 2015.
33. Shingarwade N D and Suryavanshi S C 2017 Performance Evaluation of Cloud Based Farm Automation. *International*

Conference on Computing, Communication, Control and Automation (ICCUBEA), (IEEE) pp 1–6, 2017.

34. Dubey, Ashutosh Kumar, and Ravita Chahar. "Revolutionizing agriculture: harnessing IoT for productivity, sustainability, and innovation." *2023 6th International Conference on Recent Trends in Advance Computing (ICRTAC). IEEE*, 2023.
35. Yang, Jian, Amit Sharma, and Rajeev Kumar. “IoT-based framework for smart agriculture.” *International Journal of Agricultural and Environmental Information Systems (IJAEIS),* 12.2 (2021): 1-14.

Note

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3

Digital Farming: The New Era of Agriculture-Opportunities and Challenges

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Abstract

Agriculture has its direct effects on the economy of the country, farmers' income, and the population who lives there. Therefore, an appropriate approach to do effective farming is required so that farmers have optimum results from their crops and cultivation. Digital farming enables the farmer to do so by using different tools and techniques. By using drones and sensors, farmers can spray pesticides, grid sampling can be applied to check the fertility of soil like using the global positioning system, yield monitors can be used to monitor the crop yield from time to time and feed the same data in software for later analysis and comparison, variable rate technology is also one of

the techniques to control the rate of fertilizers, etc., and different precision farming techniques can be applied. The current pandemic situations raised the requirements of digital farming and the need for relative education on digital farming. In this chapter, the opportunities and challenges of digital farming are discussed: how digital farming techniques will help the nation or farmers grow their economy in a sustainable manner and what are the real problems that become challenges for farmers to accept the concept of digital farming.

Keywords**:** Digital farming, techniques and tools, precision farming, opportunities, challenges, smart farming monitoring system services (SFMSS)

3.1 Introduction

The conventional way of doing agriculture can be characterized as a basic style of yielding. It holds the intensive use of traditional tools, manpower, and prior information on farming. The use of traditional tools, natural resources, organic fertilizers, and cultural beliefs is very usual when we talk about old ways of farming. Here farmers do farming based on the principles of mixed farming, with traditional ways and without using any artificial techniques. Traditional farming usually involves small plantations established according to local

tradition. It mainly uses organic manure. Traditional farming uses organic manures, synthetic chemicals, and fertilizers, which are usually genetically modified, to get the best out of a particular crop or crops. Here significant amounts of organic and dynamism inputs are required in this method that undermine a kind of ecology. It is still used by about 50% of the world's population. In traditional agriculture, irrigation is also done in a traditional way, in which some traditional methods of irrigation include the basin method, drain irrigation method, and strip irrigation method.

Along with having some advantages over traditional farming methods, it also has some drawbacks compared to digital farming. Some of these advantages/drawbacks are as follows:

- Crops are purer without manure: These crops are fresh and are healthier and more beneficial.
- Crops without manure can be sold at a higher price.
- Crop waste after decomposition can be used again as a fertilizer for soil.
- In traditional farming, farmers must spend about 15 h to harvest the crop, which is much more than in high-tech farming.
- Soil is used in traditional farming due to which decomposition takes longer time. In this, the crops are very

prone to soil diseases.

- Crops take longer time to harvest, so crops are becoming expensive to earn the profit.
- Pesticides are used to protect crops from pests, which make the plants not so healthy.
- Plants are not well nourished in traditional farming because the roots do not penetrate deep enough into the soil to absorb water and minerals.
- Traditional farming is heavily dependent on rainfall to water the plants. In this, the quantity of crops produced per acre is reduced, which causes starvation in the event of drought.

To overcome the drawbacks of old farming maneuvers, new technologies can be welcomed. Internet of Things (IoT) and artificial intelligence play the main role here to automate most of the farming decisions. IoT-embedded tools can be used for soil monitoring and water management [12]. The combination of IoT with cloud computing makes the farmer high-tech to reduce human-made errors by replacing human-based decisions with computer-based decisions to make economic benefits [11].

3.1.1 Digital Farming

Digital farming transmits quality agriculture information to appropriate locations, methods, and decisions to illuminate and predict farming issues across all farms. An important part of digital farming is timely access to quality information and decision making. Digital farming through digital technologies has the potential to provide the farmers with the knowledge and capabilities to meet the challenges related to farming and seize growth opportunities. Through digital farming, agricultural products can be made better, and its efficiency can be enabled. It provides an increase in value for the farmers. Digital farming is a facsimile of the application of modern information and communication technologies in the field of agriculture. Digital farming aims to form the basis of a decision-making support system for agricultural management. Digital farming provides useful information about crop growth, climate change's use in farming, and labor issues. It makes crops useful by attracting a lot of technical attention from engraining and watering to healthiness and collecting.

3.1.2 Automated Farming

Alterations in the global aging demographic have led to the adoption of automation in agricultural practices. Automation

and control system manufacturers have felt this change in the agriculture industry. With this change, the manufacturers have seen their sales increase as well. In the last few years, agricultural robots have also been included in agricultural work. Agricultural robots select the soil and crops according to their needs and reduce the need for manual labor. UAV/drone also generated the highest revenue among all the modern technologies used in smart farming. Most of the robots are used for deployment crop management. Planting seeds has become easier than a laborious process with automatic seed sowing machines in agriculture. With these automatic sowing machines, sowing can be done on the land much faster, which also saves time. Planting the seed at the correct depth and spacing the plants at the proper distance for optimum growth are two factors required for effective seeding [3]. Figure 3.1(a and b) shows the process of sowing seeds.

Apart from sowing seeds, an automated process has a watering and irrigation method that permits agriculturalists to rein when and how much water their harvests get. Weeding and pest control are also both delicate aspects of plant being looked after, and these farm duties are perfect for self-sufficient automata. The basic perception of integrating self-directed farming into farming is to increase efficiency, product yield, and quality, because of which the dependence on manual labor can be

reduced. We can say that by using autonomous farming, the farmers of the future will expend their phase doing tasks like overhauling apparatus, debugging robotic coding, investigating data, and scheduling farm operations. In contrast, the ancestors of the peasants spent most of their time in heavy labor.

(**a**)

(**b**)

Figure 3.1 (a) Manual process of sowing seeds. (b) Automated process of sowing seeds [3].

3.2 Tools and Methods Used in Digital Farming

There are several tools that can be used in digital farming to help farmers not only in the cultivation process but also in planting and many more. In the past 10 years the use of these tools has increased. The tools that are used in digital farming are discussed below.

3.2.1 GPS

The Global Positioning System (GPS) is known as a precision farming technique, where farmers can remotely control their farming in case of bad weather, darkness, downpour and fog, etc. GPS locates the position of a farmer's lands or the machines in a farmer's lands with the help of satellites. Satellites can track the farmer's lands with the help of meters that are installed on land. GPS helps to view any part of land from a remote location, and thus farmers can maintain their lands, crops, and other related things. GPS techniques also make the farmer sprinkle pesticide and fertilizers by mapping the location with their home screen. In conventional farming, Global Positioning System-based solicitations are being used mostly for farm planning, field mapping, tractor guidance,

variable rate applications, soil sampling, and yield mapping. Through Global Positioning System, farmers get to know about the small reflectivity area surroundings like rain, dust, fog, and darkness, which helps them during their work.

3.2.2 GIS

The Geographical Information System (GIS) works in combination with GPS. They both complement each other. It is a software application that is used to store and maintain information or feedback data from GPS. Thus, GPS can be used to collect information, and that information can be analyzed with the help of GIS.

3.2.3 Grid Sampling

Grid sampling is a technique where the land is divided into small chunks and the soil from these chunks are sent to the lab for analysis. Therefore, we can find which type of soil from a particular grid is good for which type of crop.

3.2.4 Variable Rate Technology

Variable rate technology is a technique that is used to control the amount of seeds, pesticides, fertilizers, and other things that

are applied in the field. Thus, with variable rates, we can control the application of these things in our fields.

3.2.5 Yield Monitors

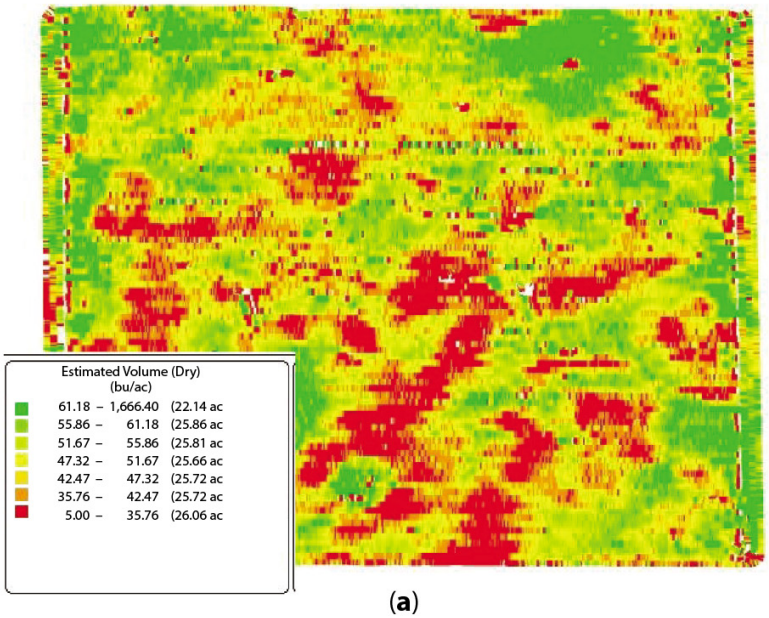
Yield monitors are the devices which are used to collect data about crops. These devices are used by farmers to improve the efficiency of their crops and to reduce cost [2]. By using these devices, farmers can track the moisture of grains and collect data about crop mass flow tons per hectare. Yield monitors are used to record the crop yield and moisture. After collecting the data, these records are displayed in the form of a map in the monitor as shown in Figure 3.2(a).

3.2.6 Remote Sensors

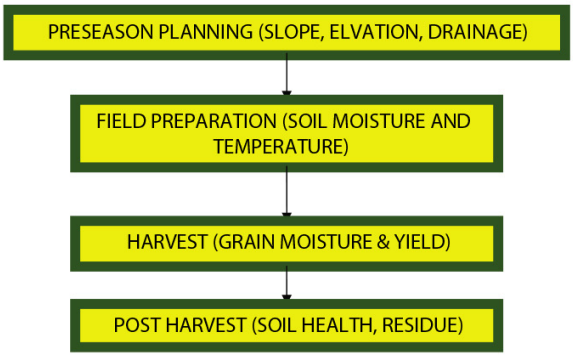
Here sensors are used to track the crop as well as the field being far away from the field. The sensors that are used can be aerial or satellite sensors. Sensors sense the changes on the basis of different colors that appear in the field such as soil color, crop development color, water, field boundaries, and so on. On the basis of colors, these analyze the status of crop- and field-related terms, and the farmer can see these records on the software or monitor screens which will be connected with that sensor. The information that is transferred by the sensor is done

by electromagnetic radiation (EMR). Some of the applications of using a remote sensor (RS) are given in Figure 3.2(b):

1. RS are used in decision making at the time of preseason planning. It includes decisions related to crop type, field operation schedule, amount of seeds for planting, etc.

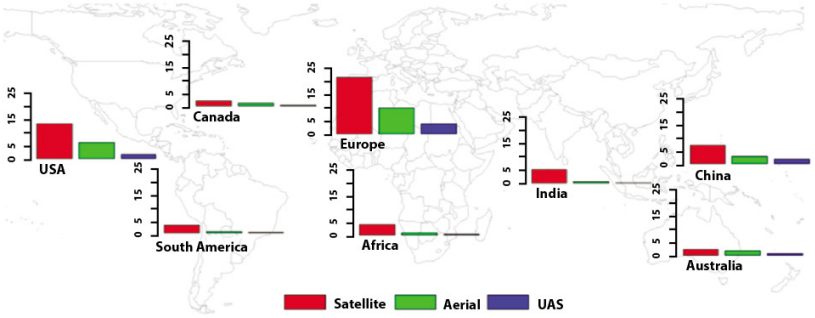

Estimated Volume (Dry)
(bu/ac)
61.18 – 1,666.40 (22.14 ac
55.86 – 61.18 (25.86 ac
51.67 – 55.86 (25.81 ac
47.32 – 51.67 (25.66 ac
42.47 – 47.32 (25.72 ac
35.76 – 42.47 (25.72 ac
5.00 – 35.76 (26.06 ac

(**a**)

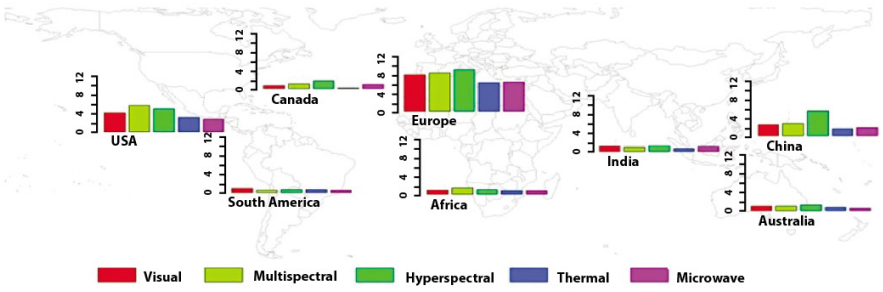

PRESEASON PLANNING (SLOPE, ELVATION, DRAINAGE)
FIELD PREPARATION (SOIL MOISTURE AND TEMPERATURE)
HARVEST (GRAIN MOISTURE & YIELD)
POST HARVEST (SOIL HEALTH, RESIDUE)

(**b**)

(**a**)


USA
Canada
Europe
South America
Africa
India
China
Australia
Satellite
Aerial
UAS

(**b**)


USA
Canada
Europe
South America
Africa
India
China
Australia
Visual
Multispectral
Hyperspectral
Thermal
Microwave

(**c**)

Figure 3.2 (a) Soybean yield field [2]. (b) Stages of production agriculture [10]. (c) Charts of spatial resolution of satellite sensors used since 1999 [10].

2. RS are used in field preparation, as crop germification takes place when the soil moisture content is according to the crop. Thus, it will help to analyze the moisture level of soil so that fertilizers can be used at the right time for better results. The land surface vegetation index [4, 10, 13, 14] is the term used to make decisions about field preparation tasks. Optical and thermal remote sensing data is used to find out that index.
3. RS are used in the stage of planting as well. Crop planting means maintaining the track of crop density that defines the emergence of crop and population of crop for knowing the harvesting logistics. If this data is available at the correct time, then the farmer can use it to do replanting at an accurate time without losing the correct time span or, we can say, without losing the correct season or time to catch the growth. It also helps to take other crop-related decisions such as herbicide, fertilizers, and pesticides used in mid-season so it will result in a good and healthy harvest.
4. RS is used in seasonal applications to find out crop stress due to pest and weed infestation. By using RS, one can see if there is any attack of pathogens on their crop or not, and if yes, then they can take precautionary measures to save their crop. For eliminating weed, herbicide sprinkling is used but

before RS technique the spray of herbicides is done all over the field, which results in wastage of herbicides and increase in cost; however, with the use of high-resolution imagery, RS can be used to find out the patches where the weed is spread and only that portion will be subjected to spraying of herbicides.

5. Harvest means with the help of RS imagery the yield can be monitored. With the help of a yield map, yield density and variability can be estimated. This facility of early prediction of yield helps decrease yield-related losses.
   Figure 3.2(c) shows the different types of satellite used from 1999 by different countries, and their use rapidly increased and spread over the globe. Table 3.1 shows the readings collected from remote sensors about yield assessment. RS is used to assess spatial and temporal data.
6. RS is also used in the post-harvest process and can be used to make decisions on residue. Crop residue management is important to continue the effectiveness of the soil of that field.

Table 3.1 Remote sensor data studies for crop yield analysis [10].

Spatial/temporal resolution	**Platform/sensor/data**	**Accuracy compared to *in situ* data**	**(**
0.02 m; three-time acquisition during early and midseason crop development	Visual sensor on board UAS	R 2 of up to 0.74	(
30 m; imagery for 2008 to 2013	Multispectral sensor; Landsat 5 and 7	R 2 = 0.14 to 0.58 for corn and 0.03 to 0.55 for soybean yield estimation	(
0.3 m; bare soil imagery	Multispectral sensor onboard a manned	R 2 = 0.52 to 0.97 for	(

Spatial/temporal resolution	Platform/sensor/data	Accuracy compared to *in situ* data	
	aircraft	corn yield estimation	

3.2.7 Auto-Guidance System

This system is used to perform symmetry while performing field operation as seed roping in a straight row and returning to the same row.

3.3 Pandemic Effects on Traditional Farming

The traditional method of agriculture involves labor and manpower from start to end. Thus, in the COVID-19 pandemic, the process of agriculture was affected badly. Around 1 billion people from all over the world get their livelihood from agriculture. They are the ones who share a large chunk of the economy also. Lockdown was the measure or preventive method used by the government to control the spread of COVID.

However, due to lockdown, the farmers become helpless to go to their field and continue the process required according to the cycle—for example, many farmers did not get access to go to the market and buy new seeds or fertilizers for their crop. The agriculture process has many steps which complete the whole process of farming, such as seeding, planting, nurturing, harvesting, and shipment to markets [5] that almost involve human labor in every step but which gets hindered due to COVID.

Low supplies and attacks of weeds and pathogens destroyed the crop, and due to impositions on transportation, herbicides and pesticides cannot be purchased and sprayed on yield at the time, thus resulting in reduced yields.

Due to traveling restrictions imposed in the nation, labor becomes unreachable to field and the production cycle gets disturbed. Reverse migration in COVID also will be a reason for disturbed agriculture at that time.

Isolation, lockdown, and self-quarantine are the key factors used during the COVID-19 spread that become useful, but due to so much restriction, the required procedure in agriculture become vanished and result into loss of yield or reduced production.

Due to COVID, the production of that year decreased, and that hiked the prices of items due to low availability and high demand as represented in Figure 3.3. The prices of many vegetables (VP) that hold vitamin C were noticed to have the highest raise due to the pandemic.

The pandemic effect on traditional farming is that the production cycle was affected by the pandemic badly, and the prices of vegetables or other agricultural products become high as shown in Table 3.2. The livelihood of many farmers gets effected, and so the economy of the nation was also deprived. Therefore, automated agriculture is needed, and its importance came to front so that possible steps of agriculture can be performed by using tools without the need of manpower.

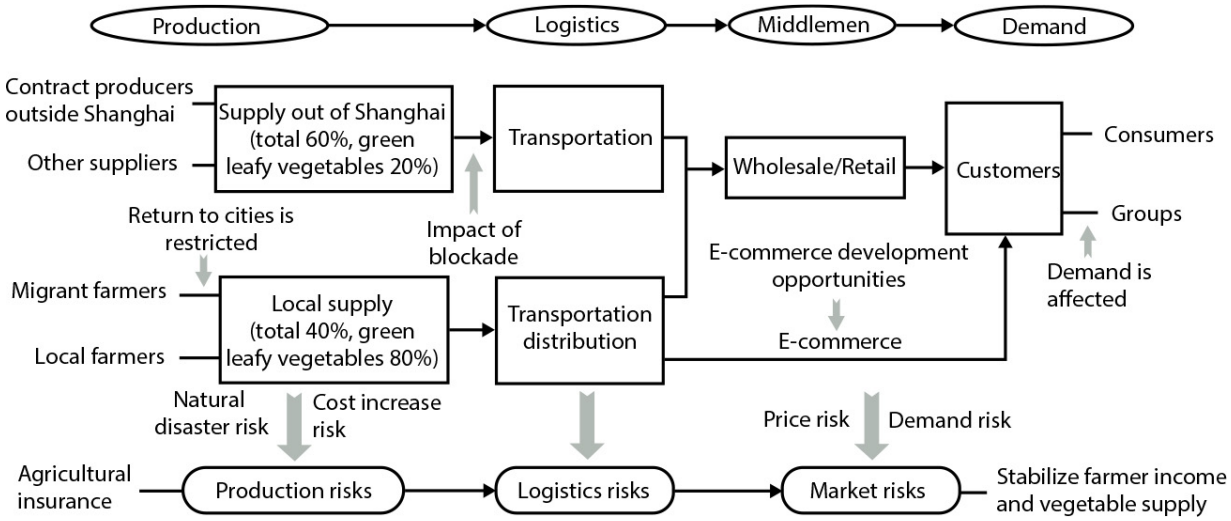


Figure 3.3 Pandemic's effects on the vegetable supply chain and agricultural insurance [7].

Table 3.2 Assessment of vegetable prices during the pandemic between 2020 and 2019 [7].

Price	Weighted vegetables			Pak Choi		
	2020	2019	Decline (%)	2020	2019	Declin (%)
Field price						
Feb	2.43	2.75	11.64	1.75	2.45	28.75
March	1.79	2.71	33.95	1.25	2.55	50.98
Wholesale price						

Price	Weighted vegetables			Pak Choi		
	2020	2019	Decline (%)	2020	2019	Declin (%)
Feb	4.39	4.59	4.36	2.35	3.38	30.47
March	4.22	4.82	12.45	2.31	3.44	32.85

3.4 New Scope at Digital Farming

Every agricultural enterprise has complex equations of inputs and outputs. As some of the agricultural inputs are directly handled by the farmer such as selection of a crop, there are some environmental aspects also that are out of the farmer's control, such as meteorological conditions, typical weather, soil, and water accessibility which are sometimes beyond the control of the farmer. The unexplained effects of digital disruption hold the promise of potential profit. Digital farming refers to a connected, i.e., knowledge-based, agricultural fabrication scheme that employs intellectual networks and data-controlling utensils. It permits the mechanization of ecological procedures in farming.

In the section above, we saw how pandemic affected the production by disturbing agriculture procedure and also affecting the production cycle, farmer's lives, and economy. Thus, searching of all possible methods that can apply to reduce the loss is used and analyzed. In this new use of technologies, we get focused to use, i.e., IoT, cloud computing, sensors, drones, and artificial robots or machines.

From Figure 3.4, we can see how the technologies can be used with farming to make it more automatic and machine-driven.

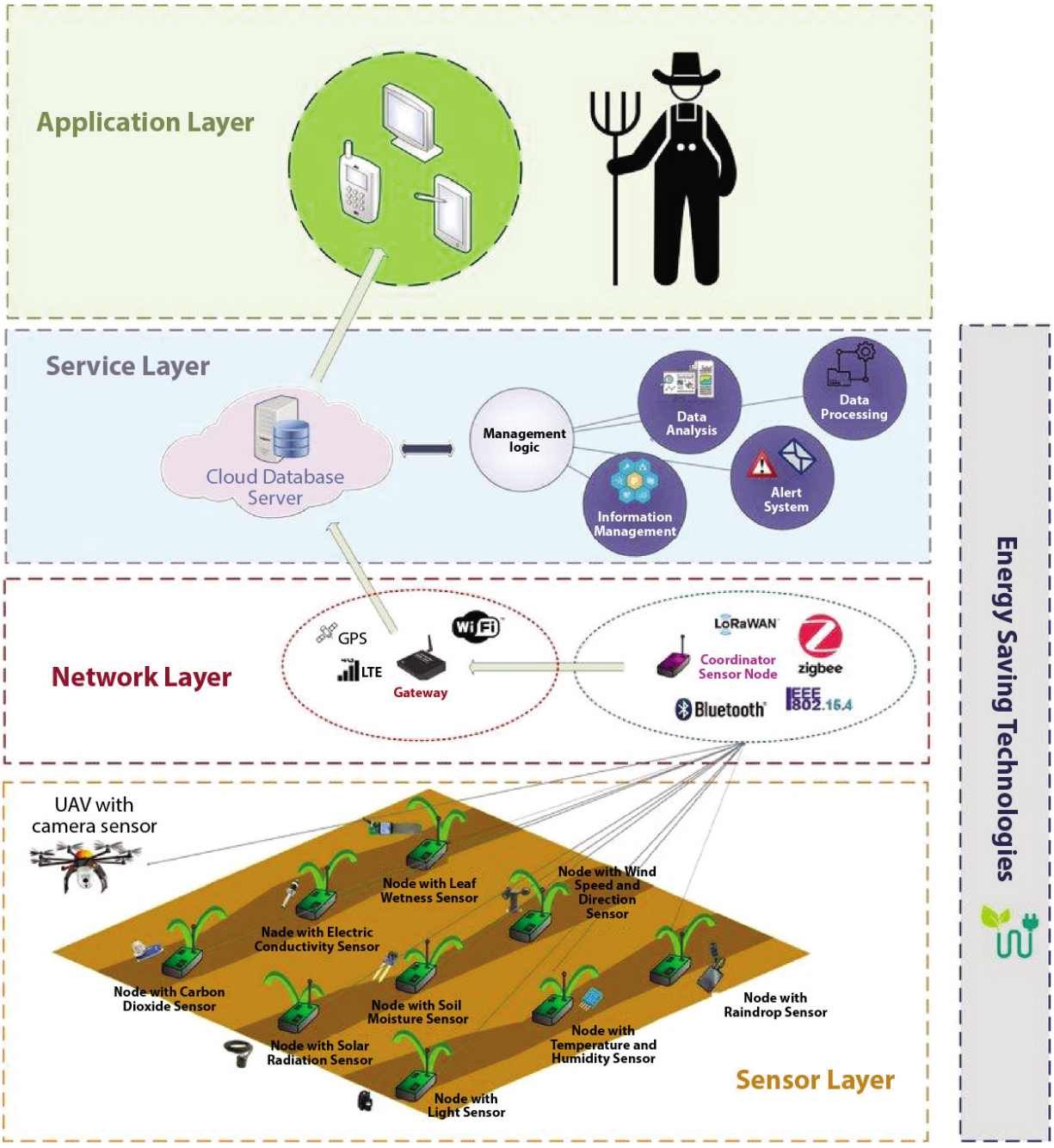


Figure 3.4 Architecture of precision farming [1].

The first layer is known as the sensor layer; the main work in precision farming is to sense the field on different factors. Sensors that can be used for sensing the pH value of soil are

known as electrochemical sensors; location sensors are used to sense the exact location of field longitude, latitude, and altitude [1]; and weather sensors are used to sense factors related to weather which influence the yield. There are factors mainly sensed by these sensors.

There is a different sensor that is used in agriculture for collection as given in Table 3.3 and the analysis of different data to be required. These are mentioned below:

- Moisture and temperature of land
- Air humidity and flow

Table 3.3 Precision crop growing sensor models [1].

Sensor type	Sensor model
Soil moisture sensor	10-HS, SY-HS-220, FC-28
Temperature sensor	LM35, SHT15, DS18B20
Humidity sensor	DHT22, DHT11
Electric conductivity sensor	DFR0300
Wind speed and direction sensor	SEN0170
Barometric pressure sensor	BMP180
Carbon dioxide sensor	CDM4161A, MHZ16
pH sensor	MCP1525
Light sensor	TSL2561, BH1750

Sensor type	Sensor model
Solar radiation sensor	6450 TSR
Thermal sensor	Thermo MAP

- Air flow and land wetness
- Rainfall
- Soil pH value

The second layer is network. In N/w layer, a communication channel is set so that sensed data can be transferred to a destination for further processing. So, we can say that IoT and cloud computing techniques worked here to transfer or save the data.

The next layer is service layer, where the actual work on data begins. The service layer is aimed to analyze the data which is sensed in the first layer and transferred by the second layer. Based on analysis, a further decision on crop plantation and management is done.

The last layer is the application layer where the visualization of decision can be done. What is the result of a particular decision

that can be viewed here? Thus, precision farming is the new scope of digital farming.

Digital farming is identical and the same with smart agriculture and digital agriculture. Therefore, we can say that digital agriculture expertise is a skill-based technologically advanced ideal which is a combination of digital trade and manufacture procedures as the tools mentioned in Table 3.4. This model pointedly challenges agriculturalists to traditional farming practices that may require upskilling and constant change. Here it is very important for farmers to adopt the mindset of digital literacy. Recognizing learning and capacity erection in digital savvy is essential to fill the capacity gap among farmers.

Table 3.4 Smart farming monitoring system services [1].

Service type	Tools	Description
Information management	MySQL Database, management logic	It defines the procedure of assembling, loading, handling, and sustaining evidence in all its forms.
Big data analytics, i.e., BDA platforms	Mahoot, IoT	It defines removing, onslaught, renovating, displaying, and conception of data with an objective to reveal significant and valuable figures that can support in originating verdict and take decisions.
Data processing	Classification algorithms	Classification of facts so as to decrease the

Service type	Tools	Description
		extent of redundant statistics
Data mining	Hadoop, Apache Spark Framework	Orderly and chronological manner of recognizing unseen outlines and evidence in a huge dataset

3.5 Challenges or Difficulties for Digital Farming

In traditional farming, it was problematic for agriculturalists to link production practices and harvest incomes with land inconsistency. In digital farming, it is possible to use more appropriate pesticides, herbicides, and fertilizers and better control the spread of those chemicals. The key advantages in digital farming are reduced costs, producing higher yields, and building an eco-friendlier farm.

In traditional farming, the farm with the farmer is very small, and it generates very few data points. Digital farming requires millions of data points to be appropriately aggregated and

analyzed; computing, storage, and processing power come at a charge. Thus, there is still some distance to be covered here for fields in traditional farming. The exact financial assessments of the total of expertise per unit land/per capita farmer and the associated reserves/yield on investment are not yet obtainable in certain contexts. This makes it particularly hard for everyone to assess technology from the point of view of economic aspects. A challenge for digital farming is also to reduce the cost of technology so that it is available and affordable to small farmers. Thus, it is natural that the farmer's ability to invest financially in technology is very limited. Keeping this in mind, it is necessary that the technology which is affordable to the farmers will be attractive. It can also provide financial returns in quick turnaround time (like harvest season). Ensuring mobility is a challenge for digital farming. Majorly digital farming in India has a better chance of being successful. This is because typical Indian farms are very small in size, and farm plots are of 1 to 2 acres. Moreover, hiring of a farming plot under various preparations is extensively experienced in India. From this, it is common to find out what crop a farmer is cultivating on a particular plot in one season and what the same farmer can transfer to another farm plot in the next season. In such a scenario, equipment that can be simply connected and reinstalled and used for multiple positions

during its lifespan would certainly be more attractive for the agriculturalist to pay and endow.

3.6 Future Scope and Benefits

There are many reasons to implement digital farming solutions in local and commercial farming. While the IoT in the world is fast-tracking the steps on data gathering and automation, significant industries like agriculture can certainly benefit through it. Soil wetness, air temperature, air moisture, and sunlight intensity can be monitored in many areas through digital technology. By collecting data through this, water use and crop yield efficiency in fields can also be improved in large and local farms. As the world's population grows over time, so will farming and food production. Therefore, the use of digital technology will enable low-cost sensors, data insights, and IoT platforms to increase efficiency and production. Advances in machinery have also expanded the scale, speed, and productivity of agricultural equipment. This has led to more productive cultivation in the land more efficiently. Seeds, irrigation, and fertilizers have also improved significantly, helping farmers to increase yields.

Following are some of the benefits of implementing digital farming solutions:

1. Water preservation: Weather forecasting and soil moisture sensors allow the use of water only when needed, which is a major part of farming.
2. To accelerate the decision-making process through digital farming, farmers can imagine fabrication levels, soil wetness, sunlight intensity, and more in real time as all these are key factors in farming.
3. Automated processes in planting, processing, and harvesting in digital farming reduce resource consumption, human error, and overall cost and lower operating costs, resulting in higher profits for farmers.
4. The quality of production can be increased with the use of digital farming. The results in relation to the analysis and treatment of production quality enable the farmers to adjust the processes to enhance the quality of the product. In addition, accurately estimating production rates by area over time can provide detailed estimates of future crop yields and farm value.
5. Through digital farming, all conservation efforts such as increasing water use and production per land unit directly impact the environmental footprint positively.
6. Many important uses of digital farming include monitoring and maintaining farm equipment to predict production rates, labor effectiveness, and failure predictions.

As we have seen in the above-mentioned benefits, farmers can actually obtain the solution or implementation through digital farming overall. It is not only a digital farming solution, but it is the innovative farming method of today. This innovative approach can address the growing concern of the global population's food consumption and environmental footprint. Agriculture in the future will all use sophisticated technologies such as robots, temperature and humidity sensors, aerial images, and GPS technology. This advanced equipment, robotic systems, and precision agriculture will help make farms more profitable, efficient, safe, and environmentally friendly. Apart from this, there should be some specific priorities for the future, namely:

1. Better data collection about digital technologies and digitization should be facilitated at the regional and population level.
2. Information about urban and rural areas should be differentiated by digital technologies so that it can be used according to urban and rural areas.
3. Involving small-scale farmers in the digital agriculture transformation process is feasible to build sustainable business models that provide digital solutions.
4. An index should be created that is based on the development of digital agriculture in the context of the cultural,

educational, and institutional dimensions of a given country.

5. Digital agriculture should study the process in terms of both the basic conditions for digitization and the availability of enablers and the potential economic, social, and environmental impacts.

References

1. A. Triantafyllou, D. C. Tsouros, P. Sarigiannidis and S. Bibi, “An Architecture model for Smart Farming,” *15th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, Santorini, Greece, pp. 385-392, 2019, doi: 10.1109/DCOSS.2019.00081.
2. B. C. Atherton, M. T. Morgan, S. A. Shearer, T. S. Stombaugh and A. D. Ward, “Site-Specific Farming: A Perspective on Information Needs, Benefits and Limitations”, *J. Soil Water Conserv.*, 54 (2), pp. 455-461, April 1999.
3. Meghan Brown, “smart-farming-automated-and-connected-agriculture”, Retrieved from https://www.engineering.com/story/smart-farming-automated-and-connected-agriculture: https://www.engineering.com/story/smart-farming-automated-and-connected-agriculture, March 15, 2018.
4. Toby, Carlson, “An Overview of the Triangle Method for Estimating Surface Evapotranspiration and soil Moisture

from Satellite Imagery", *Sensors*, 7(8), pp. 1612-1629, 2007.

5. Workie, Endashaw *et al.*, "Deciphering the impact of COVID-19 pandemic on food security, agriculture, and livelihoods: A review of the evidence from developing countries", *Curr. Res. Environ. Sustain.*, Vol. 2, 2020, doi:10.1016/j.crsust.2020.100014
6. Geipel, Jakob, Johanna Link, and Wilhelm Claupein, "Combined Spectral and Spatial Modeling of Corn Yield Based on Aerial Images and Crop Surface Models Acquired with an Unmanned Aircraft System", *Remote Sens.*, Vol 6, No. 11, pp.: 10335-10355, 2014, https://doi.org/10.3390/rs61110335
7. Hai-ying GU, Chang-wei WANG, "Impacts of the COVID-19 pandemic on vegetable production and countermeasures from an agricultural insurance perspective", *J. Integr. Agric.*, Volume 19, Issue 12, pp. 2866-2876, 2020, ISSN 2095-3119, https://doi.org/10.1016/S2095-3119(20)63429-3.
8. Sami Khanal, John Fulton, Andrew Klopfenstein, Nathan Douridas, Scott Shearer, "Integration of high resolution remotely sensed data and machine learning techniques for spatial prediction of soil properties and corn yield", *Comput. Electron. Agric.*, Vol 153, 2018, pp. 213-225, ISSN 0168-1699, https://doi.org/10.1016/j.compag.2018.07.016.
9. David B. Lobell, David Thau, Christopher Seifert, Eric Engle, Bertis Little, "A scalable satellite-based crop yield mapper",

Remote Sens. Environ., Vol 164, pp. 324-333, 2015, ISSN 0034-4257, https://doi.org/10.1016/j.rse.2015.04.021.

10. Khanal, Sami, Kushal Kc, John P. Fulton, Scott Shearer, and Erdal Ozkan, “Remote sensing in agriculture—accomplishments, limitations, and opportunities”, *Remote Sens.*, Vol 12, No. 22, pp. 3783, 2020.
11. S. Namani and B. Gonen, “Smart Agriculture Based on IoT and Cloud Computing,” *2020 3rd International Conference on Information and Computer Technologies (ICICT)*, San Jose, CA, USA, pp. 553-556, 2020, doi: 10.1109/ ICICT50521.2020.00094.
12. Suma, D. V., “Internet of Things (IoT) based Smart Agriculture in India: An Overview”, *J. ISMAC* , pp. 1-15, 2021, doi: 10.36548/jismac.2021.1.001
13. Zhang, K., Kimball, J. S., & Running, S. W., “A review of remote sensing based actual evapotranspiration estimation”, *Wiley Interdiscip. Rev.: Water*, 3(6), pp. 834–853, 2016, doi:10.1002/wat2.1168
14. Zhu, W., Jia, S., & Lv, A., “A Universal Ts-VI Triangle Method for the Continuous Retrieval of Evaporative Fraction From MODIS Products”, *J. Geophys. Res.: Atmos.*, 122(19), pp. 10206–10227, 2017, doi:10.1002/2017jd026964.

Note

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4

Challenges and Barriers to Smart Farming Adaptation: A Technical, Economic, and Social Perspective

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Abstract

Smart farming generally refers to the application of advanced technologies in the farming. It has the potential to change agriculture in a big way by making it more productive and

efficient while also being better for the environment. It basically represents the application of the latest technologies to agriculture with the goal of boosting output without sacrificing efficiency or longevity. Smart farming systems give farmers access to real-time data on things like crop development, soil quality, weather, and pest populations, allowing them to make decisions based on facts rather than gut instinct. With better resource management, lower costs, and higher yields, this technology has the potential to completely revolutionize the agricultural industry. Smart farming technology is still in its infancy, and there are many obstacles that must be overcome before it can be widely adopted. Problems with connectivity and data management are examples of technological obstacles; high expenses and a lack of knowledge among farmers are examples of economic and social hurdles. This chapter looks at the problems with smart farming from many different points of view, including technological, economic, and social. Sensors, Internet access, and effective data management systems all pose technical difficulties.

Keywords**:** Smart farming, agriculture, sensors, data analytics, machine learning, precision agriculture

4.1 Introduction

The world's population is growing, and so is the demand for food. As a result, the agricultural industry is facing a multitude of challenges, from water scarcity and soil [40] degradation to climate change and economic pressures [1]. One solution to these challenges is the adoption of smart farming, a promising approach that involves the application of advanced technologies to agriculture. Smart farming can potentially revolutionize the agricultural industry by making it more productive, efficient, and sustainable [2]. However, the adoption of smart farming is not without challenges and barriers. This chapter focuses on the challenges and barriers to smart farming adaptation from technical, economic, and social perspectives. The technical challenges of smart farming include the integration of sensors, Internet access, and data analytics systems [3]. These challenges require an understanding of information and communication technology (ICT) and its application in agriculture. The economic challenges of smart farming include the high cost of implementing smart farming technologies as well as the limited financing options available for farmers [4]. These challenges require the development of innovative financing models and cost-effective solutions for farmers. The social challenges of smart farming include the lack

of knowledge and comprehension of smart farming among farmers as well as the potential impact of technology on rural communities [5]. These challenges require the development of training and education programs that address the needs and concerns of farmers. This chapter also explores the potential impact of these challenges on the adoption of smart farming and the future of the agricultural industry [6, 7]. The implementation of smart farming technologies can lead to significant benefits, including improved resource management, reduced costs, and increased yields. However, the challenges and barriers to smart farming must be overcome to fully realize these benefits. The chapter concludes with recommendations on how governments, industry stakeholders, and farmers can work together to promote the adoption of smart farming technologies and achieve sustainable agriculture practices and food security for future generations [8, 9]. A comprehensive overview of the challenges and barriers to smart farming adaptation is explained in this chapter. It highlights the technical, economic, and social challenges that must be addressed to fully realize the potential of smart farming in the agricultural industry. The chapter also stresses the importance of collaboration and innovation in overcoming these challenges and promoting the adoption of smart farming technologies [10].

4.1.1 Definition and Importance of Smart Farming

Smart farming is an innovative approach to agriculture that leverages advanced technologies such as the Internet of Things (IoT) [36], data analytics, machine learning, and artificial intelligence (AI) to increase agricultural productivity [39], efficiency, and sustainability [11]. The importance of smart farming lies in its potential to address the many challenges facing the agricultural industry today [12]. With a growing global population, food demand is expected to increase significantly in the coming years. At the same time, climate change, soil degradation, water scarcity, and pest infestations are posing significant threats to agricultural productivity and sustainability [13, 14]. Smart farming provides a solution to these challenges by enabling farmers to make data-driven decisions, optimize resource utilization, and minimize environmental impact [15]. One of the key benefits of smart farming is precision agriculture, which involves the use of sensors [38] and data analytics to monitor crop growth, soil moisture, nutrient levels, and pest infestations in real time [16]. This information is then used to create customized management plans for individual crops or fields, which can result in significant improvements in yield and quality. Precision agriculture can also reduce the use of fertilizers and pesticides, leading to cost savings and reduced environmental

impact [17]. Another benefit of smart farming is the use of IoT-enabled livestock management systems [41], which allow farmers to monitor the health and well-being of their animals remotely. This can result in early detection of diseases and more efficient use of resources such as feed and water [18]. Smart farming can also contribute to sustainable agriculture practices by reducing waste and emissions—for example, smart irrigation systems can optimize water usage and reduce water waste, while precision fertilization can reduce the use of chemical fertilizers and their impact on the environment. Smart farming is an essential tool for modern agriculture. It enables farmers to optimize resource utilization, reduce waste, and minimize environmental impact while also improving productivity and quality [19, 20]. Governments, industry stakeholders, and farmers must work together to overcome the challenges and barriers to smart farming adoption and promote its widespread implementation [21].

4.1.2 Objectives and Scope of the Chapter

The objectives of this chapter are to examine the challenges and barriers to smart farming adoption from a technical, economic, and social perspective. This chapter will explore the issues in smart farming practices and technologies. The scope of this chapter is to provide a comprehensive analysis of the various

factors that are limiting the adoption of smart farming practices. This includes technological barriers such as the cost of equipment, data management, and connectivity issues. Economic barriers such as lack of financing and the high cost of adoption will also be explored. Furthermore, the chapter will examine social barriers such as a lack of awareness, knowledge, and training among farmers, which make it difficult for them to adopt smart farming technologies. This chapter will explore the impact that these barriers have on the agricultural industry and the potential for smart farming to address sustainability challenges and food security. The chapter will also examine case studies and examples of smart farming practices and technologies that have been implemented successfully in different parts of the world [22]. The technical aspects of smart farming, such as the use of sensors, data analytics, and machine learning, will be discussed in detail, including their potential benefits and limitations. The economic aspects of smart farming will be explored, including the potential cost savings, return on investment, and financing options for farmers. This chapter will examine the social aspects of smart farming, including the education and training necessary to effectively implement these practices. The chapter will conclude with recommendations for overcoming the challenges and barriers to smart farming adoption, including the need for government policies and

incentives, increased financing options for farmers, and greater awareness and education among farmers about the benefits and practices of smart farming. The scope of this chapter is to provide a comprehensive analysis of the challenges and barriers to smart farming adoption from technical, economic, and social perspectives [23]. This chapter will provide insights into the potential benefits of smart farming practices and technologies and explore ways to overcome the challenges that limit their widespread adoption.

4.2 Technical Challenges in Smart Farming Adaptation

Smart farming is a new technological innovation that involves the use of advanced technologies such as the Internet of Things (IoT), sensors, data analytics, and machine learning to optimize agricultural operations. However, the implementation of smart farming practices is not without its technical challenges. In this section, we will discuss some of the technical challenges in smart farming adaptation [24]. This makes it challenging for farmers to use smart farming technologies effectively, as these technologies require constant and reliable connectivity to provide accurate data. Another technical challenge in smart farming is the cost of equipment. Smart farming requires

significant investment in equipment such as sensors, drones, and data management systems, which can be expensive. This makes it difficult for small-scale farmers to adopt smart farming technologies, as the cost of equipment can be prohibitive. This challenge can be mitigated through government subsidies, financing options, and partnerships with technology providers. Data management is another technical challenge in smart farming. Smart farming generates a large amount of data that needs to be collected, stored, and analyzed in real-time. This requires robust data management systems that can handle large amounts of data and ensure data accuracy and integrity. In addition, smart farming data needs to be standardized to enable interoperability across different systems and platforms [25]. Furthermore, the use of sensors in smart farming is a technical challenge. Sensors are used to monitor various agricultural variables. To address this challenge, smart farming technologies need to be designed to account for these factors and ensure data accuracy. Lastly, smart farming also presents a challenge in terms of cybersecurity [26, 27]. As smart farming technologies involve the use of connected devices and data transmission, they are vulnerable to cyber-attacks. This can result in the loss of data, equipment damage, or even harm to crops or livestock. To mitigate this risk, smart farming systems need to have robust

cybersecurity measures in place, including data encryption, access control, and threat detection. Smart farming presents many technical challenges that need to be addressed for its widespread adoption. These challenges include connectivity, cost of equipment, data management, sensor accuracy, and cybersecurity. Addressing these challenges will require collaboration between technology providers, government agencies, and farmers to develop effective and affordable solutions that meet the needs of the agricultural industry.

4.2.1 Overview of Smart Farming Technologies

Smart farming technologies have the potential to revolutionize the agricultural industry by optimizing crop yields, reducing waste, and improving efficiency. In this section, we will provide an overview of some of the key smart farming technologies.

4.2.1.1 Precision Agriculture

Precision agriculture involves the use of sensors and data analytics to monitor and optimize crop yields. This technology enables farmers to collect real-time data on crop health, soil moisture levels, and other variables, allowing them to make data-driven decisions regarding planting, irrigation, and fertilization.

4.2.1.2 Drones

Drones are increasingly being used in agriculture to monitor crop health, track livestock, and assess the condition of fields. By providing high-resolution images and data, drones can help farmers identify areas of concern and take action before problems arise.

4.2.1.3 Robotics

Robotics is another smart farming technology that is becoming increasingly popular. Robots can be used for tasks such as planting, harvesting, and spraying crops, reducing the need for manual labor and improving efficiency. Additionally, robots can work around the clock, allowing farmers to optimize their operations and increase productivity [28].

4.2.1.4 Livestock Management

Smart farming technologies can also be used to monitor and manage livestock. Sensors and data analytics can be used to track animal health, monitor feed intake, and detect disease outbreaks, allowing farmers to take preventative measures and optimize animal health.

4.2.1.5 Aquaculture

Aquaculture is another area where smart farming technologies are being used.

4.2.1.6 Data Analytics

Data analytics is a critical component of smart farming, as it enables farmers to make data-driven decisions. By analyzing data on crop health, soil quality, weather patterns [37], and other variables, farmers can optimize their operations and maximize their yields [29]. Precision agriculture, drones, robotics, livestock management, aquaculture, and data analytics are just a few of the key technologies that are driving this transformation. As smart farming technologies continue to evolve, we can expect to see even greater improvements in efficiency, productivity, and sustainability in the agricultural sector.

4.2.2 Technical Challenges in Data Management and Connectivity

In the realm of smart farming, data management and connectivity are two of the most critical components of success. However, these areas also present some of the most significant

technical challenges. Key challenges in data management and connectivity in smart farming are covered.

4.2.2.1 Data Management

Smart farming technologies generate vast amounts of data on a continuous basis. This data must be collected, processed, and analyzed in real time to enable farmers to make data-driven decisions. However, managing this data can be challenging. Data must be accurate, reliable, and accessible to be useful, and farmers must have the necessary tools and expertise to analyze and interpret the data. Additionally, data privacy and security must be considered, as farmers need to ensure that sensitive data is protected from unauthorized access.

4.2.2.2 Connectivity

Smart farming technologies rely on Internet connectivity to function. However, connectivity can be a significant challenge in rural areas where farming is often located. Poor connectivity can result in delays in data transmission, inaccurate data collection, and reduced efficiency. Additionally, the cost of implementing connectivity infrastructure in rural areas can be prohibitive, making it difficult for farmers to access the technology they need.

4.2.2.3 Integration

Another challenge in data management and connectivity in smart farming is integrating data from different sources. Farmers may use multiple smart farming technologies, each generating data that must be combined and analyzed to provide a comprehensive picture of the farm's operations. This integration can be complex and time-consuming, requiring specialized knowledge and expertise.

4.2.2.4 Data Standards

Standardization of data is also an issue in smart farming. Different sensors and data analytics platforms may use different formats and protocols, making it challenging to combine data from multiple sources. To address this issue, industry standards are being developed, but implementing these standards can be time-consuming and require significant investment.

Data management and connectivity are essential components of smart farming, but they also present significant technical challenges. Managing large volumes of data, ensuring reliable and accurate connectivity, integrating data from multiple sources, and standardizing data are all critical components of smart farming success. Overcoming these challenges will

require collaboration between industry stakeholders, policymakers, and farmers to ensure that smart farming technologies can be used effectively to improve the sustainability, efficiency, and productivity of agriculture.

4.2.3 Role of Sensors, Internet of Things, and AI in Smart Farming

Smart farming technologies rely on a range of sensors, Internet of Things (IoT) devices, and artificial intelligence (AI) algorithms to provide real-time data on crop growth, soil quality, weather patterns, and other critical variables [30].

4.2.3.1 Sensors

Sensors are used extensively in smart farming to collect data on various aspects of the farm. These sensors can measure soil moisture, temperature, and acidity, as well as track livestock movements and monitor crop growth. The data collected by sensors is critical to enabling farmers to make data-driven decisions about irrigation, fertilization, and pest control.

4.2.3.2 IoT

IoT devices are another critical component of smart farming. These devices are used to connect sensors and other equipment to the Internet, allowing for real-time data collection and

analysis. IoT devices can also automate tasks such as irrigation and feeding, reducing the workload for farmers and improving efficiency. By connecting devices and sensors, IoT enables farmers to monitor the status of their farm from anywhere, at any time.

4.2.3.3 AI

AI is a critical tool in smart farming, enabling farmers to analyze data.

4.2.3.4 Integration

Integrating sensors, IoT, and AI is critical for the success of smart farming. By connecting sensors and other devices to the Internet, farmers can collect real-time data, and AI algorithms can analyze that data to provide insights and predictions. It can pay significant dividends in terms of efficiency, productivity, and sustainability [31].

By collecting real-time data on soil quality, crop growth, critical variables, farmers can make data-driven decisions about their operations. IoT devices can automate tasks and connect devices to the Internet, while AI algorithms can analyze data and provide insights and predictions. The integration of these technologies is essential for the success of smart farming,

enabling farmers to improve efficiency, productivity, and sustainability.

4.3 Economic Barriers to Smart Farming Implementation

Smart farming technologies have the potential to revolutionize the agricultural industry, but their implementation faces significant economic barriers. Figure 4.1 shows the one view of barriers in implimenting smart farming.

- High costs: One of the biggest economic barriers to smart farming is the high cost of implementation. Sensors, IoT devices, and other smart farming technologies can be expensive, and many farmers may not have the resources to invest in these technologies. In addition to the initial cost of equipment, there are ongoing costs associated with data management, maintenance, and upgrades.
- Lack of financing: Even if farmers are interested in adopting smart farming technologies, they may struggle to secure financing to make the necessary investments. Many lenders may view smart farming as a risky investment, particularly if farmers lack experience with these technologies. In addition, many small-scale farmers may not have the credit history or collateral necessary to secure loans.

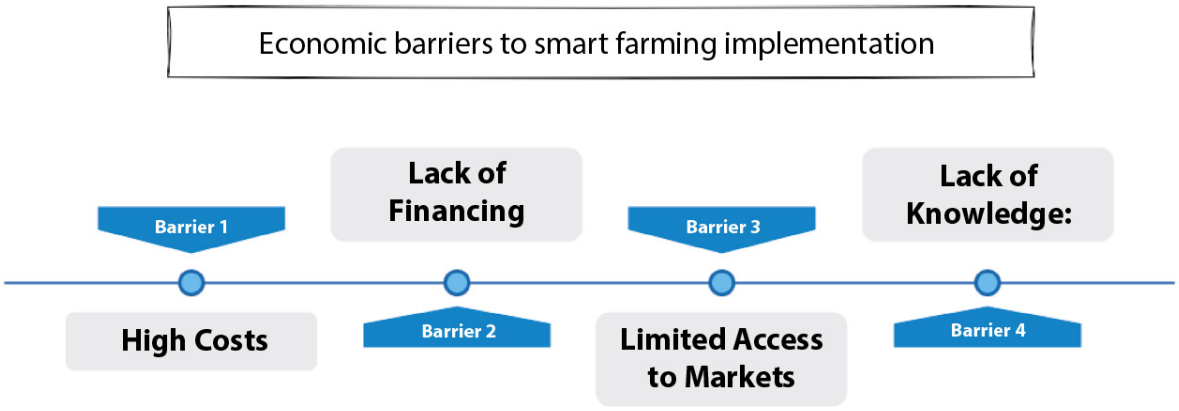

Figure 4.1 Different economic barriers in the implementation of smart farming.

- Limited access to markets: Farmers who adopt smart farming technologies may struggle to access markets that will pay a premium for sustainably produced food. While there is growing demand for sustainably produced food, farmers who adopt these technologies may not be able to sell their products at a premium, particularly if they lack the certifications necessary to prove the sustainability of their operations.
- Lack of knowledge: Many farmers may be interested in adopting smart farming technologies but lack the knowledge and skills necessary to implement these technologies effectively. This lack of knowledge can lead to inefficiencies, equipment malfunctions, and other issues that can undermine the benefits of smart farming.

Smart farming technologies face significant economic barriers, including high costs, limited access to financing, and a lack of knowledge and skills among farmers. To overcome these

challenges, governments, industry stakeholders, and financial institutions must work together to provide farmers with the resources, knowledge, and support necessary to adopt smart farming technologies. By doing so, they can help farmers increase efficiency, productivity, and sustainability, while also meeting the growing global demand for food.

4.3.1 Cost of Implementing Smart Farming Technologies

Smart farming technologies are an exciting development in the agriculture industry, promising to revolutionize the way crops and livestock are grown and managed. However, the cost of implementing these technologies can be a significant barrier for farmers, especially those who are smaller or who have limited resources. In this section, we will explore some of the costs associated with implementing smart farming technologies.

4.3.1.1 Hardware Costs

One of the most obvious costs associated with smart farming technologies is the cost of the hardware itself. This includes sensors, cameras, drones, and other equipment used to collect data and automate tasks. Depending on the type and quantity of equipment required, the cost can be substantial, ranging from a few thousand dollars to tens of thousands of dollars or more.

4.3.1.2 Data Management Costs

Once data is collected, it needs to be stored, processed, and analyzed. This requires sophisticated software and data management systems, which can also be costly. In addition, data management systems need to be maintained and upgraded, which can add to the ongoing costs of implementing smart farming technologies.

4.3.1.3 Training and Support Costs

Farmers and their employees need to be trained on how to use smart farming technologies effectively. This can involve hiring consultants or trainers to provide guidance on how to install, use, and maintain equipment, as well as how to interpret and use data. Ongoing technical support may also be necessary to troubleshoot problems and keep systems running smoothly.

4.3.1.4 Infrastructure Costs

Implementing smart farming technologies may require additional infrastructure to support the collection and analysis of data. This can include upgrading Internet connectivity, installing new power sources, and building secure data storage facilities. These infrastructure costs can add up quickly and may require significant investment.

4.3.1.5 Financing Costs

Even if farmers have the resources to implement smart farming technologies, they may still face financing costs. Loans, leases, or other forms of financing may be required to cover the upfront costs of purchasing equipment and software. However, financing costs can add to the overall cost of implementation, particularly if interest rates are high.

It is difficult to give a precise breakdown of the amount that needs to be spent in each sector for implementing smart farming technologies as it can vary depending on several factors, such as the specific requirements of the farm and the technology used. However, based on general perception, we can assume that the following approximate percentage of the total cost may be spent in each sector:

- Hardware costs: 20–40%
- Data management costs: 15–25%
- Training and support costs: 10–20%
- Infrastructure costs: 20–30%
- Financing costs: 5–10%

In Figure 4.2, the percentage cost division in the implimentation of smart farmiing is given in bar graph that gives clear visulization of cost. Again, these percentages are just an

estimation and can vary depending on the specific needs and circumstances of each farm.

Implementing smart farming technologies can be costly, and farmers must carefully consider these costs when deciding whether to adopt these technologies. However, the potential benefits of smart farming, including increased efficiency, productivity, and sustainability, may outweigh these costs in the long run. To make smart farming technologies more accessible to farmers, governments, industry stakeholders, and financial institutions must work together to provide farmers with the resources and support they need to implement these technologies effectively [32].

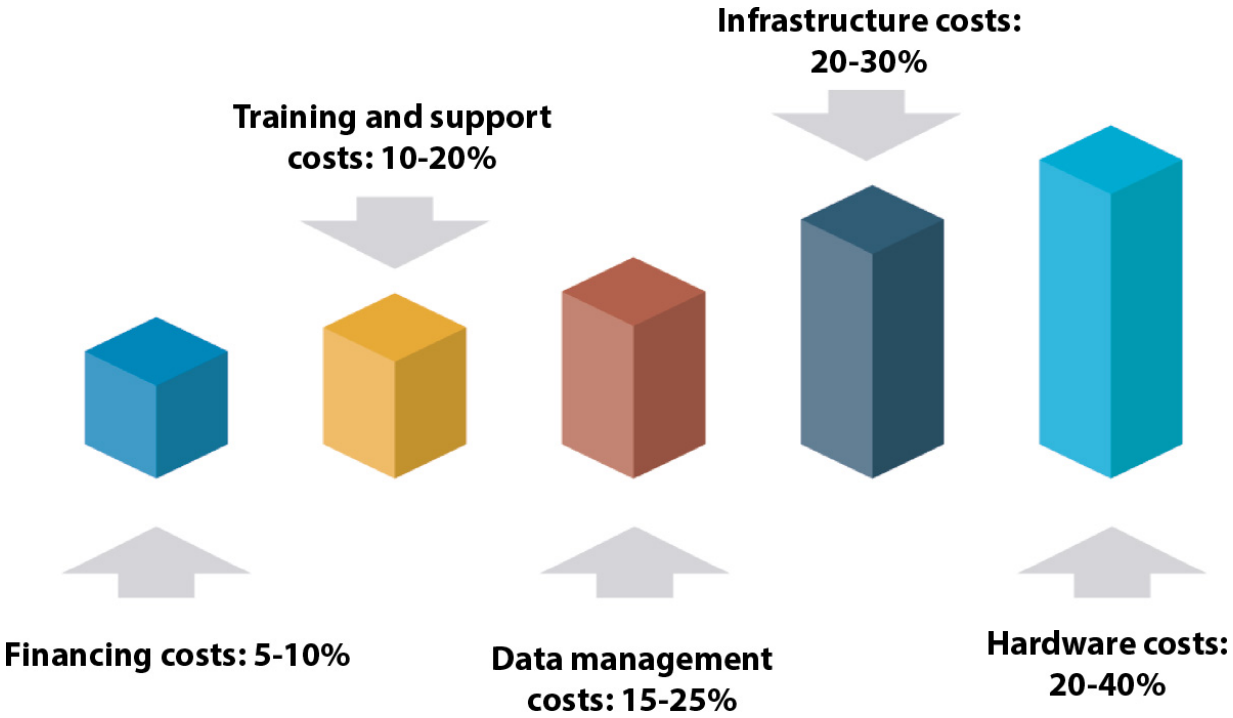

Figure 4.2 Percentage cost division in the implementation of smart farming.

4.3.2 Financing and Investment Challenges

The implementation of smart farming technologies involves significant investments in hardware, software, and infrastructure. However, financing and investment are significant challenges in the adoption of smart farming technologies. Many farmers and small-scale agricultural businesses do not have the necessary capital to invest in these technologies. Additionally, financial institutions are hesitant to finance smart farming projects due to the risks involved. The high cost of smart farming technologies is a significant barrier to their widespread adoption. The initial investment required to purchase hardware such as sensors, drones, and data analytics systems can be substantial. Additionally, the installation and maintenance of these systems require specialized expertise and resources, adding to the overall cost. Moreover, the high cost of energy and Internet connectivity is also a significant factor contributing to the cost of implementing smart farming technologies. Smart farming systems require a stable and reliable Internet connection, which may not be available in all areas, especially in remote rural areas. This lack of connectivity can result in limited access to real-time data, which is critical for smart farming. Additionally, the cost of energy required to power sensors and other devices can be a significant expense, especially in regions with limited access to cheap and reliable

energy sources [33]. Another significant financing challenge is the lack of access to capital for small-scale farmers and agricultural businesses. Financial institutions are often hesitant to finance smart farming projects due to the perceived risks involved. These risks include the uncertainty of crop yields, market prices, and weather patterns, which can impact the return on investment. Additionally, the lack of collateral and credit history of small-scale farmers can make it difficult to obtain loans. In many cases, governments and non-governmental organizations (NGOs) have initiated programs to provide financing and support for the adoption of smart farming technologies—for instance, some governments provide grants and subsidies for farmers to purchase smart farming technologies, while NGOs provide training and technical support to help farmers with implementation. The role of private sector investment cannot be ignored in financing smart farming projects. Several private companies are investing in smart farming technologies, including start-ups and established players in the agriculture industry. These companies are developing innovative solutions that are more affordable, scalable, and accessible to farmers. Financing and investment are significant barriers to the adoption of smart farming technologies. The high cost of hardware, software, and infrastructure, as well as the lack of access to capital and

financing, are significant challenges that must be addressed to promote the widespread adoption of smart farming. Governments, NGOs, and the private sector can play a significant role in overcoming these barriers by providing grants, subsidies, and technical support, and investing in innovative solutions. By addressing these challenges, smart farming can play a crucial role in increasing agricultural productivity, sustainability, and food security [34].

4.3.3 Economic Benefits and Returns on Investment

While the adoption of smart farming technologies involves some initial investments and ongoing costs, it can result in significant economic benefits and returns on investment. By utilizing real-time data on things like soil quality, weather, and pest populations, farmers can make more informed decisions about when and how to plant and fertilize crops, leading to higher yields. This increased productivity can translate into higher profits for farmers [35]. Smart farming can also lead to lower costs. Another economic benefit of smart farming is improved quality and consistency of crops. With real-time monitoring and data analysis, farmers can identify and address issues in crop development early on, resulting in higher-quality crops. This can lead to higher prices and greater demand from consumers. Smart farming technologies can also help farmers

reduce labor costs—for example, automated systems for livestock management and crop harvesting can reduce the need for manual labor, which can be costly and difficult to find in some regions. This can improve the profitability of farms while also improving working conditions for farm laborers. Investing in smart farming technologies can also have long-term economic benefits for society as a whole. By increasing the efficiency and productivity of agriculture, smart farming can help ensure food security for future generations. This can also lead to increased economic development in rural areas, as more efficient and productive farms can create new jobs and stimulate local economies. While there are some initial costs associated with implementing smart farming technologies, the economic benefits and returns on investment can be substantial.

4.4 Social Obstacles to Smart Farming Adoption

While the potential benefits of smart farming are significant, there are also social obstacles that may prevent widespread adoption. One of the main social obstacles to smart farming is the lack of knowledge and understanding of these technologies among farmers. Many farmers may be hesitant to adopt new technologies that they do not fully understand, particularly if they are not confident in their ability to use and maintain these

systems. This can be compounded by language barriers or a lack of access to training and education on smart farming technologies. Another concern is the potential impact of smart farming on employment in the agricultural sector. While automation and other technological advancements have the potential to increase productivity and efficiency, they may also lead to a reduction in the number of jobs available in agriculture. This can be a particular concern in rural areas, where agriculture is often a key source of employment. Data privacy and ownership is another social obstacle that must be addressed in order to promote the adoption of smart farming technologies. Farmers may be concerned about who has access to the data generated by these systems and how this information will be used. There may also be questions about who owns this data and how it can be shared or sold. There may be social and cultural barriers to the adoption of smart farming technologies. Some farmers may be reluctant to adopt new technologies if they perceive them as a threat to traditional farming practices or cultural values. There may also be concerns about the impact of smart farming on the environment and on the well-being of animals raised for food. To overcome these social obstacles, it is important to prioritize education and outreach efforts to ensure that farmers have the knowledge and skills needed to effectively adopt and use smart

farming technologies. This can involve providing training and support, as well as engaging with farmers to better understand their concerns and needs. Governments and industry stakeholders can also play a role in addressing concerns around data privacy and ownership by establishing clear guidelines and regulations for the use of smart farming data. It is important to recognize the potential social and cultural impacts of smart farming and work to address these concerns in a thoughtful and respectful manner. This may involve engaging with local communities to better understand their perspectives and concerns, as well as exploring ways to promote the cultural and social value of agriculture while also embracing new technologies. While there are social obstacles to the adoption of smart farming technologies, these can be overcome through education, engagement, and careful consideration of the social and cultural implications of these technologies. By working together to address these obstacles, we can promote the adoption of smart farming technologies and realize their full potential to transform the agricultural industry and promote sustainable food systems.

4.4.1 Lack of Knowledge and Understanding Among Farmers

One of the main social obstacles to the adoption of smart farming technology is the lack of knowledge and understanding among farmers. Many farmers are not aware of the benefits that smart farming can offer, or they do not have a good understanding of how it works. This lack of knowledge and understanding can lead to a reluctance to invest in the technology or to resist adopting new practices.

One of the reasons for this lack of knowledge is the rural–urban divide. Farmers in rural areas may not have access to the same educational opportunities as their urban counterparts. They may also be more isolated from technological advancements and have limited exposure to smart farming technology. This makes it more difficult to build awareness and understanding of smart farming among rural farmers. Another factor that contributes to the lack of knowledge and understanding is the generational divide. Many farmers come from generations of families who have been farming for centuries, and they may be resistant to change. They may also be more accustomed to traditional methods of farming and may be hesitant to adopt new technologies that are unfamiliar to them. To address this issue, it is important to educate farmers about the benefits of

smart farming and provide them with the necessary training and support to use the technology effectively. Governments, industry stakeholders, and educational institutions can play a key role in providing education and training programs to farmers. In addition to education and training, it is also important to address the cost barriers that may prevent farmers from adopting smart farming technology. Many small and medium-sized farmers may not have the financial resources to invest in new technology. Governments and industry stakeholders can provide incentives and subsidies to offset the costs of implementing smart farming technology. It is important to involve farmers in the development and implementation of smart farming technology. Farmers have valuable knowledge and experience that can help inform the design of new technologies and the development of new practices. By involving farmers in the process, it can help build trust and increase adoption of smart farming technology. The lack of knowledge and understanding among farmers is a significant social obstacle to the adoption of smart farming technology. By providing education and training, addressing cost barriers, and involving farmers in the development and implementation of smart farming technology, we can help build awareness and understanding of the benefits of smart farming and increase adoption rates among farmers.

4.4.2 Behavioral and Cultural Barriers to Change

The adoption of smart farming technologies faces several social obstacles, including behavioral and cultural barriers to change. These obstacles arise due to the inherent conservatism of the farming community and the reluctance to deviate from traditional methods that have been used for generations. Farmers tend to stick to methods that have worked in the past, and there is often a resistance to adopting new practices. One of the key behavioral barriers to smart farming adoption is the lack of trust in technology. Many farmers are skeptical of new technologies and are hesitant to invest in them. This skepticism stems from the fear of the unknown and the perception that technology is too complex and difficult to use. This fear is compounded by the fact that many farmers may lack the technical expertise needed to implement and manage these new technologies. Cultural barriers to change can also hinder the adoption of smart farming technologies. Traditional farming practices are often deeply ingrained in rural communities, and the idea of changing these practices can be met with resistance. This resistance may stem from a sense of pride in traditional methods, a reluctance to embrace new ideas, or a fear of losing control over the farming process. Another cultural barrier is the perception that smart farming is a threat to small-scale farmers. Some farmers may feel that

smart farming technologies are designed for large-scale operations and may not be relevant or accessible to small-scale farmers. This perception may be reinforced by the fact that smart farming technologies are often expensive and require significant investment, making them inaccessible to many small-scale farmers. There is a lack of awareness and education about the benefits of smart farming technologies among the farming community. Many farmers may not fully understand how these technologies can improve their productivity and profitability. This lack of understanding can lead to a reluctance to invest in these technologies, even if they have the potential to provide significant benefits. To overcome these behavioral and cultural barriers, it is important to engage with farmers and provide them with education and training about the benefits of smart farming technologies. This education should be tailored to the needs of the farming community and should focus on how these technologies can improve productivity, reduce costs, and increase profitability. It is also important to work with local communities and stakeholders to build trust in these technologies. This can be achieved by demonstrating the benefits of smart farming through pilot projects and case studies. Building trust with the farming community is critical to overcoming the resistance to change and encouraging adoption of new technologies. It is important to recognize the unique

needs and challenges faced by small-scale farmers. Efforts should be made to develop smart farming solutions that are accessible and affordable to small-scale farmers. This can be achieved through the development of low-cost technologies and innovative financing solutions. Overcoming the behavioral and cultural barriers to smart farming adoption is critical to realizing the potential benefits of these technologies. This can be achieved through education and engagement with the farming community, building trust through pilot projects, and developing solutions that are accessible and affordable to small-scale farmers. By addressing these obstacles, we can create a sustainable future for agriculture and ensure food security for generations to come.

4.4.3 Role of Education and Awareness Raising

Education and awareness-raising play a critical role in overcoming the social obstacles to smart farming adoption. One of the main challenges faced by farmers is the lack of knowledge and understanding of the technology. Smart farming involves the use of complex systems, such as sensors, IoT, and data analytics, that require specialized knowledge and training to use effectively. As a result, many farmers may not be aware of the benefits of smart farming, or they may not understand how to use the technology to improve their operations.

Education and awareness-raising programs can help to address this knowledge gap by providing farmers with the information and training they need to use smart farming technologies effectively. These programs can take many different forms, such as workshops, training sessions, and online resources. They can be delivered by a variety of organizations, including government agencies, industry associations, and private companies. One of the key benefits of education and awareness-raising programs is that they can help to build trust in smart farming technologies. Many farmers may be skeptical of new technologies or may be hesitant to adopt them because of a lack of understanding or a fear of the unknown. By providing farmers with accurate information and practical training, these programs can help to demystify smart farming and demonstrate its value in a tangible way. To providing information and training, education and awareness-raising programs can also help to create a culture of innovation and collaboration within the agricultural community. By bringing farmers, researchers, and other stakeholders together, these programs can foster a spirit of cooperation and encourage the exchange of ideas and best practices. This, in turn, can lead to the development of new and innovative approaches to agriculture that can help to address the challenges facing the industry. Another important role that education and awareness-

raising can play is in promoting sustainable agriculture practices. Smart farming technologies have the potential to improve the efficiency and productivity of agriculture, but they must be used in a way that is environmentally sustainable. Education and awareness-raising programs can help to ensure that farmers are aware of the environmental impact of their operations and that they have the knowledge and tools they need to implement sustainable practices. Education and awareness-raising programs can help to build a sense of community among farmers and other stakeholders in the agricultural sector. By bringing people together around a common goal of improving agriculture, these programs can help to create a sense of shared purpose and a commitment to working together to achieve common goals. This, in turn, can help to promote the adoption of smart farming technologies by fostering a sense of trust and collaboration within the agricultural community. Education and awareness-raising programs are critical to overcoming the social obstacles to smart farming adoption. By providing farmers with the information and training they need to use smart farming technologies effectively, these programs can help to build trust in the technology, promote sustainable agriculture practices, foster innovation and collaboration, and create a sense of community within the agricultural sector. As such, they are an

essential component of any strategy aimed at promoting the widespread adoption of smart farming technologies.

4.5 Environmental Considerations in Smart Farming

Smart farming, also known as precision agriculture, is an innovative approach to farming that employs advanced technology to optimize the use of resources while minimizing environmental impact. The potential benefits of smart farming, such as increased productivity, reduced costs, and improved sustainability, make it an attractive option for farmers and policymakers alike. However, there are also significant environmental considerations that must be taken into account when implementing smart farming practices. One of the key environmental benefits of smart farming is the potential for reduced chemical use. With precision agriculture techniques such as precision spraying, farmers can target specific areas of a field that require treatment, rather than spraying an entire field with chemicals. This not only reduces the amount of chemicals used but also helps to prevent overuse and the development of resistant pests and weeds. Smart farming can also help to reduce water usage through the use of sensors and other monitoring technologies. These tools allow farmers to

precisely monitor soil moisture levels and crop water requirements, which in turn enables them to apply water more efficiently. By reducing water usage, smart farming can help to conserve this valuable resource and reduce the impact of agriculture on water systems. Another significant environmental consideration in smart farming is the reduction of greenhouse gas emissions. With precision agriculture techniques such as variable rate fertilization and optimized tillage practices, farmers can reduce the amount of fertilizer and fuel used in their operations. This, in turn, reduces the amount of greenhouse gases emitted by farming equipment and practices, helping to mitigate the impact of agriculture on climate change. In addition to these direct environmental benefits, smart farming can also contribute to broader environmental goals such as biodiversity conservation and soil health. By adopting practices such as cover cropping, reduced tillage, and integrated pest management, farmers can improve soil health and biodiversity on their land. This can help to support beneficial species such as pollinators and improve the long-term sustainability of agricultural systems. Despite these potential benefits, there are also some environmental risks associated with smart farming. One concern is the potential for increased energy usage and associated greenhouse gas emissions, particularly if precision agriculture technologies are

powered by fossil fuels. However, this risk can be mitigated through the use of renewable energy sources such as solar and wind power. Another concern is the potential for smart farming to contribute to soil compaction and erosion, particularly if heavy machinery is used in the implementation of precision agriculture techniques. However, this risk can be minimized through careful planning and management of farming operations, such as through the use of controlled traffic systems and no-till practices. The environmental considerations of smart farming are complex and multifaceted. While there are significant potential benefits to adopting precision agriculture techniques, there are also some environmental risks that must be taken into account. Through careful planning and management, however, it is possible to implement smart farming practices in a way that maximizes environmental benefits while minimizing risks. By doing so, farmers and policymakers can help to ensure that agriculture remains sustainable and resilient in the face of environmental challenges.

4.5.1 Sustainable Agricultural Practices and Smart Farming

Smart farming is rapidly gaining popularity due to the advantages it offers in terms of resource efficiency and

production optimization. However, as with any technology, it is important to consider its impact on the environment and the need for sustainable agricultural practices. It is essential to ensure that smart farming is implemented in a way that is sustainable and eco-friendly. One of the primary benefits of smart farming is its potential to reduce the environmental footprint of agriculture. Smart farming techniques such as precision agriculture can reduce the use of water, fertilizer, and pesticides, thereby minimizing the impact of agriculture on the environment. Precision agriculture utilizes sensors and data analytics to optimize crop yields and resource usage. This helps to reduce the amount of waste generated and the release of harmful chemicals into the environment. Smart farming can also promote sustainable land use. By using advanced technologies, farmers can make more informed decisions about the use of land, such as where to plant specific crops and when to rotate crops. This helps to maintain soil health and reduce erosion, ensuring that land is productive and sustainable over the long term. Additionally, smart farming technologies such as vertical farming and hydroponics can be used to produce crops in urban areas, reducing the need for transportation and minimizing the carbon footprint of food production. However, by using renewable energy sources such as solar or wind power, farmers can reduce their reliance on fossil fuels and

minimize their carbon footprint. This can be achieved through the installation of solar panels, wind turbines, or other renewable energy sources on farms. Furthermore, smart farming can help to reduce food waste, another critical issue in sustainable agriculture. By providing real-time data on crop growth and quality, farmers can ensure that crops are harvested at the optimal time, reducing the amount of food that goes to waste. Additionally, smart packaging and logistics systems can help to extend the shelf life of food and reduce spoilage during transportation. Despite these benefits, it is essential to recognize that smart farming is not a silver bullet for sustainable agriculture. Implementing smart farming technologies requires resources and investment, and it is important to ensure that these technologies are accessible to all farmers, including smallholders in developing countries. Additionally, there is a need to ensure that the benefits of smart farming are not outweighed by negative environmental impacts, such as increased energy usage or the use of harmful chemicals in production. To promote sustainable agricultural practices in smart farming, it is important to develop and enforce regulations that ensure responsible use of these technologies. This includes ensuring that the use of smart farming does not harm biodiversity or contribute to soil degradation. Additionally, education and awareness-raising are

critical to promoting sustainable agricultural practices in smart farming. By providing farmers with information on the benefits of sustainable agriculture and how to implement it using smart farming technologies, they can make informed decisions that benefit both their bottom line and the environment. Smart farming has the potential to revolutionize agriculture by making it more resource-efficient, productive, and sustainable. However, it is crucial to ensure that these technologies are implemented in a way that promotes sustainable agricultural practices and minimizes their impact on the environment. By investing in renewable energy sources, reducing food waste, and promoting sustainable land use, smart farming can help to create a more sustainable and eco-friendly agriculture industry.

4.5.2 Environmental Benefits and Concerns

Smart farming has the potential to bring numerous environmental benefits to the agricultural industry. With the implementation of precision farming techniques, such as targeted fertilizer application and controlled irrigation, the usage of resources can be optimized, resulting in less waste and reduced environmental impact. Smart farming also allows for real-time monitoring of weather, soil quality, and crop development, which can aid in early detection of plant diseases and pests, thus reducing the need for pesticide use. Moreover, it

can help reduce soil erosion, minimize water usage, and decrease greenhouse gas emissions. One of the main environmental benefits of smart farming is the reduction of agricultural runoff. Excessive use of fertilizers and pesticides in traditional farming practices can lead to runoff, which pollutes nearby water sources. Smart farming technologies, such as precision irrigation and fertilization, can help reduce the amount of excess chemicals used, thus decreasing the likelihood of runoff. Smart farming can contribute to a more sustainable use of water resources. Water scarcity is a growing concern worldwide, and agriculture is one of the largest users of freshwater resources. Precision irrigation, which allows for targeted water application based on crop needs and soil moisture levels, can help reduce water waste and increase crop yields. This technology also enables farmers to track water usage and optimize irrigation scheduling, leading to more efficient use of water resources. However, there are also environmental concerns associated with smart farming. One concern is the potential for increased energy usage. Smart farming technologies, such as sensors and automated systems, require a reliable source of power, which may result in increased energy consumption. Additionally, the production and disposal of these technologies may have negative

environmental impacts, including the release of greenhouse gases and hazardous waste.

Another concern is the potential impact of genetically modified organisms (GMOs) on the environment. There is a risk of unintended consequences, such as the development of pesticide-resistant pests and the cross-pollination of GMO crops with wild plants. To address these concerns, it is important to adopt a holistic approach to smart farming that considers the environmental impacts of the entire system. This includes the production and disposal of smart farming technologies, as well as the farming practices themselves. Sustainable agricultural practices, such as crop rotation and integrated pest management, can be integrated into smart farming systems to promote environmental stewardship. Moreover, the integration of renewable energy sources, such as solar and wind power, can help reduce the environmental impact of smart farming. Additionally, the development of biodegradable and environmentally friendly materials for the production of smart farming technologies can help minimize waste and reduce the release of hazardous materials. Smart farming has the potential to bring significant environmental benefits to the agricultural industry. By addressing these concerns, smart farming can become a key contributor to sustainable agriculture practices and environmental stewardship.

4.5.3 Climate Change Mitigation and Adaptation

Climate change is one of the biggest environmental challenges of our time and it is affecting agriculture in a significant way. Agriculture contributes to climate change by emitting greenhouse gases such as carbon dioxide, methane, and nitrous oxide. On the other hand, agriculture is also affected by climate change through extreme weather events such as droughts, floods, and heat waves. Smart farming has the potential to help mitigate and adapt to climate change by reducing greenhouse gas emissions and increasing resilience to climate impacts.

4.5.3.1 Mitigating Climate Change

Smart farming can help reduce greenhouse gas emissions from agriculture by improving resource use efficiency, reducing waste, and using renewable energy sources—for instance, precision agriculture technologies such as variable rate fertilization and irrigation can reduce fertilizer and water use, which can decrease emissions of nitrous oxide and carbon dioxide. Smart farming technologies like anaerobic digestion and bioenergy can also turn farm waste into energy, thereby reducing methane emissions from waste decomposition.

4.5.3.2 Adapting to Climate Change

Smart farming can also help farmers adapt to the effects of climate change, such as increased variability in weather patterns, longer and more severe droughts, and new pest and disease pressures. Smart farming technologies like climate monitoring sensors can provide real-time data on weather patterns, allowing farmers to adjust their planting and harvesting schedules accordingly. However, there are also some environmental concerns associated with smart farming that need to be addressed—for instance, some smart farming technologies like drones and robots use batteries or fuel that may have negative environmental impacts if not disposed of properly. Furthermore, some precision agriculture technologies like GPS-guided tractors can increase soil compaction and decrease soil health, which can have long-term environmental impacts. To ensure that smart farming is environmentally sustainable, it is important to evaluate the environmental impacts of smart farming technologies and practices on a case-by-case basis. Environmental assessments can help identify potential negative environmental impacts and inform the development of strategies to mitigate them. Additionally, there is a need for sustainable agricultural practices that integrate smart farming technologies in a way that maximizes environmental benefits and minimizes environmental impacts.

Smart farming has the potential to help mitigate and adapt to climate change in agriculture, but it is important to carefully evaluate the environmental impacts of smart farming technologies and practices to ensure that they are sustainable in the long term. Smart farming can be an important tool in achieving sustainable agriculture and promoting the long-term resilience of agricultural systems in the face of climate change.

4.6 Future Prospects for Smart Farming

The integration of advanced technologies in agriculture could play a significant role in addressing the challenges associated with feeding a growing global population, while also ensuring the sustainability of agricultural practices. One of the most promising aspects of smart farming technology is the potential to increase the efficiency of agricultural production. This could lead to more precise and targeted use of fertilizers, pesticides, and other inputs, reducing waste and increasing productivity. Another promising aspect of smart farming technology is the potential to reduce the environmental impact of agriculture. Smart farming systems can help farmers reduce their carbon footprint by optimizing resource use and reducing greenhouse gas emissions—for example, precision agriculture techniques can help reduce the amount of fertilizer and pesticides that are applied to crops, reducing the amount of runoff and pollution

that enters waterways. Livestock management technologies can also help reduce emissions from animal waste and methane gas. Smart farming technology can also help address the issue of food security, particularly in developing countries where access to food is a major concern. By increasing productivity and reducing waste, smart farming systems can help increase the availability of food, while also reducing costs and increasing the profitability of farming operations. In addition, smart farming technology can help improve the quality and safety of food products by enabling better monitoring and control of production processes. The future of smart farming will likely see further integration of advanced technologies such as artificial intelligence, machine learning, and robotics. These technologies can help automate many aspects of farming, from planting and harvesting to data collection and analysis. The use of drones, for example, can help farmers monitor crop growth and health from the air, while autonomous vehicles can help reduce the labor required for tasks such as planting and harvesting. In addition to the technological advancements, the future of smart farming will also likely see increased collaboration between different stakeholders, including governments, industry players, and farmers. This collaboration will be critical to overcoming the various economic, social, and environmental challenges associated with the adoption of

smart farming technology. Governments will need to provide regulatory support and financial incentives to encourage the adoption of smart farming technologies, while industry players will need to invest in research and development to continue to improve and refine these technologies. Farmers will need to be educated and trained in the use of these technologies to ensure successful adoption and implementation. The future of smart farming technology looks promising, with the potential to address many of the challenges facing the agricultural industry. By increasing efficiency, reducing waste, and improving the sustainability of agricultural practices, smart farming technology can help ensure the availability of food for a growing global population. Continued investment in research and development, as well as collaboration between stakeholders, will be critical to ensuring the successful adoption and implementation of smart farming technology.

4.6.1 Emerging Trends and Innovations in Smart Farming

Smart farming, also known as precision agriculture, is a rapidly evolving field that incorporates advanced technologies to enhance farming practices. As smart farming technologies continue to develop, new innovations and trends are emerging, which have the potential to revolutionize the agriculture

industry. In this article, we will explore some of the emerging trends and innovations in smart farming. One emerging trend in smart farming is the use of autonomous vehicles. Autonomous tractors, drones, and other unmanned vehicles are being used for a variety of farming tasks, including planting, harvesting, and monitoring crops. These vehicles can navigate fields and complete tasks more efficiently than human labor, reducing labor costs and increasing productivity. In addition, autonomous vehicles can collect data on crop health, soil moisture, and other important factors, which can be used to improve farming practices. Another emerging trend in smart farming is the use of robotics. Robots are being developed for tasks such as planting, weeding, and harvesting crops. These robots can work autonomously and can be programmed to perform specific tasks, such as identifying and removing weeds or harvesting crops at the optimal time. Robotic systems can also collect data on crop health and other factors, which can be used to improve farming practices. Precision irrigation is another emerging trend in smart farming. This technology uses sensors and data analysis to optimize water usage in agriculture. By monitoring soil moisture levels, weather patterns, and other factors, precision irrigation systems can determine the precise amount of water needed for crops, reducing water waste and improving crop yield. This

technology is particularly important in areas where water resources are limited. Vertical farming is another emerging trend in smart farming. This technique involves growing crops in vertically stacked layers using controlled environment agriculture technology, such as LED lights, hydroponics, and aeroponics. Vertical farming allows for more efficient use of space and resources, and it can also be used to grow crops in urban areas, reducing the need for transportation and logistics. Another emerging trend in smart farming is the use of blockchain technology. Blockchain is a decentralized ledger that allows for secure and transparent record-keeping. In agriculture, blockchain can be used to track and verify the origin of crops, ensuring their quality and authenticity. This technology can also be used to track the use of pesticides and other chemicals in farming, providing greater transparency and accountability. Smart farming also has the potential to contribute to the development of smart cities. By integrating smart farming practices with urban planning, cities can develop more sustainable and resilient food systems. Urban agriculture, vertical farming, and other smart farming techniques can be used to produce fresh, healthy food locally, reducing the need for transportation and logistics. Smart farming is a rapidly evolving field that offers a range of emerging trends and innovations. These innovations have the

potential to revolutionize the agriculture industry, improving productivity, efficiency, and sustainability. Autonomous vehicles, robotics, precision irrigation, vertical farming, blockchain technology, and the integration of smart farming practices with urban planning are just a few of the emerging trends and innovations that are shaping the future of agriculture. As these technologies continue to develop, it is likely that they will play an increasingly important role in meeting the growing global demand for food and addressing the sustainability challenges facing the agricultural domain.

4.6.2 Prospects for Overcoming Challenges and Barriers

Smart farming technology has the potential to revolutionize agriculture by making it more efficient, productive, and sustainable. However, the implementation of smart farming technologies is facing numerous challenges and barriers that must be overcome. These challenges are technological, economic, and social, and they require the collaboration of all stakeholders in the agricultural industry to be resolved. Despite these challenges, there are prospects for overcoming them and achieving the full potential of smart farming. One of the primary challenges facing smart farming is the lack of connectivity and data management systems. Smart farming

technologies rely on real-time data to function properly, and without reliable connectivity and data management systems, the accuracy and usefulness of the data collected are compromised. To overcome this challenge, governments and industry stakeholders can work together to improve connectivity infrastructure in rural areas and promote the adoption of data management systems. Another significant challenge is the high cost of implementing smart farming technologies. Farmers may not have the resources or financing needed to invest in expensive equipment and systems, and this could limit the adoption of smart farming technologies. One possible solution is to create financing programs specifically designed for smart farming. Governments, financial institutions, and industry stakeholders can collaborate to offer loans, grants, or other financial incentives to support the implementation of smart farming technologies. Lack of knowledge and understanding among farmers is also a major challenge that must be overcome. Many farmers may be hesitant to adopt new technologies or practices that they are unfamiliar with, or that may disrupt their existing operations. Education and awareness-raising campaigns can help to address this challenge by providing farmers with the information they need to understand the benefits of smart farming and how it can improve their operations. Training

programs and workshops can also be provided to support farmers in the implementation and use of smart farming technologies. Cultural and behavioral barriers to change are also significant challenges that must be addressed. Farmers may be resistant to change their traditional methods and practices, or may have a preference for using their intuition and experience instead of relying on data-driven decisions. To overcome this challenge, education and awareness-raising campaigns must also address the cultural and behavioral aspects of smart farming. Farmers need to be convinced of the benefits of smart farming and understand how it can help them to achieve their goals in a more efficient and sustainable manner. Another key challenge is the need to balance economic, social, and environmental objectives in smart farming. The economic benefits of smart farming must be weighed against the social and environmental costs, and sustainable agricultural practices must be prioritized to ensure that the long-term benefits of smart farming are realized. Governments and industry stakeholders can work together to develop policies and guidelines that promote sustainable agriculture and support the adoption of smart farming technologies. Despite these challenges, there are prospects for overcoming them and achieving the full potential of smart farming. Emerging trends and innovations, such as the use of drones, robotics, and

blockchain technology, offer new opportunities for improving the efficiency and sustainability of agriculture. Partnerships between governments, industry stakeholders, and farmers can also help to overcome the challenges of implementing smart farming technologies by providing the necessary support, resources, and education needed to succeed.

The challenges and barriers facing the adoption of smart farming technologies are significant, but not insurmountable. Through collaboration, innovation, and education, governments, industry stakeholders, and farmers can work together to overcome these challenges and achieve the full potential of smart farming. The future of agriculture depends on our ability to embrace new technologies and practices that promote sustainable, efficient, and productive agriculture, and smart farming is a critical part of this future.

4.6.3 Future Outlook for Sustainable Agriculture

The future outlook for sustainable agriculture is promising, thanks to the continued development and adoption of smart farming technologies. The agricultural industry is facing unprecedented challenges, including climate change, population growth, and food security concerns. Smart farming

offers a solution to many of these problems, providing more efficient and sustainable ways of producing food.

One of the key benefits of smart farming is the ability to reduce the use of harmful chemicals and pesticides, as well as the amount of water and fertilizer required to grow crops. Precision agriculture techniques, enabled by smart farming technologies, allow farmers to monitor their fields in real-time, identifying potential issues before they become major problems. This reduces waste and helps farmers make better use of resources, resulting in a more sustainable agricultural system. Another major benefit of smart farming is the potential to increase yields and productivity. With better data on crop development, soil quality, and weather patterns, farmers can make more informed decisions about planting, irrigation, and other management practices. This can lead to higher yields and better-quality crops, improving food security and providing economic benefits for farmers. Smart farming technologies are also helping to address labor shortages in the agricultural industry. As many farmers struggle to find workers, automation and robotics are becoming increasingly important. Smart farming systems can help automate many tasks, such as planting and harvesting, reducing the need for human labor. This not only helps to address labor shortages but also reduces the risk of injury to workers. The future of sustainable

agriculture will likely involve continued innovation and development of smart farming technologies. Advances in sensors, data analytics, and artificial intelligence will enable even more precise and efficient management practices—for example, advanced analytics can help farmers identify the optimal time to plant and harvest crops, improving yields and reducing waste. There are, of course, challenges to the widespread adoption of smart farming technologies. As discussed earlier, issues such as connectivity, data management, and cost remain major barriers. However, as these obstacles are overcome, the potential for sustainable agriculture will only continue to grow. The future outlook for sustainable agriculture is bright. Smart farming technologies are providing new ways of addressing the challenges facing the agricultural industry, from climate change to food security. With continued innovation and investment, smart farming has the potential to revolutionize the way we produce food, creating a more sustainable and resilient agricultural system for future generations. The prospects for overcoming the challenges and barriers to smart farming adoption are promising, and the benefits for the environment, society, and the economy are clear.

4.7 Conclusion and Recommendations

Social obstacles to the adoption of smart farming technologies include a lack of knowledge, education, and awareness among farmers. Addressing these issues requires targeted education and training programs that provide farmers with the necessary skills and knowledge to integrate smart farming solutions into their agricultural practices. Sustainable agricultural practices and smart farming are interdependent. Therefore, future outlook for sustainable agriculture is linked with the prospects for smart farming. Implementing sustainable farming practices is essential to ensure food security, protect the environment, and mitigate climate change. Smart farming can play a crucial role in achieving these objectives by enabling farmers to manage resources efficiently and reduce waste.

Future prospects for smart farming are promising, with several emerging trends and innovations, including vertical farming, automated drones, and robotics. As these technologies continue to develop and become more accessible, they will transform the agricultural industry in unprecedented ways. To overcome the challenges and barriers to the adoption of smart farming technologies, governments, industry stakeholders, and farmers need to work collaboratively. Policies that incentivize and support the adoption of smart farming solutions should be

developed, and funding and financing mechanisms should be made available. Additionally, training and education programs must be implemented to ensure that farmers have the necessary skills and knowledge to integrate smart farming solutions into their practices. The potential benefits of smart farming are significant, and with the right policies and investment, it can address some of the most pressing challenges facing the agricultural sector. By adopting smart farming practices, farmers can increase efficiency, productivity, and profitability while also contributing to sustainable agricultural practices and mitigating the impact of climate change. Governments, policymakers, and stakeholders must support the adoption of these technologies to ensure a sustainable and secure food supply for future generations.

4.7.1 Summary of Key Points

Smart farming is a rapidly evolving field that involves the use of advanced technologies to improve agricultural production while ensuring sustainability. The adoption of smart farming technology is expected to play a significant role in meeting the growing global demand for food and addressing the sustainability challenges facing the agricultural sector. This chapter has highlighted various challenges and barriers that must be overcome for the widespread adoption of smart

farming technology. These challenges include technical challenges such as data management and connectivity, economic challenges such as the high cost of implementing smart farming technologies, and social challenges such as the lack of knowledge and understanding among farmers. Sensors, IoT, and AI are some of the technologies that have a significant role in smart farming. These technologies can provide real-time data on various aspects of agriculture, including soil quality, crop development, and pest populations, enabling farmers to make informed decisions and improve their yields. The economic benefits of smart farming include increased efficiency, productivity, and cost savings. However, the high cost of implementing smart farming technologies, the scarcity of available financing, and the long-term returns on investment are significant economic obstacles. Social obstacles to smart farming adoption include the lack of knowledge and understanding among farmers, as well as behavioral and cultural barriers to change. Education and awareness-raising programs can play a crucial role in promoting the adoption of smart farming technologies. Environmental considerations in smart farming are critical, and sustainable agricultural practices are necessary for the longterm success of the industry. Smart farming has the potential to reduce the environmental impact of agriculture by minimizing the use of water, pesticides,

and fertilizers. However, the use of technology must be carefully balanced with environmental concerns, such as the loss of biodiversity and the impact on soil health. Climate change mitigation and adaptation are essential considerations in smart farming, and sustainable agricultural practices can play a crucial role in reducing greenhouse gas emissions and increasing resilience to climate change. Despite the challenges and barriers, the future prospects for smart farming are promising, and emerging trends and innovations are expected to further improve the efficiency and sustainability of agriculture. To overcome the challenges and barriers to smart farming adoption, governments, industry stakeholders, and farmers should work together to promote the adoption of smart farming technology. This can be achieved through policy incentives, financial support, and education and awareness-raising programs. The adoption of smart farming technology has the potential to transform the agricultural industry by improving efficiency, productivity, and sustainability. However, to achieve this potential, challenges and barriers such as technical, economic, and social obstacles must be addressed, and sustainable agricultural practices must be prioritized.

4.7.2 Policy Recommendations for Promoting Smart Farming

Smart farming, the use of advanced technologies in agriculture, has the potential to revolutionize the agricultural industry and promote sustainable agriculture practices. However, there are many challenges that must be addressed to enable widespread adoption of this technology. Governments have a crucial role to play in promoting smart farming and ensuring that it is implemented in a way that is equitable and beneficial for all stakeholders. In this section, we will discuss some policy recommendations for promoting smart farming.

Firstly, governments should prioritize investment in infrastructure to support smart farming. This includes improving access to high-speed Internet, expanding sensor networks, and investing in data management and analytics systems. Without these basic infrastructure components, smart farming technologies will not be effective.

Secondly, governments should establish regulatory frameworks that support the adoption of smart farming while protecting the interests of farmers and other stakeholders. This includes addressing concerns around data privacy and ownership, as well as ensuring that smart farming technologies are accessible

to small-scale farmers who may not have the resources to invest in expensive equipment.

Thirdly, governments should provide financial incentives to farmers who adopt smart farming technologies. This can take the form of tax breaks, grants, or low-interest loans. These incentives can help to offset the high initial costs of implementing smart farming technologies and make them more accessible to farmers of all sizes.

Fourthly, governments should invest in education and training programs to improve farmers' knowledge and skills in using smart farming technologies. This can include workshops, training sessions, and demonstration farms where farmers can see the benefits of smart farming in action. Governments can also work with universities and research institutions to develop specialized degree programs in smart farming and related fields.

Finally, governments should promote collaboration and knowledge-sharing among farmers, industry stakeholders, and researchers. This can include the establishment of networks and forums where farmers can share their experiences with smart farming, as well as the creation of partnerships between farmers and researchers to develop new technologies and

techniques. Smart farming has the potential to transform agriculture and promote sustainable practices. However, governments must play an active role in promoting the adoption of this technology and ensuring that it is implemented in a way that benefits all stakeholders. By investing in infrastructure, establishing regulatory frameworks, providing financial incentives, investing in education and training, and promoting collaboration, governments can help to overcome the challenges and barriers to smart farming adoption and create a more sustainable and equitable agricultural system for the future.

4.7.3 Conclusion and Future Directions for Research

Smart farming is a rapidly evolving field that has the potential to revolutionize the agricultural industry. The incorporation of advanced technologies, such as sensors, IoT, AI, and data analytics, has enabled farmers to monitor crop growth, soil quality, pest populations, and weather conditions in real-time, enabling them to make data-driven decisions to optimize resource management and improve yields. Despite the potential benefits of smart farming, there are several challenges and barriers to its implementation, including technical, economic, and social factors. One of the main technical challenges of smart farming is data management and connectivity. The

collection and processing of large amounts of data require reliable Internet connectivity, secure storage facilities, and effective data management systems. However, many rural areas lack sufficient Internet connectivity, and farmers may not have the necessary knowledge or skills to manage the data effectively. Another challenge is the high cost of implementing smart farming technologies. Farmers may not have access to the necessary funding or financing to purchase and install the necessary equipment and software. Additionally, there may be a lack of awareness and understanding of the potential benefits of smart farming, which may discourage investment. Social factors, such as a lack of knowledge and understanding of smart farming among farmers, can also hinder its adoption. Cultural and behavioral barriers, such as resistance to change and a preference for traditional farming methods, can also impede the implementation of smart farming. However, despite these challenges, there are several policy recommendations that can promote the adoption of smart farming. Governments and industry stakeholders can invest in infrastructure and provide financial incentives to encourage the use of smart farming technologies. Education and awareness-raising programs can also help to improve farmers' understanding of the potential benefits of smart farming. In terms of environmental considerations, smart farming has the potential

to promote sustainable agricultural practices and mitigate the impact of climate change. The use of precision agriculture techniques can optimize resource management, reduce waste and pollution, and promote biodiversity. However, there are also concerns about the potential environmental impacts of smart farming, such as increased energy consumption and the potential for technological failures. Looking towards the future, there are several emerging trends and innovations in smart farming, such as the use of drones and autonomous vehicles, that have the potential to further enhance its efficiency and effectiveness.

References

1. Srisruthi, S.; Swarna, N.; Ros, G.M.S.; Elizabeth, E. Sustainable agriculture using eco-friendly and energy efficient sensor technology. In *Proceedings of the 2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, Bangalore, India, 20–21 May 2016; IEEE: Bangalore, India, 2016; pp. 1442–1446.
2. Brodt, S.; Six, J.; Feenstra, G.; Ingels, C.; Campbell, D. Sustainable Agriculture. *Nat. Educ. Knowl.*, 2011, 3, 1.
3. Obaisi, A.I.; Adegbeye, M.J.; Elghandour, M.M.M.Y.; Barbabosa-Pliego, A.; Salem, A.Z.M. Natural Resource Management and Sustainable Agriculture. In *Handbook of Climate Change*

Mitigation and Adaptation; Lackner, M., Sajjadi, B., Chen, W.Y., Eds.; Springer: Cham, Switzerland, 2022.

4. Latake, P.T.; Pawar, P.; Ranveer, A.C. The Greenhouse Effect and Its Impacts on Environment. *Int. J. Innov. Res. Creat. Technol.*, 2015, 1, 333–337.
5. Reddy, T.; Dutta, M. Impact of Agricultural Inputs on Agricultural GDP in Indian Economy. *Theor. Econ. Lett.*, 2018, 8, 1840–1853.
6. World Agriculture: Towards 2015/2030: An FAO Perspective and Summary Report; FAO: Rome, Italy, 2002; Available online: www.fao.org/3/a-y4252e.pdf (accessed on 1 August 2022).
7. Roser, M.; Ritchie, H.; Ortiz-Ospina, E. World Population Growth. 2013. Available online: https://ourworldindata.org/world-population-growth (accessed on 1 August 2022).
8. Hernández-Ochoa, I.M.; Gaiser, T.; Kersebaum, K.C.; Webber, H.; Seidel, S.J.; Grahmann, K.; Ewert, F. Model-based design of crop diversification through new field arrangements in spatially heterogeneous landscapes. A review. *Agron. Sustainable Dev.*, 2022, 42, 74.
9. Navulur, S.; Sastry, A.S.C.S.; Giri Prasad, M.N. Agricultural Management through Wireless Sensors and Internet of Things. *Int. J. Electr. Comput. Eng.*, 2017, 7, 3492–3499.

10. Ayaz, M.; Ammad-uddin, M.; Baig, I.; Aggoune, E.M. Wireless Sensor‘s Civil Applications, Prototypes, and Future Integration Possibilities: A Review. *IEEE Sens. J.*, 2018, 18, 4–30.
11. Lin, J.; Yu, W.; Zhang, N.; Yang, X.; Zhang, H.; Zhao, W. A Survey on Internet of Things: Architecture, Enabling Technologies, Security and Privacy, and Applications. *IEEE Internet Things J.*, 2017, 4, 1125–1142.
12. Tekinerdogan, B. Strategies for Technological Innovation in Agriculture 4.0. Reports; Wageningen University: Wageningen, The Netherlands, 2018.
13. Ferrandez-Pastor, F.J.; Garcia-Chamizo, J.M.; Nieto-Hidalgo, M.; Mora-Pascual, J.; MoraMartinez, J. Developing ubiquitous sensor network platform using Internet of Things: Application in precision agriculture. *Sensors*, 2016, 16, 1141.
14. Wolfert, S.; Ge, L.; Verdouw, C.; Bogaardt, M.J. Big data in smart farming—A review. *Agric. Syst.*, 2017, 153, 69–80.
15. Liakos, K.G.; Busato, P.; Moshou, D.; Pearson, S.; Bochtis, D. Machine learning in agriculture: A review. *Sensors*, 2018, 18, 2674.
16. O’Grady, M.J.; O’Hare, G.M.P. Modelling the smart farm. *Inf. Process. Agric.*, 2017, 4, 179–187.
17. Quy, V.K.; Hau, N.V.; Anh, D.V.; Quy, N.M.; Ban, N.T.; Lanza, S.; Randazzo, G.; Muzirafuti, A. IoT-Enabled Smart Agriculture:

Architecture, Applications, and Challenges. *Appl. Sci.*, 2022, 12, 3396.

18. Raj Kumar, G.; Chandra Shekhar, Y.; Shweta, V.; Ritesh, R. Smart agriculture—Urgent need of the day in developing countries. *Sustainable Comput. Inform. Syst.* 2021, 30, 100512.
19. El Nahry, A.H.; Mohamed, E.S. Potentiality of land and water resources in African Sahara: A case study of south Egypt. *Environ. Earth Sci.*, 2011, 63, 1263–1275.
20. Palombi, L.; Sessa, R. Climate-Smart Agriculture: Source Book; Food and Agriculture Organization: Rome, Italy, 2013.
21. Adamides, G.; Kalatzis, N.; Stylianou, A.; Marianos, N.; Chatzipapadopoulos, F.; Giannakopoulou, M.; Papadavid, G.; Vassiliou, V.; Neocleous, D. Smart Farming Techniques for Climate Change Adaptation in Cyprus. *Atmosphere*, 2020, 11, 557.
22. Patil, K.A.; Kale, N.R. A model for smart agriculture using IoT. In *Proceedings of the 2016 International Conference on Global Trends in Signal Processing, Information Computing and Communication*, Jalgaon, India, 22–24 December 2016; IEEE: Jalgaon, India, 2016; pp. 543–545.
23. Sisinni, E.; Saifullah, A.; Han, S.; Jennehag, U.; Gidlund, M. Industrial Internet of Things: Challenges, Opportunities, and Directions. *IEEE Trans. Ind. Inform.* 2018, 14, 4724–4734.

24. Shi, X.; An, X.; Zhao, Q.; Liu, H.; Xia, L.; Sun, X.; Guo, Y. State-of- the- Art Internet of Things in Protected Agriculture. *Sensors*, 2019, 19, 1833.
25. Elijah, O.; Rahman, T.A.; Orikumhi, I.; Leow, C.Y.; Hindia, M.N. An Overview of Internet of Things (IoT) and Data Analytics in Agriculture: Benefits and Challenges. *IEEE Internet Things J.*, 2018, 5, 3758–3773.
26. Yong, W.; Shuaishuai, L.; Li, L.; Minzan, L.; Ming, L.; Arvanitis, K.G.; Grorgieva, C.; Sigrimis, N. Smart Sensors from Ground to Cloud and Web Intelligence. *IFAC Pap. OnLine,* 2018, 51, 31–38.
27. Mekala, M.S.; Viswanathan, P. A Survey: Smart agriculture IoT with cloud computing. In *Proceedings of the 2017 International Conference on Microelectronic Devices, Circuits and Systems (ICMDCS)*, Vellore, India, 10–12 August 2017; IEEE: Vellore, India, 2017; pp. 1–7.
28. Mittal, A.; Singh, A. Microcontroller based pest management system. In *Proceedings of the Second International Conference on Systems (ICONS'07)*, Martinique, France, 22–28 April 2007; IEEE: Martinique, France, 2007; p. 43.
29. Bonneau, V.; Copigneaux, B. Industry 4.0 in Agriculture: Focus on IoT Aspects, European Commission, Digital Transformation Monitor. 2017. Available online: https://ec.europa.eu/growth/tools-

databases/dem/monitor/content/industry-40-agriculture-focus-iot-aspects (accessed on 30 December 2020).

30. King, T.; Cole, M.; Farber, J.M.; Eisenbrand, G.; Zabaras, D.; Fox, E.M.; Hill, J.P. Food safety for food security: Relationship between global megatrends and developments in food safety. *Trends Food Sci. Technol.*, 2017, 68, 160–175.
31. Chandhini, K. A Literature Study on Agricultural Production System Using IoT as Inclusive Technology. *Int. J. Innov. Technol. Res.*, 2016, 4, 2727–2731.
32. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
33. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
34. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini Robotic Face Recognition System,” *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.

35. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
36. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
37. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
38. Lang, L. GPS + GIS + remote sensing: An overview. *Earth Obs. Mag.*, 1992, 1, 23–26.
39. Batte, M.T.; VanBuren, F.N. Precision farming—Factor influencing productivity. In *Proceedings of the Northern Ohio Crops Day Meeting*, Wood County, OH, USA, 21 January 1999.
40. Chen, F.; Kissel, D.E.; West, L.T.; Adkin, W.; Clark, R.; Rickman, D.; Luvall, J.C. Field Scale Mapping of Surface Soil Clay Concentration. *Precis. Agric.*, 2004, 5, 7–26.
41. Muhammad, S.F.; Shamyla, R.; Adnan, A.; Tariq, U.; Yousaf, B.Z. Role of IoT Technology in Agriculture: A Systematic Literature Review. *Electronics*, 2020, 9, 319.

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5
Sustainable Development in Agriculture: Soil Management

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Abstract

Developing sustainable techniques for agriculture and soil management is essential for ensuring food security, protecting natural resources, and promoting sustainable livelihoods and exploring the importance of research and exploration in this field, highlighting key research questions related to soil health, water management, agricultural technologies, policy interventions, and the impacts of climate change. It emphasizes the need for interdisciplinary approaches that integrate social, economic, and environmental considerations to promote sustainable techniques for agriculture and soil management. This involves investigating the impacts of different agricultural practices on the environment and human health and exploring innovative approaches to improving agricultural productivity

while minimizing negative effects on natural resources. This will involve exploring the potential for small-scale and diversified farming systems like India to support sustainable livelihoods or assessing the effectiveness of different policy interventions by the government. It will involve field research to understand the soil quality, climate, and other factors that affect crop production and explore the potential for new approaches to farming, such as agro forestry, to improve soil health, reduce erosion, and enhance biodiversity. Based on the field research and data, an analysis will be carried out and a machine learning model will predict the crops, and manures to be added to improve soil fertility without harming the natural resources climatic conditions, and their impact during every phase. Our research may also be used to develop a country-while centralized system that will help the farmers and the government to grow the demanded crops and this will also reduce the wastage of grains in cases of overproduction. It will also uplift the farmers by providing them with an adequate sum of money for their products since they will be growing the demanded crops only. Overall, sustainable techniques for agriculture and soil management are essential for ensuring that agricultural production can meet the needs of a growing global population while protecting natural resources and supporting sustainable livelihoods.

***Keywords*:** Agriculture, artificial intelligence, centralized agriculture system, climate change, IoT, sensors, soil management, sustainable agriculture

5.1 Introduction

Sustainable development is a concept that has gained significant attention in recent years, particularly in the context of agriculture and soil management [1, 2]. With the ever-growing global population, increasing demands for food, and the degradation of natural resources, it has become imperative to find methods to meet current needs without compromising the ability of future generations to meet their own needs [3, 4]. Sustainable development in agriculture and soil management aims to address these challenges by adopting practices that are environmentally friendly, socially responsible, and economically viable. The agricultural sector plays a crucial role in ensuring food security and providing livelihoods for millions of people worldwide. However, conventional agricultural practices have often been associated with detrimental effects on the environment and natural resources. Excessive use of synthetic fertilizers and pesticides, deforestation, soil erosion, and water pollution are some of the adverse consequences of unsustainable agricultural practices. These practices not only contribute to greenhouse gas emissions but also lead to soil

degradation, loss of biodiversity, and water scarcity. To overcome these challenges, sustainable development approaches in agriculture and soil management prioritize the integration of ecological principles and social considerations into farming systems. The concept of sustainable agriculture emphasizes the use of methods that minimize environmental impact while maintaining or increasing crop productivity [5]. This involves adopting organic farming practices, agro forestry, precision farming, and integrated pest management, among others. These practices promote soil health, conserve water resources, and reduce the need for chemical inputs, ultimately leading to more resilient and sustainable farming systems.

One of the key pillars of sustainable development in agriculture and soil management is the conservation and improvement of soil quality. Soil is a vital resource that supports plant growth, regulates water availability, and stores carbon. However, unsustainable agricultural practices, such as intensive tillage and mono-cropping, can deplete soil organic matter, decrease soil fertility, and increase the risk of erosion. Soil erosion, in particular, poses a significant threat to sustainable agriculture as it leads to the loss of topsoil and the degradation of soil structure. To address these issues, sustainable soil management practices aim to enhance soil health, productivity, and resilience. These practices include the use of organic

amendments, crop rotation, cover cropping, and conservation tillage. By increasing organic matter content, improving soil structure, and reducing erosion, these practices contribute to the long-term sustainability of agricultural systems [6–8]. Furthermore, sustainable soil management also involves the implementation of soil conservation measures, such as terracing, contour plowing, and agro forestry, to prevent erosion and nutrient runoff.

In addition to environmental considerations, sustainable development in agriculture and soil management also recognizes the importance of social and economic aspects. Sustainable farming practices promote social equity by ensuring fair wages and decent working conditions for farmers and farm workers [9]. They also support rural development by creating employment opportunities and fostering vibrant local economies. Moreover, sustainable agriculture encourages the use of local and traditional knowledge, promoting cultural diversity and preserving indigenous farming practices. Economically, sustainable development in agriculture and soil management aims to ensure the long-term viability of farming systems [10]. By reducing the dependence on costly external inputs, such as synthetic fertilizers and pesticides, sustainable practices can improve farmers' profitability and resilience to market fluctuations [11]. Moreover, sustainable agriculture

promotes diversification, value addition, and market access for smallholder farmers, empowering them to participate in sustainable food systems. In conclusion, sustainable development in agriculture and soil management is a multidimensional approach that seeks to address the challenges of food security, environmental degradation, and socio-economic well-being. By adopting sustainable farming practices, such as organic farming, agroforestry, and soil conservation measures, it is possible to achieve a balance between productivity, environmental stewardship, and social equity. Emphasizing the importance of soil health, sustainable development in agriculture recognizes that the long-term sustainability of farming systems relies on the conservation and improvement of soil quality [12]. Ultimately, by embracing sustainable practices, we can ensure a resilient and prosperous future for agriculture while preserving the health of our planet and benefiting current and future generations.

5.2 Reviewing the Need for Global Food Production to be Upgraded

Studies on future global food security and production often begin with the assumption that, between 2005–2007 (the base period) and 2050, crop consumption may increase by 100–110%

while the world population would increase by 60% [13]. But "projections can frequently be lowered to a target of increasing yields, which serves as a pressing request to action for research, policy, and industry" Assessments of the future global demand for food must have their premises and presumptions checked for several reasons. The global food production system is facing unprecedented challenges due to a growing population, changing dietary preferences, climate change, and limited natural resources. The need to upgrade and transform the current food production methods has become imperative to ensure food security, environmental sustainability, and social well-being. This review examines the various factors that necessitate a shift towards an upgraded global food production system, highlighting the importance of sustainable practices, technological advancements, and policy interventions [14].

5.2.1 Growing Population and Changing Dietary Preferences

The world’s population is projected to reach 9.7 billion by 2050, posing significant challenges to global food production. As the population grows, so does the demand for food. Moreover, changing dietary preferences, particularly in emerging economies, are putting additional pressure on the food system. Rising incomes and urbanization have led to increased

consumption of resource-intensive animal products, resulting in a greater demand for feed crops and livestock production. Upgrading global food production is crucial to meet the rising demand for diverse, nutritious, and sustainably produced food [15].

5.2.2 Climate Change and Resource Limitations

Climate change poses a significant threat to global food production. Rising temperatures, erratic weather patterns, and increased frequency of extreme events, such as droughts and floods, disrupt agricultural productivity. Moreover, agriculture itself contributes to climate change through greenhouse gas emissions from livestock, synthetic fertilizers, and deforestation for agricultural expansion. Upgrading food production methods is essential to mitigate and adapt to climate change by promoting climate-smart agriculture, reducing emissions, and enhancing the resilience of farming systems. Additionally, the limited availability of land, water, and energy resources necessitates the adoption of more efficient and sustainable practices [16–18].

5.2.3 Environmental Sustainability and Biodiversity Conservation

Conventional agricultural practices have led to significant environmental degradation, including soil erosion, water pollution, deforestation, and loss of biodiversity. The use of chemical inputs, monocultures, and intensive farming methods has had detrimental effects on ecosystems and the longterm sustainability of food production. Upgrading global food production involves transitioning toward sustainable practices, such as organic farming, agroecology, and precision agriculture. These approaches minimize the use of synthetic inputs, promote biodiversity conservation, and restore ecosystem services, ensuring the resilience and health of agricultural landscapes [39–41].

5.2.4 Food Loss and Waste

Food loss and waste are major challenges in the global food system. Approximately one-third of all food produced for human consumption is lost or wasted, leading to significant economic, social, and environmental consequences. Upgrading food production involves addressing the inefficiencies in the supply chain, improving post-harvest handling and storage, and reducing consumer waste. Embracing innovative technologies, such as cold chain management, smart packaging, and

precision logistics, can help minimize food loss and waste, ensuring that the available resources are utilized efficiently to feed the growing population [19, 20].

5.2.5 Technological Advancements and Digitalization

Technological advancements play a crucial role in upgrading global food production. Digitalization, automation, and precision agriculture enable more efficient resource management, precise application of inputs, and real-time monitoring of crops. Artificial intelligence, machine learning, and remote sensing technologies provide valuable insights for decision-making, optimizing productivity, and reducing environmental impacts. Furthermore, biotechnology and genetic engineering offer opportunities to enhance crop resilience, nutritional quality, and pest resistance. Embracing these technological innovations can revolutionize global food production and make it more sustainable, productive, and resilient [21].

5.2.6 Socio-Economic Equity and Rural Development

An upgraded global food production system should prioritize socioeconomic equity and rural development. Smallholder farmers, who produce a significant portion of the world's food,

often face numerous challenges, including limited access to resources, markets, and financial services. Upgrading food production should involve empowering smallholder farmers through capacity building, improved access to credit and markets, and the promotion of inclusive value chains. Additionally, investment in rural infrastructure, education, and healthcare can contribute to sustainable rural development and poverty alleviation [21].

5.2.7 Policy Interventions and Governance

Effective policy interventions and governance mechanisms are essential for upgrading global food production. Governments, international organizations, and stakeholders need to collaborate to create an enabling policy environment that incentivizes sustainable practices, supports research and development, and promotes innovation in agriculture. Policies that prioritize agro-ecology, sustainable intensification, and inclusive agricultural value chains can drive the necessary transformation. Furthermore, international cooperation and trade agreements should prioritize fair and sustainable food systems that benefit all countries and ensure food security at a global level [22]. The need to upgrade global food production has become urgent in the face of the growing population, climate change, resource limitations, environmental

degradation, and food loss. Sustainable practices, technological advancements, and supportive policies are key pillars in the transformation of the food production system. Upgrading food production will not only ensure food security and environmental sustainability but also contribute to socio-economic equity, rural development, and improved livelihoods [23]. By embracing innovation, promoting sustainable practices, and fostering international cooperation, we can build a resilient and upgraded global food production system capable of feeding the world's population while preserving the health of the planet for future generations.

The lack of productivity development in non-agricultural businesses is stifling the growth engine of the agriculture sector. Agriculture mechanization and greater use of technology greatly enhance production per farm worker in many regions of the world, and they also correlate with larger farms. This is evident even at the level of basic technology, where the use of oxen-drawn plows, which boost labor productivity over hand hoes, enables the cultivation of bigger regions [24]. The degree of land management in Southeast Asian nations is often quite poor, notwithstanding mechanization. The adoption of yield-improving technology on farms, therefore, causes a reduction in agricultural employment as a result of the agricultural

service industry's inability to absorb the displaced farm personnel.

5.3 Soil Quality and Its Impact

Soil quality is a critical factor that directly influences the production and yield of crops. Healthy and fertile soil provides the necessary nutrients, water, and physical support for plants to grow and thrive [25]. Conversely, degraded or poor-quality soil can significantly hamper agricultural productivity. Understanding the relationship between soil quality and crop production is crucial for sustainable agriculture and ensuring global food security.

1. The quality of soil is determined by its physical, chemical, and biological properties. Physical properties include soil texture, structure, and porosity, which affect water retention, aeration, and root penetration. Chemical properties encompass soil pH, nutrient content, and availability, which directly influence plant growth and development. Biological properties involve the presence and activity of microorganisms, earthworms, and other soil organisms that contribute to nutrient cycling and soil health [25].
2. Soil fertility, a key aspect of soil quality, refers to the soil's ability to provide essential nutrients to plants in optimal

quantities and ratios. Adequate nutrient availability is crucial for plant growth, including the development of leaves, stems, roots, flowers, and fruits. Nutrients such as nitrogen (N), phosphorus (P), and potassium (K), commonly known as macronutrients, are required in relatively large quantities. Additionally, micronutrients such as iron (Fe), zinc (Zn), and manganese (Mn) are necessary in smaller amounts but are equally vital for plant health [26].

3. The presence of organic matter in soil is another vital component of soil quality. Organic matter improves soil structure, water-holding capacity, and nutrient retention. It serves as a source of slow-release nutrients, enhances microbial activity, and promotes beneficial soil organisms. Organic matter also helps prevent soil erosion, reduce compaction, and increase soil resilience to drought and other environmental stressors [27].
4. The physical structure of the soil is crucial for crop production as it influences root development, water infiltration, and aeration. Well-structured soils with good aggregation and pore spaces allow roots to penetrate deeply, accessing water and nutrients. Soil compaction, on the other hand, can restrict root growth and impair nutrient uptake. Compacted soils also have reduced water-holding capacity,

leading to poor drainage and increased risk of waterlogging [28].

5. Soil water availability is critical for plant growth and is closely linked to soil quality. The capacity of soil to hold and release water affects the ability of plants to absorb it through their root systems. Adequate soil moisture is essential for various physiological processes in plants, including photosynthesis, nutrient uptake, and transport. Soil quality influences water infiltration rates, water-holding capacity, and drainage, all of which impact plant water availability.
6. Furthermore, the biological component of soil quality plays a vital role in supporting crop production. Soil microorganisms, including bacteria, fungi, and archaea, contribute to nutrient cycling, organic matter decomposition, and disease suppression. Beneficial soil organisms, such as mycorrhizal fungi, form symbiotic relationships with plant roots, aiding in nutrient uptake and enhancing plant resilience to stress. A diverse and healthy soil microbiome contributes to overall soil fertility and plant health [28].
7. Soil quality directly impacts crop production through its influence on plant nutrition, water availability, root development, and disease suppression. Maintaining and improving soil quality is crucial for sustainable agriculture and ensuring long-term productivity. Practices such as

organic farming, cover cropping, crop rotation, and the application of organic amendments help enhance soil quality by increasing organic matter content, improving nutrient availability, and promoting beneficial soil organisms [29].

Soil quality is a critical factor that profoundly affects the production and yield of crops. Healthy soils provide the necessary nutrients, water, and physical support for plant growth, while degraded soils can significantly hinder agricultural productivity. Understanding the relationship between soil quality and crop production is essential for implementing sustainable farming practices and ensuring global food security. By prioritizing soil health and adopting soil management strategies that enhance soil fertility, structure, and biology, farmers can optimize crop production, minimize environmental impacts, and contribute to a resilient and sustainable agricultural system.

5.4 Emerging Technologies

5.4.1 AI Revolution

The industrial sector and modern industry have quickly attained productivity peaks because of the development of automation. The industrial sector was the first to be impacted

by technological advancements, and many other industries followed. In the early 1900s, it was always assumed that increasingly accurate robots would eventually replace human labor in all imaginable areas. In recent years, artificial intelligence, sometimes referred to as AI, has exceeded humans in abilities like object recognition and computer vision. With the continued advancement of computing power and learning algorithms, this situation advances significantly [30]. Figure 5.1 describing the role of AI in modern agriculture and farming for smart agree-tech. The invention of automation may be dated back to the early 1800s when it helped to fuel the industrial revolution that finally led to the current wave of technological advancements. Every imaginable business has now been affected by automation, and the burgeoning market is trading far above expectations.

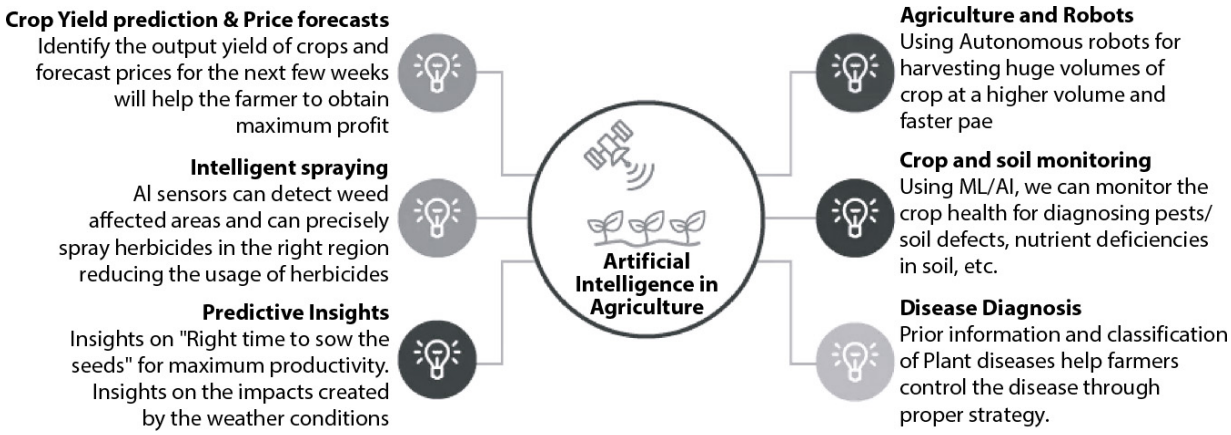

Figure 5.1 Role of AI in modern agriculture and farming [30].

The agricultural sector has always been at the forefront of technological advancements, and the emergence of artificial intelligence (AI) is revolutionizing the way farming and food production are carried out. AI, with its ability to analyze vast amounts of data, learn from patterns, and make autonomous decisions, has the potential to enhance efficiency, productivity, and sustainability in agriculture. This paragraph explores the various applications and benefits of AI in agriculture as well as the challenges and considerations that come with its implementation.

5.4.1.1 Precision Farming and Decision Support Systems

AI technologies, such as machine learning and computer vision, are transforming precision farming practices. By collecting and analyzing data from sensors, satellites, drones, and other sources, AI can provide farmers with real-time information

about crop health, soil moisture, and weather patterns. This enables precise decision-making regarding irrigation, fertilizer application, and pest management [30]. With AI-powered decision support systems, farmers can optimize resource allocation, reduce waste, and increase yields while minimizing environmental impact.

5.4.1.2 Crop Monitoring and Disease Detection

AI algorithms can analyze images and data from various sources to monitor crop growth and detect diseases at an early stage. By using computer vision techniques, AI can identify subtle changes in plant color, texture, and shape that may indicate the presence of pests, pathogens, or nutrient deficiencies [18]. This enables timely interventions and targeted treatments, reducing the need for broad-spectrum chemicals and improving overall crop health. AI-powered disease detection systems have the potential to save crops from significant losses and increase the efficiency of pest management practices.

5.4.1.3 Autonomous Farming and Robotics

AI-driven robotics and automation are revolutionizing labor-intensive tasks in agriculture. Robots equipped with AI capabilities can perform activities such as planting, harvesting,

and weeding with precision and efficiency. They can navigate through fields, identify and selectively pick ripe fruits, and remove weeds without damaging crops. Autonomous farming systems not only reduce labor costs and improve productivity but also minimize the reliance on manual labor, which is becoming increasingly scarce in many regions [31]. Moreover, by replacing traditional machinery with AI-powered robots, fuel consumption and greenhouse gas emissions can be reduced, contributing to environmental sustainability.

5.4.1.4 Supply Chain Optimization

AI technologies are transforming the agricultural supply chain, improving logistics, and reducing waste. AI algorithms can analyze data on demand, weather conditions, transportation routes, and market trends to optimize the distribution of agricultural products. This enables better inventory management, reduced spoilage, and improved delivery schedules. AI-powered predictive analytics can also help anticipate market demands, enabling farmers to make informed decisions about crop selection and production planning [32]. By optimizing the supply chain, AI enhances the efficiency and profitability of the agricultural sector while reducing its environmental footprint.

5.4.1.5 Data-Driven Farm Management

AI relies on vast data to generate insights and make informed decisions. Through the integration of AI with farm management software, farmers can collect, analyze, and interpret data related to crop performance, resource utilization, and financial metrics. This enables data-driven decision-making, allowing farmers to optimize input usage, track profitability, and assess the impact of different management practices [33]. AI-powered farm management systems provide farmers with actionable recommendations, empowering them to make informed choices that improve productivity, sustainability, and economic viability.

The AI revolution in agriculture holds immense potential for transforming the sector into a more efficient, productive, and sustainable industry. From precision farming and crop monitoring to autonomous farming and data-driven decision-making, AI technologies offer numerous benefits. However, addressing challenges related to data privacy, access, skills development, and ethical considerations is crucial for the responsible and equitable implementation of AI in agriculture. By harnessing the power of AI while considering these factors, we can leverage technology to address the global challenges of

food security, environmental sustainability, and resource efficiency in the agricultural sector.

5.4.2 Computer Vision

Computer vision, a branch of artificial intelligence (AI), is revolutionizing the agricultural industry by providing advanced imaging and analysis capabilities. With the ability to process visual data and extract valuable insights, computer vision is transforming farming practices and enabling farmers to make informed decisions about crop health, pest management, yield estimation, and more. This paragraph explores the applications and benefits of computer vision in agriculture, highlighting its potential to enhance productivity, sustainability, and profitability.

5.4.2.1 Crop Monitoring and Disease Detection

Computer vision technologies enable real-time monitoring and early detection of crop diseases and pests. By analyzing images and videos captured by drones, satellites, or ground-based cameras, computer vision algorithms can identify and classify plant diseases, nutrient deficiencies, and pest infestations [34]. This enables farmers to take timely and targeted actions, such as applying appropriate treatments or adjusting irrigation and fertilizer regimes, to mitigate crop losses. By effectively

monitoring crop health, computer vision assists in optimizing resource allocation and improving overall productivity.

5.4.2.2 Weed Identification and Management

Weed control is a critical aspect of agricultural management. Computer vision techniques can accurately identify and distinguish between crops and weeds, facilitating precise and targeted weed management strategies. By analyzing images of fields, computer vision algorithms can detect and classify different weed species, allowing farmers to implement site-specific weed control measures [35]. This reduces the reliance on herbicides, minimizes crop damage, and promotes sustainable and environmentally friendly weed management practices.

5.4.2.3 Yield Estimation and Quality Assessment

Estimating crop yield and assessing product quality is essential for planning harvest logistics and optimizing market value. Computer visionbased techniques can estimate crop yield by analyzing images of plants or fruits and using machine learning algorithms to make accurate predictions. Additionally, computer vision can assess the quality and ripeness of fruits and vegetables based on color, size, and texture [35]. This helps farmers determine the optimal harvest time, reduce post-

harvest losses, and ensure that only high-quality produce reaches the market.

5.4.2.4 Smart Irrigation and Resource Management

Water scarcity and efficient resource management are significant challenges in agriculture. Computer vision-based systems can monitor soil moisture levels, crop water stress, and plant growth patterns to optimize irrigation practices. By analyzing images and data from various sensors, computer vision algorithms can provide insights into plant water needs, enabling farmers to implement precise and efficient irrigation strategies [35]. This promotes water conservation, minimizes water waste, and enhances overall resource management on the farm.

5.4.2.5 Plant Breeding and Genetic Improvement

Computer vision plays a crucial role in plant breeding and genetic improvement programs. By capturing and analyzing detailed images of plant traits, such as leaf shape, size, and color, computer vision algorithms can assist breeders in selecting desirable characteristics. This accelerates the breeding process by automating trait measurement, reducing human bias, and facilitating the identification of plants with superior genetic traits [35]. Computer vision enables more efficient and

targeted breeding programs, leading to the development of improved crop varieties with enhanced yield, disease resistance, and nutritional value.

5.4.2.6 Farm Automation and Robotics

Computer vision technologies are driving the automation of farming operations. By combining computer vision with robotics and autonomous systems, tasks such as planting, harvesting, and sorting can be performed with precision and efficiency. Computer vision allows robots to navigate through fields, detect and handle crops, and perform tasks that were traditionally labor-intensive. Farm automation powered by computer vision not only improves productivity and reduces labor costs but also reduces the need for harmful pesticides and herbicides, thereby promoting sustainable farming practices [35].

5.4.2.7 Decision Support Systems

Computer vision provides the foundation for decision support systems that assist farmers in making informed choices. By integrating data from multiple sources, such as weather forecasts, soil analyses, and crop imagery, computer vision algorithms can generate actionable insights and recommendations. This empowers farmers to make data-driven

decisions about crop management, resource allocation, and risk mitigation [35]. Decision support systems based on computer vision contribute to improved efficiency, profitability, and sustainability in agriculture.

Computer vision is transforming agriculture by providing advanced imaging and analysis capabilities. From crop monitoring and disease detection to smart irrigation and automation, computer vision technologies offer numerous benefits, including increased productivity, reduced resource use, and improved decision-making. However, challenges related to data quality, infrastructure, affordability, and ethics need to be addressed for the widespread adoption of computer vision in agriculture. By addressing these challenges and leveraging the potential of computer vision, farmers can optimize farming practices, enhance sustainability, and contribute to the global goal of ensuring food security for a growing population.

5.5 Introduction and Theory of IoT

The technology that enables global connectivity through the Internet is commonly referred to as the “Internet of Things” (IoT). It encompasses a network of interconnected devices, such as sensors, appliances, and actuators, each with a unique

identity. The IoT not only facilitates the connection of these devices but also enables them to communicate and share data. According to the Electronic Gadgets and Network Annex-IEA-4E, it is projected that, by 2020, there will be an estimated 50 billion connected devices, which will generate substantial revenue. The IoT operates in a tiered structure, adapting to the specific problem it aims to solve. It follows a layered architecture, including functional blocks, alongside other communication models like the request–response model, publish–subscribe model, push–pull model, and exclusive pair, which contribute to the seamless operation of the IoT model. The reference architecture for IoT in agriculture encompasses various layers, including the analytics layer, device/ physical layer, end-user layer, middleware layer, network layer, and service layer. This architecture covers all physical components, infrastructure, tools, internet protocols, and data analytics methods required for the effective implementation of IoT in agriculture.

5.5.1 Perception Layer

The perception layer, also referred to as the physical or device layer or layer zero, represents the lowest level within the architecture [36]. It comprises hardware components such as sensors, actuators, microcontrollers, gateways, routers, switches, and hubs. The primary function of this layer is to

assess the capability of the devices to interact with sensors and gather data. The microcontroller device plays a vital role in networking operations, serving as a controller that regulates networks. It enables the collection of data from sensors and other devices and facilitates its transmission to the higher levels of abstraction, which constitute the subsequent layer. The primary objectives of the perception layer include data collection and transmission to facilitate further processing and analysis.

5.5.2 Network Layer

Included within this layer are the Internet and various communication technologies that play a significant role in enhancing platform communication. The integration of this layer into an agricultural framework is crucial as it enables the selection of communication technologies that are well suited for the field's size or test bed. In agriculture, several communication technologies find application, including LTE, CDMA, Global System for Mobile (GSM), Wi-Fi, ZigBee, Ultra-Wideband (UWB), Bluetooth Low Energy (BLE), Near Field Communication (NFC), and Radio-Frequency Identification (RFID). Among these, ZigBee, LoRa, and NFC are frequently utilized in the agricultural sector, chosen based on the specific problem to be addressed and the corresponding area.

Greenhouses, in particular, benefit from the implementation of NFC and Bluetooth technologies. Additionally, various protocols like HTTP, WWW, and SMTP are employed in agricultural contexts to facilitate efficient communication [36].

5.5.3 Middleware Layer

Within the IoT architecture, this particular layer is assigned the responsibility of managing devices, ensuring context awareness, promoting interoperability, facilitating portability, and addressing security-related tasks. In agricultural contexts, the most suitable middleware options are HYDRA and SMEPP due to their exceptional context-aware characteristics. These middleware solutions are well equipped to handle the unique requirements of agriculture and contribute to the seamless integration and efficient management of IoT devices in this domain [36].

5.5.4 Service Layer

In the IoT layered architecture, the cloud-assisted service layer plays a crucial role in addressing agricultural challenges by offering cloud storage and Software-as-a-Service (SaaS) solutions. This layer focuses on specific domains such as monitoring, detection, control, decision-making, and

identification to simplify sensing, actuation, and other related activities. It provides essential features including device control, discovery, monitoring, and data dissemination. By leveraging the capabilities of the cloud, this layer enables efficient storage, processing, and analysis of agricultural data, empowering farmers with valuable insights and facilitating informed decision-making. The cloud-assisted service layer serves as a fundamental component in addressing agricultural requirements within the IoT architecture [37].

5.5.5 Analytics Layer

The data processing and analysis layer in the IoT architecture assumes responsibility for handling data ingestion, reporting, mining, and learning processes [38]. These processes can be further divided into two phases:

1. Data ingestion involves the storage, cleansing, and streaming of data. This phase ensures that the collected data is properly stored and prepared for further analysis.
2. Data reporting, mining, and learning focus on utilizing data analytics techniques. Within this phase, the analytics layer employs various machine learning methodologies and data analytics algorithms. These methodologies and algorithms enable the generation of early projections and judgments

related to agricultural productivity, crop growth, disease detection, and other relevant aspects. By providing farmers with these valuable insights, monitoring their activities becomes more manageable. Furthermore, the informed decisions derived from the data analysis contribute to the development of enhanced decision support systems.

3. The data processing and analysis layer within the IoT architecture utilizes advanced data analytics techniques to process and interpret agricultural data. This layer's functionalities enable early projections and valuable insights, empowering farmers with the information needed to optimize agricultural practices and improve decision-making processes.

5.5.6 End-User Layer

The uppermost layer of the IoT model, known as the user or customer layer, serves the end users or customers of the system. This layer provides an interface that allows users to examine the system's status, handle data, and control and monitor their IoT model. In the case of agriculture, the farmer is the ultimate consumer of the agro model. This layer is designed to provide farmers with a platform that offers a user-friendly experience. Services and applications at this layer are developed by analyzing user behavior and usage patterns. The aim is to cater

to the diverse needs of customers, and therefore various intelligence tactics and data analytics studies are employed to gain insights into user behavior [59]. The user layer plays a crucial role in providing services to clients, professionals, and organizations. Platforms such as social networking, cloud computing, and mobile messaging are utilized to deliver these services. Through this layer, the entire IoT-based platform can be monitored, controlled, and operated. Many of the services created at this layer serve as a connection point between users and the underlying platforms and systems. It ensures that users have access to the necessary functionalities and capabilities to effectively utilize the IoT-based agricultural system. Figure 5.2 delineate the layered architecture of agro IoT in smart agriculture system. By focusing on the user experience and understanding user behavior, the user layer plays a vital role in bridging the gap between users and the IoT platform. It enables farmers to interact with the system, access relevant data, and make informed decisions to optimize their agricultural operations. Ultimately, the user layer aims to enhance the overall usability and accessibility of the IoT-based platform for the benefit of the end users.

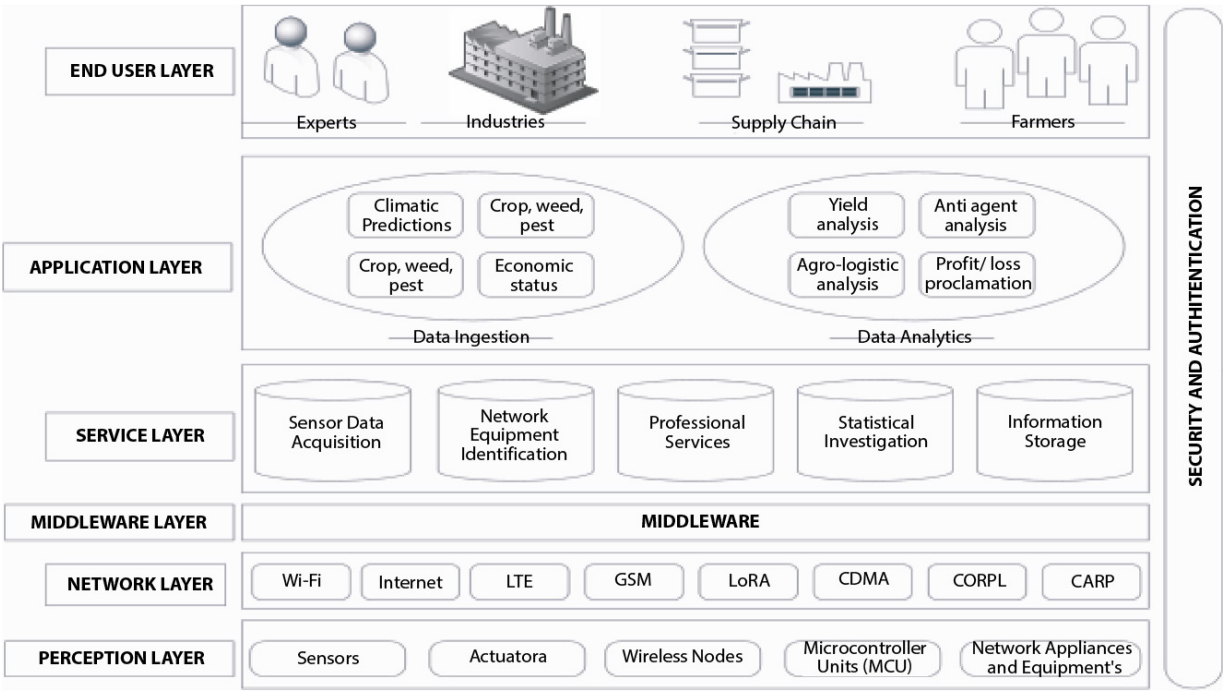

Figure 5.2 Layered architecture/ecosystem of agro-IoT [32].

5.6 Several Sensors and How They are Used in Agriculture

5.6.1 Level Sensors

To measure the composition of a liquid, a cast is utilized, which can be any agricultural area, pond, water tank, or similar structure. Data is gathered using two different approaches.

5.6.1.1 Continuous Measurements

This method involves the use of sensors that can continuously estimate the quantity of a substance and provide precise range measurements. In agricultural contexts, sensors such as

moisture, humidity, and level sensors are employed to detect water content levels in a field. Point-level sensors are utilized when the water content in the cast, such as soil or a test bed, is extremely low or in dry areas. On the other hand, continuous-level sensors are more suitable for semi-arid environments [38]. In certain cases, ultrasonic-level sensors are employed in water treatment plants. Furthermore, a low-level sensor-based fusion application has been developed for constructing farm equipment, enhancing their capabilities in data collection and analysis.

5.6.1.2 Point-Level Measurements

This approach simply indicates whether the measured value exceeds or falls below a predefined sensing point or threshold [39]. It is a binary measurement that serves to determine if a specific condition has been met or not. These measurement techniques enable the collection of data related to the liquid's composition, providing valuable insights for agricultural purposes. Continuous measurements offer more detailed and precise information, while point-level measurements offer binary indications based on threshold values. By employing these measurement approaches, farmers and agricultural practitioners can gain a better understanding of the liquid's

characteristics and make informed decisions to optimize their agricultural processes.

5.6.2 Temperature Sensors

Temperature sensors play a crucial role in measuring the temperature of the surroundings or the environment. Various types of temperature sensors are available, including semiconductor sensors, infrared sensors, resistance temperature detectors, thermocouples, and thermistors. These sensors enable accurate temperature measurements in different applications and environments. In the agricultural sector, temperature fluctuations in a field are of significant interest. Monitoring temperature variations can provide valuable insights into crop health, growth patterns, and environmental conditions [40]. To address this need, innovative solutions have been developed, such as a cost-effective infrared sensor-based field monitoring system. The use of a cheap infrared sensor in constructing a temperature-based field monitoring system has proven to be effective. This system allows farmers and agricultural practitioners to continuously monitor the temperature in their fields, providing real-time data on temperature changes. By leveraging the capabilities of the infrared sensor, the system offers a reliable and affordable solution for temperature monitoring in agriculture. With this

temperature-based field monitoring system, farmers can gain insights into temperature patterns, detect anomalies or unfavorable conditions, and make informed decisions to optimize crop management [40, 41]. By accurately monitoring temperature fluctuations, farmers can implement appropriate measures to protect crops from extreme temperatures, adjust irrigation practices, and enhance overall agricultural productivity. The integration of temperature sensors, specifically the use of inexpensive infrared sensors, in agriculture, exemplifies the application of advanced technology to address specific agricultural challenges. This innovative solution provides a practical and cost-effective means of monitoring temperature in agricultural fields, empowering farmers with valuable information for efficient crop management and decision-making [42–44].

5.6.3 Proximity Sensors

Proximity sensors are capable of detecting and locating nearby objects without requiring physical contact or exerting any physical effort. These sensors operate by emitting an electromagnetic beam into the surrounding environment and monitoring any changes in the signal. By detecting variations in the signal, proximity sensors can identify the presence or absence of objects in their proximity [45]. In the agricultural

industry, proximity sensors find various applications, including cattle grazing monitoring and fruit counting—for example, in cattle grazing, proximity sensors can be used to track the movement and location of livestock within a designated area. This information can be valuable for monitoring grazing patterns, ensuring proper pasture utilization, and managing livestock health.

Another agricultural application of proximity sensors is in fruit counting. By using proximity sensors, farmers can automate the process of fruit counting, which was traditionally done manually. This automation can significantly enhance efficiency and accuracy in yield estimation, allowing farmers to make informed decisions regarding harvesting and resource allocation [46–48]. One specific type of proximity sensor commonly used in agricultural contexts is the capacitive proximity sensor. This sensor operates based on changes in the electrical capacitance between the sensor and the object. By detecting the capacitance changes, it can determine the presence or proximity of objects without physical contact [49]. The utilization of capacitive proximity sensors has shown promising results in enhancing field harvest yield by providing accurate and real-time information on fruit ripeness, enabling farmers to optimize harvesting operations.

Proximity sensors offer a non-contact and effortless means of locating objects in their vicinity. In agriculture, these sensors have been employed for various purposes, including cattle grazing monitoring and fruit counting. The use of capacitive proximity sensors, in particular, has contributed to increased harvest yields by providing valuable insights into fruit ripeness and optimizing harvesting operations. By leveraging proximity sensors, farmers can improve efficiency, automate tasks, and make data-driven decisions to enhance agricultural productivity [50]. The variety sensors are unveiled in Figure 5.3 which are used in smart farming.

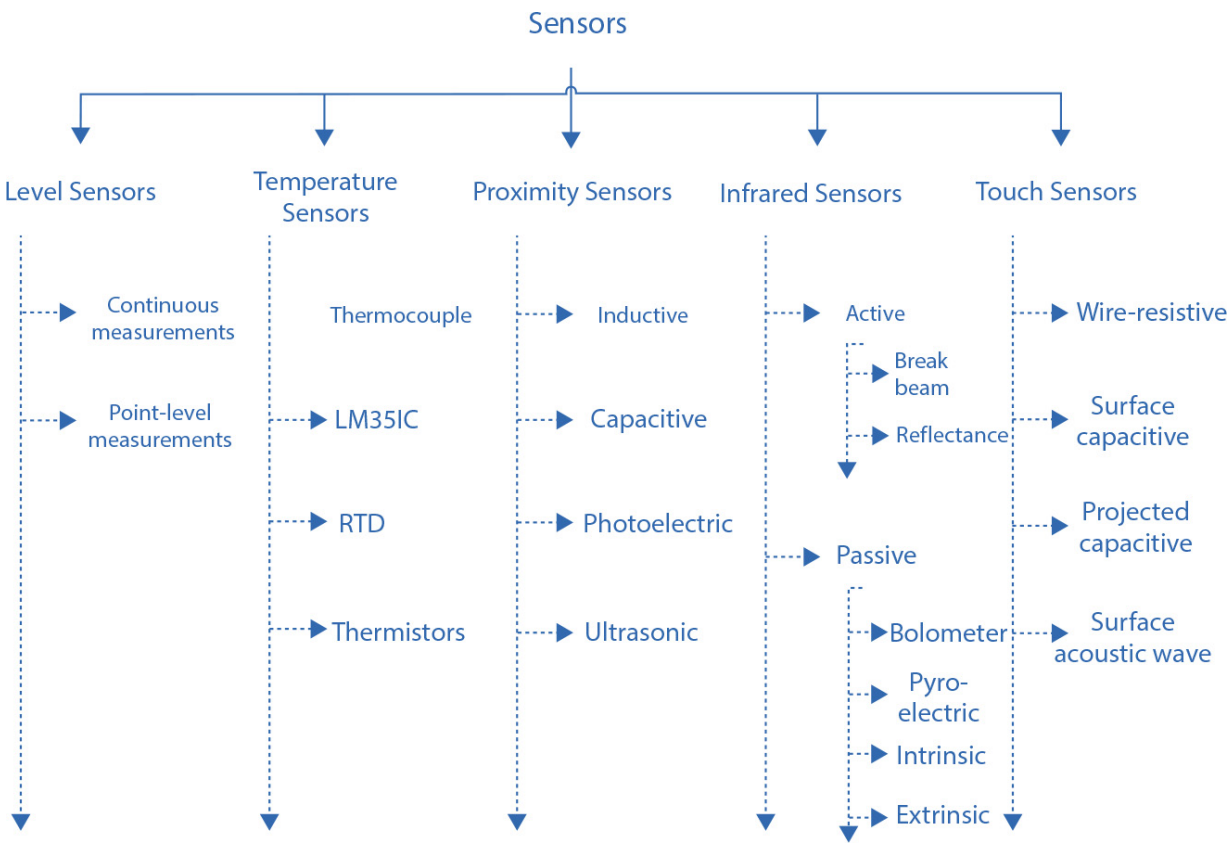

Figure 5.3 Different sensors.

5.6.4 Infrared Sensors

Infrared (IR) sensors are utilized to detect environmental changes by emitting infrared photons. These sensors employ both active and passive techniques to fulfill their functions. An active IR sensor consists of a light-emitting diode (LED) and a receiver, enabling it to both emit and detect infrared radiation. On the other hand, passive IR sensors only detect radiation and are comprised solely of a receiver module [51]. In the agricultural domain, field infrared sensors have proven to be valuable tools in various applications. They can be employed to

count the number of trees or plants in a field, enabling farmers to obtain accurate data on crop density and assess overall growth. In orchards, infrared sensors can assist in spotting rodents or other pests, allowing for timely intervention and pest management strategies [52–55]. Additionally, infrared sensors can capture satellite images of agricultural regions, providing valuable insights into crop health, vegetation patterns, and land use.

Furthermore, IR sensor-based satellite imagery has been utilized to map the findings and outcomes of unmanned aerial vehicles (UAVs) for charting agricultural drainage networks. By integrating data from IR sensors on satellites and UAVs, farmers and land managers can gain a comprehensive understanding of the drainage patterns and make informed decisions to optimize irrigation and water management practices. The combination of IR sensors and satellite imagery offers a powerful toolset for monitoring and analyzing agricultural landscapes. By leveraging the capabilities of IR sensors, farmers can obtain real-time data on crop density, pest presence, and drainage networks, enabling them to make data-driven decisions to improve agricultural productivity, optimize resource allocation, and implement effective land management strategies.

5.6.5 Touch Sensors

Tactile sensors, also known as touch sensors, are devices designed to detect and recognize physical touch or contact. These sensors offer an affordable and practical solution for various applications. In the agriculture industry, tactile sensors can be utilized to detect field invasions and monitor the presence of unwanted intrusions. There are different types of tactile sensors available, each with specific targets and detection ranges. These sensors are designed to detect and respond to physical pressure, allowing for the identification of touch or contact. By strategically placing tactile sensors in agricultural fields, farmers can monitor and detect any unauthorized entry or intrusion into their crops or farming areas. The affordability of tactile sensors makes them accessible for implementation in agricultural contexts [56]. Farmers can deploy these sensors throughout their fields, creating a network of touch-sensitive devices that provide real-time feedback on any touch or pressure applied. This information enables farmers to identify potential threats, such as animals or unauthorized human activity, and take appropriate measures to mitigate the risks and protect their crops. Tactile sensors offer a cost-effective means of recognizing touch or physical contact. In agriculture, these sensors can be employed to detect field invasions and unauthorized access. By deploying tactile sensors

strategically, farmers can enhance their monitoring capabilities and promptly respond to potential threats, ensuring the safety and security of their agricultural assets.

5.7 Centralized Agriculture System

The centralized agriculture system, also known as industrial or conventional agriculture, has been the predominant model of food production for several decades. It involves large-scale, intensive farming practices that prioritize high yields and efficiency. This system relies heavily on mechanization, synthetic inputs, and monoculture cropping. While it has contributed to increased food production and improved agricultural productivity, the centralized agriculture system has also raised concerns regarding its sustainability, environmental impact, and social implications. The primary objective of the centralized agriculture system is to maximize output by employing modern technologies and large-scale farming practices [57, 58]. It relies on the use of chemical fertilizers, pesticides, and genetically modified organisms (GMOs) to enhance crop growth and control pests. Additionally, this system often favors monocultures, where vast expanses of land are dedicated to growing a single crop variety. Monocropping maximizes efficiency in terms of mechanization, crop management, and distribution. However, the centralized

agriculture system has faced criticism for its negative environmental consequences. The excessive use of synthetic fertilizers and pesticides can result in soil degradation, water pollution, and the loss of biodiversity. The intensive farming practices associated with this system often involve clearing land, leading to deforestation and habitat destruction. Furthermore, the reliance on monocultures reduces genetic diversity and increases the vulnerability of crops to pests, diseases, and climate change.

Moreover, the centralized agriculture system has social implications that raise concerns about equity and rural livelihoods. Large-scale farms and agribusinesses dominate this system, leading to the concentration of land ownership and the displacement of small farmers. This consolidation of power can result in economic inequalities, reduced employment opportunities, and the decline of rural communities. Additionally, emphasizing high-yield crops in the centralized agriculture system may prioritize quantity over quality and overlook the cultural significance of traditional and indigenous farming practices. Along with the centralized agriculture system, alternative approaches such as sustainable agriculture and agroecology have emerged. These systems prioritize ecological sustainability, social equity, and local food systems. Sustainable agriculture promotes practices such as organic

farming, crop rotation, and integrated pest management, aiming to reduce the reliance on synthetic inputs and preserve soil health. Agro-ecology emphasizes the integration of ecological principles, traditional knowledge, and community participation to create resilient and diverse farming systems.

While the centralized agriculture system has been successful in increasing food production and meeting the demands of a growing global population, its long-term sustainability is being questioned. Concerns about environmental degradation, loss of biodiversity, and social inequalities have prompted a reevaluation of our agricultural systems [59, 60]. Transitioning to more sustainable and decentralized models of agriculture can offer viable solutions to these challenges. In conclusion, the centralized agriculture system, with its focus on high yields and efficiency, has been instrumental in meeting the global demand for food. However, it has also led to negative environmental impacts and social inequities. As we face the complex challenges of climate change, resource depletion, and food security, exploring alternative approaches that prioritize sustainability, biodiversity, and social well-being becomes increasingly important. By embracing diverse and decentralized agricultural systems, we can strive toward a more resilient, equitable, and sustainable future.

5.8 Conclusion

Given the significance of agriculture and its interconnectivity in the global trade of food, the question of “who and how food will be produced?” necessitates a comprehensive perspective on global food systems. Our investigation uncovered a wide range of farms, from small-scale holdings to large expanses exceeding 10,000 hectares. While the prices for agricultural commodities, except for specialized products, are predominantly determined by global markets, they are currently experiencing a decline. Smallholder farms, which play a critical role in ensuring nutritional security for a significant portion of the global population, will continue to be the primary source of food production in rural areas. Undoubtedly, agriculture will remain vital to rural life in developing countries. Opportunities exist to diversify agriculture to provide more nutrient-rich meals, reduce dependence on staple cereals, and incorporate a broader range of nutrient-dense crops such as pulses, vegetables, and root and tuber crops like cassava, potato, and highland banana. However, the modest profits that farmers derive from farming alone highlight their involvement in a broader food system, where they require additional income to access nutritious foods they are unable to produce themselves. In response to growing environmental challenges, increasing

temperatures, concerns about livestock welfare, human welfare, and the perspectives of both merchants and consumers, agricultural land policies and farming practices will be directed toward manufacturing and distribution methods that minimize environmental impact. While ensuring access to affordable food remains crucial, there will likely be a greater emphasis on internalizing the overall environmental costs of food production and consumption and possibly even the associated social and nutritional costs. Consequently, there will be a need for farming practices that not only yield food at prices that provide a sustainable income for producers but also respect the limitations of our planet and offer a range of environmental benefits.

References

1. G. Zavatta, Agriculture Remains Central to the World Economy. 60% of the Population Depends on Agriculture for Survival. *NetExpo. Accessed*: Nov. 9, 2014. [Online]. Available: http://www.expo2015.org/magazine/en/economy/agriculturer emains-central-to-the-world-economy.html
2. How to Feed the World 2050. Accessed: Oct. 12–13, 2009. [Online]. Available: http://www.fao.org/fileadmin/templates/wsfs/docs/expert_paper/How_to_Feed_the_World_in_2050.pdf

3. S. S. Chouhan, U. P. Singh, and S. Jain, "Applications of computer vision in plant pathology: A survey," *Arch. Comput. Methods Eng.*, vol. 27, no. 2, pp. 611–632, Apr. 2020, doi: 10.1007/s11831-019-09324-0.
4. Sawe and Benjamin Elisha, WorldAtlas. Countries Most Dependenton Agriculture. Accessed: Aug. 1, 2017. [Online]. Available: https://www.worldatlas.com/articles/countries-most-dependent-on-agriculture.html
5. FAO in India, Food and Agricultural Organisation of United Nations. Accessed: 2017. [Online]. Available: http://www.fao.org/india/fao-in-india/india-at-a-glance/en/2017
6. Employment across various sectors, NSSO 66th Nationwide Survey, *Planning Commission, Government India*, New Delhi, India, Jun. 2016.
7. F. Hussain, *Internet of Things Building Blocks and Business Models* (Springer Series in Electrical and Computer Engineering, SpringerBriefs in Electrical and Computer Engineering Book Series (BRIEFSELEC-TRIC)). Springer, 2017, doi: 10.1007/978-3-319-55405-1.
8. J. Ma, H. Yu, Y. Xu and K. Deng, "CDAM: Conservative data analytical model for dynamic climate information evaluation using intelligent IoT environment—An application

perspective," *Comput. Commun.*, vol. 150, pp. 177–184, Jan. 2020, doi: 10.1016/j.comcom.2019.11.014.

9. A. Daissaoui, A. Boulmakoul, L. Karim and A. Lbath, "IoT and big data analytics for smart buildings: A survey," *Procedia Comput. Sci.*, vol. 170 pp. 161–168, 2020, doi: 10.1016/j.procs.2020.03.021.
10. S. Puengsungwan and K. Jirasereeamornkul, "IoT based root stress detection for lettuce culture using infrared leaf temperature sensor and light intensity sensor," *Wirel. Pers. Commun.*, pp. 1–9, Mar. 2020, doi: 10.1007/ s11277-020-07219-z.
11. M. Blaxter, *Health and Lifestyles*, Routledge, 2003.
12. A.S. Blinder, Offshoring: the next industrial revolution? *Foreign Aff.*, (2006)113–128.
13. J. Bryant, Infographic to share: Ag tech on the rise, URL, https://myfarmlife.com/equipment/infographic-to-share-ag-tech-on-the-rise/, 2015.
14. D.M. Bulanon, T. Kataoka, Y. Ota, T. Hiroma, Ae—automation and emerging technologies: a segmentation algorithm for the automatic recognition of fuji apples at harvest, *Biosyst. Eng.*, 83 (2002) 405–412.
15. F.H. Buttel, O.F. Larson, G.W. Gillespie Jr., et al., *The Sociology of Agriculture*, Greenwood Press, Inc, 1990.

16. S. Caplan, B. Tilt, G. Hoheisel, T.A. Baugher, Specialty crop growers' perspectives on adopting new technologies, *HortTechnology, 24* (2014) 81–87.
17. J.X. Chen, The evolution of computing: Alphago, *Comput. Sci. Eng.*, 18 (2016) 4–7.
18. N. Chen, L. Christensen, K. Gallagher, R. Mate, G. Rafert, Global Economic Impacts Associated with Artificial Intelligence, Analysis Group, 2016.
19. X.W. Chen, X. Lin, Big data deep learning: challenges and perspectives, *IEEE Access*, 2 (2014) 514–525.
20. J.Z. Cheng, D. Ni, Y.H. Chou, J. Qin, C.M. Tiu, Y.C. Chang, C.S. Huang, D. Shen, C.M. Chen, Computer-aided diagnosis with deep learning architecture: applications to breast lesions in us images and pulmonary nodules in ct scans, *Sci. Rep.*, 6 (2016) 1–13.
21. Y. Zhang, Q. He, Y. Xiang, L. Y. Zhang, B. Liu, J. Chen and Y. Xie, "Low-cost and confidentiality-preserving data acquisition for Internet of multimedia things," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3442–3451, Oct. 2018, doi: 10.1109/JIOT.2017.2781737.
22. S. Rani, S. H. Ahmed, R. Talwar, J. Malhotra and H. Song, "IoMT: A reliable cross layer protocol for Internet of multimedia things," *IEEE Internet Things J.*, vol. 4, no. 3, pp. 832–839, Jun. 2017, doi: 10. 1109/JIOT.2017.2671460.

23. M. Usman, M. A. Jan and D. Puthal, “PAAL: A framework based on authentication, aggregation, and local differential privacy for Internet of multimedia things,” *IEEE Internet Things J.*, vol. 7, no. 4, pp. 2501–2508, Apr. 2020, doi: 10.1109/JIOT.2019.2936512.
24. A. Floris and L. Atzori, “Managing the quality of experience in the multimedia Internet of Things: A layered-based approach,” *Sensors*, vol. 16, no. 12, p. 2057, Dec. 2016, doi: 10.3390/s16122057.
25. Y. B. Zikria, M. K. Afzal and S. W. Kim, “Internet of multimedia things(IoMT): Opportunities, challenges and solutions,” *Sensors*, vol. 20, no. 8, p. 2334, Apr. 2020, doi: 10.3390/s20082334.
26. A. Nauman, Y. A. Qadri, M. Amjad, Y. B. Zikria, M. K. Afzal and S. W. Kim, “Multimedia Internet of Things: A comprehensive survey,” *IEEE Access*, vol. 8, pp. 8202–8250, 2020, doi: 10. 1109/ACCESS.2020.2964280.
27. S. H. Alsamhi, O. Ma, M. S. Ansari and Q. Meng, “Greening Inter-net of Things for greener and smarter cities: A survey and future prospects,” *Telecommun. Syst.*, vol. 72, no. 4, pp. 609–632, Dec. 2019, doi: 10.1007/ s11235-019-00597-1.
28. T. AlSkaif, B. Bellalta, M. G. Zapataa and J. M. Barcelo-Ordinas, “Energy efficiency of MAC protocols in low data rate wireless multi-media sensor networks: A comparative study,”

Ad Hoc Netw., vol. 56, pp. 141–157, Mar. 2017, doi: 10.1016/j.adhoc.2016.12.005.

29. Z. Libo, H. Tian and G. Chunyun, "Wireless multimedia sensor network for rape disease detections," *EURASIP J. Wirel.Commun. Netw.*, vol. 2019, no. 1, Dec. 2019, Art. no. 159, doi: 10.1186/s13638-019-1468-3.
30. K. E. Psannis, C. Stergiou and B. B. Gupta, "Advanced media-based smart big data on intelligent cloud systems," *IEEE Trans. Sustainable Comput.*, vol. 4, no. 1, pp. 77–87, Jan. 2019, doi: 10.1109/TSUSC.2018.2817043.
31. G. Fortino, C. E. Palau, A. Guerrieri, N. Cuppens, F. Cuppens, H. Chaouchi and A. Gabillon, "Interoperability, safety and security in IoT," in *Proc. 3rd Int. Conf., InterIoT*, in Lecture Notes of the Insti-tute for Computer Sciences, Social Informatics and Telecommunications Engineering Book Series (LNICST), vol. 242. Valencia, Spain, 2017, doi:10.1007/978-3-319-93797-7.
32. Q. Wu, C. Zhao, Y. Liang, D. Zhang and J. Hao, "Design of farmland information acquisition system based on LoRa wireless sensor network," in *Computer and Computing Technologies in Agriculture XI*. CCTA (IFIP Advances in Information and Communication Technology), vol.545. Cham, Switzerland: Springer, 2019, doi: 10.1007/978-3-030-06137-1_49.

33. D. Martín-Sacristán, J. F. Monserrat, J. Cabrejas-Peñuelas, D. Calabuig, S. Garrigas and N. Cardona, "On the way towards fourth-generation mobile: 3GPP LTE and LTE-advanced," *EURASIP J. Wirel.Commun. Netw*., vol. 2009, no. 1, Dec. 2009, Art. no. 354089, doi: 10.1155/2009/354089.
34. X. Zhao, Y. Jiao, L. Yu and C. Zhang, "Design and implementation of TD-LTE-based real-time monitoring system for greenhouse environment temperature," in *Computer and Computing Technologies in Agriculture IX*. CCTA (IFIP Advances in Information and Communication Technology), vol. 479. Cham, Switzerland: Springer, 2016, pp. 170–177, doi:10.1007/978-3-319-48354-2_18.
35. A. Yadav, R. P. Yadav and A. Alphones, "CPW fed triple band notched UWB antenna: Slot width tuning," *Wirel.Pers. Commun.*, vol. 111, no. 4, pp. 2231– 2245, Apr. 2020, doi: 10.1007/s11277-019-06983-x.
36. D. Evans, The Internet of Things How the Next Evolution of the Internet is Changing Everything, Cisco Internet Bus. Solutions Group (IBSG), San Jose, CA, USA, 2011.
37. F. Javed, M. K. Afzal, M. Sharif and B.-S. Kim, "Internet of Things (IoT) operating systems support, networking technologies, applications, and challenges: A comparative review," *IEEE Commun. Surv. Tutorials*, vol. 20, no. 3, pp.

2062–2100, *3rd Quart.*, 2018, doi: 10.1109/COMST.2018.2817685.

38. A. Musaddiq, Y. B. Zikria, O. Hahm, H. Yu, A. K. Bashir and S. W. Kim, "A survey on resource management in IoT operating systems," *IEEE Access*, vol. 6, pp. 8459–8482, 2018, doi: 10.1109/ ACCESS.2018.2808324.
39. E. Baccelli, C. Gundogan, O. Hahm, P. Kietzmann, M. S. Lenders, H. Petersen, K. Schleiser, T. C. Schmidt and M. Wahlisch, "RIOT: An open source operating system for low-end embedded devices in the IoT," *IEEE Internet Things J.*, vol. 5, no. 6, pp. 4428–4440, Dec. 2018, doi: 10.1109/JIOT.2018.2815038.
40. Y. B. Zikria, S. W. Kim, O. Hahm, M. K. Afzal and M. Y. Aalsalem, "Internet of Things (IoT) operating systems management: Opportunities,challenges, and solution," *Sensors*, vol. 19, no. 8, p. 1793, Apr. 2019,doi: 10.3390/s19081793.
41. Q.-U. Ain, S. Iqbal, S. Khan, A. Malik, I. Ahmad and N. Javaid, "IoT operating system based fuzzy inference system for home energy management system in smart buildings," *Sensors*, vol. 18, no. 9, p. 2802, Aug. 2018, doi: 10.3390/s18092802.
42. Chan, K Y. 2001 An overview of some tillage impacts on earthworm population abundance and diversity -

implications for functioning in soils. *Soil and Tillage Research* 57: 179-191.

43. R. Morais, A. Valente, C. Ser^odio, A wireless sensor network for smart irrigation and environmental monitoring: a position article, in: *5th European Federation for Information Technology in Agriculture, Food and Environement and 3rd World Congress on Computers in Agriculture and Natural Resources, EFITA/WCCA*, 2005, pp. 845–850.
44. C.D. Hergert, System and method for optimized control of moving irrigation systems, US Patent 5 (1998) 740, 038.
45. K.M. Hermann, T. Kocisky, E. Grefenstette, L. Espeholt, W. Kay, M. Suleyman, P. Blunsom, Teaching machines to read and comprehend, in: *Advances in Neural Information Processing Systems*, 2015, pp. 1693–1701.
46. G. Ismayilova, The Use of Image Labelling to Identify Pig Behaviours for the Development of a Real-Time Monitoring and Control Tool, 2014.
47. K. Januszkiewicz, M. Jarmusz, Envisioning urban farming for food security during the climate change era. vertical farm within highly urbanized areas, in: *IOP Conf. Ser.: Mater. Sci. Eng*., IOP Publishing., 2017, 052094.
48. B.F. Johnston, J.W. Mellor, The role of agriculture in economic development, *Am. Econ. Rev.*, 51 (1961) 566–593.

49. L. Jones, Driverless cars: when and where? *Eng. Technol.*, 12 (2017) 36–40.
50. H. Kagaya, K. Aizawa, M. Ogawa, Food detection and recognition using convolutional neural network, in: *Proceedings of the 22nd ACM International Conference on Multimedia*, 2014, pp. 1085–1088.
51. D. Knowler, B. Bradshaw, Farmers' adoption of conservation agriculture: a review and synthesis of recent research, *Food Policy*, 32 (2007) 25–48.
52. A.K. Kuhlemann, Food production in the anthropocene: the role of agtech in urban spaces, *Science Meets Comics*, 89 (2017).
53. Y. LeCun, Y. Bengio, G. Hinton, Deep learning, *Nature ,521* (2015) 436–444.
54. Y. LeCun, L. Bottou, Y. Bengio, P. Haffner, Gradient-based learning applied to document recognition, *Proc. IEEE, 86* (1998) 2278–2324.
55. U. Lele, S. Goswami, The fourth industrial revolution, agricultural and rural innovation, and implications for public policy and investments: a case of India, *Agric. Econ.*, 48 (2017) 87–100.
56. M.D. Levine, *Vision in Man and Machine*, McGraw-Hill, College, 1985.

57. D. Loghin, *Efficient Time-Energy Execution of Data-Parallel Applications on Heterogeneous Systems with GPU*, Ph.D. thesis. National University of Singapore, Singapore, 2017.
58. F. L€ ow, C. Conrad, U. Michel, Decision fusion and non-parametric classifiers for land use mapping using multi-temporal rapideye data, ISPRS *J. Photogramm. Remote Sens.*, 108 (2015) 191–204.
59. S.K. Lowder, J. Skoet, S. Singh, What Do We Really Know about the Number and Distribution of Farms and Family Farms in the World? Background Paper for the State of Food and Agriculture 2014, 2014.
60. B. Lyman, A Psychology of Food: More than a Matter of Taste, *Springer Sci. Business Media*, 2012.

Note

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6
Concepts of Robotics, AI, and Internet of Things (IoT) in Agriculture

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Abstract

Rising population, dwindling food supply, shifting weather patterns, and a lack of available workers are just some of the difficulties confronting the agricultural sector today. To meet these challenges, we need cutting-edge agricultural technologies that improve crop yields and resource utilization. The application of robotics, artificial intelligence, and the Internet of Things (IoT) has the potential to completely revolutionize agriculture while also assisting in the resolution of the industry's myriad problems. Possible applications of robotic AI and IoT in agriculture are explored, along with their potential effects on productivity, efficiency, and sustainability. In order to

improve crop management and irrigation, AI algorithms can examine data from sensors and cameras. IoT sensors can track soil conditions like moisture and temperature, giving farmers accurate data for making informed decisions. Furthermore, a comprehensive farming system that enhances crop growth and management can be provided by combining robotic AI and IoT. Integrating these systems can lessen the workload of farmers and help them use fewer harmful chemicals and pesticides. These chapters provide the in-depth knowledge of the processes and technological intervention necessary for addressing the need of agriculture so that interventions for plant health could be initiated timely.

***Keywords*:** IoT, agriculture informatics, AI, robotics

6.1 Introduction

Agriculture is crucial to human survival because it provides for basic needs like food, clothing, and energy. Food production and resource distribution through agriculture are among the world's oldest and most vital economic activities. However, the industry is facing numerous challenges that threaten its sustainability and long-term viability. Some of these challenges include the growing demand for food, limited resources, climate change, and labor shortages. In order to address these

challenges, the agriculture industry must turn to innovative solutions, such as robotic artificial intelligence (AI) and Internet of Things (IoT) technologies. These technologies have the potential to transform the industry by increasing productivity, improving efficiency, and promoting sustainability.

The term “information and communication technologies” (ICTs) refers to the networks, mobiles, devices, services, and applications that make it possible for a specific group of people to more easily share data, information, or knowledge with one another. Traditional forms of communication, television and video, radio, CD-ROMs, cellular phones, and other smart devices as well as a number of cutting-edge technologies such as computers and the Internet, sensors, Geographic Information Systems, satellites, and other similar technologies are all examples of what are known as converging technologies. The primary purpose of information and communication technologies is to improve the flow of information from one location to another.

6.1.1 The Implantation of Information and Commutation Technology in Agriculture Sector

An aspect of modern farming practices that is frequently disregarded is the significance of information and communication technology (ICT) as a tool that can assist

farmers in making well-informed decisions. Farmers who utilize ICT have access to the most recent discoveries that have been made in their industry. It includes everything from weather reports to information on agriculture to cutting-edge techniques for increasing yields and quality.

The widespread implementation of information and communications technologies has resulted in significant improvements to the operational efficiencies and results achieved by businesses of varying sizes. Nearly 60% of people on the planet are now able to access the Internet through their mobile devices, making mobile Internet the most popular method of Internet access worldwide. Communication has been greatly enhanced as a result of the widespread adoption of ICTs, which has also made it possible for people who previously lacked access to the necessary goods, services, and information to obtain it.

The development of new and more advanced agricultural technologies has been instrumental in the expansion of the agricultural industry across the globe. These technologies have had an impact on how crops are cultivated, harvested, and transported. E-agriculture in India has hastened agricultural and rural development by employing novel strategies to enhance preexisting information and communication

processes. This has been accomplished by improving the connectivity between farmers and consumers. It has had a revolutionary impact on smallholder farming in a number of agrarian economies because it has enabled smallholder farmers to solve some of the issues that were inherent in the methods they had been using in the past.

6.2 General Challenges Faced in the Agriculture Industry

6.2.1 Growing Demand for Food

The demand for food is increasing, driven by population growth, rising incomes, and urbanization. It is anticipated that the global population will reach 9.7 billion by the year 2050, with the number of the majority living in urban areas. This increasing demand for food puts pressure on the agriculture industry to produce more food while using fewer resources.

6.2.2 Limited Resources

Access to vital resources like water, land, and power is essential to the agricultural sector’s prosperity. However, population growth and climatic shifts are putting an increasing strain on these resources, making their scarcity even more problematic. This means that agriculture must find ways to boost food

production while cutting back on resource use. Natural resources such as arable land, fresh water, and renewable energy are crucial to the agricultural sector. These resources are finite and face increasing pressures from growing populations, urbanization, and climate change.

6.2.3 Climate Change

Increasing temperatures and shifting weather patterns pose a serious threat to agricultural output and food security as a result of climate change. The frequency of extreme weather events such as floods, droughts, and storms is increasing as a result of climate change. These events can have devastating effects on both agricultural production and infrastructure. Climate change is already having an effect on agricultural production in a number of different ways. These ways include, but are not limited to, changes in rainfall patterns, rising temperatures, and extreme weather events. These changes may result in lower crop yields, an increase in the prevalence of pests and diseases, and a decrease in the fertility of the soil.

6.2.4 Labor Shortages

The agriculture industry relies heavily on manual labor, but labor shortages are becoming increasingly common. Factors

such as aging populations, urbanization, and competition from other sectors are contributing to this problem. Labor shortages can lead to increased costs, reduced productivity, and lower yields. Agriculture is a labor-intensive industry, but the availability of labor has been decreasing due to factors such as aging populations and urbanization. This can result in labor shortages and increased labor costs.

6.2.5 Food Demand

Global food demand is expected to increase significantly in the coming decades due to population growth and changing dietary patterns. The agriculture industry must find ways to increase productivity and efficiency to meet this demand.

6.2.6 Specific Challenges in the Deployment of Technology of Robotics, AI, and IoT in Agriculture

There are concerns that could arise as a result of the implementation of AI in farms. It might appear as though every farmer ought to be utilizing artificial intelligence given all the positive effects it can have on farming practices that are kind to the environment. According to a well-known proverb, one can find happiness in ignorance:

1. It may take some time to adapt to new technological advances. Farmers need to understand that artificial intelligence (AI) is merely a more advanced variant of more fundamental technologies that are already in use for the processing, gathering, and monitoring of field data. Without a strong foundation in technology, artificial intelligence cannot perform its functions. Even for farms that already have some technological infrastructure in place, progress can be difficult to achieve for this reason.
2. The software industry also faces difficulties in core implementation of technology in actual farming processes. The incorporation of farmers should take place in stages, with the provision of fundamental technologies like a trading platform for agricultural products coming first. When farmers have adjusted to a solution that is less complicated, it makes sense to provide them with something more advanced, such as those that are powered by artificial intelligence.
3. Inadequate prior experience working with cutting-edge machinery: Agriculture in the developing world is not the same as agriculture in the developed world in places like the United States and Western Europe. Agriculture that is conducted with the assistance of artificial intelligence may be useful in certain regions, but it may be challenging to market

in regions where this kind of technology is not yet the norm. It is highly likely that farmers will require assistance in order to put it into action.

4. Technology companies that are interested in expanding into regions that have flourishing agricultural economies may find that they need to take the initiative to do so on their own. They need to do more than simply sell their products to farmers and owners of agribusinesses who are interested in experimenting with new methods for the industry.
5. Security of personally identifiable information: The use of artificial intelligence (AI) in any industry, not just agriculture, raises a number of legal questions that are rarely addressed due to the lack of clear policies and regulations surrounding the use of AI. Precision agriculture and smart farming raise a number of these questions. The ease with which a company can become a victim of malicious intent is illustrated by the fact that cyber-attacks that threaten personal data and leaks of personal data are just two examples. Unfortunately, a significant number of farms are vulnerable to these dangers.
6. The most effective combination of AI with other types of technology: Removing skepticism concerning the application of AI in agriculture, as we have gone over, artificial intelligence cannot function without a number of other technologies being in place first. These technologies include

big data, sensors, and software. In a similar vein, other technologies are unable to perform their intended functions in the absence of AI—for example, the data itself has relatively little value when considered on its own in the context of big data. The most important considerations are the manner in which it is interpreted as well as the question of whether or not it is relevant.

7. The context, the selection criteria, and the timing all play a role in determining whether or not the recommendations generated by AI based on a set of data are useful. Because of this, effective artificial intelligence technology also requires data engineers and analysts who are skilled in their craft. Let us delve even deeper into the various ways that artificial intelligence is being put to use in the agricultural industry.

Utilization of extensive data sets in the decision-making process: The application of the information that is created and gathered is where the bulk of the value lies. Utilizing data analytics in the field of agriculture can lead to significant increases in both productivity and savings. The combination of artificial intelligence and large datasets makes it possible to generate crop recommendations that are more accurate and timely. Farmers will be able to perform tasks such as watering, fertilizing, protecting crops, and harvesting with greater precision as a result of this.

6.3 Role of Robotics, AI, and IoT in Agriculture

Precision farming is another application of robotic AI. Using information and tools, precision farmers increase yield while decreasing waste and increasing productivity. Data on soil conditions, moisture levels, and weather patterns can be analyzed by robotic AI to help farmers determine the best times and locations to plant crops. Farmers may see increased income as a result of increased crop yields and decreased waste.

The shortage of workers in the agricultural sector is something that robotic AI can help alleviate. Finding qualified workers has become increasingly difficult as the average age of farmers rises and younger generations leave rural areas. By automating previously labor-intensive tasks like planting and harvesting, robotic AI helps farmers save time and money.

Artificial intelligence robots can also help agriculture become more sustainable. Farmers can lessen their environmental impact by minimizing their use of toxic chemicals and increasing the efficiency with which they produce crops. Additionally, robotic AI can help reduce soil erosion and promote soil health by precisely targeting areas that require attention.

Several studies have shown the potential benefits of using robotic AI in agriculture. One study published in *Journal of Cleaner Production* examined the use of precision agriculture technologies in grape production. The study found that using precision agriculture technologies such as robotic AI improved grape quality and reduced the use of fertilizers and pesticides, resulting in higher profits for farmers and reduced environmental impact.

Another study published in *International Journal of Agricultural and Biological Engineering* examined the use of robotic AI for tomato harvesting. The study found that robotic AI could effectively and efficiently harvest tomatoes while reducing labor costs and improving yield.

Robotic artificial intelligence has the potential to improve efficiency and productivity while simultaneously lowering the risk of injury to farm workers. According to the findings of a study that was recently published in *Journal of Agricultural Safety and Health*, employing robotic AI to perform tasks on a farm, such as harvesting and pruning, may help to reduce the likelihood of musculoskeletal injuries occurring among farm workers.

Another study published in *International Journal of Agricultural and Biological Engineering* explored the use of robotic AI in precision weed control. According to the findings of the study, the application of robotic AI for weed control has the potential to cut down on the use of herbicides, boost crop yield, and have a smaller negative impact on the environment.

Overall, these studies suggest that the use of robotic AI in agriculture has the potential to improve efficiency, increase productivity, reduce labor costs, promote sustainability, and improve worker safety. While there are still challenges to overcome, such as high initial costs and the need for specialized skills, the benefits of adopting robotic AI in agriculture are significant and can contribute to the long-term success of the industry.

Precision agriculture refers to farming that optimizes the use of inputs like water and fertilizer through the integration of technologies like robotics, AI, and the IoT. This has the potential to improve efficiency and cut down on waste, leading to higher yields and more money made.

By monitoring crop growth, soil conditions, and weather patterns, farm management systems can be developed by combining robotic artificial intelligence and the Internet of

Things. With this knowledge, farmers will be able to optimize their planting, fertilization, and harvesting practices for maximum yields at the lowest possible cost.

Planting, weeding, and harvesting are all examples of labor-intensive, repetitive tasks that can be automated with the help of robotics. Because of this, output has gone up. This has the potential to reduce labor costs and increase efficiency, freeing up time for farmers to focus on higher-value tasks like crop management.

Pest and disease management: Robotics can be used to monitor crops for pest and disease pressures and apply targeted treatments. This can reduce the use of pesticides and ultimately lead to healthier crops.

Livestock management: Robotics can be used to automate tasks such as feeding, milking, and monitoring the health of livestock. This can reduce labor costs and improve efficiency, ultimately leading to higher profitability.

6.4 Benefits of Robotic and AI in Improving Agriculture

6.4.1 Increased Efficiency

Robotic artificial intelligence systems can plant, weed, and harvest without getting tired or taking time off. As a result, productivity rises while production time and costs drop dramatically.

6.4.2 Improved Precision

Robotic AI systems can perform tasks with high precision and accuracy, such as planting seeds at the optimal depth and spacing or applying fertilizer and pesticides in a targeted manner. This can reduce waste and increase yields.

6.4.3 Reduced Labor Costs

Labor is a significant cost for many farmers, and the availability of labor is often limited in rural areas. Robotic AI systems can automate many tasks, reducing the need for manual labor and ultimately reducing labor costs.

Robotic artificial intelligence systems have the ability to monitor the state of crops, soil, and weather in real time.

Utilizing this data to optimize planting, fertilization, and harvesting practices can lead to increased yields as well as a reduction in operating costs.

6.4.4 Sustainability

Robotic AI systems allow for significant reductions in the application of potentially hazardous chemicals such as pesticides and fertilizers. They have the potential to increase productivity while simultaneously reducing waste, which could lead to more environmentally friendly farming practices.

6.5 Collecting Data and Performing Analyses With the Help of the Internet of Things

Farmers are now able to monitor and record data on a wide variety of metrics in their fields in what is referred to as “near real time” thanks to the Internet of Things (IoT) and other supplementary technologies (such as drones, GIS, and other tools). When Internet of Things (IoT) devices and software are used in conjunction with artificial intelligence (AI) farming tools, they supply farmers with data that is both more timely and accurate. If we had better information, we would be able to make more educated decisions and conduct less pointless experimentation.

Through the use of automation and robots, the need for human labor can be reduced.

One of the most common issues in farming is a lack of labor, but AI and the Internet of Things, autonomous tractors, and other forms of mechanization offer a solution. Because these systems are more accurate and, as a result, make fewer mistakes, it is possible that using them will result in cost savings. The Internet of Things (IoT), autonomous tractors, and artificial intelligence are all essential components of precision farming.

Deployment of robotics in AI-based farming: Robotics is an emerging technology that is still in its infancy compared to other technological fields. Robots are already being used in agriculture to perform tasks that were traditionally done by hand, such as harvesting crops and thinning lettuce. In farms, replacing human labor with that of robots is possible in a number of different ways. They are able to perform for longer periods of time, are more accurate, and make fewer mistakes.

High initial costs: The initial cost of implementing robotic AI and IoT systems can be high, which can be a barrier for small- and medium-sized farms.

Technical expertise: Implementing robotic AI and IoT systems requires technical expertise, which may not be readily available in some rural areas. Data management: Robotic AI and IoT systems generate huge volumes of data, which must be managed and analyzed effectively to be useful. This requires specialized skills and infrastructure.

Privacy and security: Data generated by robotic AI and IoT systems must be protected from unauthorized access and use, which can be a challenge in a connected environment.

Robotic AI technology has made significant strides in recent years, and the agriculture industry is no exception. Robots with artificial intelligence make it possible to automate labor-intensive agricultural processes such as planting, weeding, harvesting, and spraying. The capacity of this technology to increase productivity while simultaneously lowering operating costs and increasing output is enormously promising. The use of robotics and AI in agriculture will be evaluated in this article from both positive and negative perspectives.

One real-life example of robotic AI in agriculture is the Bonirob system developed by Bosch Deepfield Robotics. The Bonirob system is a fully autonomous robot that can perform a range of

tasks in the field, such as planting, weeding, and monitoring crop health.

In order to successfully navigate the field, the Bonirob system relies on a combination of sensors and machine learning algorithms, identify and classify plants, and make decisions about how to perform tasks—for example, the system can identify weeds and decide whether to remove them manually or with herbicides. It can also analyze plant health and growth patterns and provide recommendations on fertilization and irrigation.

One of the key features of the Bonirob system is its ability to learn from its environment. As the system operates, it collects data on the field conditions, plant growth, and environmental factors. This data is then used to refine the system's algorithms and improve its performance over time.

The Bonirob system has been tested in a variety of field conditions, including on farms in Germany and the United States. In these tests, the system has demonstrated the ability to perform tasks with high precision and accuracy, reducing the need for manual labor and improving yields.

One potential application of the Bonirob system is in organic farming, where the use of pesticides and herbicides is limited. By providing precise, targeted weed removal and disease monitoring, the Bonirob system can help organic farmers improve their yields and reduce labor costs.

Overall, the Bonirob system is an example of how robotic AI technology can be used to revolutionize agriculture, providing farmers with real-time data and precise, efficient tools for managing their crops.

6.6 Relationships of IoT and AI Strengthen the Agriculture Sector

The Internet of Things (IoT) is becoming increasingly popular in the agricultural industry as a way to offer real-time monitoring and control of a diverse range of operational parameters. The Internet of Things is gaining popularity in the farming industry for this very reason. The “Internet of Things” (IoT) is a network that is constructed out of physical objects in the real world, such as smartphones, automobiles, and other devices that have electronics, software, and network connectivity built into them. The Internet of Things could be of great use to the agricultural industry if it could provide real-time updates on weather conditions such as temperature and humidity. In this article, we

will discuss some of the benefits of utilizing IoT in farming as well as some of the potential drawbacks that may arise as a result of its use. Implementation of the Internet of Things can be beneficial to the agricultural industry in a number of different ways. If farmers have access to up-to-date information on the conditions of the environment, they are able to make decisions that are more informed regarding when and where to plant crops, how much water to use, and when to harvest. This may result in increased productivity, decreased waste, and increased viability over the long term.

The use of sensors to monitor the amount of moisture in the soil is one example of how the Internet of Things can be applied in agriculture. These sensors can assist farmers in optimizing their irrigation practices and reducing the amount of water that is wasted by monitoring the moisture levels of the soil in real time. As a result of this, the quality of the crop may improve, while the amount of water needed may decrease.

The installation of weather sensors is just one more illustration of this principle. Weather sensors assist farmers in making educated decisions regarding the timing of when crops should be planted and harvested by continuously gathering data on the temperature, humidity, and other aspects of the environment in

real time. It is possible for this to result in increased yields in addition to reducing the risk of crop failure.

The Internet of Things (IoT) may be able to assist farmers in lowering the amount of damage they cause to the natural environment, in addition to boosting their output and improving their efficiency. Farmers can lessen their impact on the environment and improve the health of the soil by using information to improve the efficiency of irrigation systems, cut down on the amount of fertilizer and pesticides they use, and optimize how much water they use.

However, there are also several challenges associated with the adoption of IoT technology in agriculture. One challenge is the cost of implementing and maintaining IoT systems. The cost of sensors, connectivity, and data storage can be prohibitively expensive for small-scale farmers.

The unreliable Internet connections in rural areas are another difficulty. Many farms are in outlying areas where there is spotty or no access to the web, making it difficult to transmit and receive data from IoT sensors.

Furthermore, the sheer volume of data generated by IoT sensors can be overwhelming for farmers. Without the proper

tools and infrastructure in place to analyze and interpret data, IoT technology may not provide meaningful insights for farmers.

Despite these challenges, there are several solutions that can help farmers overcome these obstacles and reap the benefits of IoT technology in agriculture.

One solution is to collaborate with other farmers or agricultural organizations to share the cost of implementing IoT systems. This can help reduce the financial burden on individual farmers and increase access to IoT technology.

Another solution is to use low-cost sensors and connectivity options, such as LoRaWAN or Sigfox, which can be more affordable and suitable for rural areas with limited connectivity.

In addition, data analysis and interpretation tools, such as machine learning algorithms, can help farmers make sense of the vast amounts of data generated by IoT sensors. These tools can provide insights on crop growth patterns, disease outbreaks, and optimal irrigation levels, allowing farmers to make informed decisions and improve their overall efficiency.

Finally, it is important to address data privacy and security concerns associated with IoT technology in agriculture. Farmers need to ensure that their data is protected from cyber threats and that they have control over how their data is shared and used.

6.7 Benefits of IoT in Agriculture

Real-time monitoring and analysis of data: Devices connected to the Internet of Things are able to provide data on a variety of environmental factors, including the temperature and moisture content of the soil. This information can be utilized to make informed decisions regarding irrigation, fertilization, and pest control, which will ultimately lead to improvements in yields while simultaneously leading to a reduction in costs.

Devices connected to the Internet of Things have the potential to assist farmers in more effectively managing resources like water, fertilizer, and energy, thereby enabling the operations of the farmers to run more smoothly—for example, soil moisture sensors can be put to use in order to optimize irrigation schedules, which, in turn, help to cut down on the amount of water that is lost to waste.

Increased productivity can be achieved by automating a variety of tasks using devices that are connected to the Internet of things (IoT). Some examples of these tasks include monitoring and controlling irrigation systems and adjusting the temperature and humidity in greenhouses. It is possible that this will lower labor costs while simultaneously increasing productivity.

Better decision-making: The data collected by Internet of Things devices can be used to make more informed choices about planting, fertilizing, and harvesting, which will ultimately result in higher yields and lower costs. These benefits can be realized in agriculture.

IoT devices can be used to track the movement of crops and animals, providing valuable information that can be used to improve food safety and supply chain management. This information can be used to improve food safety and supply chain management. This leads to an increase in the level of traceability.

New business models: The use of robotic AI and IoT in agriculture can also enable new business models, such as farm-as-a-service or data-driven farming. These models can help

smaller farmers access the benefits of these technologies without having to make significant capital investments.

In the world of agriculture, the use of technology has long been a gamechanger. The latest technology trend that is transforming the industry is the Internet of Things (IoT). The use of IoT in agriculture allows farmers to collect data on environmental factors, such as soil moisture, temperature, and other key metrics in real time. With this data, farmers can make informed decisions about resource management and improve crop yield while reducing waste.

6.8 The Role of AI in Tomorrow's Farming

Technologies based on artificial intelligence benefit agriculture by enhancing the output of conventional farming methods and freeing them from constraints. The term "artificial intelligence" (AI) is used to describe the process of designing and building machines with cognitive abilities similar to those of the human brain but with the capacity to handle much more information at once. While artificial intelligence (AI) is traditionally associated with computer science, its application in the agricultural sector should extend beyond this (Jha *et al.*, 2019). Using AI, engineers have created an assortment of implements that have been field-tested and finetuned for use in agriculture.

A number of agricultural field-steps have been successfully developed by them, such as soil testing, weeding, agricultural pesticide control, treating diseased crops, insufficient irrigation purposes for satisfying harvest requirements, as well as post-harvest operations like storage management and preservation parameter optimization. The quantity and quality of agricultural products produced by farmers have both improved (Talaviya *et al.*, 2020).

In contrast, artificial intelligence can be applied to farming to lessen the negative impact of unsustainable methods like excessive pesticide use, inefficient water management, and fertilizer runoff. Artificial intelligence (AI) could be used to address both of these issues (Jha *et al.*, 2019). Multiple AI systems have been proposed and developed by scientists for use in plantations in the past (Bannerjee *et al.*, 2018).

With the help of AI-based technologies, we can reduce the amount of manual labor needed to achieve our goals. Because of their ability to gather and analyze massive amounts of data, from official websites to real-time field data, AI-based devices can swiftly respond to questions that humans cannot answer.

Problems that would normally take a lot of time and effort to solve can then be proposed.

Smart farming with AI, IoT, and robotics would need sophisticated human ingenuity to develop. Because AI requires training with the biological skills of the farmer and *vice versa* (Talaviya *et al.*, 2020), trained farmers will also need to be trained in these AI technologies.

6.8.1 How Artificial Intelligence (AI) Can Help the Farming Industry

The introduction of AI into any sector begins with machine learning. It is preferable to supply the necessary information in a machine-readable format and return the processed result in an accessible format to humans' language. As it processes the input data, the AI-based machine should be able to pull relevant information from the directed databases. The ability to interpret real-time parameters is important in cases where the AI needs to use time-sensitive information to reach a conclusion. When deciding what to plant and when, it is important to keep the upcoming season's forecast in mind.

Chatbots are computers programmed to simulate human conversation in order to provide remote assistance to people in non-technical fields, such as farming. Unmanned aerial vehicles (UAVs) are commonly used by government and institutional agricultural officers to geolocate potential threats to the fields,

including the spread for forest fires, pest invasions, pathogen attacks, and many others (Talaviya *et al.*, 2020).

Neuro-fuzzy logic, fuzzy logic, expert systems, and artificial neural networks (ANNs) are four methods for problem solving (Jha *et al.*, 2019). The use of ANNs has become the standard method for developing AI-based technologies. An ANN is a machine that can learn and make decisions in the same way that a human can. Axons and synapses in neurons allow for the transmission of electric signals throughout the brain. Silva, Almeida, and delta-bar-delta are just some of the algorithms employed. Algorithms like these allow the machine to perform an internal task, which is what sets them apart from regular computer programs (Jha *et al.*, 2019). The machine's ease of use for farmers and other stakeholders depends on the creation of a hardware–software interface. What we call "embedded systems" are software-powered hardware components.

6.8.2 Selecting Seeds and Plants With the Help of AI

The key to optimizing yields and ensuring uniformity in production is selecting seeds with high vigor and good germination and seedling emergence rate (TeKrony&Egli, 1991). Farmers usually rely on past experience when making the best seed selection decisions, as conducting experiments to fine-tune

seed selection in a lab is both time-consuming and errorprone. Artificial intelligence (AI) technical devices are able to analyze historical data to a greater extent than a typical farmer can, giving them a deeper understanding of how different seed varieties react to varying weather conditions, disease resistance, etc.

Intelligent Farming Equipment and Computerized Systems

Machine learning-based phenotypic analysis and automated seed imaging were used to develop a phenotyping platform called SeedGerm. The main algorithm of SeedGerm has been developed, and it has capabilities like background removal, feature extraction, germination detection, and trait measurements. The SeedGerm system consists of a transparent plastic box and an overhead camera.

SeedGerm's seed imaging module allows for high-throughput imaging and the removal of background. Humidity and temperature sensors from the environment are integrated into the system as well. SeedGerm can rate the viability of seedlings by measuring morphological changes and scoring germination. Certificates of germination may be issued based on these criteria (Colmer *et al.*, 2020).

6.8.3 Use of Artificial Intelligence in Farming

Growth chambers require specific levels of light, humidity, oxygen, and carbon dioxide, all of which can be supported by sensors and embedded systems. They are able to make instantaneous adjustments to the crop environment in accordance with optimized parameters derived from the most recent crop growth data (Lakhiar *et al.*, 2018). Trace Genomics is a technology company that analyzes soil samples from farms to extract DNA and determine the total number of microbes present. In order to analyze the soil data, machine learning methods are used. Finally, the farmer would be given recommendations that are grounded on the biological and chemical properties of the soil sample and are supported by historical data. This could be useful in determining which crop would thrive best on a given plot of land.

Growing crops calls for a reliable and nearby source of water. Because freshwater is so scarce, it is crucial that farmers only water their crops as much as they need to. Thus, by recording real-time data on soil moisture content and weather conditions, AI technologies could regulate the amount of water needed and automate the supply and cessation of the water (Talaviya *et al.*, 2020). Kumar *et al.* (2014) enumerated a few of these automated irrigation methods that rely on AI. Mahmood *et al.* (2016) state

that excessive pesticide use endangers human health and biodiversity by increasing the risk of eutrophication in waterways. In addition to enhancing crop coverage, the “Rover” invention by Facchinetti *et al.* (2021) allows for up to 55% reductions in the amount of pesticide typically sprayed.

6.8.4 A Predictive Model for Yield Using Artificial Intelligence

Prediction models are one of the most promising artificial intelligence (AI) techniques for adoption by farmers because they promise to increase both yield and profit. Examining the soil’s nutrient content, crop data, and weather conditions are all prerequisites to making a yield prediction. In 2020, Van Klompenburg *et al.* conducted a comprehensive literature review on yield prediction models.

6.8.5 Implementing AI for Weed and Pest Management

Partel *et al.* (2019) describe a method for developing automated machines that could specifically detect weeds and spray them with agrochemicals; this would reduce the waste of weedicide and crop exposure to agrochemicals.

Bannerjee *et al.* (2018) mention several programs for pest management, such as Pasqual and Mansfield, SMARTSOY, and CORAC.

6.8.6 AI-Powered Inventory Management and Sales Promotion

The quality of agricultural products sold to consumers depends on how they are stored before they are sold. In order to keep these goods fresh for longer after harvest, a number of sensors for use in storage facilities have been developed. Farmers today only have a rudimentary grasp of the traditional markets for their products, but market data can accurately analyze current market trends, pricing decisions, and consumer consumption information to suggest for the next crop cycle to farmers (Talaviya *et al.*, 2020).

The ability of farmers to make more informed decisions regarding irrigation and fertilization is among the most significant advantages brought about by the implementation of IoT in agricultural settings. Traditional farming methods frequently involve the application of significant quantities of water and fertilizer, which can result in significant amounts of waste and significant damage to the environment. Nevertheless, farmers are now able to monitor the levels of soil moisture in

real time thanks to the Internet of Things (IoT), which enables them to adjust their irrigation practices as necessary.

Farmers now have the ability to monitor soil temperature, which is an essential component in crop growth and can be tracked using IoT. Monitoring the temperatures at which the soil is kept can assist farmers in determining the optimal times to plant and harvest their crops, as different types of produce have varying temperature requirements for optimal growth. With this knowledge, farmers can improve crop yields by optimizing their planting and harvesting practices to get the most out of their crops.

The Internet of Things has the potential to assist farmers in recognizing the preliminary symptoms of plant stress, such as nutrient deficiencies and diseases. This has the potential to be an enormous windfall for agriculturalists. With the help of sensors connected to the Internet of Things (IoT), farmers are now able to monitor the state of their crops in real time, which gives them the opportunity to take corrective action before the damage is irreversible.

Farmers are able to make better, more informed decisions about crop management with the assistance of connected devices that can monitor the weather and relay that

information to them in real time. One area that could benefit enormously from the utilization of these data is that of “precision agriculture.” Now that farmers have access to these data, they are able to make decisions regarding crop management that are more informed and can mitigate the effects of crop losses that are caused by the weather.

As more farmers adopt these technologies, we can anticipate seeing an agriculture industry that is more efficient, sustainable, and profitable. This industry will be able to meet the requirements of a growing population while also preserving our planet for the generations that will come after us. While there may be initial costs associated with the adoption of these technologies, the long-term benefits are clear. By investing in innovation and embracing new technologies, the agriculture industry can continue to provide the food and resources that our world depends on while also protecting the environment and promoting sustainability.

Furthermore, the adoption of robotic AI and IoT technologies in agriculture can also bring about a new era of economic growth, creating job opportunities for highly skilled workers in areas such as data analytics, robotics engineering, and software development. This can help address the labor shortage issue in

the agriculture industry and boost economic development in rural areas.

Moreover, by adopting sustainable agricultural practices and using technologies that reduce waste and conserve resources, the agriculture industry can play a critical role in mitigating the impact of climate change. By reducing greenhouse gas emissions and improving carbon sequestration, the industry can contribute to global efforts to combat climate change.

6.9 Robots in Agriculture—Perceptions and Pros, Cons

The use of robots in agriculture is currently most beneficial during the harvesting process; however, there is potential for their application in other agricultural tasks, such as weeding. Robots that are used in agriculture are referred to as agricultural robots.

The use of automation in agriculture has enabled farmers to save time and money while also contributing to the overall development of the sector as a whole. Some of the technological advances that have been put to use in farming include the mobile robot, the flying robot, the forester robot, and Demeter.

Farmers in even the most impoverished nations are expressing an interest in robots that can assist them in tending to their crops, harvesting their crops, and providing care for their livestock. Because of the complexity of the programming, interaction between humans and agricultural robots is essential.

The end effector, the manipulator, and the gripper are all components of the mechanical design. The design of the manipulator takes into account a variety of factors, including the task at hand, the required motions, and the cost-effectiveness of the task. “End effector” is the name given to the tool that is located at the very tip of the robotic arm on an agricultural robot, which is used for a wide variety of different agricultural tasks.

6.9.1 Farming Equipment With Automated Functions and the Positive Effects

As a result of their increased precision, machines are now able to accurately detect higher-quality products, and the field capacity of each round can now be played to its fullest potential. Robots will never require rest because they will never become ill or tired.

Because they do not require a driver's seat, controls, or cab, robots can be utilized in a diverse range of agricultural applications. They are also highly skilled at avoiding and navigating around obstacles such as trees, rocks, ponds, and other man-made and natural structures.

On farms, the quantity of pesticides used could be reduced by as much as 80% if robots were used instead of pesticides. It is possible that in the not-too-distant future, robots will be able to take on additional responsibilities. It is possible that manufacturing and repairing robots will result in an increase in demand for human labor.

There are numerous ways in which agricultural robots such as the Merlin Robot Milker, Rosphere, Harvest Automation, Orange Harvester, Lettuce Bot, and Weeder can be of assistance to farmers. These are just some of the examples of agricultural robots that are currently available. The industry of horticulture is yet another example of a possible use case. The RV 100, which was created by Harvest Automation Inc., is one example of a product that has a use in the horticultural industry.

During the process of handling and organizing the potted plants, the RV100 is utilized to create space between the plants, collect them, and then consolidate them. By utilizing RV100 for

this objective, production costs can be cut, placement precision can be improved, and autonomous operation can be achieved both outside and inside.

There is a vast potential for increasing agricultural productivity through the utilization of robots, and there are increasingly more instances of robots of varying kinds being used on farms. The concept of utilizing technology like robotics in agricultural settings is a relatively new one.

Without the need for human intervention, it is likely that robots will carry out agricultural tasks such as spraying, mechanical weed control, and fruit picking in the future. As a consequence of this, farmers will be able to have a smaller impact on the environment, work more precisely and effectively, and take a more nuanced approach when it comes to the management of individual plants.

Each new day sees the field of robotics expand to incorporate new subfields of study. Because robots have the potential to take over jobs previously performed by humans, they are bringing with them more effective and lucrative solutions. Robots are also capable of being programmed to carry out tasks that humans would find either challenging or potentially harmful to their own health.

The use of robots can protect workers from the dangers of handing chemicals and operating high-pressure spray systems, which would otherwise be necessary. A farm that uses them can reduce its use of pesticides by as much as 80%, which is another benefit.

The fully automated fruit harvesters: Before any action can be taken, a number of considerations, including the size and color of the fruit that is to be picked, need to be made. Both the driverless tractor/sprayer and the robotic sheep-shearing equipment have a long way to go before they can fully replace human labor.

Robots are capable of performing a variety of horticultural tasks, including pruning, wedding, spraying, and monitoring, amongst others. The term “livestock robotics” refers to the practice of employing robots in agricultural settings, more specifically for the purpose of automating mundane tasks such as milking, washing, and castrating livestock. By utilizing these robots, labor-intensive tasks that were previously performed by humans, such as spraying for weeds or bracken, can be mechanized and automated without putting humans in danger. As a result of the increased use of robots in agriculture, there is a decreased demand for human labor, an increase in the overall

quality of the fresh produce, and a decrease in the cost of production.

Rice farmers usually plant their seeds by hand in their fields. Robotic fruit pickers can be found working in orchards, while robotic milkers can be found in barns or factories milking cows. Depending on the tasks that they are designed to carry out, robots are programmed with a variety of different sets of instructions. In contrast to the robotic milker and fruit picker, the rice planter is only capable of carrying out a single job at a time.

Activities such as spreading manure or fertilizers, dispensing heavy chemicals or drugs, and dispensers of such substances are all included in the purview of the deployment of unmanned options. These activities and dispensers fall under the purview of the deployment of unmanned options. The sensors, cameras, software, and other technology that are integrated into agricultural robots are at the cutting edge of their fields.

6.10 Complications Associated With Robots Used in Agriculture

It is costly to search for ways to address energy problems, jobs in the agricultural industry could be at risk, and the

introduction of robots into farming could potentially alter the cultural or emotional appeal of the industry. All of these issues are interconnected. Because the work requires intelligence and quickness, it is sometimes possible for human operators to be replaced by robots in agricultural settings. It is possible that this will make people more open to the idea of robots in industries other than agriculture.

6.10.1 Difficulties and Limitations to be Conquered

The use of robotics in agriculture has the potential to confer a great number of benefits; however, there are also a great number of challenges and restrictions that must be overcome. The high initial investment required to get started with these technologies is one of the most difficult obstacles to overcome. Large-scale implementation of these technologies is further complicated by the fact that rural areas often suffer from a dearth of necessary infrastructure.

Another problem is finding people with the right skills to run and repair the robots. This presents a significant challenge for technologically inexperienced farmers and nations lacking in technical expertise.

6.10.2 Successful Applications and Other Case Studies

Despite these challenges, robotics have been widely implemented in agricultural settings around the world with great success. As an illustration, harvesting and soil analysis are two examples of common applications for robotic technology in Japan. Both of these applications have led to increased productivity and decreased costs. In Europe, weeding, pruning, and planting are all being done with the assistance of robotic systems, and the results have been quite impressive.

Recent research carried out in Australia has provided evidence of the benefits that robotics can bring to agricultural settings. Using a robot system to perform common farming tasks like planting and harvesting can increase productivity by as much as 50% when compared to traditional methods. In addition to increased crop yields, the farmer also saw a significant decrease in labor costs.

6.10.3 Societal and Ethical Implications

As the use of robotics in agriculture increases and becomes more common, it is crucial to think about the social and ethical implications of this technology. Some worry that this will lead to more industry consolidation, which will have negative effects

on farming communities and force farmers to find new employees.

While these are legitimate worries, it is important to keep in mind that robotics could end up being a huge boon for both farmers and consumers. Increased food production and long-term sustainability may result from the use of robotics in agriculture due to the potential for greater efficiency, lower costs, and higher crop yields.

One of the main benefits brought by the use of robotics in agriculture is increased productivity. Automation of repetitive tasks, such as planting and harvesting, can greatly increase productivity on farms and give farmers more time for other endeavors. Furthermore, robots can work around the clock, providing steady, reliable, and productive labor. The widespread use of harvesting robots and soil analysis robots in Japan has resulted in a number of positive outcomes, two of which are an increase in output and a reduction in input costs. Automated systems are showing promise for performing labor-intensive tasks in Europe, including weeding, pruning, and planting, among other things.

Recent research carried out in Australia shed light on the benefits that robotics can bring to agricultural settings. The

high cost of introducing robots into agricultural settings is one of the biggest obstacles. A lack of supporting infrastructure and technical knowledge could also make it challenging for farmers to adopt this technology. Because of this, the widespread use of this technology is more challenging.

Nevertheless, many successful applications of robotics in agricultural settings have been made despite these challenges—for instance, in recent years, drones have become increasingly popular for use in agricultural settings. Information gathered by drones about the health of crops and soil is invaluable to farmers. The use of robots to assist with agricultural tasks like planting, harvesting, and crop health monitoring has also contributed to an improvement in agricultural output and product quality.

Last but not least, it is crucial to consider the potential ethical and social consequences of expanding robotics use in farming. There is a danger that human workers will be replaced by robots as a result of the increased automation of the food supply chain despite the fact that robots can improve food production and the sustainability of agricultural practices.

In summary, the agriculture industry is facing several challenges, but with the adoption of innovative solutions such

as robotic AI and IoT technologies, these challenges can be overcome. The benefits of adopting these technologies are numerous, ranging from increased productivity, reduced labor costs, and improved sustainability. By investing in innovation and embracing new technologies, the agriculture industry can continue to provide food and resources to meet the needs of a growing population while preserving the environment for future generations.

It is crucial for governments, private sector, and the agriculture industry as a whole to work together to promote the adoption of these technologies and sustainable agricultural practices. Policies and initiatives that incentivize the adoption of these technologies can go a long way in promoting their use and ensuring that they are accessible to all farmers, including those in developing countries.

In addition, education and training programs can help farmers learn how to use these technologies effectively, thereby increasing their adoption and maximizing their benefits. Such programs can also help farmers understand the importance of sustainable agricultural practices and the role they can play in addressing climate change.

Arguments for and against the use of agricultural robots: There are many different biosystems where robotic technology has naturally spread, including agriculture, forestry, greenhouses, horticulture, and fisheries, to name just a few of them. It is gradually taking the place of the conventional methods that were once utilized to achieve the same goals.

6.11 Conclusion

The agriculture industry has a critical role to play in meeting the food and resource needs of a growing population while also preserving the environment for future generations. The adoption of robotic AI and IoT technologies, along with sustainable agricultural practices, can help the industry overcome its challenges and pave the way for a more efficient, sustainable, and profitable future. By working together and investing in innovation, we can ensure that the agriculture industry continues to play a vital role in our world, providing the food and resources that we all depend on. The widespread use of agricultural robotics could dramatically alter our perspectives on farming and food production. Despite the difficulties and restrictions, the benefits of enhanced productivity, reduced expenses, and higher harvest yields make it a worthwhile investment for farmers. If we have the necessary resources and knowledge, we can continue to

develop novel applications for robotics in agricultural settings. As a result, we can look forward to a more efficient and environmentally friendly food production system.

The use of robotics in agriculture is not a threat, but rather a valuable tool that can help farmers increase productivity, reduce costs, and enhance harvests. This is because the increased use of robots in agriculture has led to higher crop yields. Maintaining our investment in R&D will ensure that agricultural robotics' benefits will be realized for many years to come.

References

1. S.M. Pedersen, S. Fountas, H. Have, B.S. Blackmore, Agricultural robots— System analysis and economic feasibility, *Precis. Agric.*, 7 (4) (Oct. 2006) 295–308,doi:10.1007/s11119-006-9014-9.
2. M.G. Lampridi, *et al.*, A Case-Based Economic Assessment of Robotics Employment in Precision Arable Farming, *Agronomy*, 9 (4) (Apr. 2019) 175, doi:10.3390/agronomy9040175.
3. G.T. Aguiar, G.A. Oliveira, K.H. Tan, N. Kazantsev, D. Setti, Sustainable Implementation Success Factors of AGVs in the Brazilian Industry Supply Chain Management, *Procedia*

Manuf., 39 (2019) 1577–1586, doi:10.1016/j.promfg.2020.01.284.

4. K. Konolige, M. Agrawal, FrameSLAM: from Bundle Adjustment to Real-Time Visual Mapping, *IEEE Trans. Robot.*, 24 (5) (Oct. 2008) 1066–1077,doi:10.1109/TRO.2008.2004832.
5. M. Cummins, P. Newman, Appearance-only SLAM at large scale with FAB-MAP 2.0, *Int. J. Robot. Res.*, 30 (9) (Aug. 2011) 1100–1123, doi:10.1177/0278364910385483.
6. M.J. Milford, Gordon.F. Wyeth, SeqSLAM: visual route-based navigation for sunnysummer days and stormy winter nights, in: *2012 IEEE International Conference on Robotics and Automation*, St Paul, MN, USA, May 2012, pp. 1643–1649,doi:10.1109/ICRA.2012.6224623.
7. P. Neubert, N. Sunderhauf, P. Protzel, Appearance change prediction for long-term navigation across seasons, in: *2013 European Conference on MobileRobots, Barcelona*, Catalonia, Spain, Sep. 2013, pp. 198–203, doi:10.1109/ECMR.2013.6698842.
8. R. Giubilato, *et al.*, RelocalizationWithSubmaps: multi-Session Mapping for Planetary Rovers Equipped With Stereo Cameras, *IEEE Robot. Autom. Lett.*, 5 (2) (Apr. 2020) 580–587, doi:10.1109/LRA.2020.2964157.
9. J.P. Underwood, G. Jagbrant, J.I. Nieto, S. Sukkarieh, Lidar-Based Tree Recognition and Platform Localization in

Orchards: lidar-Based Tree Recognition and Platform Localisation, *J. Field Robot.*, 32 (8) (Dec. 2015) Art.no. 8, doi:10.1002/rob.21607.

10. C. Kester, H. Griepentrog, R. Hörner, Z. Tuncer, A survey of future farm automation A descriptive analysis of survey responses, in: *Precision agriculture '13*, Wageningen, Jul. 2013, pp. 785–972.
11. Online article by HEBA SOFFAR, PUBLISHED MAY 11, 2016, https://www.online-sciences.com/robotics/agricultural-robots-advantages-and-disadvantages/
12. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. "Plant Disease Identification Using Convolution Neural Networks." In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
13. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. "Solutions Using Machine Learning for Diabetes." *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
14. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, "Image Processing Based Intelligent Mini Robotic Face Recognition System," *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.

15. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
16. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
17. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485.
18. Talaviya, T., Shah, D., Patel, N., Yagnik, H., & Shah, M. (2020). Implementation of artificial intelligence in agriculture for optimisation of irrigation and application of pesticides and herbicides. *Artif. Intell. Agric.*, 4, 58-73.
19. Jha, K., Doshi, A., Patel, P., & Shah, M. (2019). A comprehensive review on automation in agriculture using artificial intelligence. *Artif. Intell. Agric.*, 2, 1-12.

Notes

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7

Data Analytics in Agriculture: Predictive Models and Real-Time Decision-Making

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Abstract

The agricultural sector increasingly uses data analytics to help with decision-making. Predictive models and real-time decision-making are essential components of data analytics effectively utilized in agriculture. An overview of these ideas, their significance in agriculture, and their implementations in the sector are given in this chapter. The chapter begins by discussing the difficulties in gathering and organizing

agricultural data and methods for doing so successfully. The discussion of predictive models then goes in-depth, covering the many kinds employed in agriculture, their uses, and instances of successful predictive modelling in agriculture. The necessity of real-time decision-making in agriculture, the methods and techniques employed, and successful instances are all covered in this section. In addition, the chapter covers the advantages, difficulties, and approaches for incorporating predictive models and real-time decision-making in agriculture. Lastly, the possible uses and ramifications of new trends and technology for the agricultural sector and society are examined. A case study on precision farming offers an illustration of how data analytics have been successfully used in agriculture. The relevance of data analytics in agriculture and its potential advantages for the sector are emphasized throughout this chapter.

***Keywords*:** Data analytics, agriculture industry, decision-making, crop health, weather conditions, environmental impact, water usage, predictive models

7.1 Introduction

Using data analytics in agriculture, choices regarding crop management, resource optimization, and other farming

practices are made by gathering, processing, and analyzing data. Agriculture has always been a data-driven business, but the recent years have rapidly increased data analytics tools and technology, improving crop yields, resource efficiency, and environmental sustainability [1]. Using statistical models and machine learning algorithms, data analytics in agriculture often entails gathering data from several sources, including soil sensors, weather stations, and drones [2, 3]. The objective is to conclude the data that may be used in decision-making concerning crop management practices such as planting, irrigation, fertilization, controlling pests and diseases, and other areas. Finding patterns and trends in the data that would be difficult or impossible to find using conventional approaches is one of the main advantages of data analytics in agriculture [4]—for instance, machine learning algorithms may be built to forecast when crops will likely face water stress and to detect trends in soil moisture levels. This enables farmers to optimize irrigation and preserve water resources [5, 6]. Creating predictive models for crop production forecasting, spotting possible pest and disease outbreaks, and maximizing fertilizer and pesticide usage are all aspects of data analytics in agriculture. Statistical techniques and machine learning algorithms are frequently used in these models to analyze massive volumes of data and find patterns and connections that

could be challenging for people to see. For farmers and the sector, data analytics in agriculture can revolutionize how crops are grown and managed, boosting productivity, sustainability, and profitability [7].

Two essential elements of data analytics in agriculture are real-time decision-making and predictive models, which can significantly enhance crop management and resource efficiency [8]. Farmers can estimate crop yields, anticipate outbreaks of pests and diseases, and manage fertilizer and irrigation more effectively thanks to predictive models. These models can assist farmers in making knowledgeable decisions about crop management that can boost yields and decrease waste by analyzing historical and real-time data on weather patterns, soil conditions, and other environmental factors [9, 10].

Real-time decision-making enables farmers to make decisions immediately based on real-time data from sensors and other sources [11]—for instance, farmers can swiftly change irrigation rates if soil moisture levels decline to avoid water stress and preserve crop health. Farmers may further optimize resource utilization by using real-time decision-making to ensure that resources like water and fertilizer are applied precisely when and where they are required [12, 13]. It is impossible to exaggerate the value of predictive models and

real-time decision-making in agriculture. Farmers can maximize crop yields, cut waste, and preserve resources by using data analytics to make knowledgeable decisions regarding crop management [14, 15].

These advantages help farmers become more profitable while promoting more ecologically friendly and sustainable farming methods. In the end, predictive models and real-time decision-making can assist in ensuring a more secure and consistent food supply for an expanding global population. Predictive models and real-time decision-making can help farmers reduce risks related to unexpected weather patterns and other environmental factors and enhance crop management and resource efficiency [16]—for instance, farmers may take measures to lessen the effects of drought on their crops by using more effective irrigation practices or modifying planting dates using predictive models to foresee probable drought conditions [17].

Farmers can react more swiftly to unforeseen occurrences like sudden weather changes or insect outbreaks using real-time decision-making. Farmers may make informed judgments on how to respond to these occurrences by getting real-time data from sensors and other sources, perhaps saving substantial harm to their crops. However, there are several difficulties in

applying real-time decision-making and predictive modeling in agriculture [18]—for instance, creating precise prediction models necessitates using high-quality data, which might be challenging to come by in particular areas or for some crops. Similarly, making real-time decisions requires access to accurate and timely data, which can be difficult in places with poor connections or defective data-collecting equipment [19].

Predictive models and in-the-moment decision-making are the main topics of this chapter's investigation of data analytics' application in agriculture. The chapter will review several data gathered in agriculture and the difficulties in analyzing and understanding that data. Additionally, it looks at some of the technology and techniques being utilized in agriculture for data analytics, such as sensors, drones, and machine learning algorithms. The use of predictive models in agriculture to anticipate crop yields, spot pests and diseases, and maximize irrigation and fertilizer usage will be covered in this chapter. The difficulties of incorporating intricate environmental aspects into prediction models for agriculture and the requirement for high-quality data will be discussed. Finally, the chapter examines how real-time decision-making transforms agriculture, allowing farmers to make decisions on the fly based on real-time data from sensors and other sources.

7.2 Data Collection and Management in Agriculture

Many data sources are used in agriculture to collect data for analysis and are briefly discussed.

7.2.1 Soil Sensors

Solectron sensors are electrical devices that assess soil characteristics such as moisture, temperature, pH, and nutrient levels. These sensors can be buried in the ground at different depths depending on the crop and the soil conditions being tracked [20]. The wireless link that soil sensors often have enables real-time data transfer to a centralized database or a mobile device. To improve irrigation practices, soil moisture monitors, in particular, are frequently utilized in agriculture. These sensors gauge the water content of the soil and may be used to calculate when and how much irrigation is required. Using soil moisture monitors, farmers can prevent both over- and under-watering, which can result in stressed crops and lower yields, as well as waterlogging and other soil-related issues [21]. Crop management can also be improved by using soil temperature sensors. The temperature impacts how plants grow and develop, and the soil temperature may be a key determinant of when to plant and harvest crops. In addition to

helping farmers choose the optimal time to sow their crops, soil temperature sensors may assist them with other management practices like fertilization and insect control. The soil's acidity or alkalinity, which can significantly influence crop development and nutrient availability, can be measured with soil pH sensors [22]. Farmers may modify their management techniques to ensure that crops have access to the nutrients they require for optimal development by monitoring the pH of the soil. The soil sensors are a crucial piece of equipment for precision farming and may assist growers in maximizing crop yields while reducing waste and environmental impact. Farmers may choose wisely regarding irrigation, fertilization, and other crop management practices using real-time data from soil sensors, resulting in more sustainable and practical agriculture [23].

7.2.2 Weather Stations

Weather stations are instruments that monitor and keep track of various meteorological factors, including temperature, humidity, precipitation, wind, and solar radiation. These devices are often put in open spaces on the farm or at neighboring weather stations to gather information on the local weather [24]. Weather stations are crucial equipment in agriculture for monitoring weather conditions that may impact

crop growth and development. Temperature and humidity, for instance, can influence crop growth and blooming dates, while precipitation can influence soil moisture and plant development. Wind speed and direction can also impact pollination and seed dissemination, while sun radiation can affect photosynthesis rates. Farmers may choose wisely when to sow and harvest crops and when to use irrigation and other crop management techniques by gathering data from weather sensors—for instance, if a farmer anticipates a heat wave, they might modify irrigation schedules to guarantee that crops get enough water to prevent heat stress [25].

Additionally, by using data from weather stations to create prediction models, farmers may better foresee weather-related threats like drought, frost, or flooding. By accessing reliable weather forecasts, farmers can decide wisely regarding crop management practices, such as whether to fertilize or use pesticides. A crucial tool for precision agriculture, weather stations help farmers maximize crop production while reducing waste and environmental impact. Farmers may decide on crop management techniques in a way that is more sustainable and effective by gathering and analyzing meteorological data [26].

7.2.3 Drones

Unmanned aerial vehicles (UAVs), or drones, are miniature aircraft that may be remotely piloted or fly independently. Drones with cameras and sensors can be used in agriculture to gather high-resolution photographs and information on plant density, crop health, and other elements that may affect agricultural yields. Drones can take overhead pictures of fields and give farmers precise details on the condition and growth of their crops, enabling them to make educated decisions regarding crop management practices like fertilization and irrigation. Drones, for instance, can help farmers optimize irrigation schedules and consume less water by identifying field sections experiencing water stress [27].

Drones may have sensors that detect several agricultural metrics, including biomass, leaf area, and chlorophyll content, and provide visual footage. These sensors can give farmers more in-depth knowledge of the condition and development of their crops, enabling them to improve their fertilization and pest management procedures. Drones may also track crop development over time, helping farmers see trends and decide on crop management strategies with more excellent knowledge. Farmers can modify planting density and other practices to maximize agricultural yields, for instance, by tracking the

growth rates of plants over time [28]. Drones are a crucial tool for precision farming because they enable farmers to gather high-resolution information on the health and development of their crops, resulting in more productive and sustainable agriculture.

7.2.4 Satellite Imagery

Another crucial data source in agriculture is satellite imaging, which may give farmers in-depth knowledge of crop health, growth patterns, and other elements affecting yields. A bird's eye view of agricultural fields may be obtained from orbiting satellite photos, allowing for a thorough examination of crop health and environmental conditions. Numerous agricultural metrics, such as vegetation density, chlorophyll content, and water stress, may be tracked using satellite photography. Farmers can determine patterns in crop growth and health and choose the best crop management techniques by examining satellite photos over time [29].

Crop types and land use patterns may be mapped using satellite photography, which can aid farmers in optimizing crop rotation and other management techniques—for instance, by examining satellite photos, farmers might spot areas vulnerable to erosion or other soil-related difficulties and act to fix them. Additionally,

satellite imaging may track environmental factors like temperature, precipitation, and soil moisture that may influence crop development. Farmers may make wise judgments regarding irrigation and other management techniques thanks to access to real-time data on environmental conditions, resulting in more productive and sustainable agriculture [30].

An essential tool for precision agriculture, satellite imagery enables farmers to monitor crop health and growth on a large scale and decide on crop management practices. Farmers may maximize crop yields while reducing waste and environmental impact by fusing satellite photography with additional data sources like meteorological information and soil sensors [31].

7.2.5 Farm Machinery

Another agricultural data source that might offer important insights into crop management techniques is farm machinery. Many modern agricultural equipment come with sensors and other monitoring tools that may gather information on various crop factors, including soil moisture, plant density, and crop health—for instance, sensors that track soil moisture levels and offer information on soil compaction and other elements affecting crop yields can be added to tractors and other tillage

equipment. Planting equipment can also be fitted with sensors that assess plant spacing and seed placement to optimize planting density and minimize seed waste [32].

Farm machinery may be used to precisely and carefully apply fertilizers, insecticides, and other inputs and gather data on crop metrics. Farmers may apply inputs just where required, decreasing waste and environmental effects by employing variable rate technology and other precision application techniques. The information gathered by agricultural equipment may be utilized to create prediction models that can assist farmers in foreseeing possible issues and selecting the best crop management techniques [33]— for instance, farmers can forecast the likelihood of drought and modify irrigation schedules as necessary by analyzing data on soil moisture levels. Farm equipment is a crucial tool for precision agriculture, enabling farmers to gather information on crop attributes and decide on crop management strategies. Farmers may maximize crop yields while reducing waste and environmental effects by merging data from farm machines with other data sources such as meteorological data and satellite photography [33].

7.2.6 Pest and Disease Monitoring Systems

Systems for monitoring pests and diseases are one form of data source in agriculture that aids farmers in quickly identifying and responding to pest and disease outbreaks. These systems monitor crop health and look for indications of pest and disease infestations using a range of technologies, including sensors, cameras, and prediction algorithms—for instance, some systems for monitoring pests and diseases employ sensors to gauge environmental factors like temperature and humidity, which may be used to measure the possibility of outbreaks of pests and diseases. Other systems employ cameras and image processing software to look for malformations or discolorations in plant leaves as indicators of pest and disease infestations [34].

Farmers may utilize the information gathered by these monitoring systems to act swiftly and efficiently if a pest or disease epidemic is discovered—for instance, farmers may apply pesticides precisely where required, minimizing waste and adverse effects on beneficial insects and other non-target creatures. These monitoring systems can track the long-term spread of pests and illnesses and identify outbreaks of such things. Farmers can choose crop management practices, such as

rotating crops or selecting resistant cultivars, by carefully considering data on pest and disease populations.

Pest and disease monitoring systems are crucial for precision agriculture because they enable farmers to quickly and effectively identify pest and disease outbreaks [35]. Farmers may maximize crop yields while reducing the use of pesticides and other inputs by integrating data from these systems with information from other sources, such as meteorological data and soil sensors.

7.2.7 Market Data

Market information is a crucial data source for agriculture, assisting farmers in selecting, pricing, and marketing crops. Market data includes information on supply and demand, price patterns, and other elements that may affect agricultural activities' financial success. Farmers, for instance, might identify crops with high demand by analyzing market data and then modify their production methods appropriately [36]. They can also utilize market data to determine competitive pricing and accurate reflections of the state of the market.

Market information may also be used to locate new markets for agricultural goods, such as specialized exports or locally

produced or organic goods. Farmers may boost their profitability and expand their enterprises by seeing these opportunities and adapting their production practices accordingly [37]. Market data may also be used to monitor business trends and spot innovations or methods that can aid farmers in streamlining their processes. Farmers may maintain competitiveness and adjust to shifting market conditions by adapting to recent trends and technologies.

7.2.8 Crop Records

Crop records are a crucial source of information in agriculture that helps farmers monitor the progress of their crops over time and choose the best crop management techniques. These records may contain details about crop yields, input usage, planting and harvest dates, and other elements affecting crop performance. Farmers can see trends in crop performance and alter their production techniques as necessary by keeping thorough crop records—for instance, producers might explore potential causes, such as soil nutrient deficits or insect infestations, and take appropriate measures if crop yields in a specific area are regularly lower than projected [38].

Crop records may also be used to minimize waste and maximize the utilization of inputs. Farmers can find chances to

use fewer inputs, including herbicides and fertilizers, without reducing crop yields by monitoring input use over time. This can lower expenses and lessen the adverse effects of agricultural activities on the environment. Crop records can also be utilized to meet legal obligations, such as those about environmental protection and food safety. Farmers may demonstrate compliance with these rules and lower their risk of fines or legal action by keeping thorough records of input use and crop management techniques.

The crop records are a crucial tool for precision agriculture because they let farmers monitor crop performance over time and choose effective crop management techniques. Farmers may increase crop yields while consuming fewer resources and having a minor environmental effect by merging crop records with other data sources like meteorological data and soil sensors [39].

7.2.9 Livestock Records

In animal husbandry, livestock records are an essential data source that helps farmers better manage their herds and decide on breeding, health, and performance. Data on individual animals, such as age, breed, sex, weight, and reproductive history, may be included in livestock records. Herds of flock-

level statistics, such as average daily growth, feed efficiency, and death rates, may also be included. Farmers can follow the performance of individual animals and spot possible production or health concerns by keeping thorough livestock records—for instance, by monitoring feed intake and weight increase, farmers may spot animals that are not doing as well as the rest of the herd and take remedial action, like changing feed rations or giving them medical care.

Farmers can track breeding success and improve genetic selection by using livestock records. Farmers may find superior genetics and choose breeding techniques with more knowledge if they keep track of breeding history and performance data. Over time, this may result in increased herd or flock productivity. Additionally, livestock data can be utilized to meet legal requirements for the care and health of animals. Farmers may demonstrate compliance with these rules and guarantee the health and welfare of their animals by keeping thorough records of vaccines, treatments, and other animal health interventions [40].

Animal producers rely heavily on livestock records since they help them manage their animals more skillfully and decide on breeding, health, and performance. Farmers may maximize animal output and health while minimizing input consumption

and environmental effects by merging livestock records with other data sources such as weather data and soil sensors.

These are just a few of the numerous data sources utilized in agriculture to gather information for analysis. Collecting and analyzing data from several sources is essential to thoroughly understand crop health, resource usage, and other factors affecting crop yields and profitability [41].

7.3 Challenges in Collecting and Managing Agricultural Data

The management and collection of agricultural data are fraught with difficulties. Agricultural data can be of varying quality and consistency and may contain mistakes or inconsistencies—for instance, improperly calibrated soil sensors might produce erroneous data, and regional microclimates could affect meteorological data. Making correct judgments requires ensuring data consistency and quality. Agricultural data could be kept in several places and forms, making it challenging to access and integrate. Additionally, some data sources might not be usable due to infrastructure issues or technological restrictions.

Sensitive information, such as financial and personal data, may be present in agricultural data and must be safeguarded. To prevent data breaches and protect the interests of farmers and other stakeholders, it is crucial to ensure the privacy and security of agricultural data. There may be several owners of farming data, including farmers, researchers, and agribusinesses. A complicated problem might arise when deciding who owns the data and how it can be shared [42]. After gathering and managing agricultural data, making informed judgments is the first step. For data to be meaningful, it must be appropriately analyzed and interpreted, which requires specialized knowledge and equipment. Farmers, academics, legislators, and other stakeholders must work together to find solutions to these problems. While legal and regulatory frameworks can assist in preserving data privacy and ownership, investments in data infrastructure, standardization, and training can help increase data quality and accessibility. Overall, overcoming these obstacles is crucial if agricultural data reach their full potential for enhancing productivity, profitability, and sustainability in the sector [43]. Table 7.1 depicts the challenges of collecting data and strategies.

7.4 Strategies for Effective Data Collection and Management

Farmers and other stakeholders can employ a variety of ways to gather and handle agricultural data efficiently. To guarantee that the appropriate data is being captured, clearly identify the data needs and objectives. This will guarantee that the data acquired is valuable and usable and will assist in focus data gathering activities. Standardize the data gathering techniques and formats to ensure the data obtained is accurate and comparable from various sources. This will make it simpler to integrate and analyze the data.

Table 7.1 Challenges for collecting data and strategies.

Challenges in collecting and managing agricultural data	Implications	Strategies for overcoming challenges
Agricultural data can be of varying quality and consistency and may contain mistakes or inconsistencies.	Misleading insights and inaccurate decision-making	Ensuring data consistency and quality, properly calibrating sensors, and accounting for regional microclimates
Data could be kept in several places and forms, making it challenging to access and integrate.	Difficulties in accessing and integrating data	Investments in data infrastructure, standardization, and training to increase data quality and accessibility

Challenges in collecting and managing agricultural data	Implications	Strategies for overcoming challenges
Sensitive information, such as financial and personal data, may be present in agricultural data and must be safeguarded.	Risk of data breaches and privacy violations	Establishing legal and regulatory frameworks to protect data privacy and ownership and implementing security measures to prevent data breaches
Data ownership among farmers, researchers, and agribusinesses may be disputed.	Disputes over data ownership and sharing	Collaboration and negotiation among stakeholders to establish data ownership and sharing agreements

Challenges in collecting and managing agricultural data	Implications	Strategies for overcoming challenges
Proper analysis and interpretation of data require specialized knowledge and equipment.	Inability to extract meaningful insights from data	Providing training and support to farmers and other stakeholders in data analysis and interpretation

Using sensors, drones, and other technologies to automate data collecting can increase precision and save labor costs. However, ensuring the automated systems are regularly maintained and calibrated is crucial. To arrange and keep data in one place, use data management software. This will simplify accessing and analyzing the data and contribute to data security. Inform employees about the best practices for data gathering, software, and analysis. This will guarantee accurate data collection and efficient data analysis [44].

With other stakeholders, establish data-sharing agreements to guarantee that data is shared correctly and morally. This will make it possible to use data to understand better and make better decisions. Analyze data requirements often and change technology and data-gathering procedures as required. This will aid in ensuring that information is gathered effectively and efficiently. By implementing these techniques, farmers and other interested parties may better collect and manage agricultural data, empowering them to make decisions that will increase agriculture's productivity, profitability, and sustainability [42].

7.5 Predictive Models in Agriculture

7.5.1 Types of Predictive Models

Various kinds of predictive models can be used in agriculture. Linear regression establishes a link between environmental factors like temperature, rainfall, and agricultural production. Predicting crop yields will assist farmers. AUC, AR, and FPI's effects on yield may be determined using regression analysis. Rice was chosen for investigation since it is India's most widely grown crop [45]. AI turns field data into crop management in digital agriculture. Early detection through digital agricultural models reduces expenses. Ineffective models might backfire.

Before implementing recommendations, farmers typically spot-check their crops to verify models [46]. An effective drought index is necessary to appropriately measure agricultural drought, which may threaten food security and social protection. This article utilized the random forest algorithm (RF) to examine six frequently used drought indicators for agricultural drought assessment [47]. The proposed work reviews various AI techniques [fuzzy logic (FL), artificial neural network (ANN), genetic algorithm (GA), particle swarm optimization (PSO), artificial potential field (APF), simulated annealing (SA), ant colony optimization (ACO), artificial bee colony algorithm (ABC), harmony search algorithm (HS), bat algorithm (BA), cell decomposition (CD), and firefly algorithm (FA)] in agriculture, focusing on expert systems, robots developed for agriculture [48]. Fuzzy logic and multiclass support vector machine (SVM) methods are used to design and monitor model parameters such as ideal temperature, humidity, and soil wetness in the greenhouse farm at Modakkurichi near Erode [49]. Sustainable agricultural landscape management requires precise soil maps and GIS soil data. Digital soil mapping has increasingly employed machine learning (ML) models with little data for varied settings [50].

- Regression models: Regression models may be used to forecast results and are used to find associations between

variables. Regression models, for instance, may be utilized to predict agricultural yields based on meteorological information, soil moisture levels, and other variables.

- Decision trees: A decision tree is a model representing decisions and potential outcomes in a tree-like graph. The chance of certain occurrences, such as crop illnesses or insect outbreaks, may be predicted using these models.
- Random forests: These are a form of model that mixes several decision trees to increase the precision of predictions. Various outcomes, like agricultural yields, insect outbreaks, or animal output, may be predicted using these models.
- Artificial neural networks: These models imitate the composition and operation of the human brain. These models may examine large data sets and anticipate outcomes based on the data's patterns and correlations.
- Support vector machines: These are a form of model that may be applied to classify data into several categories. These models, for instance, may be used to categorize agricultural diseases according to symptom patterns and other variables.
- Ensembles models: Models known as "ensembles" integrate many models to increase the precision of predictions. Various outcomes, like agricultural yields, insect outbreaks, or animal output, may be predicted using these models.

The particular application and data type will determine the best prediction model. A suitable model must be chosen for the task at hand, and model accuracy must be constantly assessed to ensure the model makes accurate predictions.

7.6 Applications of Predictive Models in Agriculture

Tools that use a variety of statistical methods, such as data mining, predictive modeling, and machine learning, to analyze a variety of current and historical agricultural, biological, climate, and hydrological data from various sources to make predictions about future events are known as predictive analytics tools in agriculture [51]. Predictive models have many applications in agriculture, as shown in Figure 7.1.

- Yield prediction: Crop yields may be predicted using predictive models based on weather information, soil characteristics, and other variables. Farmers that use this information may plan for the planting, harvesting, and sale of their crops.
- Disease and pest prediction: Based on weather patterns, crop history, and other variables, predictive models may be used to forecast the risk of crop diseases and pest outbreaks.

Farmers that use this knowledge can proactively stop or control epidemics.

- Livestock production: Using predictive models, it is possible to forecast cattle production based on elements like genetics, feed, and environmental variables. Farmers can control herd health and optimize breeding and feeding practices using this information.

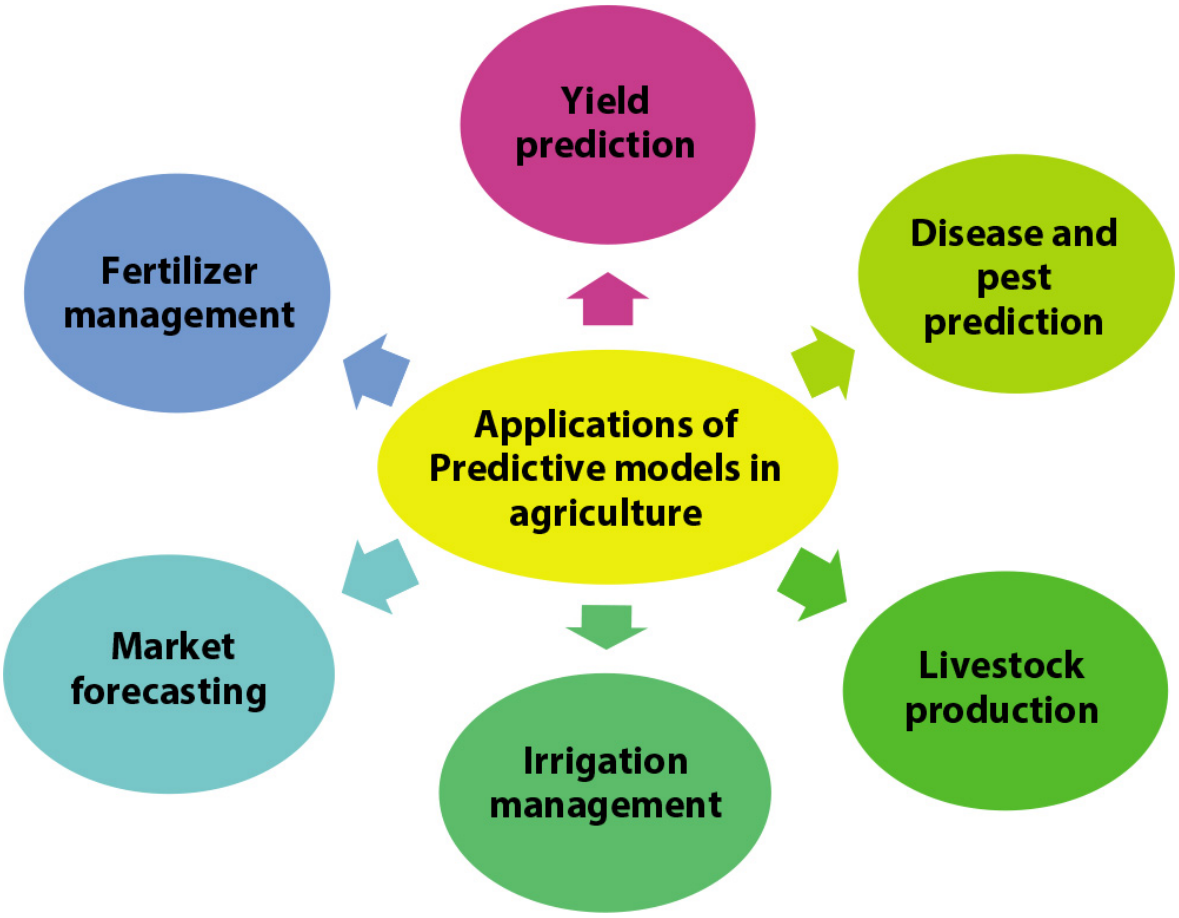

Figure 7.1 Applications of predictive models in agriculture.

- Irrigation management: Based on meteorological information, soil type, and other criteria, predictive models may forecast soil moisture levels and water requirements. Farmers that use this information may optimize irrigation schedules and save water.
- Fertilizer management: Based on the findings of soil tests, the kind of crop, and other variables, predictive models may be used to forecast nutrient requirements. Farmers that use this information can reduce nutrient runoff and optimize fertilizer application rates.

- Market forecasting: Based on historical data, consumer habits, and other variables, predictive models may be used to estimate market demand and pricing. Farmers that use this knowledge may choose crops, set production goals, and develop effective marketing plans.

These are but a handful of the numerous uses for predictive models in agriculture. Farmers may increase agricultural production, profitability, and sustainability by employing these models to make data-driven choices.

7.7 Real-Time Decision-Making in Agriculture

7.7.1 Importance of Real-Time Decision-Making in Agriculture

In agriculture, real-time decision-making is essential because it enables farmers to react swiftly to shifting circumstances and make wise choices that may influence crop yields, animal health, and profitability. Crop production and livestock health may both be significantly impacted by the weather. In reaction to meteorological occurrences like storms, droughts, or excessive temperatures, farmers might make decisions regarding planting, irrigation, and animal housing. Outbreaks of pests and diseases may spread swiftly and have a disastrous effect on animals and crops. Farmers can be informed of pests

or illnesses *via* real-time monitoring systems, which enables them to take prompt action to stop the spread and lessen the damage. Market circumstances are subject to rapid fluctuation, which affects how profitable crops and livestock [52].

Farmers can choose the best crops, set the right production goals, and use effective marketing techniques by using real-time market data. Effective resource management is essential since resources like water, fertilizer, and energy are frequently scarce in agriculture. Farmers can maximize resource utilization and reduce waste thanks to real-time data on soil moisture, nitrogen levels, and other variables. Real-time information on labor availability and productivity may assist farmers in better managing their labor resources and making educated decisions on recruiting, training, and scheduling. Agriculture demands a substantial amount of labor input. Farmers may increase their operations' effectiveness, productivity, and profitability while reducing risks and guaranteeing sustainable practices by making real-time choices based on data [53].

7.7.2 Tools and Technologies for Real-Time Decision Making

Several tools and technologies can aid farmers in making real-time decisions in agriculture.

• **Sensor networks**

Sensor networks are linked sensors that can continuously track changes in physical and environmental circumstances. Sensor networks can be used in agriculture to monitor conditions impacting plant growth and development, such as soil moisture, temperature, humidity, and nutrient levels. Sensors can track meteorological conditions and soil moisture levels to optimize irrigation schedules. Farmers may be able to minimize water waste and increase agricultural yields. Farmers can optimize fertilizer application rates and timing by using sensors to monitor soil nutrient levels and offer real-time data. By doing this, farmers may maintain crop yields while using less fertilizer and spending less money. By spotting symptoms of stress, sickness, or pest infestation, sensors may be used to keep an eye on the health of plants. This can help farmers promptly address problems and prevent yield loss [54].

Sensors can monitor weather conditions, air quality, and water quality to ensure compliance with environmental regulations and minimize ecological impact. Sensors can monitor animal health and behavior, such as tracking the location and activity of grazing livestock. This can help farmers optimize feed and grazing resources and detect early signs of illness or injury. By using sensor networks to collect and analyze real-time data,

farmers can make more informed decisions about their operations, leading to improved productivity, sustainability, and profitability [55].

• Satellite imagery

Images of the Earth's surface taken by satellites circling the globe are called satellite imagery. Satellite imaging may be used in agriculture to track weather patterns, crop health, and soil moisture across broad areas. Crop growth and health may be tracked using satellite photography over a wide region. Farmers may use this to spot stressed or infested regions and take prompt action to remedy issues before they worsen. Based on vegetation indices, plant strength, and health indicators, satellite data may be used to calculate agricultural yields. Farmers may use this information to forecast crop yields and plan harvest and marketing. By estimating soil moisture levels using satellite data, farmers may cut water wastage and improve irrigation schedules. Weather patterns like storms and droughts, which may influence crop production, can be tracked using satellite photography. Decisions on resource management, planting, and harvesting may be made using this knowledge. Monitoring changes in land use and land cover, such as deforestation and urbanization, may be done using satellite photography. Decisions concerning land use planning

and natural resource management may be made more intelligently with this knowledge by policymakers and farmers. Farmers may make better decisions about their operations and increase productivity, sustainability, and profitability by utilizing satellite images to gather and analyze data [29].

• Mobile applications

Mobile applications, sometimes called “apps,” are computer programs created specifically to operate on portable electronics like smartphones and tablets. Mobile applications may be used in agriculture to gather data, analyze it, let farmers communicate and work together, and give real-time data to help with decision-making. Data collection on several aspects, including agricultural yields, soil moisture levels, and meteorological conditions, may be done via mobile apps. These statistics can help farmers manage and allocate resources in an educated way. Crop pests and diseases may be detected and managed using mobile applications. Farmers may use the applications to identify pests or illnesses, monitor their progress, and choose the best treatments.

Real-time market data, such as commodity pricing and patterns in supply and demand, can be provided through mobile applications. Farmers may use this information to decide when

and where to sell their goods by using it as a guide. Farm operations may be managed *via* mobile applications, including inventory management, work scheduling, and equipment maintenance. Farmers that do this can streamline their processes and cut expenses. Farmers may exchange knowledge and best practices using mobile apps to enhance communication and collaboration. This can help farmers gain new information and skills, increase their output, and become more profitable. Farmers may make better decisions about their operations and increase productivity, sustainability, and profitability by utilizing mobile applications to gather and analyze data [56].

• Decision support systems

Decision support systems (D.S.S.) are computer programs that give decision-makers access to data and analytical tools to help them with complicated decision-making processes. By giving them access to pertinent data, models, and analytical tools, D.S.S. may assist farmers and other stakeholders in the agricultural industry in making more informed decisions. Farmers can employ decision support systems to assist them in making decisions on crop selection, planting times, fertilization, irrigation, and pest control. To offer the best crop management techniques, the computers may analyze data from various

sources, including weather stations, soil sensors, and satellite photography. The utilization of resources like water, electricity, and fertilizers may be optimized by farmers with the use of decision support systems. The systems may analyze crop needs, weather variables, and resource availability data to suggest resource allocation and management.

Systems for supporting decision-making can be used to identify and control hazards in agricultural operations. The systems' weather, soil, and market data analysis enables them to identify possible dangers early and suggest ways to mitigate them. Systems that enable decision-making may be used by farmers to make financial decisions, including debt management, investment planning, and crop insurance. To produce financial estimates and suggestions, the computers may analyze data on market pricing, crop yields, and production costs. Systems of decision support can assist decision-makers in developing and assessing agricultural policies. To evaluate the prospective effects of policy changes, the systems may analyze data on economic, environmental, and social issues [57].

• Livestock monitoring systems

Technology-based tools called livestock monitoring systems assist farmers in keeping an eye on the health and welfare of

their livestock. These systems gather information on numerous facets of animal behavior and physiology using a variety of sensors and other devices. Methods for monitoring livestock can be used to spot early indications of sickness or illness in livestock. The apparatus may monitor and notify farmers of any irregularities in vital indicators, including body temperature, heart rate, and breathing rate. Farmers who receive this early warning can move quickly to treat the animal and stop the sickness from spreading. Systems for keeping an eye on livestock can be used to track an animal's ability to reproduce. The tools help farmers to manage breeding programs and improve breeding efficiency by tracking estrus cycles and detecting pregnancy.

The consumption of feed by animals may be tracked using livestock monitoring devices. The devices can monitor feed intake and identify any alterations in feeding behavior, which may indicate an upcoming sickness or other health issues. Livestock monitoring systems may monitor environmental factors, including temperature, humidity, and ventilation. This may assist farmers in ensuring that their animals are living in the best possible circumstances, which can enhance their well-being and output. Systems for monitoring livestock can be used to locate animals on a farm. This can assist farmers in reducing

the danger of theft or loss, monitoring animal movement, and optimizing grazing patterns [58].

• Automated equipment

The term “automated equipment” describes technical advancements that enable jobs or processes to be completed with little or no human involvement. Automated machinery can be utilized in agriculture to increase output, effectiveness, and precision. Crops may be harvested with little to no human involvement using automated harvesting machinery like combine harvesters. These tools can identify and harvest ripe crops, lowering labor costs and raising output. Crops may be watered precisely and effectively using automated irrigation systems, such as drip irrigation or center pivot systems. These systems can deliver the proper quantity of water at the right time, decreasing water waste and boosting crop yields.

In dairy farming, cows can be milked automatically using milking systems. These systems enhance efficiency and save labor costs using sensors and robots to locate and connect milking cups to the cow’s udder. Nutrient concentrations, pH, and other parameters of soil samples may be determined using automated soil testing equipment. These devices can deliver precise and fast data, enabling farmers to choose the best

methods for soil management, including fertilization. Crops may be planted accurately and effectively with automated seeding tools like precision planters and seed drills. These devices can plant seeds at the proper depth and spacing to increase agricultural yields and decrease the need for manual labor. Farmers may boost output, save labor costs, and enhance efficiency by adopting automated equipment. These technological advancements can also increase agricultural operations' precision, resulting in better crop yields and profitability [59].

7.8 Integration of Predictive Models and Real-Time Decision Making in Agriculture

7.8.1 Benefits of Integrating Predictive Models and Real-Time Decision-Making

Integrating predictive models and real-time decision-making in agriculture can bring several benefits, as shown in Figure 7.2 and discussed below:

- Increased productivity: Predictive models may assist farmers in choosing the correct times and methods for planting, fertilizing, irrigating, and harvesting their crops. Farmers can react swiftly to changes in the weather, insect infestations, or other factors that might impact crop production by using

real-time decision-making. Farmers may optimize their agricultural operations and boost output by employing prediction models and real-time decision-making.

- Cost savings: Using predictive models, farmers may use fewer inputs like water, pesticides, and fertilizers to produce their crops. Making real-time decisions may help farmers save money, increase productivity, and prevent costly errors. Farmers may save expenses and increase profitability by combining predictive modelling with real-time decision-making.
- Increased precision: Predictive models may give farmers precise and trustworthy information on crop yields, soil characteristics, weather patterns, and other elements influencing farming operations. Farmers may respond quickly to changes in these elements and modify their agricultural practices using real-time decision-making. Farmers may increase the precision of their farming operations and make better judgments by merging predictive models with real-time decision-making.
- More effective risk management: Predictive models can assist farmers in spotting possible hazards like crop failures or insect infestations before they happen. Farmers can respond swiftly to these hazards and lessen their effects using real-time decision-making. Farmers may enhance risk

management techniques and defend their crops from potential dangers by incorporating predictive models and real-time decision-making.

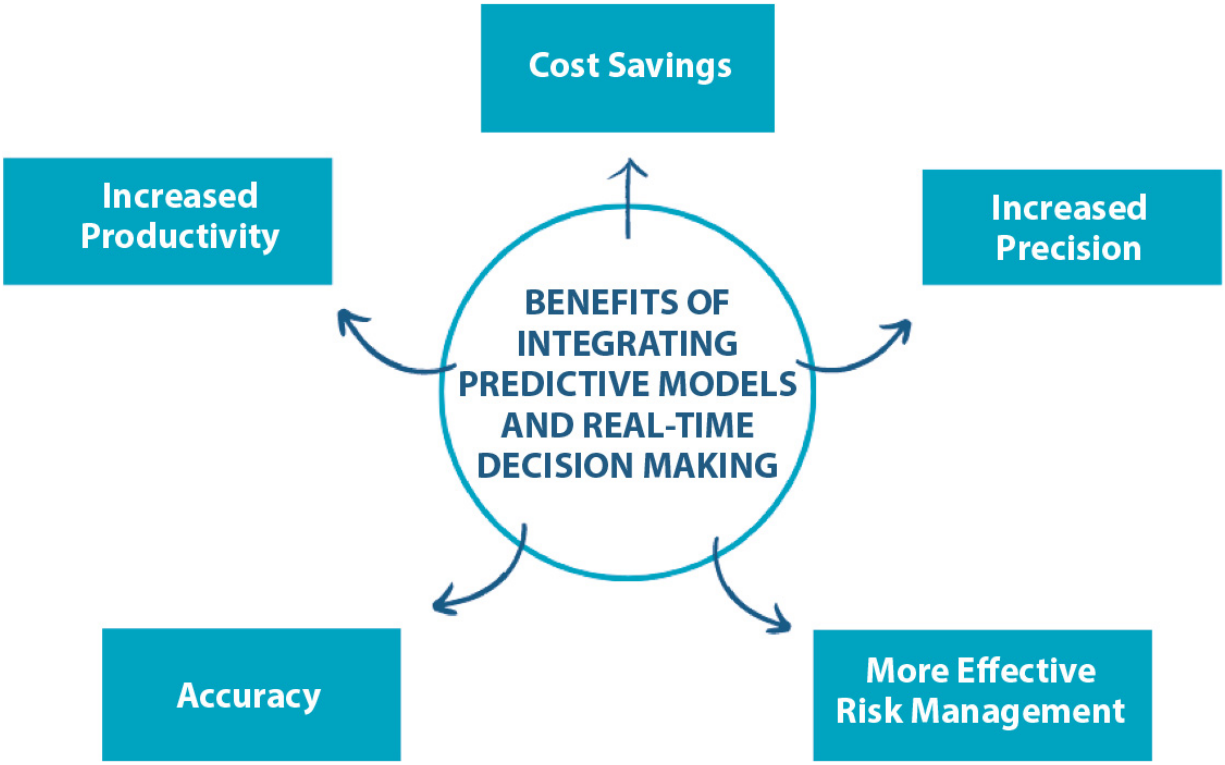

Figure 7.2 Benefits of integrating predictive models and real-time decision making.

Agriculture may gain from integrating predictive models with real-time decision-making in several ways, including increased production, lower costs, more accuracy, and better risk management. Farmers can optimize their agricultural operations and increase profitability using these tools and technology [59]. The main benefits of integration are shown in Figure 7.2.

7.8.2 Challenges of Integration

There are many advantages to using predictive models and real-time decision-making in agriculture, but there are also some potential drawbacks for farmers and other agricultural stakeholders. The main challenges of integration are shown in Figure 7.3.

- Data availability and quality: High-quality and fast data are essential for making precise predictions and in-the-moment judgments. However, due to the complexity and variety of farming operations and the high cost of collecting and managing data, data quality and availability can be a barrier in agriculture.
- Technical proficiency: Data analytics, machine learning, and other cutting-edge technologies are required to integrate predictive models and real-time decision-making tools. However, many farmers and agricultural stakeholders do not have the means or technical know-how to deploy and use these technologies.

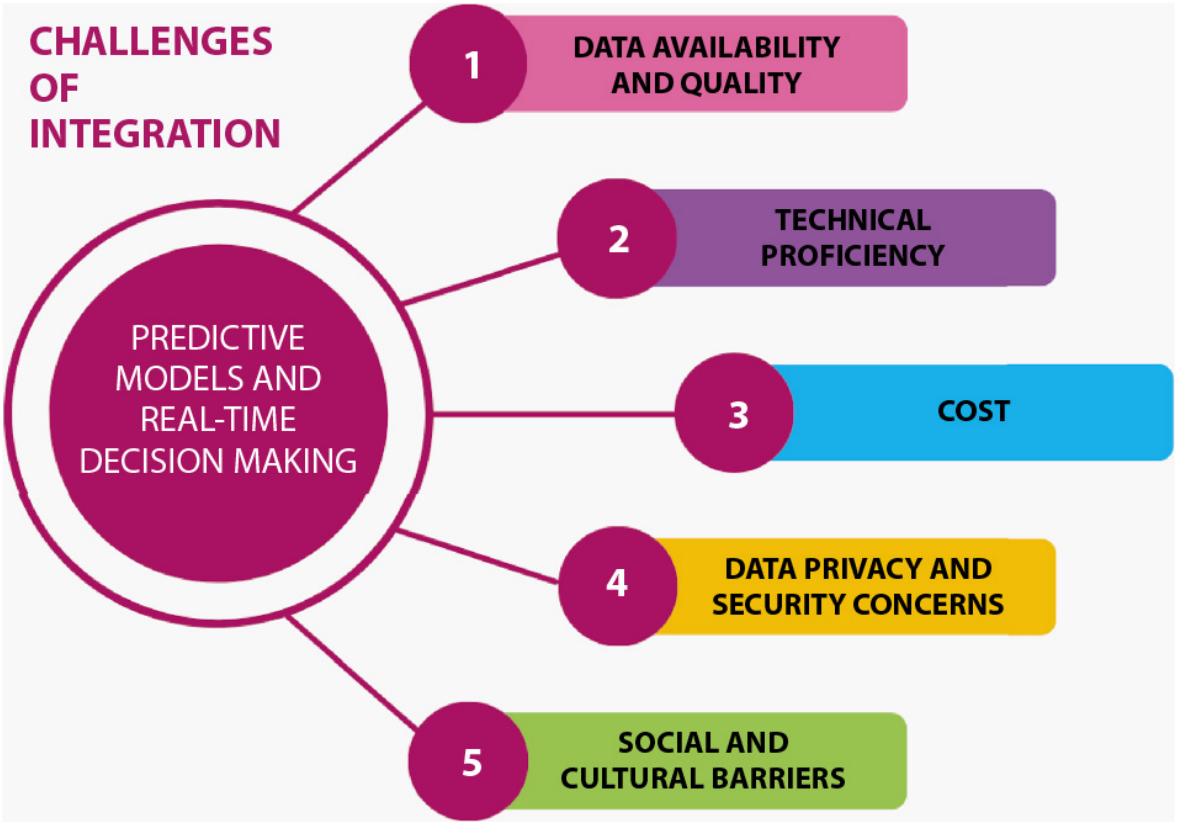

Figure 7.3 Challenges of integration.

- Cost: For many farmers and agricultural stakeholders, the price of integrating predictive models and real-time decision-making tools can be a significant obstacle. Some people, particularly small-scale farmers and those living in underdeveloped nations may find the expense of technological equipment, software, and knowledge excessive. Using predictive models and real-time decision-making tools raises data privacy and security concerns. Farmers and other agricultural stakeholders need to ensure that private information, such as records of farms and livestock, is shielded from unauthorized access or usage.

- Social and cultural barriers: Due to cultural and social factors, farmers and other stakeholders may resist adopting new technology and data-driven decision-making. Some farmers could be unwilling to alter their conventional agricultural methods or might not have confidence in data-driven decision-making techniques.

These issues must be addressed, and solutions developed to integrate predictive models with real-time decision-making in agriculture. This can entail putting money into technological know-how and data infrastructure, protecting the privacy and security of data, and working with farmers and stakeholders to encourage innovative methods and technology [60].

7.8.3 Strategies for Successful Integration

To successfully integrate predictive models and real-time decision-making in agriculture, farmers and agricultural stakeholders can adopt the following strategies:

- A clear definition of the business objectives and identifying the areas where predictive models and real-time decision-making may be most advantageous are the first steps in successful integration. This entails determining the metrics

and key performance indicators that must be tracked and examined to make better judgments.

- Invest in data infrastructure: A dependable, strong data infrastructure that can gather, handle, and analyze data in real-time is necessary for effective integration. This involves spending money on cloud computing infrastructure, data management systems, and analytics tools.
- Ensure data availability and quality: Timely and accurate data are essential for precise forecasting and immediate decision-making. Farmers and agricultural stakeholders must engage in data quality assurance procedures, including data cleansing and validation, to guarantee that data is available when required.
- Develop your technical knowledge: Technical proficiency in data analytics, machine learning, and other cutting-edge technologies is needed to integrate predictive models and real-time decision-making tools. Farmers and agricultural stakeholders can spend money on seminars and training programs to get the essential technical skills or collaborate with outside specialists to get help.
- Addressing data privacy and security concerns is important, given the usage of predictive algorithms and real-time decision-making tools. Implementing data security measures like encryption and access restrictions and adhering to

pertinent data protection laws are requirements for farmers and agricultural stakeholders.

- Encourage an innovative and experimental culture: A culture of invention and experimentation is necessary for successful integration. Farmers and other agricultural stakeholders should support experimenting with cutting-edge methods and technology and be receptive to criticism and ongoing development.
- Collaboration and involvement with stakeholders, such as farmers, agronomists, and researchers, are essential to integration. Farmers and other agricultural stakeholders can participate in community gatherings, seminars, and forums to exchange information, encourage the use of new technology, and promote best practices.

Farmers and agricultural stakeholders may successfully incorporate predictive models and real-time agricultural decision-making and reap the rewards of data-driven decision-making using these tactics [61].

7.9 Future of Data Analytics in Agriculture

7.9.1 Emerging Trends and Technologies

The field of predictive analytics and real-time decision-making in agriculture is rapidly evolving, and several emerging trends and technologies are shaping the future of this field.

The Internet of Things uses sensors and other devices to gather and send real-time data. IoT technology in agriculture may be used to track soil moisture, crop growth, and other essential metrics, empowering farmers to make better decisions. Artificial intelligence and machine learning technologies are being utilized increasingly to create more precise prediction models and decision-making frameworks. These technologies can generate predictions and find patterns in massive amounts of data, assisting farmers in streamlining their processes and increasing yields.

Blockchain technology benefits supply chain management and food traceability by providing a safe and transparent means to store and distribute data. Blockchain technology may be applied to agriculture to trace the provenance and quality of animals and crops, enhancing food safety and transparency.

Drones and autonomous vehicles are increasingly utilized in agriculture to do activities including planting, harvesting, and crop monitoring. These innovations can increase productivity, lower labor costs, and give farmers real-time crop growth and health information. Precision agriculture uses data analytics and technology to optimize agricultural practices based on specific field circumstances, such as planting, fertilizing, and irrigation. Farmers may increase yields while reducing their environmental impact by adapting these practices to specific areas and crops. Predictive analytics and real-time decision-making in agriculture are anticipated to grow more complex and data-driven as these and other new technologies advance. Farmers and other agricultural stakeholders can remain ahead of the curve and continue to spur innovation and expansion by implementing these techniques and methods [62].

7.9.2 Potential Applications of Data Analytics in Agriculture

The potential applications of data analytics in agriculture are vast and wide-ranging. Some of the key areas where data analytics can be applied in agriculture include:

- Crop management: Data analytics may be used to improve procedures for planting, watering, fertilizing, and harvesting

crops. Farmers may boost crop yields and decrease waste by using data from weather stations, soil sensors, and satellite imaging to help them decide when and how to do these chores.

- Animals management: Data analytics may also be used to improve procedures for taking care of animals, including feeding, breeding, and disease control. Farmers may detect possible difficulties early and take corrective measures to lessen the impact on their herds by monitoring aspects like animal behavior and health.
- Data analytics may improve supply chain management in agriculture, lowering waste and increasing efficiency. Farmers and other stakeholders may make more educated decisions about when and where to sell their products. This will increase revenues and decrease waste by tracking crop yields, transit schedules, and market pricing data.
- Agriculture-related hazards, such as crop failures, insect infestations, and weather disasters, may be managed using data analytics. Farmers may detect possible dangers early and take precautions to lessen their impact by analyzing data from weather stations, crop records, and pest monitoring systems.
- Environmental management: Agricultural environmental elements, including soil quality, water utilization, and

pesticide use, may be managed using data analytics. Farmers may reduce their influence on the environment while still maximizing their harvests by monitoring data from sources, including soil sensors, weather stations, and satellite imaging.

The management and optimization of agriculture have the potential to be revolutionized, resulting in more effective and sustainable agricultural practices and better outcomes for farmers, consumers, and the environment [63].

7.9.3 Implications for Agriculture Industry and Society

Data analytics in agriculture has important effects on society and the agricultural sector. Farmers may streamline their processes using data analytics and boost productivity and efficiency. Higher yields and cheaper costs for farmers may follow, contributing to meeting the rising demand for food. Real-time information on crop health, weather, and other crucial elements may be given to farmers using data analytics. This can aid farmers in choosing the best times to sow, irrigate, and harvest their crops. By maximizing fertilizer and pesticides, lowering water use, and minimizing waste, data analytics may assist farmers in lessening their environmental effects.

Data analytics can help improve food safety by enabling farmers to track the production and distribution of their crops and identify potential contamination risks. Data analytics can give consumers more information about how their food is produced, including where it comes from and how it was grown. This can increase transparency and help build trust between consumers and the agriculture industry. Data analytics can help drive economic growth in rural areas by creating new jobs and opportunities in the agriculture industry.

The implications of data analytics in agriculture are significant and far-reaching and can potentially transform how food is produced and consumed. By helping farmers monitor the production and distribution of their crops and spot possible contamination hazards, data analytics can assist in increasing food safety. Thanks to data analytics, consumers may learn more about how their food is produced, including where it comes from and how it was grown. This can improve transparency and support the development of consumer and industry trust. By generating new employment and possibilities in the agriculture sector, data analytics may contribute to economic growth in rural regions [64].

7.10 Concluding Remarks

Since data analytics can increase production, decrease waste, and minimize hazards, it is becoming increasingly crucial in the agricultural sector. Agriculture has many data sources, such as soil sensors, weather stations, drones, satellite imaging, farm equipment, pest and disease monitoring systems, market data, crop records, and livestock records. Due to concerns with data ownership and privacy, as well as data volume, diversity, and quality, collecting and maintaining agricultural data can be difficult.

In agricultural data analytics, predictive models are a crucial tool that may be used for various tasks, including crop production prediction, disease outbreak detection, and fertilizer and pesticide usage optimization. Real-time decision-making is critical to agriculture since it enables farmers to make better choices about activities like insect management, harvesting, and irrigation. Sensor networks, satellite imaging, mobile apps, decision support systems, livestock monitoring systems, and automated equipment are examples of tools and technology for real-time decision-making.

Using predictive models and real-time decision-making together can enhance agricultural outcomes, but it can also be difficult

owing to data quality and compatibility challenges. Building partnerships and collaborations, investing in data infrastructure, and emphasizing user-centered design are all strategies for effective integration. Crop management, livestock management, supply chain management, risk management, and environmental management are potential uses of data analytics in agriculture.

7.11 Future Directions for Research and Practice

The field of data analytics in agriculture is constantly evolving, and there are several directions that research and practice may take in the future. Researchers may create new methods and procedures for gathering, storing, and analyzing data to address the demand for higher data quality and accessibility in agriculture. Although agricultural prediction models are currently in use, there is an opportunity for development. Future research focuses on developing new models to better forecast agricultural yields, disease outbreaks, and other crucial factors. More thorough data analytics programs are required to combine information from many sources, including satellite imaging, soil sensors, and weather stations. Future studies may concentrate on creating fresh approaches to fusing various data

sources. Agriculture relies heavily on real-time decision-making, but improvement is still possible. The main emphasis of future study is the development of new tools and technologies that better enable real-time decision-making. Data analytics can help solve some of the sustainability problems agriculture is experiencing. Research in the future could concentrate on creating new techniques for using data analytics to enhance sustainability in agriculture. Although there are now numerous uses for data analytics in agriculture, other services could remain undiscovered. Future studies could investigate the new benefits of data analytics in agriculture.

References

1. Kaur Channi, H., *et al.*, Agricultural waste assessment for the optimal power generation in the Ludhiana district, Punjab, India. *Mater. Today: Proc.*, 2022. 50: p. 700-708.
2. Elijah, O., *et al.*, An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges. *IEEE Internet Things J.*, 2018. 5(5): p. 3758-3773.
3. Shu, X., *et al.*, Sustainability Assessment of Energy Storage Technologies Based on Commercialization Viability: MCDM Model. *Sustainability*, 2023. 15(6): p. 4707.
4. Channi, H.K., Energy Resources and Their Consumption, in *Applications of Nature-Inspired Computing in Renewable*

Energy Systems, 2022, I.G.I. Global. p. 267-276.
5. Ward, F.A. and M. Pulido-Velazquez, Water conservation in irrigation can increase water use. *Proc. Natl. Acad. Sci.*, 2008. 105(47): p. 18215-18220.
6. Orbak, Â.Y., *et al.*, Mathematical Model Assisted Six-Sigma Approach for Reducing the Logistics Costs of a Pipe Manufacturing Company: A Novel Experimental Approach. *Mathematics*, 2023. 11(3): p. 621.
7. Ash, A., *et al.*, Boosting the productivity and profitability of northern Australian beef enterprises: Exploring innovation options using simulation modelling and systems analysis. *Agric. Syst.*, 2015. 139: p. 50-65.
8. Koteish, K., *et al.*, AGRO: A smart sensing and decision-making mechanism for real-time agriculture monitoring. *J. King Saud Univ. - Comput. Inf. Sci.*, 2022. 34(9): p. 7059-7069.
9. Kitchen, N.R., Emerging technologies for real-time and integrated agriculture decisions. *Comput. Electron. Agric.*, 2008. 61(1): p. 1-3.
10. Kim, S., H. Kim and Y. Park, Early detection of vessel delays using combined historical and real-time information. *J. Oper. Res. Soc.*, 2017. 68(2): p. 182-191.
11. Marios, S. and J. Georgiou. Precision agriculture: Challenges in sensors and electronics for real-time soil and plant

monitoring. in *2017 IEEE Biomedical Circuits and Systems Conference (BioCAS)*. 2017. IEEE.

12. Saini, A., *et al.*, Designing and Modeling of Water Level Indicator. *Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol.*, 2017. 2(6): p. 374-377.
13. Jatin, *et al.*, selection of water purifier with TOPSIS using impartial preferences by entropy technique. *Mater. Today: Proc.*, 2022. 50: p. 1389-1396.
14. Sharma, H. and A. Gupta, The objectives of waste management in India: a futures inquiry. *Technol. Forecast. Soc.*, 1995. 48(3): p. 285-309.
15. Kumar, R., *et al.*, Review of Lubrication and Cooling in Computer Numerical Control (C.N.C.) Machine Tools: A Content and Visualization Analysis, Research Hotspots and Gaps. *Sustainability*, 2023. 15(6): p. 4970.
16. Cao, X., *et al.*, Water resources efficiency assessment in crop production from the perspective of water footprint. *J. Clean. Prod.*, 2021. 309: p. 127371.
17. Mishra, A. and V. Desai, Drought forecasting using stochastic models. *Stochastic Environ. Res. Risk Assess.*, 2005. 19: p. 326-339.
18. Beeri, O. and A. Peled, Geographical model for precise agriculture monitoring with real-time remote sensing. *ISPRS J. Photogramm. Remote Sens.*, 2009. 64(1): p. 47-54.

19. Махортова, Н.В., Data Collecting System of Condition Box Units of Rolling Stock. *East.-Eur. J. Enterp. Technol.*, 1(2): p. 54-56.
20. Channi, H.K. and R. Kumar, The Role of Smart Sensors in Smart City, in *Smart Sensor Networks: Analytics, Sharing and Control*, U. Singh, *et al.*, Editors. 2022, Springer International Publishing: Cham. p. 27-48.
21. Bains, P.S., R.K. Jindal and H.K. Channi, Modeling and designing of automatic plant watering system using Arduino. *Int. J. Sci. Res. Sci. Technol. (IJSRST)*, 2017. 3(7): p. 676-680.
22. Shanmugapriya, P., *et al.*, Applications of remote sensing in agriculture-A Review. *Int. J. Curr. Microbiol. Appl. Sci.*, 2019. 8(01): p. 2270-2283.
23. Lamb, R.L., Fertilizer use, risk, and off-farm labor markets in the semi-arid tropics of India. *Am. J. Agric. Econ.*, 2003. 85(2): p. 359-371.
24. Hubbard, K.G., N.J. Rosenberg and D.C. Nielsen, Automated weather data network for agriculture. *J. Water Resour. Plann. Manage.*, 1983. 109(3): p. 213-222.
25. Leelavinodhan, P.B., *et al.* Energy-neutral weather stations for precision agriculture: challenges and approaches. in *2020 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor)*. 2020. IEEE.

26. Venäläinen, A. and M. Heikinheimo, Meteorological data for agricultural applications. *Phys. Chem. Earth. Parts A/B/C*, 2002. 27(23-24): p. 1045-1050.
27. Kim, J., *et al.*, Unmanned aerial vehicles in agriculture: A review of perspective of platform, control, and applications. *IEEE Access*, 2019. 7: p. 105100-105115.
28. van der Merwe, D., *et al.*, Drones in agriculture. *Adv. Agron.*, 2020. 162: p. 1-30.
29. Yang, C., High resolution satellite imaging sensors for precision agriculture. *Front. Agric. Sci. Eng.*, 2018. 5(4): p. 393-405.
30. Siesto, G., M. Fernández-Sellers and A. Lozano-Tello, Crop classification of satellite imagery using synthetic multitemporal and multispectral images in convolutional neural networks. *Remote Sens.*, 2021. 13(17): p. 3378.
31. Bacco, M., *et al.*, The digitisation of agriculture: a survey of research activities on smart farming. *Array*, 2019. 3: p. 100009.
32. Waleed, M., *et al.*, Classification of agriculture farm machinery using machine learning and internet of things. *Symmetry*, 2021. 13(3): p. 403.
33. Dyer, J. and R. Desjardins, The impact of farm machinery management on the greenhouse gas emissions from Canadian agriculture. *J. Sustain. Agric.*, 2003. 22(3): p. 59-74.

34. Dong, Y., *et al.*, Automatic system for crop pest and disease dynamic monitoring and early forecasting. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, 2020. 13: p. 4410-4418.
35. Gao, D., *et al.*, A framework for agricultural pest and disease monitoring based on internet-of-things and unmanned aerial vehicles. *Sensors*, 2020. 20(5): p. 1487.
36. Kumar, R., *et al.*, A New Joint Strategy for Multi-Criteria Decision-Making: A Case Study for Prioritizing Solid-State Drive. *Int. J. Comput. Commun. Control.*, 2022. 17(6).
37. Ranjan, N., *et al.*, Investigation of Fused Filament Fabrication-Based Manufacturing of ABS-Al Composite Structures: Prediction by Machine Learning and Optimization. *J. Mater. Eng. Perform.*, 2022: p. 1-20.
38. Ngo, V.M., *et al.*, A big data smart agricultural system: recommending optimum fertilisers for crops. *Int. J. Inf. Technol.*, 2023: p. 1-17.
39. Dupuis, A., C. Dadouchi and B. Agard, Methodology for multi-temporal prediction of crop rotations using recurrent neural networks. *Smart Agric. Technol.*, 2023. 4: p. 100152.
40. Alshehri, M., Blockchain-assisted internet of things framework in smart livestock farming. *Internet Things*, 2023. 22: p. 100739.
41. Amini, M. and A. Rahmani, Agricultural databases evaluation with machine learning procedure. *Aust. J.*

Eng.Appl. Sci., 2023. 8(2023): p. 39-50.
42. Devare, M., *et al.*, Governing agricultural data: Challenges and recommendations. *Towards Responsible Plant Data Linkage: Data Challenges for Agricultural Research and Development*, 2023: p. 201.
43. King, C.D., *et al.*, Farmers' attitudes towards agricultural plastics– Management and disposal, awareness and perceptions of the environmental impacts. *Sci. Total Environ.*, 2023. 864: p. 160955.
44. Perosa, B., P. Newton and R.F.B. da Silva, A monitoring, reporting and verification system for low carbon agriculture: A case study from Brazil. *Environ. Sci. Policy*, 2023. 140: p. 286-296.
45. Agyeman, P.C., *et al.*, Prediction of the concentration of antimony in agricultural soil using data fusion, terrain attributes combined with regression kriging. *Environ. Pollut.*, 2023. 316: p. 120697.
46. Romero-Gainza, E. and C. Stewart, AI-Driven Validation of Digital Agriculture Models. *Sensors*, 2023. 23(3): p. 1187.
47. Zarei, A.R., M.R. Mahmoudi and M.M. Moghimi, Determining the most appropriate drought index using the random forest algorithm with an emphasis on agricultural drought. *Nat. Hazard.*, 2023. 115(1): p. 923-946.

48. Wakchaure, M., B. Patle and A. Mahindrakar, Application of A.I. Techniques and Robotics in Agriculture: A Review. *Artif. Intell. Life Sci.*, 2023: p. 100057.
49. Thangavel, K.D., *et al.*, Prediction of factors for Controlling of Green House Farming with Fuzzy based multiclass Support Vector Machine. *Alex. Eng. J.*, 2023. 62: p. 279-289.
50. Adeniyi, O.D., *et al.*, Digital Mapping of Soil Properties Using Ensemble Machine Learning Approaches in an Agricultural Lowland Area of Lombardy, Italy. *Land*, 2023. 12(2): p. 494.
51. Nishant, P.S., *et al.* Crop yield prediction based on Indian agriculture using machine learning. in *2020 International Conference for Emerging Technology (INCET)*. 2020. IEEE.
52. Neethirajan, S., The Significance and Ethics of Digital Livestock Farming. *AgriEngineering*, 2023. 5(1): p. 488-505.
53. Mendes, J.d.J., *et al.*, Adoption and impacts of messaging applications and participation in agricultural information-sharing groups: an empirical analysis with Brazilian farmers. *J. Agribus. Dev. Emerg. Econ.*, 2023.
54. Cama-Pinto, D., *et al.*, A Deep Learning Model of Radio Wave Propagation for Precision Agriculture and Sensor System in Greenhouses. *Agronomy*, 2023. 13(1): p. 244.
55. Romano, E., *et al.*, The Correlation between Proximal and Remote Sensing Methods for Monitoring Soil Water Content in Agricultural Applications. *Electronics*, 2023. 12(1): p. 127.

56. Hanselmann, T., *et al.*, Mobile applications (apps) for agriculture in Germany-a comparative market analysis. *Landtechnik*, 2023. 78(1).
57. Kyrgiakos, L.S., *et al.*, Assessing efficiency differences in a common Agriculture Decision Support System-A comparative analysis between Greek and Italian durum wheat farms. *Int. J. Food Syst. Dyn.*, 2023. 14(1): p. 22-45.
58. Mishra, S. and S.K. Sharma, Advanced contribution of IoT in agricultural production for the development of smart livestock environments. *Internet Things*, 2023. 22: p. 100724.
59. Shutske, J.M., K.J. Sandner and Z. Jamieson, Risk Assessment Methods for Autonomous Agricultural Machines: A Review of Current Practices and Future Needs. *Appl. Eng. Agric.*, 2023: p. 0.
60. Othman, S., *et al.*, Artificial intelligence-based techniques for adulteration and defect detections in food and agricultural industry: A review. *J. Agric. Food Res.*, 2023: p. 100590.
61. Valle-García, J.D., *et al.*, Integration of Eco-Friendly Biological and Nanotechnological Strategies for Better Agriculture: A Sustainable Approach. *Agricultural and Environmental Nanotechnology: Novel Technologies and their Ecological Impact*, 2023: p. 647-674.
62. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using

Convolution Neural Networks." In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.

63. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. "Solutions Using Machine Learning for Diabetes." *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
64. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, "Image Processing Based Intelligent Mini Robotic Face Recognition System," *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.

Note

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8

Examining the Role of IoT and AI in Revolutionizing Agriculture: A Smart Farming Approach

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Abstract

Farming is really important for the world. It involves growing and raising things like plants and animals that people use for food. Farmers sell their products in markets, but farming has

problems like changing weather, bad soil, soil erosion, not enough jobs, and trouble getting products to where they are needed. Today people are using smart technology, like artificial intelligence (AI) and the Internet of Things (IoT), to make farming better. With IoT, sensors collect information about plants, animals, and fields, so farmers do not have to do everything by hand. Drones with sensors can fly over fields to see what is needed. AI looks at all this data and helps solve problems. There are different kinds of sensors used, like optical sensors for plants, electrochemical sensors for soil nutrients, yield monitor sensors for measuring grain, airflow sensors for soil air, mechanical soil sensors for soil properties, and location sensors to track vehicles and animals with GPS. All this data gives farmers a real-time picture of their crops, and AI suggests the best solutions. This is called “smart farming.”

***Keywords*:** Internet of Things (IoT), artificial intelligence (AI), agriculture, smart farming, GPS, technology, farm management, sustainable agriculture, sensor

8.1 Introduction

Agriculture is crucial for our livelihood and also plays a key role in preserving the environment. According to fiscal policy statements, the Indian agriculture sector is expected to grow by

3.5% in the fiscal year 2022–2023. However, agriculture faces challenges globally, such as reduction in available land due to industrial activities, population growth, and development projects.

Farmers today use various chemical fertilizers and pesticides to accelerate crop growth. However, the excessive use of these chemicals can negatively impact soil nutrition, leading to a decline in soil fertility and potential future issues for crop growth. Similarly, inadequate or excessive irrigation can hinder crop development.

Transportation is a critical aspect. If a farmer cannot transport their harvested crops to the market promptly, all their hard work goes to waste, causing suffering for both the farmer and the people who depend on those yields. Proper transportation is essential for optimizing the use of harvested crops.

Livestock also require careful attention, as people naturally have an emotional connection to them. Additionally, various products obtained from cattle have a good market value in society. It is vital to address these challenges to ensure sustainable agriculture and the well-being of both farmers and consumers.

Cattle make up about 96% of mammals, but they contribute less than 30% as reported by the National Accounts Statistics (NAS) in 2020. Using a combination of AI and IoT technology can boost cattle production. Climate change is a crucial factor in agriculture, and AI can predict future weather conditions, helping farmers take better care of their crops. AI also detects various agricultural field properties, including soil moisture, air flow, soil nutrition, water levels, pest levels, and equipment errors.

Machine learning models powered by AI analyze the data collected by IoT sensors strategically placed in the fields. This large amount of data is used to train the model, providing valuable insights into the field conditions. If the AI model detects anomalies or deviations, immediate notifications are sent to farmers through messages [22]. Figure 8.1 described the role of AI and IOT in agriculture.

By combining AI and IoT, farmers can reduce their physical workload and save valuable time. This technology not only supports farmers in their field activities but also enables them to focus on other essential tasks. The synergy of AI and IoT in agriculture proves to be a significant asset in enhancing productivity and efficiency.

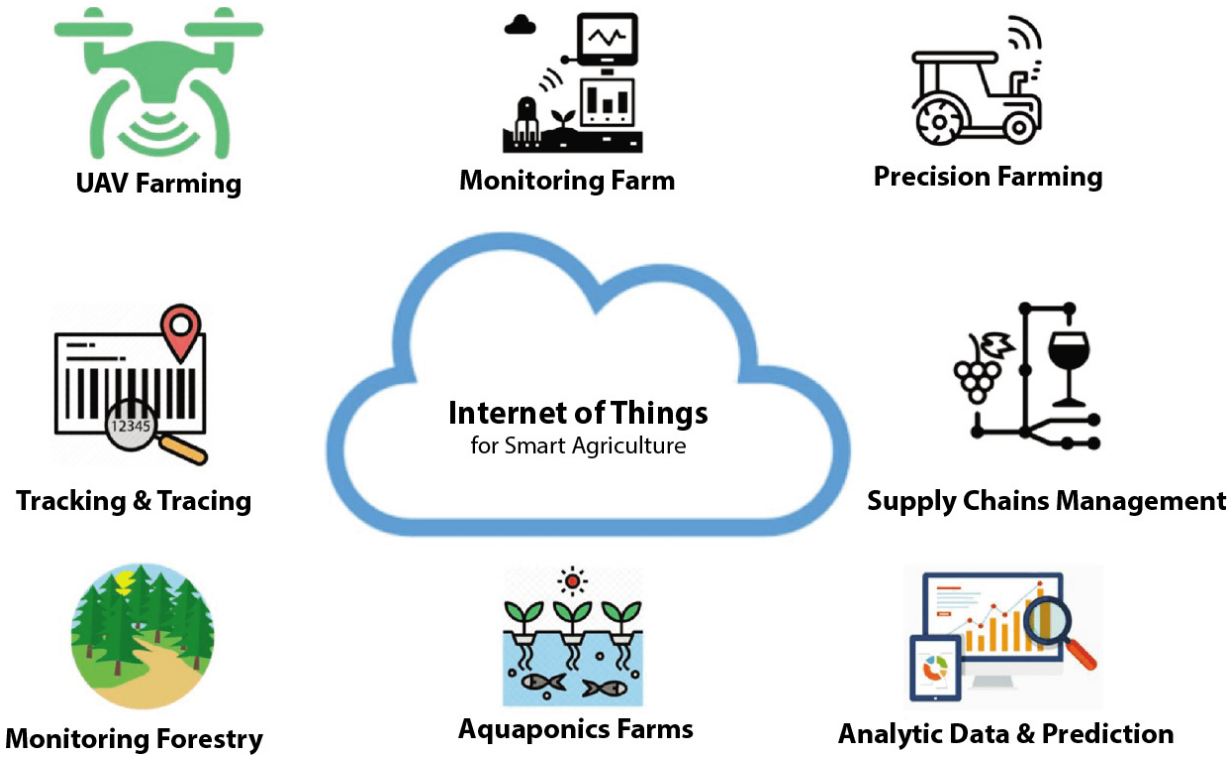

Figure 8.1 AI and IOT in agriculture [21].

8.2 IoT in Agriculture

There are many positive impacts of using IOT (Internet of Things) in agriculture. This makes farming more efficient, productive, and sustainable. To increase yields, reduce waste, save resources, improve the quality and quantity of yields, increase transparency, and so on, IOT is implemented in agriculture [1–3]. Sensors play a vital role and they are fundamental for IOT technology. These sensors are connected in the ARDUINO UNO board for connecting and reading the collected data from the crops. Then, these data collected are analyzed to optimize crop growth and improve agricultural

practices. Some of the important sensors used by IOT for smart farming are soil moisture sensors, weather monitoring sensors, livestock monitoring sensors, crop monitoring sensors, automated irrigation systems, and drones equipped with sensors and cameras used to gather data [4]. In this context the physical layout of the Arduino board parts for sensor connections shown in Figure 8.2. These sensors help in collecting data from the surrounding environment and used to monitor the wide range of parameters in farming such as soil moisture, temperature and humidity, light intensity, weather conditions, pest and disease detection, and livestock monitoring [2].

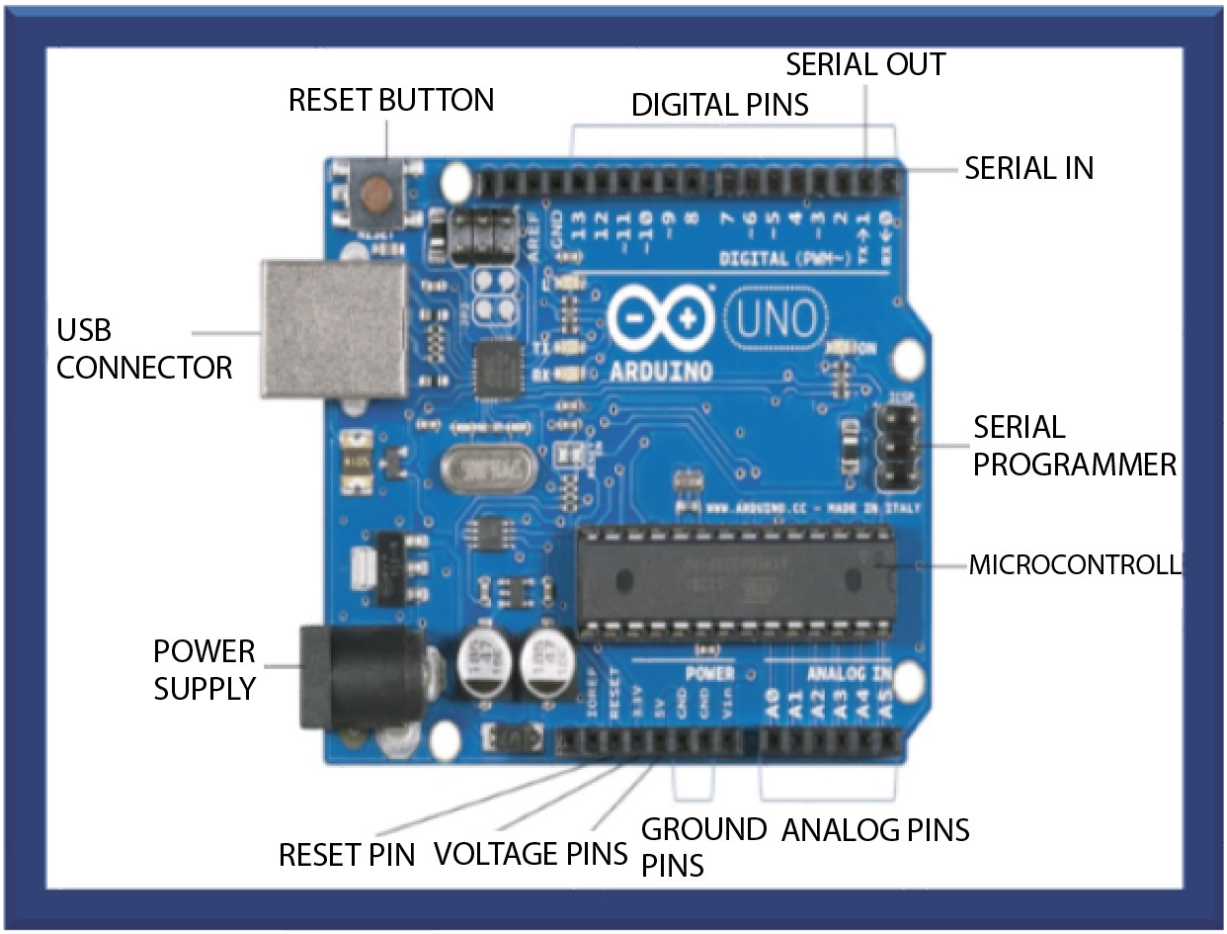

Figure 8.2 Arduino board parts for sensor connections [2].

8.2.1 Soil Moisture

In the past, farmers depended on manual measurements for soil moisture monitoring, resulting in a lack of up-to-date information and an inability to detect variations in soil moisture across different parts of their fields. This led to time-consuming and inaccurate data, causing issues such as overwatering or underwatering crops, resource wastage, limited yields, and uneven crop growth with yield variability.

However, with the adoption of IoT (Internet of Things), farmers can now collect real-time and accurate data on soil moisture levels in various parts of their fields. This technology enables them to optimize their irrigation practices by avoiding overwatering or underwatering, leading to higher yields and reduced water usage. The use of IoT in soil moisture monitoring provides a more efficient and effective approach, contributing to better water conservation and improved crop outcomes for farmers [5].

8.2.2 Temperature and Humidity

In the past, farmers faced challenges with temperature and humidity monitoring as they relied on manual measurements. This resulted in a lack of up-to-date data and made it difficult to detect fluctuations in temperature and humidity across different parts of their fields. The consequences included prolonged and imprecise data, making it challenging for farmers to identify potential issues such as pests, diseases, or harmful weather conditions that could harm crops and lead to irregular crop growth.

With the application of IoT (Internet of Things), farmers can now gather real-time and accurate data on temperature and humidity levels in various parts of their fields. This

technological improvement enhances their irrigation and pest management systems. Farmers can take proactive measures to address potential issues before crops are damaged, and they receive alerts when data levels deviate from the optimal range, enabling them to make the right decisions promptly. Figure 8.3 is the representation of Column graph for decreased water usage in agriculture after using IOT sensors. The use of IoT in temperature and humidity monitoring significantly improves the overall efficiency of farming practices and helps safeguard crops from potential threats.

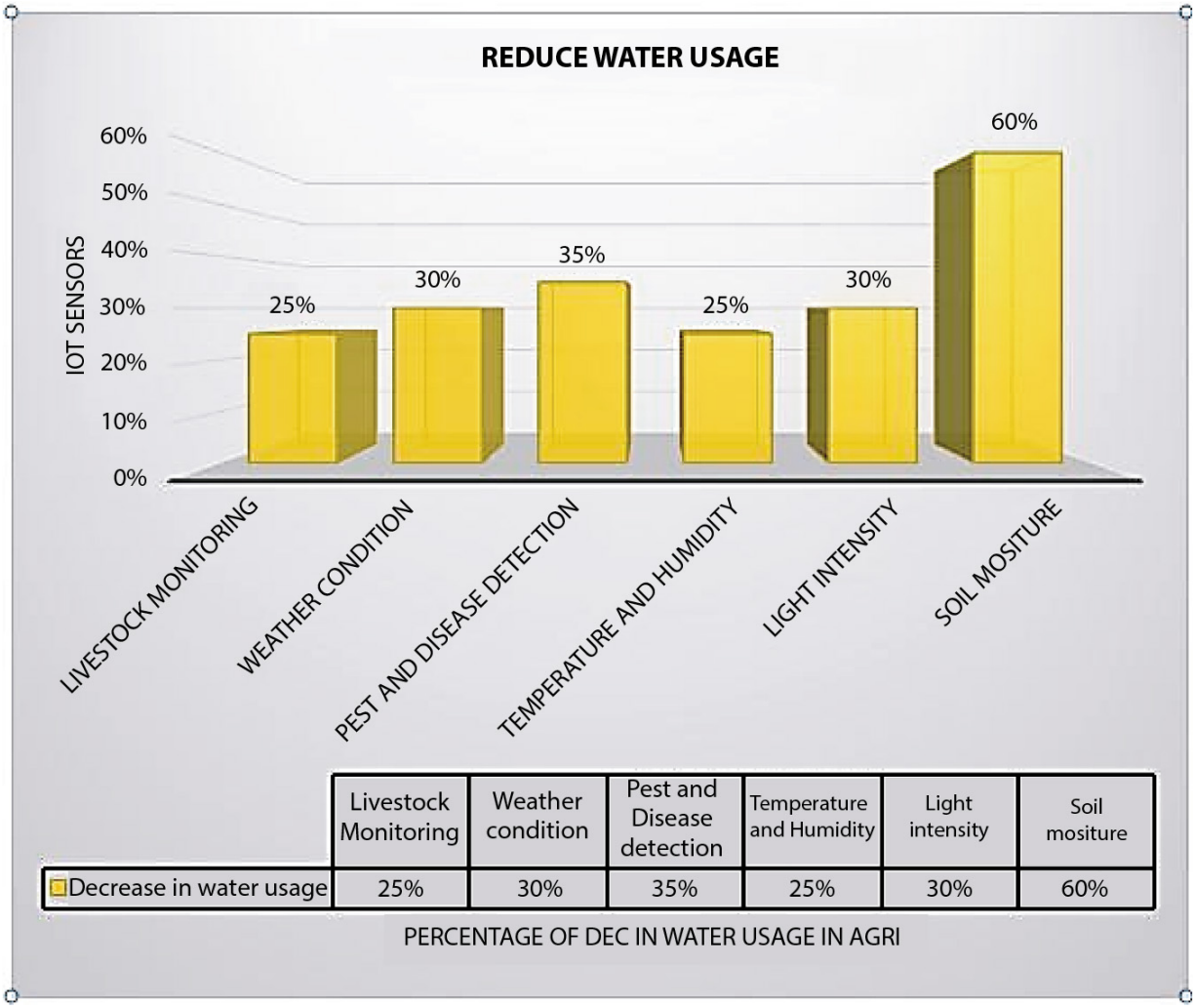

Figure 8.3 Column graph for decreased water usage in agriculture after using IOT sensors.

8.2.3 Light Intensity

In the past, farmers faced challenges in collecting hand-measured data on light intensity levels. This process was time-consuming, and the data obtained was not regularly updated or accurate. As a result, farmers found it difficult to optimize the growth conditions for their crops, leading to lower yields and a decrease in the quality and quantity of crops [7].

In today's agriculture, with the implementation of IoT (Internet of Things), farmers can now collect real-time data on light intensity. This advancement allows farmers to optimize the growth conditions for their crops. If there is a lack or excess of light outside the optimal range, artificial lighting systems or strategic plant positioning can be employed to provide the necessary light conditions for the crops. This ensures that crops receive the right amount of light, contributing to improved growth and ultimately enhancing both the quality and quantity of the harvest. The use of IoT in light intensity monitoring is a valuable tool for modern agriculture, promoting efficient and effective crop management.

8.2.4 Weather Conditions

Earlier farmers relied on local weather forecasts report or predictions on weather made by past experiences, so it was challenging for farmers to get exact data and weather data for a wide range of area. After executing IOT, an up-to-date picture of weather conditions is provided for the entire farm, such as temperature, humidity, precipitation, and wind speed. This enables farmers to take proactive steps to protect crops, planting, and irrigation and to reduce crop losses.

8.2.5 Pest and Disease Detection

In the last few decades, farmers primarily relied on visual inspections to identify pests and diseases in crops. Manual data collection was the standard practice, lacking centralized data storage and analysis. This slow and unreliable process led to farmers purchasing expensive pesticides and chemical treatments to control pests, resulting in environmental impact and compromised crop health. Figure 8.4 reflects the Bar graph for increase in agriculture using IOT sensors.

With the advent of IoT (Internet of Things), sensors and cameras can now be deployed throughout the farm, offering continuous monitoring of crops and real-time detection of pest and disease outbreaks. This technology facilitates the quick access and analysis of large amounts of data. As a result, farmers can implement targeted treatments, reducing the dependence on broad-spectrum pesticides and chemicals. This not only improves the efficiency of pest and disease management but also contributes to environmental sustainability and the production of healthier yields. The use of IoT in agriculture represents a significant advancement in optimizing crop protection practices [13].

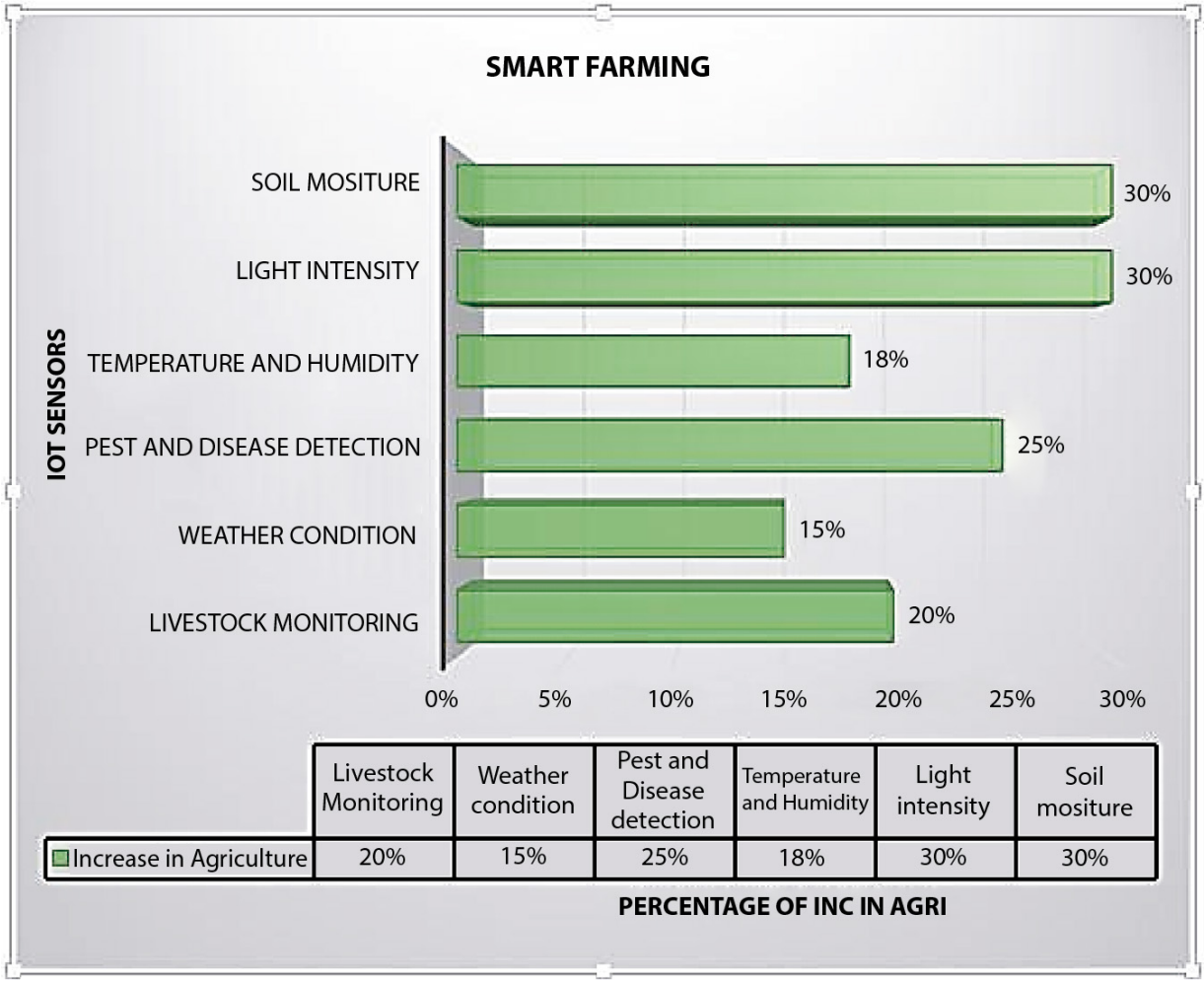

Figure 8.4 Bar graph for increase in agriculture using IOT sensors.

8.2.6 Livestock Monitoring

Farmers before relied on visual observations to monitor the health and behavior of their livestock, which could be limited in scope.

Overseeing livestock required physical presence; this was difficult for famers in case of large-scale farming operations. With IOT, sensors and cameras can be deployed throughout the

farm, providing a timely watch of livestock behavior, health, and welfare [8]. Figure 8.5 is the presentation of the various IOT sensors used for smart farming. IOT has made it easier to identify the pattern and detect potential health issues quickly, enabling early intervention and prevention of livestock diseases. It reduces the need of generalized medications and improves animal welfare [6, 20].

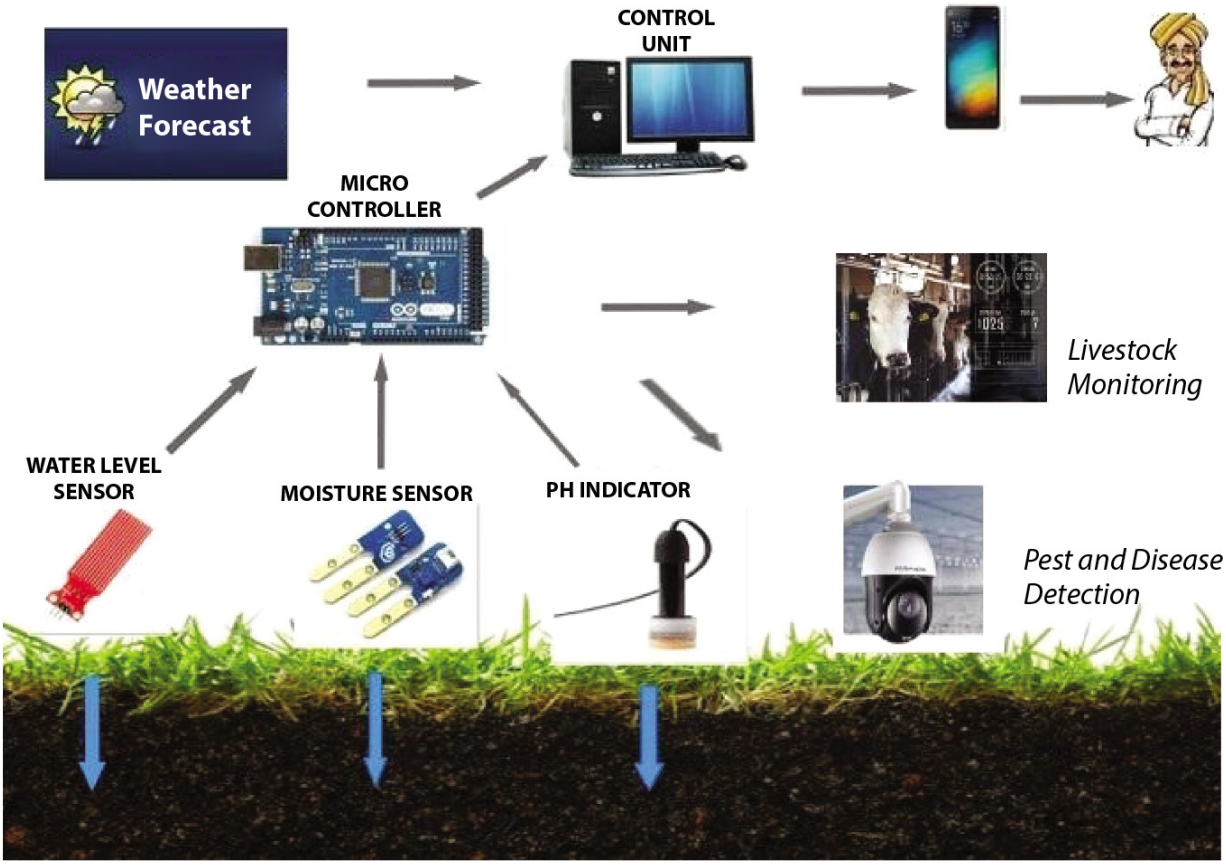

Figure 8.5 IOT sensors used for smart farming.

8.3 Major Advantages of Using IoT in Agriculture

- Data-driven decision making: IoT utilizes sensors to collect and transmit data on plant growth, weather conditions, and soil moisture. This empowers farmers to make informed decisions about planting, irrigation, harvesting, and pesticide application, ultimately enhancing field productivity and reducing waste.
- Comprehensive farm management: IoT sensors, when combined with drones, offer a holistic view of farming operations. Farmers can monitor soil fertility, temperature, humidity, weather patterns, and even the health of livestock. This real-time information enables quick responses to changes in the agricultural environment, allowing farmers to take proactive measures against potential issues.
- Cost efficiency: The implementation of IoT in agriculture is cost-efficient. The data collected by IoT sensors helps prevent the overuse of water, fertilizers, and pesticides. This reduction in resource expenditure contributes to increased profitability for farmers.
- Environmental impact: Agriculture often has significant environmental consequences, including field waste and the use of harmful chemicals. IoT technology promotes sustainability by enabling more precise resource management, reducing the environmental impact of farming practices.

- Remote farm management: IoT allows farmers to remotely manage and monitor their farms. This reduces the need for physical labor and provides the flexibility for farmers to work from home or any location. This not only enhances efficiency but also accommodates the evolving needs of modern farming practices.

8.4 Major Disadvantages of Using IoT in Agriculture

- High implementation cost: The initial cost of implementing IoT technology can be expensive, particularly for smallscale farmers who may lack the necessary resources to invest in the required infrastructure and equipment.
- Technical complexity: IoT systems can be technically complicated, requiring specialized skills and knowledge for understanding and maintenance. Farmers may find it challenging to manage these systems efficiently without investing in additional staff or undergoing training.
- Privacy and security concerns: The data collected and transmitted by IoT devices can pose privacy and security issues if not adequately protected. Safeguarding sensitive information becomes crucial to prevent unauthorized access or misuse of the collected data.

- Dependency and skill shift: Increased reliance on IoT technology may lead to farmers becoming less skilled in traditional farming practices. Dependency on technology could pose challenges if the technology fails or is not readily available in the market.
- Energy consumption and environmental impact: IoT devices require electricity to operate, and their production and disposal have environmental impacts. Farmers need to be aware of the full life cycle impact of IoT systems on the environment, considering factors such as energy consumption and proper disposal methods to minimize environmental harm.

8.5 AI in Agriculture

Artificial intelligence is a field of computer science engineering that focuses on creating machines to perform tasks with human intelligence such as analyzing, understanding, and solving problems. AI began to evolve in the 1950s with the development of the first electronic computers. The research began to explore to simulate human thought process and behavior [9].

The origins of AI in the field of study can be traced back to the Dartmouth Conference in 1956, where an AI camera was introduced. This conference brought together experts from

various disciplines to explore the possibility of creating intelligent machines. In the 1960s and 1970s, researchers focused on designing decision-making processes in specific fields, but these systems were limited in their capabilities and could not learn from experience. By the use of artificial intelligence Crop diagnosis cycle is represented in Figure 8.6. The 1980s and 1990s saw the development of machine learning algorithms, allowing computers to learn from data without explicit programming. This opened up new possibilities for AI, including natural language processing, computer vision, and robotics.

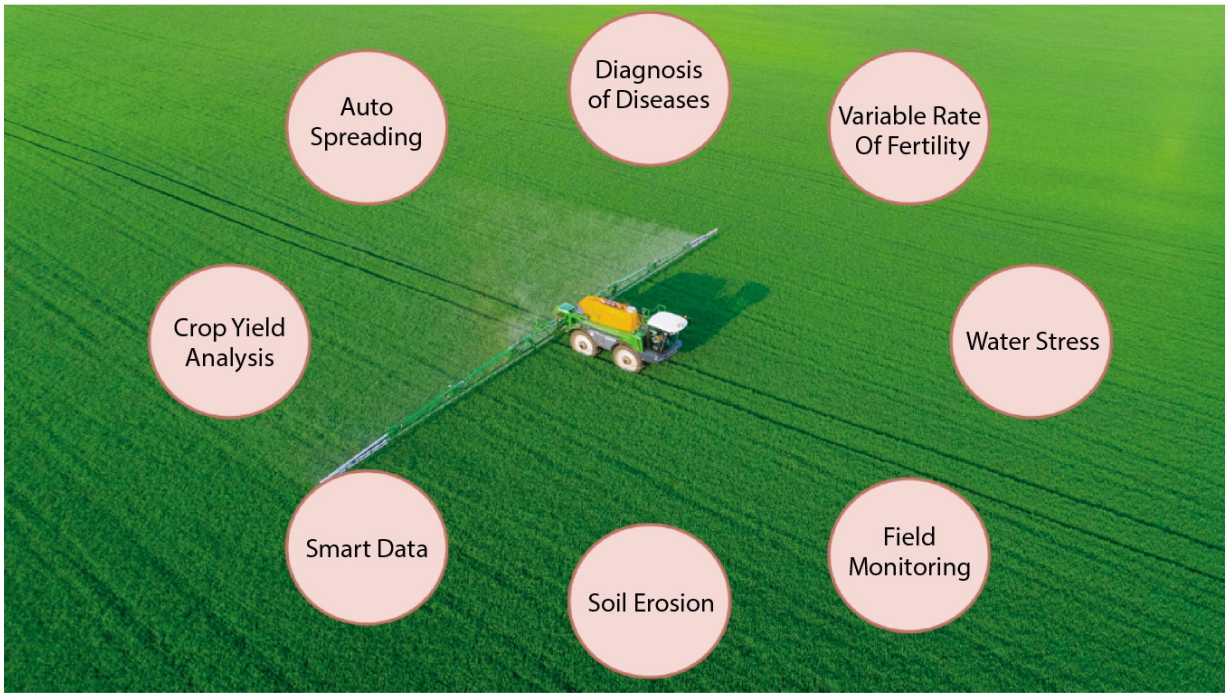

Figure 8.6 Crop diagnosis cycle using AI.

In the 2000s and beyond, with vast amounts of data and improved computing power, AI advanced significantly, and this trend is expected to continue. AI is now widely applied in voice assistants, self-driving cars, medical diagnosis, and financial analysis and across various sectors like healthcare, finance, marketing, manufacturing, education, transportation, agriculture, and entertainment.

In agriculture, AI is implemented to boost productivity, cut costs, and enhance sustainability. It aids farmers in decision-making, reducing waste, and increasing yields. AI transforms crop growth and livestock management by enabling precision agriculture, crop monitoring, disease detection, climate

prediction, and more. Various tools such as sensors, robots, computer vision, and machine learning are employed to support farmers, generating large volumes of structured and unstructured data daily, including weather patterns, soil reports, research findings, and imaging, through drones and cameras.

AI plays a crucial role in soil management, establishing relationships between soil composition and various layers. Weed control is another vital aspect, as uncontrolled weed growth can result in 50% loss of yield. AI weed detection systems analyze precise spray amounts in laboratories, reducing costs and preventing crop damage. In Figure 8.7, we demonstrate the tasks done by drones in the field of agriculture.

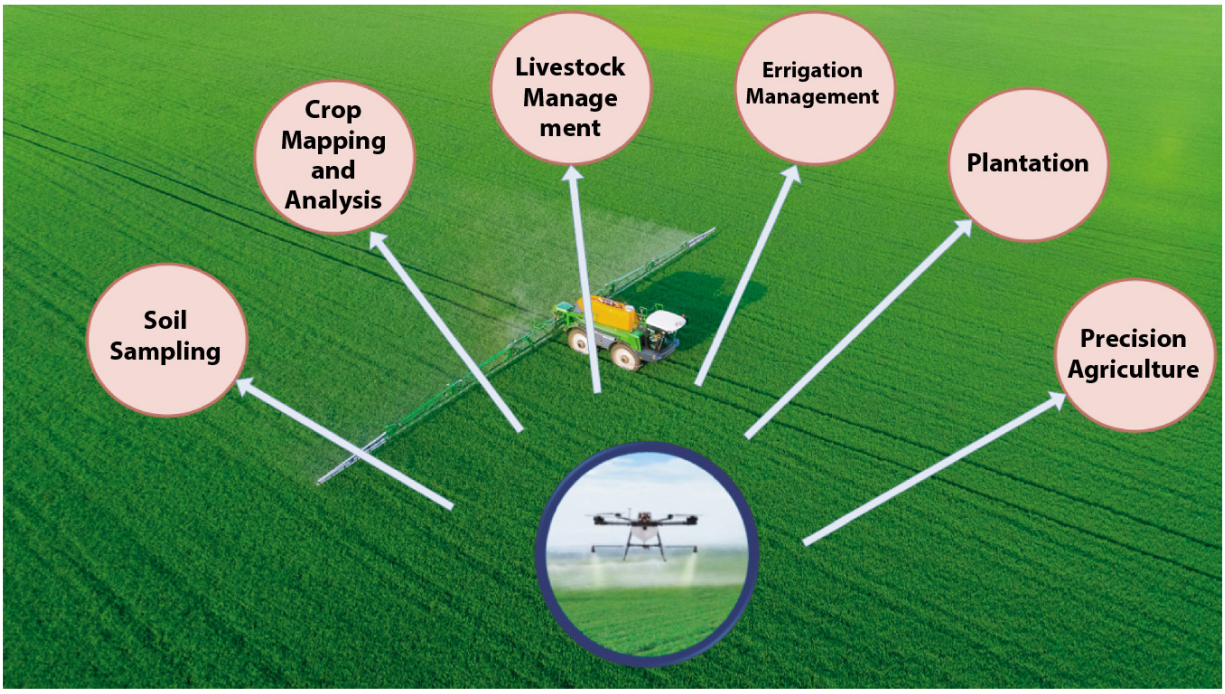

Figure 8.7 Task done by drone.

While significant progress has been made with robots in agriculture, the solutions for complex agricultural environments are still evolving. AI may have some disadvantages, but its overall impact helps farmers protect soil quality and enhance the overall quality of agriculture.

8.5.1 Equipment Used for AI in Agriculture

There are several types of equipment and technologies by AI used in agriculture. Some of them are the following [10]:

- Drones and UAVs: They are used to collect data on crops. They are equipped with sensors and cameras to monitor crops and livestock.

- Sensors and IOT devices: They can be used to collect data on environmental factors. They are installed in fields, greenhouses, and livestock facilities to monitor and provide real-time data for decision making.
- GPS and GIS mapping: These are technologies used to map fields, identify areas of high productivity, and optimize resource use.
- Robotics and automation: This is used to automate tasks such as seeding, weeding, and harvesting.
- Satellite imagery: This is used to monitor crops and livestock over vast areas. It provides insights into crop health, soil moisture level, and other environmental factors.
- Machine learning algorithms: These can be used to analyze the data collected from various sources and provide insights into crop growth, pest disease management, and resource management.
- Automated machinery: Self-driving tractors, robotics harvesting, and other equipment are programmed to operate anonymously. It improves efficiency and accuracy and reduces labor costs. AI is revolutionizing agriculture by providing farmers with new tools to manage their resources more effectively, increase productivity, and improve sustainability.

8.6 Benefits of Drones in Agriculture

Drones, also known as unmanned aerial vehicles (UAVs), play a significant role in agriculture by integrating various sensors and devices for smart farming innovations. These devices are connected to drones to perform a range of tasks, from observing crops to analyzing the collected data. The implementation involves connecting different types of sensors to enhance agricultural practices. Drones can efficiently handle tasks such as fertilizing or sprinkling weed control.

In the context of agriculture, IoT sensors are deployed to collect observational data, which is then analyzed by AI systems. This integration makes it easier for farmers to identify errors in the agricultural field or pinpoint issues leading to suboptimal crop yields. The use of AI in conjunction with drones addresses the challenge of analyzing large volumes of data, providing farmers with valuable insights and facilitating informed decision-making. This combination of technologies helps streamline agricultural processes, optimizes resource usage, and contributes to overall efficiency in modern farming practices.

8.6.1 Precision Farming

Precision farming is aimed to optimize crop production and reduce waste in agriculture using technologies like sensors, GPS

mapping, and data analytics. It is also known as precision agriculture.

By using this technique, farmers can collect and analyze the real-time data about their crops, like soil condition, weather pattern, and other factors, which can improve the growth of the crop and yield. It is also used to predict the time to harvest and amount of water and fertilizers to supply to the crops [11]. It is mainly used to approach modern agriculture by using advanced technologies to promote crop production and environmental impacts.

8.6.2 Crop Management

AI is used in crop management which can detect the disease in plants using powered tools, and it analyzes overall crop condition. Weather pattern, soil conditions, and previous yield data of crops are examined by AI to predict how much yield will be produced. It helps the farmers for making decision about planting, harvesting, and fertilizing.

AI is used in irrigation by analyzing soil moisture, weather condition, and the crop's water requirements which enhance the irrigation schedule and reduce the water waste. It prevents the risk of overwatering and underwatering the crops. AI can

analyze the soil nutrition level, crop nutrition requirements, and fertilizer application rate to control fertilizer use and reduce waste. It can help the farmers to reduce the environmental impact on fertilizers. AI can reduce the use of pest by analyzing data like pest population, weather condition, and crop susceptibility. AI reduces the use of harmful chemicals which will affect the environment and increases the efficiency and sustainability of agriculture.

8.6.3 Disease and Pest Detection

AI is a powerful tool for predicting and detecting disease and pest infections in crops. AI alerts farmers to take actions before the damage becomes very severe to cause potential problems in crops [13].

Example: AI uses machine learning algorithms for disease detection in crops, which is a plant village project, and identified the signs of disease. By uploading the images of the crops in the platform, the AI system can provide the diagnosis and recommend appropriate treatments. It is also used in pest detection in crops. By analyzing data on weather patterns, pest behavior, and plant health, AI can predict when and where pests are likely to strike and help the farmers to take preventative measures. A company named “Ceres Imaging”

uses aerial images and machine learning algorithms to identify areas of crops that are at risk of pest infection. This helps the farmers to take action before the pests cause significant damage and reduce the crop yield [12].

8.6.4 Weed Management

AI indeed plays a crucial role in weed management, offering precise and efficient identification and targeting of weeds in agricultural fields. By analyzing images of crops, AI can distinguish between crops and weeds. This capability allows farmers to specifically target weeds for herbicide application, reducing the overall use of chemicals and minimizing the environmental impact.

In the realm of robotic weeders, AI is employed to control these devices, automating the process of weed removal. This not only reduces the need for manual labor but also decreases the reliance on herbicides. AI algorithms can be trained to recognize weed growth patterns and predict their spread, enhancing the effectiveness of weed control strategies.

Furthermore, AI assists farmers in selecting the most effective and efficient weed management strategies based on various factors such as crop type, location, and other relevant

parameters. This personalized approach, driven by AI, contributes to improving the overall sustainability of agriculture. By precisely targeting weeds and optimizing resource usage, AI aids in making weed management practices more environmentally friendly and sustainable.

8.7 Main Part of AI in Agriculture

The role of AI in agriculture encompasses crucial aspects such as data collection, analysis, and decision-making, offering valuable insights to farmers. Here is a breakdown of these components:

- Data collection:
  - Sources: Data is collected from various sources like sensors, drones, and satellites. This information includes details on soil quality, weather conditions, crop health, and more.
  - Ethical use: Farmers need to ensure that data is collected and utilized in a useful and ethical manner, respecting privacy and ethical considerations.
- Data analysis:
  - AI algorithms: AI algorithms are employed to analyze the collected data, identifying patterns and making predictions.
  - Accuracy and unbiased analysis: It is crucial that the data being analyzed is accurate and unbiased to ensure reliable insights.
- Decision making:
  - Informed decisions: AI aids in making informed decisions based on the analyzed data, leading to increased efficiency

and reduced labor costs.

    - Human–AI collaboration: Decision-making in agriculture ideally involves a collaboration between human expertise and AI-powered insights. In Figure 8.8, we demonstrate the use of AI in agriculture to monitor the soil, crops, problems, livestock, and drones through a pie chart.
- Task automation:
    - Labor cost reduction: AI is utilized to automate tasks, reducing time-consuming and labor-intensive activities. Robots, for example, can handle tasks like seedling and harvesting, reducing labor costs.

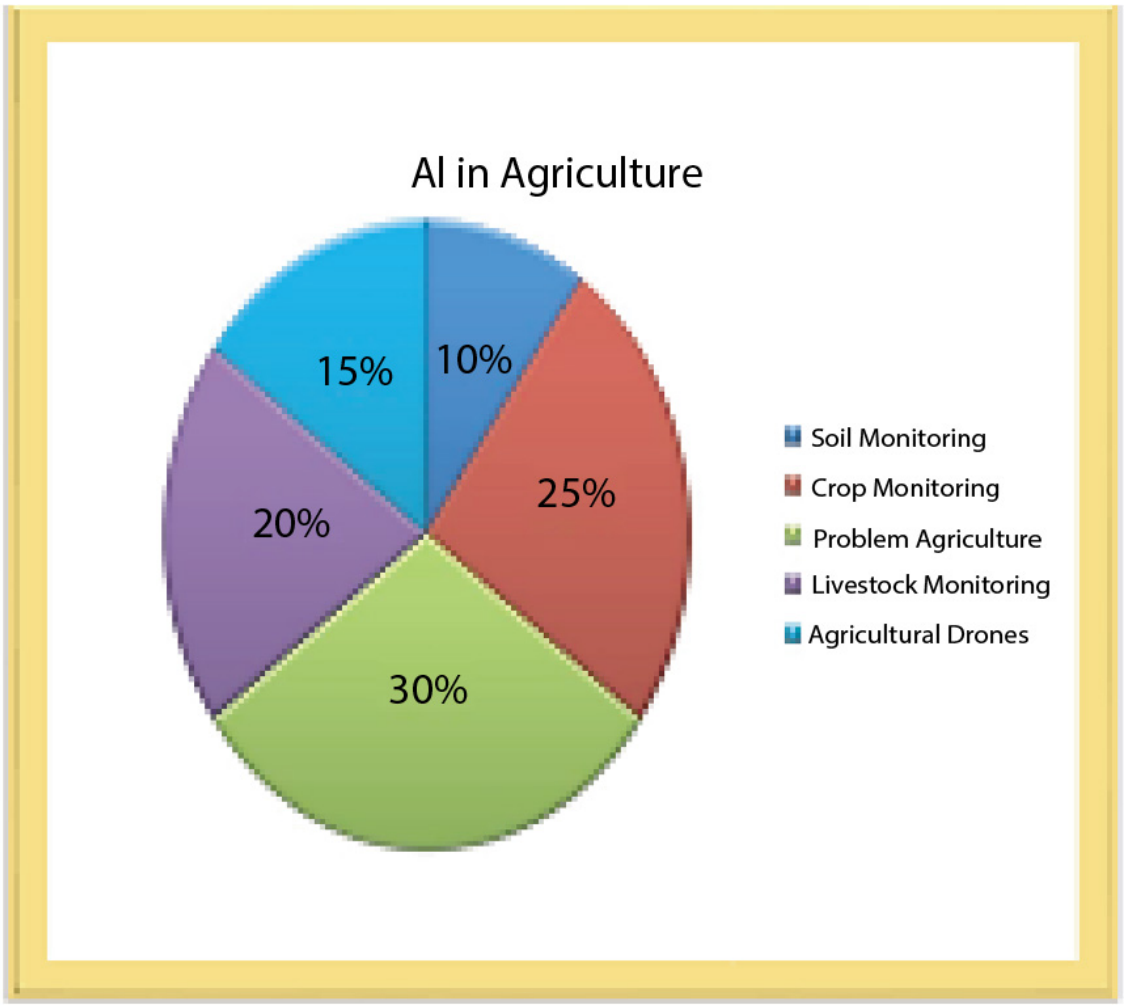

Figure 8.8 Pie chart of AI in agriculture.

- Autonomous farming:
  - Integration with machinery: AI is integrated with farming machinery such as tractors and harvesters, enabling autonomous farming processes like planting, spraying, and harvesting.
  - Predictive maintenance: AI can predict when farm machinery requires maintenance or repair based on data analysis of performance and usage patterns, contributing to equipment maintenance.

In summary, AI in agriculture plays a pivotal role in optimizing processes, reducing costs, and making more informed decisions, ultimately contributing to increased efficiency and sustainability in farming practices.

AI can analyze market data and provide farmers the required demand and supply chain management. This can help the farmers make informed decisions about which crops to plant and when to sell them.

8.8 Effects of AI in Agriculture

Positive effects

- It helps the farmers to upgrade their operations.
- It helps to increase the efficiency and reduce the implementation cost in agriculture.
- It increases productivity.
- It reduces labor cost.
- It enhances the quality and safety of food.
- It improves the yield and quality of crop.
- It improves supply chain management.

Negative effects

- Using AI leads to job displacement by machines instead of humans who rely on jobs for their livelihood.
- AI algorithms can be biased if the data used to train is not diverse and inclusive, which leads to unequal access to resources and opportunities for farmers from marginalized communities.
- It requires high initial cost and complex implementation.
- It has dependence on stable Internet connectivity.
- There is limited data availability in some regions.
- There are concerns about data privacy and security.

8.9 Combination of AI and IoT

The combination of AI and IOT in the field of agriculture gives the very potential outcome to the farmers and to this society. The technology benefits the field by doing precision farming, livestock monitoring, reduction of wastage, automated machinery, predictive analysis, etc. Initially, the IOT sensors are fixed over different places like in depth of the soil for predicting the moisture content from which the present quality of the soil can be understood, and those information will be send to the data control room or data logger; for calculating the temperature of soil or air (greenhouse gases), a temperature

sensor that will collect the data can be fixed in the soil and leaves; to measure how much light the crops receive can be monitored by fixing light sensors; the pH level—like acidic or alkaline—can be identified by fixing pH sensors in the soil; it can also monitor the nutrient level in soil for its optimum growth. Cattle can be monitored by analyzing its temperature, food intake level, milk production quantity, location tracking, rumination behavior, and other activities by fixing temperature sensors, activity sensors, GPS sensors, rumination sensors, milk sensors, etc. Here the pressure, temperature, flow rate, and vibrating level of the equipment can also be monitored with the help of sensors [14]. The above Figure 8.9 is a pictorial representation of the process of AI with IOT in an agricultural field.

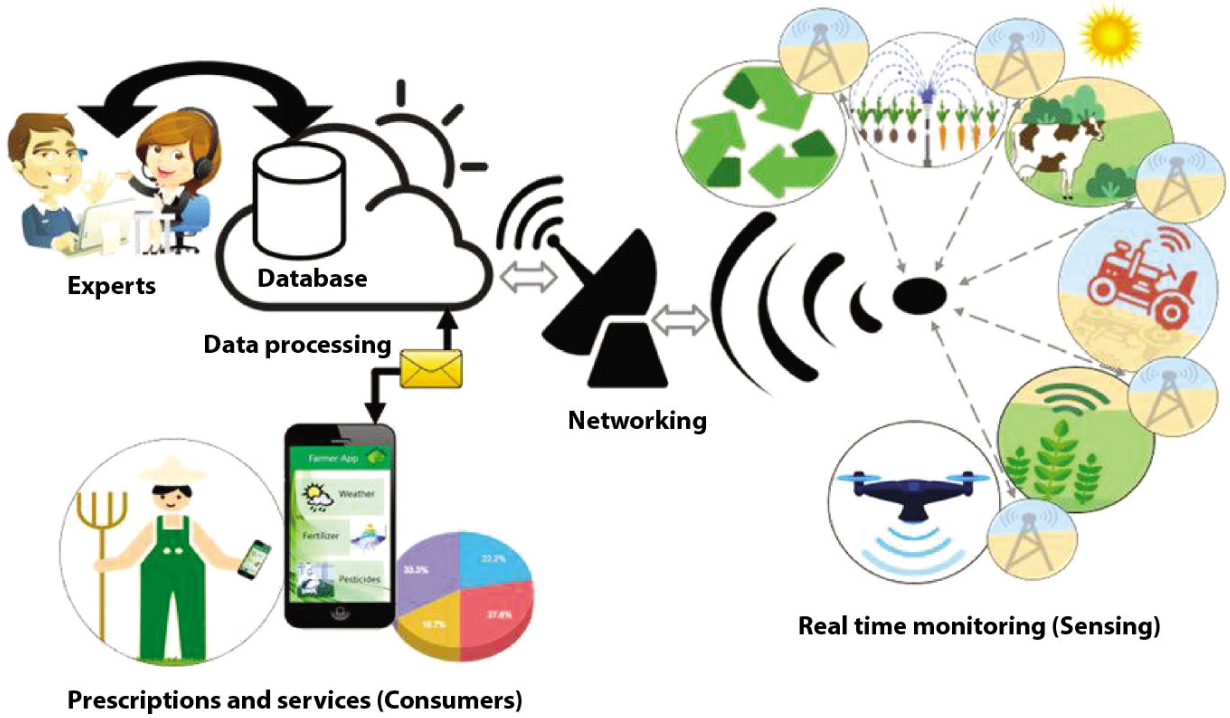

Figure 8.9 Process of AI with IOT in an agricultural field.

The collected data in agriculture undergoes a process of transfer and analysis for further insights. Here is a breakdown of the steps involved:

- Data collection and transmission:
  - Analog to digital conversion (ADC): Data collected in analog format is converted to digital format using ADC (analog-to-digital converter).
  - Digital Mux (multiplexer): A Digital Mux selects one input signal from multiple sensors and sends it to an output, facilitating the transfer of data to a central control system.
- Wireless sensor networks (WSNs):

    - Definition: WSNs are groups of sensors communicating wirelessly with each other and a central control system.
    - Application: In agriculture, WSNs are utilized to monitor conditions in various field or greenhouse areas, transmitting data to a central control system for analysis [15, 16].
- AI-powered predictive analysis:
    - Algorithmic options
    - Regression analysis: Analyzes relationships between variables to predict future trends; useful for predicting crop yields, commodity prices, and other multifactorial variables.
    - Decision trees: Makes decisions based on a series of criteria; useful for analyzing complex datasets and predicting future events.
    - Neural networks: Mimics the structure of the human brain for complex data analysis; applicable to tasks like image recognition and natural language processing.
    - Random forest: Builds decision trees and combines their results for more accurate predictions; suitable for various applications such as stock market analysis and predictive maintenance.
    - Support vector machines (SVMs): Used for classification and regression analysis; particularly beneficial for

datasets with many variables and identifying less obvious patterns.

- Data analysis:
  - Application of algorithms: The data in the control system is analyzed using one or more of these algorithms, depending on the nature of the analysis required.

In summary, the data collected in agriculture goes through a sophisticated process of conversion, transmission, and analysis, leveraging AI algorithms to provide valuable insights and predictions for informed decision-making in farming practices.

8.10 Automatic Irrigation

Farmers face challenges when irrigating large fields because they need to monitor the process closely. This is difficult for farmers with extensive land. To ensure proper irrigation, soil moisture sensors measure the soil's moisture content. An algorithm controls the irrigation system, maintaining a constant moisture level. When the levels drop, the system activates until the desired moisture level is reached. Weather data predicts water needs, adjusting irrigation based on conditions. Evapotranspiration combines evaporation and plant transpiration, calculated by an algorithm using weather data, crop type, soil, and system efficiency. Crop coefficient (Kc)

considers a crop's water needs, calculated by an algorithm based on crop type, growth stage, and weather. The irrigation system adjusts water application accordingly. Various algorithms, including Fuzzy Logic, cater to specific agricultural irrigation needs by handling uncertainty and imprecision in data.

Fuzzy Logic is commonly used in agriculture for automatic irrigation as it models complex relationships between soil moisture, weather, and plant growth. It determines the optimal irrigation schedule based on current soil moisture and weather conditions. Machine learning algorithms, such as random forest and support vector machines, learn from historical data to predict soil moisture and establish optimal irrigation schedules. Control Theory, a mathematical framework, controls water application based on desired soil moisture levels and adjusts schedules in real time using algorithms. Rule-based systems use if–then rules for automatic irrigation by determining schedules based on weather and soil conditions.

In practical application, the control room analyzes moisture levels, initiates irrigation as needed, and notifies the farmer. This allows uninterrupted field work, saving the farmer's energy and valuable time. Figure 8.10 represents the Connection between a physical farm and a virtual farm.

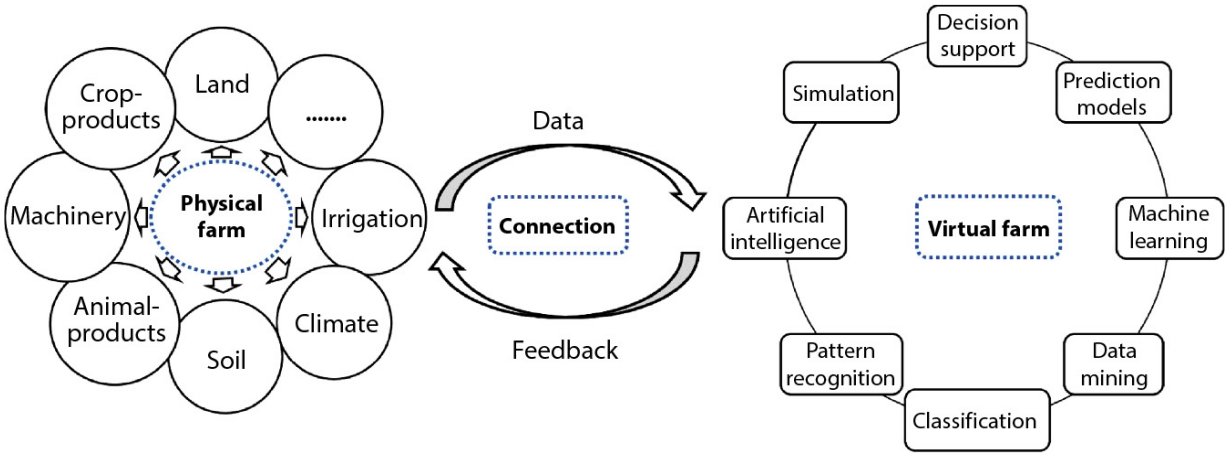

Figure 8.10 Connection between a physical farm and a virtual farm.

8.11 Crop Health Monitoring

Automated pest detection is an AI algorithm that can be trained to detect the presence of pests in crops by analyzing images captured by cameras or drones. These algorithms can recognize patterns and features specific to certain pests and identify their presence in the crops. Early detection allows farmers to take appropriate measures before the infestation spreads. Environmental sensors are IoT-enabled environmental sensors that can monitor temperature, humidity, and other factors that affect pest growth and reproduction [18]. Data from these sensors can be analyzed using AI algorithms to predict the likelihood of pest outbreaks and to identify optimal conditions for pest control. Predictive modeling is an AI algorithm that can be trained using historical data on pest outbreaks and crop health to predict when and where pests are likely to appear.

These models can help farmers take proactive measures, such as applying pesticides or using integrated pest management techniques, to prevent infestations. Automated pest control is an IoT-enabled pest control system that can be programmed to automatically release pesticides or other control measures when pests are detected [17]. This can reduce the need for human intervention and ensure that pests are controlled as quickly and efficiently as possible. Crop monitoring is also an AI and IoT technology that can be used to monitor crop health and growth, allowing farmers to detect early signs of stress or damage caused by pests. Figure 8.11 displays the IOT in agriculture market size, 2021 to 2030 (USD billion).

By catching these issues early, farmers can take action to prevent pest damage before it becomes severe. By these algorithms, a farmer can obtain all the details and start to take care of that diseased crop, which will increase the yield of the product.

8.12 Supply Yields to the Demanded Areas

The main drawback in an agricultural field is that the harvested crops are not transported properly to the market. The delay in time may lead to the wastage of those crop yields. Hence, to reduce this impact, the yield should be transported to the

nearer demanded areas than the distant market. It can be made possible by analyzing the yield demanded areas and indicating that here is a presence of harvested products they are in need of. For this, several methods will be used for analysis like sensor-based monitoring in which IoT sensors can be deployed on agricultural fields and storage facilities to monitor various quality parameters such as temperature, humidity, and soil moisture. The data collected by these sensors can be analyzed using AI algorithms to detect any anomalies and deviations from quality standards [18]. Computer vision is an AI-based computer vision technology that can be used to remotely assess the quality of agricultural products by analyzing images and videos of the products. Computer vision algorithms can detect defects, discoloration, and other quality issues by analyzing color, texture, and other visual attributes of the products. Machine learning algorithms can be trained on data collected from sensors and other sources to detect patterns and anomalies that may indicate quality issues. By analyzing data from multiple sources, machine learning algorithms can provide a comprehensive view of the quality of agricultural products and identify potential issues before they become serious problems [19]. Predictive analytics algorithms can be used to forecast quality issues based on historical data and current trends. By using these algorithms, growers and

producers can take proactive measures to prevent quality issues before they occur. Overall, by all these methods, the producer can easily do his work and the consumer also buys a quality product after the analysis of the supplying product.

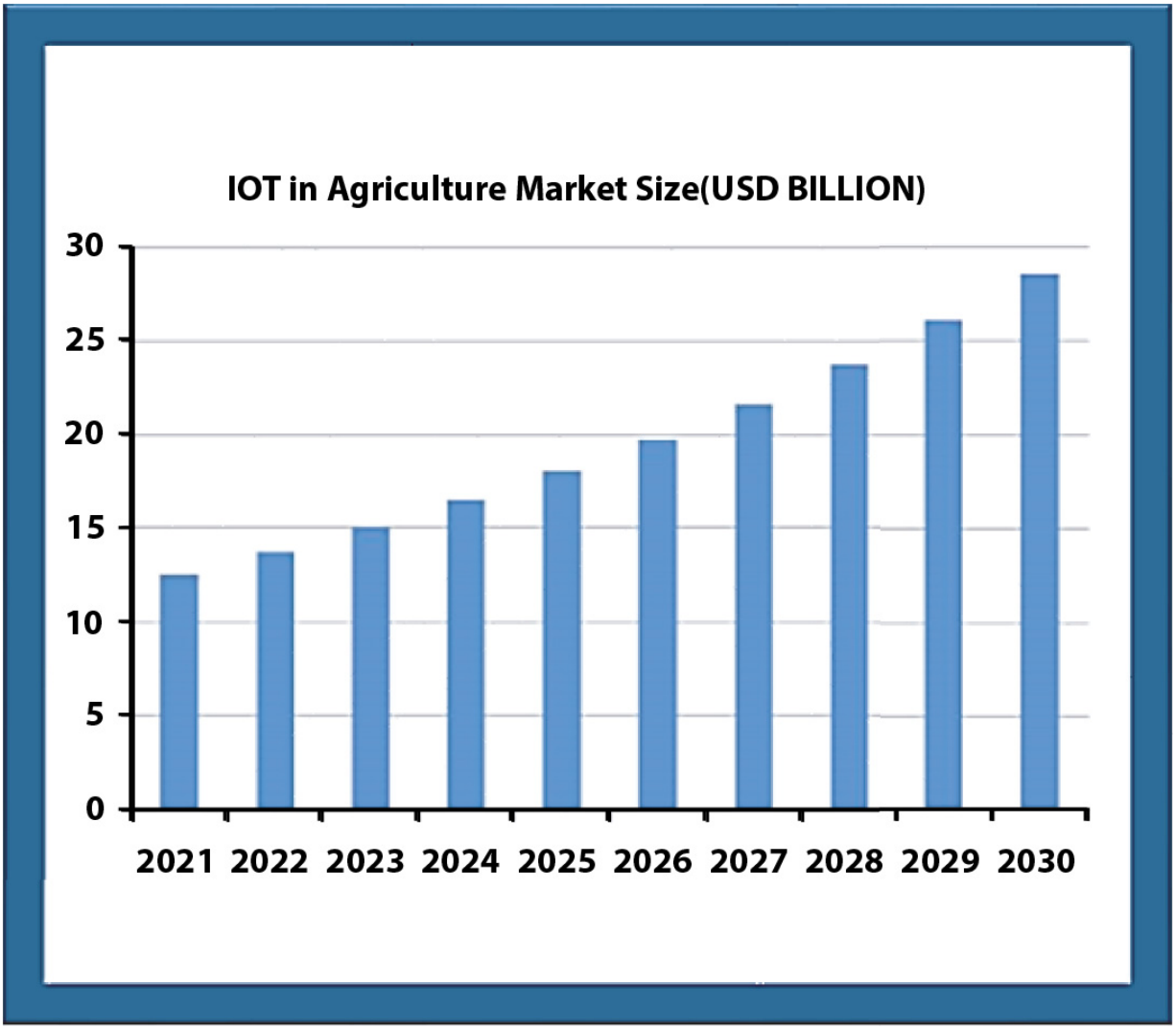

Figure 8.11 IOT in agriculture market size, 2021 to 2030 (USD billion).

8.13 Conclusion

In conclusion, agriculture stands as a vital lifeline for human survival, and the integration of AI and IoT technologies promises a substantial positive impact. The challenges faced by farmers in remote locations can be alleviated through the collaborative efforts of AI and IoT, which analyze entire fields and promptly notify farmers *via* messages. This eliminates the need for frequent physical presence, saving the farmers' time, costs, and efforts. The affordability of sensors further facilitates its widespread adoption.

The implementation of this technology not only reduces the burden on farmers but also enhances field productivity. In the rapidly advancing world of 2022–2023, where agriculture contributes approximately 18.3% to the nation's gross value added (GVA) according to the Ministry of Statistics and Programme Implementation (MoSPI), this modern technology has the potential to significantly elevate the contribution of agriculture in India. By embracing this combined technological approach, the agricultural sector can play an even more substantial role in both the nation's development and global trade markets.

References

1. Zhang, L., Yang, J., Cao, Q., & Li, Z. (2020). Smart agriculture based on IoT and AI: A survey. *Comput. Commun.*, 156, 251-262.
2. Liu, Z., Xie, X., Lu, X., & Zeng, P. (2020). Design and implementation of a smart farming system using IoT and AI technologies. *Comput. Electron. Agric.*, 170, 105231.
3. Ray, P. P. (2019). IoT-based smart agriculture: A review. *Agric. Res.*, 8(4), 425-432. doi: 10.1007/s40003-019-00407-1
4. Yan, G., Chen, J., Yang, S., & Zhang, X. (2015). IoT-based smart agriculture: Toward sustainable farming. *IEEE Commun. Mag.*, 53(5), 72-78. doi: 10.1109/MCOM.2015.7110482
5. Chatterjee, P., & Prasad, A. (2019). Real-time soil moisture monitoring using IoT for precision agriculture. *J. Agro Inf.*, 9(2), 13-22. doi: 10.17700/ jai.2019.9.2.454
6. Rodrigues, J. J., de Souza, P. R., Mateus, R., & Saleem, K. (2019). An IoT-based temperature and humidity monitoring system for industrial applications. *Sensors*, 19(4), 849. doi: 10.3390/s19040849
7. Al-Ali, A. R., Arar, S. S., Al-Ali, A. R., & Masa’deh, R. E. (2021). Development of a smart light intensity monitoring system for greenhouse environments. *Sensors*, 21(8), 2813. doi: 10.3390/s21082813

8. Damasceno, J. C., de Souza, E. P., Peixoto, M. F., & Lima, J. P. (2016). A livestock monitoring system based on IoT and wireless sensor networks. *IEEE Lat. Am. Trans.*, 14(4), 2023-2029. doi: 10.1109/TLA.2016.7771555
9. Lobos-Morales, J., Ochoa, A., Cartes, D., Mellado, M., & Reyes, F. (2021). AI and IoT for smart agriculture: A systematic literature review. *Comput. Electron. Agric.*, 181, 105972. doi: 10.1016/j.compag.2020.105972
10. Qin, Z., Zhang, Y., Zhu, L., Wang, J., & Zhang, Y. (2020). UAVs-based high-throughput phenotyping in crop breeding programs. *Front. Plant Sci.*, 11, 1010. doi: 10.3389/fpls.2020.01010
11. Sørensen, C. G., Pesonen, L., Bochtis, D. D., Vougioukas, S. G., & Suomi, P. (2011). Functional requirements for a future farm management information system. *Comput. Electron. Agric.*, 76(2), 266-276.
12. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Front. Plant Sci.*.
13. Ferentinos, K. P. (2018). Deep learning models for plant disease detection and diagnosis. *Comput. Electron. Agric.*, 145, 311-318.
14. Solanki, V., & Makwana, R. (2013). Intelligent agriculture system with weather monitoring. In *2013 Nirma University*

International Conference on Engineering (NUiCONE) (pp. 1-5). IEEE.

15. Bandyopadhyay, S., Saha, A. K., Maulik, U., & Deb, D. (2011). Wireless sensor network data integration for an agricultural decision support system using K-means clustering. *J. Indian Soc. Agric, Stat.*, 65(1), 53.
16. Sharma, P., & Singh, N. (2016). Design and development of precision agriculture system using wireless sensor network. *Int. J. Comput. Appl.*, 138(10).
17. Joshi, A., Kale, S., Chandel, S., & Pal, D. K. (2015). Likelihood of adoption of IoT solutions in Indian agriculture. IoT and cloud computing technologies for agriculture and related sectors, 13.
18. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
19. Meena, G., & Jain, S. (2020). Internet of Things (IoT) based smart irrigation systems: a review. *Artif. Intell. Agric.*, 4, 50-61.
20. Bhatnagar, R., Prasad, M., & Gupta, R. (2016). Wireless sensor networks for agriculture: The state-of-the-art in practice and future challenges. *Comput. Electron. Agric.*, 118, 66-84.

21. Vu Khanh Quy, Nguyen Van Hau, Dang Van, Nguyen Minh Quy, Nguyen Tien Ban , Stefania Lanza, Giovanni Randazzo and Anselme Muzirafuti (2022) “IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges”, published in Applied Scienses, MDPI, *Appl. Sci.* 2022, 12(7), 3396; https://doi.org/10.3390/app12073396.
22. Santosh Kumar Srivastava, Dr. Yogesh Kumar Sharma, Dr. Sheo Kumar, “Using Of WEKA Tool In Machine Learning: A Review” Published in *Int. J. Adv. Sci. Technol.*, Vol. 29, No. 6, (2020), pp. 8604-8614, ISSN: 2005-4238 IJAST.

Note

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9
Smart AgriTech: Sensors & Networks

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Abstract

This chapter offers a comprehensive overview of IoT's impact on agriculture, reshaping farming practices. IoT, the interconnection of physical devices *via* the Internet, promises to revolutionize how we interact with technology. The chapter explores IoT's application in agriculture, highlighting its benefits, including enhanced efficiency, waste reduction, and improved crop yields. It also acknowledges challenges like high infrastructure costs and limited rural connectivity. Sensor-based IoT's crucial role is emphasized, showcasing how sensors gather vital data on crops, soil, weather, and livestock, optimizing farming methods. The importance of choosing suitable network and communication protocols, tailored to farm locations, is also discussed. Real-world case studies from diverse countries, including India, Thailand, China, Malaysia,

the Philippines, and the United States, demonstrate IoT's transformative impact on global farming practices. In conclusion, the chapter envisions a promising future for IoT in agriculture, driven by advancements in technologies like artificial intelligence and machine learning, promising increased productivity and sustainability in the agricultural sector.

***Keywords*:** Agriculture, artificial intelligence, communication, IOT, network, sensor, security

9.1 Introduction

The IoT has transformed many industries. It converted our normal home into a smart home [1, 2] and a normal city into a smart city [3, 4]. It has a great impact on healthcare [5, 6], and modern autonomous vehicles [7, 8] are also based on IoT. Besides that, it has a great impact on the energy sector to generate green energy [9, 10]. The Internet of Things has also made a huge contribution in the field of agriculture [11–13] by leveraging sensors, networks, and communication protocols to enhance efficiency, productivity, and profitability. IoT applications in agriculture have gained significant attention in recent years due to the pressing need for sustainable and innovative agricultural practices to face basic needs. With the

rising population, climate change, and resource depletion, IoT in agriculture presents a significant opportunity to optimize resource utilization, reduce waste, and mitigate environmental impact. Farooq *et al.* [14] nicely represent how IoT helps different areas of agriculture.

The agricultural fields face various challenges, such as unpredictable weather conditions, soil degradation, pest and disease infestations, water scarcity, and labor shortages, which can significantly affect crop yield and quality. IoT solutions in agriculture can address these challenges by providing real-time insights and automated monitoring of crop and soil conditions, enabling farmers to make decisions based on the collected data and optimize their operations. Moreover, IoT in agriculture can enhance food safety, traceability, and transparency by enabling stakeholders to track and monitor the entire supply chain from farm to fork.

The use of IoT in the agricultural industry is likely to rise in the upcoming years since it is expanding quickly. The use of IoT in the agricultural market is anticipated to increase from $11.4 billion in 2021 to $18.1 billion by 2026, at a CAGR of 9.8%, according to a study [49]. The paper also emphasizes the rising need for IoT solutions in smart irrigation systems, livestock monitoring, and precision farming.

Precision farming, which uses IoT sensors and analytics to optimize crop output and quality while reducing resource consumption, is a primary advantage of IoT in agriculture. IoT for agriculture can reduce water usage in smart farms by up to 30%, according to an online resource [50]. Moreover, IoT in agriculture can help reduce the use of pesticides and fertilizers by providing real-time monitoring of pest and disease outbreaks and soil nutrient levels, enabling farmers to apply targeted treatments only where needed.

Another area where IoT in agriculture can make a significant impact in livestock management. IoT sensors can provide real-time monitoring of animal health, behavior, and feed intake, enabling farmers to detect and prevent diseases, optimize feed usage, and improve animal welfare. According to a report by Mordor Intelligence, the global livestock monitoring market size is projected to reach $2.5 billion by 2027 at a CAGR of 12.67% [51].

One further intriguing use of IoT in agriculture is smart irrigation systems. Farmers can optimize irrigation timing and use less water by using IoT sensors to monitor crop water needs, weather conditions, and soil moisture levels. Grand View Research [52] estimates that the size of the worldwide smart

irrigation market will reach 8.69 billion USD by 2027, growing at a CAGR of 16.8%.

IoT adoption in agriculture, however, is not without difficulties. Data security and privacy are one of the major issues since the gathering and analysis of vast volumes of data can lead to worries about unauthorized access and exploitation. Moreover, the diversity of IoT devices and communication protocols can create interoperability issues, making it difficult to integrate different IoT systems and platforms. Other challenges include network connectivity, energy efficiency, and maintenance costs.

To address these challenges, various initiatives and standards have been developed to ensure the secure and interoperable deployment of IoT in agriculture. For instance, the Agricultural Industry Electronics Foundation (AEF) has developed the ISOBUS standard, which enables the communication and compatibility of different agricultural machinery and sensors. In addition, the Open Ag Data Alliance (OADA) has created an open data architecture for agriculture that allows stakeholders to share data in a safe and transparent manner.

There are several current research projects aiming at enhancing the effectiveness as well as the viability of IoT systems for farming in addition to the breakthroughs and

difficulties highlighted above. For instance, researchers are creating new types of sensors that can deliver more precise and current information on the moisture in the soil, levels of nutrients, and crop health. This information may assist farmers in making better decisions on irrigation, fertilization, and insect management. Efforts are also being made to combine Internet of Things (IoT) devices with other cutting-edge innovations like robots and artificial intelligence (AI), which might lead to the development of more accurate and self-sufficient farming techniques.

Moreover, the COVID-19 pandemic has highlighted the importance of digital technologies such as IoT in agriculture. The pandemic has disrupted global food supply chains, leading to shortages and price hikes in many regions. In response, governments and organizations are investing in digital technologies to improve the resilience and efficiency of food systems—for example, the European Union has launched a “farm to fork” strategy, which aims to promote sustainable and digital agriculture practices across Europe. Similarly, the United States Department of Agriculture (USDA) has launched several initiatives to support the development and usage of IoT technologies in agriculture.

The contribution of IoT to encouraging sustainable agricultural practices is another crucial factor to take into account. There is increasing demand in the sector to adopt more environmentally friendly and sustainable practices because farming is a major source of emissions of greenhouse gases and environmental deterioration. IoT technology enables precision agricultural techniques that maximize resource utilization and minimize waste, which can assist in lessening the environmental effect of agriculture. IoT technologies, for instance, can assist farmers in better monitoring and managing water consumption, lowering the danger of shortages and contamination. By enabling more focused and exact applications, IoT can also contribute to a decrease in the usage of toxic pesticides and fertilizers.

IoT technology development and adoption for agriculture have the ability to completely change the sector and solve some of its most pressing problems. However, a number of technological, governmental, and moral challenges need to be resolved in order to fully enjoy the advantages of IoT. To make sure that IoT is implemented in such a way that is sustainable, egalitarian, and advantageous for everyone, interaction with beneficiaries like farmers, technology providers, legislators, and consumers will be essential.

9.2 IoT in Agriculture

In order to escape the impending food crisis brought on by the population growth, we will need to use cutting-edge technologies such as AI or IoT. A new technology called IoT enables the real-time collection and sharing of data between physical objects, sensors, and systems through the Internet. The healthcare sector, logistics, production, and smart cities are just a few of the industries where the Internet of Things has already found application. However, the use of IoT in agriculture is among its most exciting applications. IoT in agriculture uses sensors, networks, robotics, communication protocols, unmanned aerial vehicles, and robots to track and manage a variety of agricultural operations, including soil moisture, crop development, irrigation, the health of livestock, and machinery performance [15]. The potential advantages, difficulties, and restrictions of the IoT for farming are highlighted in this section's introduction.

9.2.1 Benefits of IoT in the Agricultural Sector

Numerous advantages of IoT in agriculture may assist growers and other stakeholders improve their operations, boost production, and save expenses. The capacity to gather and analyze real-time data from numerous sources, including the

soil, the climate, and crop sensors, is one of the key advantages of IoT in agriculture. Based on the actual circumstances in the field, this data may be used to make smart choices regarding time to plant, irrigate, fertilize, or harvest crops. Farmers may be able to increase their yields, cut down on waste, and conserve freshwater and other resources because of this.

Monitoring and controlling livestock performance is another advantage of IoT in agriculture. Animals can have IoT sensors connected to them to monitor their behavior, eating habits, and vital indications like body temperature and heart rate. By using this information, one can identify early indications of disease, damage, or stress and take appropriate preventative or therapeutic steps. As a result, farmers may be able to decrease the likelihood of disease outbreaks, enhance animal welfare, and boost both the quantity and quality of their output.

IoT in agriculture can also help farmers to improve the use of farm machines and equipment, such as tractors, harvesters, and irrigation systems. IoT based sensors can be attached to these devices to monitor their performance, detect malfunctions, and schedule maintenance. This can help farmers to reduce downtime, save energy, and prolong the lifespan of their equipment.

By providing real-time information regarding the position, quality, and number of agricultural products, IoT for agriculture can also serve to enhance supply chain management. Utilizing this information helps improve logistics, cut waste, and guarantee on-time client delivery.

9.2.2 Limitations and Challenges of IoT in Agricultural Farming

Despite its potential benefits, IoT in agriculture also faces several challenges and limitations that need to be addressed for its widespread adoption. One of the main challenges of IoT in agriculture is the cost of implementation. IoT sensors, networks, and communication protocols can be expensive to install and maintain, especially for small-scale farmers or in remote areas with limited connectivity. The cost-effectiveness of IoT in agriculture needs to be carefully evaluated on a case-by-case basis, taking into account the specific needs, resources, and constraints of each farm or region.

IoT in agriculture has additional challenges related to data security and privacy. IoT-enabled systems in agriculture to collect and transmit sensitive data, such as crop yields, livestock health, and machinery performance. This data can be vulnerable to cyberattacks, data breaches, or misuse, which can

have serious consequences for farmers and consumers. The security and privacy of IoT systems in agriculture need to be ensured by using robust encryption, authentication, and access control mechanisms, as well as by raising awareness and education about cyber threats and best practices.

Connectivity is another challenge of IoT in agriculture. Many rural areas still lack adequate network infrastructure, which can limit the coverage and reliability of IoT systems. IoT technologies that can work in low-bandwidth or offline environments, such as satellite, mesh, or edge networks, need to be explored and optimized for agricultural applications.

The interoperability of IoT-enabled systems for agricultural use is also a crucial factor. The term “interoperability” describes the straightforward communication and collaboration between various devices, programs, and systems. Due to the variety of sensors, networks, as well as protocols for communication used by numerous players in agriculture, including farmers, equipment makers, and data analysts, interoperability is essential. No matter where they come from or what technology they use, it is crucial to make sure that these various components can successfully connect and share data.

The lack of interoperability can lead to several challenges, such as data silos, duplication of effort, and inefficiencies in data collection and analysis—for example, if a farmer has different sensors from different manufacturers, it may be difficult to integrate the data from these sensors into a single platform for analysis. This can result in data silos, where data is stored in separate systems and is not easily accessible or usable. It can also lead to duplication of effort, where the same data is collected and analyzed multiple times by different stakeholders, leading to inefficiencies and increased costs.

To address these challenges, there are several efforts underway to promote interoperability in IoT systems in agriculture. One approach is to develop common data standards and protocols that enable different devices and systems to communicate effectively—for example, the Open Connectivity Foundation (OCF) is a global consortium that is developing open standards for IoT devices and systems to promote interoperability. Similarly, the Agricultural Industry Electronics Foundation (AEF) is developing standards for agricultural machinery and equipment to ensure that they can communicate and exchange data effectively.

Another approach is to develop middleware solutions that enable different systems and devices to communicate and

exchange data without requiring extensive integration efforts—for example, the IoT Platform for Agri-Food (IoF2020) is an EU-funded project that is developing a middleware solution for IoT systems in agriculture. Regardless matter where they come from or what technology they use, this platform enables various Internet of Things (IoT) gadgets and systems to connect and share data without any issues.

In addition to these technical approaches, there are also efforts to promote interoperability through policy and regulation—for example, the European Union's General Data Protection Regulation (GDPR) includes provisions that require data controllers and processors to make sure that personal information is gathered and processed in such a way that promotes interoperability and data portability.

Overall, interoperability is an important consideration in IoT systems in agriculture, and efforts to promote interoperability are essential to ensure that these systems can achieve their full potential. As the use of IoT-based technology in agriculture continues to grow, it is likely that interoperability will become an increasingly important area of focus and that efforts to promote interoperability will continue to evolve and mature over time.

9.3 Utilization of Sensor-Based IoT Devices in the Agricultural Sector

There are many areas in the agricultural sector where IoT can be used successfully few of them are Precision farming, livestock tracking, and greenhouse management. IoT applications for farming, mobile applications, and sensor-based technologies are the three categories into which applications for farming have been categorized.

9.3.1 Precision Farming

Precision farming is a method for employing technology to improve the accuracy and productivity of agriculture [16]. It entails tracking and tracing, farm management systems, soil structures, insect & disease of crops monitoring, irrigation monitoring, and calculating the best times to plant and harvest [17]. By giving farmers the opportunity to monitor and manage numerous areas of their farms, IoT has significantly changed how farming is done [18].

Climate Condition Monitoring

Climate condition monitoring involves using IoT sensors to monitor the weather and environmental conditions. This information is used to take decisions about crop management,

including irrigation, fertilization, and pest control. The sensors used in climate condition monitoring measure temperature, humidity, wind speed, and rainfall [19]. Figure 9.1 shows the IoT-enabled devices for climate condition monitoring.

Soil Patterns

Soil patterns can be monitored using IoT sensors to provide information about soil quality, moisture levels, and nutrient content [20], and using this information time for fertilization, irrigation, and planting schedule [21]. Soil sensors can be placed at different depths to provide a more accurate picture of soil conditions [22]. Figure 9.2 gives an idea of soil pattern monitoring using IoT, where different sensors are used to measures surface temperature, soil moisture etc.

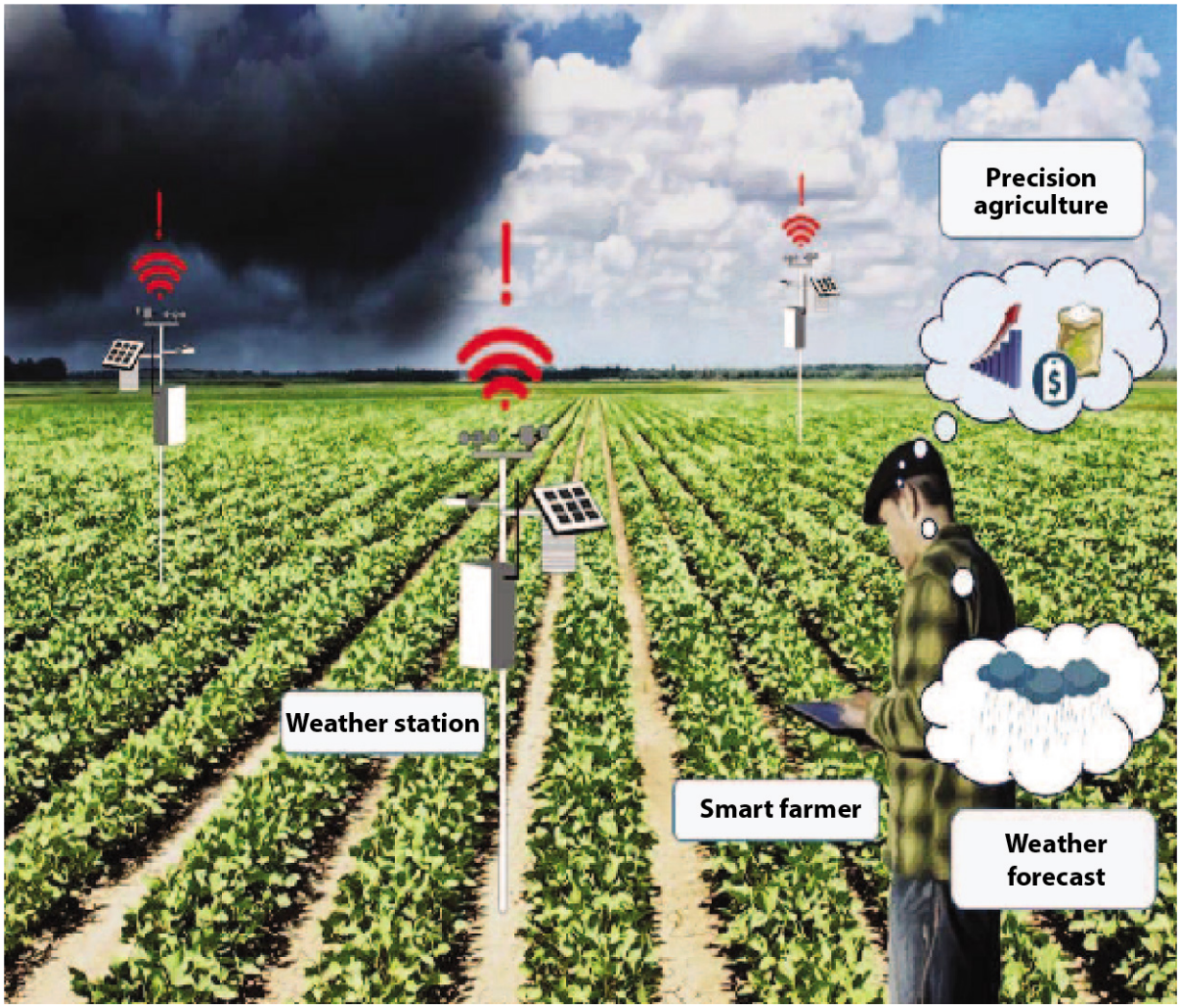

Figure 9.1 IoT-based climate condition monitoring for agri-farm (source: Tenzin *et al.* [53]).

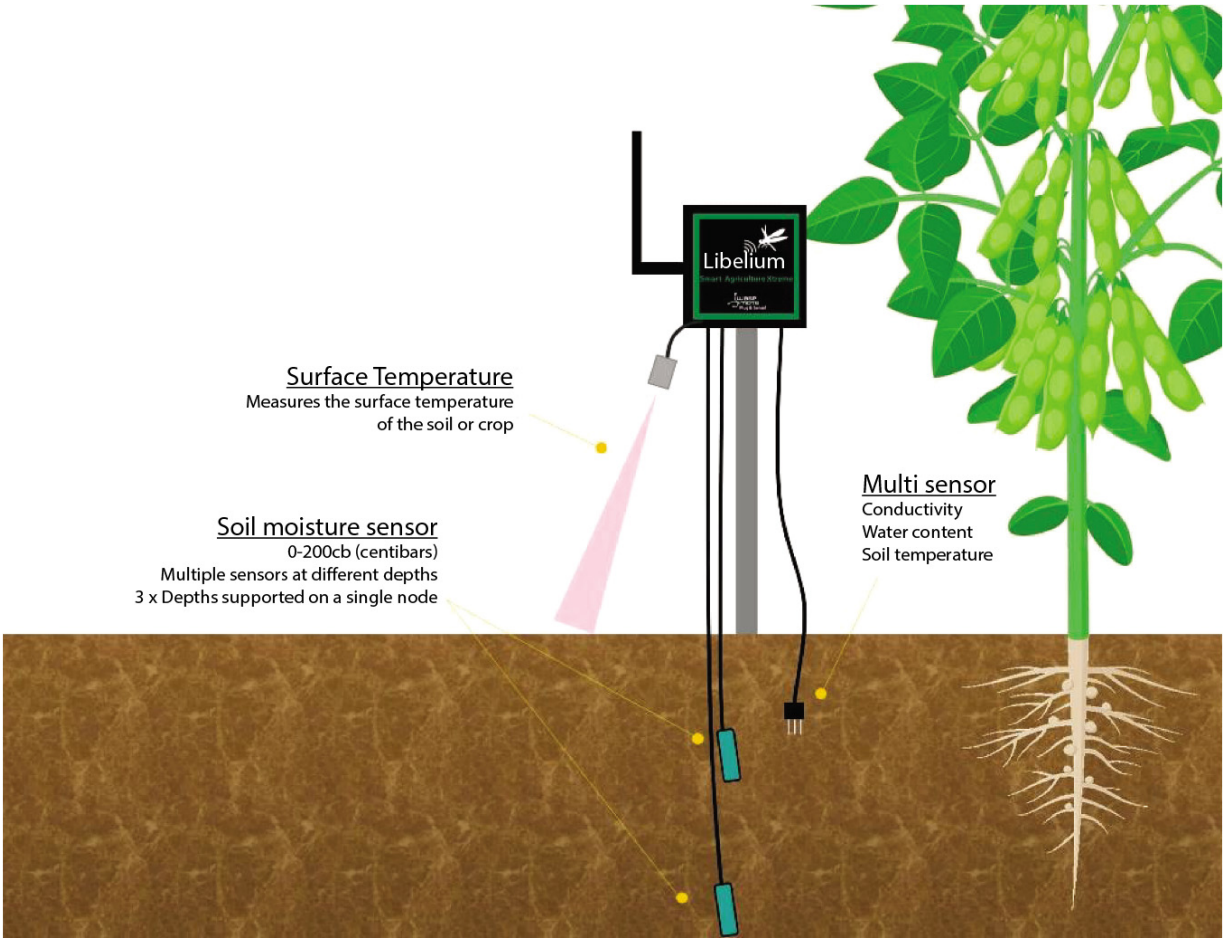

Figure 9.2 IoT-based soil pattern monitoring system (source: www.manxtechgroup.com).

Monitoring of Pests and Crop Diseases

IoT-based sensors are used to monitor the early signs or symptoms of plant disease. This will help the farmers to take the precaution to prevent crop losses [23]. IoT-based sensors will be able to monitor temperature changes, changes in humidity, as well as other environmental conditions which indicate the presence of pests or diseases.

Irrigation Monitoring System

IoT-based efficient sensors will be used to check the moisture levels in soil as well as it will be able to control water supply systems automatically [24]. This can help farmers save water and reduce costs. Sensors can be placed at different depths in the soil to provide a more accurate picture of moisture levels. Figure 9.3 gives an idea about smart irrigation using IoT.

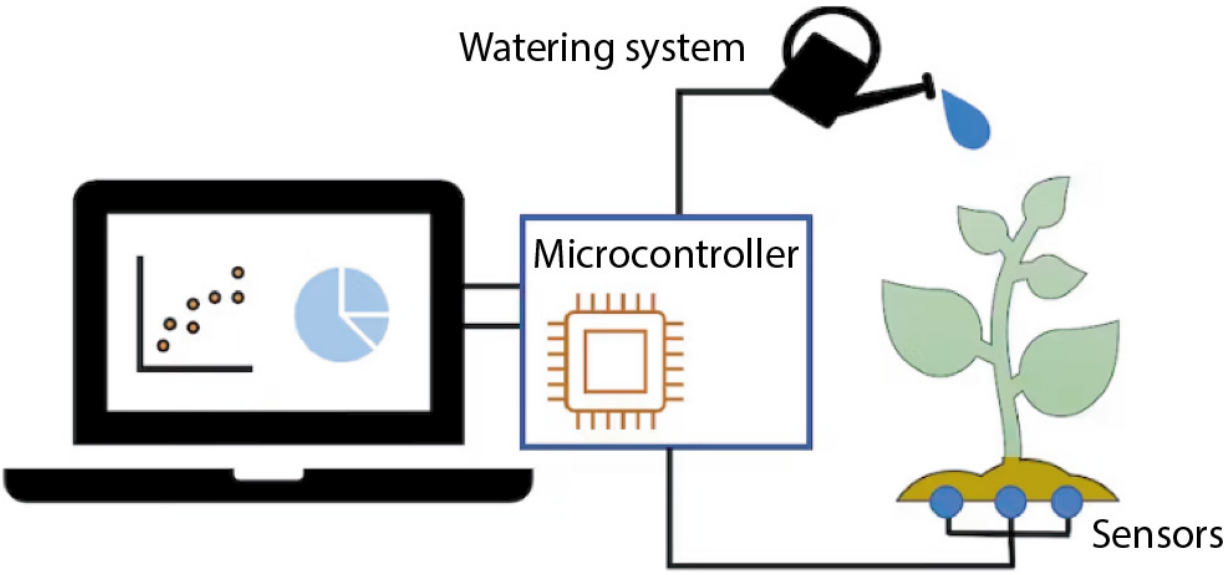

Figure 9.3 IoT-enabled devices for irrigation monitoring (source: Bing).

Determine the Suitable Time to Plant and Harvest

IoT-enabled sensors are useful for checking the growth and development of crops. Which will help to find out a suitable time to plant and harvest crops. The Sensors will be able to detect temperature changes, changes in humidity, and other environmental conditions [25].

Tracking and Tracing

IoT sensors can be used to track and trace agricultural products from farm to consumer. This can help improve food safety and reduce waste. Sensors can be used in temperature monitoring and humidity of the products during storage and transportation [26].

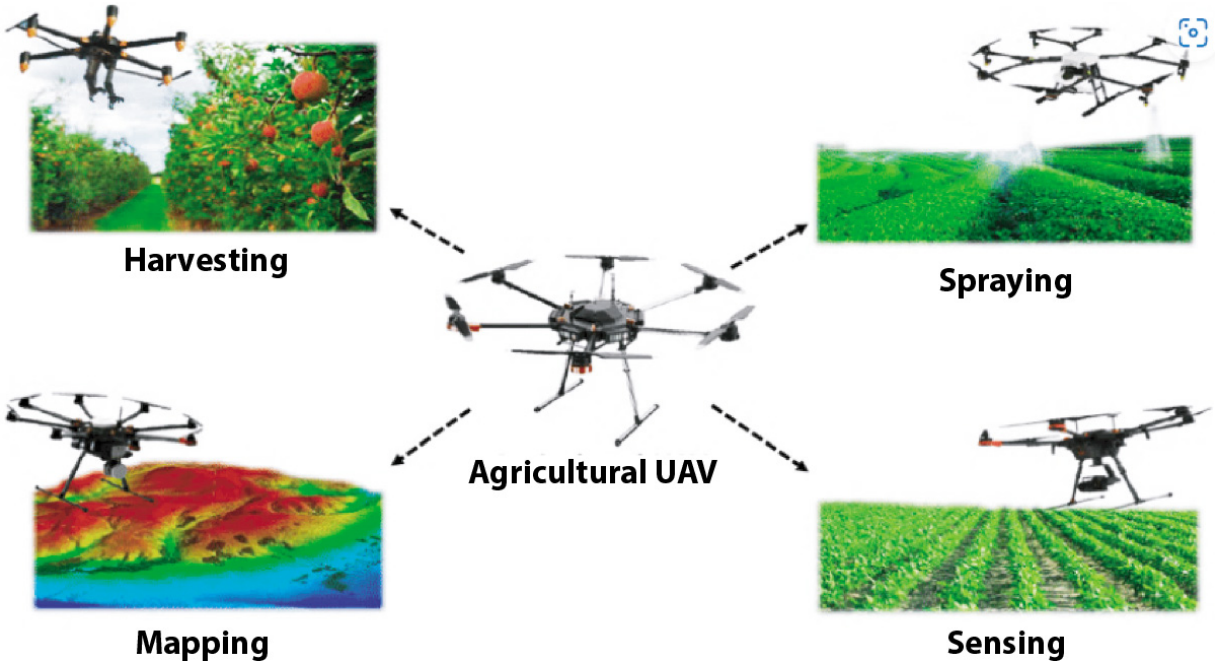

Figure 9.4 IoT-enabled drones for smart agriculture (source: Kim *et al.* [60]).

Farm Management System

IoT sensors can be used to create a farm management system that provides real-time information about crop and livestock conditions. This can help farmers make informed decisions about resource allocation and crop management [27].

Agricultural Drones

Agricultural drones equipped with IoT sensors can be used to monitor crops, soil conditions, and weather patterns. Drones can collect data more efficiently and accurately than traditional methods, reducing the need for manual labor [28]. Figure 9.4 shows use of drones in various fields of agriculture.

9.3.2 Greenhouse Monitoring

Greenhouse monitoring involves using IoT sensors to monitor water management, plant growth, and climate conditions. Decisions about temperature management, fertilization, and irrigation may be made using this information [29]. The development of smart greenhouse using IoT are presented in Figure 9.5.

Water Management

IoT-based sensors will be used to track the moisture levels in the soil and control watering systems automatically. This can help farmers save water and reduce costs. Sensors can be placed at different depths in the soil to provide a more accurate picture of moisture level.

Plant Monitoring

IoT sensors can be applied to check the growth and health conditions of different plants in greenhouses. From this information, decisions concerning fertilization and control of pests can be taken. Sensors are able to measure the changes in temperature as well as humidity and other environmental factors which affect the plant growth [30].

Climate Monitoring

Different sensors are used to monitor humidity, temperature, and other environmental factors in greenhouses. This information will help to take decisions about temperature control and ventilation [31].

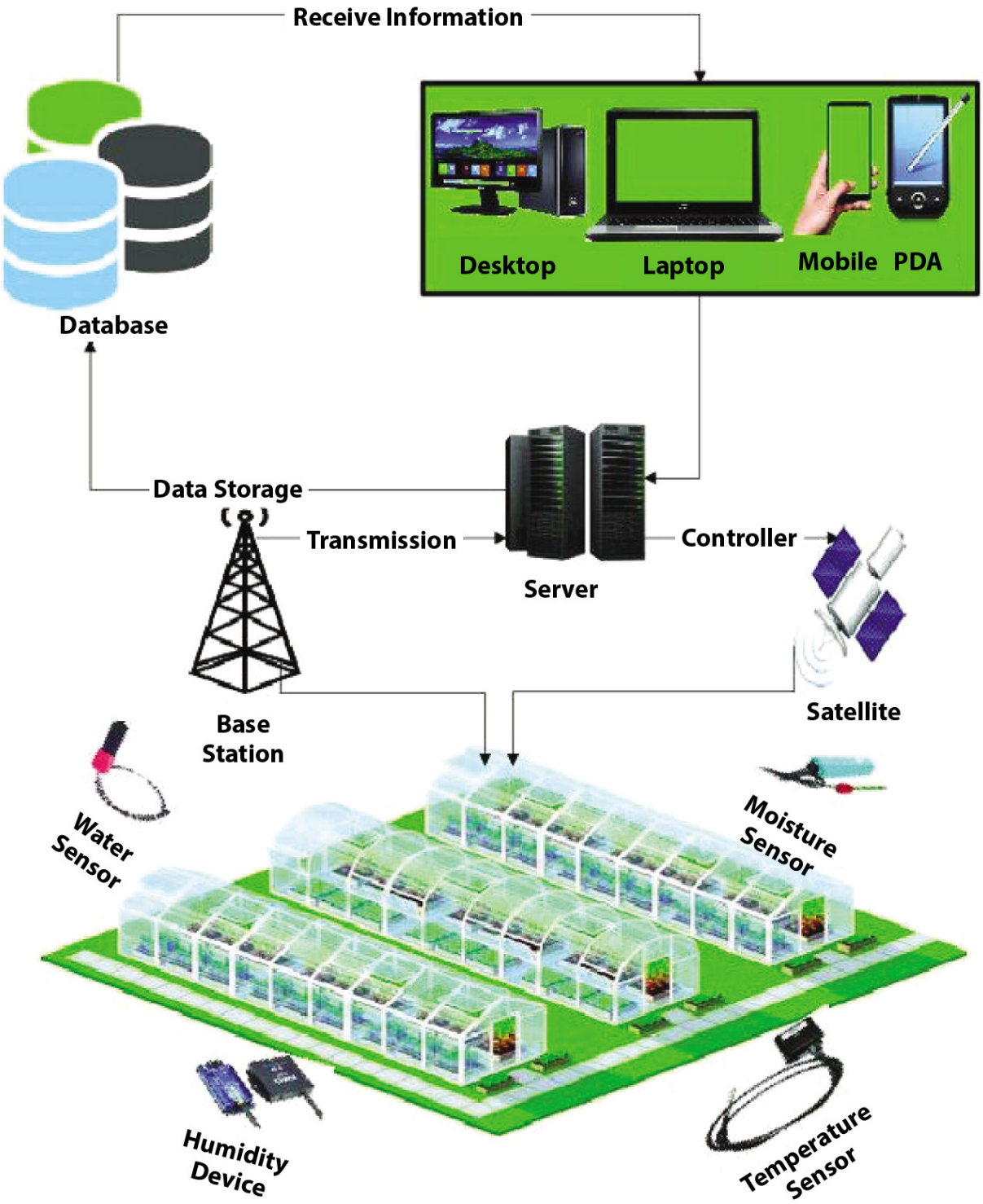

Figure 9.5 IoT in greenhouse monitoring (source: Farooq *et al.* [14]).

9.3.3 Livestock Monitoring

In livestock monitoring, different sensors are used to track the health and behavior of livestock. These sensors can monitor various parameters like the temperature of the body, heart rate, rumination, and physical activity. Livestock monitoring will help the farmers to detect health issues in their animals, which can improve their overall herd health and productivity [32]. There are different kinds of livestock sensors that are already available, which are discussed below. An IoT-enabled smart livestock monitoring system is presented in Figure 9.6.

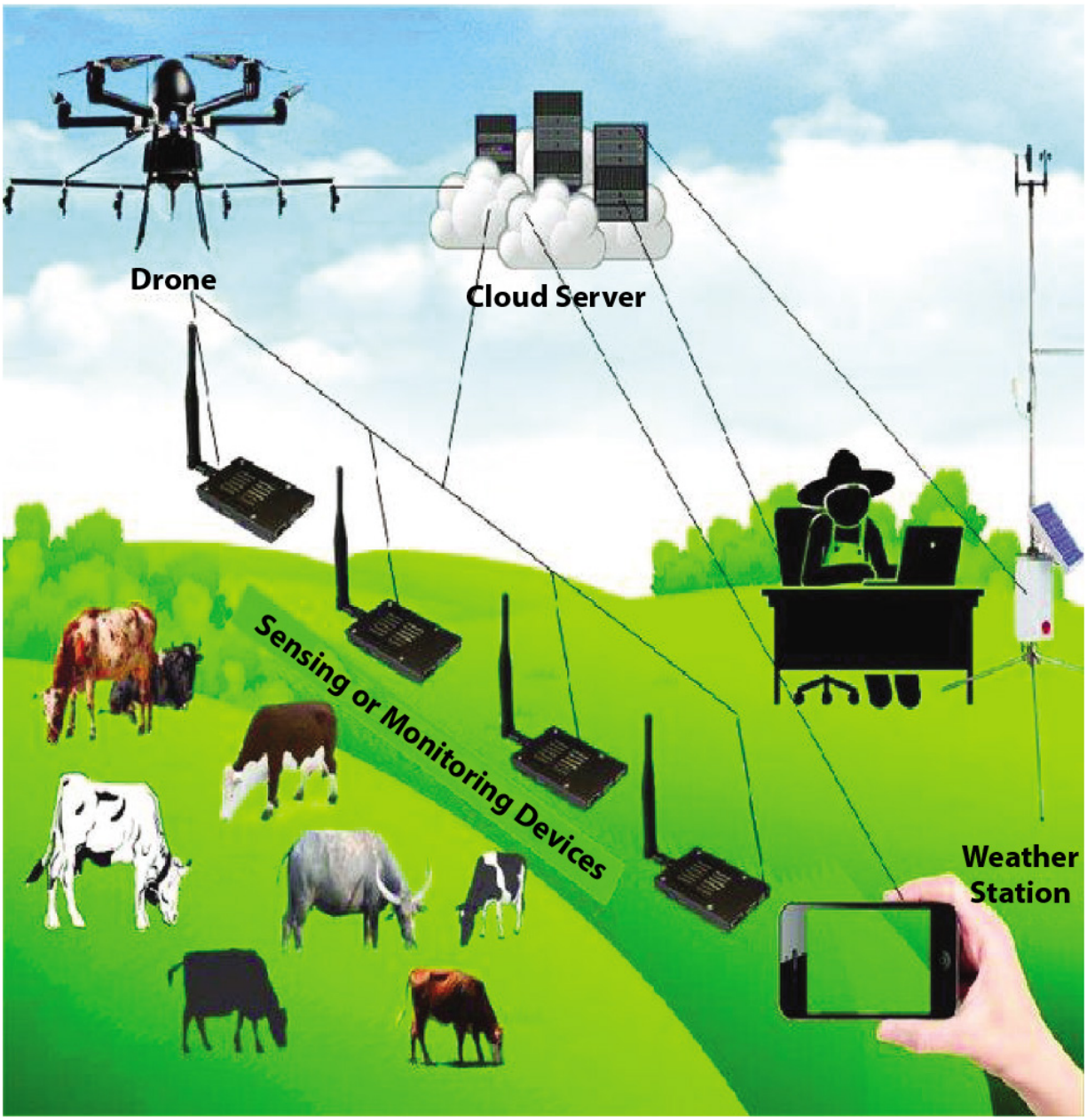

Figure 9.6 IoT-enabled devices for livestock monitoring (source: Farooq *et al.* [14]).

Animal Body Temperature Monitoring Sensors

Temperature of the body is an important indicator to check the health condition of an animal. Temperature monitoring sensors can be placed inside the animal's ear, rectum, or under the skin

to monitor the animal's temperature. These sensors can detect even slight changes in the animal's body temperature, which can indicate an illness or infection. These sensors can also be used to detect heat stress in animals [33].

Heat Stress Level Sensors

Heat stress is a common problem in livestock, especially in hot and humid environments. High heat stress can cause a decrease in milk production, fertility, and growth rate. Use of heat stress level sensors will help to monitor the temperature and humidity in animal's environment. These sensors will also help to monitor the body temperature and behavior of the animals to detect signs of heat stress.

Sensors to Recognize the Gesture

Physical gesture recognition sensors will be useful to monitor the movement and behavior of animals. These sensors can detect changes in an animal's posture, gait, and body position. They can be used to detect lameness, which is a common health issue in livestock. These sensors can also be used to monitor the animal's activity level, which can provide insight into their overall health and well-being.

Rumination Sensors

Rumination sensors are used to monitor the chewing activity of livestock. These sensors can detect the frequency and duration of the animal's chewing, which can indicate their level of feed intake and digestion. These sensors can also detect changes in rumination patterns, which can indicate health issues such as digestive problems.

Heart Rate Sensors

Heart rate sensors are used to monitor the heart rate of livestock. These sensors can be placed on the animal's body or in its ear to detect changes in heart rate. Changes in heart rate can indicate health issues such as stress or illness.

GPS-Based Monitoring Sensors

GPS-based monitoring sensors can be used to track the location and movement of livestock. These sensors can be attached to the animal's collar or ear tag and can provide real-time location data. Wireless technology-enabled GPS monitoring systems can be used to detect when an animal is in distress or has strayed from their designated area [34].

Livestock monitoring systems can be integrated with IoT platforms to provide farmers with real-time data on their

animals. These systems can send alerts to the farmer's mobile phone or computer when there is a change in an animal's behavior or health status. Livestock monitoring systems can also be used to improve the overall efficiency of farming operations by reducing labor costs and increasing productivity. The use of Internet of Things in various fields of agriculture is shown in Figure 9.7.

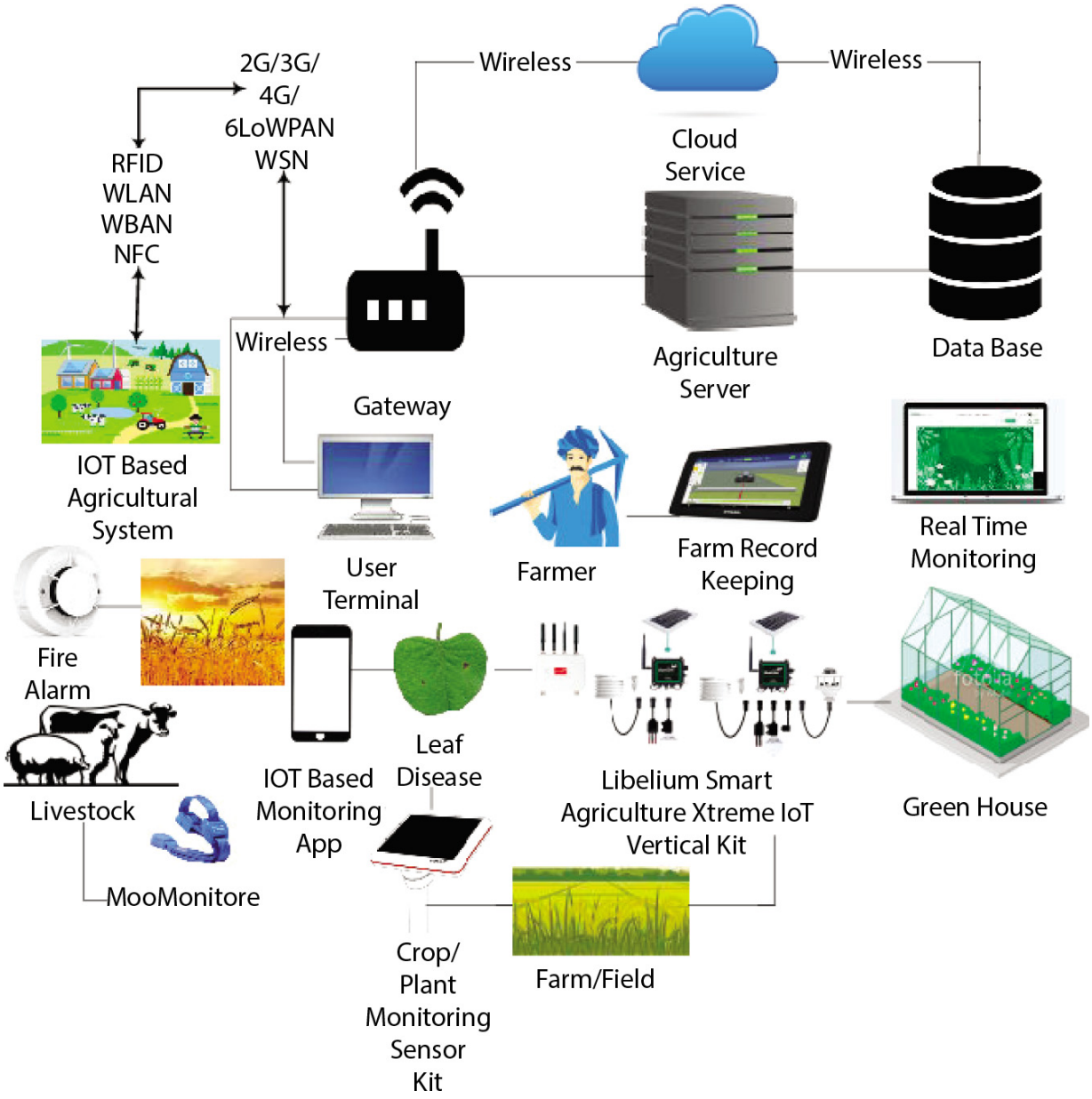

Figure 9.7 Application of IoT in different fields of agriculture (source: Farooq *et al.* [14]).

9.3.4 Agricultural Solutions by Using Smartphones

Smartphones have become an essential tool for farmers in managing their farming operations. There are several smartphone applications available that can provide farmers

with real-time data on their crops and livestock. These applications can also be used to manage farm operations, such as scheduling tasks, tracking expenses, and monitoring weather conditions. In Figure 9.2 some mobile apps used for smart agriculture are highlighted. Different kinds of mobile apps used in agriculture are presented in Figure 9.8.

Smartphone Agricultural Applications

Agriculture Information Management Apps	Farm Manager, Agriapp, Bayer TurafXpert, Irrigo-crops, agriculture irrigation and cost, Fertilizer removal by crop, Viewpoint FieldView, OneSoil Scouting-Satellite Field Monitoring.
Agriculture Calculator Apps	Blend Calculator, Fertilizer Removal by crop, Farm Calculators, Simple NPK Calculator, Corn N Rate Calculator, N Price Calculator, Fertilizer Calculator n2f, Fertilizer Mixing Calculators, Blend Calculator, Fertilizer Cost Calculator, Illinois Manure Calculator.
News and Weather Information Apps	Crop Monitor, AgWeb News & Markets, Farm Futures, Agri Zone: All in oneagri app, RealAg Markets, Ag Mobile, Farm Journal Television, Brownfield mobile 2.0, Argus Publications, Gavilon Grain, Commodity Classic 2019, Farm Progress, Farming Solution, CVA Coop, FarmWeekNow, The Weather Channel, AccuWeather, Weather Underground, Weather by Weather-Bug Real Time Forcast & Alert.
Tank Mix Apps	Tank Mixing Ag app. TankMix from DuPont, OnMRK, Agronote Farming Expenses.
Record Keeping	Pest Detector, AgDNA Prime, Farm At Hand.
Soil Sampling	Soil Sampler, Ag PhD Soils, SOILapp, Civil Tests Mini, Soil Sampling with GPS.
Precision Ag Apps	Explorer for ArcGIS, Field NET Mobile from Lindsay, Manna Irrigation Intelligence, SoilInfo app, Farm Progress Growing Degree Days, From the Field from Pioneer, Seeds Planting, Ag PhD Soybean Diseases.
Area Apps	Field navigator, GPS Fields Area Measure, Planmeter-GPS Area Measure Land Survey on Map.
Commodity Pricing Apps	Farm Futures, FarmPartner, Crop prices.

Figure 9.8 Mobile apps used for smart agriculture (source: Farooq *et al.* [14]).

9.3.5 IoT Devices and Sensors in Agriculture

Everything needs to be automatic in today's contemporary society in order to use less manpower and time. A sensor is a type of gadget that may meet this condition by receiving the same input from the physical world as is present and responding. Sensing devices can be programmed by the user to carry out certain tasks without the need for human intervention. Major IoT sensors include motion detectors, PIRs, soil moisture sensors, temperature and humidity sensors, barometric pressure sensors, ultraviolet (UV) sensors, pH sensors, and gas sensors.

9.4 Networks for IoT in Agriculture

Networks serve as the backbone of IoT systems in agriculture, allowing sensors and other devices to communicate with each other and with the cloud. In this section, we will discuss various network technologies used in IoT agriculture, their suitability for different applications, and network-related issues that need to be addressed.

9.4.1 Wireless Networks

Wireless networks are a popular choice for IoT agriculture systems due to their flexibility, scalability, and cost-effectiveness. These networks use radio waves to transmit data between devices and a central gateway, which then sends the data to the cloud for storage and analysis. Some examples of wireless network technologies used in IoT agriculture include:

Wi-Fi

Wi-Fi networks are commonly used in indoor environments such as greenhouses, where there is reliable power and internet connectivity. Wi-Fi allows for high-speed data transfer and is well-suited for streaming video and audio data from cameras and microphones. The details related to Wi-Fi are already described in earlier section.

Zigbee

Zigbee is a low-power wireless network technology that is ideal for connecting small, battery-powered sensors and devices. The details about Zigbee are described in earlier section.

LoRaWAN

LoRaWAN is a low-power, long-range wireless network technology that is ideal for applications that require low data rates and extended range, such as asset tracking and soil moisture monitoring. The details about LoRaWAN are described in earlier section.

9.4.2 Cellular Networks

Cellular networks are another popular choice for IoT agriculture systems, especially in areas where wireless networks are not available or reliable. Cellular networks use radio waves to transmit data between devices and a central server, which then sends the data to the cloud. Some examples of cellular network technologies used in IoT agriculture include:

2G/3G/4G LTE

2G, 3G, and 4G LTE are cellular network technologies that are widely used for IoT applications due to their reliability, coverage, and high data rates. These networks are ideal for applications such as precision farming, where real-time data transmission is critical [38].

9.4.3 Satellite Networks

Satellite networks are used in IoT agriculture systems where terrestrial networks are not available or reliable. These networks use satellites to transmit data between devices and a central server, which then sends the data to the cloud. Some examples of satellite network technologies used in IoT agriculture include:

Globalstar

Globalstar is a satellite network that provides low-cost, low-data-rate communication for IoT applications. Globalstar is ideal for applications such as asset tracking and remote monitoring of agricultural equipment.

Iridium

Iridium is a satellite network that provides global coverage and low-latency communication for IoT applications. Iridium is well-suited for applications such as precision farming and livestock monitoring.

9.4.4 Network-Related Issues

When it comes to IoT systems in agriculture, network-related issues such as coverage, capacity, and latency can have a

significant impact on the system's overall performance.

Network Coverage

One of the most significant challenges facing IoT agriculture systems is ensuring adequate network coverage. In rural areas, network coverage can be spotty, which can result in connectivity issues and data loss. This is particularly true for wireless networks, as their range is limited by their transmission power and the terrain. In order to address this issue, some IoT agriculture systems use satellite networks, which provide wider coverage and are not limited by terrain or distance. However, satellite networks are more expensive than wireless or cellular networks, and they also tend to have higher latency, which can affect real-time data transmission.

Network Capacity

IoT agriculture systems generate large amounts of data, which can put a strain on network capacity. This is particularly true for systems that rely on wireless or cellular networks, which have limited bandwidth. In order to address this issue, some IoT agriculture systems use low-power wide-area networks (LPWANs), which are specifically designed to handle low-bandwidth data transmissions over long distances. LPWANs are

less expensive than cellular networks, but they are also slower and have lower data rates.

Network Latency

Latency is the amount of time it takes for data to be transmitted from the sensor to the cloud and back. In IoT agriculture systems, latency can be a critical factor, particularly in real-time applications such as irrigation systems or weather monitoring. Network latency can be influenced by a variety of factors, including network coverage, network capacity, and network congestion. In order to reduce latency, some IoT agriculture systems use narrowband Internet of Things (NB-IoT), which is a low-power WAN technology designed to provide low-latency connectivity for IoT devices.

9.4.5 New Network Technologies

As IoT agriculture systems become more complex and generate more data, there is a need for new network technologies that can handle the increased load. In recent years, several new network technologies have emerged that are specifically designed for IoT applications, including narrowband IoT (NB-IoT), low-power wide-area networks (LPWANs), and 5G networks.

Narrowband IoT (NB-IoT)

Narrowband IoT is a low-power wide-area network technology developed by Huawei and it is specifically designed for IoT devices [39]. It operates on a narrowband frequency, which enables it to provide better coverage and lower power consumption than traditional cellular networks. NB-IoT is ideal for IoT agriculture systems that require low-latency connectivity and low power consumption.

Low-Power Wide-Area Networks (LPWANs)

LPWANs are a type of wireless network that is specifically designed to handle low-bandwidth data transmissions over long distances. LPWANs use a variety of technologies, including LoRaWAN and Sigfox, to provide lowpower, long-range connectivity for IoT devices. LPWANs are ideal for IoT agriculture systems that require long-range connectivity and low power consumption.

5G Networks

5G networks are the latest generation of cellular networks, and they are designed to provide high-speed connectivity for a wide range of applications, including IoT devices. 5G networks offer significantly higher data rates than previous generations of

cellular networks, which makes them ideal for IoT agriculture systems that require high-bandwidth data transmissions.

Therefore, the choice of network technology depends on the specific IoT application and its requirements—for example, if real-time data transfer is necessary, cellular networks may be more suitable, while if low-power, long-range connectivity is required, LPWANs or NB-IoT may be preferred. Additionally, the cost of deploying and maintaining the network must also be considered.

One of the main challenges in network deployment for IoT agriculture is network coverage. Many farms are located in rural areas, where network coverage may be limited or non-existent. This can be addressed by utilizing network extenders or repeaters to increase coverage, or by using satellite networks. Another solution is to use mesh networks, where nodes or devices within the network act as relays to transmit data between devices, thereby increasing the range and coverage of the network.

Network capacity is another important issue in IoT agriculture, especially when dealing with large amounts of data from multiple devices. This can be addressed by utilizing edge computing, which involves processing data locally on the device

or at a nearby edge device, rather than transmitting all data to a centralized cloud server. This reduces the amount of data transmitted over the network and improves overall system efficiency.

Latency is also an important consideration in IoT agriculture, especially in applications that require real-time data transfer—for example, in precision agriculture, real-time data on soil moisture levels, temperature, and humidity is critical for making timely decisions about irrigation and fertilization. To minimize latency, it is important to choose a network technology with low latency and to design the system with latency in mind.

New network technologies, such as NB-IoT, LPWANs, and 5G networks, are rapidly emerging and hold promise for IoT agriculture. It provides long-range connectivity with low power consumption. LPWANs, such as LoRa and Sigfox, also provide low-power, long-range connectivity, but at slower data rates. 5G networks offer high-speed, low-latency connectivity, which could be particularly useful for real-time applications such as precision agriculture and livestock monitoring.

In conclusion, network technologies are critical components of IoT systems in agriculture, and the choice of network

technology depends on the specific IoT application and its requirements. Network-related issues, such as coverage, capacity, and latency, must be carefully considered when designing and deploying IoT systems in agriculture. New network technologies, such as NB-IoT, LPWANs, and 5G networks, hold promise for addressing some of these issues and enabling new IoT applications in agriculture.

9.5 Communication Protocols Used in Agriculture

Using a variety of sensors and devices, the Internet of Things, or IoT, in agriculture collects and transmits data regarding soil water content, temperature, humidity, and various other environmental factors. However, for these devices to work together efficiently and effectively, they need to communicate with one another using specific communication protocols. Communication protocols are the set of rules that govern the transmission of data between IoT devices and systems. In this section, we will examine some of the communication protocols that are commonly used in IoT agriculture.

Bluetooth

Bluetooth is a short-range wireless communication protocol that is widely used in IoT devices for its minimum energy consumption and low cost. It is commonly used for devices that are located in close proximity to one another, such as those in a smart irrigation system. Bluetooth is also easy to configure and can be used to create a mesh network, which is a network of devices that are interconnected and can relay data to one another [40].

One of the benefits of Bluetooth in IoT agriculture is that it enables real-time data transmission between devices—for example, a farmer can use a Bluetooth-enabled sensor to monitor the soil moisture levels in their field and receive the data in real-time on their smartphone or tablet. However, Bluetooth has a limited range of around 10 meters, which makes it unsuitable for larger farms.

Wi-Fi

Wi-Fi is a popular communication protocol that is widely used in homes and offices. It is also used in IoT agriculture to connect sensors and devices to the internet. Wi-Fi offers high-speed data transmission and a relatively long range of up to 50 meters

indoors and up to 500 meters outdoors. It also provides a high level of security through its encryption protocols.

Wi-Fi can be used to connect sensors and devices to a central hub, such as a gateway or a cloud-based platform. This enables farmers to monitor their crops and livestock remotely and receive alerts in real-time. However, Wi-Fi can be expensive to implement on larger farms, and it requires a stable power supply [35].

ZigBee

It is a low-power based wireless communication protocol that is commonly used in IoT agriculture. It offers a range of up to 100 meters and can be used to create a mesh network of devices. ZigBee is particularly useful for sensors and devices that require low power consumption, such as those that monitor soil moisture or temperature.

One of the benefits of ZigBee is that it provides a high level of security through its encryption protocols. It is also easy to configure and can be used to create a self-healing network, which means that if one device fails, the network can automatically reroute the data through other devices [36].

LoRaWAN

It is a long-range, low-power communication protocol that is commonly used in IoT agriculture. It has a range up to 10 km in rural areas and 2 km in urban areas. LoRaWAN is particularly useful for sensors and devices that are located in remote areas, such as those that monitor water quality or air pollution.

One of the benefits of LoRaWAN is that it offers low power consumption, which means that devices can run on batteries for several years. It also provides a high level of security through its encryption protocols. However, LoRaWAN has a limited data transmission rate, which makes it unsuitable for applications that require real-time data transmission [37].

MQTT

MQTT is a lightweight communication protocol that is widely used in IoT agriculture. It offers a small data footprint and is particularly useful for sensors and devices that generate a large amount of data, such as those that monitor weather conditions or crop growth. MQTT is also easy to configure and can be used to create a publish-subscribe model, which enables communication between many devices without requiring a dedicated connection between each device.

In an MQTT system, there are two types of entities: brokers and clients. The broker is the central hub that manages all the incoming and outgoing messages, while clients are the end devices that publish or subscribe to topics. A topic is a string that identifies a particular data stream, such as temperature readings from a sensor in a particular field.

When a client publishes a message to a topic, the broker receives the message and forwards it to all other clients that have subscribed to the same topic. This allows multiple clients to receive the same data without the need for point-to-point connections between each device.

MQTT is also useful for low-bandwidth or unreliable network connections, as it includes features such as message queuing, persistent storage, and reconnection. This allows devices to send and receive messages even when the network connection is weak or intermittent. Additionally, MQTT supports various quality of service (QoS) levels, which determine the reliability of message delivery.

Overall, MQTT is an efficient and reliable communication protocol that is widely used in IoT agriculture applications. Its small data footprint, publish-subscribe model, and support for low-bandwidth networks make it ideal for use in agricultural

environments where data collection and transmission can be challenging. Figure 9.9 shows a diagram of an MQTT system in agriculture, where temperature sensors publish data to an MQTT broker, which forwards the data to a data analytics platform for analysis and decision-making.

In this system, the MQTT broker acts as a central hub that receives data from weather sensors in different fields. The sensors publish data to different topics, such as temperature, humidity, or rainfall. The MQTT broker then forwards the data to a data analytics platform, which uses the data to provide insights into weather conditions in different fields. These insights can help farmers make informed decisions about crop management, such as irrigation, fertilization, or pesticide application.

In summary, MQTT is a lightweight and efficient communication protocol that is well-suited for IoT agriculture applications. It publish-subscribe model, support for low-bandwidth networks, and reliability features make it a popular choice for connecting sensors and devices in agricultural environments.

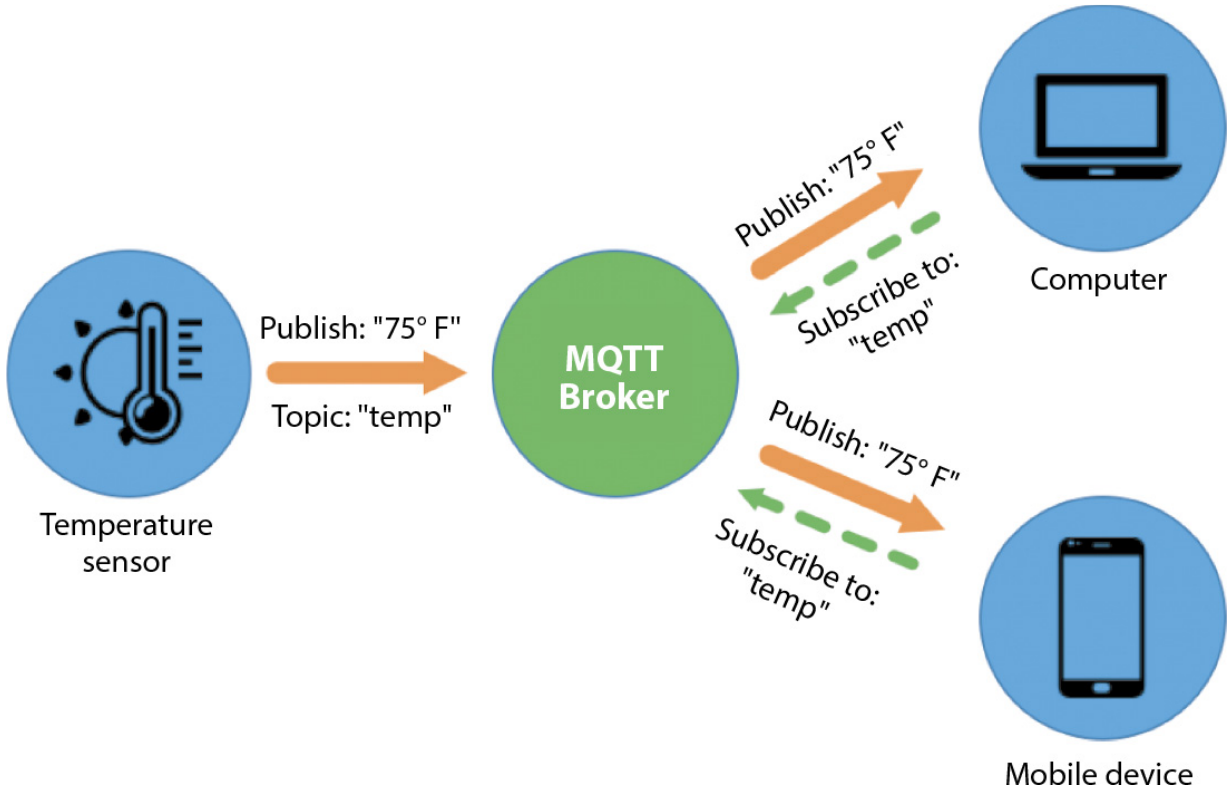

Figure 9.9 Diagram of an MQTT system in agriculture.

Thread

Thread is a wireless communication protocol that is designed for Internet of Things (IoT) enabled devices. It uses the IPv6 network protocol and is built on the IEEE 802.15.4 standard, making it suitable for connecting a large number of devices over a long range. Thread is particularly useful for agriculture applications that require a large number of sensors and devices to be connected, such as precision irrigation systems and soil moisture monitoring.

NB-IoT

NB-IoT is a device based on WAN technology and this device has very low power consumption. Details of NB-IOT are described in earlier section.

Sigfox

Another LPWAN technology created for Internet of Things applications is Sigfox. It operates on its own network and provides long-range, low-power connectivity for devices that transmit small amounts of data [41]. Sigfox is particularly useful for agricultural applications that require simple, low-cost connectivity, such as tracking assets or monitoring water levels. Generally, the farmer uses Sigfox to find the whereabouts of the livestock and boost output [42].

In summary, the selection of communication protocol in IoT agriculture depends on the specific application and requirements. It is important to consider factors such as data transmission rate, energy consumption, compatibility, range, and cost when choosing a protocol. MQTT, Thread, NB-IoT, and Sigfox are just a few examples of the many communication protocols available for IoT agriculture.

9.6 Case Studies and Examples of IoT Applications in Agriculture

In this section, we will explore some examples of how IoT technology is being used in agriculture to solve problems, increase efficiency, and improve outcomes. We have done a case study of 6 countries which is presented below.

India

India aims to improve its agricultural field using IoT policies. These policies monitor the soil conditions, temperature, and pests to alert the farmers. In order to use IoT to change the digital environment, the MeitY published a policy on it in 2015 [43].

Thailand

To improve agricultural output, Thailand's NECTEC (National Electronics and Computer Technology Centre) focuses on goods made from rubber, rice, sugar, and cassava [44]. An IoT-driven irrigation control system was created by Thammasat University to let growers pre-set the times for watering cycles. A pesticide aerial drone that can cover a maximum of 8 acres in a single fly is one of the main products produced by FarmD Asia, an effective farm service provider.

China

In 2016, China unveiled its 13th 5-year plan, which includes IoT integration for agricultural. Multiple goods, technologies, and applications were used in the initiative, which was first launched in eight provinces. In contrast to cellular networks, Huawei's NB-IoT App offers affordable agricultural solutions that address the problem of dispersed agricultural data [45].

Malaysia

The objectives of Malaysia's agricultural strategy were to increase crop growth and reduce poverty. The Mi-MSCANT PH sensor was created by the Malaysian Institute of Microelectronic Systems (MIMOS) to collect environmental data [46]. Additionally, MIMOS developed a framework for agriculture that uses IoT technology and forges connections between suppliers, dealers, and agricultural producers.

Philippine

The Philippines is gathering more data on various agricultural situations through remote sensing methods and satellite photos. The University of South-eastern Philippines (USeP) suggested employing IoT technology to monitor agricultural heat stress. The research team is developing a system that effectively

combats heat stress and enables farmers to maximize yields with a minimum of inputs [47].

USA

To meet the demands for food and energy, the USA has invested millions of dollars in developing new agricultural technology [48]. Sensing technologies are being developed by the National Institute of Food and Agriculture for use in smart farming techniques. The US Department of Agriculture (USDA) is also engaged in a project to tackle problems with water management and provide fresh approaches.

9.7 The Future of IoT in Agriculture

The area of IoT in agriculture is constantly evolving, with new technologies and trends emerging that are transforming the way farming is done. This section focuses on the future of IoT in agriculture, exploring some of the emerging technologies and trends that will shape this field's future.

9.7.1 Emerging Technologies

Artificial Intelligence

AI has the power to make a revolution in agriculture sector. By using machine learning algorithms [61], AI systems can analyze

numerous data collected by IoT devices and provide insights and recommendations to farmers—for example, AI algorithms can help farmers to take better decision regarding farming.

Artificial intelligence (AI)-based systems [54–59] can be utilized to develop methods of farming that are more efficient and sustainable. By analyzing data on soil composition, water usage, and other factors, AI systems can assist farmers in maximizing resource usage, minimizing waste, and minimizing environmental effects.

Edge Computing

A decentralized computing approach known as edge computing involves putting computation and data storage closer to the gadgets and sensors that generate it. This approach reduces the latency and bandwidth requirements of IoT systems, making them more efficient and responsive. In agriculture, edge computing can be used to process data from sensors in real time, allowing farmers to make quick decisions about crop management and irrigation.

Edge computing can also enhance data security and privacy in IoT systems. By processing data locally, edge computing reduces the need for data to be transmitted over the internet, minimizing the risk of interception and unauthorized access.

Blockchain

Blockchain works as a distributed database technology, permitting transactions that are safe, transparent, and tamper-proof. In agriculture, blockchain can be used to create a transparent and trustworthy supply chain, tracking the movement of goods from farm to consumer. By recording the origin, quality, and other relevant information of agricultural products on a blockchain, consumers can be assured of their authenticity and quality.

Blockchain can also be used to streamline payments and reduce transaction costs in agricultural supply chains. By eliminating intermediaries and automating payment processes, blockchain can make supply chains more efficient and cost-effective.

Digital Twins

Digital twins are virtual models of physical systems that can be used to simulate and optimize their performance. In agriculture, digital twins can be used to model and simulate the growth and development of crops, allowing farmers to make planting decisions that are well-informed.

Digital twins can be used to optimize resource allocation in agricultural systems. By modelling the effects of different

variables such as weather patterns and soil conditions, digital twins can help farmers optimize their use of resources such as water and fertilizer.

9.7.2 Ethical and Societal Implications

The implementation of Internet of Things or IoT in agriculture has significant ethical and societal implications that must be considered. One of the big concerns is its impact on employment. As IoT systems become more sophisticated, they may replace human labor in some farming activities, leading to job losses in the agricultural sector.

Another concern is the environmental impact of IoT in agriculture. While IoT systems can help farmers optimize their use of resources, they also require energy to operate and may contribute to the production of electronic waste. The long-term sustainability of IoT systems in agriculture must be carefully considered to ensure that they do not exacerbate environmental problems.

Finally, there are concerns about social justice and equity in the adoption of IoT in agriculture. Small-scale farmers and those in underdeveloped countries may not have the provisions or infrastructure to adopt IoT technologies, creating a digital

divide between farmers who have access to these technologies and those who do not.

9.8 Conclusion

The deployment of IoT systems in agriculture has the potential to revolutionize the way we produce, manage, and distribute food. IoT technologies enable farmers and other stakeholders to monitor their operations more closely, collect valuable data, and optimize their decision-making processes. By connecting devices and systems, IoT in agriculture enables stakeholders to gather real-time information, automate processes, and reduce costs, leading to increased efficiency, productivity, and profitability.

While the benefits of IoT in agriculture are clear, several challenges remain. Data privacy and security, connectivity, interoperability, and standardization are among the issues that must be addressed to ensure the widespread adoption of IoT in agriculture. Moreover, IoT technologies must be accessible and affordable to small farmers and marginalized communities to prevent the creation of a digital divide in the agricultural sector.

The future of IoT in agriculture is bright, with emerging technologies such as AI, edge computing, blockchain, and digital

twins offering new opportunities for farmers to optimize their operations further. AI and edge computing can help farmers to process large amounts of data faster and make more informed decisions, while blockchain can enable secure and transparent supply chains. Digital twins, which are virtual replicas of physical objects or systems, can allow farmers to simulate and optimize their operations before implementing them in the real world.

However, the deployment of these technologies must be accompanied by a consideration of their ethical and societal implications. The implementation of IoT enabled devices in agricultural farming must be done in a way that is sustainable, equitable, and inclusive. It must not exacerbate existing social and economic inequalities or harm the environment. Moreover, it must respect the rights and autonomy of farmers and other stakeholders and involve them in the decision-making process.

In conclusion, the future of IoT in agriculture is promising, with IoT technologies enabling stakeholders to collect data, automate processes, and optimize their decision-making processes. However, the deployment of IoT in agriculture must be done in a way that is responsible, sustainable, and inclusive, taking into account the interests and needs of all stakeholders. With proper planning and execution, IoT in agriculture can contribute to the

transformation of the agricultural sector and the creation of a more sustainable and equitable food system.

References

1. W. Li, T. Logenthiran, V. T. Phan and W. L. Woo. A novel smart energy theft system (SETS) for IoT-based smart home. *IEEE Internet Things J.*, vol. 6, no. 3, pp. 5531–5539, Jun. 2019, doi: 10.1109/JIOT.2019.2903281.
2. D. Shin, K. Yun, J. Kim, P. V. Astillo, J. N. Kim and I. You. A Security Protocol for Route Optimization in DMM-Based Smart Home IoT Networks. *IEEE Access*, vol. 7, pp. 142531–142550, 2019, doi: 10.1109/ACCESS.2019.2943929.
3. J. An et al. Toward global IoT-enabled smart cities interworking using adaptive semantic adapter. *IEEE Internet Things J.*, vol. 6, no. 3, pp. 5753–5765, Jun. 2019, doi: 10.1109/JIOT.2019.2905275.
4. F. Cirillo, D. Gómez, L. Diez, I. Elicegui Maestro, T. B. J. Gilbert and R. Akhavan. Smart city IoT services creation through large-scale collaboration. *IEEE Internet Things J.*, vol. 7, no. 6, pp. 5267–5275, Jun. 2020, doi: 10.1109/ JIOT.2020.2978770.
5. M. Ammad et al. A Novel Fog-Based Multi-Level Energy-Efficient Framework for IoT-Enabled Smart Environments. *IEEE Access* vol. 8, pp. 150010–150026, 2020, doi: 10.1109/ACCESS.2020.3010157.

6. C. K. Metallidou, K. E. Psannis and E. A. Egyptiadou. Energy Efficiency in Smart Buildings: IoT Approaches. *IEEE Access*, vol. 8, pp. 63679–63699, 2020, doi: 10.1109/ACCESS.2020.2984461.
7. F. Kiani et al. Adaptive Metaheuristic-Based Methods for Autonomous Robot Path Planning: Sustainable Agricultural Applications. *Appl. Sci. (Switzerland)*, vol. 12, no. 3, Feb. 2022, doi: 10.3390/app12030943.
8. V. K. Quy, V. H. Nam, D. M. Linh, N. T. Ban and N. D. Han. Communication Solutions for Vehicle Ad-hoc Network in Smart Cities Environment: A Comprehensive Survey. *Wirel. Pers. Commun.*, vol. 122, no. 3. Springer, pp. 2791–2815, Feb. 01, 2022. doi: 10.1007/s11277-021-09030-w.
9. K. S. Patle, R. Saini, A. Kumar and V. S. Palaparthy. Field Evaluation of Smart Sensor System for Plant Disease Prediction Using LSTM Network., *IEEE Sens. J.*, vol. 22, no. 4, pp. 3715–3725, Feb. 2022, doi: 10.1109/JSEN.2021.3139988.
10. A. Vangala, A. K. Das, N. Kumar and M. Alazab. Smart Secure Sensing for IoT-Based Agriculture: Blockchain Perspective. *IEEE Sens. J.*, vol. 21, no. 16, pp. 17591–17607, Aug. 2021, doi: 10.1109/JSEN.2020.3012294.
11. B. Citoni, F. Fioranelli, M. A. Imran and Q. H. Abbasi. Internet of Things and LoRaWAN-Enabled Future Smart Farming.

IEEE Internet Things Mag., vol. 2, no. 4, pp. 14–19, Feb. 2020, doi: 10.1109/iotm.0001.1900043.

12. R. Kumar, R. Mishra, H. P. Gupta and T. Dutta. Smart Sensing for Agriculture: Applications, Advancements, and Challenges. *IEEE Consum. Electron. Mag.*, vol. 10, no. 4, pp. 51–56, Jul. 2021, doi: 10.1109/MCE.2021.3049623.
13. Y. C. Chang and Y. H. Lai. Campus Edge Computing Network Based on IoT Street Lighting Nodes. *IEEE Syst. J.*, vol. 14, no. 1, pp. 164–171, Mar. 2020, doi: 10.1109/JSYST.2018.2873430.
14. M. S. Farooq, S. Riaz, A. Abid, K. Abid and M. A. Naeem. A Survey on the Role of IoT in Agriculture for the Implementation of Smart Farming. *IEEE Access*, vol. 7. Institute of Electrical and Electronics Engineers Inc., pp. 156237–156271, 2019. doi: 10.1109/ACCESS.2019.2949703.
15. E. Said Mohamed, A. A. Belal, S. Kotb Abd-Elmabod, M. A. El-Shirbeny, A. Gad and M. B. Zahran. Smart farming for improving agricultural management. *Egypt. J. Remote Sens. Space Sci.*, vol. 24, no. 3. Elsevier B.V., pp. 971–981, Dec. 01, 2021. doi: 10.1016/j.ejrs.2021.08.007.
16. A. Khanna and S. Kaur. Evolution of Internet of Things (IoT) and its significant impact in the field of Precision Agriculture. *Comput. Electron. Agric.*, vol. 157. Elsevier B.V., pp. 218–231, Feb. 01, 2019. doi: 10.1016/j. compag.2018.12.039.

17. Popović, T., Latinović, N., Pešić, A., Zečević, Ž., Krstajić, B., & Djukanović, S. Architecting an IoT-enabled platform for precision agriculture and ecological monitoring: A case study. *Comput. Electron. Agric.*, 140, 255-265, 2017.
18. Institute of Electrical and Electronics Engineers, *2016 13th Annual IEEE International Conference on Sensing, Communication, and Networking (SECON)*: 27-30 June 2016, London, UK.
19. J. M. Talavera et al. Review of IoT applications in agro-industrial and environmental fields. *Comput. Electron. Agric.*, vol. 142. Elsevier B.V., pp. 283– 297, Nov. 01, 2017. doi: 10.1016/j.compag.2017.09.015.
20. Morais, C. M. D., Sadok, D., & Kelner, J. An IoT sensor and scenario survey for data researchers. *J. Braz. Comput. Soc.*, *25*, 1-17, 2019.
21. Bodake, K., Ghate, R., Doshi, H., Jadhav, P., & Tarle, B. Soil based fertilizer recommendation system using Internet of Things. *MVP J. Eng. Sci.*, *1*(1), 13-19, 2018.
22. X. Zhang, J. Zhang, L. Li, Y. Zhang and G. Yang. Monitoring citrus soil moisture and nutrients using an IoT based system. *Sensors (Switzerland)*, vol. 17, no. 3, Mar. 2017, doi: 10.3390/s17030447.
23. Giordano, S., Seitanidis, I., Ojo, M., Adami, D., & Vignoli, F. IoT solutions for crop protection against wild animal attacks.

In *2018 IEEE international conference on Environmental Engineering (EE)* (pp. 1-5). IEEE.

24. IEEE Education Society. Malaysia Chapter and Institute of Electrical and Electronics Engineers, *2017 International Conference on Electrical, Electronics and System Engineering (ICEESE): 9-10 Nov. 2017*.
25. T. H. F. Khan and D. S. Kumar. Ambient crop field monitoring for improving context based agricultural by mobile sink in WSN. *J. Ambient Intell. Hum. Comput.*, vol. 11, no. 4, pp. 1431–1439, Apr. 2020, doi: 10.1007/ s12652-019-01177-6.
26. Satyanarayana, G. V., & Mazaruddin, S. D. Wireless sensor based remote monitoring system for agriculture using ZigBee and GPS. In *Conference on Advances in Communication and Control Systems (CAC2S 2013)* (pp. 110-114). Atlantis Press.
27. Gardašević, G., Veletić, M., Maletić, N., Vasiljević, D., Radusinović, I., Tomović, S., & Radonjić, M. The IoT architectural framework, design issues and application domains. *Wirel. Pers. Commun.*, *92*, 127-148, 2017.
28. *Annual IEEE Computer Conference*, IEEE International Conference on Computer Science and Education 9 2014.08.22-24 Vancouver, and ICCSE 9 2014.08.22-24 Vancouver, *9th International Conference on Computer Science & Education (ICCSE),* 2014 22-24 Aug. 2014, Vancouver, Canada.

29. C. A. Gonzalez-Amarillo et al. An IoT-Based traceability system for greenhouse seedling crops. *IEEE Access*, vol. 6, pp. 67528–67535, 2018, doi: 10.1109/ACCESS.2018.2877293.
30. B. Windsperger et al. Greenhouse gas emissions due to national product consumption: from demand and research gaps to addressing key challenges. *Int. J. Environ. Sci. Technol.*, vol. 16, no. 2, pp. 1025–1038, Feb. 2019, doi: 10.1007/s13762-018-1743-6.
31. D. O. Shirsath, P. Kamble, R. Mane, A. Kolap and R. S. More. IoT based smart greenhouse automation using Arduino. *Int. J. Innov. Res. Comput. Sci. Technol.*, vol. 5, no. 2, pp. 234–238, 2017.
32. L. W. Turner, M. C. Udal, B. T. Larson and S. A. Shearer. Monitoring cattle behavior and pasture use with GPS and GIS. *Can. J. Animal Sci.*, vol. 80, no. 3, pp. 405–413, 2000.
33. X. Hi, X. An, Q. Zhao, H. Liu, L. Xia, X. Sun and Y. Guo. State-of-the-art Internet of Things in protected agriculture. *Sensors*, vol. 19, no. 8, p. 1833, 2019.
34. Saravanan, K., & Saraniya, S. Cloud IOT-based novel livestock monitoring and identification system using UID. *Sens. Rev.*, *38*(1), 21-33, 2017.
35. Y.-J. Lai, W.-H. Kuo, W.-T. Chiu and H.-Y. Wei, Accelerometer-assisted 802.11 rate adaptation on mobile WiFi access.

EURASIP J. Wirel. Commun. Netw., vol. 2012, no. 1, p. 246, 2012

36. Xiao-liang, C., & Zhi-dong, D. Construction of large-scale wireless sensor network using ZigBee specification. *J. Commun.*, *29*(11), 158-164, 2008.
37. J. Dias and A. Grilo, Multi-hop LoRaWAN uplink extension: Specification and prototype implementation. *J. Ambient Intell. Hum. Comput.*, pp. 1–15, Feb. 2019
38. X. Feng, F. Yan and X. Liu, Study of wireless communication technologies on Internet of Things for precision agriculture. *Wirel. Pers. Commun.*, vol. 108, no. 3, pp. 1785–1802, 2019
39. NB-IoT Apps Enable Agricultural Digitalization. Accessed: Jun. 18, 2019. [Online]. Available:https://e.huawei.com/en/publications/global/ict_insights/201806041630/ecosystem/201808170841?source=corp_comm
40. L. Ruiz-Garcia, L. Lunadei, P. Barreiro and I. Robla, A review of wireless sensor technologies and applications in agriculture and food industry: State of the art and current trends., *Sensors*, vol. 9, no. 6, p. 4728, 2009.4750
41. A. Pitì, G. Verticale, C. Rottondi, A. Capone and L. L. Schiavo, The role of smart meters in enabling real-time energy services for households: The italian case. *Energies*, vol. 10, no. 2, p. 199, 2017.

42. A. Llaria, G. Terrasson, H. Arregui and A. Hacala, Geolocation and monitoring platform for extensive farming in mountain pastures. in *Proc. IEEE Int. Conf. Ind. Technol. (ICIT)*, Mar. 2015, pp. 2420–2425.
43. IOT Policy in India. Accessed: Jun. 20, 2019. [Online]. Available: https://meity.gov.in/sites/upload_files/dit/files/Draft-IoT-Policy(1).pdf
44. Food and Agriculture. Accessed: Jun. 17, 2019. [Online]. Available: https://www.nectec.or.th/en/research/
45. NB-IoT Apps Enable Agricultural Digitalization. Accessed: Jun. 18, 2019. [Online]. Available:https://e.huawei.com/en/publications/global/ict_insights/201806041630/ecosystem/201808170841?source=corp_comm
46. IoT Blooms in Malaysian Agro-Sector. Accessed: Jun. 17, 2019. [Online]. Available: https://mit-insights.my/iot-blooms-in-malaysian-agro-sector/
47. Philippine Agricultural and Food Policies. Accessed: Jun. 20, 2019. [Online]. Available: https://www.ifpri.org/publication/philippineagricultural-and-food-policies-0
48. Developing Sensing Technologies for Smart Farming Practices in an Internet-of-AG-Things World. Accessed: Jun. 19, 2019. [Online]. Available:

https://reeis.usda.gov/web/crisprojectpages/1013254-developingsensing-technologies-for-smart-farming-practices-in-an-internet-of-agthings-world.html

49. Agriculture IoT Market by Hardware, Application (Precision Farming, Precision Forestry, Precision Live Stock, Precision Aquaculture, Smart Green House), Farm Size, Production Stage, and Geography (2021-26). Accessed: May 9, 20223. [Online]. Available: Agriculture IoT Market Share, Scope & Industry Analysis 2026 (marketsandmarkets.com)
50. IoT for agriculture to reduce water consumption in smart farms by 30%. Accessed: May 9, 2023. [Online]. Available: https://futureiot.tech/iot-for-agriculture-to-reduce-water-consumption-in-smart-farms-by-30/
51. Livestock monitoring market - growth, trends, covid-19 impact, and forecasts (2023 - 2028). Accessed: May 9, 2023. [Online]. Available: Livestock Monitoring Market Size & Share Analysis - Industry Research Report - Growth Trends (mordorintelligence.com)
52. Irrigation Automation Market Size, Share & Trends Analysis Report By Automation Type, By Component (Controllers, Sensors, Valves, Sprinklers), By Irrigation Type, By Application, And Segment Forecasts, 2020 – 2027. Accessed: May 9, 2023, [Online]. Available: Irrigation Automation

Market Size & Share Report, 2020-2027 (grandviewresearch.com)

53. S. Tenzin, S. Siyang, T. Pobkrut and T. Kerdcharoen, Low-cost weather station for climate-smart agriculture. *2017 9th International Conference on Knowledge and Smart Technology (KST)*, Chonburi, Thailand, 2017, pp. 172-177, doi: 10.1109/KST.2017.7886085.
54. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
55. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
56. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini Robotic Face Recognition System,” *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.
57. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.”

In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.

58. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
59. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
60. Kim, J.; Kim, S.; Ju, C.; Son, H.I. Unmanned Aerial Vehicles in Agriculture: A Review of Perspective of Platform, Control, and Applications. *IEEE Access*, 2019, 7, 105100–105115. [CrossRef]
61. Kumar, Amit, and Bikash Kanti Sarkar. A case study on machine learning and classification. *Int. J. Inf. Decis. Sci.*, 9, no. 2 (2017): 179-208.

Note

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10
Internet of Things in Agriculture: Sensor, Network and Communication Protocol

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Abstract

This chapter presents an overview of the Internet of Things (IoT), a transformative technology that is reshaping our interaction with the world. It covers essential aspects of IoT, including its physical network architecture, characteristics, enabling technologies, challenges, and system levels. The chapter also delves into sensor technology, sensor interfacing techniques, and various types of networks employed in IoT deployments. Communication protocols relevant to IoT and

their applications in agriculture are discussed as well. The introduction highlights the fundamental attributes of IoT, such as its capability to connect physical objects to the Internet, reliance on sensors and data collection devices, and utilization of data analytics for informed decision-making. Furthermore, the chapter explores the integral components of an IoT system, encompassing sensors, gateways, cloud servers, and user interfaces. At the end, this chapter aims to provide a comprehensive understanding of IoT and sensors that can be implemented in the field of agriculture.

***Keywords*:** IoT enabling technologies, communication protocols for IoT, sensor technology for agriculture, wireless sensor network

10.1 Introduction

The term Internet of Things (IoT) is a transformative evolution of the Internet that creates a global communication infrastructure, seamlessly connecting humans and machines. It is reshaping multiple facets of our daily lives, including healthcare services, manufacturing processes, agricultural practices, and mining operations. By facilitating the seamless integration of emerging artificial intelligence (AI) technologies, IoT is opening up new possibilities for advanced functionalities

and innovative solutions [1]. In this context, it becomes crucial to delve into the various aspects of IoT. This chapter provides a comprehensive exploration of its fundamental concept, brief history, features, communication protocols, and challenges where it finds applications especially toward precision agriculture. Moreover, it also addresses vital concerns related to IoT enabling technologies and sensor networks that offer insights into the functional perspective of IoT.

Let us see how Internet has accelerated the pace of technological development in recent history. A quick summary is presented below:

1990s:

- In the 1990s, the groundwork for IoT was laid with the advancement of technologies like the Internet and wireless communication.
- The concept of “ubiquitous computing” emerged, envisioning a world where computing devices seamlessly integrate into everyday objects and environments.
- The “World Wide Web” has come into the picture and about 3 lakhs computers are connected to the Internet and the foundation for interconnected devices and data exchange was being established.

2000s:

- Kevin Ashton coined the term “Internet of Things” in 1999, and the concept gained recognition and attention in the early 2000s.
- The 2000s witnessed significant advancements in RFID (radio frequency identification) technology, which became a key enabler of IoT.
- The Auto-ID Center, founded in 1999, played a crucial role in developing RFID technology and establishing the technical framework for IoT.

2010s

- The 2010s marked the explosive growth of IoT, driven by advancements in connectivity, sensors, and cloud computing.
- The number of connected devices soared, encompassing consumer applications like smart homes, wearables, and connected cars.
- Industrial IoT (IIoT) gained prominence, revolutionizing sectors like manufacturing, logistics, and energy with improved efficiency and automation.
- Standards and protocols, such as MQTT and IPv6, emerged to facilitate communication and interoperability among IoT devices.

2020s

- The 2020s continue to witness the proliferation of IoT, with exponential growth in the number of connected devices and the expansion of IoT applications across industries.
- Edge computing and AI have gained prominence, enabling real-time data processing

and analysis at the edge of the network, closer to the data source.
- Security concerns around IoT have become increasingly prominent, leading to efforts to address vulnerabilities and establish robust security frameworks [2].
- 5G technology has emerged, promising faster and more reliable connectivity, which will further accelerate the growth and capabilities of IoT.

Overall, the history of IoT has evolved from the initial conceptualization of interconnected devices to the widespread integration of IoT technology in various domains. Each decade has seen significant advancements in connectivity, sensor technology, data analytics, and security, paving the way for the IoT-driven world we live in today. The future of IoT holds even greater potential as technology continues to advance and innovative use cases emerge.

10.2 IoT in Agriculture

IoT has tremendous applications, as illustrated in Figure 10.1. However, in this section, we discuss the application in

agriculture. In the field of agriculture, IoT is employed at various stages of the production chain, offering valuable solutions. It facilitates the continuous monitoring of crop conditions, soil moisture levels, and environmental factors. This data enables farmers to make informed decisions about irrigation, fertilization, and pest control. By leveraging IoT in agriculture, farmers can enhance productivity, optimize resource utilization, and minimize environmental impact [3]. Here are some key applications of IoT in agriculture:

Precision farming: IoT sensors and monitoring systems can be used to collect real-time data on soil moisture level, temperature of soil, humidity throughout the day, and nutrient levels. This information allows farmers to optimize irrigation, fertilization, and crop protection, leading to improved crop yields, reduced resource wastage, and cost savings.

Livestock monitoring: IoT devices can track and monitor the health, location, and behavior of livestock. Sensors embedded in wearable devices or smart collars can provide real-time data on factors like body temperature, heart rate, and feeding patterns. This helps farmers detect diseases, monitor animal well-being, and optimize feeding and breeding practices [4].

Smart greenhouse: IoT sensors and automation systems can be used to create smart greenhouses. Sensors measure parameters such as temperature, moisture content, light intensity, and various gas levels, allowing for precise control of environmental conditions for crops [5].

Crop monitoring and management: IoT enables remote monitoring of crops, even across large agricultural fields. Drones equipped with cameras and sensors can capture images and collect data on crop health, growth rates, and pest infestations [6, 7].

Supply chain optimization: IoT can be used to track and monitor the movement of agricultural products throughout the supply chain. Sensors and RFID tags can provide real-time information on location, temperature, and humidity, ensuring proper storage conditions and reducing spoilage [8]. This improves logistics, traceability, and quality control, enhancing overall supply chain efficiency.

Automated irrigation: IoT-based irrigation systems can monitor soil moisture levels and weather conditions to determine precise irrigation requirements. Automated valves and sprinklers can be controlled remotely, ensuring optimal water usage and reducing water waste. This not only conserves water

resources but also improves crop health and minimizes the risk of overwatering or underwatering [9].

Pest and disease management: IoT devices can detect and identify pests and diseases early on, allowing for timely interventions. Wireless insect traps, smart traps, and image recognition systems can monitor pest populations and provide alerts to farmers [10, 11].

Farm equipment management: IoT can improve the management and maintenance of farm equipment. Sensors and connectivity solutions can track the usage, location, and performance of machinery, ensuring timely maintenance and optimizing operational efficiency.

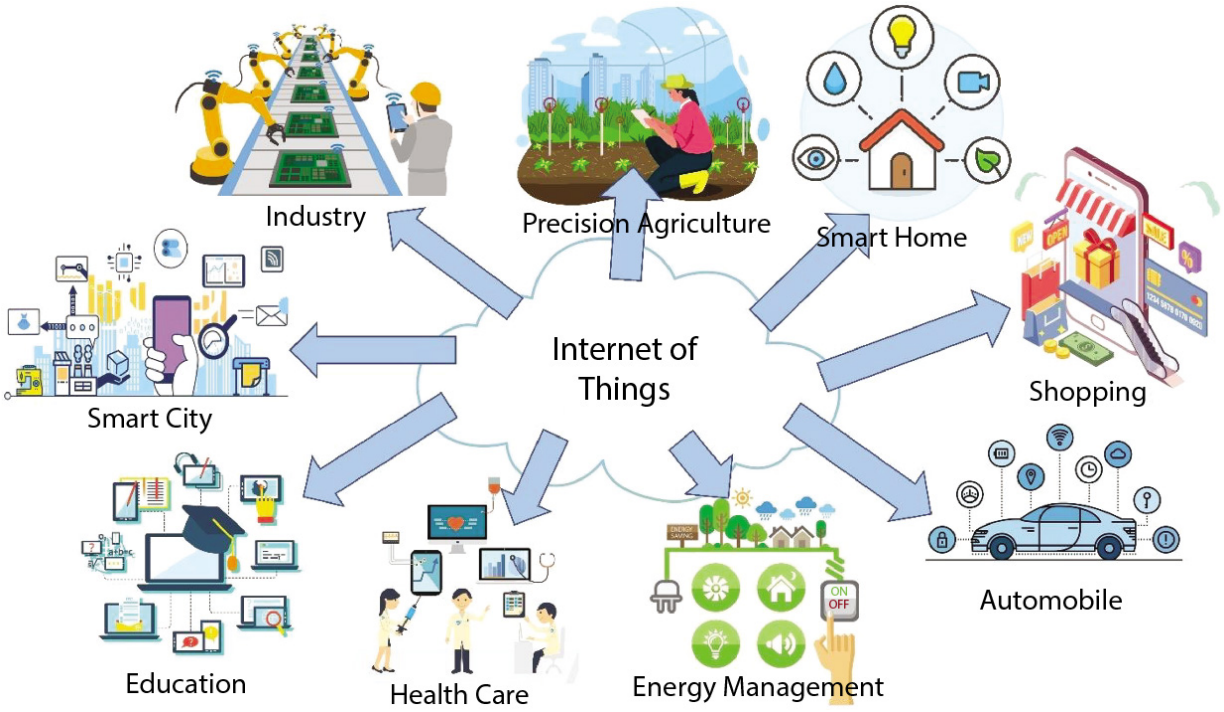

Figure 10.1 Application of Internet of Things in the modern world.

Weather monitoring and forecasting: IoT weather stations can collect real-time data on temperature, humidity, wind speed, and rainfall. Combined with advanced analytics and machine learning, this information can be used to generate accurate weather forecasts for specific farm locations. Farmers can make informed decisions about planting, harvesting, and other farm activities based on reliable weather predictions.

Agricultural analytics: IoT-generated data can be analyzed to gain valuable insights and optimize farming practices. Advanced analytics, combined with AI and machine learning algorithms, can process large datasets to provide actionable

recommendations on crop rotation, pest control strategies, resource allocation, and yield optimization [12].

10.3 Physical Design of IoT Architecture

The physical design of an IoT system refers to the arrangement and deployment of physical components, devices, and infrastructure required to enable communication, data collection, and interaction within the IoT ecosystem [13]. Here are some key aspects of the physical design of IoT. Figure 10.2 shows the scenario that expresses the physical architecture of IoT.

10.3.1 IoT Devices

Sensors and actuators: These are the fundamental components that interact with the physical world, collecting data (e.g., temperature, humidity, motion) or performing actions based on commands received from the network.

Embedded systems: Many IoT devices consist of embedded systems with microcontrollers or specialized chips that handle data processing, communication protocols, and device management.

10.3.2 Connectivity

Wireless technologies: IoT devices typically communicate wirelessly using various protocols like Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRaWAN, or cellular technologies (e.g., 3G, 4G, 5G) depending on the range, power consumption, and data requirements [14].

Wired connections: In some cases, IoT devices may utilize wired connections, such as Ethernet or Power over Ethernet (PoE), for higher data rates or when power supply is readily available.

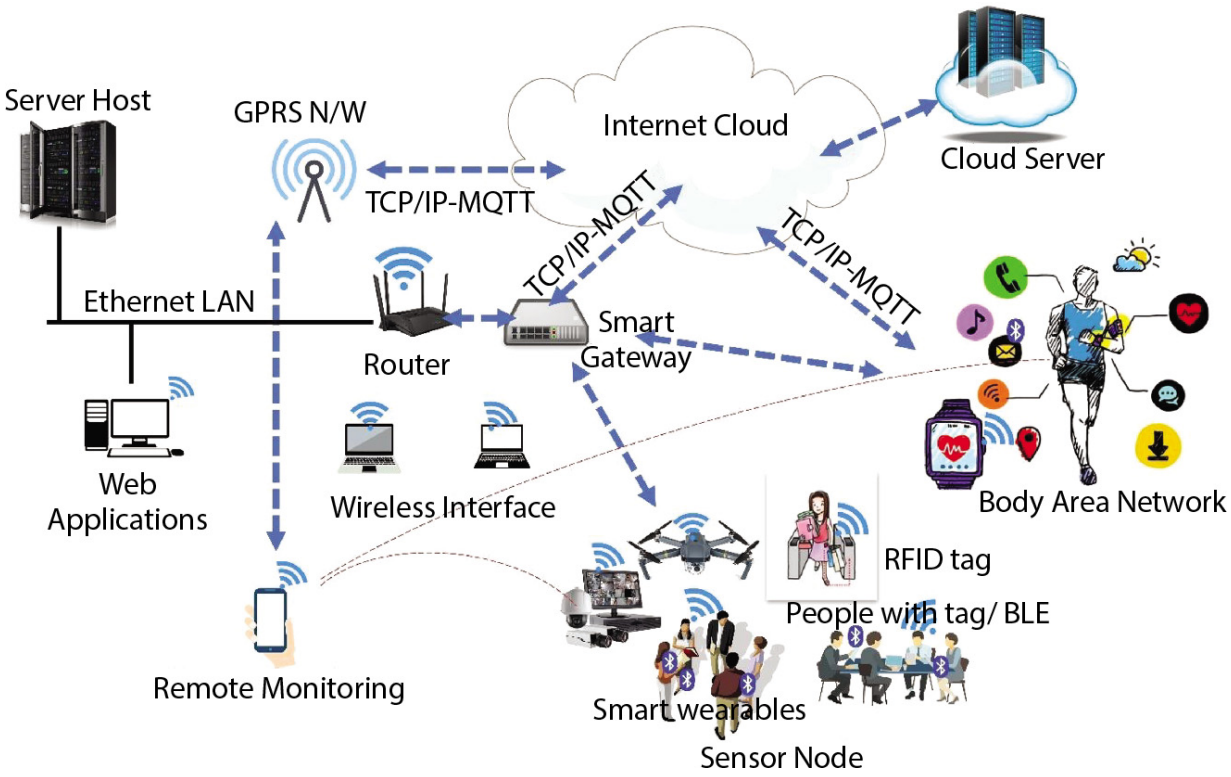

Figure 10.2 Physical design architecture of Internet of Things.

10.3.3 Gateways

Edge gateways: These devices serve as intermediaries between IoT devices and the network. They aggregate data from multiple devices, perform initial processing, and transmit it to the cloud or other parts of the network. Cloud gateways: Cloud gateways facilitate secure communication between edge devices and cloud platforms, enabling seamless data transfer, device management, and remote control.

10.3.4 Network Infrastructure

Local area network (LAN): IoT devices and gateways are connected through a LAN, which may include routers, switches, and access points to establish a reliable and secure network within a limited area.

Wide area network (WAN): For larger-scale deployments, IoT devices may utilize WAN connections to connect across different locations or communicate with cloud services. This can involve VPNs (virtual private networks), cellular networks, or satellite communication.

10.4 IoT Network Levels and Stages

Let us understand how sensor data goes through various stages to enable effective data collection, processing, and analysis in IoT system.

Sensing: Capturing data from the physical environment using sensors. Data acquisition and preprocessing: Converting raw sensor data into a usable format through acquisition and preprocessing techniques.

Communication: Transmitting the preprocessed data from IoT devices to gateways or central hubs securely.

Data storage and management: Storing the transmitted data in databases or cloud storage and ensuring data integrity and retrieval.

Data processing and analysis: Extracting meaningful insights from the collected data using algorithms and statistical methods.

Visualization and presentation: Presenting the analyzed data in a userfriendly format for easy comprehension and decision-making.

Decision-making and action: Making informed decisions and triggering actions based on the insights derived from the analyzed data.

These various stages in an IoT based system shown in Figure 10.3 represent a cyclical process, ensuring a continuous flow of data, analysis, and action. Each stage plays a critical role in harnessing the potential of IoT devices, enabling data-driven insights, and driving efficient decision-making and optimization.

Figure 10.3 Network-level and stages of IoT-based system.

10.5 Characteristics of IoT

The Internet of Things (IoT) is characterized by several distinct features that define its nature and capabilities. These characteristics collectively contribute to the transformative potential and widespread adoption of IoT in various domains. Let us explore these characteristics in a descriptive and comprehensive manner.

Connectivity: The IoT is built on the foundation of extensive connectivity, enabling seamless communication and data exchange between interconnected devices, sensors, and systems. It encompasses a diverse range of connectivity options such as wired, wireless, and cellular networks, facilitating real-time data transmission and interaction.

Interoperability: Interoperability is a fundamental characteristic of IoT, allowing different devices, platforms, and systems to seamlessly work together. Standardized protocols and interfaces enable interoperability, enabling devices from

various manufacturers to communicate and share data, fostering a collaborative and cohesive IoT ecosystem.

Sensor-driven: IoT relies on an extensive network of sensors that capture and collect data from the physical world. These sensors detect and measure a wide range of parameters such as temperature, humidity, pressure, motion, and more, providing real-time insights into the environment and enabling informed decision-making.

Data-driven: Data is at the core of IoT, with a massive volume of data generated by connected devices and sensors. The collection, analysis, and utilization of this data drive actionable insights, predictive analytics, and informed decision-making. The IoT ecosystem thrives on the ability to process and derive valuable information from the vast amounts of data produced.

Scalability: IoT exhibits exceptional scalability, allowing for the seamless addition and integration of a large number of devices and sensors into the network. It can accommodate the exponential growth of connected devices without compromising the overall performance, ensuring flexibility and adaptability to varying demands and expanding deployments.

Real-time processing: IoT systems are designed to process and analyze data in real-time or near real-time. This capability

enables immediate response, timely decision-making, and automated actions based on the incoming data. Real-time processing allows for proactive measures, predictive maintenance, and improved operational efficiency across multiple domains.

Automation and control: IoT empowers automation and control capabilities, enabling devices and systems to operate autonomously based on predefined rules, algorithms, or user-defined settings. Through remote monitoring, management, and control, IoT facilitates the automation of tasks, enhancing efficiency and reducing manual intervention.

Ubiquitous connectivity: IoT transcends traditional boundaries, creating a globally connected network that is pervasive and accessible across various environments. It facilitates ubiquitous connectivity, enabling devices to seamlessly communicate and exchange data regardless of geographical location, creating a seamless and integrated digital fabric.

Security and privacy: As the IoT ecosystem grows, ensuring robust security and privacy mechanisms becomes paramount. IoT systems must implement secure protocols, encryption, authentication, access controls, and privacy measures to protect

sensitive data, mitigate cyber threats, and safeguard user privacy.

Value creation: One of the most significant characteristics of IoT is its potential to generate substantial value across industries and sectors. By leveraging the data, insights, and automation capabilities, IoT enables new business models, improved efficiency, enhanced customer experiences, optimized resource management, and innovation in areas such as healthcare, transportation, manufacturing, agriculture, and smart cities [36].

These characteristics collectively make IoT a transformative and powerful technology paradigm, revolutionizing industries, enhancing decision-making, and enabling the creation of smart, connected ecosystems. With its inherent capabilities, IoT holds immense potential to drive innovation, economic growth, and societal advancements in the digital era.

10.6 IoT Enabling Technology

IoT enabling technologies are the foundational components that enable the connectivity, communication, and interaction between IoT devices and systems. These technologies provide the infrastructure and capabilities required for the

implementation and functioning of IoT solutions. Here are some key IoT enabling technologies.

10.6.1 Wireless Communication

Wireless communication technologies such as Wi-Fi (IEEE 802.11), Bluetooth (IEEE 802.15.1), Zigbee (IEEE 802.15.4), and cellular networks (e.g., LTE, 5G) employ specific protocols to enable IoT devices to establish connections and exchange data. Wi-Fi utilizes protocols like TCP/IP for reliable data transmission over local area networks (LANs). Bluetooth uses protocols like L2CAP, RFCOMM, and GATT to enable short-range communication between devices. Zigbee employs the Zigbee protocol stack, including the Zigbee application layer, network layer, and physical layer, for low-power, low-data-rate wireless communication in IoT networks [15]. Cellular networks utilize protocols like GSM, CDMA, or LTE for wide-area coverage and high-speed data transmission. These protocols define how data is packaged, transmitted, and received, ensuring efficient, secure, and standardized communication between IoT devices and other components in the network [16].

10.6.2 Cloud Computing

Cloud computing in IoT relies on various network protocols to facilitate communication between IoT devices and cloud

platforms. Protocols like HTTP (Hypertext Transfer Protocol) enable IoT devices to send requests and receive responses from cloud-based services. MQTT (Message Queuing Telemetry Transport) is a lightweight publish-subscribe protocol commonly used for IoT device-to-cloud and device-to-device communication. CoAP (Constrained Application Protocol) is a protocol designed for resource-constrained IoT devices and supports efficient data transfer over UDP (User Datagram Protocol). These protocols provide reliable, secure, and efficient data exchange between IoT devices and cloud servers, allowing for scalable storage, processing, and analytics in cloud-based IoT systems.

10.6.3 Embedded Systems

Embedded systems in IoT devices often rely on specific protocols for communication and interaction with sensors, actuators, and other components. Protocols like I2C (Inter-Integrated Circuit) and SPI (Serial Peripheral Interface) are used for low-speed, short-distance communication between microcontrollers and peripheral devices. UART (universal asynchronous receiver–transmitter) enables serial communication between devices. CAN (controller area network) is a protocol widely used in automotive and industrial IoT applications for real-time communication between

microcontrollers and sensors/actuators. These protocols define the electrical signaling, data formats, and transmission rules necessary for seamless integration and interaction between embedded systems and peripheral devices. Arduino and Raspberry pi family are most frequently used and adopted embedded computing platforms. Some of the members are illustrated in Figure 10.4.

10.6.4 Edge Computing

Edge computing in IoT involves the use of protocols that enable communication and data processing at the network edge. MQTT, CoAP, and AMQP (Advanced Message Queuing Protocol) are commonly used for lightweight and efficient messaging between edge devices, gateways, and cloud platforms. Protocols like OPC UA (Open Platform Communications Unified Architecture) and Modbus facilitate interoperability and communication between edge devices and industrial systems. Edge computing also utilizes protocols like MQTT-SN (MQTT for Sensor Networks) and LWM2M (Lightweight M2M) for constrained IoT devices that have limited processing power and memory. These protocols ensure efficient and secure data transfer, device management, and control between edge devices and other components in the IoT architecture.

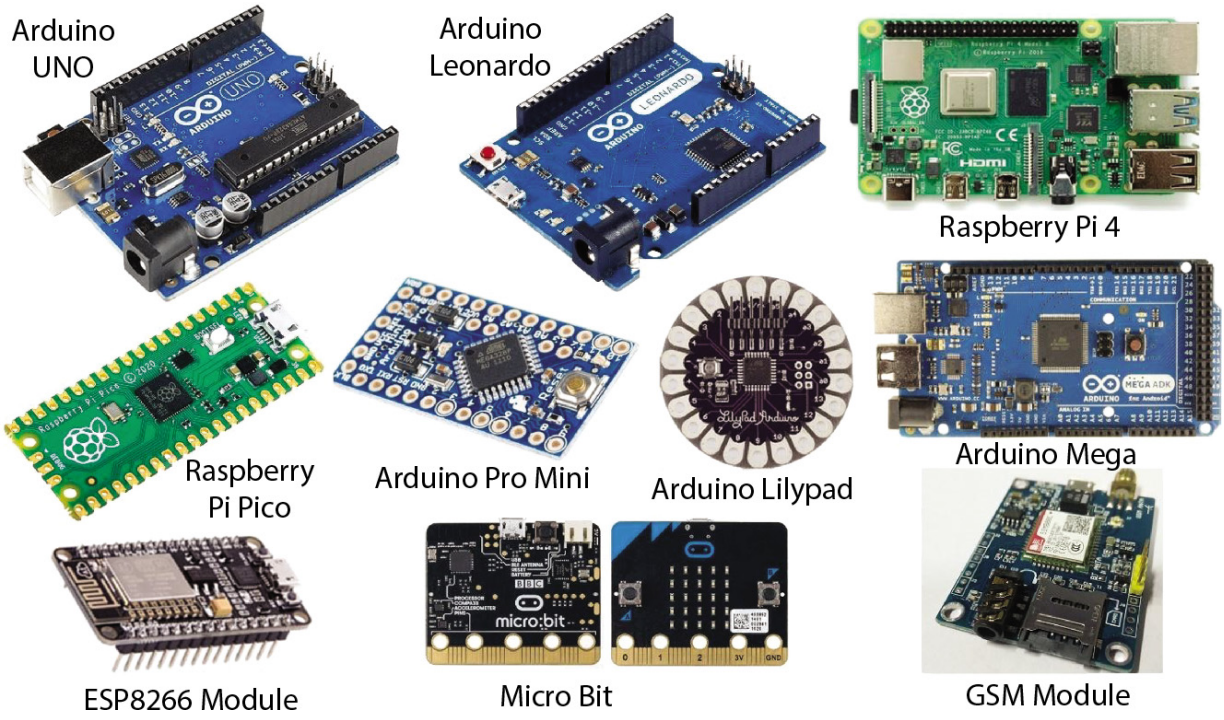

Figure 10.4 Embedded system computing devices.

10.6.5 Artificial Intelligence and Machine Learning

Artificial intelligence (AI) and machine learning (ML) have surfaced as transformative technologies capable of reshaping diverse industries, among them agriculture. AI involves crafting intelligent systems capable of emulating human intelligence, whereas ML concentrates on the capacity of computer systems to acquire knowledge and enhance performance through experience without explicit programming. The domain of AI encompasses a broad spectrum of techniques and methodologies aimed at enabling machines to execute tasks that conventionally demand human intelligence. In contrast, ML, a subset of AI, concentrates on algorithms and statistical models

that empower computers to learn from data and enhance their performance progressively. This iterative learning process empowers ML models to continually enhance their forecasts and decision-making capabilities [17, 35].

10.6.6 Blockchain Technology

Blockchain technology is a groundbreaking innovation that has the potential to revolutionize industries across the globe. It is a decentralized and distributed ledger system that ensures secure and transparent transactions among multiple participants. At its core, a blockchain is a chain of blocks, with each block containing a set of transactions. When a new transaction occurs, it is bundled into a block along with other pending transactions. Each block is linked to the previous block through a cryptographic hash, forming an unbroken chain of blocks. This cryptographic hash ensures the integrity of the data and prevents any unauthorized changes to the blockchain. The consensus mechanism holds significant importance in facilitating concurrence among network participants. Proof-of-work (PoW) and proof-of-stake (PoS) stand as two extensively employed consensus mechanisms. In the PoW mechanism, miners must unravel intricate mathematical puzzles to authenticate transactions and introduce fresh blocks to the chain. On the other hand, PoS determines the likelihood of

generating a new block in accordance with participants' investment in the network [18].

10.6.7 Remote Sensing Technology

Remote sensing plays a crucial role in enabling the Internet of Things (IoT) by providing essential data for monitoring and decision-making. By using sensors and imaging technologies, remote sensing gathers information about the physical world, which can be integrated into IoT systems. This data can be collected from satellites, aircraft, or ground-based sensors and transmitted to IoT platforms for analysis and action. Remote sensing enables IoT by providing real-time or near-real-time data on various parameters such as weather conditions, air quality, soil moisture, and vegetation health. This data can be used to monitor and control IoT devices and systems, optimizing their performance and efficiency—for example, in smart agriculture, remote sensing data on crop health and soil conditions can be combined with IoT-enabled irrigation systems to precisely control water usage based on actual needs. Remote sensing also aids in environmental monitoring, infrastructure management, and disaster response within IoT ecosystems. By integrating remote sensing with IoT, organizations and individuals can make informed decisions, improve resource

management, enhance operational efficiency, and create more sustainable and resilient systems [19].

10.7 IoT Challenges

The IoT holds immense promise in revolutionizing our lives and industries. However, it also presents several significant challenges that need to be addressed for its successful adoption and implementation. These challenges span various domains, including technical, security, privacy, scalability, interoperability, and ethical concerns [20]. Let us explore these challenges one by one.

Security and privacy: With the vast number of interconnected devices and data being transmitted, IoT systems are susceptible to security breaches and privacy threats. Weak authentication mechanisms, inadequate encryption, and vulnerabilities in devices can expose sensitive data and create entry points for cyberattacks. Robust security measures, such as encryption, access controls, authentication protocols, and regular security updates, are essential to protect IoT systems from unauthorized access, data breaches, and privacy violations [21].

Interoperability and standards: IoT encompasses a wide range of devices, platforms, and protocols, leading to interoperability

challenges. Incompatibility between devices and systems from different manufacturers hampers seamless communication and data exchange. The lack of common standards and protocols creates fragmentation and complexity, making integration and scalability difficult. The development and adoption of open standards and protocols are crucial for achieving interoperability and ensuring seamless connectivity among diverse IoT devices and platforms. Scalability and infrastructure: With the rapid proliferation of interconnected devices, the issue of scalability becomes increasingly prominent. Internet of Things (IoT) ecosystems must efficiently manage the immense surge of data and effectively accommodate the expanding array of devices. This necessitates the establishment of a resilient and expandable foundation encompassing factors such as network bandwidth, cloud computing assets, and the prowess of edge computing. Upgrading existing infrastructure and deploying robust and reliable networks are essential to accommodate the scalability demands of IoT [22, 23].

Data management and analytics: IoT generates massive volumes of data, often in real time, requiring efficient data management and analytics capabilities. Processing, storing, and analyzing this data pose significant challenges in terms of computational power, storage capacity, and data processing

techniques. Advanced analytics tools, machine learning algorithms, and edge computing solutions are crucial to extract meaningful insights, perform real-time analytics, and make informed decisions based on the collected data.

Energy efficiency: IoT devices frequently function with constrained power resources, which might include batteries or energy harvesting mechanisms. Enhancing energy efficiency and prolonging the operational lifespan of IoT devices poses a significant and imperative hurdle. Low-power communication protocols, efficient algorithms, and energy harvesting techniques are employed to minimize power consumption and enable long-term operation of IoT devices without frequent battery replacements.

Ethical and social implications: The proliferation of IoT raises ethical and social concerns. Privacy implications arise from the collection and utilization of personal data, leading to questions about consent, data ownership, and transparency. Ethical considerations surrounding data usage, data sharing, and potential biases in AI algorithms need to be addressed. Balancing technological advancements with ethical and societal implications is crucial to ensure the responsible and ethical deployment of IoT.

Integration with legacy systems: Many organizations have existing infrastructure and legacy systems that need to be integrated with IoT solutions. Integrating IoT with legacy systems can be complex, as it requires retrofitting or upgrading existing systems to enable interoperability and data exchange. Legacy system compatibility and seamless integration pose challenges in terms of cost, compatibility, and system stability [21, 23].

Navigating regulatory and legal landscapes: The swift integration of IoT has surged ahead of the formulation of appropriate regulatory and legal structures. The dynamic evolution of IoT technologies and their multifaceted applications presents a complex terrain for managing concerns encompassing data privacy, security, accountability, and adherence. Crafting well-defined regulatory frameworks and robust standards to grapple with these intricacies emerges as a pivotal requisite, fostering the conscientious and ethical deployment of IoT while safeguarding the rights and stakes of both individuals and entities. Addressing these challenges requires a collaborative effort from industry stakeholders, policymakers, researchers, and technology providers, investing in research and development, fostering innovation, establishing robust security measures, promoting interoperability standards, and raising awareness [21].

10.8 IoT Communication Protocol

The Internet of Things (IoT) relies on IoT protocols to enable efficient and secure communication between interconnected devices. These protocols ensure interoperability, security, and optimization in data transmission. They establish a standardized language for devices to exchange information, promoting compatibility among diverse devices and systems. IoT protocols incorporate security mechanisms to protect data integrity and privacy. They also optimize bandwidth usage, latency, and power consumption, allowing devices with limited resources to communicate effectively. IoT protocols drive the connectivity and collaboration of IoT applications, from smart homes to industrial automation and healthcare monitoring. Popular IoT protocols include MQTT, CoAP, HTTP, AMQP, and XMPP.

10.8.1 MQTT Protocol

MQTT (Message Queuing Telemetry Transport) is a simple and efficient way of sending messages in IoT applications. It works like this: Devices send messages about specific things, and other devices receive these messages if they are interested. This system has three main parts: devices that send messages, devices that receive messages, and something called a broker

that helps manage the messages [24]. Let us take a closer look at each part:

Publishers: Publishers act as creators or senders of messages to the MQTT broker. They are responsible for producing MQTT packets and sending them to specific topics. Publishers could include sensors, IoT devices, applications, or any source that generates data and wants to share it with other devices or applications.

Subscribers: Subscribers are like receivers—they can be devices or apps that get messages from the MQTT broker. They choose certain topics they want to know about and wait for messages to show up on those topics. Subscribers could be things like IoT devices, apps, or anything that wants to use the shared data or act based on it.

MQTT broker: The MQTT broker works as a middleman between those who send messages and those who receive them. It gets the messages from senders (publisher) and sends them to the right receivers (subscriber) according to the topics they are interested in. The broker also keeps track of who wants what and makes sure messages get to the right places. MQTT brokers also make sure messages are sent as requested by the senders and that they get delivered reliably.

10.8.2 CoAP Protocol

The Constrained Application Protocol (CoAP) serves as a lightweight and effective approach tailored for IoT devices that have limitations in resources. CoAP's primary function is to provide a straightforward and energy-efficient way for IoT devices to communicate among themselves and with backend servers. It adopts a client–server pattern, where CoAP clients initiate requests to CoAP servers for accessing and managing resources. CoAP is particularly helpful for IoT because it addresses the constraints of IoT devices, such as low power, limited processing capabilities, and intermittent connectivity [24]. It achieves this by using a compact binary format, reducing overhead, and employing UDP as the transport protocol. These features make CoAP suitable for resource-constrained devices and enable them to communicate effectively even in constrained network environments. Here is an overview of the architecture design of CoAP:

Client: The client in the CoAP architecture initiates requests to interact with resources or services available on the server. The client can be a sensor, actuator, or any IoT device that wants to retrieve or manipulate data from the server. The architecture is illustrated in Figure 10.5(b).

Server: The server hosts resources or services that the clients can access. It responds to client requests and provides the requested information or performs the desired operations. The server can be a cloud-based service, a gateway, or any device with network connectivity that exposes its resources.

10.8.3 HTTP (Hypertext Transfer Protocol) for IOT

HTTP (Hypertext Transfer Protocol) is a protocol used for communication and data transfer over the Internet. While HTTP is widely used for web browsing and accessing websites, it is not typically used as the primary protocol for IoT (Internet of Things) applications. However, there are cases where HTTP can be utilized within an IoT context. In some IoT scenarios, where lightweight and low-power devices are not required, HTTP can be employed for data communication between IoT devices and servers. This approach allows for easy integration with existing web-based systems and infrastructure. It enables IoT devices to send data to a server using HTTP requests, and the server can respond with HTTP responses, allowing for real-time monitoring, control, and data exchange.

10.8.4 Advanced Message Queuing Protocol (AMQP)

Advanced Message Queuing Protocol is a messaging protocol that enables efficient and reliable communication between

applications, services, and systems shown in Figure 10.5(c). It is designed to facilitate the exchange of messages in a distributed environment, supporting scenarios where multiple entities need to exchange information asynchronously. AMQP provides a standardized approach for queuing, routing, and delivering messages, making it a robust and flexible protocol for various messaging applications. At the core of AMQP is the concept of a message broker, which acts as an intermediary between message senders (producers) and receivers (consumers). The broker receives messages from producers and routes them to the appropriate consumers based on predefined rules and message properties. This decoupling of producers and consumers allows for asynchronous communication and ensures reliable delivery even in the face of network disruptions or varying workloads. AMQP defines a rich set of features, including message queuing, message persistence, message acknowledgment, and message routing. It supports different messaging patterns such as point-to-point, publish–subscribe, and request–response. The protocol provides various quality-of-service levels to control message delivery guarantees, allowing applications to prioritize performance, reliability, or a balance between the two [25].

10.8.5 Extensible Messaging and Presence Protocol (XMPP)

XMPP (Extensible Messaging and Presence Protocol) is an open and XML-based communication protocol designed for real-time messaging, presence information, and collaboration. The architecture is shown in Figure 10.5(d). It operates on a decentralized client–server architecture and is widely used for building messaging applications, chat systems, and collaborative platforms. XMPP utilizes XML for message encoding and provides various features for secure and reliable communication, including authentication, encryption, presence, and subscription. In the context of agriculture, XMPP plays a vital role in facilitating machine-to-machine (M2M) communication [25]. It enables seamless data exchange between sensors, actuators, and control systems deployed in a farm environment. Sensors such as temperature sensors, humidity sensors, and soil moisture sensors are connected to an XMPP server through XMPP client modules. These sensors continuously collect data and publish it to the XMPP server using XMPP protocol standards. The XMPP server acts as a central hub, relaying sensor data to subscribed entities, such as the farm owner or automated control systems. This real-time data allows for monitoring and analysis of environmental conditions in the farm. Actuators, such as irrigation systems and

motors, can also be controlled through XMPP. The control systems or farm operators send commands to the XMPP server, which then forwards them to the appropriate actuators.

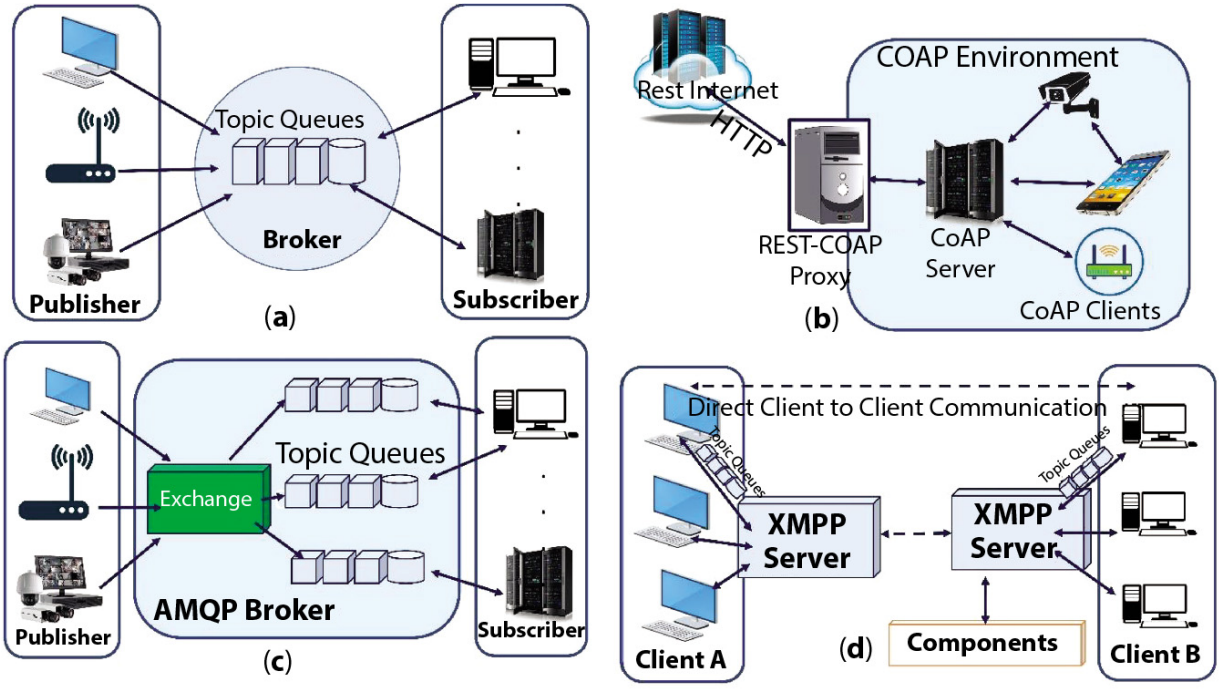

Figure 10.5 IoT communication protocol architecture. (a) MQTT, (b) CoAP, (c) AMQP, and (d) XMPP.

10.9 Sensors for IoT-Based Smart Agriculture

As previously mentioned, a sensor is a tool or device that identifies and sense physical or chemical occurrences, translating them into an electrical or digital signal. Within the agricultural realm, sensors hold significant importance by collecting information related to different environmental aspects and crop states. This empowers farmers with valuable

insights to make informed choices and enhance their farming methods. By monitoring these parameters, farmers can take proactive measures to enhance crop yield, conserve resources, and minimize risks. Sensors that can be utilized in agriculture and farming are illustrated in Figure 10.6. Let us now understand in detail the sensors that will help the farmers.

10.9.1 Soil Moisture Sensor

Soil moisture monitoring is highly useful and crucial in agriculture, as it provides vital information about the water content in the soil, which directly impacts crop growth and yield. Soil moisture sensors in smart agriculture revolutionize the way farmers monitor and manage irrigation practices. Soil moisture sensors can be implanted at different depths in the soil, depending on the crop's root zone and water requirements. They can be embedded directly in the field or placed in pots for container gardening. The data collected from the sensors can be transmitted wirelessly to a central system, where it can be analyzed and used to make informed irrigation decisions.

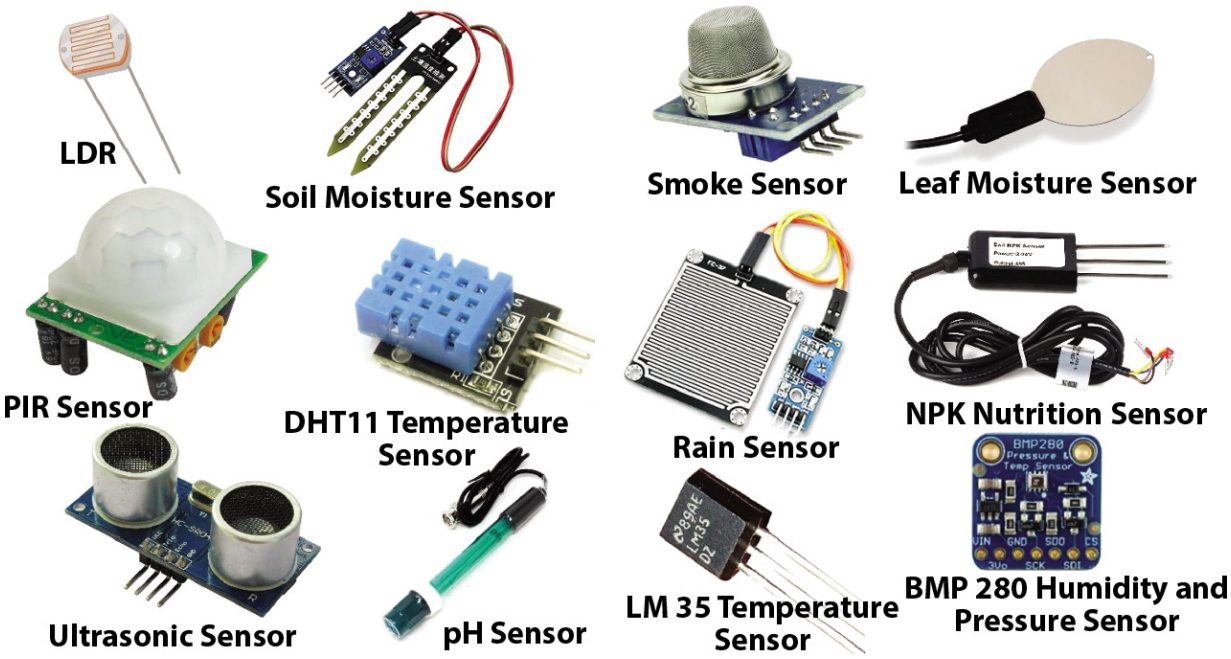

Figure 10.6 Various types of sensors for IoT agriculture application.

10.9.1.1 Resistive-Type Sensor

A resistive-type soil moisture sensor is a widely used agricultural and irrigation sensor that measures soil moisture content based on changes in electrical resistance. The sensor has two parts that go into the soil. It checks how hard it is for electricity to move between these parts. It works by sending a little bit of electricity through these parts and seeing how much goes through. When the soil is less wet, it is harder for the electricity to move, and this tells us how much moisture is in the soil. These types of sensors are liked because they are easy to use, not expensive, and can work with different small computers and controllers. However, they have limitations. The direct contact of the electrodes with the soil can lead to

corrosion over time, affecting accuracy and durability. Additionally, these sensors may face challenges in highly saline or sandy soils and require regular calibration to accommodate variations in soil types and sensor aging.

10.9.1.2 Capacitive-Type Soil Moisture Sensor

A capacitive-type soil moisture sensor utilizes the principle of changing dielectric constant with varying moisture levels in the soil. The sensor consists of two buried electrodes that form a capacitor, and the capacitance between them is influenced by the moisture content. Dry soil has a lower dielectric constant and therefore lower capacitance, while moister soil has a higher dielectric constant and higher capacitance. The sensor includes a probe with electrodes, a control circuit, and output pins. The control circuit generates an oscillating signal applied to the electrodes, and the resulting capacitance is measured. This value is then converted into a corresponding moisture level using calibration curves or formulas specific to the sensor. Capacitive sensors offer advantages over resistive sensors, including higher accuracy and reliability, as they are less prone to corrosion and physical damage. They are not in direct contact with the soil, which mitigates issues caused by soil conductivity variations or electrode corrosion. Capacitive sensors are also

suitable for a wider range of soil types, including sandy and saline soils, ensuring more accurate results [26].

10.9.1.3 Tensiometer

Tensiometers are a particular kind of sensor for soil moisture. They work by measuring how hard the soil pulls water. Tensiometers have a special cup made of porous ceramic that goes into the soil. This cup is connected to a hollow tube filled with water. The ceramic cup acts as a semi-permeable membrane that allows water to move through it based on the soil's moisture content. When the soil is dry, water moves out of the ceramic cup into the soil, creating a negative pressure or tension within the hollow tube. This tension is measured using a pressure gauge or a pressure transducer. The tension reading indicates the soil moisture level, with higher tensions corresponding to drier soil and lower tensions corresponding to wetter soil. In summary, resistive sensors offer simplicity and affordability, while capacitive sensors provide improved accuracy and versatility. Tensiometers excel in measuring soil moisture tension directly. The choice among these sensors depends on the specific requirements of the application, soil conditions, and budget considerations.

10.9.2 pH Sensor

The pH level plays a pivotal role in soil and water systems as it governs numerous chemical and biochemical processes within the field. It serves as a measure of soil's acidity or alkalinity. The study of soil or water pH holds immense significance in agriculture as it directly impacts the availability of plant nutrients by influencing their chemical forms and reactions. Consequently, soil pH directly affects the productivity of crops and the overall yield. While the typical range of soil pH extends from 1 to 14, the most suitable pH range for most agricultural crops falls between 5.5 and 7.5. Nevertheless, certain crops have adapted to flourish even in soil pH levels beyond this optimal range. Traditional farmers rely on various methods to assess water and soil pH, including plant indicators, basic soil tests, and shared knowledge within farming communities. They observe indicators like plant growth patterns and coloration to gauge soil pH suitability for specific crops. Local experts and experienced farmers provide guidance based on their past knowledge. However, these traditional techniques may lack precision. Modern methods, such as pH meters and soil testing kits, offer more accurate results. It is important to understand the working concepts of pH sensors in water and soil separately to grasp their reliability and effectiveness [27].

10.9.2.1 Water pH Sensor

The working concept of a water pH sensor involves the use of a glass electrode and a reference electrode immersed in the water sample. The glass electrode contains a special membrane that responds to changes in hydrogen ion concentration, which is indicative of pH. When the glass electrode touches the water, it triggers an electrochemical reaction. This reaction creates a voltage that shows the pH level of the water. The reference electrode is there to give a steady reference point, which helps make the pH measurements accurate. The pH sensor typically has a pH-sensitive glass electrode, a reference electrode, and output pins. Interfacing the water pH sensor with a microcontroller or controller involves connecting the output pins, usually in the form of an analog or digital signal, to appropriate input pins of the microcontroller. The microcontroller can then read the analog voltage or digital signal and convert it into a pH value using calibration curves or formulas specific to the pH sensor.

10.9.2.2 Soil pH Sensor

Soil pH sensors employ a slightly different working concept compared to water pH sensors. Soil pH sensors generally use a combination electrode or a series of electrodes to measure the pH of the soil. The combination electrode typically consists of a

glass electrode and a reference electrode. However, to measure soil pH accurately, the soil pH sensor also includes an additional electrode, called the soil electrode or ground electrode, which makes direct contact with the soil. The ground electrode measures the electrical potential difference caused by the chemical reactions occurring between the soil and the electrode. This potential difference is converted into a pH value using calibration curves or formulas specific to the soil pH sensor. Although specific models may have different pin configurations, the general interfacing process remains similar. The sensor typically consists of three main pins: power, ground and signal output pin. The power pin is connected to the positive supply voltage, usually ranging from 3.3 to 5 V, and the ground pin is connected to 0 V. A signal pin is used for transmitting the pH measurement data to the microcontroller or other interfacing devices.

10.9.3 Soil Nutrient Sensor

Nitrogen (N), phosphorus (P), and potassium (K) are crucial nutrients for crop production and are commonly applied as fertilizers in agriculture. However, the excessive use of these nutrients has significant economic and environmental consequences. Researchers shows that only a small portion of fertilizers, such as nitrogen and phosphorus, is actually utilized

by crops, while a substantial amount is lost through processes like leaching, volatilization, and erosion. This loss not only leads to inefficient resource utilization but also contributes to environmental degradation. To address this issue, it becomes imperative to develop efficient and accurate methods for detecting nutrient compounds in the soil. Nutrient sensors generally have three pins: VCC, GND, and signal output, and the connection is similar to a pH sensor. Furthermore, various nutrient sensors are expressed below.

10.9.3.1 Soil Electrical Conductivity and Salinity Sensor

Electrical conductivity (EC) sensors are employed for assessing the soil's electrical conductivity, which indirectly reflects nutrient levels and soil fertility. These sensors typically consist of two or more electrodes that are inserted into the soil. When an electrical current is applied to the soil through the electrodes, the conductivity of the soil determines how easily the current flows. Electrical conductivity is influenced by the presence of ions in the soil, including nutrients like potassium, calcium, and magnesium. The sensor measures the resistance or impedance of the soil and converts it into an electrical signal that corresponds to the soil's conductivity. To measure soil conductivity, the electrodes are inserted into the soil at a specific depth and a known voltage is applied across them. The

resulting current is measured and used to calculate the conductivity. Some EC sensors also measure soil temperature to account for its influence on conductivity.

10.9.3.2 Ion-Selective Electrode Sensor

Soil nutrient sensors play a crucial role in smart agriculture by providing real-time data on the nutrient levels in the soil. These sensors come in different types and employ various working principles to measure the concentration of nutrients. One common type is the ion-selective electrode sensor, which detects specific ions such as nitrate, phosphate, and potassium in the soil. This sensor utilizes a membrane that selectively allows the passage of target ions, generating an electrical signal proportional to their concentration. This sensor is designed to measure the concentration of nitrate ions (NO_3^-) in soil or water, providing valuable information for optimizing fertilizer application and managing nutrient levels in agricultural practices.

10.9.3.3 Optical Soil Nutrient Sensors

Soil nutrient optical sensors utilize optical techniques to measure the concentrations of various nutrients in the soil. These sensors employ different measurement techniques such as absorption spectroscopy, fluorescence spectroscopy, and

reflectance spectroscopy. In absorption spectroscopy, sensors emit light of specific wavelengths into the soil and measure the amount of light absorbed by the soil, which is proportional to the concentration of the targeted nutrient. Fluorescence spectroscopy sensors excite compounds in the soil with light, causing them to emit fluorescence at specific wavelengths. The intensity and wavelength of the emitted fluorescence provide information about nutrient concentrations. Reflectance spectroscopy sensors measure the reflectance of light from the soil surface across different wavelengths, carrying information about the soil's chemical composition, including nutrient concentrations. By analyzing the measured signals and comparing them to calibration curves or statistical models, these sensors estimate nutrient levels accurately. Soil nutrient optical sensors offer a non-destructive and efficient method for assessing nutrient content in the soil, aiding in precise nutrient management and optimizing agricultural practices.

10.9.4 Temperature and Humidity Sensors

IoT provides a variety of thermal sensors that can be utilized in agriculture.

10.9.4.1 DHT11 Sensor

The DHT11 sensor is a popular choice in smart agriculture for monitoring temperature and humidity levels. Its working principle is based on a digital signal output that corresponds to the temperature and humidity readings. The sensor consists of a humidity-sensing component and a temperature-measuring component, both of which are calibrated and integrated into a single chip. The functionality of the DHT11 sensor involves the utilization of a thermistor to gauge temperature and a capacitive humidity sensor to measure relative humidity. As temperature changes, the thermistor's resistance changes, and the humidity sensor's capacitance fluctuates based on humidity levels. These alterations are then transformed into digital signals by an internal analog-to-digital converter (ADC) present in the sensor. The DHT11 sensor provides temperature measurements with an accuracy of ±2°C and humidity measurements with an accuracy of ±5%. Widely used as a digital temperature and humidity sensor, the DHT11 is easy to connect with microcontrollers or other devices. It features a four-pin configuration for easy connection. Two pins are dedicated for power and ground. The data pin is used for bidirectional communication between the sensor and the microcontroller [28].

10.9.4.2 BME280 Sensor Module

The BME280 sensor is a versatile environmental sensor that combines temperature, humidity, and barometric pressure measurements. It utilizes advanced MEMS (micro-electro-mechanical systems) technology to accurately sense these parameters. The sensor operates on the principle of capacitance changes caused by variations in temperature and humidity, which are then converted into digital signals by the built-in ADC (analog-to-digital converter). The temperature sensor measures the voltage changes resulting from temperature variations, while the humidity sensor detects changes in capacitance caused by the absorption of water vapor. The pressure sensor utilizes a piezoresistive element to measure changes in atmospheric pressure. These analog signals are converted into digital values using on-chip ADCs and are processed internally to provide accurate measurements of temperature, humidity, and pressure. The BME280 sensor typically comes in a four-pin configuration. Two pins are dedicated to power and ground, respectively. The SDA pin is used for bi-directional data communication with the microcontroller using the I2C (interintegrated circuit) protocol, and SCL (serial clock) is used to synchronize the data communication between the sensor and the microcontroller. Keeping in mind that the four-pin configuration of the BME280 sensor corresponds to the I2C

interface, if the sensor is employed in SPI mode, it will feature supplementary pins meant for SPI communication. These include CSB (Chip Select), SDO (Serial Data Out), SDI (Serial Data In), and SCK (Serial Clock). When integrating the BME280 sensor with a microcontroller, ensure proper connections by attaching the VCC pin to the suitable power voltage source, the GND pin to the ground, and the SDA and SCL pins to their corresponding I2C pins on the microcontroller. The microcontroller communicates with the sensor by sending and receiving data through the SDA and SCL pins using the I2C protocol.

10.9.4.3 LM35 Sensor Module

The LM35 is a precision temperature sensor widely used in various applications, including agriculture. It operates on the principle of proportional output voltage with respect to temperature. The LM35 delivers a precise output voltage that correlates directly with the celsius temperature. It possesses three pins: VCC, GND, and Vout. The VCC pin links to the power supply (typically 5 V), the GND pin connects to the ground, and the Vout pin produces the analog voltage output proportionate to the temperature. The microcontroller reads the analog voltage from the sensor and converts it into a temperature value. TMP 36 sensor also works similar to LM35 but having different temperature sensing range. The LM35 temperature

sensor's accuracy, simplicity, and low cost make it a popular choice for temperature monitoring in agriculture. Its linear output voltage, directly proportional to temperature, simplifies temperature conversion and facilitates precise temperature measurements. The sensor's wide operating temperature range and minimal self-heating effect enhance its suitability for diverse agricultural applications.

10.9.5 Leaf Wetness Sensor

Leaf wetness sensors are crucial instruments utilized for identifying the existence of moisture on plant surfaces, especially leaves. They find extensive use in various environmental applications [29]. They serve various purposes, including studying the effects of controlled wetting on vegetation, stress of plants, determining optimal planting times for crops, and monitoring conditions conducive to fungal development. Leaf wetness sensors come in different types, such as external sensors, moisture detection sensors, and pole-mounted sensors. External sensors assess the duration of leaf surface wetness, while moisture detection and pole-mounted sensors detect surface moisture levels. A leaf moisture sensor is made up of a circuit that is chemically treated and coated with gold. This circuit changes how much it resists electricity based on how wet the surface becomes. The usual features of leaf

wetness sensors include a grid for sensing leaf wetness, a low-voltage circuit for stimulating it, and a circuit for measuring conductivity. When the grid becomes wet, it short-circuits, activating the relay circuit, which is reset when the grid dries.

10.9.6 Gas/Smoke Sensors

Gas sensors play a crucial role in agriculture by monitoring and detecting various gases in the agricultural environment. They provide real-time data on gas concentrations to ensure the safety and well-being of plants, animals, and workers. Commonly used gas sensors in agriculture include carbon dioxide (CO_2) sensors, methane (CH_4) sensors, ammonia (NH_3) sensors, and volatile organic compound (VOC) sensors. CO_2 sensors are essential for maintaining optimal CO_2 levels in controlled environment agriculture (CEA) systems like greenhouses. Methane sensors detect the presence of methane gas emitted by livestock, while ammonia sensors monitor ammonia levels. VOC sensors help identify volatile compounds released by plants or harmful substances in the air. MQ series smoke detectors are utilized in agriculture to detect smoke and gas hazards, enabling timely actions to prevent fires, protect crops, and maintain air quality. MQ smoke sensors operate based on a tin dioxide (SnO_2) semiconductor sensing element that undergoes changes in electrical conductivity proportional

to the concentration of the target gas in the air. These sensors provide analog and digital output signals for gas concentration measurement and detection thresholds [30].

MQ-2 gas sensor: This sensor is used to detect flammable gases like LPG, propane, butane, and smoke. In agriculture, it can be used for detecting gas leaks, such as those from cooking stoves, heaters, or gas-powered agricultural equipment. This helps ensure safety and prevent potential accidents.

MQ-4 gas sensor: The MQ-4 sensor is designed to detect flammable gases, particularly methane (CH_4). In agriculture, it can be used for detecting methane leaks from biogas plants, manure storage facilities, or underground pipelines.

MQ-5 gas sensor: This sensor is primarily used for detecting natural gas (methane), but it can also detect other gases such as LPG, hydrogen, and alcohol vapor.

MQ-7 gas sensor: The MQ-7 sensor is designed specifically for detecting carbon monoxide gas. In agriculture, it can be used to monitor CO levels in enclosed spaces where combustion processes occur, such as greenhouses, livestock barns, or grain storage facilities.

MQ-135 air quality sensor: While primarily used for air quality monitoring, the MQ-135 sensor can also detect various gases, including ammonia, benzene, smoke, and other harmful substances.

MQ-137 ammonia sensor: In agriculture, the MQ-137 ammonia sensor can be used to monitor and control ammonia levels in livestock facilities, such as poultry houses, pig barns, or cattle sheds.

10.9.7 Light-Dependent Resistor (LDR)

LDR (light-dependent resistor), also referred to as a photocell or photoresistor, serves as a sensor utilized for sensing the intensity level of light. In agriculture, LDRs find various applications related to monitoring and controlling light conditions for plant growth. The working principle of an LDR is based on the variation of its electrical resistance with changes in light levels. It consists of a semiconductor material that exhibits higher resistance in the dark and lower resistance in the presence of light. When exposed to light, photons excite the electrons in the semiconductor material, allowing current to flow more easily, thus reducing the resistance. LDRs can be used in greenhouse or indoor farming setups to measure ambient light levels. By interfacing LDRs with a control system, the

artificial lighting can be adjusted based on the desired light conditions for plant growth. This ensures that plants receive the optimal amount of light for photosynthesis and growth.

10.9.8 Rain Sensor

In agriculture, rain sensors are used for monitoring rainfall levels in fields and gardens to determine irrigation needs. By detecting rainfall, farmers can adjust their watering schedules, accordingly, preventing over- or under-irrigation. These sensors also aid in weather monitoring and can be integrated into automated agricultural systems for efficient water management. In agriculture, resistive type rain sensors based on PCB (Printed Circuit Board) technology are commonly used to detect and measure rainfall. These sensors utilize a combination of conductive traces and resistive materials on a PCB to sense the presence and intensity of rain. The PCB-based resistive rain sensor consists of several components, including the conductive traces, resistive material, and contact pads. The conductive traces are designed in a specific pattern on the PCB, forming a grid or matrix. A layer of resistive material is applied or coated onto the surface of the PCB, covering the conductive pathways. When raindrops land on the sensor’s surface, they create a path for conducting electricity between the traces through the resistive material. This leads to a change in

resistance across the sensor, which can be measured and interpreted to determine the occurrence and intensity of rainfall.

10.9.9 MEMS Microphone Sensor

One concept for using MEMS microphones in pest and insect detection is to develop an acoustic monitoring system that analyzes insect sounds to identify and track pest populations. The system would consist of an array of sensitive microphones strategically placed in agricultural fields or areas prone to pest infestations. The microphones would continuously capture the acoustic signals generated by insects, such as their mating calls, wing beats, or other distinct sounds. These signals would then be processed and analyzed using advanced algorithms and machine learning techniques to differentiate between various insect species and their activities. By monitoring the acoustic signatures of pests, the system could detect and quantify the presence and density of specific insect populations in real-time. This information would enable farmers and pest control professionals to implement targeted pest management strategies, such as deploying insect traps, applying specific insecticides, or adjusting irrigation and cultivation practices.

10.9.10 Ultrasonic Sensor

Ultrasonic sensors can be utilized for rat and reptile movement detection in agriculture to enhance pest management practices. These sensors work by emitting sound waves at high frequencies that are outside the range of what humans can hear. The ultrasonic sensor emits a burst of ultrasonic waves, typically above 20 kHz, into the surrounding environment. These waves are generated by a piezoelectric transducer within the sensor [31, 37]. The emitted ultrasonic waves travel in a cone-shaped pattern away from the sensor. They propagate through the air, bouncing off objects in their path. If a rat is present within the detection range of the sensor, the emitted ultrasonic waves will encounter the rat's body. The ultrasonic waves will then be reflected back towards the sensor. The same piezoelectric transducer that generated the ultrasonic waves now acts as a receiver. It detects the reflected waves as they return and converts them back into electrical signals. By calculating the time it takes for ultrasonic waves to go from the sensor to the rat and back, the sensor can figure out how far away the rat is. The microcontroller uses the speed of sound in the air (which is about 343 meters per second at 20 degrees Celsius) to calculate the distance to the rat. It uses the formula Distance = (Speed of Sound × Time) /2. Based on the analyzed data, the sensor determines whether there is a rat within the

detection range. If the calculated distance falls within a predetermined threshold, it indicates the presence of a rat. The ultrasonic sensor provides an output signal indicating the detection of the rat. This output can trigger various actions, such as activating an alarm, initiating pest control measures, or activating deterrent devices [32].

10.9.11 PIR Motion Sensor

The working concept of a PIR (Passive Infrared) motion sensor involves detecting changes in infrared radiation within its field of view. The sensor consists of a pyroelectric sensor that detects infrared radiation emitted by objects in its surroundings [33]. When the PIR motion sensor is powered on, it goes through an initialization phase where it calibrates itself to the ambient infrared radiation levels. This helps establish a baseline and reduces false triggers. The PIR motion sensor has two pyroelectric sensors that detect infrared radiation. These sensors are divided into segments or zones. When something warm, like a person or an animal, moves in front of the sensor, it gives off infrared radiation. The sensor has parts that can detect this radiation. If the PIR motion sensor notices a change in the infrared radiation in any of these parts, it sends a signal out through the OUT pin. This signal is typically a digital HIGH (3.3V or 5V) when motion is detected and remains HIGH for a

specific duration (often adjustable) before returning to LOW. After detecting motion, the PIR motion sensor usually has a built-in delay mechanism to prevent continuous triggering. During this delay period, the sensor ignores any additional motion. After the waiting time is over, the sensor is prepared to detect movement once again.

10.10 Sensor Interfacing and Control

The primary focus of this section is on the process of connecting sensors to embedded computing boards. It is important to recognize that the field of sensors is extensive and diverse, encompassing various types and applications. However, to ensure clarity and conciseness, we will specifically concentrate on the fundamental concepts and practical aspects of sensor interfacing with Arduino platforms. By understanding the principles and techniques you will gain the essential foundation to connect a wide range of sensors to these popular microcontroller and single-board computer platforms. This knowledge will empower you to explore and experiment with different sensors, opening up exciting opportunities for innovative projects and applications.

The Arduino Uno is a commonly used microcontroller board built around the ATmega328P microcontroller. It is widely

employed in a range of electronics projects and offers a user-friendly platform for experimenting and creating interactive systems..

10.10.1 Feature of Arduino UNO Board

The Arduino Uno board consists of several key components, including:

Microcontroller: The core of the Arduino Uno is the ATmega328P microcontroller. It operates at a speed of 16 MHz and possesses 32KB of flash memory for program storage, 2KB of SRAM for data storage, and 1KB of EEPROM for retaining data even when power is turned off. Figure 10.7 shows the detailed description of Arduino uno r3 version.

Digital I/O pins: The Arduino Uno offers a sum of 14 digital input/output pins, designated from 0 to 13. These pins have the flexibility to be set as inputs or outputs and are utilized to link with digital elements like sensors, LEDs, and motors.

Analog input pins: The Arduino Uno comes with 6 analog input pins, named A0 to A5. These pins are capable of reading analog voltage levels, enabling the Arduino to connect with sensors that produce analog outputs. Power pins: The board has several power pins, including 5V and 3.3V outputs for supplying power

to external components, and GND pins for connecting the ground.

Serial communication: The Arduino Uno includes an integrated USB-toserial converter, enabling communication between the board and a computer. This enables programming the board and exchanging data with the computer using serial communication. The Arduino board supports standard C/C++ syntax and concepts, such as variables, conditionals, loops, functions, and arrays. Further, the Arduino Uno includes a reset button for restarting the program, an LED (usually connected to pin 13) that can be used for basic output testing, and a power jack for external power supply.

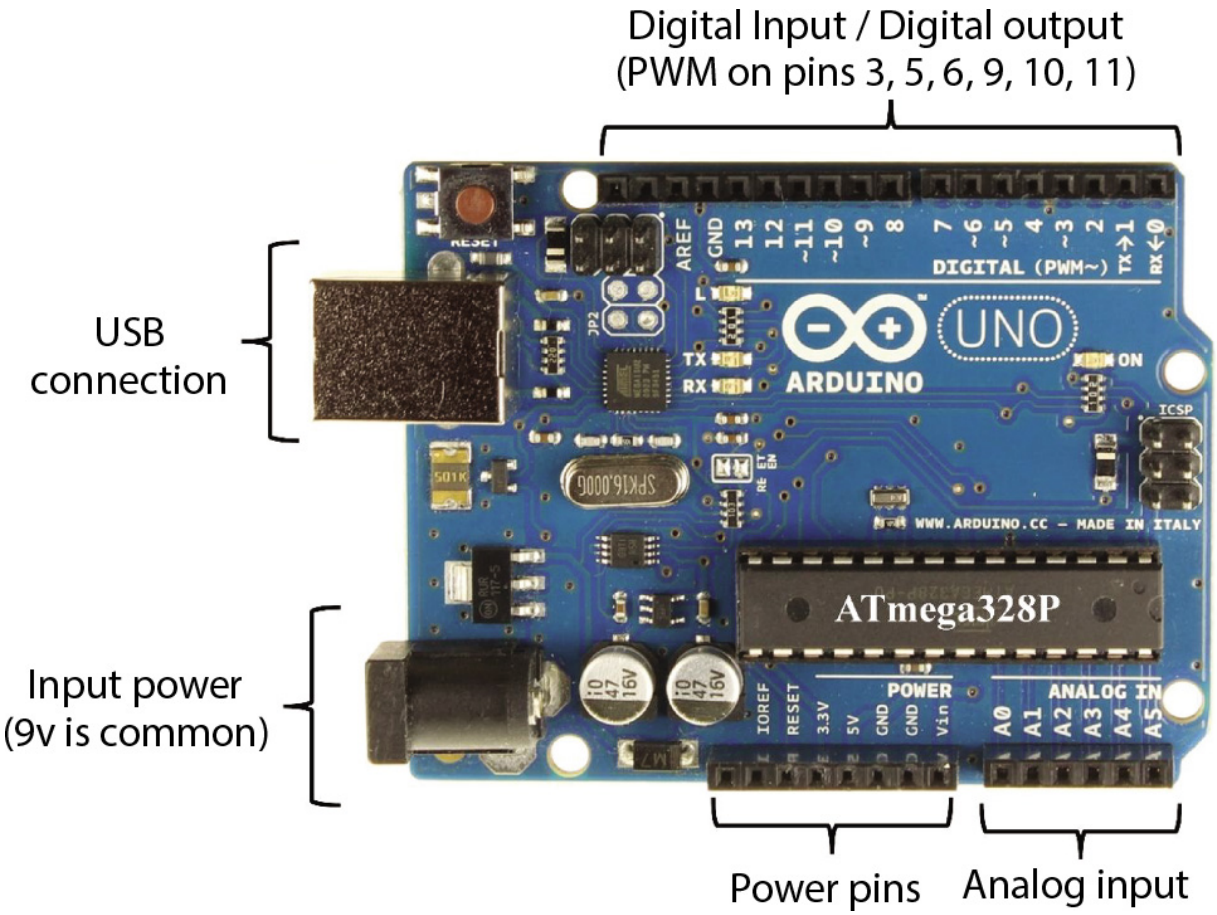

Figure 10.7 Pin description of Arduino UNO Revision 3.

10.10.2 Interfacing for Soil Moisture Sensor

The given circuit demonstrates the functionality of an automatic water system controlled by soil moisture levels. The soil moisture sensor is connected to analog pin (A0) of the Arduino board. The motor is connected to digital pin 13, while the red and green LEDs are connected to digital pins 8 and 9 respectively. The Arduino board is also connected to the 16 × 2 LCD display module. The LCD module uses pins 2, 3, 4, 5, 6, and 7 for communication with Arduino board. The power supply

and ground connections are established to provide the necessary voltage and reference for the components. The given circuit is created and simulated on Tinkercad online platform shown in Figure 10.8. The Arduino reads the analog value from the moisture sensor to determine the moisture level in the soil. If the moisture level falls below a predefined threshold, the Arduino activates the water pump by setting a digital pin to HIGH. If the moisture level is equal to or above the threshold, the Arduino deactivates the water pump and turns on the green LED HIGH to indicate sufficient moisture. The LCD display shows the moisture level and the system status. This setup ensures that the water pump is automatically turned on when the moisture level decreases below the specified threshold, helping to maintain adequate moisture in the soil [34, 40].

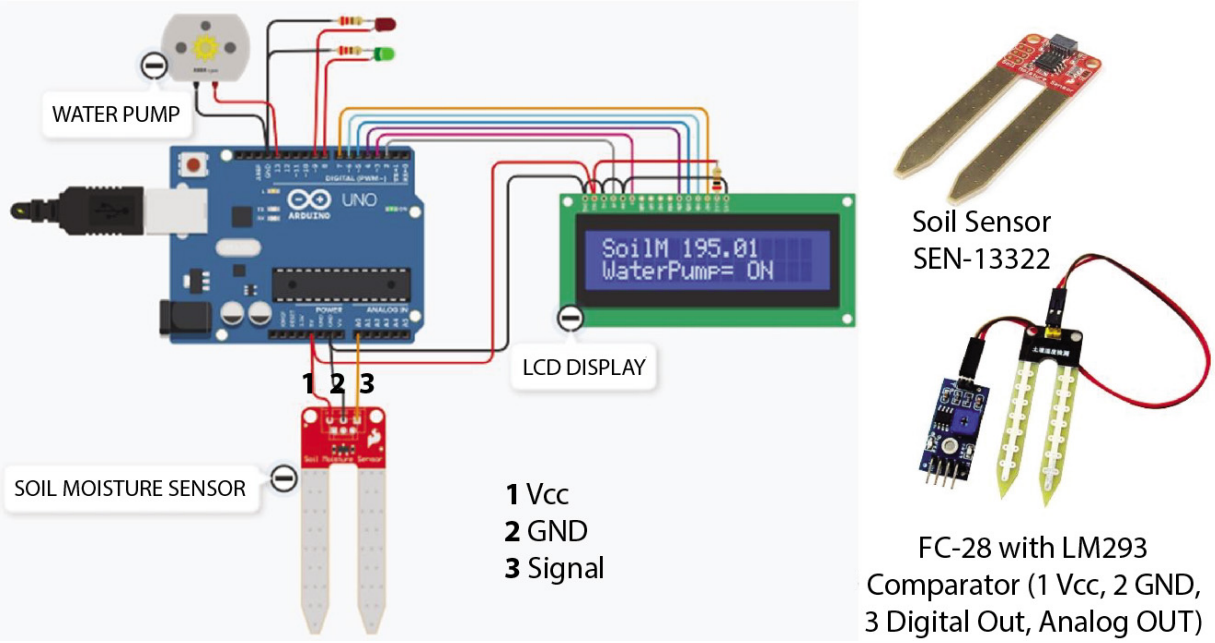

Figure 10.8 Interfacing of soil moisture sensor using Arduino UNO board.

Program Code for Soil Moisture Sensor System

```
#include <LiquidCrystal.h>
const int moistureSensorPin = A0; // Define pin numbe
components
const int motorPin = 13;
const int redLedPin = 8;
const int greenLedPin = 9;
int percentValue = 0;
LiquidCrystal lcd(2, 3, 4, 5, 6, 7); // Initialize th
void setup() {
   Serial.begin(9600); // Initialize the serial commu
   lcd.begin(16, 2); // Display startup messages
   lcd.print("__Automated__");
   lcd.setCursor(0, 1);
   lcd.print("__Irrigation__");
   pinMode(motorPin, OUTPUT); // Set pin modes for co
   pinMode(redLedPin, OUTPUT);
```

```
    pinMode(greenLedPin, OUTPUT);
    delay(2000); // Delay for startup visual effect
    lcd.clear(); // Clear the LCD display and set labe
    lcd.print("SoilM = ");
    lcd.setCursor(0, 1);
    lcd.print("WaterPump = ");
}
void loop() {
    int value = analogRead(moistureSensorPin); // Read
    moisture sensor
    float soilMoisture = value * 500.0 / 1023.0; // Co
    soil moisture percentage
    lcd.setCursor(6, 0);
    lcd.print(soilMoisture); // Display soil moisture
    lcd.setCursor(11, 1);
    if (soilMoisture < 240) // Check soil moisture and
    accordingly
{
    digitalWrite(motorPin, HIGH); // Turn on water pum
    turn off green LED
    digitalWrite(redLedPin, HIGH);
    digitalWrite(greenLedPin, LOW);
    lcd.print("ON ");
 } else {
    digitalWrite(motorPin, LOW); // Turn off water pum
    turn on green LED
    digitalWrite(redLedPin, LOW);
    digitalWrite(greenLedPin, HIGH);
    lcd.print("OFF");
 }
 Serial.print("Soil Moisture Level: "); // Print soil
 monitor
 Serial.println(soilMoisture);
}
```

Code Explanation of Soil Moisture Sensor

This section explore the step wise step function of programming code of automatic irrigation circuit based on soil moisture sensor level

1. The code begins with including the library LiquidCrystal.h, which is required to control the Liquid Crystal Display (LCD) module.
2. The next lines define constants for various pins used in the circuit. motorSensorPin represents the analog pin A0, motoPinr represents the digital pin 13, redLedPin represents pin 8, and greenLedPin represents pin 9.
3. The variable percentValue is declared but not used in the provided code.
4. A LiquidCrystal class object is made and set up with the pin numbers for the LCD (RS = 2, E = 3, D4 = 4, D5 = 5, D6 = 6, D7 = 7). This links the LCD screen to the Arduino board.
5. In the setup() function, communication through serial connection is initiated at a speed of 9600 baud. This sets up a link between the Arduino board and a computer.
6. The LCD module is initialized with the dimensions 16x2, and two lines of text are displayed: “Automated” on the first line and “Irrigation” on the second line.

7. The motorPin, redLedPin, and greenLedPin pins are set as outputs using pinMode() function to control the connected components.
8. A delay of 2000 milliseconds (2 seconds) is introduced using delay() function.
9. The LCD is cleared, and two lines of text are printed: “SoilM = “ on the first line and “WaterPump = “ on the second line.
10. In the loop() function, the moisture level is read from the analog pin motorSensorPin using analogRead() function. The value is stored in the variable value.
11. The moisture level is then calculated by mapping the analog reading (value) from the range 0–1,023 to the range 0–500. The formula value * 500.0 / 1,023.0 is used to convert the reading to a percentage value.
12. The LCD cursor is set to position (6,0), and the moisture level is printed on the LCD using lcd.print().
13. The LCD cursor is set to position (11,1), and an if-else statement is used to determine the state of the water system based on the moisture level. If the moisture level is below 240, it is considered low, and the motor and the red LED are turned on while the green LED is turned off. The text “ON” is printed on the LCD. Otherwise, if the moisture level is equal to or above 240, it is considered sufficient, and the motor and

the red LED are turned off while the green LED is turned on. The text “OFF” is printed on the LCD.

14. The level of moisture is displayed on the serial monitor using the Serial.print() and Serial.println() functions, which are useful for debugging or keeping track of the situation.

10.10.3 Interfacing for Motion Sensor

The circuit described in this section uses an Arduino board, a PIR motion sensor, and an LED to detect and indicate motion. Let us take an example to understand it better. Visualize a setup in which the PIR motion sensor is connected to digital pin 9 (pirSensor), and the LED is attached to digital pin 2 (ledPin) on the Arduino board, as depicted in Figure 10.9. Whenever the PIR motion sensor detects any movement, the LED will turn on, and a message will be shown on the serial monitor. In the absence of motion, the LED will remain off. This arrangement enables the creation of a basic motion detection system using the Arduino board, PIR sensor, and LED [39].

Now, Let us understand the concept of the program. In the setup() function, you set the pinMode for pirSensor as INPUT and for ledPin as OUTPUT. This arrangement sets up the pins to collect motion information from the PIR sensor and manage the LED output. Inside the loop() function, you use

digitalRead(pirSensor) to obtain the motion status from the PIR sensor, and this value is stored in the motionState variable. When the PIR sensor detects motion and motionState is HIGH, the LED is activated by setting ledPin to HIGH. Additionally, the message “Motion detected!” gets displayed on the serial monitor. Conversely, if no motion is detected and motionState is LOW, the LED is turned off by setting ledPin to LOW. To create a brief pause before reading the motion status again, the delay(10) function introduces a delay of 10 milliseconds between each iteration of the loop.

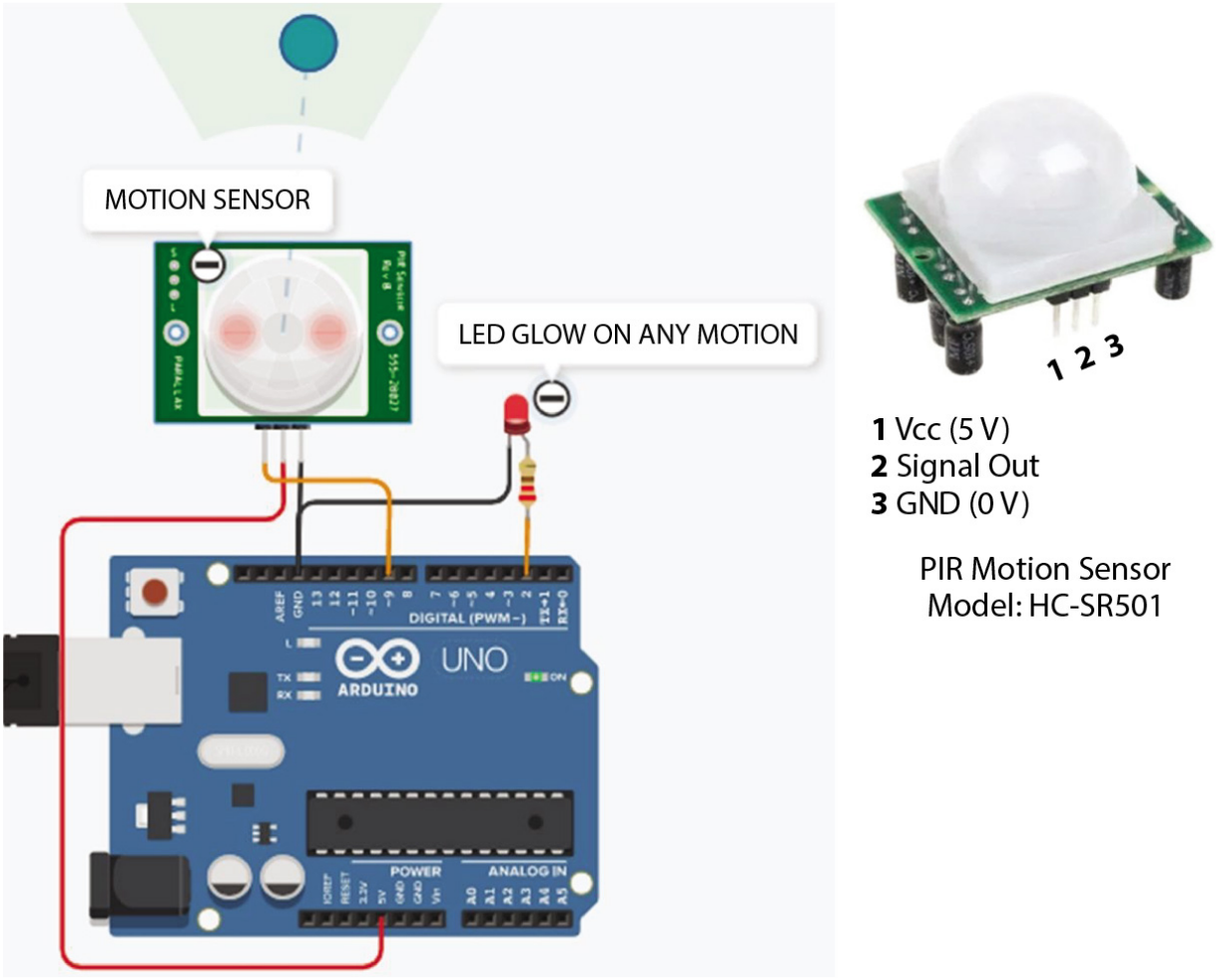

Figure 10.9 Interfacing of PIR sensor for motion detection using Arduino UNO board.

Program Code for PIR Motion Sensor

```
const int pirSensor = 9; // Define pin numbers for PI
const int ledPin = 2;
void setup()
{
   pinMode(pirSensor, INPUT);
   pinMode(ledPin, OUTPUT); // Set PIR sensor pin as
   pin as output
   Serial.begin(9600); // Initialize serial communica
}
```

```
void loop()
{
   int motionState = digitalRead(pirSensor); // Read
   PIR sensor
   if (motionState == HIGH) // Check if motion is det
   {
   digitalWrite(ledPin, HIGH); // Turn on the LED and
   Serial.println("Motion detected!");
   }
   else
   {
   digitalWrite(ledPin, LOW); // Keep turn off the LE
   }
   delay(10); // Introduce a small delay to prevent r
}
```

10.10.4 Interfacing for LDR Sensor

To interface an LDR (Light Dependent Resistor) sensor for an automatic street light circuit, an Arduino board is commonly used. The circuit is shown in Figure 10.10. The LDR sensor is connected to an analog pin (A0) of the Arduino, and a voltage divider circuit is created by adding a resistor in series with the LDR. A digital pin (pin 8) of the Arduino is connected to a relay module for controlling the light. A relay is an electromagnetic switch that allows the control of high-power electrical devices using a lowpower signal. The Arduino code reads the analog value from the LDR sensor and compares it to a predefined threshold. If the light level falls below the threshold (Light <

500), indicating darkness, the LED or relay is activated to turn on the light. Otherwise, if the light level is above the threshold (Light > 500), the light remains off.

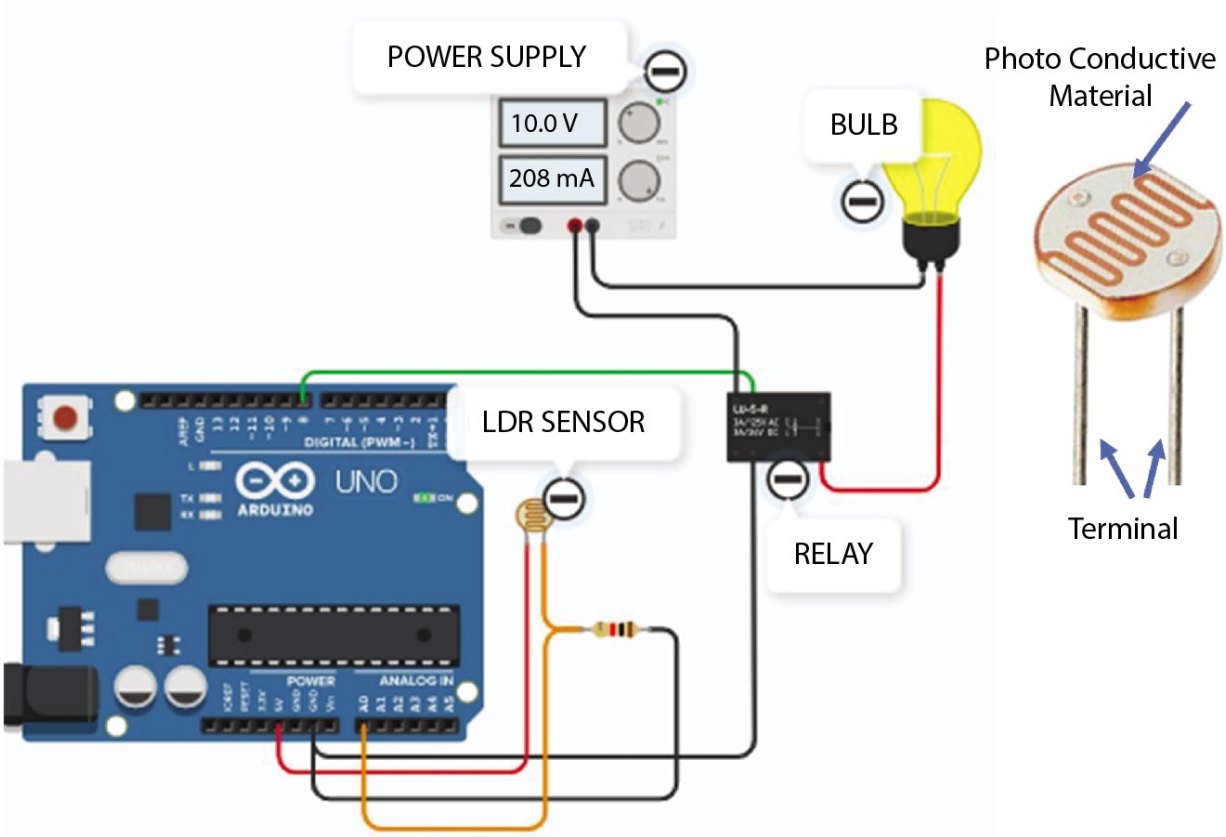

Figure 10.10 Interfacing of light dependent resistor as automatic light control using Arduino UNO board.

Program Code for LDR Sensor

```
int Light = 0; // Declare a variable ‘Light’ to store
void setup()
{
   pinMode(A0, INPUT); // Set A0 pin as input to read
   intensity
```

```
    Serial.begin(9600); // Initialize serial communica
    pinMode(8, OUTPUT); // Set pin 8 as output to cont
}
void loop()
{
    Light = analogRead(A0); // Read analog light inten
    pin
    Serial.println(Light); // Print the light intensit
    monitor
    if (Light <= 500) // Check if the light intensity
{
     digitalWrite(8, HIGH); // Turn on the LED connect
    }
else
    {
     digitalWrite(8, LOW); // Turn off the LED connect
    }
    delay(10); // Introduce a small delay to avoid rap
}
```

10.10.5 Interfacing for Ultrasonic Sensor

The schematic diagram portrayed in Figure 10.11 depicts the connections between the Arduino, LCD module, and ultrasonic sensor (HC-SR04), enabling the measurement of distance using the ultrasonic sensor and presenting the outcomes on the LCD module. The LCD module gets linked to the Arduino through six data pins (D4-D7), which connect to pins 4, 5, 6, and 7 of the Arduino. Furthermore, the control pins (RS, E) link to pins 1 and 2 of the Arduino, correspondingly. For the ultrasonic sensor,

there exist four pins: VCC, GND, trig, and echo. The VCC pin hooks up to the 5V power supply from the Arduino, while the GND pin associates with the Arduino's ground (GND). In terms of connections, the trig pin of the sensor connects to pin 9 of the Arduino, while the echo pin establishes a link with pin 10.

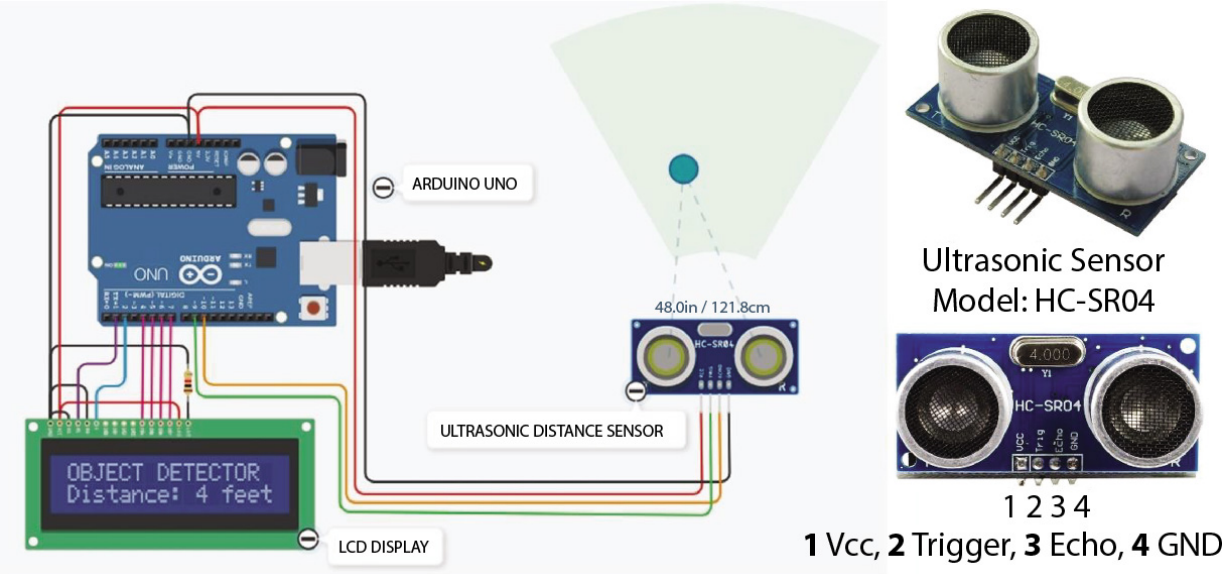

Figure 10.11 Interfacing of ultrasonic sensor for object detection using Arduino Uno board.

To measure the distance, the Arduino sends a trigger signal to the trig pin of the ultrasonic sensor. This trigger signal activates the sensor, which then emits ultrasonic waves. The ultrasonic waves travel towards an object in the vicinity, and if they encounter an obstacle, they bounce back. The sensor's echo pin receives the reflected waves. The Arduino determines the duration it takes for the waves to travel to the object and back by measuring the time between the trigger and echo signals.

This time interval is directly linked to the distance between the sensor and the object. Using this time measurement, the Arduino calculates the distance using the known speed of sound in the air and the propagation time. This distance value can be displayed on an LCD, communicated through serial communication, or used for further processing within the Arduino. By continuously repeating this process in the main loop, the circuit can effectively detect and measure the distance to objects in front of the ultrasonic sensor, enabling object detection applications such as obstacle avoidance, distance measurement, or presence detection [38].

Program Code for Ultrasonic Sensor

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(1, 2, 4, 5, 6, 7); // Initialize th
pin connections
const int trigPin = 9;
const int echoPin = 10; // Define pins for the ultras
long duration;
int distanceCm, distanceFeet; // Assign variables to
measurements
void setup() {
   lcd.begin(16, 2); // Initialize the LCD display
   pinMode(trigPin, OUTPUT); // Set pins as output an
   ultrasonic sensor
   pinMode(echoPin, INPUT);
}
void loop() {
```

```
    digitalWrite(trigPin, LOW); // Generate a trigger
    delay(20); // Provide delay to respond the system
    digitalWrite(trigPin, HIGH);
    delay(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH); // Measure the
    echo pulse
    distanceCm = duration * 0.0345 / 2; // Calculate d
    and feet
    distanceFeet = (distanceCm / 2.54) / 12;
    lcd.setCursor(0, 0);
    lcd.print("OBJECT DETECTOR "); // Display the dist
    lcd.setCursor(0, 1);
    lcd.print("Distance: ");
    lcd.print(distanceFeet);
    lcd.print(" feet");
    delay(10); // Introduce a small delay before the n
}
```

10.10.6 Interfacing for Smoke Sensor

The circuit diagram to demonstrate the interfacing of smoke sensor is illustrated in Figure 10.12. It contains Arduino board, an LCD display, a red LED, a green LED, a buzzer, and a smoke sensor. The LCD display is connected to digital pins 5, 6, 8, 9, 10, and 11 on the Arduino board. The digital pin 3 connects to the red LED, the green LED is connected to digital pin 2, and the horn (buzzer) is linked to digital pin 4. Meanwhile, the smoke sensor establishes its connection with the analog pin (A0). To power the Arduino board, an appropriate power source is used.

In the code, the analogRead() function is utilized to retrieve the analog value from the smoke sensor. If the analog value is above a predefined threshold, indicating the presence of smoke, the red LED is turned on, the buzzer emits a tone, and the LCD displays relevant messages. Otherwise, if the analog value is below the threshold, indicating no smoke, the green LED is turned on, the buzzer is turned off, and the LCD displays “NO SMOKE” and “ALL OK” messages alternately with delays.

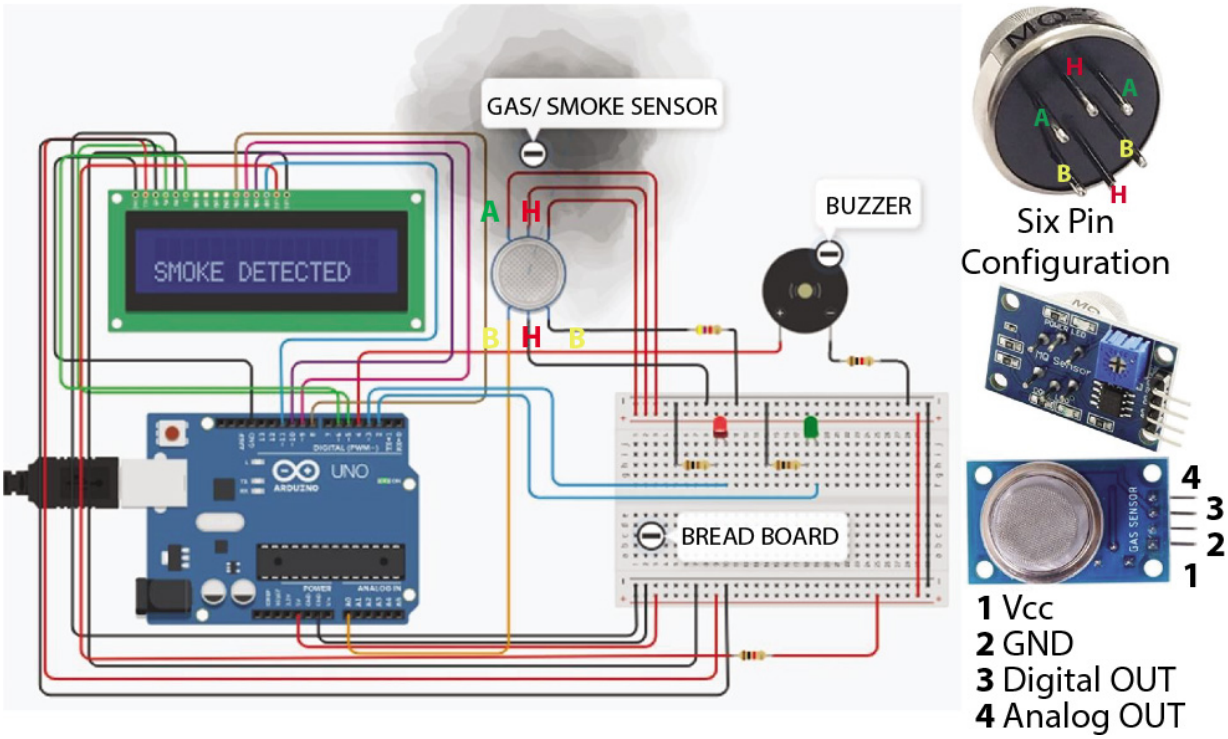

Figure 10.12 Interfacing of gas sensor using Arduino Uno board.

Program Code for Smoke Sensor

```
#include <LiquidCrystal.h>
```

```
// Initialize the LCD with pin connections (RS, E, D4
LiquidCrystal lcd(5, 6, 8, 9, 10, 11);
// Define pin numbers for red LED, green LED, horn, a
int redLight = 3;
int greenLight = 2;
int horn = 4;
int smoke = A0;
int sensorLimit = 380; // Define threshold value for
void setup()
{
   pinMode(redLight, OUTPUT); // Set pin modes for LE
   smoke sensor
   pinMode(greenLight, OUTPUT);
   pinMode(horn, OUTPUT);
   pinMode(smoke, INPUT);
   Serial.begin(9600); // Initialize serial communica
   lcd.begin(16, 2); // Initialize the LCD display
}
void loop()
{
   int analogValue = analogRead(smoke); // Read analo
   sensor
   Serial.print(analogValue); // Print analog value t
   Serial.print("\n");
    if (analogValue > sensorLimit) // Check if the an
    sensor threshold
   {
    digitalWrite(redLight, HIGH); // Turn on red LED,
    and activate horn
    digitalWrite(greenLight, LOW);
    tone(horn, 1000, 10000);
    lcd.clear();
    lcd.setCursor(0, 1);
    lcd.print(" BE CAREFUL"); // Display "BE CAREFUL"
    LCD
    delay(1000);
```

```
      lcd.clear();
      lcd.setCursor(0, 1);
      lcd.print(“EMERGENCY!! FIRE “); // Clear LCD and
      “EMERGENCY!! FIRE” message
      delay(1000);
     }
     else
     {
      digitalWrite(greenLight, HIGH); // Turn on green
      LED, and deactivate horn
      digitalWrite(redLight, LOW);
      noTone(horn);
      lcd.clear(); // Display “NO SMOKE” message on LCD
      lcd.setCursor(5, 0);
      lcd.print(“ NO SMOKE”);
      delay(1000);
      lcd.clear(); // Clear LCD and display “ALL OK” me
      lcd.setCursor(5, 1);
      lcd.print(“ALL OK”);
      delay(1000);
  }
```

10.10.7 Interfacing for TMP36 Temperature Sensor

The provided circuit and code is aimed at monitoring and displaying temperature values using a TMP 36 temperature sensor and three controlling LEDs based on predefined temperature ranges. In the setup() function, the Arduino board is configured to read temperature values from analog pin (A0) and initiate Serial communication for data output. Additionally, digital pins 2, 3 and 4 are set as outputs to control three LEDs.

Moving to the loop() function, the code starts by assigning a reference temperature of 40°C to the variable ‘RefTemp’. It then reads the analog value from the temperature sensor connected to pin (A0), maps this value to a Celsius temperature range between -40°C and 125°C, and calculates the Fahrenheit temperature. The temperature values are printed on the serial monitor for observation. Based on the temperature readings, the code uses a series of if statements to determine the appropriate LED states. Each LED represents a temperature range, and they are controlled accordingly using the digitalWrite() function. Finally, there is a delay of 1 second before the loop restarts. This process continues indefinitely, allowing for continuous temperature monitoring, LED control, and serial output. The circuit setup is shown in Figure 10.13.

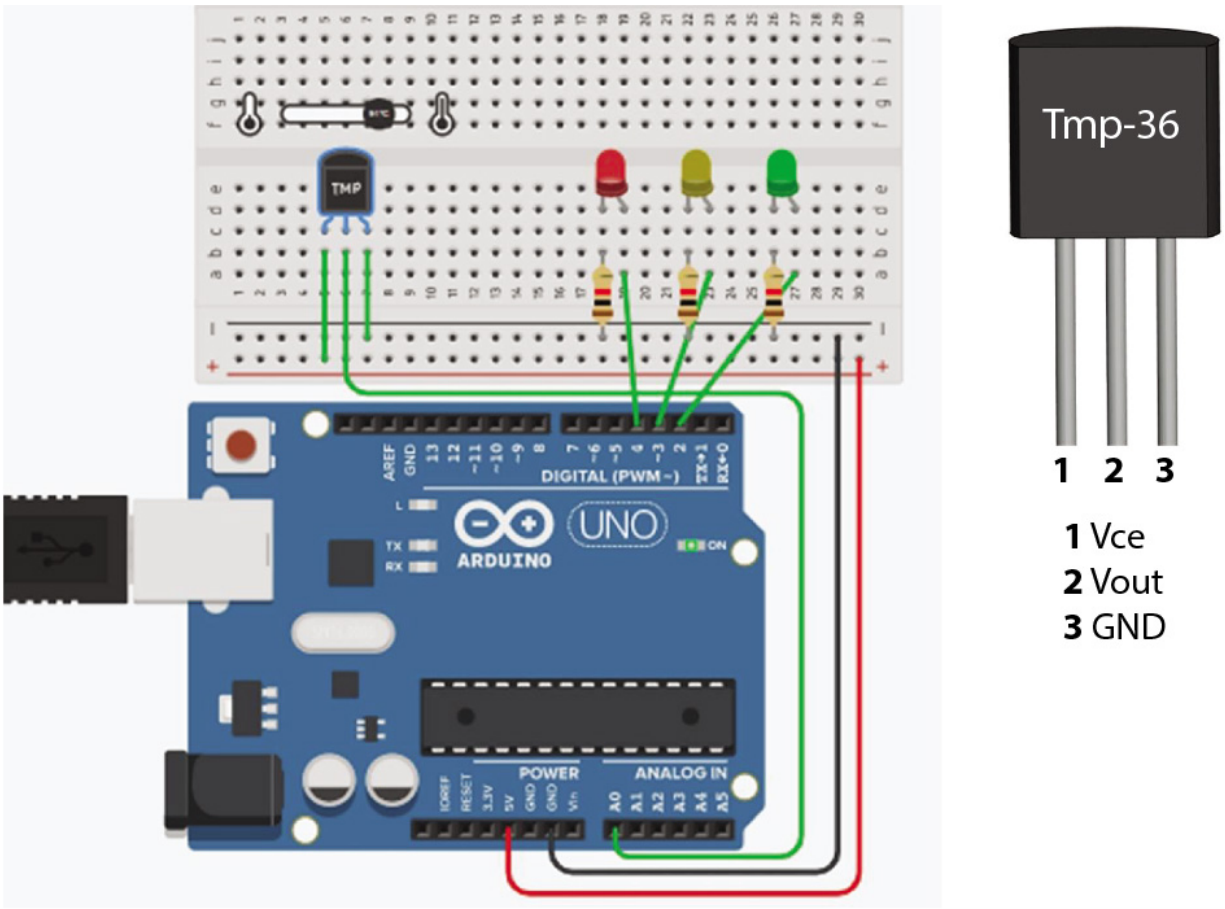

Figure 10.13 Interfacing of TMP36 sensor for temperature measurement using Arduino Uno board.

Program Code For TMP 36 Sensor

```
// Declare variables to store reference temperature,
readings
int RefTemp = 0;
int Celsius = 0;
int Fahrenheit = 0;
void setup()
{
   pinMode(A0, INPUT); // Set pin A0 as input for tem
   Serial.begin(9600); // Initialize serial communica
   pinMode(4, OUTPUT); // Set pins 4, 6, and 9 as out
```

```
    pinMode(6, OUTPUT);
    pinMode(9, OUTPUT);
}
void loop() //
{
    RefTemp = 40; // Set a reference temperature for c
    // Read analog value from temperature sensor and c
    and Fahrenheit
    Celsius = map(((analogRead(A0) - 20) * 3.04), 0, 1
    Fahrenheit = ((celsius * 9) / 5 + 32);
    // Print temperature readings in both Celsius and
    monitor
    Serial.print(celsius);
    Serial.print(“ C, “);
    Serial.print(Fahrenheit);
    Serial.println(“ F”);

    // Control LEDs based on temperature ranges
    if (celsius < RefTemp) {
     digitalWrite(2, LOW);
     digitalWrite(3, LOW);
     digitalWrite(4, LOW);
    }
    if (celsius >= RefTemp && celsius < RefTemp + 20)
     digitalWrite(2, HIGH);
     digitalWrite(3, LOW);
     digitalWrite(4, LOW);
    }
    if (celsius >= RefTemp + 20 && celsius < RefTemp +
     digitalWrite(2, HIGH);
     digitalWrite(3, HIGH);
     digitalWrite(4, LOW);
    }
    if (celsius >= RefTemp + 40 && celsius < RefTemp +
     digitalWrite(2, HIGH);
     digitalWrite(3, HIGH);
```

```
      digitalWrite(4, HIGH);
     }
     if (celsius >= RefTemp + 60) {
      digitalWrite(2, HIGH);
      digitalWrite(3, HIGH);
      digitalWrite(4, HIGH);
     }
     delay(1000); // Introduce a delay before the next
  }
```

10.11 Conclusion

In conclusion, this chapter furnishes an insight into the realm of IoT (Internet of Things) within the domain of agriculture. It delves into the elemental composition of IoT architecture, encompassing devices, connectivity, gateways, and network infrastructure. It also explores the levels and stages of IoT networks, highlighting the characteristics and enabling technologies such as wireless communication, cloud computing, embedded systems, edge computing, artificial intelligence, machine learning, and blockchain technology. The chapter addresses the challenges associated with IoT and introduces various communication protocols suitable for IoT applications. Additionally, it presents a range of sensors used in IoT-based smart agriculture, covering soil moisture, pH, soil nutrient, temperature, humidity, leaf wetness, gas/smoke, light dependent resistor (LDR), rain, MEMS microphone, ultrasonic, and PIR

motion sensors. The chapter concludes with discussions on sensor interfacing and control, specifically focusing on the features of the Arduino UNO board and providing examples of interfacing for different types of sensors.

References

1. Ramasamy, L.K. and Kadry, S., *Blockchain in the Industrial Internet of Things*, IOP Publishing, 2021, 978-0-7503-3663-5, 2021.
2. Hameed, S., Khan, F.I. and Hameed, B., Understanding Security Requirements and Challenges in Internet of Things (IoT): A Review, *J. Comput. Networks Commun.*, 2019, 9629381, 2019
3. Elijah, O., Rahman, T.A., Orikumhi, I., Leow, C.Y. and Hindia, M.H.D.N., An Overview of Internet of Things (IoT) and Data Analytics in Agriculture: Benefits and Challenges, *IEEE Internet Things J.*, 5, 2844296, 2018.
4. K., S. and S., S., Cloud IOT based novel livestock monitoring and identification system using UID, *Sens. Rev.*, 38, SR-08-2017-0152, 2018.
5. Rayhana, R., Xiao, G. and Liu, Z., Internet of Things Empowered Smart Greenhouse Farming, *IEEE J. Radio Freq. Identif.*, 4, 2984391, 2020.

6. Wu, J., Crop Growth Monitoring System Based on Agricultural Internet of Things Technology, *J. Electr. Comput. Eng.*, 2022, 8466037,2022.
7. Mohanraj, I., Ashokumar, K., and Naren, J., Field Monitoring and Automation Using IOT in Agriculture Domain, *Procedia Comput. Sci.*, 93, j.procs.2016.07.275, 2016.
8. Pongnumkul, S., Chaovalit, P., and Surasvadi, N., Applications of Smartphone-Based Sensors in Agriculture: A Systematic Review of Research, *J. Sens.*, 2015, 195308, 2015.
9. Risso, A., *et al.*, Zero standby power crop water-stress detector leading to the optimization of water usage and yield, *Sci. Rep.*, 12, s41598-022-16419-5, 2022.
10. Nayagam, M.G., *et al.*, Control of pests and diseases in plants using IOT Technology, *Meas.: Sens.*, 26, 100713, 2023.
11. Matthiessen, J.N. and Kirkegaard, J.A., Biofumigation and Enhanced Biodegradation: Opportunity and Challenge in Soilborne Pest and Disease Management, *Crit. Rev. Plant Sci.*, 25, , 07352680600611543, 2006.
12. Field, H.L. and Long, J.M., *Introduction to Agricultural Engineering Technology*. Cham: Springer International Publishing, 2018.
13. Mohd Aman, A.H., Yadegaridehkordi, E., Attarbashi, Z.S., Hassan, R., and Park, Y.-J., A Survey on Trend and

Classification of Internet of Things Reviews, *IEEE Access*, 8, 3002932, 2020.

14. Sikimic, M., Amovic, M., Vujovic, V., Suknovic, B., and Manjak, D., An Overview of Wireless Technologies for IoT Network, in *2020 19th International Symposium INFOTEH-JAHORINA (INFOTEH)*, 2020, 9066337, 2020.
15. Wang, N., Zhang, N., and Wang, M., Wireless sensors in agriculture and food industry—Recent development and future perspective, *Comput. Electron. Agric.*, 50, j.compag. 2005.09.003, 2006.
16. Kogias, D.G., Michailidis, E.T., Tuna, G., and Gungor, V.C., Realizing the Wireless Technology in Internet of Things (IoT), in *Emerging Wireless Communication and Network Technologies*, Singapore: Springer Singapore, 2018, 978-981-13-0396-8_10, 2018.
17. Sharma, A., Jain, A., Gupta, P., and Chowdary, V., Machine Learning Applications for Precision Agriculture: A Comprehensive Review, *IEEE Access*, 9, 3048415, 2021.
18. Abdelmaboud, A., *et al.*, Blockchain for IoT Applications: Taxonomy, Platforms, Recent Advances, Challenges and Future Research Directions, *Electronics*, 11, 11040630, 2022.
19. Saiz-Rubio, V. and Rovira-Más, F., From Smart Farming towards Agriculture 5.0: A Review on Crop Data Management, *Agronomy*, 10, 10020207, 2020.

20. Shafique, K., Khawaja, B.A., Sabir, F., Qazi, S., and Mustaqim, M., Internet of Things (IoT) for Next-Generation Smart Systems: A Review of Current Challenges, Future Trends and Prospects for Emerging 5G-IoT Scenarios, *IEEE Access*, 8, 2970118, 2020
21. Frustaci, M., Pace, P., Aloi, G., and Fortino, G., Evaluating Critical Security Issues of the IoT World: Present and Future Challenges, *IEEE Internet Things J.*, 5, 2767291, 2018.
22. Bhuiyan, M.N., Rahman, M.M., Billah, M.M., and Saha, D., Internet of Things (IoT): A Review of Its Enabling Technologies in Healthcare Applications, Standards Protocols, Security, and Market Opportunities, *IEEE Internet Things J.*, 8, 3062630, 2021.
23. Lohiya, R. and Thakkar, A., Application Domains, Evaluation Data Sets, and Research Challenges of IoT: A Systematic Review, *IEEE Internet Things J.*, 8, 3048439, 2021.
24. Al-Masri, E., *et al.*, Investigating Messaging Protocols for the Internet of Things (IoT), *IEEE Access*, 8, 2993363, 2020.
25. Bahashwan, A.A.O. and Manickam, S., A Brief Review of Messaging Protocol Standards for Internet of Things (IoT), *J. Cyber Secur. Mobil.*, 8, jcsm2245, 2018.
26. Hirsch, C., Bartocci, E., and Grosu, R., Capacitive Soil Moisture Sensor Node for IoT in Agriculture and Home, in

2019 IEEE 23rd International Symposium on Consumer Technologies (ISCT), 2019, 8901012, 2019.

27. Akhter, F., Siddiquei, H.R., Alahi, M.E.E., Jayasundera, K.P., and Mukhopadhyay, S.C., An IoT-Enabled Portable Water Quality Monitoring System With MWCNT/PDMS Multifunctional Sensor for Agricultural Applications, *IEEE Internet Things J.*, 9, 3069894, 2022.
28. Majumdar, P., Mitra, S., and Bhattacharya, D., IoT for Promoting Agriculture 4.0: A Review from the Perspective of Weather Monitoring, Yield Prediction, Security of WSN Protocols, and Hardware Cost Analysis, *J. Biosyst. Eng.*, 46, /s42853-021-00118-6, 2021.
29. Gupta, S., Huang, C.H., Singh, G.P., Park, B.S., Chua, N.-H., and Ram, R.J., Portable Raman leaf-clip sensor for rapid detection of plant stress, *Sci. Rep.*, 10, 20206, 2020.
30. Liu, L., *et al.*, Heteronanostructural metal oxide-based gas microsensors, *Microsyst. Nanoeng.*, 8, s41378-022-00410-1, 2022.
31. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.

32. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. "Solutions Using Machine Learning for Diabetes." *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
33. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, "Image Processing Based Intelligent Mini Robotic Face Recognition System," *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.
34. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. "Mining of data through various soft computing techniques." In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
35. Goyal, Mohit, and Durgesh Srivastava. "A behaviour-based authentication to internet of things using machine learning." *Design and Development of Efficient Energy Systems* (2022): 245-263.
36. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
37. Escolà, A., *et al.*, Performance of an Ultrasonic Ranging Sensor in Apple Tree Canopies, *Sensors*, 11, s110302459, 2011.

38. Gómez Álvarez-Arenas, T., *et al.*, Ultrasonic Sensing of Plant Water Needs for Agriculture, *Sensors*, 16, s16071089, 2016.
39. Yang, D., Xu, B., Rao, K., and Sheng, W., Passive Infrared (PIR)-Based Indoor Position Tracking for Smart Homes Using Accessibility Maps and A-Star Algorithm, *Sensors*, 18, s18020332, 2018.
40. Ahmed, S.M., Kovela, B., and Gunjan, V.K., IoT Based Automatic Plant Watering System Through Soil Moisture Sensing—A Technique to Support Farmers' Cultivation in Rural India, 2020, 978-981-15-3125-5_28, 2020.

Note

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11
An Irrigation System: Design, Implementation, and Benefits

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Abstract

A smart irrigation system is a technology-driven approach to manage irrigation in agricultural or landscaping applications. To maximize water use and make sure plants get the proper quantity of water at the right time, it entails integrating sensors, data analytics, and automation. An increase in population growth and a high demand in food lead to overall requirements of water resources since agriculture consumes 85% of the world's available freshwater resources, and this will continue to rule water usage as the world's population grows and food demand rises. For sustainability of water, there is a need to develop strategies encompassing technological, agronomic, managerial, and institutional improvements. There are several water-saving solutions available, ranging from simple to highly complicated. Traditional irrigation methods can be automated

using various sensors and controllers. These automated methods not only saves water but can monitor water flow as well. In this chapter, an author explores various smart irrigation techniques. This chapter also covers the usage of different sensors and controllers. This paper aims to perform a thorough analysis of the existing research on intelligent irrigation systems. It will be focused on systems that employ artificial intelligence methods in both urban and rural agriculture to identify crop requirements in terms of soils, water, and other requirements.

Keywords**:** Smart irrigation, sensors, GPS, controllers, artificial intelligence, cloud

11.1 Introduction

Agriculture and farming contribute significantly to the GDP (gross domestic product) of emerging and developed countries. It is thus important that existing farming methods be improved for better and quality yield. This would not only aid in the growth of the entire world but will also help to address issues like climate change and ailments like drought. Better technology increases productivity, preventing situations like starvation and malnutrition. The technology should be

inexpensive so that its influence can reach billions of people throughout the world.

Farming and agriculture contribute significantly to GDP (gross domestic product) in both developing and developed countries. Thus, improving and optimizing the present farming technology is critical. Along with promoting the healthy, sustainable growth of people, plants, and animals, it will also make it easier to deal with serious global issues like climate change and illnesses like drought.

A smart city incorporates information and communication technology (ICT) to improve the overall quality and efficiency of living by the use of smart infrastructure, smart transportation, smart energy, smart healthcare, smart government, smart education, and smart agriculture [1]. Smart irrigation is one of the most important aspects of our environment when discussing smart cities and IoT. In order to automate this procedure, technology has suggested several IoT-based techniques and methods. Intelligent irrigation techniques will enable intelligent agronomists to better understand land requirements. This will result in developing effective practices with respect to soil, plant, and land conditions. By avoiding unsuitable farming circumstances, agricultural production is increased significantly [2]. This chapter discusses various smart

irrigation systems. It also discusses the use of controllers and sensors in these systems. It will also give a general review of irrigation systems used in urban and rural agriculture that are based on artificial intelligence.

11.2 Smart Irrigation Composition

The smart irrigation unit consists of various modules as shown in Figure 11.1.

Sensors and Controllers

For collecting data, different types of sensors and controllers are available in the market which can be configured with different threshold values depending upon the environment and soil conditions [3]. These sensors are collecting and sending data at regular intervals to the farmers. The various sensors and controllers are shown in Figure 11.2.

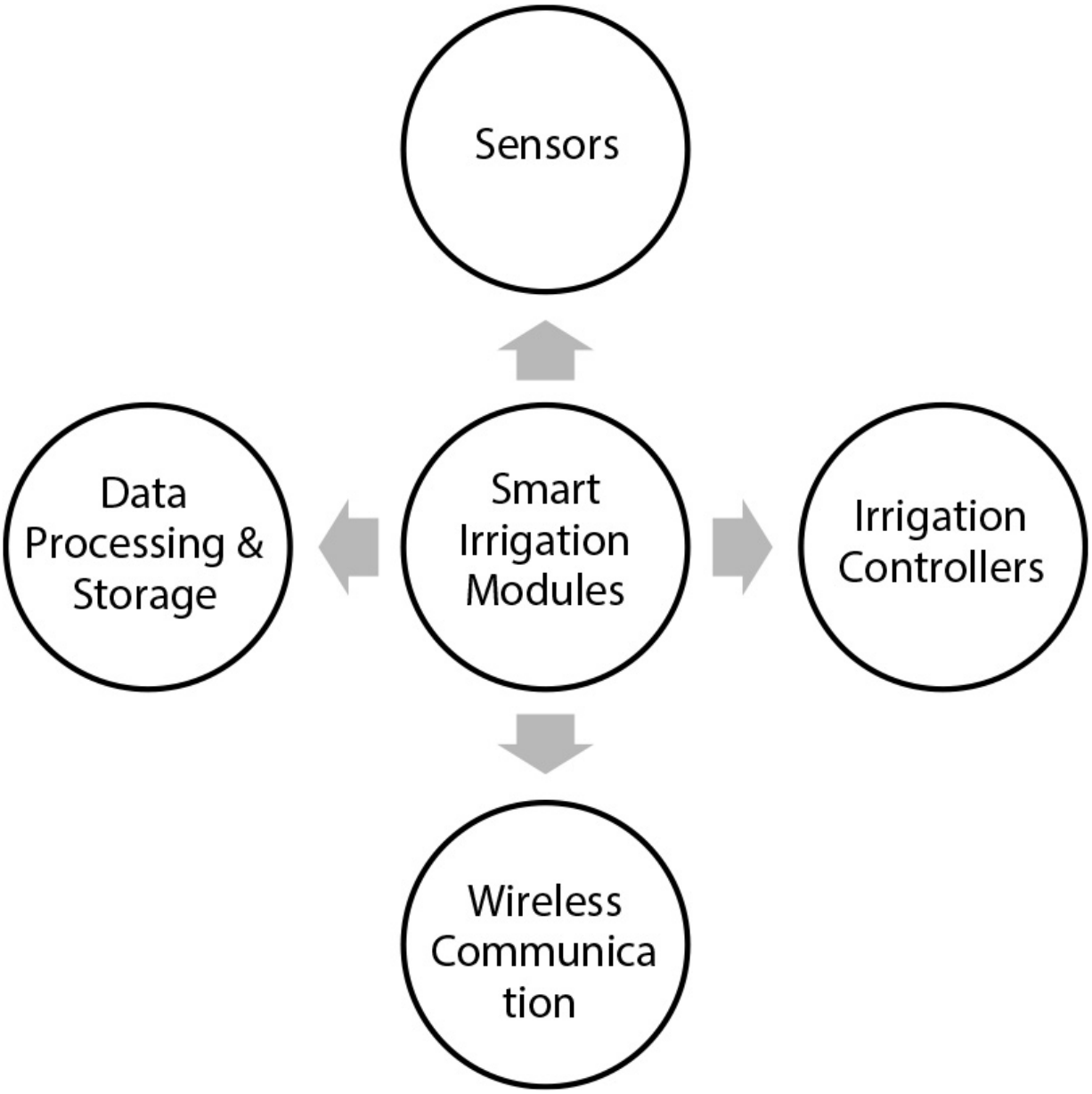

Figure 11.1 Smart irrigation modules.

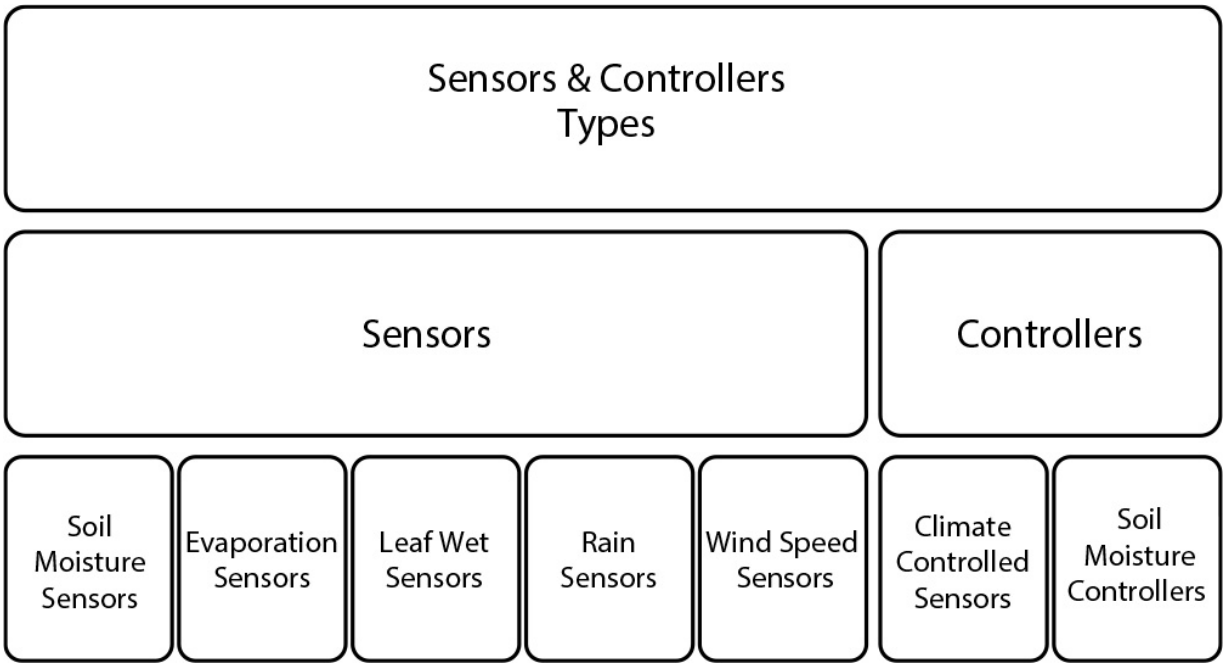

Figure 11.2 Types of sensors and controllers.

Soil Moisture Sensors and Controllers

One of the important sensors used in irrigation is the soil moisture sensor. The sensor determines the moisture content in the soil. The measurement is precise as the probe is inserted at the same depth as a root of the plant. One of the available soil moisture sensors in the market is shown in Figure 11.3.

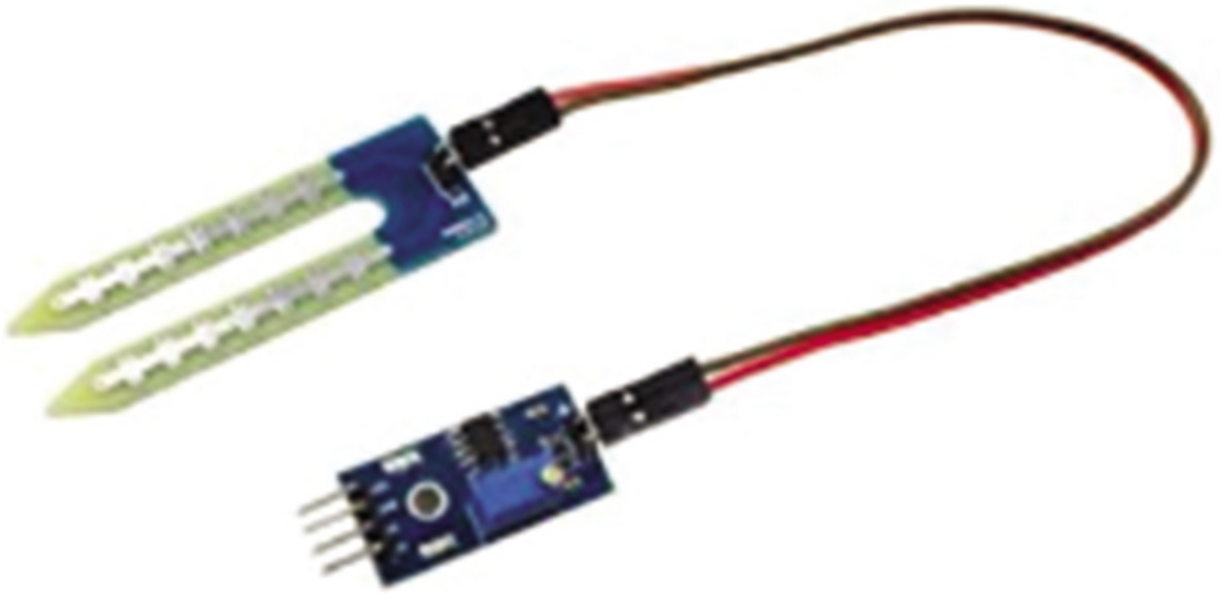

Figure 11.3 Soil moisture sensor (source: https://www.amazon.in/xcluma-Moisture-Humidity-Arduino-Respberry/dp/B071LH5Z9D).

The controllers can be used in combination with moisture sensors which generate periodic alarms. The controllers play a management role based on the threshold values [4]. If the moisture content exceeds or falls below the threshold, the controller opens or closes the water valves, respectively.

There are different varieties of soil moisture sensors available in the market based on the soil conditions.

Evaporation Sensor and Leaf Wetness Sensor

Weather monitoring equipment includes evaporation sensors. Its value represents the soil surface evaporation. The length of irrigation may be changed by gathering this climate-related

data. This encourages localized water circulation [5]. Significantly less irrigation water may be used using evaporation sensors. A RS-EVA-N01-2 evaporation sensor can be seen in Figure 11.4.

Figure 11.4 (a) Evaporation sensors (source: https://embsys.in/evaporation-sensor) and (b) leaf wet sensors (source: https://www.tradeindia.com/products/leaf-wetness-sensor-c5347781.html).

Leaf wet sensors monitor the moisture content of leaves. The data can be analyzed to find whether the plants lack water and the need to water the soil.

Rain Sensor

Rain sensors are used to make efficient use of rain water to avoid water wastage. During rain, these sensors can send signal

to the controller to automatically close the valve to avoid misuse of water resources. Based on the functionalities, large varieties of rain sensors are available. The most prevalent kind of rain sensor now in use uses a tiny cup or basin to collect water; once a certain amount has been gathered, the weight of the cup stops the irrigation cycle. The cup has to be cleaned on a regular basis to avoid the deposition of debris on it. One of the available rain sensors is shown in Figure 11.5.

Wind Speed Sensors

Wind speed sensors can block watering when the wind speed is too high as shown in Figure 11.6.

This tool consists of wind cups attached on an axis. The speed of the wind can be determined by the rotation of the cups by airflows. The sensors embedded on the axis can be used to determine the wind speed.

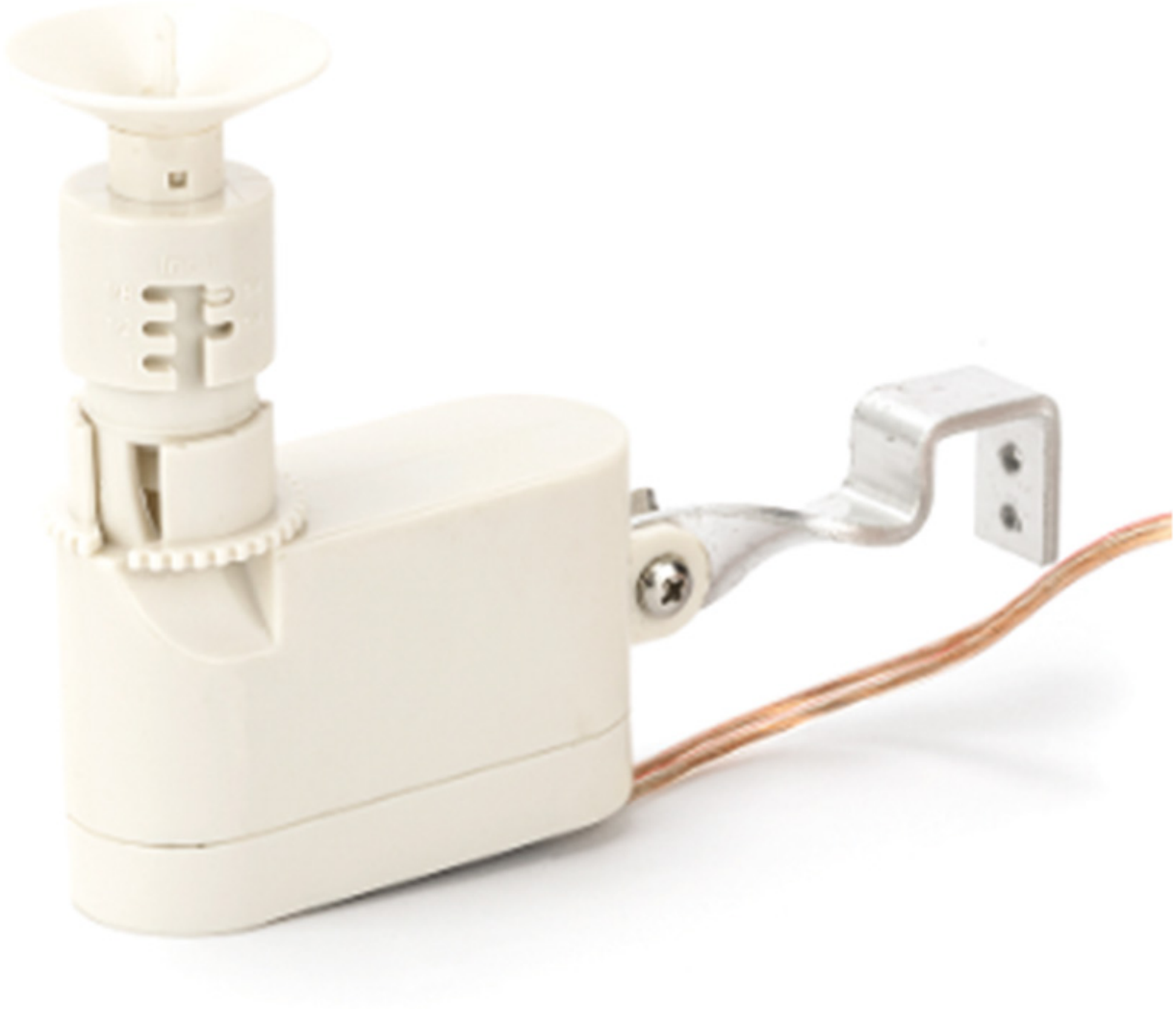

Figure 11.5 Rain sensors (source: https://www.irrigation.org/SWAT/Water-Efficient-Products/Rain-Sensors/SWAT/Water-Efficient-Products/Rain-Sensors.aspx).

Figure 11.6 Wind speed sensors (source: https://www.renkeer.com/wind-sensor-working-principle/).

Climate-Based Controllers

Climate-based sensors use meteorological data to adjust the irrigation schedule as shown in Figure 11.7. This weather-related data is shared with the controller to adjust the irrigation

schedule. The controller receives this weather-related data to modify the watering schedule. The most typical types of climate-based controllers are (a) signal-based controllers, which determine the ET value for a grass surface at the site using meteorological data from a publicly accessible source. The controller then receives the ET data over a wireless connection; (b) historic ET controllers which utilize a pre-set water consumption curve that is based on past water usage patterns in various locations. It is possible to modify the temperature and solar radiation curve; and (c) on-site weather measurement controllers that compute continuous ET readings and water using the meteorological information gathered there.

Wireless Communication Module

The most common protocol for communication is Wi-Fi. This benefit of the Wi-Fi system may be used to monitor and track the state of crops in real time, along with the current status of the crop and any requirements for the crop that may exist. IoT-based systems are used to collect data from the sensors and share it on cloud for further processing and analyses of the data. There are two main categories of wireless communication: long-range and short-range protocols. Long-range network protocols can transmit data over long distances from one device to another as LPWA, LoRaWAN [6], Sigfox [7],

and NB-IoT [18]. The short-range protocols include Bluetooth and near-field communication (NFC)-enabled devices like ZigBee.

Figure 11.7 Climate-based controller (source: https://extension.okstate.edu/fact-sheets/smart-irrigation-technology-controllers-and-sensors.html).

Data Processing

The data collected from sensors like soil moisture, rainfall, and temperature will help to make decision about the crops need. If the any of the data collected value is less than the threshold, the command can be given to the controller to pump water to the crops. This will also take rainfall sensor weather forecasting info into consideration before allowing the watering or not. The data is further stored to make an analysis so that, based on the previous irrigation schedule, the next irrigation cycle can be defined.

11.3 Advantages of Intelligent Irrigation Systems

There are numerous benefits of smart irrigation system. The features can be explored by incorporating various sensors and controllers with the traditional system to make it more cost-effective and eco-friendly. Some of the benefits are listed in Figure 11.8.

- Water conservation: A smart irrigation system ensures to optimize soil moisture level by using moisture sensors. It also ensures that plants get the right quantity of water without overwatering or underwatering; a smart irrigation system may minimize water waste. This can save a lot of water,

especially in places where water is expensive or scarce. Weather prediction can also save a lot of water.

- Cost savings: A smart irrigation system can assist in lowering water bills as well as maintenance and repair expenses related to conventional irrigation systems by minimizing water waste and optimizing watering schedules. Based on local meteorological information, it may identify when a farm or piece of land needs to be watered. You will be able to better manage your irrigation and landscaping demands, and you will be able to drastically reduce water waste, which will lower your water costs.

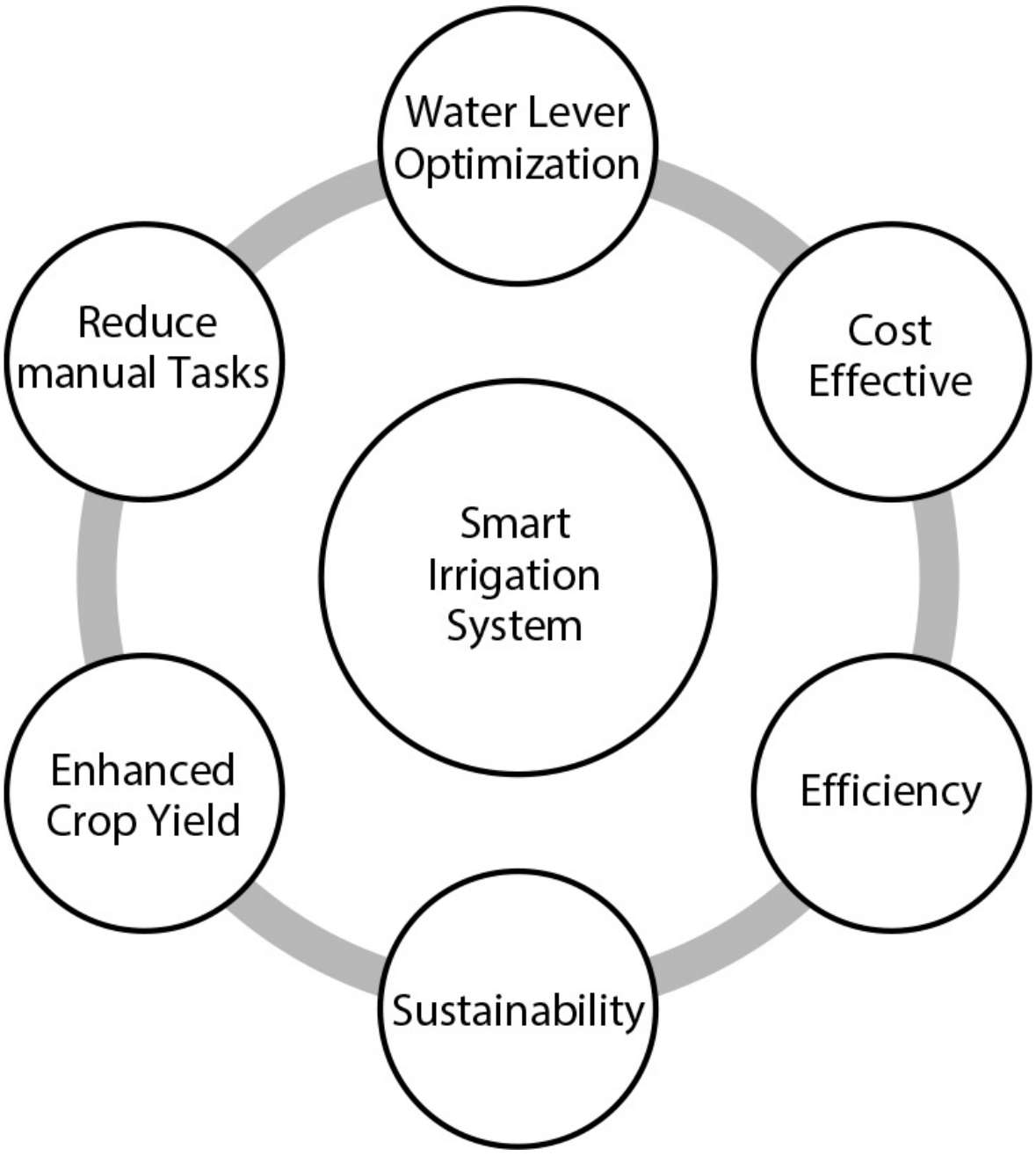

Figure 11.8 Benefits of smart irrigation systems.

- Maintain plant health: By ensuring that plants receive the proper amount of water at the appropriate time, a smart irrigation system may encourage healthier development and lower the risk of pests and diseases.

- Increased efficiency: The manual work of farmers can be reduced by remotely controlling the water schedules. This will not only reduce the time and efforts of the farmers but can also be used to maintain and control the soil moisture level for increased yield.
- Sustainability: A smart irrigation system may support more sustainable farming practises and help save the environment by decreasing water waste and encouraging resource efficiency.
- Reduce manual tasks: All the tasks performed are automatic in nature. It will automatically adjust to the need of the land, thus leading to lower manual tasks. The schedules of timely watering the plant can be set and followed. It results in better quality yield. The smart irrigation methods will reduce runoff of waters and nutrients from the soil.

11.4 Irrigation Methods

There are numerous irrigation methods available to cater the water need of plants for their optimal growth as discussed in Figure 11.9. It is an artificial way of providing water to the agricultural land in order to supplement the rain water. These methods are mostly used where the rainfall is not adequate. Through these methods, the soil moisture can be maintained so that plants can be grown in favorable conditions.

Drip Irrigation

It is the most widely used and favorable method for vegetable, fruit, and flower growth. In this method, water is allowed to fall drop by drop over the roots of the plants using a dripper. Through this technique, an optimal level of moisture can be maintained in the soil.

Figure 11.10 gives us an idea of drip irrigation where water pipes are added across the plant. The water will drip through these pipes in a fixed interval of time [8]. This method has the benefit of minimizing nutrient loss and reducing leaching. Soil erosion and weed growth are also less, but the cost of a drip setup is more costly and installation has to be properly done for its effectivity.

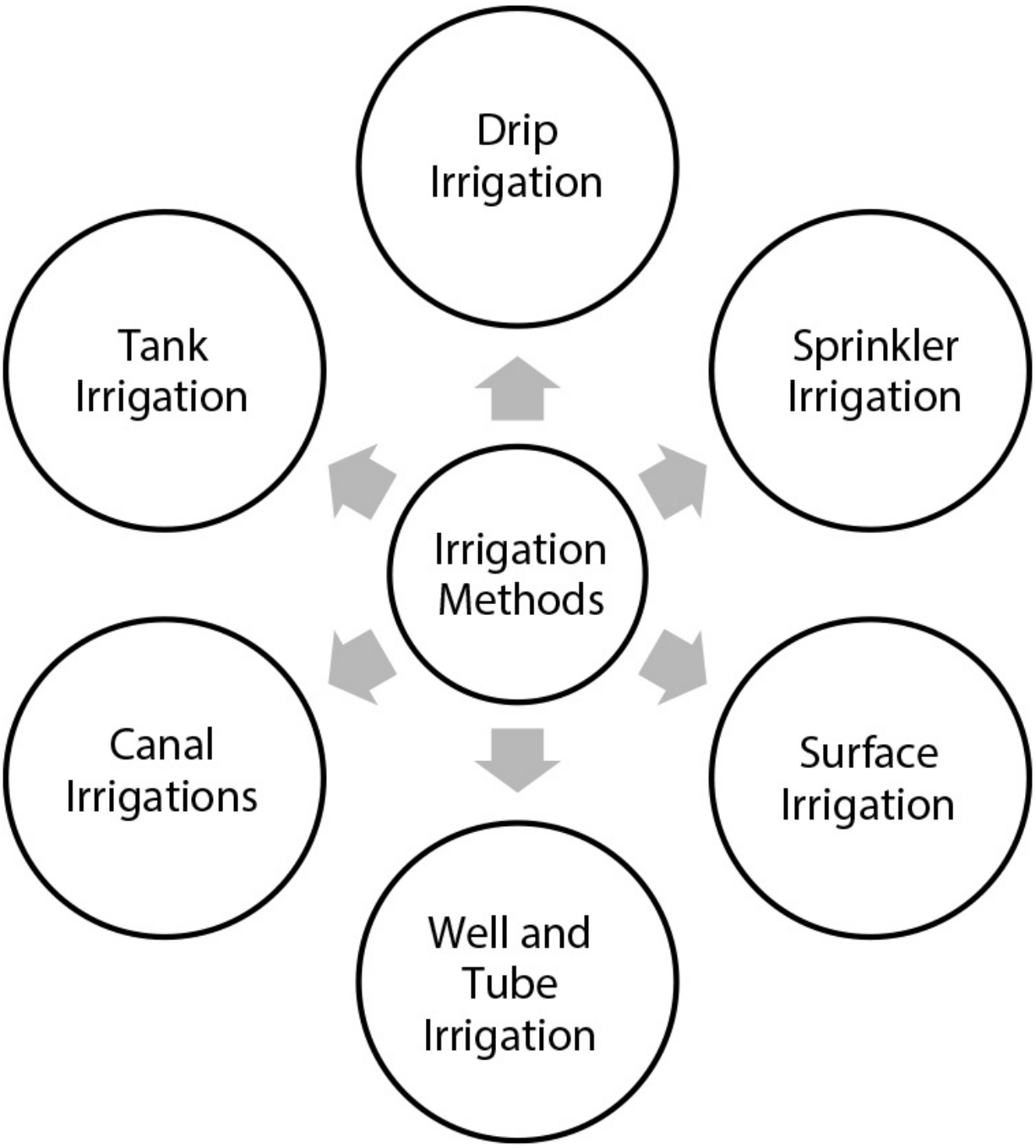

Figure 11.9 Irrigation methods.

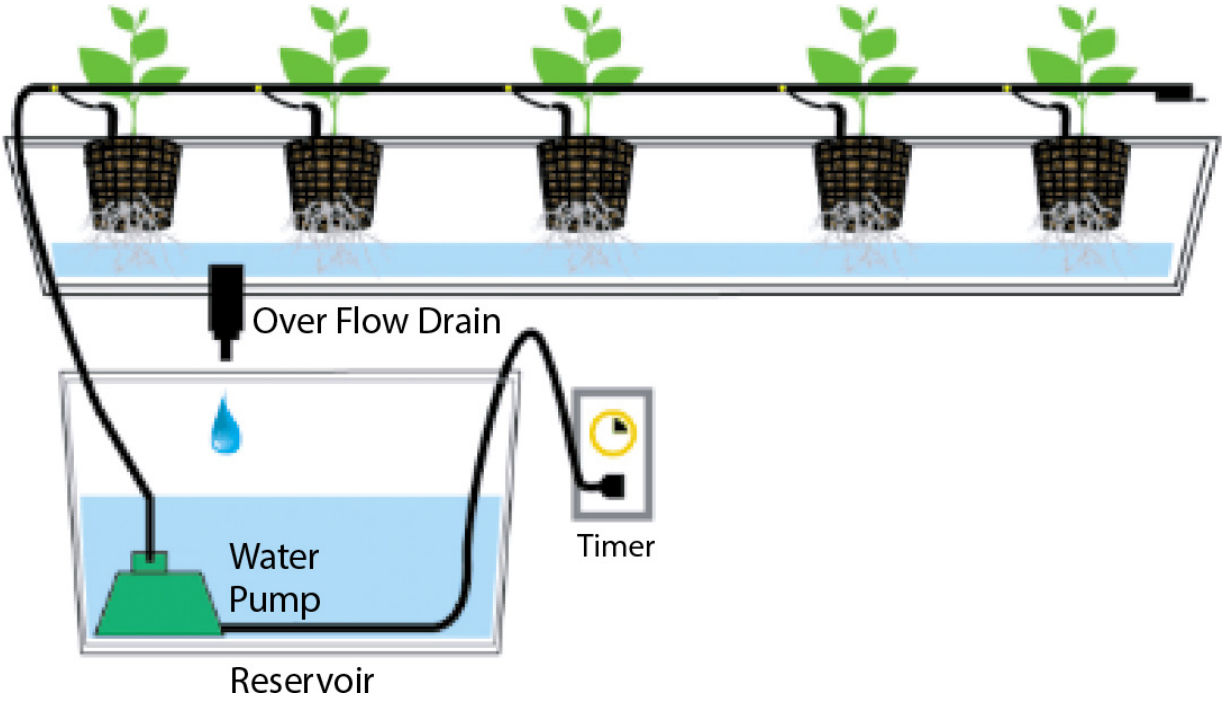

Figure 11.10 Drip irrigation smart system (source: https://thestempedia.com/project/drip-irrigation-sytem/).

Sprinkler Irrigation

In this method, sprinklers are used to spray water into the air and let them fall over the ground as shown in Figure 11.11. This resembles a rainfall. The water is flown through a nozzle under pressure. It is suitable for lands that have uneven landscapes. This method can be used on all types of crops and vegetables. It can save water as a timer can be used for spraying water and can be used to cover large areas. This method involves an initial setup cost and may be susceptible to wind and high temperature.

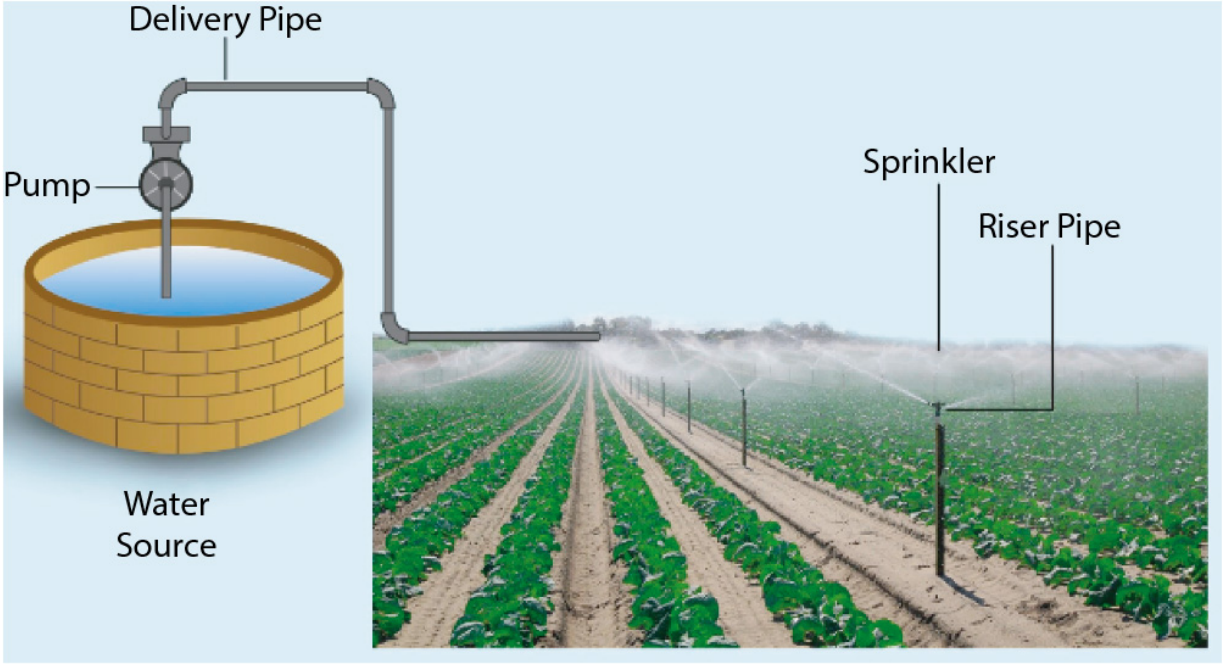

Figure 11.11 Sprinkler irrigation system [9].

Tank Irrigation

This ancient irrigation method is used in states like Karnataka, Odisha, Rajasthan, and Kerala. This involves constructing artificial reservoirs of mud banks across small streams to store rain water. The connected channels can be used to fed water to the crops. These are most popular in areas where it is difficult to dig wells due to stony grounds. The cost involved in the construction of tanks is less, and these have a longer life, but in summers the tanks can dry up due to high temperature. Carrying water from the tank to the agriculture land may involve some initial cost as shown in Figure 11.12.

Figure 11.12 Tank irrigation (source: http://jainpipe.com/irrigation/Filter/Jain_Fertilizer_Tank_Silver.htm).

Tube Well Irrigation

In places which are rich in ground water, a tube well can be installed near the agricultural land. The tube well runs on electricity and can be used to irrigate large areas as shown in Figure 11.13. This is a popular irrigation method used in states like Punjab, UP, and Bihar. This method of irrigation can give a constant supply of water to the areas. It can be installed quickly in a small area. The quality of water is good as it includes all essential minerals and chemicals of ground water. Excessive use may lead to depletion of ground water. Installation of submersibles is required, and the cost of electricity is also there.

Canal Irrigation

This type of irrigation usually exists in areas that are rich in fertile soil and have a good source of water. Thus, canal irrigation likely exists in areas of UP, Punjab, and Haryana. These can be built by digging deep grounds or artificially using stones and bricks as shown in Figure 11.14.

Surface Irrigation

A proper path can be dug on the land surface to circulate water to flow from one end of the land to another end. This method is most commonly used, and it also utilizes rain water properly. This method is cost-effective, but it is highly inefficient. This method can be utilized efficiently by proper management and in a controlled manner. It can often lead to problems like waterlogging and deep drainage.

Figure 11.13 Tube well irrigation (source: https://irrigation.assam.gov.in/frontimpotentdata/popular-systems-of-irrigation).

Figure 11.14 Canal irrigation (source: https://www.drishtiias.com/images/uploads/1581943559_image4.png).

11.5 Review on Smart Irrigation Systems

Megalingam and Gedela proposed a cost-effective solar power smart irrigation system for farmers in 2018 that can pump water automatically when the moisture content of the soil is below critical level [10]. This can be determined by the moisture sensor planted in the field. This proposed approach helps Indian farmers overcome the current energy crisis. This device conserves water by minimizing water losses and

electricity by lowering the grid power demand. This project uses a solar-based automated water pump system, which is economical and eco-friendly, instead of using diesel-based water pumps that are more expensive and harm the environment as shown in Figure 11.15.

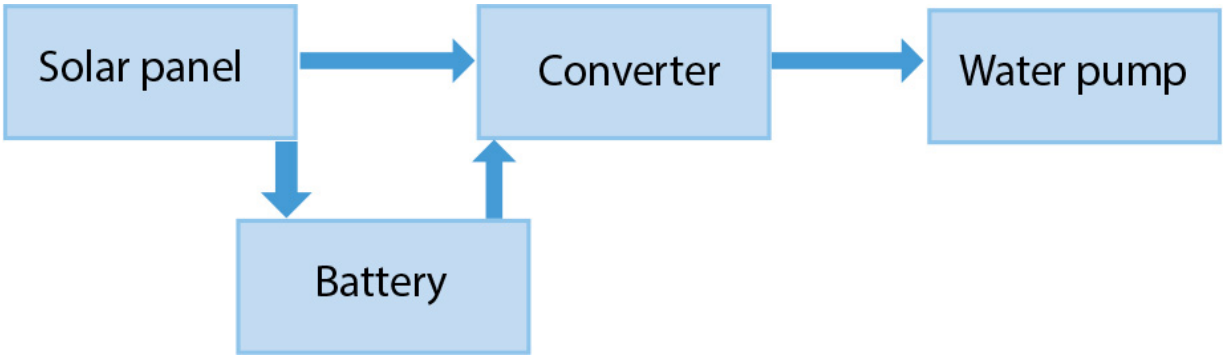

Figure 11.15 Block diagram of a solar panel pumping module.

An IoT-based smart irrigation system was proposed by Sreyash *et al.* This system comprises of soil and temperature sensors, which sense the soil conditions and send data to Think Speak. After this, the farmer can decide whether to switch on or off the water pump [11]. The goal of this suggested solution is to create a productive irrigation system that will enable the pump to switch on and off when the presence of ground moisture is recognized. The basic components of these systems are wireless sensors, a soil moisture valve, and short message services (SMS). In these systems, the farmer must manually run the

pump and must turn the pump off as the soil becomes more wet. That may become chaotic at times.

In 2018, Mishra D. and his colleagues suggested an automated system that merely makes use of sensors to measure how much water should be supplied to the crops using soil moisture sensors. It would be impossible to effectively monitor and manage the system with just one sensor [12].

There is an open-source technology-based smart irrigation prediction system that predicts field irrigation needs based on ground characteristics such as soil moisture, soil temperature, and ambient variables, as well as weather forecast data from the Internet [13]. The sensor nodes engaged in ground and environmental sensing incorporate crop field, soil moisture, relative humidity, and soil and air temperature. The system uses intelligent algorithm that consider various parameters like sensor data, UV, and weather predictions. The system has been set up and deployed. It collects all the data from the sensor nodes and then shares it over the cloud. A real-time monitoring and decision support system that analyzes the collected data and make forecasting is constructed.

A prototype for measuring the quantity of soil moisture and temperature is presented by Darshna *et al.* in their study [14]. A

range is set in the system for the soil temperature and moisture. These setting can be changed with types of crop, environment, and soil conditions. The proposed solution can be switched on or off whenever the environmental conditions change from the settings. In high temperature or when the moisture content in the soil is low, it will automatically switch on the system. However, this approach fails to align with the weather conditions to determine the predicted rainfall that would have saved the water supplied to the soil.

Feng [15] likewise highlights the use of sensor networks and an Internetbased irrigation system for agricultural areas. They set up a wireless network of sensor nodes using routing protocols to collect timely data on a regular basis. The data is analyzed and can be used to set up an initial schedule for watering the plants using the controllers. The authors deploy local servers that save all the data into a database that is linked with the Internet. However, the authors do not take the benefits of cloud computing (such as high availability, pay-as-you-go, metered service, flexibility, and scalability) that may be utilized to enhance the irrigation system. They instead employ a database that is hosted on a local server that is linked to the Internet.

Moreover, the concept for a home automation system put out by Agrawal *et al.* [16] uses readily available, affordable, and

energy-efficient components including raspberry pi, Arduino microcontrollers, XBee modules, and relay boards. A smart drip irrigation system is created using solenoid valves and ultrasonic sensors. A notification or a command can be transferred to water the plants in a big agricultural setup or in small areas. The authors of ISD2017 Cyprus do not, however, emphasize how they automate the irrigation in their study. Therefore, the above-mentioned benefits of cloud computing are not utilized by that solution.

The authors Zaier *et al.* [17] offer a completely automated and wireless irrigation control system that does not rely on judgements about the time and volume of watering. A wireless network of sensor nodes is set up in the agricultural area where one node will act as a master node and the other nodes behave like slaves. The slaves will collect all the data and forward it to the master. The master node will further forward the data to the host computer. Their irrigation control and monitoring program is installed in the farm's host computer, which keeps track of the conditions of all the crops and operates the valves using timers and thresholds. There are approximately 14 farms that have the smart irrigation system installed. However, the processing of data for this approach does not utilize cloud computing.

In contrast to this, a model proposed by Pedro [18] collects data from the environment and stores it to determine how much water is required for irrigation. In this set up, Arduino One module is used for collecting data from all sensors. Through online services, this information is transmitted to a cloud database. This cloud-based setup will determine whether a crop needs water in a predetermined length of time and, if so, send a command to the Arduino module to open the valve for a specified amount of time.

A sophisticated cloud-based control system and a mobile app has been proposed by S Iyer [19] in order to monitor and manage the state. The issue of long-distance communication has also been addressed. However, the suggested system is complex since it makes use of several nodes and networks. The system as a whole would be impacted by a single flaw or malfunction in a minor component.

11.6 Smart-Based Irrigation Techniques for Urban and Rural Areas

Meher *et al.* [20] conducted research on edge-cloud-based IoT's plant irrigation and water logging systems. For use in making decisions, this technology sends sensor data to the cloud. This system requires a client response before watering can begin, as

opposed to employing an automatic irrigation technique. It also lacks a machine-learning-based automated decision-making process. Even so, this technique can help manage water shortages in the country.

An irrigation decision support model was created by Goldstein *et al.* [21] and Vij *et al.* [22] using ML techniques and a ready-made irrigation data collection. They used several ML algorithms and assessed their accuracy in making irrigation choices. In contrast to the requirement for a ready data source, our work aims to create a model that gradually learns the watering requirements of any plants. Before making informed selections, our approach just requires a few manual irrigations first. Additionally, because our model is entirely dynamic and based on the analysis of data gradually gathered from the environment, it is easily adaptable to various plants with various irrigation circumstances.

The proposed irrigation model offers two more methods for figuring out how to water the plant [23]. In the first approach, the model learns the watering requirements without any pre-prepared data being given to it. In order to learn how to use this option, the system must first be trained several times using manual irrigations in accordance with the agronomists' predetermined instructions. After a sufficient number of

manual irrigations, the system gathers the required data and trains itself, and the suggested model starts to understand the plant's watering requirements. It would not need human irrigation any more after that since it can decide to water itself using machine learning techniques and the data it has acquired. In the second approach, machine learning techniques are used to enable the model to decide on irrigation on its own without the need for any pre-prepared data. The suggested model in each option improves its learning process progressively as more precise irrigation locations are given to the algorithm.

An artificial intelligence-based smart irrigation method uses real soil moisture data for watering the field automatically [24]. This system analyzes weather data to irrigate fields based on rainfall pattern and soil conditions. It uses prediction algorithms that give accurate results based on real-time data. The proposed efficient system is tested in a closed environment, and results with 80% accuracy have been achieved .

11.7 Barriers of Smart Irrigation Systems

A smart irrigation system has a number of advantages, but there are certain limitations also associated with these systems. The limitations are listed in Table 11.1.

Table 11.1 Barriers of smart irrigation systems.

High power consumption of devices	**The devices require electricity for running devices like water pump, circuitry, and sensors.**
Devices exposed to harsh environments	Different sensors are embedded on the ground and crop surface to collect information. These are exposed to heat, rain, and other dangers
Weak communication signals	The collected data from sensors are transmitted to the cloud for further analysis and prediction. Sometimes, due to weak signal strengths, the devices would not be able to cater accurate readings
Network connectivity issues	The Wi-Fi module is used for transmitting collected data from sensors to the controller. There can be Internet issues in rural or terrain lands

High power consumption of devices	**The devices require electricity for running devices like water pump, circuitry, and sensors.**
Poor IoT knowledge	The farmers are not literate enough to understand the usage and working of devices and are aware about the applications of IoT.
Security and privacy issues	The hardware devices are prone to security attacks. The hackers can collect all the readings and can even change the reading which might result to efficiencies
Short battery life	The sensors used have a short life span. They become ineffective after some time and need to be replaced with new ones.

11.8 Conclusion

This chapter gives an overview on smart irrigation systems. It also emphasizes the significance of adopting sustainable and technologically advanced irrigation strategies to address the challenges posed by way of water shortage and climate change. It also explores different styles of irrigation systems, such as floor irrigation, sprinkler irrigation, and drip irrigation, highlighting their respective strengths and limitations. It can be seen that by using technologies like artificial intelligence, smart sensors, controllers, and available applications, the crop yield can be increased and optimal use of water resources can be made.

By minimizing water wastage, enhancing crop productiveness, and making sure of food security, these systems play a pivotal function in fostering sustainable agricultural development. Additionally, the accountable control of water resources *via* properly designed irrigation structures facilitates mitigation of the damaging outcomes of droughts, preserving atmosphere fitness, and keeping biodiversity.

References

1. Mohanty, S.P. *et al.*, “Everything you wanted to know about smart cities: The Internet of things is the backbone”, *IEEE Consum. Electron. Mag.*, vol. 5, no. 3, pp. 60–70 , 2016.
2. Fan, T., Chen, Y, “A scheme of data management in the Internet of Things. 2010”, *2nd IEEE Int. Network Infrastructure Digial Content*, pp. 110–114 , 2010.
3. P. A. Tenezaca Sari, C. D. Piedra Garcia, A. S. Godoy Mendia, D. F. Merchan Piedra, E. F. Patiño Zaruma and I. P. Cedillo Orellana. “Smart irrigation system for smart farming,” in *26th International Conference on Information Systems Development*. UCLan Chipre, INSPIRE Interdisciplinary Science Promotion & Innovative Research Exploration, 2017
4. [online] Available: https://www.renkeer.com/smart-irrigation-technology-and-system/
5. [online] Available: https://extension.okstate.edu/fact-sheets/smart-irrigation-technology-controllers-and-sensors.html
6. Miles, B., Bourennane, E. B., Boucherkha, S., & Chikhi, S., “A study of LoRaWAN protocol performance for IoT applications in smart agriculture.”, *Comput. Commun.*, vol.164, pp.148-157, 2020

7. C. Fathy and H. M. Ali, “A Secure IoT-Based Irrigation System for Precision Agriculture Using the Expeditious Cipher,” *Sensors*, vol. 23, no. 4, p. 2091, Feb. 2023, doi: 10.3390/s23042091
8. N. N. Saxena., “A Literature Review on Drip Irrigation System.” *Int. J. Mod. Agric.*, vol. 10, no.1, pp. 690 –;696, 2021
9. E. S. Mohamed and N. T. Abdo, “Overview of Pumping Systems with Focus on Deferrable Irrigation Loads, 2017.
10. Rajesh Kannan Megalingam and Gedela, V. V., “Solar Powered Automated Water Pumping System for Eco-Friendly Irrigation”, *International Conference on Inventive Computing and Informatics (ICICI)*, 2017
11. Shreyash , Srashti Sharma , Subhash Nagar , Shivani Priyadardshani, “ Smart Irrigation Systems”, *Int. J. Electr. Electron. Comput. (IJECC)*, vol. 6,no. 5, pp.1-5, 2021
12. Mishra, Dweepayan, *et al.* “Automated Irrigation System-IoT Based Approach.” *3rd International Conference On Internet of Things: Smart Innovation and Usages (IoT-SIU)*. IEEE, 2018.
13. A. Goap, D. Sharma, A.K. Shukla, C.R. Krishna, “An IoT based smart irrigation management system using Machine learning and open source technologies”, Elsevier, vol. 155, pp.41-49, 2019
14. Darshna, S. *et al.*, “Smart Irrigation System”, *IOSR J. Electron. Commun. Eng. Ver. II*, vol. 10, no. 3, pp. 2278–2834, 2015

15. Zhang Feng, “Research on water-saving irrigation automatic control system based on internet of things”, *Int. Conf. Electr. Inf. Control Eng*, pp. 2541–2544, 2011
16. Agrawal, N., Singhal, S, “: Smart drip irrigation system using raspberry pi and Arduino”. *Int. Conf. Comput. Commun. Autom.* pp. 928–932, 2015.
17. Zaier, R. *et al.*, “Design and implementation of smart irrigation system for groundwater use at farm scale.”, In: *Proceedings of 2015 7th International Conference on Modelling, Identification and Control (ICMIC 2015)*, Institute of Electrical and Electronics Engineers
18. Priscila, C., “Smart irrigation system for smart farming., *26th international conference on information systems development (isd2017 cyprus)*, 2017
19. Iyer, Saikumar and Patro, PankajKumar and Kapadia, Rajdeep and Das, Abhishek and Cheriyan, Sanish and Ansari, Namrata, “IoT based Cost-effective Centralised Smart Irrigation System using LoRa”, *Proceedings of the 3rd International Conference on Advances in Science & Technology (ICAST)*, 2020
20. C. P. Meher, A. Sahoo and S. Sharma, “IoT based Irrigation and Water Logging monitoring system using Arduino and Cloud Computing,” *2019 International Conference on Vision*

Towards Emerging Trends in Communication and Networking (ViTECoN), Vellore, India, 2019, pp. 1-5

21. A. Goldstein, L. Fink, A. Meitin, S. Bohadana, O. Lutenberg, G. Lavid, “Applying machine learning on sensor data for irrigation recommendations: revealing the agronomist’s tacit knowledge.”, *Precis. Agric.* ,vol. 19, no. 3, pp. 421-444,2018
22. A. Vij, S. Vijendra, A. Jain, S. Bajaj, A. Bassi, A. Sharma, “IoT and Machine Learning Approaches for Automation of Farm Irrigation System”, *Procedia Comput. Sci.*, vol. 167, pp. 1250 -;1257, 2020.
23. K. Cagri Serdaroglu, C. Onel and S. Baydere, “IoT Based Smart Plant Irrigation System with Enhanced learning,” *2020 IEEE Computing, Communications and IoT Applications (ComComAp)*, Beijing, China,pp. 1-6, 2020
24. Savita *et al.* “Autonomous Crop Irrigation System using Artificial Intelligence”, *Int. J. Eng. Adv. Technol. (IJEAT)*, vol. 8 no.5, 2019

Note

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12
Implementation of Smart Irrigation Systems

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Abstract

The economic development of smart agriculture and watering relies heavily on manufacturing industries that utilize Internet of Things (IoT) technologies. Water scarcity has become a global issue, particularly in remote and desert regions. A smart irrigation system is necessary to optimize water usage, monitor agriculture, and lower energy costs. This chapter proposes a real-time monitoring and auto-watering system that employs predictive mathematical models based on IoT technology to

regulate the water rate required for optimal crop growth efficiently. It detects soil moisture levels and automatically waters crops when they fall below a predetermined threshold. This system conserves water and promotes interoperability among various sensing data streams to support large-scale agricultural analytics. By reducing water consumption by over 75%, enhancing irrigation efficiency, increasing production, and decreasing maintenance and operator employee costs, this proposed system offers a viable solution to water challenges in smart agriculture. The proposed irrigation system mainly benefits farmers in remote and desert areas, where water and labor costs are scarce. In addition, the system also ensures interoperability among various sensing data streams to support large-scale agricultural analytics. Overall, the proposed system provides a sustainable and efficient solution to the problem of water scarcity in agriculture, promoting sustainable farming practices and supporting economic development in the industry.

Keywords**:** Smart irrigation, Internet of Things (IoT), auto-watering system, irrigation efficiency, sustainable solutions, predictive mathematical models

12.1 Introduction

In recent years, effective management of irrigation systems has become increasingly important worldwide, primarily due to water resource scarcity caused by climate changes, high climatic temperatures, and the adverse impact of human activities on the environment [1]. The issue of the availability of potable water has become a global problem affecting human lives in many countries, especially in remote desert areas. The Sultanate of Oman faces a major challenge in providing and managing potable water resources to ensure optimal population growth and vegetation, as it contains vast desert areas with limited potable water resources [2]. Therefore, countries should reduce the drainage of water for drinking and irrigation through the application of automatic irrigation, which provides many advantages, including watering as needed and a specific tool for water flow, as well as reducing the labor needed to manage farms, watering and cutting water when not needed efficiently compared to manual systems, and reduce water runoff and nutrients. However, automatic irrigation systems have some defects, including the costs of purchasing and installing equipment, its installation and expertise to maintain it, and the need to ensure its reliability due to human errors during installation [3].

In order to address these problems, the automatic irrigation system must be characterized by long operating capacity and efficiency, which improves the process when needed, which helps reduce water consumption and thus reduce the cost of irrigation and increase profits. Auto-irrigation aims to provide the needed amount of water for optimal growth and decide the proper timing for crop irrigation, which will achieve the highest performance [4]. Technological advances in the manufacture of equipment and the development of information and communication technology helped to develop automated irrigation methods that require less water for irrigation compared to manual techniques. Introducing modern smart equipment in the irrigation equipment industry improves crop production by reducing excess water in the field, which helps in better crop growth and reduces water consumption. Therefore, the issue of improving water use efficiency and reducing energy use through irrigation by smart, automated methods has become important areas of research. Creating and developing solutions based on an independent architecture is ideal for processing and analyzing the data that we collect through sensors for agricultural parameters to support real-time monitoring. This will facilitate interoperability between homogeneous and non-homogeneous sensor data streams to support large-scale agricultural analyses [5].

The demand for precision farming has increased recently due to the growing need to produce food with high-quality crops at a minimum cost and reduce environmental pollution. The emergence of various wireless sensor network technologies helped to find optimal solutions in the agricultural field for collecting, processing, and analyzing relevant agricultural information and activities. However, we must remember that different characteristics of farms require different solutions to suit, for example, different soil types, types of fertilizers, water quantity requirements, and ambient weather conditions [6].

The proposed automatic irrigation model is suggested to solve the mentioned challenges. It utilizes a mathematical model based on the land's nature and climatic conditions, such as temperature, pressure, and humidity. The proposed model helps manage and monitor the crops' needs efficiently and can easily and quickly be modified to adapt to climatic conditions. Using sensors ensures efficient water use and reduces water consumption and the energy required for irrigation. The automated irrigation system uses a solar panel to reduce the need for an electrical power source, and there is no need for labor to turn the motor on and off. The existing traditional systems require a connection to the internet to send/receive the external data storage to manage/control the crop's needs.

The proposed approach can automatically manage and control crops' needs without Internet connectivity. It is embedded in the field and can be easily updated to accommodate new conditions. An efficient irrigation system is essential for managing water resources, reducing water consumption and energy costs and ensuring optimal crop growth. Advancements in industry tools, information technology, and communication have enabled the development of intelligent irrigation methods that use wireless sensor network technologies to collect, process, manage, and analyze relevant agricultural information and farming activities [7].

The suggested approach automatically controls and handles the crop's necessities without an Internet connection. It is lodged throughout the field and can be conveniently updated to accustom to foreign conditions. In order to manage water resources, limit water and energy expenditures, and allow for optimum crop growth, an efficient irrigation system is required. Technological development in equipment and communications helped produce modern irrigation methods integrated with wireless sensors in order to collect and analyze field data and to manage the various agricultural activities.

This chapter proposes a solution for developing automated irrigation systems that can collect, analyze, and process field

data in real time using distributed field-scale sensors to support large-scale agricultural analyses. The main contributions of the proposed approach include reducing water consumption in irrigation, enhancing farm productivity, managing and monitoring crop needs efficiently, reducing expenses needed to manage farms, using an internal wireless network that covers several adjacent fields, generating the energy needed for self-operation by solar panels, and simulating irrigation time and crop needs.

12.2 Literature Survey

Most modern irrigation systems are characterized as self-operating and distribute water to crops based on a schedule to ensure that the crop is given its need of water, which reduces waste of water, reduces costs, and provides ideal irrigation conditions to ensure crop health. Many studies reviewed the benefits and disadvantages of modern irrigation systems, showing that they helped ration water use, improved production, and reduced irrigation and labor costs.

A work that uses a hybrid model consisting of K-nearest neighbors, long short-term memory, gradient boosting-based tree, and Spearman's rank correlation for decision-making using observation parameters was proposed by Vianny DM *et*

al. [1] and implemented in banana crops. The results displayed 31.4% optimization in water requirement in 2020, leading to a decrease in both freshwater usage and energy wastage. A mathematical prediction model for real-time monitoring and automatic watering based on IoT technologies has been proposed by Yousif and Abdalgader [2]. This system efficiently controls the amount of water crops need, providing them with an optimal water level while conserving water resources. The system ensures that diverse sensing data streams can support largescale agricultural analytics. The mathematical model is integrated into the Arduino system, which senses the soil's moisture level and automatically waters the crops if it falls below the pre-defined threshold value. It reduces water consumption by over 70% and increases production through irrigation optimization, improving the watering system's efficiency.

Additionally, it reduces water and energy consumption as well as maintenance costs. The authors, Alves RG *et al.*, have introduced a digital twin irrigation system [3] that utilizes the FIWARE platform of the Internet of Things. The system employs a discrete event simulation model within Siemens Crop to imitate the behavior of an irrigation system based on a particular scenario application. The platform is divided into layers that collect, aggregate, and process data on soil, weather,

and irrigation prescriptions daily. This proposed system enables evaluating of the IoT platform and various irrigation systems before their implementation in the field, thereby enhancing farm operations and reducing water usage. By enabling farmers to collect and evaluate data on soil, weather, and crops, the system provides multiple irrigation strategies for optimal results.

Abdalgader and Yousif [4] developed an agricultural irrigation management model based on sensor-enabled architecture, which is implemented in Oman. This model utilizes sensor technology to monitor and control dynamic environmental factors, such as soil moisture level and other weather conditions to manage irrigation needs. The study by Saddik *et al.* [5] also proposed a real-time evaluation mixed embedded model to enhance the precision of agriculture based on different metrics. The model has utilized machine learning approaches to manage various sensor technologies to optimize crop production and reduce water waste rate. In addition, Rahaman and Azharuddin [6] reviewed machine learning techniques for simulating and predicting agriculture models using wireless sensor networks. This approach helps farmers decide the needs of their crops and environment based on real-time sensing data.

Another study by Lakshmi *et al.* [7] utilized a smart irrigation model using an IoT-coupled precision sensor for modern agriculture applications. The proposed model employed machine learning algorithms to control and enhance water use and amount, which improves crop products. The study by Harun AN *et al.* [8] improved an IoT indoor farming system for remotely controlling and monitoring Brassica chinensis crops' growth. The system used the light sensors method to measure and monitor environmental factors, such as carbon dioxide range, temperature, humidity, and leaf area. They proposed an IoT-embedded irrigation system based on a pulse-width-modulated actuator. They showed that the deployed light spectrum intensified effectively, which uses the water efficiency in all parts of the physiology of *Brassica chinensis* crop. The amount of leaf photosynthesis and the chlorophyll production are ideal.

Farooq H *et al.* [9] and his team explored the latest research on agriculture systems using IoT from 2015 to 2020. They explored and analyzed different features, including results, scope, and methodology, which helped to provide an inclusive summary. The study raised a set of questions to discuss the effectiveness of each study by analyzing the scope, methodology, and results and by identifying problems. The study also identified the characteristics of the platforms needed to create an efficient

automatic irrigation system, which is the beginning for researchers to launch experiments in the future. In their study, Touil *et al.* [10] thoroughly examined the current irrigation systems, both traditional and smart technologies, to assess their impact on water conservation, crop production, and quality. By analyzing relevant research papers, they discovered that the effectiveness of soil moisture sensors in improving water use efficiency is contingent upon the farmers' volumetric moisture content threshold value. Rain sensors, soil moisture sensors, and evapotranspiration controllers effectively preserved crop growth and quality while conserving water, with savings ranging from 7% to 50%, 20% to 92%, and 20% to 71%, respectively. In contrast, the conventional controlled deficit irrigation technique was found to decrease crop production by up to 25%, with water savings of only up to 13%.

The research gaps in automatic irrigation systems include the following:

- Standardization: There needs to be more standardization regarding the communication protocols, data formats, and interfaces used in different irrigation systems. This is one of the challenges that make integrating different systems and technologies difficult and, as a result, delay the new advanced development and efficient systems.

- Data management: As a result of the increasing amount of data being collected from different sensors and other monitoring systems, there is a need for advanced technologies to manage the data efficiently. Data management include several processes, such as collecting and storing, analyzing data, and then visualizing them.
- Robustness: More advanced hardware and software must be developed to manage and control automatic irrigation systems. Thus, we should produce irrigation devices that are robust, flexible, and make it easy to diagnose device failures and other environmental conditions.
- Economic feasibility: More studies are needed to discuss how to propose effective automatic irrigation systems with low prices by reducing the cost of installation and maintenance. This will help farmers with small agriculture to widely implement the modern irrigation system and increase crop production.
- Human factors: Installing and operating modern automatic irrigation systems require training workers to operate and manage the system efficiently. The training includes installing, controlling, data sensing, and maintenance when an error occurs.

Therefore, to fill the current research gaps, we need to conduct more research on how to implement and develop efficient

irrigation systems based on machine learning approaches. The mathematical models will help to stimulate and predict the factors that affect crop growth, such as water requirements, weather conditions, and soil types. More cooperation between researchers and industry experts should also be made to develop automated irrigation systems that are efficient and sustainable [11].

One possible solution is using mathematical models derived from the land's nature and climatic conditions, such as temperature and humidity, to manage and monitor crop needs efficiently. Using sensors can also help ensure efficient water use, reduce water consumption, and minimize the energy required for irrigation. Renewable energy sources can also reduce the need for labor to operate and run the motor, thus minimizing energy costs [12].

12.3 Irrigation System Design

Irrigation systems typically consist of several layers and components that work together to deliver water to crops efficiently [13]. The layers and components of an irrigation system shown in Figure 12.1 may include the following:

- Data source layer: The water source is where the water comes from, such as a well, pond, lake, river, or municipal water supply. The data source layer is responsible for sensing agricultural parameters using different sensors installed in the soil and surrounding environment.
- Data transmitter layer: This is designed to transmit the sensing water data from the soil sensor to the processing layer. These sensors are connected to a sensor node and a built-in solar energy source. The sensors transmit the collected data to the data processing layer using communication channels like Wi-Fi. Using an integrated solar panel to generate the needed power will reduce power consumption and help charge batteries.

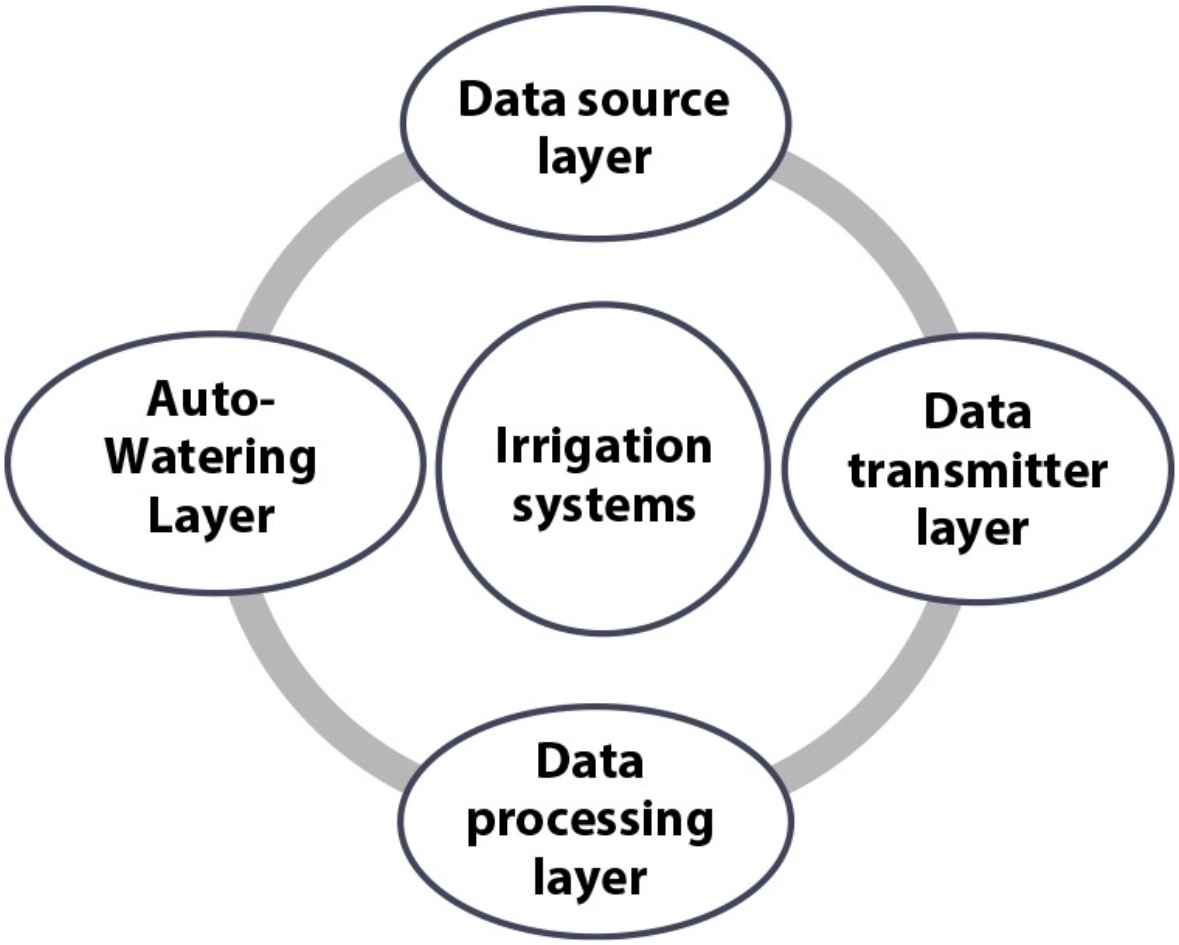

Figure 12.1 Irrigation system layers and components.

- Data processing layer: This layer helps to analyze the collected data and provides real-time information to farmers who can remotely monitor the agricultural status and parameters using IoT communication technologies.
- Auto-watering layer: This layer is used to water the crop with the needed water automatically without human involvement. This design ensures that crops get the right amount of water, which helps their growth and health.

There are many types of irrigation equipment, such as drip irrigation through pipes or sprinklers or other irrigation methods. Therefore, irrigation systems must include an efficient control system to manage irrigation scheduling and water flow by introducing timers, remote sensors, and other automated devices. Hence, there is a need to design an automatic irrigation system that controls water levels through a mathematical model to ensure that crops receive the appropriate amount of water, which increases production and maintains crop health.

12.4 Irrigation System Implementation Benefits

Irrigation is one of the important agricultural tasks to ensure that crops receive the water required to promote growth and increase production. With the great progress in industries and communications technology, modern irrigation systems have become more complex because they include sensors and advanced watering. Advances in modern machine learning algorithms and Internet of Things (IoT) technologies have also contributed to improving water use efficiency and improving crop production [14]. Hence, there is a need to study irrigation systems' classification schemes based on multiple criteria such as soil type, sensors, machine learning algorithm, type of communication networks, and the Internet of Things.

This chapter aims to explore these classifications in detail and their benefits while reviewing their main advantages.

12.4.1 Sensor Types

The irrigation system's efficiency is greatly affected by the type of sensors used to read and send data to the treatment site, including the following:

- Soil moisture sensors: To measure the degree of moisture in the soil for scheduling of irrigation time and the required amount of water.
- Leaf moisture sensors: To measure the crop leaves' moisture and monitor the leaf intensity moisture. It helps to identify if the plants are growing normally or infected with a disease such as a fungal infection.
- Rain sensors: To collect and detect rainfall data, it automatically stops if irrigation is unnecessary.
- Flow sensors: These sensors measure the strength of water flow in the pipes and detect leakage points or any other problems to prevent wasting water.
- Evaporation sensors: These sensors measure the percentage of water lost through the evaporation and transpiration process and help adjust the irrigation indicators according to evaporation rates and field needs.

- Weather sensors: Weather sensors consist of several types, such as temperature, humidity, atmospheric pressure, wind direction, and speed, of which the farmer need to know to control crop growth and water requirements.

12.4.2 Machine Learning Methods

According to a conducted survey of literatures about the machine learning methods that are utilized or have applications in agriculture, following is a summary of them:

- K-nearest neighbors: It is a supervised learning classifier that is used to classify the current data into similar categories and to predict the output for unseen data.
- Gradient boosting-based tree: This method uses successive steps in the learning process to create decision trees with different categories to improve the accuracy of the predicted output.
- Long short-term memory: This method is a type of deep learning method using recurrent neural network that used time-series data and order dependence to capture the longterm dependencies between the data.
- Spearman's rank correlation: This method measures the Pearson correlation between two variables that indicates the strength of the relationship.

12.4.3 IoT Methods

Most modern irrigation systems used to transmit data and read field data depend on using different sensors that are controlled remotely using many Internets of Things methods to improve their efficiency and effectiveness [15]. Here is an explanation of some of the most common types:

- Wireless sensor networks (WSNs): Wireless sensor networks are distributed over the agricultural field by several sensors that communicate wirelessly and are controlled by a central console. Sensors help manage the field, read data, schedule irrigation times, and control water flow and quantity, such as measuring soil moisture, temperature, atmospheric pressure, air humidity, and other environmental factors.
- Cloud-based irrigation systems: Cloud-based irrigation systems depend on controlling a set of remotely read and stored sensors using cloud computing systems to manage and regulate the irrigation process: where sensors measure environmental conditions, and data is sent via wireless networks to the cloud, where it is stored, analyzed, and then processed to determine optimal irrigation schedules and adjust the water flow. Irrigation schedules can also be reset and modified remotely according to field needs *via* the web or a mobile application.

- Mobile sensor networks: Portable sensor networks are installed on autonomous vehicles controlled by a control panel, such as drones or robots. It collects field, vegetation, and weather data, such as soil moisture, temperature, and other environmental factors. Sensors read the field data and transmit it wirelessly to a central control unit for processing, which then adjusts the irrigation schedules based on the data and controls water flow.
- Precision irrigation: Precision irrigation depends on installing sensors that target specific field areas to control the water flow rates and reach the crops directly through drip irrigation or pipe sprinklers, which help to ration water use in the required quantities, improve crop growth, and increase production quantities.
- Digital twins: To properly study the field conditions, a virtual replica is created, and its performance is simulated by applying different irrigation scenarios and then improving the system's performance and outputs. One of the benefits of using digital twins is helping farmers study and simulate field and crop data and make critical decisions about irrigation systems and their scheduling to reduce water waste and improve crop production.

Table 12.1 Comparison of IoT transmission technologies.

Transmission technology	Data rate (bandwidth)	Transmission range	Power consump
LoRa	0.3–50 kbps	2–5 km	1.5 μA
WiFi	20 MHz, up to 160 MHz	Indoor: 30–45 m, outdoor: 20–100 m	Varying f a few milliamp to a few hundred milliamp
ZigBee	2–5 MHz	Indoor: 9–15 m, outdoor: several hundred meters	Transmit mA and receive 2 mA

There are several ways to read field data by sensors and connect to the irrigation system wirelessly after determining

the system requirements, such as range, bandwidth, and power consumption percentage, as shown in Table 12.1.

Modern irrigation systems can utilize any communication network, such as Wi-Fi, LoRa, and Zigbee. Wi-Fi can be used with irrigation systems that require transmitting large amounts of data, while the LoRa network is used with systems spread over a wide area. Finally, the Zigbee network is used with irrigation systems for small fields that require low power and have intermittent data transmission.

12.4.4 Land Types

Understanding the type of land is crucial for determining the most suitable irrigation system because different types of land have varying water retention capacities, soil types, and topographies [16].

Using the wrong irrigation system on a particular type of land can result in poor crop production, water wastage, and increased costs—for instance, using flood irrigation on sandy soil with low water retention can lead to water wastage, while better options for clay soils can retain water for extended periods. Therefore, it is important to know the type of land to

implement the appropriate irrigation model that fits the needs of this land type.

The next section will explain some types of irrigation systems based on the land type.

- Field irrigation systems: Field irrigation systems are used on large open farms where various crops are grown on a large scale. Several irrigation methods can be used, depending on the need of the field, the type of soil used, and the amount of water available. Irrigation can be used by immersion, where the water is opened to the field until it is completely flooded, or by sprinkler irrigation, and watering is done here using sprinklers to spray water according to specific times and in limited quantities. Furthermore, drip irrigation uses pipes that deliver water directly to the crop sites.
- Greenhouse irrigation systems: Greenhouse irrigation systems are used in closed farms and greenhouses that are used to grow crops and provide certain environmental conditions according to the type of crops and are under the supervision and control of farmers. Here different irrigation systems can be used, including sprinkler, drip, or watering systems that adopt tidal properties to provide the crops with the water and other nutrients that they need.

- Landscape irrigation systems: These irrigate homes or small public gardens, lawns, and other ornamental landscapes. These systems use two main types of irrigation—drip irrigation and irrigation using sprinklers—with sensors to control and monitor the water levels and adjust the irrigation schedule according to the need.

Machine learning approaches are used to develop automated decision-making systems, which could be implemented to develop automatic irrigation models to improve water use efficiency and irrigation scheduling. However, it requires significant data inputs that may be costly to implement. Table 12.2 presents a comparison of irrigation technologies.

On the other hand, an Internet of Things network of interconnected devices is used to communicate in real time. It is used to monitor and control irrigation and remotely access data, which requires significant investment in hardware and software. The use of sensors also has some advantages and disadvantages, such as soil moisture sensors being used to accurately measure water content in the soil. However, installing and maintaining many types of sensors, such as weather sensors, including temperature, pressure, rainfall, and humidity, may be expensive.

Table 12.2 Comparison of irrigation technologies.

Author/year	ML method	IoT methods	Se
Vianny DM *et al.* 2020 [1]	Hybrid model consisting of K-nearest neighbors, gradient boosting-based tree	ZigBee	So se wa mo se (H RS ter se (T hu se (H
Yousif and Abdalgader 2022 [2]	Mathematical model/linear regression	WiFi/LoRa	So se ter se hu se

Author/year	ML method	IoT methods	Se
Alves *et al.* 2023 [3]	Discrete event simulation model	LoRa	Wa soi
Abdalgader and Yousif 2022 [4]	Estimation mathematical model based on the reference ETo value obtained using Penman–Monteith (PM-ETo) and Hargreaves–Samani (HS-ETo)	LoRa, ZigBee, and WiFi	Te pr hu se mo se ra se we se sp di se lev
Saddik *et al.* 2021 [5]	Mixed system Hardware/Software Co-design approach	Embedded implementation	Se im re

Author/year	ML method	IoT methods	Se
		based on OpenMP	
Lakshmi *et al.* 2022 [7]	Intelligent IoT sensor with a coupled precision irrigation model	LoRa	Wa cr pr
Harun AN *et al.* [8]	(IoT) monitoring system for growth optimization of Brassica chinensis	WiFi, ZigBee	Mo co va ar
Farooq H *et al.* 2020 [9]	Smart farming based on IoT	LoRa, ZigBee, WiFi	Te pr hu se mo se ra se we

Author/year	ML method	IoT methods	Se
			se sp di se le
Touil *et al.* 2022 [10]	Review conventional and smart irrigation technologies		So m ra an vo m co

12.5 Irrigation System Benefits

Irrigation systems are used to deliver water to crops or landscapes, which introduces many benefits, including the following:

- Increasing crop production: Using automatic irrigation systems allows to control and manage the irrigation amount

and timing that give crops the needed amount and increase crop production.

- Constant water supply: The automatic irrigation systems supplied the crops with a fixed amount of water based on the prescheduled times, regardless of the surrounding weather conditions.
- Water conservation: The automatic irrigation systems manage the water levels efficiently and reduce water resource waste.
- Other factors essential to improve irrigation systems include reducing labor costs, improving soil structure, and increasing flexibility to allow farmers to manage water use and adjust irrigation amounts.
- Therefore, the automatic irrigation system efficiently manages the water resources and reduces the water waste rate, which increases crop production.

12.6 Results and Discussion

This section reviews the experimental implementation and conditions. It is implemented in a field in Sohar, Oman. Figure 12.2 shows the steps for implementing the proposed irrigation system. The weather station is installed with the needed sensors to capture real data based on weather and crop conditions. The weather station has different sensors, such as temperature,

humidity, pressure, soil moisture, leaf wetness, wind speed, direction, and rain level. Then, the sensors collected data and transmit it using IoT methods, such as LoRa, WiFi, and ZigBee.

Figure 12.3 presents a block diagram of the proposed irrigation system containing an Arduino-embedded system. In this experiment, the Arduino-embedded system was programmed to control the water valve automatically based on the proposed mathematical model. It meets the crop's needs based on various environmental factors such as temperature, precipitation, and humidity, which optimizes water resources and reduces water wastage.

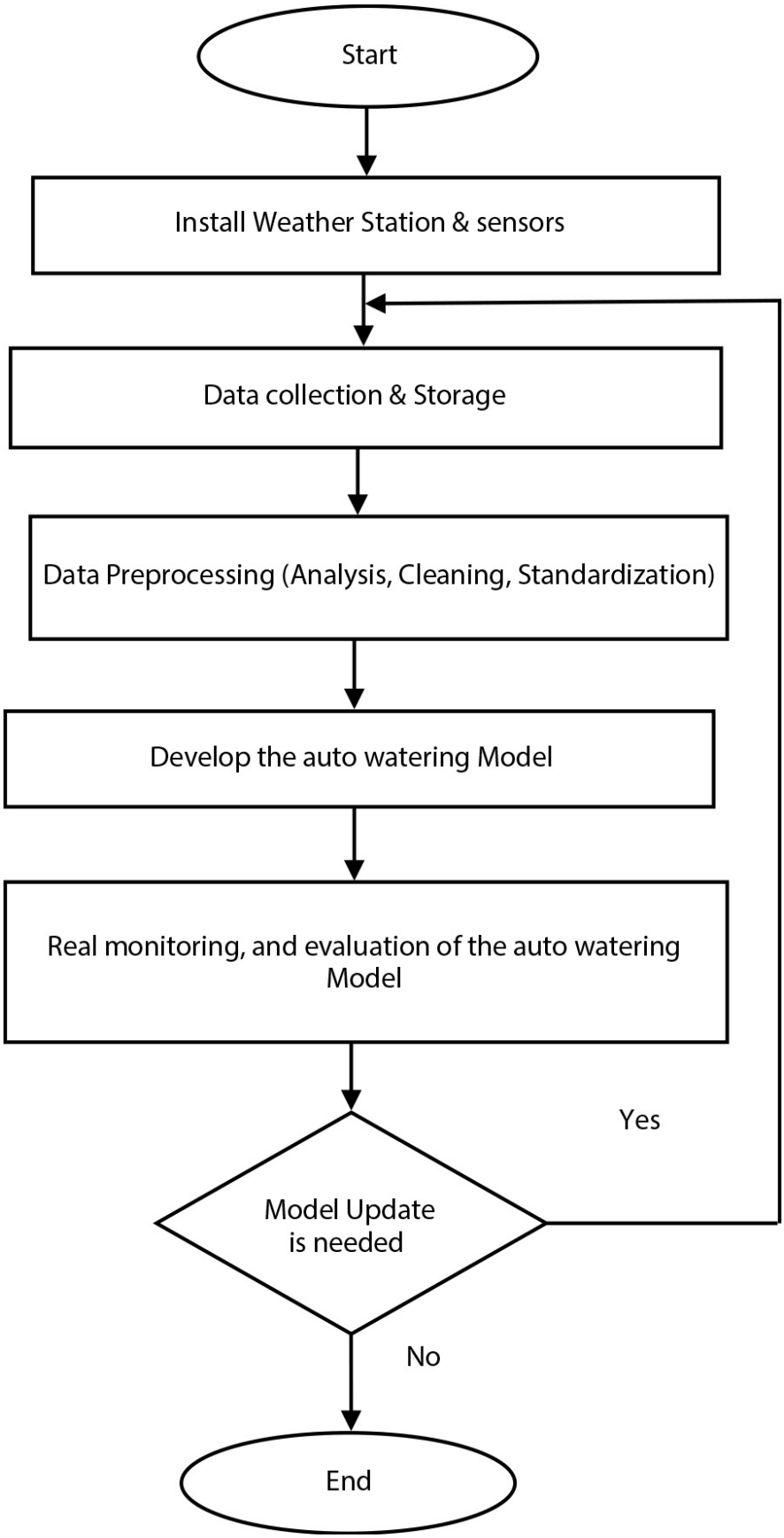
Start
Install Weather Station & sensors
Data collection & Storage
Data Preprocessing (Analysis, Cleaning, Standardization)
Develop the auto watering Model
Real monitoring, and evaluation of the auto watering Model
Model Update is needed
Yes
No
End

Figure 12.2 Flow chart of implementing the step of the auto-watering model.

The proposed automatic irrigation system ensures the optimal quantity of water for the crops each time, which will enhance crop growth and reduce waste watering. The Arduino-embedded system is easily modified and customized according to each crop's requirements. The essential power source of the system is provided using an integrated solar panel. The system can also operate without sunlight as the battery storage component stores the generated power and makes it available at night.

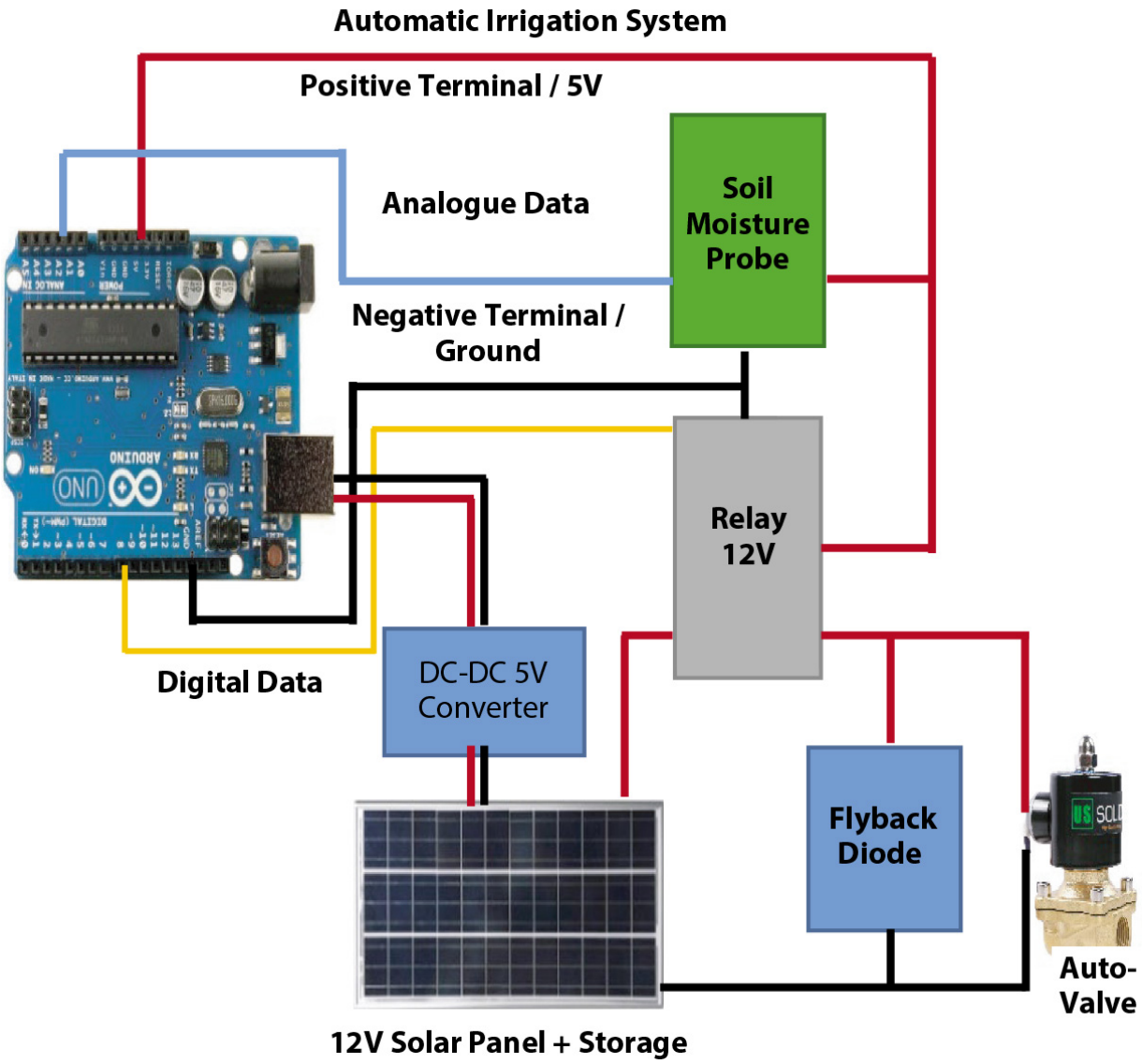

Figure 12.3 Block diagram of the proposed irrigation system.

The soil moisture probe functions by being inserted into the soil and sending data to the Arduino-embedded system concerned with the moisture level. It then utilizes the data to determine if the crop requires water. If the crop does, the system appropriately decides the quantity of water to provide. This ensures that the crop receives the right amount of water to

meet its requirements while reducing water wastage and energy consumption. This automatic irrigation system provides an efficient and sustainable method for watering crops, making it the ideal choice for anyone looking for an affordable, eco-friendly solution.

Once the data is collected, it is subjected to several processes that are cleaned, standardized, and analyzed to determine the amount of water required by the crops under various harsh conditions. The experiments are designed to compare the differences between manual and automatic irrigation system efficiencies. Figure 12.4 illustrates the results from the data collected about the soil moisture level in both irrigation systems.

Overall, the automatic irrigation system conserves water by providing the correct amount of water as required instead of manual irrigation methods. Several statistical methods are deployed to describe the statistical analysis results [17], including minimum and maximum values, median, different quartile statuses, mean, variance, and standard deviation, as presented in Table 12.3.

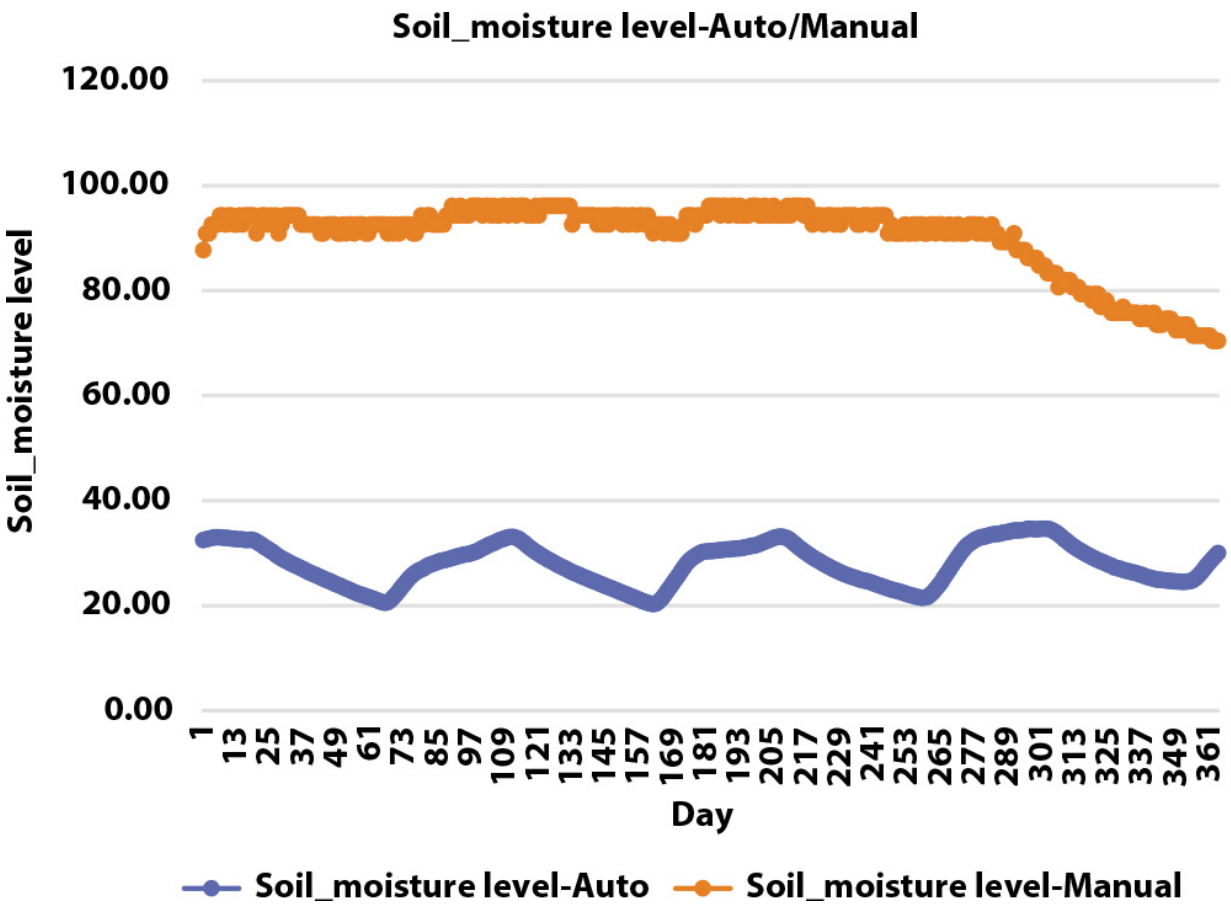

Figure 12.4 Soil moisture level in automatic and manual irrigation methods.

Table 12.3 Descriptive statistics analysis results.

Statistic	Soil moisture level, auto	Soil moisture level, manual
No. of observations	365	365
No. of missing	0	0

Statistic values	**Soil moisture level, auto**	**Soil moisture level, manual**
Minimum	20.250	69.444
Maximum	34.625	96.154
Mean	27.911	90.234
Standard deviation (*n*)	3.988	6.963

Furthermore, a mathematical model is developed based on linear and multi-regression models that explain the most impactful factors with their base value. This mathematical model is embedded in the Arduino system, which controls the required water rate based on environmental factors. Using an automatic model demonstrates that it saves more than 75% of water, which is a significant improvement over the manual system.

In conclusion, this experiment's agricultural system is a comprehensive approach to optimize water usage in harsh weather conditions. The implementation of IoT methods, statistical analysis, and mathematical modeling, combined with the Arduino system's control, can significantly reduce water usage, thus enhancing the efficiency of agriculture in the region.

12.7 Conclusion

An irrigation system is a mechanism that delivers water to crops without manual intervention, providing the ideal watering conditions to promote crop health and development. Auto-watering and irrigation systems have been shown to increase crop production, conserve water, and reduce labor costs. Various studies have proposed and implemented different approaches, such as a hybrid model of K-nearest neighbors, gradient boosting-based tree, long short-term memory, and Spearman's rank correlation for decision-making based on observation parameters; a mathematical prediction model for real-time monitoring and automatic watering based on IoT technologies; and a digital twin irrigation system that utilizes the FIWARE platform of the Internet of Things.

These approaches leverage sensor technology, machine learning algorithms, and IoT technology to optimize water usage, reduce water and energy consumption, improve crop production and quality, and enhance farm operations. The effectiveness of soil moisture sensors in improving water use efficiency depends on the farmers' volumetric moisture content threshold value. Rain sensors, soil moisture sensors, and evapotranspiration controllers effectively preserved crop growth and quality while conserving water, with savings ranging from 50% to 75%.

Future directions for the smart irrigation system include integrating sustainable irrigation methods and modern technologies in production to increase efficiency and productivity. The following are some future directions:

- Develop more automatic irrigation systems by integrating them with modern technology in manufacturing sensor systems, such as sensors for the early detection of pests and diseases in crops and their early treatment.
- Develop irrigation systems that rely on machine learning algorithms to simulate historical irrigation and weather data and then create mathematical models to predict optimal irrigation schedules, adjust these schedules according to Crop

needs, and predict Crop growth, which helps improve water flow efficiency and increase crop productivity.

- Develop irrigation systems using drone image processing to analyze the captured high-resolution images and then make the appropriate decision to increase or decrease irrigation rates. Drone image processing application helps extract vital information about crop health and growth rates and detect diseases and pests in early bunches through image processing. This information also helps make critical decisions regarding irrigation scheduling, fertilization times, and pest control, leading to increased field production and healthy crops.
- Lastly, VR/AR technology has not yet been widely used in irrigation systems. It is possible to consider it as a new direction in the future. One potential application of VR/AR in irrigation systems is for training purposes. Farmers or irrigation technicians could practice irrigation techniques in a virtual environment using VR/AR headsets. It can show the position of sensors, valves, and other components in real time, which could help farmers and technicians identify any issues with the system quickly and efficiently.

References

1. Vianny DM, John A, Mohan SK, Sarlan A, Ahmadian A. Water optimization technique for precision irrigation system using IoT and machine learning. *Sustain. Energy Technol. Assess.*, 1;52:102307, 2022. https://doi.org/10.1016/j.seta.2022.102307
2. Yousif, J.H. and Abdalgader, K. Experimental and mathematical models for real-time monitoring and auto watering using IoT architecture. *Computers*, 11(1), p.7, 2022. https://doi.org/10.3390/computers11010007
3. Alves RG, Maia RF, Lima F. Development of a Digital Twin for smart farming: Irrigation management system for water saving. *J. Clean. Prod.*, 7:135920, 2023. https://doi.org/10.1016/j.jclepro.2023.135920
4. Abdalgader K, Yousif JH. Agricultural Irrigation Control using Sensorenabled Architecture. *KSII Trans. Internet Inf. Syst.*, 1;16(10), 2022.
5. Saddik A, Latif R, Elhoseny M, El Ouardi A. Real-time evaluation of different indexes in precision agriculture using a heterogeneous embedded system. *Sustain. Comput.: Inf. Syst.*, 1;30:100506, 2021. https://doi.org/10.1016/j.suscom.2020.100506
6. Rahaman MM, Azharuddin M. Wireless sensor networks in agriculture through machine learning: A survey. *Comput.*

Electron. Agric.,1;197:106928, 2022. https://doi.org/10.1016/j.compag.2022.106928

7. Lakshmi GP, Asha PN, Sandhya G, Sharma SV, Shilpashree S, Subramanya SG. An intelligent IOT sensor coupled precision irrigation model for agriculture. *Meas.: Sens.*, 1;25:100608, 2023. https://doi.org/10.1016/j.measen.2022.100608
8. Harun AN, Mohamed N, Ahmad R, Ani NN. Improved Internet of Things (IoT) monitoring system for growth optimization of Brassica chinensis. *Comput. Electron. Agric.*, 1;164:104836, 2019. https://doi.org/10.1016/j.compag.2019.05.045
9. Farooq H, Rehman HU, Javed A, Shoukat M, Dudley S. A review on smart IoT based farming. *Ann. Emerg. Technol. Comput. (AETiC)*, pp :2516-0281, 2020. DOI: 10.33166/AETiC.2020.03.003
10. Touil S, Richa A, Fizir M, Argente Garcia JE, Skarmeta Gomez AF. A review on smart irrigation management strategies and their effect on water savings and crop yield. *Irrig. Drain.*, 2022. https://doi.org/10.1002/ird.2735
11. Bwambale E, Abagale FK, Anornu GK. Smart irrigation monitoring and control strategies for improving water use efficiency in precision agriculture: A review. *Agric. Water Manage.*, 1;260:107324, 2022. https://doi.org/10.1016/j.agwat.2021.107324

12. Yin H, Cao Y, Marelli B, Zeng X, Mason AJ, Cao C. Soil sensors and plant wearables for smart and precision agriculture. *Adv. Mater.*, 33(20):2007764, 2021. https://doi.org/10.1002/adma.202007764
13. Obaideen K, Yousef BA, AlMallahi MN, Tan YC, Mahmoud M, Jaber H, Ramadan M. An overview of smart irrigation systems using IoT. *Energy Nexus*, 25:100124, 2022. https://doi.org/10.1016/j.nexus.2022.100124
14. Okura F, Budiasa IW, Kato T. Exploring a Balinese irrigation water management system using agent-based modeling and game theory. *Agric. Water Manage.*, 274:107951, 2022. https://doi.org/10.1016/j.agwat.2022.107951
15. Sinha BB, Dhanalakshmi R. Recent advancements and challenges of Internet of Things in smart agriculture: A survey. *Future Gener. Comput. Syst.*, 2022 Jan 1;126:169-84. https://doi.org/10.1016/j.future.2021.08.006
16. Zhao P, Ma M, Hu Y, Wu W, Xiao J. Comparison of international standards for irrigation with reclaimed water. *Agric. Water Manage.*, 2022 Dec 1;274:107974. https://doi.org/10.1016/j.agwat.2022.107974
17. Mirás-Avalos JM, Gonzalez-Dugo V, García-Tejero IF, López-Urrea R, Intrigliolo DS, Egea G. Quantitative analysis of almond yield response to irrigation regimes in

Mediterranean Spain. *Agric. Water Manage.*, 2023 Apr 1;279:108208. https://doi.org/10.1016/j.agwat.2023.1082

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13
Smart Plant-Based Wastewater Treatment for Agricultural Irrigation Systems

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Abstract

Untreated wastewater contains a substantial amount of physical, chemical, and biological toxins which create several critical issues for the agricultural production system. Therefore, an efficient waste treatment process should be followed. Several plant-based natural treatment methods have been adopted to make wastewater useful for agricultural irrigation. Semi-aquatic and aquatic plants are capable to reclaim highly contaminated wastewater like chemical industry wastewater. The microorganism developed in plant roots can efficiently remove different hazardous heavy metals, bacteria, and other toxic substances from the raw wastewater. Different types of

plants such as water hyacinth, soft rush, watercress, water lilies, etc., are generally utilized as natural water filters. Unlike a typical wastewater treatment plant which uses chemicals and electromechanical equipment, natural plant-based wastewater treatment needs low maintenance cost and consumes very low energy. However, the main challenge of such a natural system is the selection of the plant types considering local climate adaption capability, nutrient absorption rate, oxygen transfer capacity, biomass production, etc. In this case, algorithms such as artificial neural networks (ANNs) are useful to make the optimal decision. Therefore, the present chapter mainly focuses on artificial intelligence (AI) application for the development of plant-based wastewater treatment and discusses the various challenges.

Keywords**:** Agriculture, artificial intelligence, irrigation, plant, wastewater

13.1 Introduction

13.1.1 Importance of Wastewater Treatment

Clean and pure water is a big necessity of human life [1]. Groundwater and surface water are the two primary sources of water. However, with the rapid growth of population and

climate changes, fresh water resources are getting shrunk day by day. Besides this, in the heavy industrial cities, the ground water and surface water are getting contaminated by industrial effluents [2]. Along with these, wastewater is discharged from various resources, based on which it can be categorized as domestic wastewater, industrial wastewater, municipal wastewater, and agricultural wastewater [3]. Therefore, it is getting important to adopt wastewater treatment to meet the necessity. Wastewater treatments are used to remove different pollutants, heavy metals, organic, and inorganic components and improve the quality of water for reuse in different sectors [4, 5].

13.1.2 Steps of Conventional Wastewater Treatment

Conventional wastewater treatment comprises physical, chemical, and biological processes to remove the pollutants from the wastewater [6]. The different steps of the conventional wastewater treatment processes consist of preliminary treatment, primary treatment, secondary treatment, tertiary treatment, and sludge treatment [7]. The preliminary treatment is the initial stage which can remove stones, sticks, and other large garbage though a screening process. In this initial step, grit chambers can be utilized for heavy particles (i.e., gravel, sands) settlement [8]. The second stage, primary treatment,

involves floatation and sedimentation to eliminate suspended matters as well as different solid particles [9]. In the secondary treatment, different types of organic matter are broken down and converted into water, carbon dioxide, and new cell mass by microorganisms [10]. Hence, this process is also known as biological treatment process. Activated sludge process, rotating biological contactors, and trickling filters are examples of this biological reclamation practice [11].

Tertiary treatment is the advanced treatment which involves filtration, nutrient removal, and disinfection [12]. Filtration uses sand filters to eliminate very fine suspended pollutants [13]. The nutrient removal process involves chemical precipitation and membrane bioreactors to remove different nutrients like nitrogen, potassium and phosphorus [14]. The disinfection step kills different pathogens by ozonation, ultraviolet irradiation, chlorination, etc. Sludge treatment techniques involve thickening, digestion, and dewatering processes [15].

13.1.3 Aquatic Plant-Based Wastewater Treatment

Aquatic plant-based wastewater treatment is one of the popular wastewater treatments also known as constructed wastelands or phytoremediation [16]. It uses the natural abilities of aquatic

plants to uptake the hazardous contaminants from highly polluted water bodies [17]. Therefore, this natural treatment process does not need a highly mechanical and energy-intensive process [18]. Hence, this process reduces energy consumption and associated operational costs for wastewater treatment [19]. Along with these, the aquatic plant-based method provides a natural habitat to various animal species [20].

Aquatic plants are of a large number of varieties such as reeds, cattails, bulrushes, water hyacinths, duckweeds, water lettuce, etc. [21]. These plants remove pollutants from different graded wastewater by absorption, filtration, and microbial degradation [5]. However, the performances of different plants vary from one another depending upon the wastewater quality [22]. In this aspect, real-time monitoring of the different aquatic plants is required to maintain the treated water quality within the recommended level for utilization in different purposes like drinking, agricultural irrigation, etc. Therefore, this chapter critically discusses the importance of artificial intelligence in aquatic plant-based wastewater treatment system for agricultural irrigation.

13.1.4 Objectives and Scopes

The chapter is structured with some useful objectives:

- To highlight the use of wastewater in agricultural purpose
- To present aquatic plant-based wastewater treatment technologies for agricultural purpose
- To highlight the benefits of aquatic plant-based wastewater treatment technologies
- To assess the potential limitations of aquatic plant-based wastewater treatment technologies
- To analyze the importance of AI in aquatic plant-based wastewater treatment

The scope of the present chapter includes a review of the relevant literature, including peer-reviewed scientific studies, technical reports, and policy documents. The chapter also discusses case studies and examples from around the world to illustrate the potential benefits and challenges of using AI in aquatic plant-based wastewater treatment for sustainable agriculture. Furthermore, the importance of stakeholder engagement and community education in endorsing the successful application of AI in a wastewater purification system has been also elaborated in this chapter.

13.2 Water Quality Parameters for Agricultural Irrigation

Quality of water is one of the utmost important factors which have an impact on soil health, crop growth, productivity, and footprints [23]. Regional soil types, variety of crops as well as other management factors have influence on different water quality parameters [24]. Important physical, chemical, and biological parameters of wastewater are documented here.

13.2.1 Physical Parameters

Important physical parameters include total suspended solids (TSS), turbidity, color, and odor [25]. The TSS refers to all suspended materials which are unable to pass through any filter [26]. Generally, a spectrophotometer is used to measure TSS. Turbidity takes place due to the presence of suspended materials which include both organic and inorganic matters, planktons, etc. It is generally measured by using a turbid meter and expressed by the unit nephelometric turbidity units (NTU). Other important parameters are color and odor [27]. These parameters are normally characterized by their intensity and hedonic tone (i.e., nastiness or pleasantness) [28].

13.2.2 Chemical Parameters

Different types of wastewater contain a wide range of chemical pollutants [29]. However, the variety and concentrations of different chemical parameters depend on locality and composition [30]. Some common chemical parameters include pH, biochemical oxygen demand (BOD), chemical oxygen demand (COD), total dissolved solids (TDS), different nutrients, heavy metals, organic compounds, oil, grease, chlorine residuals, etc. [31]. Water acidity or alkalinity is generally measured by water ph. The BOD indicates the oxygen amount needed by different microorganisms for organic matter decomposition within water [32]. The COD measures the required oxygen quantity for the chemical oxidization of both organic and inorganic compounds. The TDS includes salts, minerals, and different dissolved organic as well as inorganic compounds. Nutrients indicate the presence of nitrogen, potassium, and phosphorus in water. One of the most hazardous chemical pollutants is heavy metal concentration in water. Heavy metals include iron (Fe), lead (Pb), mercury (Hg), Cadmium (Cd), Chromium (Cr), etc. High concentrations of heavy metals, oil, and grease can cause severe water as well as environmental pollution [33].

13.2.3 Biological Parameters

The biological contaminants of wastewater include total coliforms, fecal coliform, biological indicator organism, chlorophyll-a, etc. [34]. The presence of coliforms indicates harmful pathogenic microorganisms for human health. Common biological indicator organisms are *E. coli* and enterococci. The chlorophyll-a parameter indicates the presence of algae in wastewater. A high concentration of chlorophyll-a in wastewater can cause eutrophication in receiving water bodies [35].

13.2.4 Treated Wastewater Suitability for Agricultural Irrigation Application

Several factors like concentration of different water quality parameters, treatment processes, regional rules and regulations, irrigation practices, variety of crops, and frequency of monitoring can have influences on the suitability of treated wastewater for agricultural irrigation [5]. Table 13.1 shows the recommended level of different physical, chemical as well as biological water quality parameters for irrigation application [5]. Wastewater purification also depends on the species of aquatic plants [36]. Regional policies fix the crop management, irrigation practice, water quality requirements, and buffer zones. The selection of an irrigation method is also a vital

subject in this regard [37]. Generally, drip irrigation is preferred over sprinkler irrigation to minimize the risk of contamination in edible plant parts [38]. Crop selection plays an important role in this issue. The sensitivity of crop to wastewater varies from one to another. Trees are less sensitive to water quality parameters in comparison with root crops or leafy vegetables. Regular inspection on water quality parameters is of utmost importance to ensure the ongoing suitability of wastewater for agricultural crop irrigation [39].

Table 13.1 Water quality guidelines for agricultural crop irrigation [5].

Type of contaminants	Water quality parameters	Units	Recommended level of parameters	
			FAO	USEPA
Physical	TSS	mg/l	-	30
	Turbidity	NTU	-	2
Chemical	pH	-	6.5–8.50	6.50–8.40
	BOD	mg/l	40	10
	TDS		2,000	450
	COD		120	-
	Nitrogen (N)		30	-

Type of contaminants	Water quality parameters	Units	Recommended level of parameters	
			FAO	USEPA
	Phosphorus (P)		2	-
	Cd		0.01	0.01
	Cr		0.10	0.10
	Pb		2	5
	Fe		5	5
	Manganese (Mn)		0.20	0.20
	Arsenic (As)		-	0.10
	Chloride		1,100	70

Type of contaminants	Water quality parameters	Units	Recommended level of parameters	
			FAO	USEPA
	Ec	dS/m	3	0.70
	Oil and grease	mg/l	10	-
Biological	Fecal coliform	MPN/100 ml	200	23
	Total coliform		400	-
	E. coli		200	-

*FAO, Food and Agriculture Organization; USEPA, United States Environmental Protection Agency.

13.3 Wetland Method for Wastewater Treatment

13.3.1 Working Principle of Wetland for Wastewater Treatment

A wetland supports plant growth and the interactions between soil, water, plants, and microorganisms [5]. It can be constructed with sand, gravel, and aquatic plants. Wetland plants adsorb and filter out organic matters, suspended solids, and sediments with the help of its root systems [40]. Beneath the plant root zone, different types of microorganisms like fungi, bacteria, etc., break down the organic contaminants into simpler forms through several processes like nitrification, denitrification, and aerobic and anaerobic decomposition as well as phosphorus removal [41]. A wetland system needs continuous monitoring (i.e., accumulated solids removal, control of invasive plant species, and water quality monitoring) for longterm optimal functioning [42].

13.3.2 Types of Wetland

Wetlands are of several types. These can be categorized as detailed in the following subsections.

13.3.2.1 Natural Wetland

Natural wetlands like marshes, bogs, and swamps remove different physical, chemical, and biological contaminants through sedimentation, adsorption, filtration, and microbial degradation [5].

13.3.2.2 Constructed Wetland

Constructed wetlands (CW) can be constructed with layers of sand, gravel, and wetland plants [5]. These are mainly of two types: (a) surface flow (SF) wetlands and (b) subsurface flow (SSF) wetlands. The SF wetlands consider horizontal wastewater flow through vegetation cover [43]. The SF system eliminates several physical, chemical, and biological contaminants by sedimentation, adsorption, filtration, and microbial activity. In the SSF system, wastewater flows either horizontally or vertically [44]. Wastewater can be purified in the SSF wetlands by nutrient uptake and biological degradation. Another special type of CW is reed bed system where tall and dense reeds remove pollutants by sedimentation, mechanical filtration, and microbial degradation [45].

13.3.2.3 Hybrid Wetland

Some wetlands can be combined with other treatment methods (like activated sludge process and anaerobic digestion) to maximize the pollutant removal efficiency. These types of wetlands are known as hybrid wetlands [46].

13.3.3 Aquatic Plants for Wetland Treatment

There are several types of aquatic plants for wastewater treatment. The selection of plant species depends on regional climate, wastewater characteristics, local conditions, and treatment goals. This chapter has documented some common types of aquatic plants for wastewater treatment [47].

13.3.3.1 Reeds

Reeds are known as *Phragmites australis*. These are tall and robust plants. Reeds remove heavy metals from wastewater by sediment stabilization and filtration [48].

13.3.3.2 Cattails

Cattails are *Typha* spp. These are widely used wetland plants. Cattails improve water quality by removing organic loads and various nutrients [49].

13.3.3.3 Water Hyacinth

Water hyacinth is also known as *Eichhornia crassipes*. These are floating plants which have abilities to remove hazardous heavy metals and organic contaminants from different types of wastewater [50].

13.3.3.4 Duckweed

Duckweeds are *Lemna* spp. These are small floating plants. Duckweeds are effective to remove several chemical pollutants from wastewater [51].

13.3.3.5 Water Willow

Water willow is also known as *Justicia americana*. These are submerged plants whose roots are habitat for some beneficial microbes. These are effective in sediment stabilization [52].

Some other types of aquatic plants include water lettuce, sagittaria, water smartweed, pickerelweed, etc. [53–56]. Advice from local experts is essential to determine suitable aquatic plant species for a particular wastewater treatment scheme.

13.4 Importance of Artificial Intelligence in Wastewater Treatment

13.4.1 Different Artificial Intelligence Models

There are different types of machine learning models in AI. Those can be used for aqua plant-based wastewater treatment [57]. The machine learning models can be linear and nonlinear types. If a model consists of a linear combination of weights or coefficients, then it is called a linear model— for instance, $w_0 + w_1x_1 + \ldots + w_nx_n = y$ is a linear model. Here $\{x_1, x_2, \ldots x_n\}$ are input, $\{w_1, w_2, \ldots w_n\}$ are weights or coefficients, and y is the output of the model. For nonlinear models, the output is not the linear combination of weights. Linear regression is an example of linear model and logistic regression, and deep learning models are an example of nonlinear model. In the following, we will discuss about some important types of machine learning models.

13.4.1.1 Regression Model

Regression model describes the relationship between one or more independent and dependent variables. It is used to predict either the dependent variable when independent variables are known or the effect of an independent variable on the dependent variable. There may be different types of

regression models such as linear regression, logistic regression, and polynomial regression [58].

13.4.1.2 Support Vector Machine (SVM)

It is a supervised vector machine learning algorithm. It can be used for both classification and regression problems. SVM mainly works by finding an optimum hyperplane and segregating the dataset into two classes. SVM can be of two types: linear SVM and nonlinear SVM [59].

13.4.1.3 Artificial Neural Network (ANN)

It is a nonlinear model. It replicates the human brain structure. The basic component is artificial neurons or nodes. ANN consists of multiple layers. The components are one input layer, one or more hidden layer, and one output layer. Each layer consists of multiple artificial neurons. Each neuron is interconnected with each other from one layer to another, and each of the connections has its individual weight [60].

13.4.1.4 Deep Learning Model

It uses ANN model to perform the computation. There are different types of deep learning algorithms such as convolutional neural network (CNN), long short term memory

networks (LSTM), and recurrent neural network (RNN). Each of these algorithms is used for different types of applications such as CNN which is mainly used for image processing object detection. LSTM is used for time series prediction, and RNN is used for natural language processing [61–63].

13.4.2 Application of Artificial Intelligence for Wastewater Treatment

Wastewater has a very complex composition with the existence of different concentrations of pollutants. Due to this complex structure, wastewater treatment process faces fluctuation as a result. To overcome this problem, the modern wastewater treatment process focuses on AI technology.

13.4.2.1 Challenges of Aquatic Plant-Based Wastewater Treatment

13.4.2.1.1 Selection of Aquatic Plant

Different aquatic plant species have different capacities to absorb the pollutants. Furthermore, a high variation of climatic condition such as drought, flood, and frost may affect the survival of aquatic plants. Therefore, the selection of proper aquatic plants for wastewater treatment based on the pollutant rate in the wastewater and the local climatic condition is a

challenging one [64]. The wrong selection of aquatic plants may degrade the efficiency of the treatment process.

13.4.2.1.2 Nutrient Presence

A high concentration of certain pollutant and the absence of specific pollutants may affect optimal plant growth. Therefore, proper nutrient balance is required for an efficient wastewater treatment process.

13.4.2.1.3 Hydraulic Loading Rate

Hydraulic loading rate denotes the rate at which wastewater is discharged at the surface, denoted as volume per unit area. The high volume of wastewater with a high flow rate may affect the plant uptake rate which ultimately degrades the treatment process [65].

13.4.2.1.4 Maintenance of Plant Growth

Aquatic plants have a high growth rate. Thus, it needs proper maintenance such as periodic harvesting and disease monitoring.

13.4.2.1.5 Various Lifespan of Aquatic Plants

Different aquatic plant species have different lifespans, or the performance of some aquatic plants may decline after a period of time. Therefore, regular monitoring is needed for consistent treatment efficiency [5].

13.4.2.2 Application of Artificial Intelligence

These aforementioned challenges can be easily addressed by using the artificial intelligence (AI) tools in the aquatic plant-based wastewater treatment. However, for the successful implementation of AI in aquatic plant-based wastewater treatment, accurate data collection, secure data sensors, and high accurate models trained over large-scale data are required [66]. In the following subsections, we will discuss some of the significant contributions of AI in aqua plant-based wastewater.

13.4.2.2.1 Data Analysis

A large amount of data related to water quality and plant health quality can be gathered using different types of sensors. The gathered data can be analyzed by using AI algorithms to provide valuable insights for monitoring the treatment process [67].

13.4.2.2.2 Decision Support System

AI supports to build a decision support system that can assist the operators to make important decisions. The decision support system may leverage AI models to analyze heterogeneous data such as weather conditions, water nutrients, and plant performance [67].

13.4.2.2.3 Plant Growth Monitoring

Plant growth can be optimized through leveraging the AI models, which considers data from various sources. This process can help to maximize the pollutant removal capabilities of plants.

13.4.2.2.4 Energy Consumption

AI can be used to optimize energy usage in aquatic plant-based wastewater treatment by controlling the pumping process, temperature, lighting intensity, and aeration process.

13.4.2.2.5 Contaminant Removal Optimization

AI can be effective in the removal of contaminants by analyzing different real-time data such as water quality and plant health. AI models can be leveraged to remove the pollutant through

different operations such as maintaining the nutrient dosing rate and hydraulic retention rates.

13.4.2.2.6 Predictive Maintenance

The efficiency of the wastewater treatment system can be increased through employing the early prediction ability using AI. Here AI can be used to determine the early detection of equipment failure, plant disease, etc.

13.5 Economic Feasibility of Smart Plant-Based Treatment for Agricultural Irrigation

13.5.1 Cost Analysis of Smart Treatment System for Irrigation

The cost analysis involves the assessment of several factors. These factors are summarized in the following subsections.

13.5.1.1 Initial Investment

This category includes the purchasing cost of smart system equipment and their installation cost. Smart system equipment include sensors, communication devices, actuators, etc.

13.5.1.2 Operational Cost

Operational cost includes cost of water and energy consumptions. The smart system can effectively save the potential cost of water and energy consumption through an optimized irrigation system.

13.5.1.3 Maintenance Cost

This category includes the cost of regular maintenance activities like sensor calibration, software updates, and equipment inspection. Repair and replacement costs are also considered under maintenance cost.

13.5.2 Comparison of Smart System Cost With Conventional Method

Cost comparison varies depending on the specific requirements of technology and its application. Generally, a smart system requires some initial investments. Therefore, initial investment may be higher in a smart system as compared to conventional method. However, depending upon some particular requirements, the traditional method can also demand some extra initial investment [68]. In case of operational cost, a smart system can be capable of saving potential cost in terms of energy consumption, automation as well as remote monitoring.

However, a smart system includes some potential costs of software updates, maintenance, and data handling. Furthermore, a smart system reduces the labor cost significantly compared to the conventional method. A smart system may have a higher initial cost but may offer more long-term cost savings in terms of improved efficiency, optimized resource usage, and reduced energy consumption with a higher return of investment (ROI) in comparison with the conventional system.

13.5.3 Economic Viability of Smart System Implementation in Rural Agriculture

The economic viability of smart system implementation in rural agriculture depends on several factors like implementation cost, operational efficiency, risk mitigation, data-driven decision making, scalability, and adaptability with access in financing and support. The implementation cost may vary depending upon the complexity of the agricultural application. A smart system enables farmers to make instant decisions with real-time data [69]. This can lead to better management of natural resources with better productivity and lower footprints. However, it is crucial to have skillful farmers or laborers in the successful implementation of smart wastewater treatment systems in rural areas [70]. Along with this, the availability of

government subsidies and financing options play a crucial role in smart system implementation [71]. Therefore, an assessment of local conditions and thorough analysis of cost–benefit are challenging steps in smart AI-enabled system implementation in rural areas.

13.6 Challenges Associated with Artificial Intelligence Application in Wastewater Treatment

13.6.1 Social, Technical, and Regulatory Challenges

Several technical, social, and regulatory challenges are associated with the application of artificial intelligence in wastewater treatment. The challenges included are discussed in the following subsections.

13.6.1.1 Quality and Availability of Data

Reliable and comprehensive data are really challenging in wastewater treatment. Accurate AI prediction is not possible without huge amounts of high-quality data.

13.6.1.2 Compatibility and Integration of Data

Wastewater treatment systems have different data sources and legacy systems which may not be compatible with modern AI.

Data integration from various sources can be complex and time-consuming.

13.6.1.3 Model Development and Validation

Model development and validation is really a challenging task with low-quality data. Particularly, the aquatic plant performance data are really limited.

13.6.1.4 Explainability and Interpretability

The AI models for wastewater treatment have to be explainable and interpretable to achieve confidence from engineers, operators, and other stakeholders.

13.6.1.5 Regulatory and Ethical Considerations

Security and privacy as well as data protection are of utmost importance in wastewater treatment processes. It is very crucial to ensure that the developed AI models are unbiased and do not carry existing inequalities.

Along with these, implementation cost and resource limitations are also critical challenges in AI-based smart wastewater system applications for agricultural irrigation.

13.6.2 Strategies and Solutions for Addressing Challenges

Probable fruitful strategies and solutions are addressed here. These include the items discussed in the following subsections.

13.6.2.1 Quality and Availability of Data

Installation of sensors, real-time monitoring systems, and IoT systems in several locations of the wastewater treatment system can be helpful in the collection of water quality parameter data and other relevant variables.

13.6.2.2 Pre-Processing and Integration of Data

A robust data pre-processing pipeline can be established to normalize, clean, and integrate data from several resources. Different data integration methods can be employed to combine data into a unified format.

13.6.2.3 Model Development, Training, and Validation

Several AI techniques like machine learning (i.e., regression, clustering, and classification), deep learning (i.e., neural networks), and hybrid models can be explored to address these challenges.

Adoption of such types of strategies may be helpful in addressing the key challenges and may lead to a comparatively more efficient wastewater treatment system for agricultural application.

13.7 Conclusion

An aquatic plant-based wastewater treatment system acts as a natural filter for wastewater treatment. It uses plant roots for the absorption of heavy metals and different undesirable organic loads as well as harmful biological contaminants from different types of wastewater. Hence, it is a low-cost treatment method compared to other modern methods. Therefore, this method can be easily acceptable for developing as well as under-developed countries. However, the performance of the method in terms of removal efficiency of different pollutants varies from one location to another and one type of wastewater to another. Along with these, several factors like hydraulic loading rate and hydraulic retention time as well as type of aquatic plants affect the performance of aquatic plant-based wastewater treatment systems. Hence, the real-time monitoring of treatment systems using artificial intelligence is of utmost importance. Different artificial intelligence models like SVM, ANN, RNN, etc., can be used to monitor aquatic plant growth, optimize pollutant removal, and make necessary decision

timely. However, the AI-based smart wastewater system has some limitations in agricultural applications. Some specific technical and social as well as regulatory challenges need to be addressed with a large amount of good-quality data. Along with these, rural agricultural farmers and other stakeholders need to be familiar with smart AI-based technology before implementation.

References

1. Biswas, A., Duary, S., Islam, A., Bhattacharjee, S., Water footprint and productivity of lettuce with non-conventional water resources. *BFIJ*, 15(4), 1-4, 2023.
2. Parde, D., Patwa, A., Shukla, A., Vijay, R., Killedar, D.J., Kumar, R., A review of constructed wetland on type, treatment and technology of wastewater. *Environ. Technol. Innov.*, 21, 101261, 2021.
3. Biswas, A., Mailapalli, D.R., Raghuwanshi, N.S., Treated municipal wastewater to fulfil crop water footprints and irrigation demand–a review. *Water Supply*, 21(4), 1398-1409, 2021.
4. Kumar, S. and Dutta, V., Constructed wetland microcosms as sustainable technology for domestic wastewater treatment: an overview. *Environ. Sci. Pollut. Res.*, 26(12), 11662-11673, 2019.

5. Biswas, A., Constructed wetlands: A possible solution for municipal wastewater treatment – a review. *Int. J. Ecol. Environ. Sci.*, 2(2), 41-48, 2020a.
6. Biswas, A., Application of membrane bio-reactor for municipal wastewater treatment: a review. *J. Pharmacogn. Phytochem.*, 9(4), 892-899, 2020b.
7. Ghimire, U., Nandimandalam, H., Martinez-Guerra, E., Gude, V.G., Wetlands for wastewater treatment. *Water Environ. Res.*, 91(10), 1378-1389, 2019.
8. Ali, M., Rousseau, D.P., Ahmed, S., A full-scale comparison of two hybrid constructed wetlands treating domestic wastewater in Pakistan. *J. Environ. Manage.*, 210, 349-358, 2018.
9. Licciardello, F., Milani, M., Consoli, S., Pappalardo, N., Barbagallo, S., Cirelli, G., Wastewater tertiary treatment options to match reuse standards in agriculture. *Agric. Water Manage.*, 210, 232-242, 2018.
10. Fountoulakis, M.S., Daskalakis, G., Papadaki, A., Kalogerakis, N., Manios, T., Use of halophytes in pilot-scale horizontal flow constructed wetland treating domestic wastewater. *Environ. Sci. Pollut. Res.*, 24, 16682-16689, 2017.
11. Biswas, A., Vermi-biofiltration for removal of chemical and biological pollutants from municipal wastewater for agriculture – a review. *Pollut. Res.*, 40(2), 457-460, 2021.

12. Díaz-Garduño, B., Pintado-Herrera, M.G., Biel-Maeso, M., Rueda-Márquez, J.J., Lara-Martín, P.A., Perales, J.A., ... Martín-Díaz, M.L., Environmental risk assessment of effluents as a whole emerging contaminant: Efficiency of alternative tertiary treatments for wastewater depuration. *Water Res.*, 119, 136-149, 2017.
13. Huang, G., Pan, F., Fan, G., Liu, G., Application of heterogeneous catalytic ozonation as a tertiary treatment of effluent of biologically treated tannery wastewater. *J. Environ. Sci. Health, Part A*, 51(8), 626-633, 2016.
14. Hafiz, M., Hawari, A.H., Alfahel, R., Hassan, M.K., Altaee, A., Comparison of nanofiltration with reverse osmosis in reclaiming tertiary treated municipal wastewater for irrigation purposes. *Membranes*, 11(1), 32, 2021.
15. Pei, M., Zhang, B., He, Y., Su, J., Gin, K., Lev, O., ...Hu, S., State of the art of tertiary treatment technologies for controlling antibiotic resistance in wastewater treatment plants. *Environ. Int.*, 131, 105026, 2019.
16. Liu, J., Yi, N. K., Wang, S., Lu, L.J., Huang, X.F., Impact of plant species on spatial distribution of metabolic potential and functional diversity of microbial communities in a constructed wetland treating aquaculture wastewater. *Ecol. Eng.*, 94, 564-573, 2016.

17. Wang, Q., Cao, Z., Liu, Q., Zhang, J., Hu, Y., Zhang, J., ... Chen, Q., Enhancement of COD removal in constructed wetlands treating saline wastewater: intertidal wetland sediment as a novel inoculation. *J. Environ. Manage.*, 249, 109398, 2019.
18. Zhao, X., Chen, J., Guo, M., Li, C., Hou, N., Bai, S., Constructed wetlands treating synthetic wastewater in response to day-night alterations: Performance and mechanisms. *Chem. Eng. J.*, 446, 137460, 2022.
19. Sossalla, N.A., Nivala, J., Reemtsma, T., Schlichting, R., König, M., Forquet, N.,....Escher, B.I., Removal of micropollutants and biological effects by conventional and intensified constructed wetlands treating municipal wastewater. *Water Res.*, 201, 117349, 2021.
20. Al-Ajalin, F.A.H., Idris, M., Abdullah, S.R.S., Kurniawan, S.B., Imron, M.F., Effect of wastewater depth to the performance of short-term batching-experiments horizontal flow constructed wetland system in treating domestic wastewater. *Environ. Technol. Innov.*, 20, 101106, 2020.
21. Du, L., Zhao, Y., Wang, C., Wu, Z., Zhou, Q., Effects of plant on denitrification pathways in integrated vertical-flow constructed wetland treating swine wastewater. *Ecotoxicol. Environ. Saf.*, 201, 110752, 2020.
22. Anh, B.T.K., Van Thanh, N., Phuong, N.M., Ha, N.T.H., Yen, N.H., Lap, B.Q., Kim, D.D., Selection of suitable filter materials

for horizontal subsurface flow constructed wetland treating swine wastewater. *Water Air Soil Pollut.*, 231, 1-10, 2020.
23. Malakar, A., Snow, D.D., Ray, C., Irrigation water quality—A contemporary perspective. *Water*, 11(7), 1482, 2019.
24. Shoushtarian, F. and Negahban-Azar, M., Worldwide regulations and guidelines for agricultural water reuse: a critical review. *Water*, 12(4), 971, 2020.
25. Biswas, A., Mailapalli, D.R., Raghuwanshi, N.S., Treatment technologies for removal of physical contaminants from municipal wastewaters – a review. *Res. J. Chem. Environ.*, 24(11), 140-147, 2020.
26. Mainardis, M., Cecconet, D., Moretti, A., Callegari, A., Goi, D., Freguia, S., Capodaglio, A.G., Wastewater fertigation in agriculture: Issues and opportunities for improved water management and circular economy. *Environ. Pollut.*, 296, 118755, 2022.
27. Kesari, K.K., Soni, R., Jamal, Q.M.S., Tripathi, P., Lal, J.A., Jha, N.K., ...Ruokolainen, J., Wastewater treatment and reuse: a review of its applications and health implications. *Water Air Soil Pollut.*, 232, 1-28, 2021.
28. Ustaoğlu, F., Tepe, Y., Taş, B., Assessment of stream quality and health risk in a subtropical Turkey river system: A combined approach using statistical analysis and water quality index. *Ecol. Indic.*, 113, 105815, 2020.

29. Takeuchi, H. and Tanaka, H., Water reuse and recycling in Japan—History, current situation, and future perspectives. *Water Cycle*, 1, 1-12, 2020.
30. Ezugwu, C.K., Onwuka, O.S., Egbueri, J.C., Unigwe, C.O., Ayejoto, D.A., Multi-criteria approach to water quality and health risk assessments in a rural agricultural province, southeast Nigeria. *HydroResearch*, 2, 40-48, 2019.
31. Singh, A., A review of wastewater irrigation: Environmental implications. *Resour. Conserv. Recycl.*, 168, 105454, 2021.
32. Natasha, Shahid, M., Khalid, S., Murtaza, B., Anwar, H., Shah, A.H., ...Niazi, N.K., A critical analysis of wastewater use in agriculture and associated health risks in Pakistan. *Environ. Geochem. Health*, 1-20, 2020.
33. Ofori, S., Puškáčová, A., Růžičková, I., Wanner, J., Treated wastewater reuse for irrigation: Pros and cons. *Sci. Total Environ.*, 760, 144026, 2021.
34. Ungureanu, N., Vlăduț, V., Voicu, G., Water scarcity and wastewater reuse in crop irrigation. *Sustainability*, 12(21), 9055, 2020.
35. Saleh, I.A., Zouari, N., Al-Ghouti, M.A., Removal of pesticides from water and wastewater: Chemical, physical and biological treatment approaches. *Environ. Technol. Innov.*, 19, 101026, 2020.

36. Ahmad, H.R., Sabir, M., Zia ur Rehman, M., Aziz, T., Maqsood, M.A., Ayub, M.A., Shahzad, A., Wastewater irrigation-sourced plant nutrition: concerns and prospects. *Plant Micronutrients: Deficiency and Toxicity Management*, 417-434, 2020.
37. Lyu, S., Wu, L., Wen, X., Wang, J., Chen, W., Effects of reclaimed wastewater irrigation on soil-crop systems in China: A review. *Sci. Total Environ.*, 813, 152531, 2022.
38. Al-Mefleh, N.K., Talozi, S., Abu Naser, K., Assessment of treated wastewater reuse in drip irrigation under different pressure conditions. *Water*, 13(8), 1033, 2021.
39. Tabatabaei, S.H., Nourmahnad, N., Golestani Kermani, S., Tabatabaei, S.A., Najafi, P., Heidarpour, M., Urban wastewater reuse in agriculture for irrigation in arid and semi-arid regions-A review. *Int. J. Recycl. Org. Waste Agric.*, 9(2), 193-220, 2020.
40. Sharma, R., Vymazal, J., Malaviya, P., Application of floating treatment wetlands for stormwater runoff: A critical review of the recent developments with emphasis on heavy metals and nutrient removal. *Sci. Total Environ.*, 777, 146044, 2021.
41. Langergraber, G. and Dotro, G., *Wetland technology: practical information on the design and application of treatment wetlands*. IWA publishing, 2019.

42. Hammer, D.A., (Ed.), *Constructed wetlands for wastewater treatment: municipal, industrial and agricultural*, CRC Press, 2020.
43. Khan, N.A., El Morabet, R., Khan, R.A., Ahmed, S., Dhingra, A., Alsubih, M., Khan, A.R., Horizontal sub surface flow Constructed Wetlands coupled with tubesettler for hospital wastewater treatment. *J. Environ. Manage.*, 267, 110627, 2020.
44. Yuan, C., Huang, T., Zhao, X., Zhao, Y., Numerical models of subsurface flow constructed wetlands: review and future development. *Sustainability*, 12(8), 3498, 2020.
45. Jehawi, O.H., Abdullah, S.R.S., Kurniawan, S.B., Ismail, N.I., Idris, M., Al Sbani, N.H., ...Hasan, H.A., Performance of pilot Hybrid Reed Bed constructed wetland with aeration system on nutrient removal for domestic wastewater treatment. *Environ. Technol. Innov.*, 19, 100891, 2020.
46. Ren, B., Wang, T., Zhao, Y., Two-stage hybrid constructed wetland-microbial fuel cells for swine wastewater treatment and bioenergy generation. *Chemosphere*, 268, 128803, 2021.
47. Sudiarto, S.I.A., Renggaman, A., Choi, H.L., Floating aquatic plants for total nitrogen and phosphorus removal from treated swine wastewater and their biomass characteristics. *J. Environ. Manage.*, 231, 763-769, 2019.
48. Cooper, P.F. and Hobson, J.A., Sewage treatment by reed bed systems: The present situation in the United Kingdom. In

Constructed wetlands for wastewater treatment, CRC Press, 153-171, 2020.

49. Miller, G., Use of artificial cattail marshes to treat sewage in Northern Ontario, Canada. In *Constructed wetlands for wastewater treatment*, CRC Press, 636-642, 2020.
50. Madikizela, L.M., Removal of organic pollutants in water using water hyacinth (Eichhornia crassipes). *J. Environ. Manage.*, 295, 113153, 2021.
51. Li, X., Wu, S., Yang, C., Zeng, G., Microalgal and duckweed based constructed wetlands for swine wastewater treatment: A review. *Bioresour. Technol.*, 318, 123858, 2020.
52. Sas, E., Hennequin, L.M., Frémont, A., Jerbi, A., Legault, N., Lamontagne, J., ...Pitre, F.E., Biorefinery potential of sustainable municipal wastewater treatment using fast-growing willow. *Sci. Total Environ.*, 792, 148146, 2021.
53. Gaballah, M.S., Ismail, K., Beltagy, A., Zein Eldin, A.M., Ismail, M.M., Wastewater treatment potential of water lettuce (Pistia stratiotes) with modified engineering design. *J. Water Chem. Technol.*, 41, 197-205, 2019.
54. López-Ocaña, G. and Magaña-Flores, A., Domestic wastewater treated with Sagittaria latifolia in constructed wetlands. *DYNA*, 90(226), 27-35, 2023.
55. Barma, M.D. and Dasgupta, S., A Review on potential of Freshwater macrophytes for Phytoremediation of

wastewater. *Ind. J. Appl. Pure Bio.* Special Vol, 2(1), 200-206, 2021.

56. Hariyadi, H., The Ability of Water Hyacinth, Heartshape False Pickerelweedand Water Lettuce as Chlorine Phitoremediation in Water. *Int. J. Life Sci. Earth Sci.*, 3(1), 30-37, 2020.
57. Zhao, L., Dai, T., Qiao, Z., Sun, P., Hao, J., Yang, Y., Application of artificial intelligence to wastewater treatment: A bibliometric analysis and systematic review of technology, economy, management, and wastewater reuse. *Process Saf. Environ. Prot.*, 133, 169-182, 2020.
58. Safeer, S., Pandey, R.P., Rehman, B., Safdar, T., Ahmad, I., Hasan, S.W., Ullah, A., A review of artificial intelligence in water purification and wastewater treatment: Recent advancements. *J. Water Process Eng.*, 49, 102974, 2022.
59. Zhu, J., Jiang, Z., Feng, L., Improved neural network with least square support vector machine for wastewater treatment process. *Chemosphere*, 308, 136116, 2022.
60. Jawad, J., Hawari, A.H., Zaidi, S.J., Artificial neural network modeling of wastewater treatment and desalination using membrane processes: A review. *Chem. Eng. J.*, 419, 129540, 2021.
61. Pisa, I., Santin, I., Morell, A., Vicario, J.L., Vilanova, R., LSTM-based wastewater treatment plants operation strategies for

effluent quality improvement. *IEEE Access*, 7, 159773-159786, 2019.

62. Guo, Z., Du, B., Wang, J., Shen, Y., Li, Q., Feng, D., ...Wang, H., Data-driven prediction and control of wastewater treatment process through the combination of convolutional neural network and recurrent neural network. *RSC Adv.*, 10(23), 13410-13419, 2020.
63. Wongburi, P. and Park, J.K., Prediction of sludge volume index in a wastewater treatment plant using recurrent neural network. *Sustainability*, 14(10), 6276, 2022.
64. Alnawajha, M.M., Kurniawan, S.B., Imron, M.F., Abdullah, S.R.S., Hasan, H.A., Othman, A.R., Plant-based coagulants/flocculants: characteristics, mechanisms, and possible utilization in treating aquaculture effluent and benefiting from the recovered nutrients. *Environ. Sci. Pollut. Res.*, 29(39), 58430-58453, 2022.
65. Rezania, S., Kamyab, H., Rupani, P. F., Park, J., Nawrot, N., Wojciechowska, E., ...Cabral-Pinto, M.M., Recent advances on the removal of phosphorus in aquatic plant-based systems. *Environ. Technol. Innov.*, 24, 101933, 2021.
66. Bhagat, S.K., Tung, T.M., Yaseen, Z.M., Development of artificial intelligence for modeling wastewater heavy metal removal: State of the art, application assessment and possible future research. *J. Clean. Prod.*, 250, 119473, 2020.

67. Kamali, M., Appels, L., Yu, X., Aminabhavi, T.M., Dewil, R., Artificial intelligence as a sustainable tool in wastewater treatment using membrane bioreactors. *Chem. Eng. J.*, 417, 128070, 2021.
68. Ilyas, M., Kassa, F.M., Darun, M.R., Life cycle cost analysis of wastewater treatment: A systematic review of literature. *J. Clean. Prod.*, 310, 127549, 2021.
69. Yahya, N. and Yahya, N., Agricultural 4.0: Its implementation toward future sustainability. *Green Urea: For Future Sustainability*, 125-145, 2018.
70. Hrynevych, O., Blanco Canto, M., Jiménez García, M., Tendencies of precision agriculture in Ukraine: Disruptive smart farming tools as cooperation drivers. *Agriculture*, 12(5), 698, 2022.
71. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
72. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.

73. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini Robotic Face Recognition System,” *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.
74. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
75. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
76. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
77. Alam, M.F.B., Tushar, S.R., Zaman, S.M., Gonzalez, E.D.S., Bari, A.M., Karmaker, C.L., Analysis of the drivers of Agriculture 4.0 implementation in the emerging economies: Implications towards sustainability and food security. *Green Technol. Sustain.*, 1(2), 100021, 2023.

Note

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14

Drones in Agriculture: Mapping, Monitoring, and Decision-Making

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Abstract

Drones have revolutionized agriculture and farming operations, providing benefits such as boosting crop yields, saving time, and increasing efficiency and overall profitability. They can collect data on a variety of topics, including crop yields, soil quality, and weather, providing farmers with accurate mapping of existing problems and reliable data to develop solutions. Drones have the capability to conduct remote assessment of field crops, reducing labor and producing results much more swiftly than conventional approaches. They can precisely spray crops with liquid pesticides, fertilizers, and herbicides, enabling farmers to know exactly where to apply them when necessary. Furthermore, drones are experiencing growing utilization in agricultural insurance and evaluation domains. To ensure the widespread adoption of this cutting-

edge technology, spreading knowledge about it and making it accessible to end users are essential. The adoption of cognitive solutions will determine the future potential of drones as a tool for precision agriculture. Employing drones in agriculture represents the upcoming technological breakthrough that will assist the industry in meeting the evolving and expanding demands of the future, improving risk and damage evaluations and revolutionizing disaster preparedness and response strategies for both at-risk farmers and national food stability.

***Keywords*:** Drone, agriculture, technology, precision, monitoring, mapping, NDVI

14.1 Introduction

Dynamic remotely operated navigation equipment, DRONE for short, provides aerial surveillance. The name “drone” comes from the buzzing sound it makes, like that of a male honey bee. Originally developed for military and police operations where manned aircraft were impractical or too risky to use, unmanned aerial vehicles (UAVs) or remotely piloted vehicles (RPVs) are now gaining popularity in various industries for mapping, monitoring, and decision-making purposes. As a promising technology, drones have the potential to revolutionize many sectors such as education, agriculture,

weather forecasting, healthcare, defense, and disaster management [4].

Drones are compact, lightweight aerial vehicles with the ability to fly at significant heights and equipped with a wide range of navigation capabilities, recording equipment, cameras including RGB (red, green, and blue), infrared, multispectral cameras, and other sensors. These functionalities make drones a valuable instrument in precision farming, as they are capable of capturing real-time, high-resolution images of crop conditions at a low cost. By using remote sensing indices and other physiological estimations, coupled with the flying capacity of drones, they can provide an easy-to-use instrument for crop analysis and management [17]. Utilizing drones in agriculture can assist in managing adverse weather conditions, increase productivity, and enable precision farming operations for crop yield management and quality. Crop monitoring and health assessment stand out as one of the foremost applications of drone technology in precision agriculture. This involves leveraging artificial intelligence (AI) algorithms and computer vision to analyze qualitative field images. Convolutional neural networks (CNNs) have the capability to capture weed-infested areas, recognize specific crops in need of additional irrigation, and assess plant stress levels during various stages of growth

[8]. This enables more accurate and effective crop management practices.

Drone technology is rapidly being adopted in agriculture, providing high-tech solutions to various farming operations. With the ongoing improvement in sensors capable of gathering precise data and advancements in different software, the possibility of autonomous aerial vehicles collecting data and carrying out tasks could become a reality. Agricultural drones are currently accessible for purchase or lease, and companies such as Precision Hawk provide comprehensive packages that comprise both hardware components and analytical software. India has even used drones to control locust attacks, becoming the first country in the world to do so after finalizing protocols and obtaining statutory approvals. With drone technology, farmers can achieve more with less, improving the quality of their crops and executing a rapid digital plan for safeguarding crops [3].

14.2 Opportunities

There are numerous opportunities for drones in agriculture, including the following:

- Precision farming: By employing drones, farmers can gather crucial data regarding soil conditions, crop health, and weather patterns. This information empowers them to make a data-driven approach, ultimately optimizing their crop yields.
- Crop monitoring and management: By equipping drones with sensors and cameras, farmers can monitor their crops effectively and identify early indicators of diseases or stress. This enables a prompt action to be taken before the issue escalates and affects a larger area of the crop.
- Irrigation management: Drones have the capability to generate 3D maps of crop fields, identifying specific areas that require more or less water and enabling farmers to optimize their irrigation systems.
- Livestock monitoring: Drones can be used to monitor livestock, check for signs of illness or injury, and identify potential issues with fencing or facilities.
- Crop spraying: Drones can be equipped with pesticide or herbicide sprayers, enabling farmers to spot-treat problem areas and reduce the amount of chemicals used overall.
- Seed planting: Drones can effectively plant seeds in expansive fields, reducing reliance on manual labor and significantly enhancing the efficiency of the planting process.

Overall, drones offer farmers a cost-effective and efficient way to manage their crops and improve their yields while also reducing the environmental impact of traditional farming practices.

14.3 Drone Mapping for Transforming Agriculture

Drone mapping is a valuable tool for agriculture as it provides farmers with detailed and up-to-date information about their crops and land. By using drones equipped with cameras and sensors, farmers can create accurate maps of their fields and obtain information about crop health, plant density, water stress, and soil quality. The data gathered from drone mapping can be used to optimize crop management strategies and improve overall yields— for example, by analyzing the 3D maps created by drones, farmers can identify areas of the field with lower plant density and take steps to improve seeding rates or adjust fertilizer application to improve plant growth. Additionally, drone mapping can be utilized for pinpointing sections within the field with potential issues such as pest infestations or disease outbreaks, allowing for targeted treatment rather than blanket applications of pesticides or herbicides. This not only saves time and money but also

reduces the environmental impact of crop management practices [7].

Using drones for monitoring crops and acreage on a large scale offers several advantages, including the ease and effectiveness of capturing real-time video and creating time-based animations to visualize crop growth as it unfolds. In contrast to using satellite or plane imagery, which can be expensive and lack precision, drones equipped with near-infrared (NIR) sensors can provide a bird's eye view of the overall farm health by determining plant health based on light absorption.

Hyper-spectral and multispectral band imagery have become increasingly popular tools for creating Normalized Difference Vegetation Index (NDVI) maps in agriculture. These maps can differentiate between various types of vegetation and soil, identify stressed plants, and differentiate between various crop types and growth stages. Figure 14.1 shows the ranges of NDVI for different crop stages. The data of NDVI, integrated with additional indices such as Crop Water Stress Index (CWSI) and Canopy-Chlorophyll Content Index (CCCI), can offer valuable observations regarding the health of crops. Figure 14.2 shows the plant health conditions by using light sensors. By analyzing crop health and soil conditions with the help of infrared, multispectral, and hyperspectral sensors, drones can provide

precise and accurate information to farmers. The NDVI mapping tool operates on the principle that crop leaves reflect more NIR light than visible light. When plants experience stress or dehydration, they exhibit reduced reflection of NIR light, while the reflection of visible light remains relatively constant. Figure 14.3 shows the evidence of rice field drone mapping for determination of NDVI. By mathematically combining the signals, it becomes possible to differentiate between healthy and unhealthy plants. This information can be used to address crop health issues early, estimate crop yields accurately, and optimize agricultural practices [1].

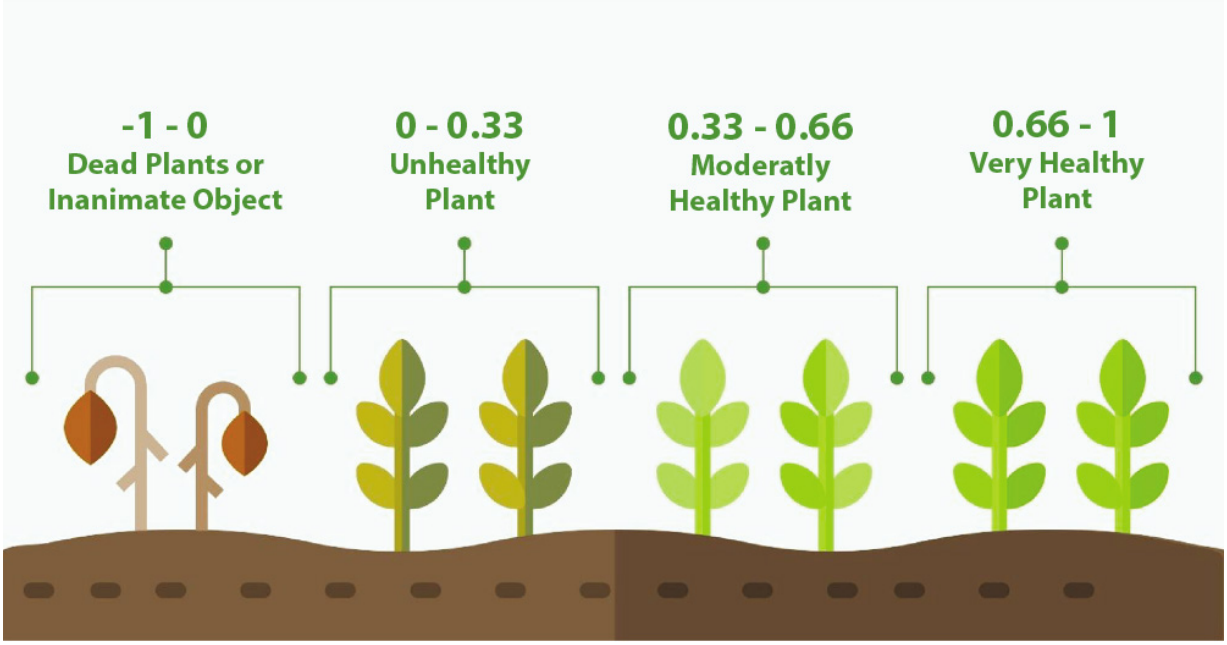

Figure 14.1 NDVI range.

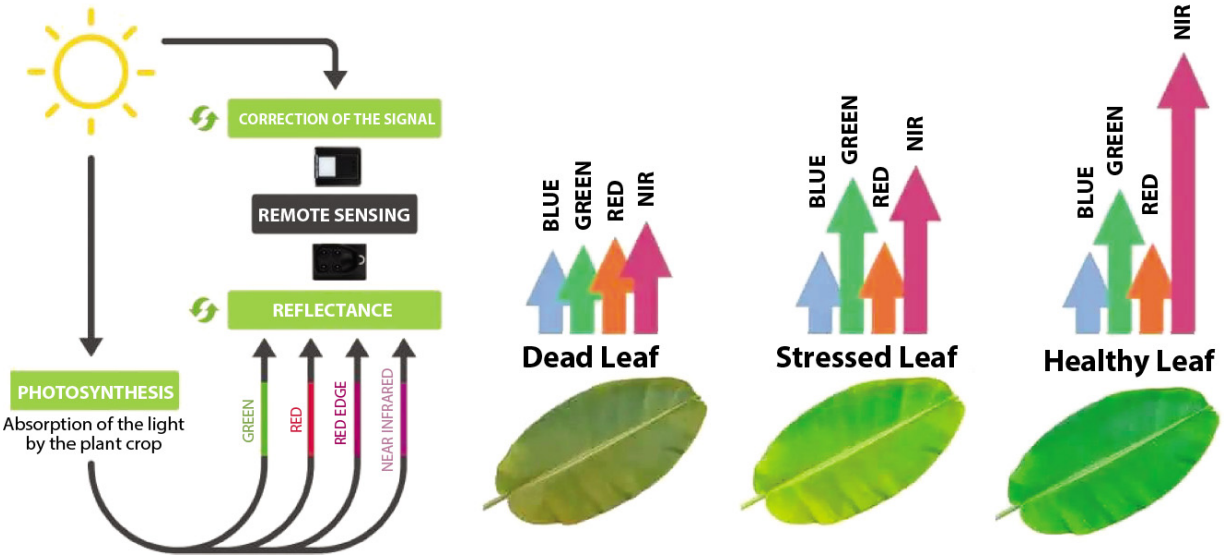

Figure 14.2 Plant health through light sensors.

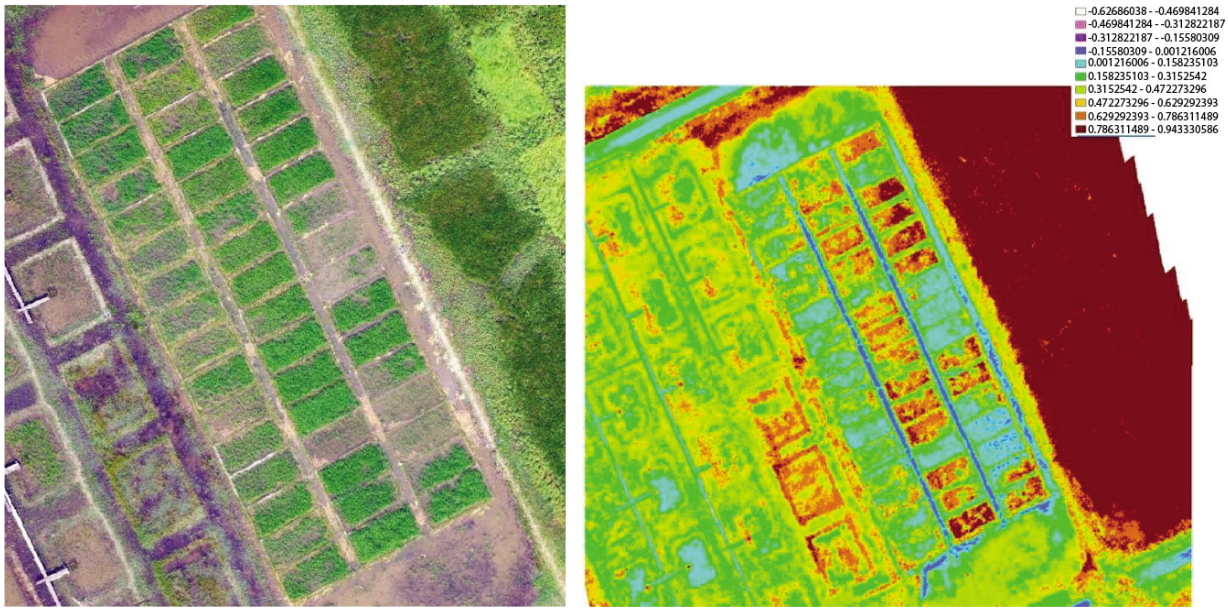

Figure 14.3 Rice field drone mapping for NDVI determination.

The use of drones in the agricultural insurance and assessment sector has become more common, particularly in forensics of insurance claims. Imagery of drones proves beneficial in accurately assessing and calculating the extent of damage. Skymet Weather Services is one such company that employs drones to offer agricultural surveying services to insurance companies and state governments in Gujarat, Madhya Pradesh, Maharashtra, and Rajasthan in India.

14.4 Significance of Drone Monitoring in Precision Agriculture

Agriculture is an area where drones show great potential in addressing major challenges. Drone monitoring in agriculture

involves using drones equipped with various sensors to gather data and information about crops, soil, weather, and other aspects of farming. This data is then analyzed to make informed decisions that can optimize farm productivity and efficiency [18]. Some examples of drone monitoring in agriculture are discussed below.

14.4.1 Crop Health Assessment

Drones equipped with specialized sensors, such as infrared, near-infrared, thermal, and multi-spectral cameras, can assess crop health by detecting changes in chlorophyll levels and identifying areas of stress or disease. This information can be used to target treatments to specific areas of the field, reducing costs and improving yields—for instance, NIR sensors can detect plant stress up to 10 days prior to its manifestation. Drones can also collect data on the rates of sunlight absorption and transpiration, enabling the early identification of issues and prompt resolution. Figure 14.4 shows the status of crop health by drone mapping and monitoring with respect to temperature.

Inadequate crop monitoring presents a significant hurdle in agriculture, but drones can effectively address and overcome this challenge. Time series animations created by drones can

illustrate the crop's development and expose production inefficiencies, enabling improved crop management. By determining the most suitable index for a particular crop, farmers can develop specific prescriptions and classifications for more effective field management. Indeed this approach can optimize inputs like fertilizers, enhance irrigation efficiency, and lead to more effective water resource management.

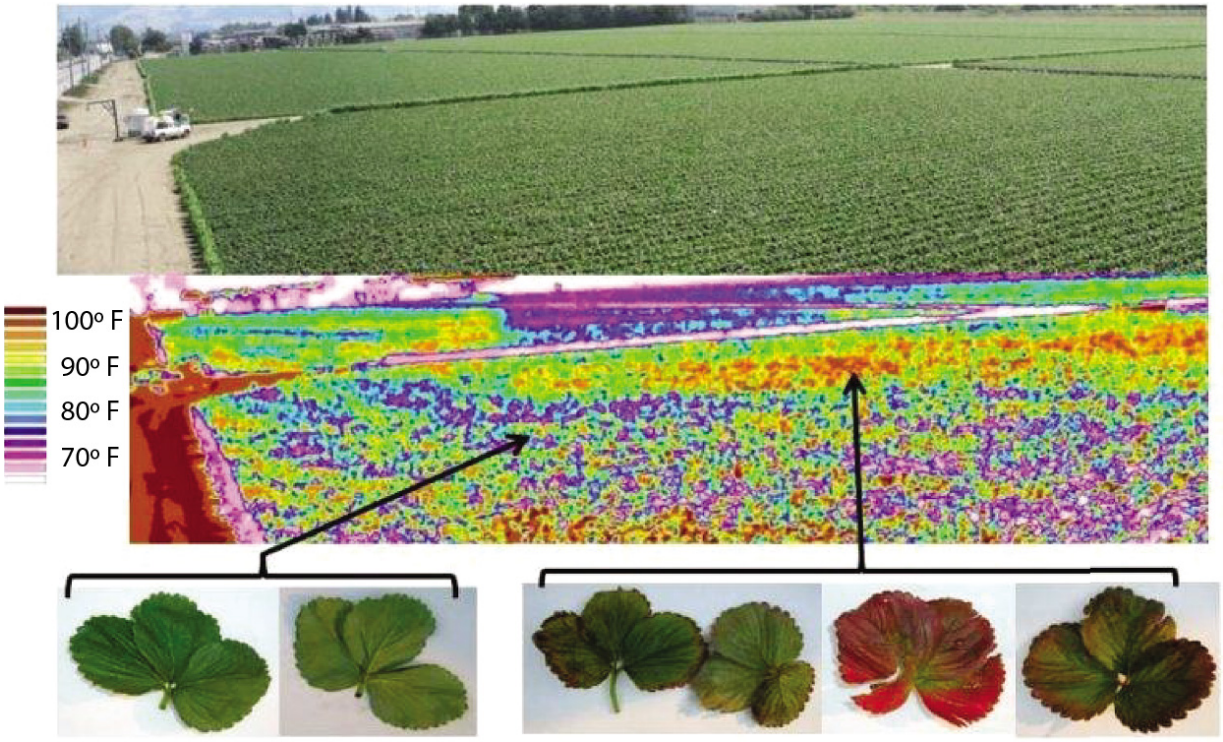

Figure 14.4 Crop health monitoring.

Devices mounted on drones can aid in monitoring plant changes and assessing crop health by scanning through visible and NIR light. This allows farmers to detect diseases early and take appropriate measures to prevent them from spreading.

Nowadays, numerous startups are providing services for monitoring and assessing crop health in agriculture through the utilization of drones.

14.4.2 Irrigation Management

Drones have the capability to precisely gauge the water status of plants and evapotranspiration with a high level of spatial details, making them an ideal tool for irrigation management of specific sites. Additionally, information collected by drones can be utilized to validate and calibrate remote sensing energy balance (RSEB) models. This technology can be employed to ascertain the status of plant water for a variety of crops, and integrating it with an approach to assess chlorophyll fluorescence as an indicator of water use efficiency (WUE) on a crop level would be an exciting challenge. Using sensors, drones can identify areas of a field that require improvements in water management [10]. It can pose challenges due to the large area it covers. However, drones outfitted with thermal imaging devices have the capability to detect irrigation issues and areas experiencing insufficient or excessive levels of moisture. This information can be used to optimize crop layout, improve drainage, and avoid water pooling, which can harm crops. As water and irrigation problems can be costly and reduce crop yields, drone surveying can help identify issues before they

become significant. Overall, drones are highly useful for precision agriculture and water irrigation management as well as planning, monitoring, and evaluating watershed management.

14.4.3 Livestock Monitoring

Thermal imaging cameras mounted on drones offer an efficient way to monitor livestock, providing farmers with valuable information about animal behavior and health at less time and labor costs. Figure 14.5 shows the real-time livestock monitoring by drone technology enables farmers to keep track of their livestock more frequently, allowing for quick detection of any injured or missing animals. This technology enables farmers to keep track of their livestock more frequently, allowing for the quick detection of any injured or missing animals. In addition, drones can monitor animals giving birth, eliminating the need for farmers to constantly patrol their herds. Figure 14.6 shows thermal cameras arm drone for cattle scouting, thermal imagery aids in identifying potential predators targeting livestock, granting farmers a valuable advantage in safeguarding their livestock. Thermal imagery aids in identifying potential predators targeting livestock, granting farmers a valuable advantage in safeguarding their livestock. This can help farmers to identify potential issues

early, improving animal welfare and reducing costs associated with disease and injury.

Figure 14.5 Real-time livestock monitoring by drone.

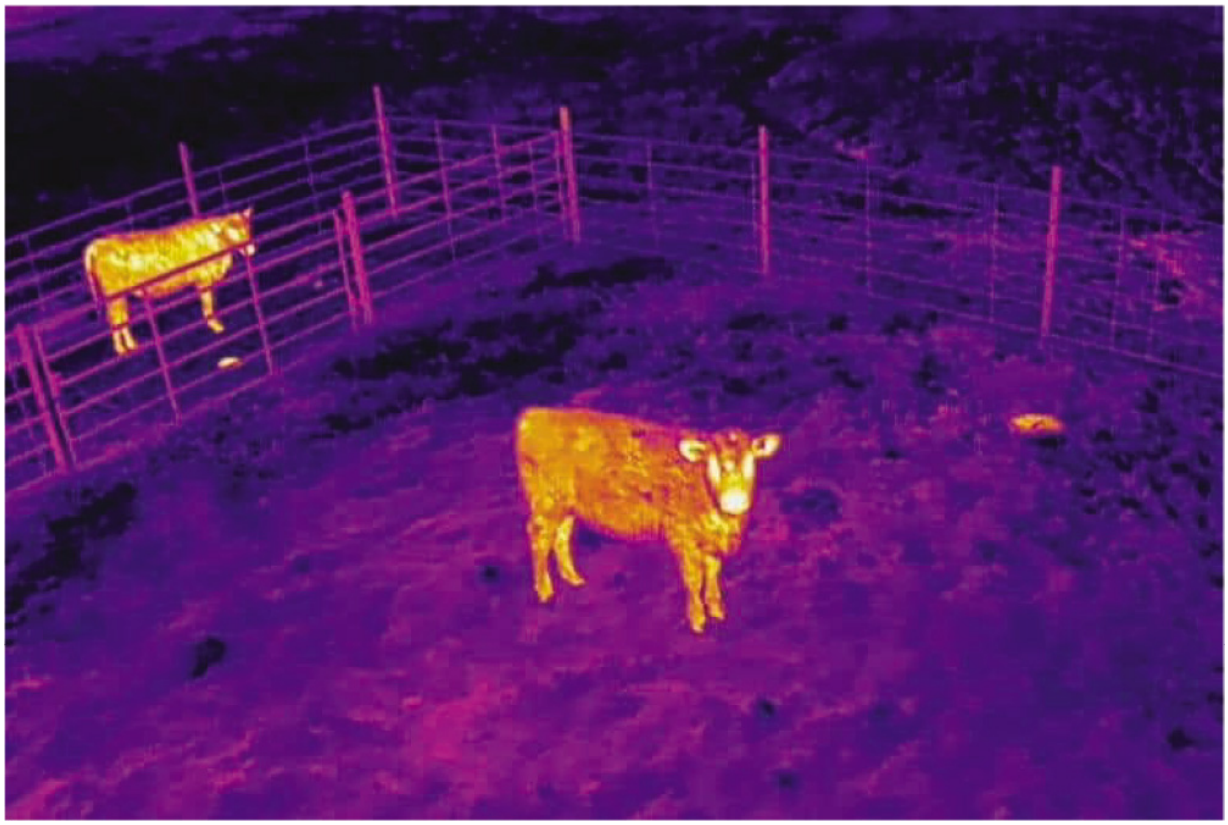

Figure 14.6 Thermal cameras arm drone for cattle scouting.

14.4.4 Weather Monitoring

Drones can be used to collect weather data, such as temperature, humidity, and wind speed, in real-time. The data gathered from drones can be utilized to make well-informed decisions regarding sowing, harvesting, and other farm operations. Overall, drone monitoring in agriculture offers a cost-effective, efficient, and environmentally friendly way to gather data and make informed decisions about farm management.

14.5 Leveraging Drone for Real-Time Decision-Making in Agriculture

Drones equipped with sensors and imaging capabilities are a valuable tool in precision farming as they provide high-resolution data and imagery that can be analyzed in real time to inform decision making. Through the application of drone surveying and mapping, farmers can make technology-driven decisions based on real-time data, avoiding reliance on outdated images or relying on guesswork. It is true that agriculture drones can provide valuable information for farmers to make informed decisions [5, 6]. Some of the data that can be collected by agriculture drones include the following:

- Crop health: Drones equipped with multispectral or hyperspectral sensors can capture images of crops that can be used to analyze their health and identify potential issues such as nutrient deficiencies or pest infestations.
- Crop type and distribution: By analyzing drone imagery, farmers can identify different crop types and their distribution across the field. This information can help them optimize planting patterns and make better use of their land.
- Crop life cycle: By continually monitoring crops throughout their growth cycle, drones can assist farmers in making well-informed decisions about the optimal time for harvesting,

applying fertilizers or pesticides, and identifying specific areas within the field that may necessitate irrigation or fertilizer application. This data can be utilized to optimize irrigation schedules and enhance fertilizer application efficiency, resulting in reduced waste and cost savings.

- Field mapping: Drones equipped with Global Positioning System (GPS) technology can create detailed maps of the farm, including topography and soil type, which can be used to optimize planting patterns and irrigation. Additionally, drones have the capability to estimate yield data and offer valuable information for weather analysis. This can help farmers make informed decisions about predicting weather patterns that may impact their crop yields.

The data gathered *via* drones can be integrated with additional data sources and advanced analytical tools to offer farmers actionable insights. Overall, the data collected by agriculture drones can help farmers maximize their yield while minimizing the environmental impact of their operations and optimize their use of resources, resulting in more efficient and sustainable farming practices with better efficiency to revolutionize the way farmers approach agriculture [11–16].

14.6 Common Usages of Agricultural Drones

The integration of drones in agriculture has revolutionized the industry by streamlining time-consuming tasks and cutting down on costs. As the industry continues to develop and advance technologically, we can expect to see further evolution in the application of drones [2]. There are presently primary uses of drones in agriculture, which are outlined below.

14.6.1 Soil and Field Analysis

Drones can be utilized to collect essential information on soil quality at various phases of a crop cycle. By obtaining maps of the soil by field surveying shows in Figure 14.7, farmers can identify any potential issues such as nutrient deficiencies, moisture stress, soil compaction, or dead zones. This encompasses observations at the start, middle, and end of the crop cycle. By obtaining maps of the soil, farmers can identify any potential issues such as nutrient deficiencies, moisture stress, soil compaction, or dead zones. This information can help farmers make better decisions about planting, crop management, and soil remediation. In addition, ongoing monitoring of soil quality can help farmers optimize water usage and ensure that crops are receiving the appropriate levels of nutrients. By using drones to collect this data, farmers

can save time and resources while also gaining more accurate and detailed information about their soil.

14.6.2 Seed Planting

Drone planting is an emerging technology that is not yet widely used, but some companies are exploring its potential. These companies are innovating specialized setup that enable drones to deploy seed pods accurately into well prepared soil shows in Figure 14.8. These companies are innovating specialized setups that enable drones to deploy seed pods accurately into well-prepared soil. Some drone startups are using unmanned aircraft to deliver tree seeds, herbicides, fertilizers, and water to aid reforestation and replanting projects—for example, Drone Seed is using this technology to deliver up to 57 pounds of payload per aircraft per flight, reducing the need for costly and time-intensive on-the-ground planting. This same technology can also be applied to a variety of farms, reducing overall planting times and labor costs.

Figure 14.7 Field surveying.

Figure 14.8 Seed planting.

14.6.3 Crop Spraying

Spraying agricultural fields using drones offers precise and accurate application of liquids, reducing the overuse of chemicals, which preserves the environment and leads to cost savings. Drone models with different capacities, ranging from 5 to 20 L, are available for pesticides, fungicides, herbicides, and nutritional sprays. The process eliminates the direct interaction of humans with pesticides, which reduces the risk of casualties. Drones can cover 1 to 2 acres of land in just 15–20 min, spraying with high precision and saving 80–90% of the water used in traditional methods. Agricultural drones have the capability to ascend vertically up to 20 m (60 ft) and cover horizontal distances of up to 3 km, making it possible to spray even the tallest crops with any density. Real-time scanning of the ground ensures even coverage, and aerial spraying using drones is five times more rapid compared to conventional machinery [9]. Additionally, drones can be programmed to regulate the height above the ground surface that varies based on the terrain, allowing for efficient atomization of liquids during the spraying process. Additionally, drones can be programmed to regulate the height above the ground surface varies based on the terrain, allowing for efficient atomization of liquids during the crop spraying process shows in Figure 14.9.

Fertilization and spraying are essential for maintaining high crop yields, but traditional methods involving manual labor,

vehicles, or airplanes can be inefficient, costly, and burdensome. By contrast, using drones equipped with a sizable tank containing pesticides or fertilizers offers enhanced safety and cost-effectiveness. Drones can be programmed to operate automatically and follow specific routes and schedules—for instance, in the case of a fungal outbreak in specific crop sections, drones can be employed for the spot-treatment of the issue, preventing it from spreading to the entire farm. Previously, spraying of crops at a specific area was difficult and time-consuming and required spraying the entirety of the acreage, resulting in the inefficient utilization of resources and in ecological impact. However, with the use of drones, this objective can be achieved more efficiently using fewer resources while minimizing the environmental impact.

Figure 14.9 Crop spraying.

14.6.4 Crop Scouting

Crop scouting is a useful application of drones in agriculture, allowing farmers to monitor their crops continuously and efficiently. By using high-resolution images, drones can detect anomalies in agricultural lands, such as diseases, insects, weeds, and crop stress. This enables farmers to identify trouble spots and opportunities for better crop management. Time series animations created from drone surveys can show changes in crop growth over time, revealing areas that require attention. Additionally, drones equipped with both RGB and infrared sensors can generate multispectral images to identify specific clusters of infected plants and provide targeted

remedies. With drone crop scouting, farmers can save time and improve crop yields by detecting problems early and implementing timely solutions.

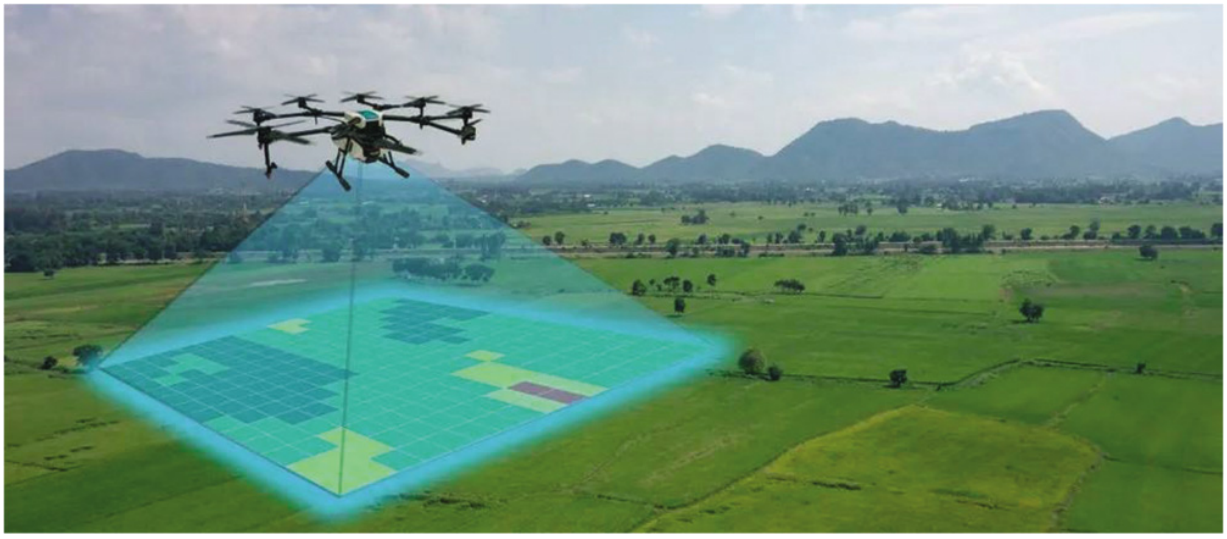

Figure 14.10 Land management.

14.6.5 Land Management

Land management decisions are crucial for optimizing agricultural productivity. Drones can be a valuable tool for land management by providing accurate data for visualization and processing into a digital surface model (DSM) that represents height or altitude shows in Figure 14.10. Drones can be a valuable tool for land management by providing accurate data for visualization and processing into a digital surface model (DSM) that represents height or altitude. This elevation data can assist in making informed choices regarding replanting,

addressing drainage problems, and determining optimal seeding rates.

Accurate plant density determination and identification of canopy gaps are crucial for effective plantation management, enabling yield prediction and maximizing output through refilling gaps of planting. Drones can deliver precise plant counts and detect instances of plant mortality across hundreds of hectares daily, enabling farmers to make informed decisions. Moreover, images captured by drones can be processed to create accurate 3D maps for initial soil analysis, aiding in seed planting planning and providing data for effective nitrogen and irrigation level management. Drones can support land management decisions to optimize agricultural productivity.

14.6.6 Tracing Plantation Lines

In order to prevent erosion and enhance natural irrigation on sloping fields, crops like sugarcane require planting lines along the contours of the terrain. Traditionally, this process of mapping contour lines is time-consuming and requires laborious tractor work that can take several days. However, using drones to generate contour maps provides a quicker and safer alternative. By capturing highly precise elevation data of the land, drones eliminate the need for extensive surveying

methods and expedite the process by up to 75%. Some startups have even developed drones equipped with planting systems that significantly reduce the planting expenses by 85%. This type of system use pods containing seeds with nutrients, which are shot into the upper layer of the soil, delivering all the essential elements required for plant growth.

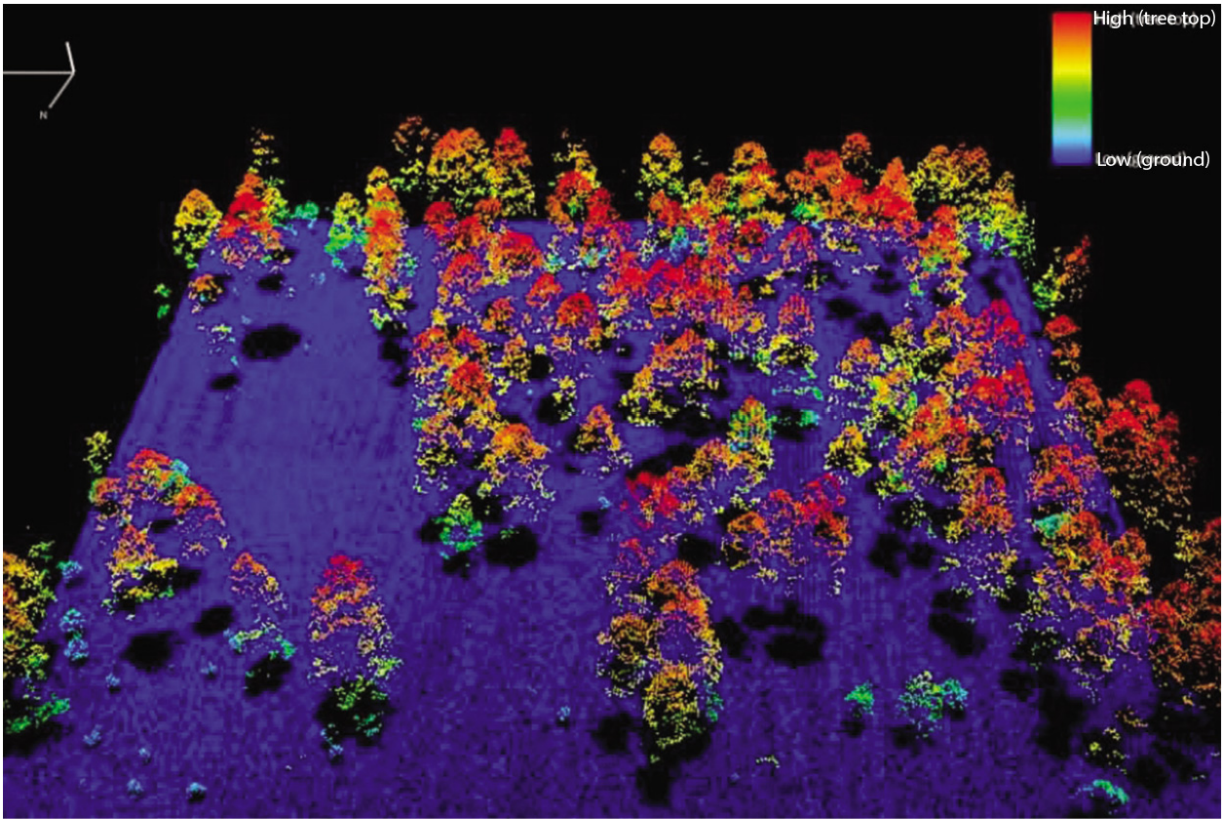

Figure 14.11 Biomass estimation.

14.6.7 Biomass Estimation

Drones can be used to collect data that, when combined with accurate terrain models, can provide precise measurements of

biomass and species heights in monoculture plantations. This is particularly advantageous because drones are relatively inexpensive and can rapidly collect data, thus overcoming the time and cost constraints commonly linked to Light Detection and Ranging (LiDAR) technology and field inventories. Figure 14.11 shows the LiDAR image of biomass estimation.

14.6.8 Integrated GIS Mapping

Drones have the capability to create accurate, geo-tagged 2D maps and 3D models of farmland, which can be analyzed in more detail with a Geographic Information System (GIS). Compared to satellite imagery, using drones is a more cost-effective solution and results in higher-resolution photos. The use of multi-spectral images taken by drone cameras allows for a combination of hyper-spectral imagery and 3D scanning methods that can establish a comprehensive geospatial information system for extensive agricultural land areas. This system can provide guidance throughout the plant's life cycle and allows for control over when, where, and how images are collected.

14.7 Challenges

While drones in agriculture offer numerous benefits, there are also several challenges that need to be addressed. Some of the challenges include the following:

- Regulations and permits: Drone usage is regulated by government agencies, and obtaining necessary permits and following regulations can be time-consuming and costly.
- Data management: Drones capture a large amount of data, which can pose difficulties in terms of management and analysis. Developing effective management and analysis systems is essential for successful drone use in agriculture.
- Technology limitations: Drones are limited in terms of their battery life and range, which can restrict their use for largescale agriculture operations.
- Weather conditions: Drones can be sensitive to weather conditions such as high winds, rain, and fog, which can impact their ability to fly and collect data.
- Privacy and security concerns: Drones can capture images and data that can raise privacy concerns for landowners and the public. Additionally, data security is crucial to ensure that sensitive information is not accessed or misused.
- Cost: The purchasing and maintaining cost of drones can be a significant challenge for smaller farmers and may require

collaboration and investment from multiple stakeholders.

Addressing these challenges requires ongoing research and development, collaboration between stakeholders, and effective policy frameworks that balance the benefits of drone use with privacy and security concerns.

14.8 Summary and Scope

Drones are unmanned aerial vehicles that are revolutionizing the agriculture industry. They offer farmers and agricultural researchers the ability to capture high-resolution aerial images of their fields, gather data on crop health and growth, and identify areas in need of attention. With drones, farmers can circumvent the laborious task of conventional crop scouting and continuously monitor their lands. Drones can be used in various agricultural applications, including crop mapping and analysis, yield monitoring, land management, and tracing plantation lines. Additionally, drones are a cost-effective alternative to traditional methods such as LiDAR and field inventories, making them accessible to farmers and researchers with limited resources. The scope of drones in agriculture is vast, and they have the potential to transform the way farmers manage their lands, increase crop yields, and reduce environmental impact. Agriculture is predicted to be

responsible for approximately 80% of all commercial UAVs, with an expected $80 billion economic impact within the next decade, as stated by the Association for Unmanned Vehicle Systems International (AUVSI). This highlights the importance of raising awareness about the advantages of utilizing this sophisticated drone technology in the precision agriculture sector.

14.9 Conclusion

The application of drone technology in agriculture has revolutionized the way farmers approach their work. With the ability to map, monitor, and make decisions based on high-resolution data, drones have enabled farmers to increase efficiency, reduce costs, and improve crop yields. From crop scouting to land management and tracing plantation lines, drones have demonstrated their significance as an essential asset in modern farming. As technology continues to advance, the possibilities for the use of drones in agriculture are endless, and it is essential to keep exploring new ways to leverage this technology to address the needs of a growing population with a changing environment. Farmers can use drones to conduct a precise analysis of their farms, forecast crop yield accurately, and promptly respond if crops are not performing as expected. Employing precise drone technology can enhance farm

productivity optimization and profitability through real-time field data analysis while also preserving the ecosystem. Adapting cognitive solutions is crucial for the future of farming, and drones have great potential as a tool for precision agriculture.

References

1. Chen, G., Zheng, Y., Cheng, T., Guo, S., Li, X., Liu, J., & Wang, C. (2021). A review of multispectral remote sensing with drones in agriculture. *Precis. Agric.*, 22(2), 387-409.
2. Liu, X., Liu, H., Chen, D., & Chen, X. (2021). A review of drone applications in precision agriculture. *J. Cleaner Prod.*, 285, 125446.
3. Shafique, M., Rizwan, M., & Hussain, M. (2021). Drones in agriculture: A review of recent advances and future prospects. *Biosyst. Eng.*, 207, 151-167.
4. U. B. Zalavadiya & R. N. Vasoya (2020). Significance of drones in precision agriculture. *Agric. Environ. E-Newsletter*, 1(2), 36-40.
5. Zhang, Q., Cheng, T., Li, M., Li, X., & Wang, C. (2020). Drones in agriculture: A review of current status and future perspectives. *Precis. Agric.*, 21(1), 1-21.
6. Huang, X., Yang, W., & Liu, X. (2019). Applications of drones in agriculture: A review. *J. Biosyst. Eng.*, 44(3), 297-308.

7. Clercq, M., Vats, A. and Biel, A. (2018). *Agriculture 4.0: The Future of Farming Technology*. https://www.worldgovernmentsummit.org/observer/reports/2018/agriculture-4.0-the-future-of-farming-technology
8. Dharmaraj, V. and Vijayanand, C. (2018). Artificial Intelligence (AI) in Agriculture. *Int. J. Curr. Microbiol. Appl. Sci.*, 7(12): 2122-2128.
9. Guan, J., Lee, W. S., & Chong, H. Y. (2018). Drones in agriculture: A review of applications and economic potential. *Int. J. Remote Sens.*, 39(5), 1549-1570.
10. Bonneau, V. and Copigneaux, B. (2017). Industry 4.0 in agriculture: Focus on IoT aspects. https://ec.europa.eu/growth/tools-databases/dem/monitor/content/industry-40-agriculture-focus-iot-aspects
11. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
12. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.

13. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini Robotic Face Recognition System,” *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.
14. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
15. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
16. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
17. Matese, A., Toscano, P., Di Gennaro, S. F., Genesio, L., Vaccari, F. P., Primicerio, J., ... & Zaldei, A. (2015). Intercomparison of UAV, aircraft, and satellite remote sensing platforms for precision agriculture. *ISPRS J. Photogramm. Remote Sens.*, 108, 104-115.

18. Zalavadiya, U. B., Kapupara, P. J. and Sabhadiya, H. R. (2024). Harvesting Tomorrow: A Closer Look at Information Technology in Agriculture. *Farm Chronicle*, 03(01), 20-25..

Note

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15

Investigation of Banana Plant Disease Detection Using Transfer Learning

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Abstract

Plant disease may severely reduce food supplies, potentially wiping off species variety. Early crop disease identification, using reliable or automated detection methods, can boost food output while decreasing financial losses. Deep learning has significantly improved the identification precision of picture classification and object recognition systems in recent years. As a result, in this research, a pretrained convolutional neural network (CNN) model is used to swiftly identify plant illnesses. The focus of the suggested model was on enhancing the hyper parameters of well-known pre-trained models like Mobile Net,

ResNet-152, VGG-19, and Inception v3. The suggested model makes use of the ResNet152 architecture and intricate hidden layers tuned using random search tuning methods on the real-time dataset of banana leaf capture. Classification precision, recall, accuracy, and F1_score were used to assess the suggested model's efficiency. There was also a comparison with other cutting-edge investigations. According to the findings, ResNet152 performed 98.3% better in classification accuracy tests than the most advanced models.

Keywords**:** Plant disease, deep learning, CNN, ResNet152, banana leaves, random search tuning techniques, classification

15.1 Introduction

Agricultural production, as a major actor in the global economy, is the primary source of food, money, and employment. Agriculture generates 18% of India's GDP and 53% of job opportunities as it does in other low- and middle-income nations with a large farmer population. This industry makes the greatest contribution to economic growth. As a result, the impact of plant illness and pest infections on agriculture may have an impact on the global economy by reducing the quality of food output.

Agriculture is regarded as the primary source of revenue in the majority of developing countries. Banana production is a key sector of the worldwide agriculture-business since bananas are rich in minerals including calcium, iron, and other nutrients. Because this crop has so many nutrients and is regarded as an instant energy booster, people from all over the world consume it. Around 15% of the world's production of bananas, according to Wikipedia, is exported for consumption in western nations. According to banana production and export estimates, India accounts for around 25.7% of worldwide banana output, with the Philippines, Indonesia, and Brazil contributing for approximately 20%. Up to 100% of the nation's banana production and export will be lost if a banana tree becomes diseased due to disease or other climate-related factors. Bananas are typically affected by five main illnesses, including 256*256 banana leaf spot, banana Panama wilt, banana black Sigatoka, banana yellow Sigatoka, and banana healthy. Figure 15.1 displays a few examples of the dataset's pictures.

The introduction of algorithms for machine learning and neural network methodologies has resulted in substantial advances in research in plant disease diagnosis. This has rendered it easier to transmit an image's original features *via* automatic classification and extraction of features. In addition, the accessibility of data and technology that enables advanced

deep-learning structures with fewer components has helped the move from traditional to deep-learning platforms. The ability of convolution neural networks to distinguish and categorize images by extracting complex lowlevel information has recently gained a lot of attention. As a result, because they yield better results, CNNs are preferred to replace more traditional techniques in automated plant disease diagnosis [1]. The DL model will probably perform best when trained on a large dataset. Despite the good outcomes of earlier research, there is always opportunity for improvements to the assortment of image collections. Compared to genuine image materials acquired from real agricultural fields, models developed on publicly available data lack variety in information as well as environment.

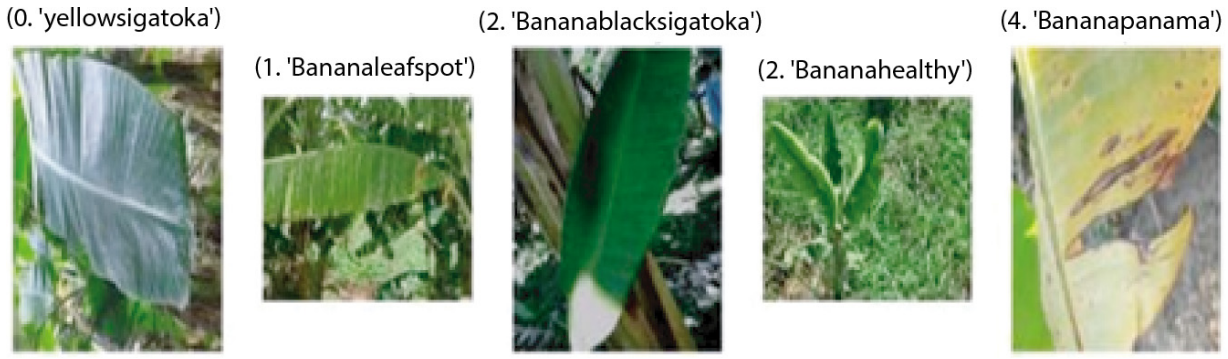

Figure 15.1 Sample input images from real-time data set for four diseases of banana leaf.

15.2 Literature Survey

The authors of [2] described a corn leaf function optimal control framework for improving maize qualities in a complex setting. Robust Alex net was employed during this research for precise extracting features. Increased usability and accuracy are among the benefits. The job lacks real-time diagnostics and difficulty assessment. The researchers were suggested by [3] sensing difficulties in rice images such as noise, blurring picture edges, low high detection, and higher noise interference. The results showed that precision had increased while time complexity had decreased. One limitation was that the identification of illness and dynamic scanners of huge paddy cultivation was not carried out.

The researchers of [4] presented an illness detection approach that is web-enabled designed for effective disease detection *via* compressed sensing. The research proposal was endorsed by the researchers to use both modeling as well as novel analyses for something like a wide range of plants. SVM was used for categorization, yielding an overall accuracy of 98.5. A classification algorithm that is dependent on evolutionary algorithms and a support vector machine was proposed by the researchers of [5]. According to the results, the suggested task has a detection accuracy of 95.71% and a classification accuracy of 97.6%.

For the farmers' convenience, the author proposed [6] an efficient plant strain detection mechanism that is automated. The author developed a web-based tool to identify pomegranate diseases based on color and morphology. With an accuracy of 82%, the K-means method is used for classification and segmentation using SVM. A framework employs a camera to record images of the leaves in the natural environment as well as a gadget that must learn to classify the leaf as healthy or unhealthy. This article [7] concentrated on leaf crop diseases. The author created a method to identify plant illnesses in fresh leaves using transfer learning methods like deep learning, using CNN as a trait representation and SVM as a classification engine.

This paper [8] focuses on developing quite a scheme for smart agriculture, specifically the initial detection of illnesses in leaf tissue, and provides farm owners with solutions. The primary contributors are the innovative pixel replacement fragmentation and the double trying to extract characteristic methods for improving classification. This paper [9] illustrates the effectiveness of the structure by evaluating variables such as memory space, safety handling overhead, in live time using a WMSN testbed.

A purpose of this research [10] is to inform farmers about cutting-edge innovations for reducing plant leaf illnesses. Because tomato is a common vegetable, methods utilizing machine learning and image analysis with a precise automated system have been recognized for identifying crop disease in tomato plants. The suggested system [11] processes leaf images and extracts texture information using imaging techniques and IoT. Located somewhere at a station, the GLCM features are classified using the RFC method and analyzed by agriculture experts who provide solutions.

The author's technique [12] proposes a computerized approach for agro diagnostic evaluation that uses feature extraction and a classifier to categorize distinct leaf sample images from various crop varieties. Each crop health phase has its own one description when applying the proposed technique. The grape leaf-trained devices operated flawlessly when tested in other crops. When ambiguous test data can be attributed to one or more criteria, a novel technique based on conflict resolution across all systems determines the correct state. A comprehensive research of the various ways utilized in recognizing plant sickness utilizing AI-based machine learning algorithm methodologies has been undertaken in this approach [13]. The use of deep learning (DL) as a tool in computer vision has grown in prominence.

This paper [14] investigates the literature on machine learning techniques used in agriculture, with a focus on disease and pest classification, detection, and prediction, with a focus on tomato plants. By supporting the creation of strategies that will allow the company to use organic practices and chemical goods while preserving and improving harvest production and quality, this survey intends to enhance smart agriculture and precision agriculture.

The author [15] suggested an automated approach for diagnosing diseases that provides farmers with quick and accurate diagnostic tests for agricultural diseases.

Technology of leaf disease in plant sensing systems is essential for speeding up crop diagnostic tests. Using computer training and image handling, the above-mentioned article discusses a method for detecting leaf illness.

This framework can accept a leaf image as input. In conventional agriculture, manually monitoring crops is a labor-intensive, slowly moving, and error-prone process. Continuous plant monitoring helps minimize the financial and production losses brought on by plant diseases. The novel approach increases precision farming significantly by automating the realworld challenge of crop disease diagnosis using several

models. Fully automated pesticide sprayers, leaf recognition using the YoloV3 CNN model, disease spread using image analysis, illness diagnosis using InceptionV3, and picture capture automation using Raspberry Pi and a smart line tracking robot are all discussed in the paper [16].

This article [17] goes on to discuss how to use the Web of Things (IoT) to monitor an electric vehicle power efficiency. An electric vehicle is obviously entirely reliant on the energy source provided by a power supply. The energy delivered to the vehicle, on the other hand, is progressively lowering, resulting in framerate drops. The system is capable of recognizing the degraded performance of the battery and sending alert notification to the user for further action, according to the findings of the tests. This method is advantageous in public transport and V2G systems. The suggested technique will facilitate city living while enhancing urban planning. A method for sensing diseases in plants is described in this paper [18]. Illness is detected on the yields different. The detection of plant diseases system presented here is simple as well as computationally efficient, requiring less time for forecast than other deep having to learn strategies. The accuracy of various leaf and plant diseases is calculated and presented in this paper.

This report's [19] purpose is to provide a computational modeling (ML) strategy for early clinical threat that varies depending on elements from the IOT that can be immediately felt in crop fields. The life cycles of plant diseases are tightly correlated with environmental factors. Plant diseases can be predicted to occur using the environmental factors in the harvest field. Given that the propagation of the illness and environmental variables have a linear connection, crop field environmental parameters can predict the existence of plant diseases using the IOT. The suggested model is used to predict blister blight in plant leaves in order to assess the efficiency of the proposed remedy.

Detection and classification of viral infections that have afflicted six cereal grains over the previous 5 years as well as the most prevalent and harmful illnesses influencing wheat crop yields were carried out [20]. According to the results of this survey, transfer learning is one of the most widely used and effective training methods for hyper spectral images, and it provides the most effective strategy for diagnosing early-stage illnesses.

The author investigates recent breakthroughs in diagnosing diseases of plants by analyzing plant leaves with machine vision and methods for image processing [21]. Canny-edge

detection is used to extract edge features, grid movement is used to extract hue information, and the local binary pattern (LBP) is used for texture analysis. The collected features were then concatenated to create a merged feature vector, which was utilized to train the SVM and ANN techniques for machine learning. The design of these machine learning techniques allows the proposed decision-making support model to recognize and classify distinct leaf seedlings. Inside this article, the author originally proposed a transfer learning approach built on fully connected convolutional networks to identify diseases in plants, which we anticipate to execute on network edge with increased computing capabilities.

The author proposed [22] a light deep neural networks approach to minimize model length and computational expense as well as adjust the convolutional network and reduce input size. The proposed models are trained using a range of image sizes in order to determine the appropriate size for the input image. Through trials, the effectiveness of the suggested algorithms on real-world datasets is assessed; the findings indicate that the offered methods may accurately identify plant illness while requiring little computer work.

The author [23] proposed an Internet of Things plan. Ground sensors are used to gather tree data in the form of photographs

at various intervals to aid in the study of the state of ground agricultural water. These instruments can also detect crop diseases and ground micronutrients. Temperature sensors are also used. Groundwater detectors are used to continually monitor the water quality. All information will be sent to the gardener through IoT. Binarization thresholding is used to categorize images. During harvesting, the robot does image classification and identification.

The study [24] offers a coconut disease categorization and detection method powered by artificial intelligence (AI) for sustainable farming. The technique mentioned here tries to identify oil palm tree ailments in the context of precision agriculture in order to improve crop yield. To begin, the suggested method makes use of a noise reduction method based on median filtering. The Naive Bayes fuzzy swarm segmentation is then used to identify the damaged leaf regions. Another application for the capsule required for progress is as a feature representation. Using the GRC model, the author wants to diagnose illnesses in coconut plants. During the experimental examination, the proposed technique was applied, and the findings were excellent.

The planned [25] plan uses classification approaches to track farmland elements like as humidity and temperature, soil

moisture content, and plant diseases. A Raspberry Pi is used in the system to automate jobs and identify ailments. The farmers may view the sensor data *via* a Web page. Any diseases identified in the plants are instantly sent to the grower. A research [26] discovered that segmentation was occasionally employed with Mask CNN to detect diseases in apple crop leaves. Three separate Mask R-CNN networks are trained and evaluated for the purposes of object recognition, classification, and sickness diagnosis. The proposed method is trained and evaluated using the author's segmentation masks. Researchers discovered that a prototype using a ResNet model backbone works well on the job, especially when recognizing extremely small rust disease particles on the leaf.

In this article, pre-trained models for convolutional neural network-based plant disease diagnostics are presented [27]. The study concentrated on the high-dimensional parameters of recognized models that had been trained. The tests made use of the well-known Plant Village data set of various diseased plant types classified into 38 distinct categories. The model's efficacy was assessed using classifying precision, specificity, sensitivity, and F1 score. A comparison with similar cutting-edge investigations was also carried out. The trials show that DenseNet-121 outperformed state-of-theart models in classification accuracy by 99.81%.

For enhanced agricultural productivity, early disease detection and control are essential. Plant disease can be diagnosed by the texture, color, and spots on the leaves, which differ from healthy leaves. The traditional way of viewing the leaves necessitates knowledge. As a result of the article [28], farmers who lack the knowledge or resources to hire an expert can benefit from the development of leaf illness identification methods like transfer learning. In this study, illnesses affecting maize, tomatoes, and potatoes are identified using ResNet learning model. The performance of ResNet model is compared to that of various deep learning models built and trained from scratch. ResNet model had the best performance for detecting plant diseases, scoring 98.7%. The program can recognize 16 different classes of plant diseases.

This research [29] presents a high-precision transfer learning model that may give farmers and agricultural organizations a mobile solution for detecting rice leaf diseases on the spot. The amount of illness samples is balanced in the current experiment using a generative adversarial network. When tested on a GAN-augmented dataset, the presented model beats paradigm classification architectures with an average cross-validation accuracy of 98.79%. Without GAN augmentation, the framework is also tested on three different datasets, with benchmark results of 98.38%. In this work [30] using deep

learning, features of potato plants are identified using the MobileNet architecture. In order to employ transfer learning, the base layers are frozen, and the top 23 levels—including the new classifier layer—are trained only. The model is then given more training to improve performance. This improves model performance and results in 99.83% accuracy in image categorization of infected illness on potato foliage. The experimental findings show that this approach is feasible on portable devices.

It would be ideal if rice plant illnesses could be detected automatically. There are several methods for identifying rice plant illnesses that have been proposed. Recent research [31] suggests that utilizing CNN models in such scenarios might be highly beneficial. In this study, we examined and trained various CNN models to enhance accuracy by combining unique training and learning methodologies. The most advanced large-scale designs are built using the baseline and transfer learning approaches. These models are created and tested using data from various sources. When compared to other CNN designs and literature, the ResNet50 architecture achieved the highest accuracy of 97.5%.

The author [32] investigates computer vision approaches that can assist plant owners in swiftly and automatically identifying

these leaf abnormalities in their plants. In our studies, the simulations produce highly accurate results on a dataset of around 4,000 images. In our study, the transfer learning-based model outperformed the other two techniques in terms of accuracy. Furthermore, a mobile Android application based on this framework will allow consumers to easily see the status of their plants in real time.

Grapes diseases of the leaves have a significant influence on the development of the grape industry and the productivity of grape harvests. It is critical to diagnose grape diseases in their early stages in order to prevent disease growth and its consequences as well as to preserve the grape industry's continuous expansion and production. However, it is crucial to spot grape leaf disease in its early stages and challenging to do so. Therefore, in this work [33], machine learning algorithms are employed to distinguish between several disease classifications properly and identify grape leaf disease early. For the classification of leaf diseases, enhanced KNN and a neural network-based categorization model are also constructed. The qualities of a high-quality histogram and an expanded histogram can give boundary, structural, pattern, and discriminative information. The generated high-quality gradient-based features are subsequently used in the classification process. The proposed models significantly

enhance categorization precision. The suggested models are tested for accuracy using the Plant Village dataset, a publicly available dataset. In terms of classification accuracy, the proposed models outperform a variety of current classification strategies.

According to the author [34], specialists have significantly increased the efficiency and precision of object recognition and identification systems as a result of recent advances in deep learning. Deep learning techniques were used to construct conventional neural network models for recognizing and diagnosing plant leaf disease in basic photos of affected and healthy plants.

Four models based on deep learning are used in this article to identify sickness in leaves. For training and testing, a new dataset on plant diseases was employed. There are 38 unique classes, each with basic images of healthy and ill plant leaves. This trained algorithm is also used to evaluate online photos of greenery. The successful conclusion of the model makes it an excellent tool for early warning systems and a strategy that might be utilized to develop a viable integrated plant disease detection system. We found that the MobileNet model, which has accuracy in training and validation of 99.07 and 97.52%, is a great match for the new plant ailments in the data set.

A significant issue in agriculture for raising crop yield and productivity is identifying nutrients in leaves that are either excessively abundant or insufficient. Researchers have created a number of image processing techniques for locating and minimizing deficiency in crops [35]. Many of these types of detection techniques are costly, ineffective, and threaten crop immutability if they do not identify the nutritional value of a leaf immediately. As prediction techniques advance, erroneous predictions that lower crop quality and productivity result. The greenness approach's analysis of nutrient range is used to identify and classify nitrogen nutrients in groundnut leaves. The median filter is used in pre-processing to reduce the maximum peak signal-to-noise ratio value. Using color features, the proportion of green in groundnut leaves was shown to be closely related to nitrogen intake. The suggested convolution neural network-based holding vector network model then detects and categorizes the three distinct nutritional ranges. The CNN-based HVN model developed has a 95% training accuracy and a 92% validation accuracy, according to the results.

Recent advances in computer vision, aided by deep learning, have enabled the capture of plant leaves and their usage as a foundation for discriminating between plant species. This study [36] developed a novel method for categorizing various plant

species. The approach was created in order to diagnose and cure a variety of tomato leaf diseases. Deep learning models were created to identify tomato leaf diseases in the Plant Village dataset. The training accuracy of our model was 96.37%, which is considered good.

Among emerging countries, agriculture is the main industry with a significant percentage of GDP. Low yield is caused by natural disasters, a lack of rainfall, incorrect fertilization, and the failure to detect illnesses in time. Furthermore, typical illness detection procedures will not produce reliable findings. As a result, in this study [37], a system is created employing digital image processing to identify plant illness from the leaf since plant disease is reflected on its leaf. Preprocessing, segmentation, and extraction of features are the three pipelining methods used in the DIP methodology. The sick leaf is used to provide texture features, color features, and form features. The acquired characteristics are then used to train an SVM classifier to label various illnesses. Through cross-validating, experiments on tomato plant images obtained from conventional data sets yielded an accuracy of 86.6%.

This study [38] investigates the possibility of detecting illnesses in jasmine plants using a machine learning approach based on Image Processing that includes five critical phases. The methods

comprise picture acquisition, input image preliminary processing, segmentation of pre-processed images, feature extraction from segmented images, and illness classification. This study, among other things, focuses on identifying four prevalent illnesses in jasmine plants. There is a need for an effective pre-processing method that results in increased accuracy in identifying jasmine plant illnesses, even if pre-processing chores can have an impact on disease classification performance. Therefore, in order to overcome the obstacles presented by picture data, cleaning of the image data is required before segmentation.

To prevent these challenges, this research focuses on picture scaling and shifting photo color spaces, among other things. The image acquisition process is carried out with the use of real-time image sets obtained from several jasmine crop cultivations. Images of sick jasmine leaves are captured with a typical Android smartphone or tablet camera. The median filter is used to the source pictures for contrast enhancement, followed by bidirectional filtration. The pre-processed images are utilized to diagnose ailments in jasmine crops by segmentation and feature extraction. The purpose is to develop efficient processing methods for enhancing the accuracy of recognizing diseases in jasmine harvests. This study describes a

hybrid image noise reduction and image scaling pre-processing method.

Former Leaf, a transformer-based leaf disease detection model, as well as two optimization methodologies to increase model performance, is introduced in this paper by the author [39]. To be more explicit, we provide the least significant attention pruning approach for selecting the most significant attention heads in each transformer model layer. It has the ability to reduce model size by up to 28% while increasing evaluation time by 15% and improving accuracy by about 3%. In addition, we apply sparse matrices–matrix multiplication in the model to determine matrix correlation. The model's complexity is reduced, resulting in a 10% reduction in training time while keeping equal performance. The results of the cassava leaves diseases data assessment show that our approach outperforms state-of-the-art models in the majority of cases.

This paper [40] proposed a low-cost solution for classifying apples and recognizing apple disease using a neural network. This approach takes use of visual cues like color and texture. The study made use of a multi-layer perceptron neural network, with the output being the selected classifications and the input being the image attributes. The accuracy of the proposed technique was examined using various topologies of

single-layer and two-layer neural network architectures after the network had been trained with 60% of the photographs and the rest of the pictures were kept for testing. Based on the research, a two-layer architecture with eight cells in the first layer and eight cells in the second layer generated the best results.

A deep learning approach for weed detection was developed in this article [41] utilizing a dataset of multispectral photos of sunflowers taken at various times of day. After being trained on photographs of crops during their emergence and subsequent development phases, a deep learning system dubbed U-Net was evaluated on images of crops in their ultimate form, when chemical treatments may be administered. Using conditional random fields to achieve superior segmentation in terms of intersection over union enhanced the U-Net results significantly. The recommended approach employing channels as input achieved the best mean on pictures with a resolution of 512 *512 pixels. The results suggest that the technique and input employed provide a viable plan for distinguishing plants from weeds.

In this study, an automated approach for detecting olive trees and computing crop yield was applied [42]. The proposed method consists of two steps: a regression-based crop

estimation model followed by a deep learning-based categorization model. During the classification stage, foreground tree information is gathered using an enhanced segmentation approach, namely, the K-mean clustering methodology, and a super-feature vector composed of statistical and geometric attributes is synthesized.

Following that, the retrieved characteristics are used to predict the projected crop output. Furthermore, the proposed technique is confirmed using Google Maps satellite pictures of olive farms. In contrast to current approaches, the suggested method added originality and accuracy, exceeding the others with an overall classification accuracy of 98.1%.

15.3 Transfer Learning

Innumerable real-world apps employ image processing [43, 44] machine learning (ML) [45] approaches to generate perhaps the best data available to customers. Transfer learning (TL), one of the ML categories, has gained a lot of attention from the academic community in recent years. Traditional ML [46] methods operate under the assumption that a model will train and test samples from a constrained data distribution. These traditional approaches anticipate target tasks without difficulty and are applied to small data distributions. However, this

problem might be remedied using TL. TL is known for its ability to link extra testing and training samples, resulting in speedier output with more efficient outcomes.

In research based on images, CNN are frequently employed. They are excellent at identifying subtle, complex qualities in images. On the other hand, deep CNN [47, 48] layers are challenging to train because to the expensive computational process. Various academics have developed transfer learning-based approaches to address such challenges. VGG19, ResNet152, MobileNet, and Inception are examples of popular transfer learning models. These models are trained using the ImageNet data set, which has many classifications. Because visual characteristics like edges and contours are ubiquitous across datasets, such models may be trained with any dataset.

15.4 Methodology

CNN models work well with picture databases for object identification and classification. Despite the advantages of CNNs, there are still several drawbacks, including the extensive training period and the need for large datasets. Model training is more difficult since deep CNN models are needed to extract low-level and complicated attributes from images. The previously mentioned concerns can be addressed via transfer

learning methodologies. Transfer learning makes use of pre-trained networks to apply parameters of the model learned on one data to another.

In image-based research, deep-learning algorithms are commonly employed. They are particularly good at extracting complex properties from images. Deep convolution neural networks layers, on the other hand, are challenging to train because of the computational expense of the method. To overcome such issues, the suggested model was built by transferring learning-based methodologies. These models are trained using the capture dataset, which has many classifications. Because visual characteristics like corners and shapes are ubiquitous across data, such algorithms may be trained with real-time capture image. As a result, the transfer learning strategy was discovered as being the most appropriate and robust model for picture categorization. We selected color photographs from the real-time collected dataset for our experiments since they matched well with the transfer learning models. Because we employed several pre-trained network models that need varied input sizes, the photos were downscaled to $n*n$ pixels as a consistent format.

Plant disease files include photos of sick and healthy plants as well as the classifications for each sample. If we treat the

banana plant as a class, all images of both healthy and sick plant samples will be grouped together. To classify the target image, just the properties of the input image were used. The implementation is based on datasets from banana diseases such as banana black sigatoka sickness. For our experimental investigation, we separated the collected data into training, testing, and validation samples. The pre-trained models were trained using 80% of the capture dataset, with the remaining 20% utilized for testing and validation.

Following training with sets of disease samples from various plant classes, a single illness sample is retrieved as input, and the result of the testing phase will classify the illness's precise label from among the four categories mapped under that specific class. As a result, multiple-class classification and multi-label categorization are mutually incompatible, whereas multi-class classification recognizes each category inside a class as a separate class.

Because the model may be trained by freezing the outermost or innermost layers, transfer learning allows for more accurate tasks. The algorithm's settings may be kept and altered by freezing the layers for feature extraction and classification. In this study, the performance of multiple transfer learning models is compared to deep convolutional neural networks in

order to improve recognition and classification accuracy while decreasing time complexity. Figure 15.2 depicts the structural organization. The studies included pre-trained CNN models including VGG19, Mobile Net, Inception v3, and ResNet152 in addition to the real-time capture dataset.

Any machine learning model training activity requires hyper parameter adjustment or optimization. The learning process cannot derive a model's hyper parameters from the available datasets. They are, nonetheless, critical in controlling the learning process itself. The mathematical description of machine learning models yields these hyper parameters. The weights obtained while training a linear regression model, for example, are parameters, but the learning rate in gradient descent is a hyper parameter. The success of a model on a dataset is heavily reliant on good tuning, or determining the ideal combination of model hyper parameters.

Grid search, randomized search, Bayesian optimization, and more approaches are available for hyper parameter optimization. Today we will look at the randomized search technique and implementation. Data scientists configure the model hyper parameters to regulate the model's implementation. Once data scientists have determined the values of a model's hyper parameters, they may be considered

model settings. These values must be changed for each task because the best hyper parameters for one dataset won't be the best across all datasets. Grid search and randomized search are two strategies often used in hyper parameter tuning. Grid search searches through every possible combination of the hyper parameter values supplied. In contrast to grid search, randomized search does not test all specified parameter values. Rather, a predetermined number of parameter settings are drawn from the defined distributions. If all parameters are supplied as a list, sampling without replacement is done. Sampling with replacement is used if at least one parameter is given as a distribution. The sample mechanism in a randomized search can be specified ahead of time. For each hyper parameter, you may provide either a distribution of potential values or a list of discrete values. The transfer learning model has the benefit of learning quicker than models developed from scratch, as well as the option to freeze model layers and train the last layers for more exact categorization. Initially, some hyper parameter standardizations for various pre-trained models were conducted. Table 15.1 shows the characteristics of the hyper parameter modification.

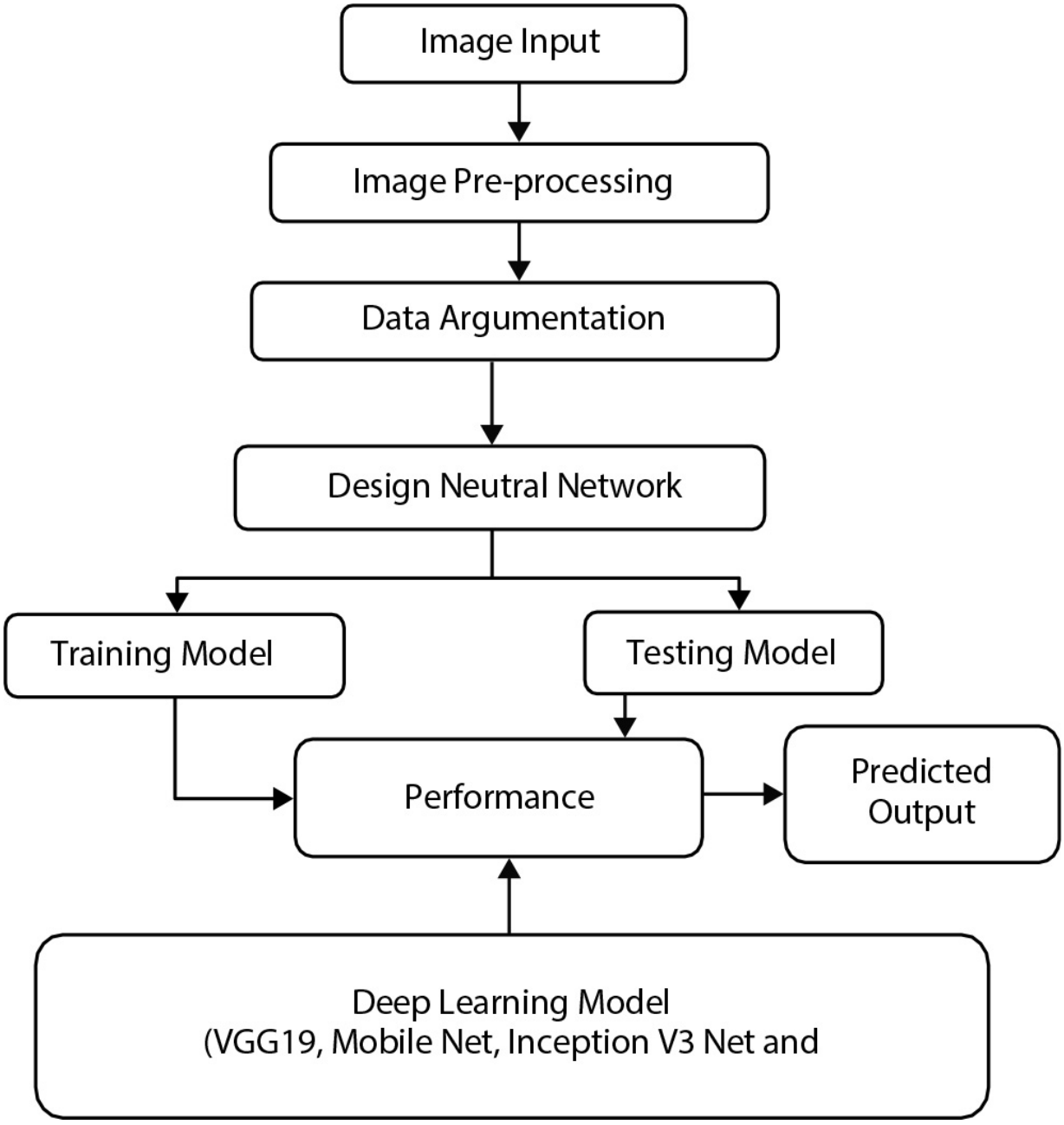

Figure 15.2 Deep-learning-based overall framework for plant disease detection.

This study problem's data source provides picture data of roughly 3012 both sick and healthy banana leaf photographs taken from various parts of South India, including southern Tamil Nadu outside Madurai, Dindigul, and other regions. Table

15.2 shows the photographs gathered in various resolutions utilizing high-resolution mobile phones, VGA cameras, and DSLR camera systems. The processed photos were enhanced with rotation, vertical and horizontal flipping, zoom intensity, and rescaling in both clockwise and anti-clockwise orientations as shown in Figure 15.3. Because the photos were not duplicated but rather improved, no actual copies of the upgraded images were preserved, although they were temporarily utilized in the training process. This augmentation technique reduces overfitting and model loss while increasing the model's robustness, which enables the model to more accurately categorize images of real-world plant diseases.

Table 15.1 Specification of hyper parameter.

S. no.	Hyper parameter	Specification
1	Optimizer	Stochastic gradient descent (SGD)
2	Activation	Softmax
3	Epochs	30
4	Learning rate	0.0001
5	Output classes	4

Table 15.2 Collection of captured banana leaf images data.

Banana leaf disease	No. of images
Banana healthy	492
Banana leaf spot	419
Banana yellow Sigatoka	588
Banana black Sigatoka	1,123
Banana Panama wilt	390
Total	3,012

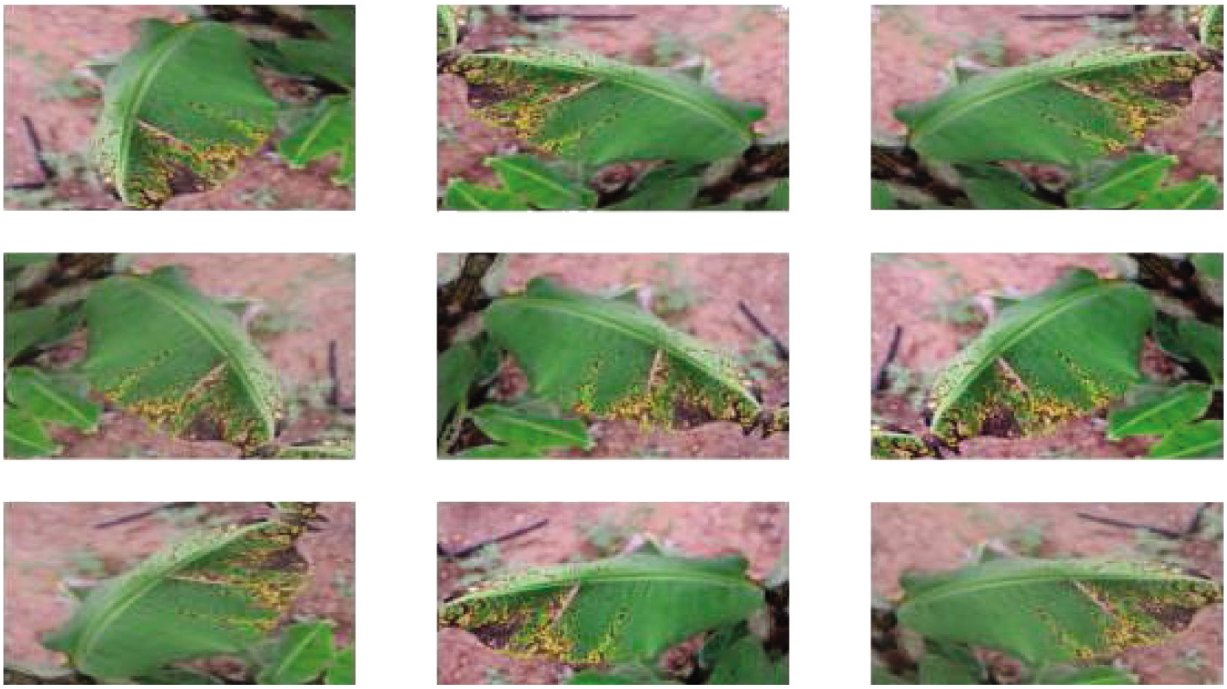

Figure 15.3 Data augmentation of banana leaf image.

15.5 Result and Discussion

The input size for VGG19, Mobile Net, Inception v3, and ResNet152 is 224* 224. Despite the dataset's vastness, which consists of over 4,000 images of banana leaf diseases, the images are accurate representations of real-world images that farmers have collected using a variety of image-collection methods, including high-definition cameras and mobile devices. Furthermore, a large dataset is prone to overfitting. Overfitting normalization strategies, such as data augmentation after preprocessing, were developed to remedy this. The augmentation method not only reduces overfitting and model loss but also increases the model's resistance, allowing it to

recognize genuine plant pathogen pictures more accurately. Most cutting-edge models require days or weeks to train and adapt, even when trained on highend GPU computers. It takes time to train and construct a model from the ground up. The batch size for training the model was set to 32, and for validation and testing the model photographs for each disease, initial and fine turning 10 epochs were chosen.

The transfer learning model has the benefit of learning quicker than models developed from scratch as well as the ability to freeze layers of the model and train the last layers for more accurate categorization. Firstly, we standardized specific hyper parameters for various pre-trained models in the transfer learning process. The optimization of these models utilized the RMSprop optimizer. Our trials effectively addressed concerns about overfitting and performance decline, as evidenced by the convergence of the output graph after a few iterations. This component of the study employed cutting-edge deep-learning algorithms and the transfer learning approach to identify plant illnesses. The real-time data was used to train the previously learnt deep CNN networks on the capture dataset. For our study, each system was normalized using a learning rate of 0.0001 and a dropout rate of 0.2. The data set was segmented into three parts: testing, validation, and training.

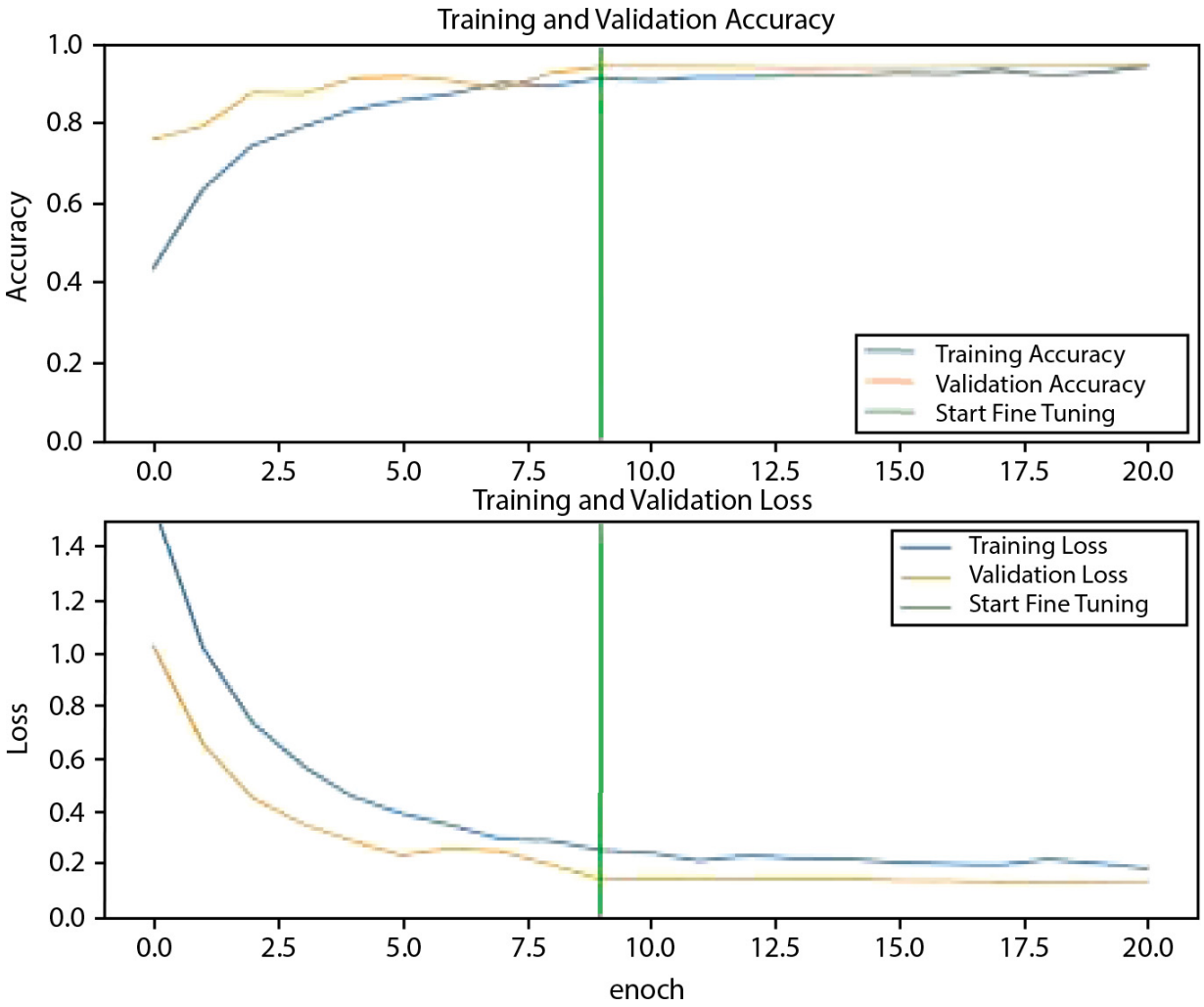

Figure 15.4 Recognition accuracy and loss for VGG19.

The VGG19, Mobile Net, Inception v3, and ResNet152 models were trained using 80% of the collected data. Each model was ran for 30 epochs, and our model began to converge with high accuracy after 10 epochs. Figure 15.4 depicts the recognition accuracy of the VGG19 model. Figure 15.5 depicts the log loss of the Inception v3 approach. The Inception v3 model's training and testing accuracy were 95.71% and 94.87%, respectively. Using the same dataset, the VGG-19 model was evaluated in a

subsequent experiment. After hyper parameter standardization, the model was trained on 80% of the same dataset, with 10% used for testing and the remaining 10% used for testing and validation. Figure 15.5 reveals that the model recognition accuracy reached around 78% in the first 10 epochs, then gradually increased to the peak training and testing recognition accuracy of 91.3% and 92.5%, which was lower than the Inception V3 model.

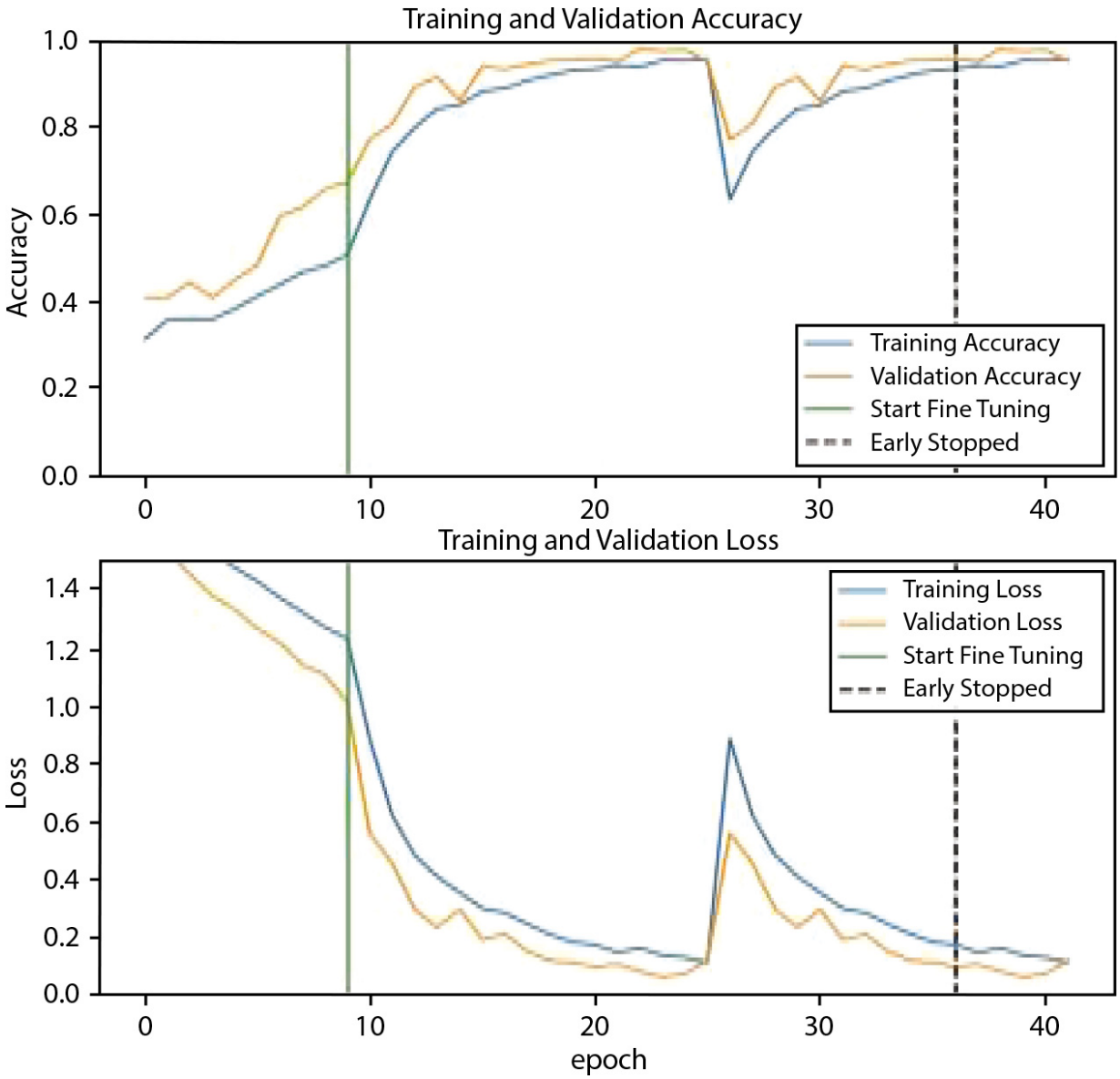

Figure 15.5 Recognition accuracy and loss for Inception Net.

The Mobile Net model was employed in the third trial. The same technique was used to analyze model loss and recognition accuracy, and the resulting graphs for recognition accuracy, validation, and training loss are displayed in Figure 15.6. The training and testing accuracy of this model were 94.83% and 93.89%, respectively. It outperformed the models Inception V3

and VGG-19 (Figure 15.7). After standardizing the hyper parameters, the final experiment was performed with ResNet152, which has 152 layers with three dense blocks and a transition layer between each dense block. Figure 15.7 depicts the training and validation accuracy/loss graphs for 30 epochs. The greatest accuracy attained in the testing phase following training and testing accuracy was 98.11% and 98.12%, while the maximum validation loss computed was 0.0154%. Table 15.3 displays a performance comparison analysis for the pre-trained network model trials. Table 15.4 shows a computational time analysis of various deep learning model. Figure 15.8(a–d) shows a confusion matrix of deep learning models ResNet152, Mobile Net VGG19, and Inception v3.

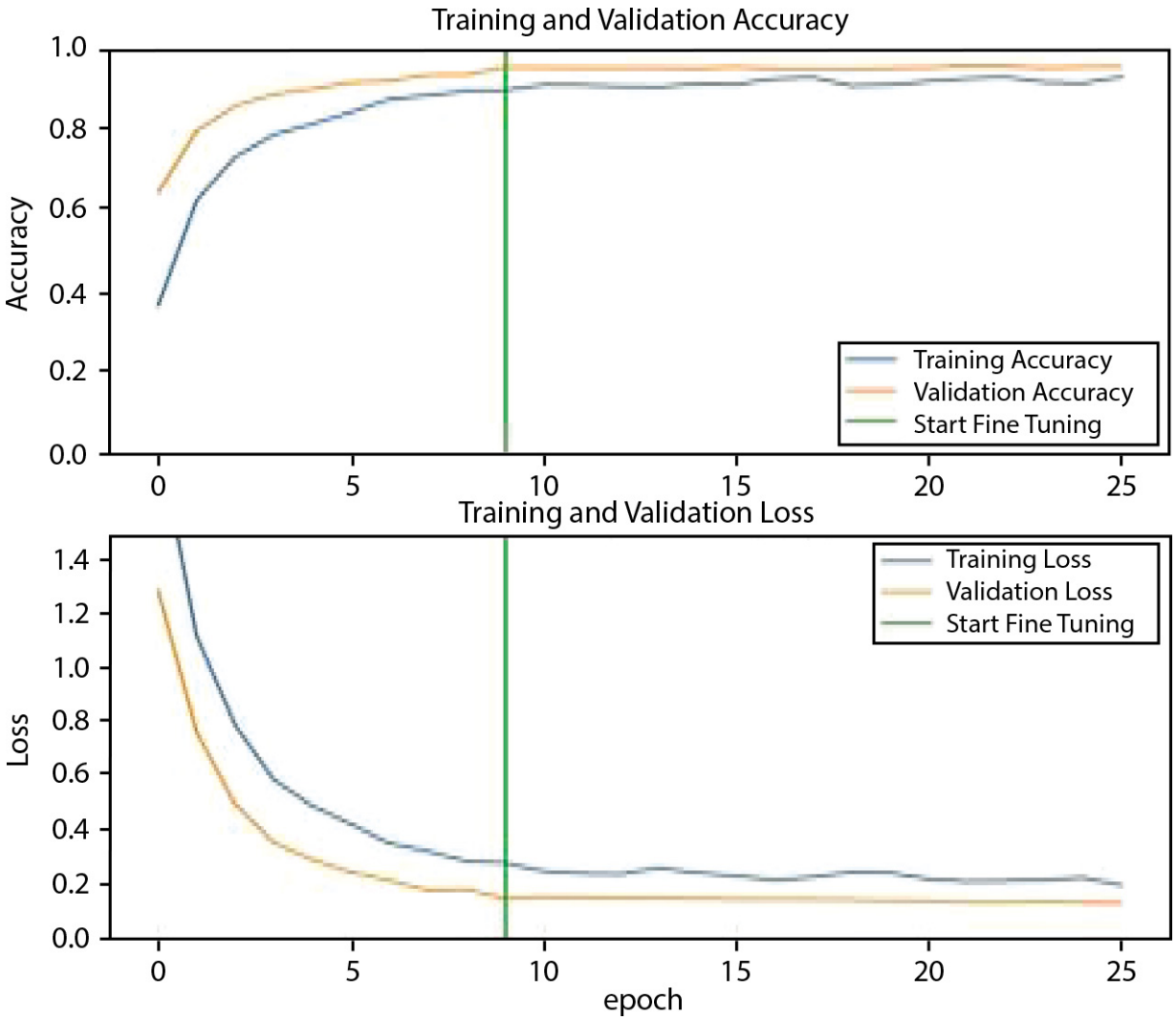

Figure 15.6 Recognition accuracy and loss for Mobile Net.

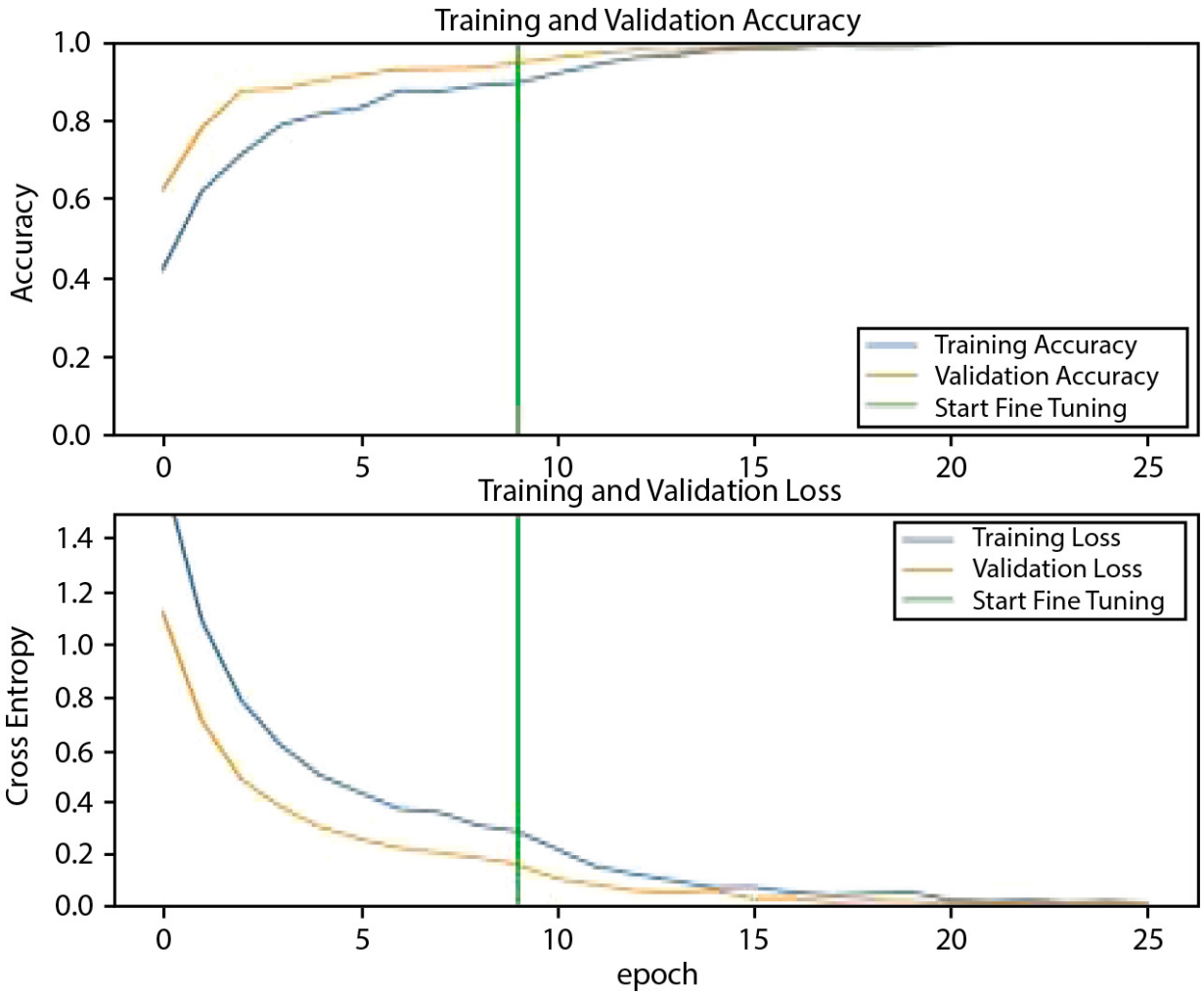

Figure 15.7 Recognition accuracy and loss for ResNet152.

Table 15.3 Performance comparison analysis for the pre-trained network models.

Model	Training accuracy (%)	Testing accuracy (%)
VGG19	91.3	92.5
InceptionV3	95.7	94.8

Model	Training accuracy (%)	Testing accuracy (%)
Mobile Net	94.8	93.8
ResNet152	98.3	98.1

Table 15.4 Computational time for various models.

Model	Computational time	
	Training time (min)	Testing time (min)
VGG19	4.851	2.482
Mobile Net	3.785	2.23
Inception v3	3.268	2.11
ResNet152	2.225	1.68

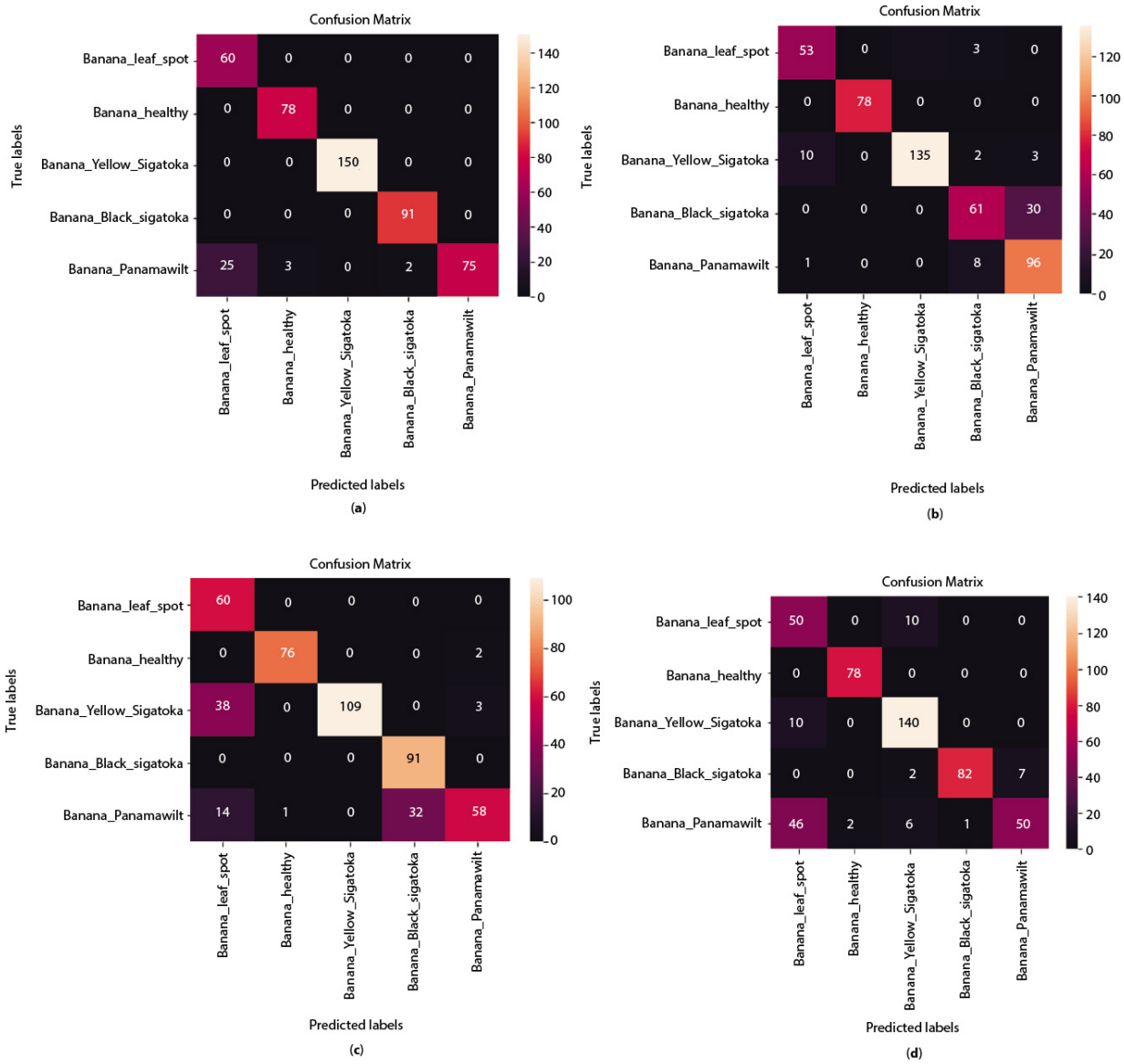

Figure 15.8 Confusion matrix of models: (a) ResNet152 model, (b) Mobile Net model, (c) VGG 19 model, and (d) Inception v3.

15.6 Conclusion

Banana leaf diseases lead to a drop in banana production quality, and diagnosing these diseases is extremely useful and provides a practical, simple, and suitable technique to increase banana output. The current research investigated multiple

transfer learning models suitable for correctly categorizing banana leaf disease. Using transfer learning approaches, the classification accuracy, sensitivity, and F1 score were utilized to standardize and assess state-of-the-art convolutional neural networks. In a performance comparison of multiple pre-trained architectures, ReseNet152 outperformed Mobile Net, VGG-19, and Inception V3. Because it featured fewer trainable parameters and reduced processing complexity, the ResNet-152 model looked to be straightforward to train. ResNet-152 is consequently more suited for identifying plant illnesses when a new plant disease must be added into the model due to its shorter training needs. The suggested model received an F1 score of 98.6% and a classification accuracy of 93.8%.

References

1. Karthik, R.; Hariharan, M.; Anand, S.; Mathikshara, P.; Johnson, A.; Menaka, R. Attention embedded residual CNN for disease detection in tomato leaves. *Appl. Soft Comput.*, 2019, 86, 105933.
2. M. Lv, G. Zhou, M. He, A. Chen, W. Zhang and Y. Hu, “Maize Leaf Disease Identification Based on Feature Enhancement and DMS-Robust Alexnet,” in *IEEE Access*, vol. 8, pp. 57952-57966, 2020, doi: 10.1109/ ACCESS.2020.2982443.

3. G. Zhou, W. Zhang, A. Chen, M. He and X. Ma, “Rapid Detection of Rice Disease Based on FCM-KM and Faster R-CNN Fusion,” in *IEEE Access*, vol. 7, pp. 143190-143206, 2019, doi: 10.1109/ACCESS.2019.2943454.
4. Aasha Nandhini S; Hemalatha R; Radha S; Indumathi K; “Web Enabled Plant Disease Detection System for Agricultural Applications using WMSN”, Springer’s *Wirel. Pers. Commun.*, Published online, pp. 1-16, December 2017.
5. Singh, Vijai and A. K. Misra. “Detection of plant leaf diseases using image segmentation and soft computing techniques.” *Inf. Process. Agric.*, (2016).
6. Bhange, Manisha and H. A. Hingoliwala. “Smart Farming: Pomegranate Disease Detection Using Image Processing.” *Procedia Comput. Sci.*, 58 (2015): (280-288).
7. Karthickmanoj, R., Sasilatha, T. and Padmapriya, J., 2021. Automated machine learning based plant stress detection system. *Mater. Today: Proc.*, 47, pp.1887-1891.
8. Varshney, D., Babukhanwala, B., Khan, J., Saxena, D. and Singh, A.K., 2022, May. Plant Disease Detection Using Machine Learning Techniques. In *2022 3rd International Conference for Emerging Technology (INCET)* (pp. 1-5). IEEE.
9. Karthickmanoj, R., Padmapriya, J. and Sasilatha, T., 2021. A novel pixel replacement- based segmentation and double

feature extraction techniques for efficient classification of plant leaf diseases. *Mater. Today: Proc.*, 47, pp.2048-2052.

10. Nandhini, S.A. and Radha, S., 2017. Efficient compressed sensing-based security approach for video surveillance application in wireless multimedia sensor networks. *Comput. Electr. Eng.*, 60, pp.175-192.
11. Harakannanavar, S.S., Rudagi, J.M., Puranikmath, V.I., Siddiqua, A. and Pramodhini, R., 2022. Plant Leaf Disease Detection using Computer Vision and Machine Learning Algorithms. *Global Transitions Proc.*
12. Pantazi, X.E., Moshou, D. and Tamouridou, A.A., 2019. Automated leaf disease detection in different crop species through image features analysis and One Class Classifiers. *Comput. Electron. Agric.*, 156, pp.96-104.
13. Kumar, T.A., Rajmohan, R., Pavithra, M., Ajagbe, S.A., Hodhod, R. and Gaber, T., 2022. Automatic face mask detection system in public transportation in smart cities using IoT and deep learning. *Electronics*, 11(6), p.904.
14. Jackulin, C. and Murugavalli, S., 2022. A comprehensive review on detection of plant disease using machine learning and deep learning approaches. *Meas.: Sens.*, p.100441.
15. Domingues, T., Brandão, T. and Ferreira, J.C., 2022. Machine Learning for Detection and Prediction of Crop Diseases and Pests: A Comprehensive Survey. *Agriculture*, 12(9), p.1350.

16. Zamani, A.S., Anand, L., Rane, K.P., Prabhu, P., Buttar, A.M., Pallathadka, H., Raghuvanshi, A. and Dugbakie, B.N., 2022. Performance of Machine Learning and Image Processing in Plant Leaf Disease Detection. *J. Food Qual.*, 2022.
17. Reddy, B.A., Krishna, G.S., Saraswathi, K.P., Sadhvika, I., Udutalapally, V. and Das, D., 2022, April. dScout: Unmanned Ground Vehicle for Automatic Disease Detection and Pesticide Atomizer. In *2022 IEEE 7th International conference for Convergence in Technology (I2CT)* (pp. 1-6). IEEE.
18. Themozhi, G., Prabha, A., Radhakrishnan, P. and Manigandan, K., 2021. Battery Monitoring And Smart Charging Using Iot For Electrical Vehicle Applications. *Int. J. Aquat. Sci.*, 12(3), pp.593-601.
19. Joshi, K., Awale, R., Ahmad, S., Patil, S. and Pisal, V., 2022. Plant Leaf Disease Detection Using Computer Vision Techniques and Machine Learning. In *ITM Web of Conferences* (Vol. 44, p. 03002). EDP Sciences.
20. Liu, Z., Bashir, R.N., Iqbal, S., Shahid, M.M.A., Tausif, M. and Umer, Q., 2022. Internet of Things (IoT) and Machine Learning Model of Plant Disease Prediction–Blister Blight for Tea Plant. *IEEE Access*, 10, pp.44934-44944.
21. Waldamichael, F.G., Debelee, T.G., Schwenker, F., Ayano, Y.M. and Kebede, S.R., 2022. Machine Learning in Cereal Crops Disease Detection: A Review. *Algorithms*, 15(3), p.75.

22. Ale, L., Sheta, A., Li, L., Wang, Y. and Zhang, N., 2019, December. Deep learning based plant disease detection for smart agriculture. In *2019 IEEE Globecom Workshops (GC Wkshps)* (pp. 1-6). IEEE.
23. Suhag, S., Singh, N., Jadaun, S., Johri, P., Shukla, A. and Parashar, N., 2021, June. IoT based soil nutrition and plant disease detection system for smart agriculture. In *2021 10th IEEE International Conference on Communication Systems and Network Technologies (CSNT)* (pp. 478-483). IEEE.
24. Maray, M., Albraikan, A.A., Alotaibi, S.S., Alabdan, R., Al Duhayyim, M. and Al-Azzawi, W.K., 2022. Artificial intelligence-enabled coconut tree disease detection and classification model for smart agriculture. *Comput. Electr. Eng.*, 104, p.108399.
25. Somasundareswari, D., KH, M.S. and Santhiya, A., 2022. Design and Implementation of Smart Farming System. *Int. J. Progressive Res. Sci. Eng.*, 3(05), pp.184-187.
26. Storey, G., Meng, Q. and Li, B., 2022. Leaf disease segmentation and detection in apple orchards for precise smart spraying in sustainable agriculture. *Sustainability*, 14(3), p.1458.
27. Eunice, J., Popescu, D.E., Chowdary, M.K. and Hemanth, J., 2022. Deep Learning-Based Leaf Disease Detection in Crops

Using Images for Agricultural Applications. *Agronomy*, 12(10), p.2395.

28. M. S. Arshad, U. A. Rehman and M. M. Fraz, “Plant Disease Identification Using Transfer Learning,” *2021 International Conference on Digital Futures and Transformative Technologies (ICoDT2)*, Islamabad, Pakistan, 2021, pp. 1-5, doi: 10.1109/ICoDT252288.2021.9441512.
29. Kathiresan, G., Anirudh, M., Nagharjun, M. and Karthik, R., 2021, May. Disease detection in rice leaves using transfer learning techniques. In *J. Phys.: Conf. Ser.*, (Vol. 1911, No.. IOP Publishing.
30. Mishra, S., Singh, A. and Singh, V., 2021, March. Application of MobileNet-v1 for Potato Plant Disease Detection Using Transfer Learning. In *2021 Workshop on Algorithm and Big Data* (pp. 14-19).
31. Agrawal, M. and Agrawal, S., 2023. Rice plant diseases detection using convolutional neural networks. *Int. J. Eng. Syst. Modell. Simul.*, 14(1), pp.30-42.
32. Mimi, A., Zohura, S.F.T., Ibrahim, M., Haque, R.R., Farrok, O., Jabid, T. and Ali, M.S., 2023. Identifying selected diseases of leaves using deep learning and transfer learning models. *Mach. Graph. Vis.*, 32(1), pp.55-71.
33. Shantkumari, M. and Uma, S.V., 2023. Grape leaf image classification based on machine learning technique for

accurate leaf disease detection. *Multimedia Tools Appl.*, 82(1), pp.1477-1487.

34. Binnar, V. and Sharma, S., 2023, January. Plant Leaf Diseases Detection Using Deep Learning Algorithms. In *Machine Learning, Image Processing, Network Security and Data Sciences: Select Proceedings of 3rd International Conference on MIND 2021* (pp. 217-228). Singapore: Springer Nature Singapore.
35. Janani, M. and Jebakumar, R., 2023. Detection and classification of groundnut leaf nutrient level extraction in RGB images. *Adv. Eng. Software*, 175, p.103320.
36. Bandyopadhyay, S., Sarkar, A., paria, A.C.O.M.A., and Jana, B., 2023, January. Deep Learning Approaches for Analysis and Detection of Tomato Leaf Diseases. In *Artificial Intelligence: First International Symposium, ISAI 2022*, Haldia, India, February 17-22, 2022, Revised Selected Papers (pp. 265-273). Cham: Springer Nature Switzerland.
37. Narla, V.L. and Suresh, G., 2023, January. Multiple Feature-Based Tomato Plant Leaf Disease Classification Using SVM Classifier. In *Machine Learning, Image Processing, Network Security and Data Sciences: Select Proceedings of 3rd International Conference on MIND 2021* (pp. 443-455). Singapore: Springer Nature Singapore.

38. Padmapriya, D. and Prema, A., 2023, January. Detecting Diseases in Jasmine Plants Using Proposed Image Pre-processing Algorithm. In *Advancements in Interdisciplinary Research: First International Conference, AIR 2022*, Prayagraj, India, May 6–7, 2022, Revised Selected Papers (pp. 389-399). Cham: Springer Nature Switzerland.
39. Thai, H.T., Le, K.H. and Nguyen, N.L.T., 2023. FormerLeaf: An efficient vision transformer for Cassava Leaf Disease detection. *Comput. Electron. Agric.*, 204, p.107518.
40. Azgomi, H., Haredasht, F.R. and Motlagh, M.R.S., 2023. Diagnosis of some apple fruit diseases by using image processing and artificial neural network. *Food Control*, 145, p.109484.
41. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. “Plant Disease Identification Using Convolution Neural Networks.” In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
42. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. “Solutions Using Machine Learning for Diabetes.” *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
43. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, “Image Processing Based Intelligent Mini

Robotic Face Recognition System," *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.

44. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. "Mining of data through various soft computing techniques." In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
45. Goyal, Mohit, and Durgesh Srivastava. "A behaviour-based authentication to internet of things using machine learning." *Design and Development of Efficient Energy Systems* (2022): 245-263.
46. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
47. Sahin, H.M., Miftahushudur, T., Grieve, B. and Yin, H., Segmentation of Weeds and Crops Using Multispectral Imaging and Crf-Enhanced U-Net. Available at SSRN 4329500.
48. Khan, U., Maqsood, M., Gillani, S., Durrani, M.Y., Mehmood, I. and Seo, S., 2023. A deep learning-based framework for accurate identification and crop estimation of olive trees. *J. Supercomput.*, 79(2), pp.1834-1855.

Note

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16

Exploring Dataset for Apple Leaf Disease Detection: A Focus on Horticulture

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Abstract

This study focuses on examining a dataset intended for the identification of apple leaf diseases with a particular emphasis on its possible horticultural utility. Images of apple leaves infected with a variety of maladies, such as apple scab, cedar apple rust, and powdery mildew, are being analyzed and processed as part of this study. To extract significant characteristics and patterns that might help in the detection of damaged leaves, the dataset is evaluated using a variety of computer vision techniques and algorithms. The results of this

study are anticipated to aid in the creation of an automated system that is accurate and dependable for spotting apple leaf diseases. Farmers and agronomists may use this system in the field to monitor and control disease outbreaks. In the end, this research may help apple producers and customers by increasing the productivity and sustainability of the industry.

***Keywords*:** Dataset, apple leaf disease, machine learning, deep learning, segmentation, accuracy, identification

16.1 Introduction

The productivity and quality of the crop can be greatly impacted by a number of illnesses that might affect apple leaves [1]. These diseases can result in plant death, defoliation, diminished fruit size and quality, and discolored leaves. The most prevalent ailments that affect apple leaves are apple scab [2], cedar apple rot [3], powdery mildew [4], and fire blight [5]. One of the most harmful diseases of apple trees is known as apple scab, which is brought on by a fungus called Venturia inaequalis. The disease mostly impacts the leaves, turning them yellow or brown and causing olive-green dots to appear. Fruit size and quality may be impacted by the afflicted leaves falling off too soon. Apple trees can also get cedar apple rust, another fungus-borne disease. Later, orange-brown markings appear on

the undersides of the leaves from the yellow spots that were initially present on the leaves [6].

The fungus Podosphaera leucotricha, which causes powdery mildew, leaves with a white powdery coating. Apple trees can succumb to a severe bacterial disease called fire blight, which is brought on by the bacterium Erwinia amylovora [7]. For apple production to be sustainable, prompt detection and management of apple leaf diseases are crucial. The conformist method of disease identification involves an expert looking at the leaves. However, this procedure is time-consuming and prone to mistake. As a result, creating automated systems for disease control and detection in agriculture is becoming increasingly important [8]. The detection of plant diseases, such as those that affect apple leaves, has shown tremendous promise for computer vision approaches and algorithms. These techniques utilize statistical or deep learning algorithms to analyze digital photographs of leaves, extract pertinent information, and locate damaged leaves. Numerous studies have demonstrated how well computer vision-based systems work for spotting diseases in apple leaves early [9]. Convolutional neural networks (CNNs), for instance, have been employed by researchers to accurately detect the presence of apple disease and powdery mildew in digital photographs of leaves [10].

The crop's quality and output can be impacted by a variety of apple leaf diseases. The following are a few of the most typical diseases of apple leaves:

1. A fungus called apple scab damages apple trees' leaves, fruit, and twigs. The upper surfaces of the leaves develop darkbrown or olive-green patches, and the fruit develops tiny brown lesions [11].
2. It is a fungus that causes cedar apple rust, which also affects apple trees. Both small, cup-shaped structures and yellow-orange dots can be seen on the leaves' undersides [11].
3. The leaves, fruit, and shoots of apple trees are all impacted by the fungal disease known as powdery mildew. The leaves start to get a white, powdered coating on their surface.
4. The fruit, flowers, and foliage of apple trees are all impacted by the bacterial disease known as "fire blight" [11]. The sections that are damaged have a wilted, darkened, or burned appearance.
5. Apple trees' fruits are susceptible to the fungal disease known as "black rot" [11]. The fruit may develop black, sunken lesions as a result, and it may even turn into mummified fruit.
6. Apple trees are susceptible to the fungal disease known as "alternaria leaf spot," which damages the leaves. On the

leaves, it manifests as tiny, elliptical black patches. Figure 16.1 shows the different categories of apple leaf disease.

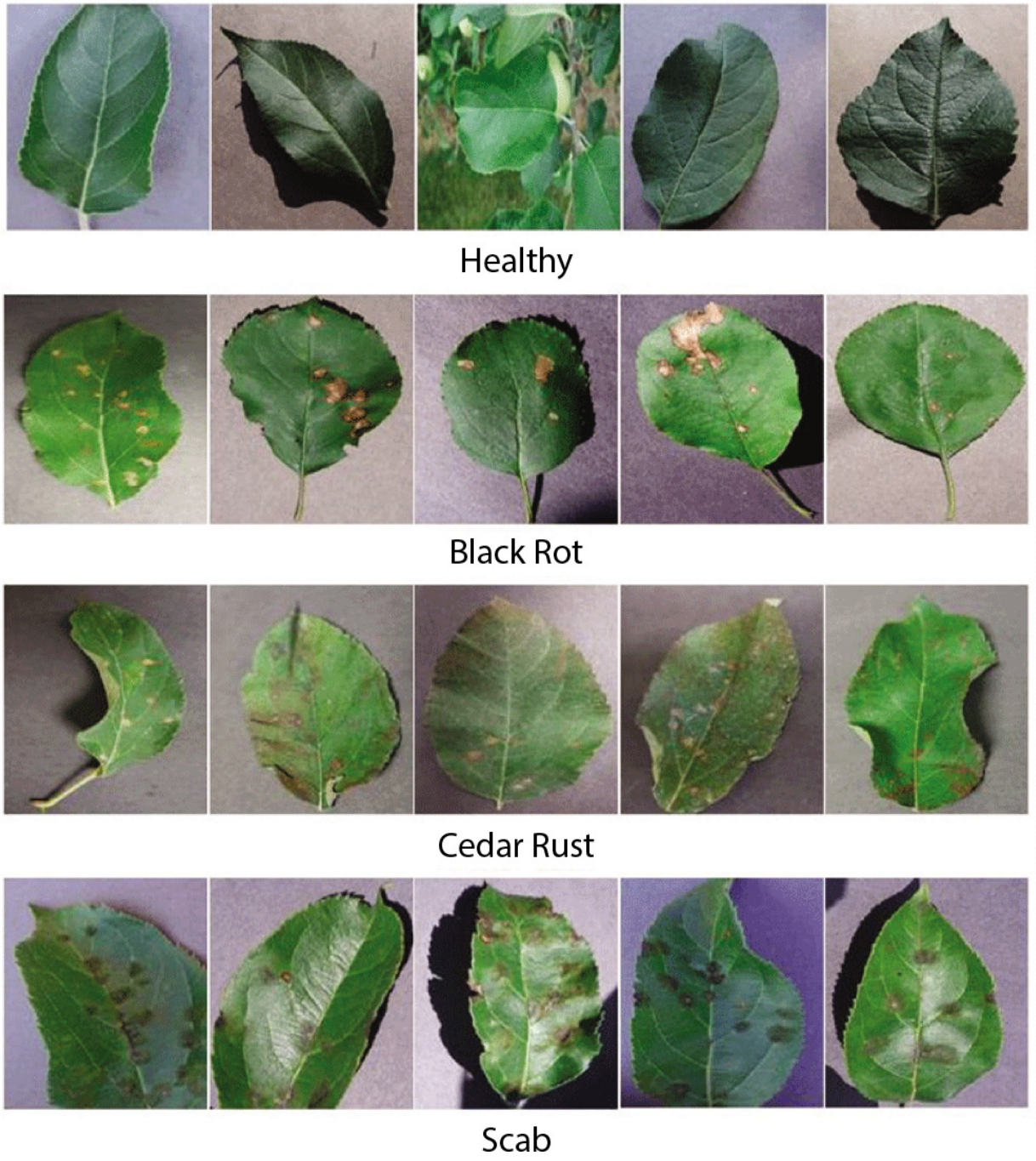

Figure 16.1 Various types of apple leaf disease.

In addition to lowering the fruit quality, these diseases can cost apple growers considerable sums of money. The prevention and management of these diseases depend heavily on effective

disease management techniques, such as early detection, good cleanliness, and the use of fungicides. The figure above shows different categories of apple leaf disease.

16.1.1 Segmentation and Machine Learning Techniques for Detection of Apple Foliage Disease

Segmentation is the process of locating and removing important items from a picture [12]. For the purpose of identifying apple leaf disease, segmentation techniques may be employed to distinguish the diseased from the healthy leaf portions. This can help with the proper categorization of the illness. Following are a few segmentation methods that have been employed to find apple leaf disease:

- Thresholding: Thresholding, a fundamental segmentation technique, involves converting an image into a binary image by changing all pixels above and below a predefined threshold value into white and black, respectively [13]. This method has been applied to find apple scab in leaf photos. Figure 16.2 illustrates the thresholding segmentation.
- Edge-based segmentation: This method segments an image by identifying its edges and utilizes those edges to segment the picture [14]. This method has been used to find apple scab and cedar apple rust in pictures of leaves as shown in Figure 16.3.

- Machine learning-based segmentation: This method automatically segments the picture based on the characteristics that have been taken out of the image. Using this technique, apple scab, cedar apple rust, and powdery mildew have all been located in leaf images.

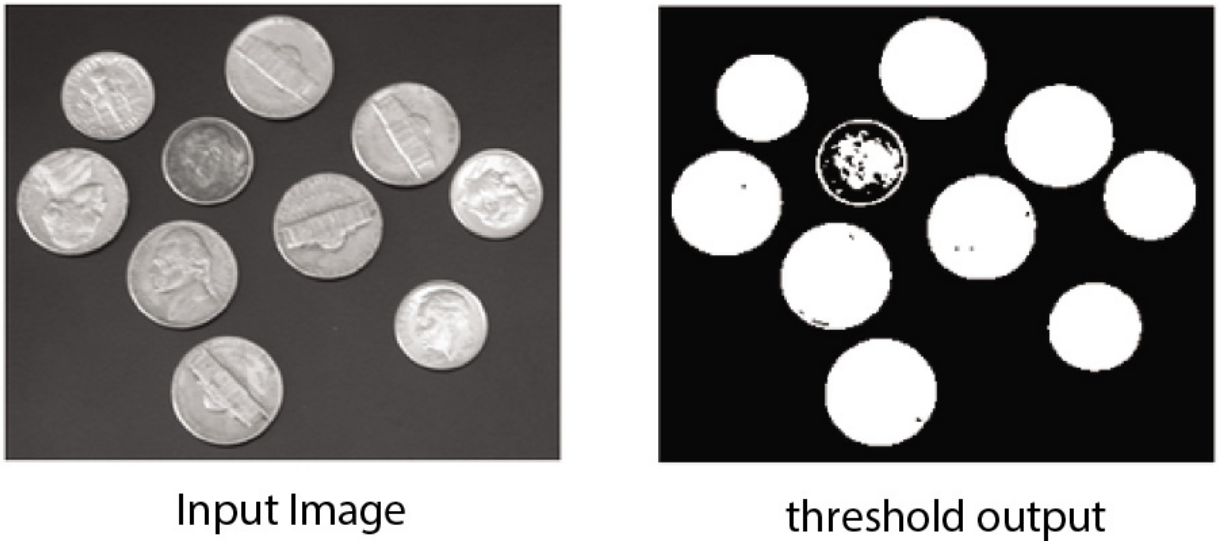

Figure 16.2 Thresholding segmentation.

Figure 16.3 Edge-based segmentation.

Machine learning methods may be used to identify apple leaf disease in addition to segmentation. The following are some frequently used machine learning techniques:

SVMs (support vector machines): A class of supervised learning methods called SVMs can be applied to classification issues [15]. Apple leaf disease has been detected using SVMs by first training the system on a collection of labeled photos and then applying it to categorize fresh, unlabeled images as elaborate in Figure 16.4.

Random forests: They are an ensemble learning method that may be utilized for categorization problems. In order to identify apple leaf disease, random forests have been utilized. Before being used to classify the recent, unlabeled photographs in Figure 16.5, the algorithm is trained on a library of labeled pictures.

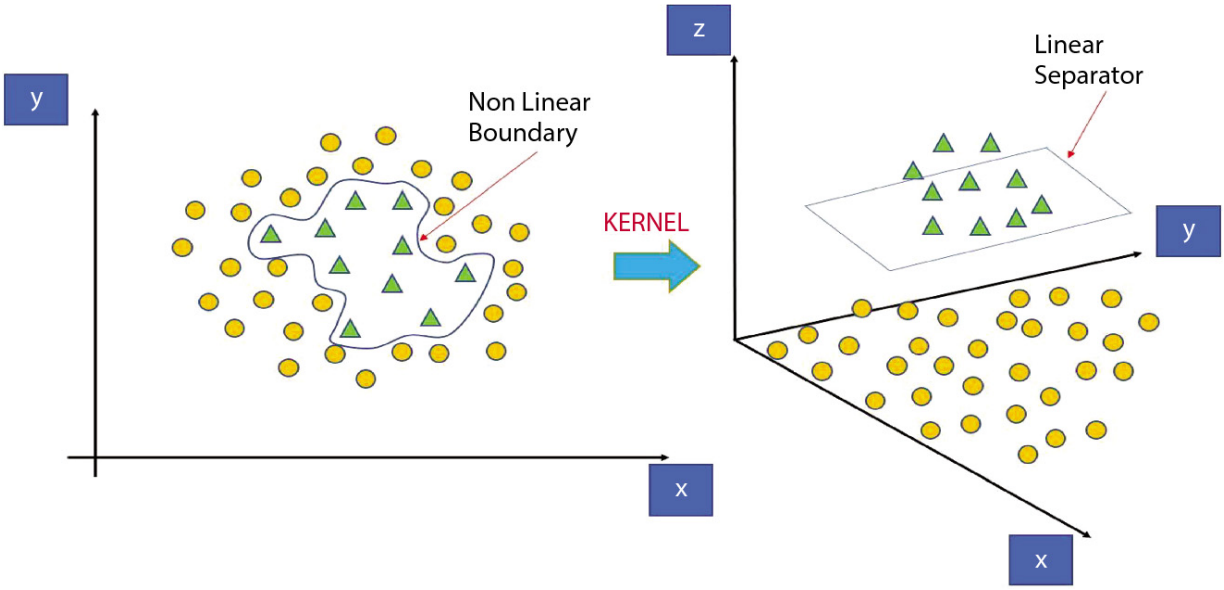

Figure 16.4 Process of support vector machine.

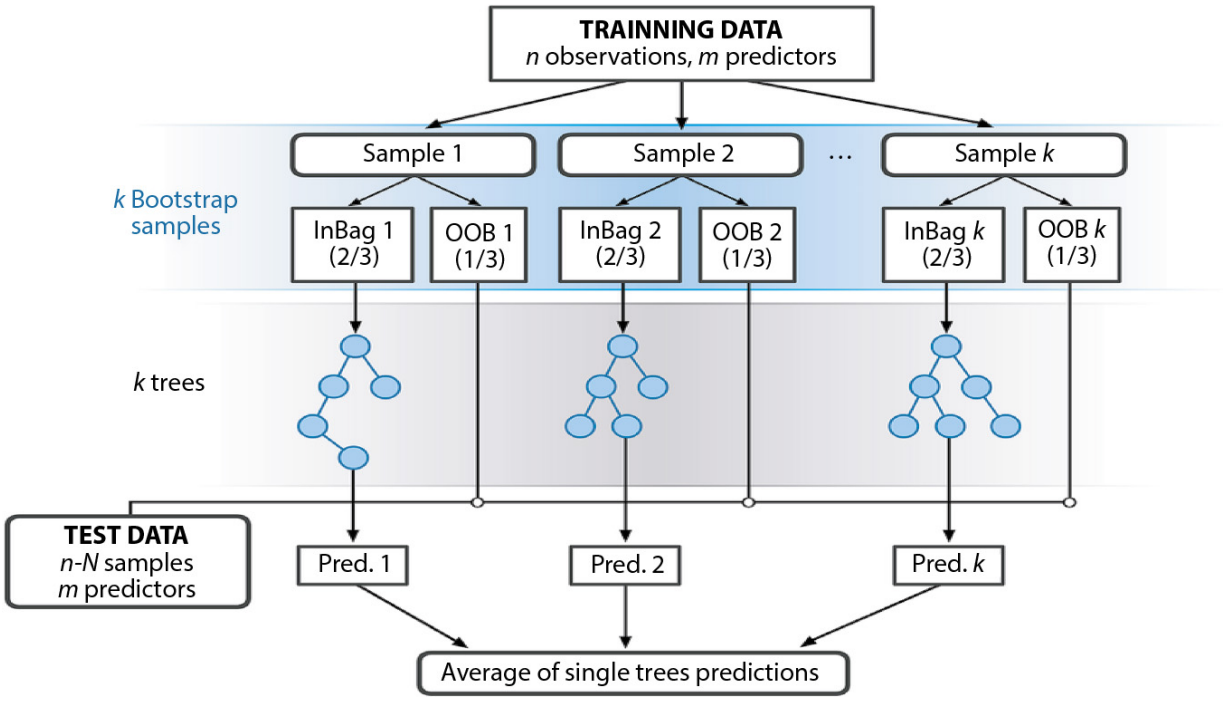

Figure 16.5 Process of random forest machine learning.

KNN (K-nearest neighbors): KNN is an approach to artificial intelligence that is simple to use but effective for classification

tasks [16]. By first training KNN on a batch of labeled photos and then applying it to the classification of fresh, unlabeled images, apple leaf disease has been detected using this method.

16.1.2 Deep Learning Method for Apple Leaf Disease Detection

The science of computer vision has seen a major transformation because of deep learning techniques [17], which have also demonstrated significant promise in the detection of numerous plant diseases [18], including those that affect apple leaves. Convolutional neural networks (CNNs), for example, can properly identify images by learning complicated properties from them using deep learning models. We evaluate several methods put forth in this paper to achieve this goal and discuss the solicitation of deep learning algorithms for detecting apple leaf disease.

The following stages are typically included in deep learning algorithms for identifying apple leaf disease:

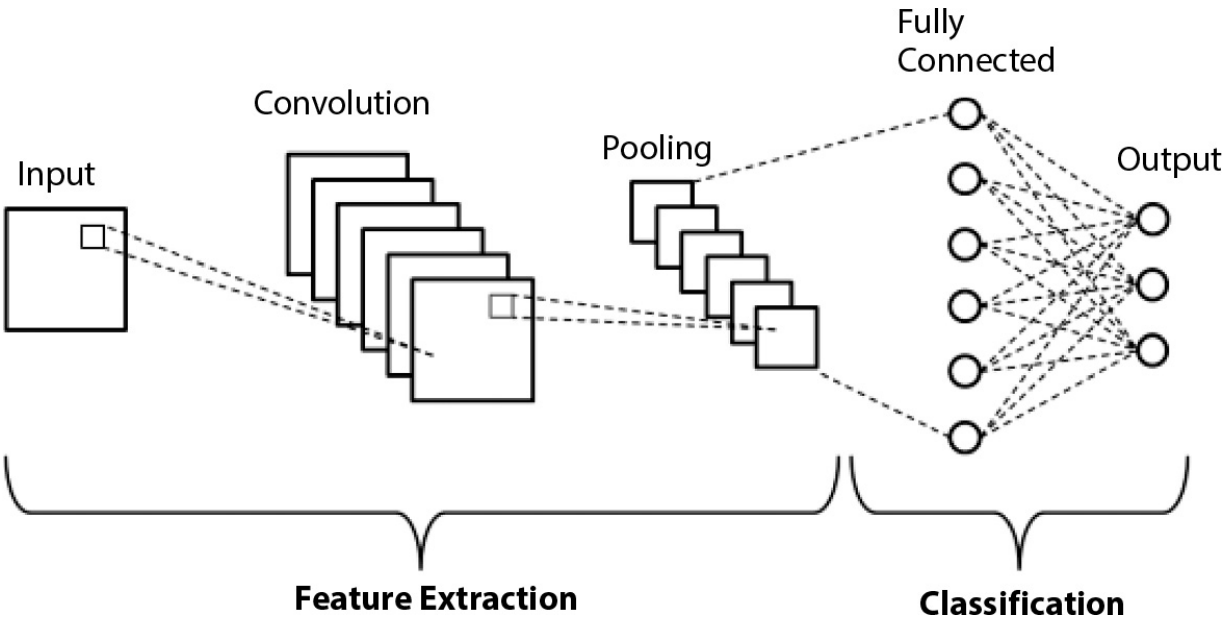

Figure 16.6 Working of a convolutional neural network.

Image acquisition: Accurate disease detection of apple leaves requires high-quality photographs. Digital cameras and drones are only two of the methods that can be used to capture the photographs [19].

Data preprocessing: The obtained photos could have a lot of noise, distracting background objects, or other artefacts that make it difficult to accurately detect diseases. Images can be preprocessed using methods like noise reduction, picture segmentation, and morphological treatments to enhance their quality.

CNN architecture: A subtype of deep learning algorithms called CNNs allows for autonomous learning and feature extraction

from large datasets of tagged images. Apple leaf disease detection has been successfully carried out using CNNs [20]. Several CNN architectures, including VGGNet, ResNet, and Inception, can be employed for the detection of apple leaf disease as shown in Figure 16.6. Multiple layers make up the CNN architecture, which uses the preprocessed pictures to learn features. The performance of the model can be impacted by the selection of hyper parameters and the number of layers.

Training and evaluation: A dataset of preprocessed photos that have been classified as healthy or diseased is used to train the CNN model. On a different test set, the model's performance is assessed using measures like accuracy, precision, and recall.

16.1.3 Contribution

The main contribution of this research is the evaluation of datasets that are currently available for the detection of apple leaf disease with a focus on horticulture application. This article probably gives a summary of the numerous datasets that may be used to identify apple leaf disease, together with information on their size, accuracy, and composition. It can also go through the particular difficulties that the horticulture sector has in identifying and controlling plant diseases.

Research questions

1. What are the main features of the dataset that is used to identify apple leaf diseases in horticultural settings, and how do these features affect how well disease detection algorithms perform?
2. What are the best machine learning algorithms for identifying certain apple leaf diseases in horticulture contexts, and how accurate are they in comparison to established disease detection techniques?
3. What are the results of this dataset review's implications for future research on the recognition of apple leaf disease and what are the most promising lines of inquiry in this field?

Study structure: Following the introduction provided in Section 16.1, Section 16.2 outlines past investigations carried out by various scholars. Section 16.3 discusses the techniques for identifying apple leaf disease, and Section 16.4 investigates the significance of datasets and numerous datasets that are employed in this process. The conclusions stated in the preceding section are discussed in section 16.5. The research comes to a close in Section 16.6.

16.2 Literature Review

Machine learning models built utilizing photo datasets have been the main emphasis in the field of apple leaf disease identification. As an illustration, in a study by [21], a deep learning model was taught to identify the illness apple scab using a library of pictures of apple leaves. The model has a high accuracy rate, and the study suggests that it may be used to help treat ailments in the real world. A machine learning model was created by [22] for the identification of illnesses in apple leaves using a collection of hyperspectral photos of apple leaves. The study indicated that integrating hyperspectral imaging with machine learning methodology improved the accuracy of illness identification when compared to employing only conventional visual imaging techniques. In a related work, [23] utilized a dataset of multispectral photographs of apple leaves to create a machine learning model for the identification of powdery mildew illness. The study discovered that the model had good accuracy rates and might be utilized as a low-cost, non-destructive way for locating illnesses in the field. In a study that was published in 2019, Liu *et al.* investigated the application of transfer learning to raise the detection precisions of machine learning models for illnesses that affect apple leaves. When compared to building models from scratch, the

study indicated that transfer learning from previously trained models increased the accuracy of illness identification. To identify the fire blight illness, a machine learning model was created [24] using a dataset of hyperspectral pictures of apple leaves. The study discovered that when compared to conventional visual imaging techniques, hyperspectral imaging enables more precise illness identification. An artificial intelligence (AI) model was created in a study by [25] to recognize a number of ailments that harm apple leaves, including as powdery mildew, cedar apple rot, and apple scab. Using a library of pictures of apple leaves, this model was trained. The study discovered that deep learning and data augmentation techniques enhanced the data's quality and increased the accuracy of illness detection. A machine learning model was created by [26] for the diagnosis of illnesses in apple leaves using a dataset of multispectral photographs of apple leaves. According to the research, combining spectral and textural signals improved the accuracy of illness identification compared to utilizing simply spectral data.

16.3 Apple Leaf Disease Detection Mechanism

Inspecting apple leaves visually, analyzing them in a lab, and using technology are just a few of the approaches available to identify illnesses in apple leaves. Here are some illustrations:

Visual examination: In this technique, the leaves are visually inspected for signs of disease, such as spots, discolorations, malformations, or lesions. On the basis of their visual traits, seasoned growers and researchers can recognize frequent illnesses. Visual inspection though could miss minute or early indications and takes time and experience.

Laboratory analysis: This procedure entails gathering leaf samples and sending them to a lab for evaluation. To determine the presence and kind of disease-causing agents, such as bacteria, fungus, viruses, or nematodes, the laboratory can carry out a variety of tests, such as molecular, biochemical, or histological assays. Although expensive and time-consuming, laboratory analysis can produce precise and trustworthy results.

Technology-based approaches: This technique uses sensors, cameras, or other tools to automatically identify and keep track of diseases in real time. In order to analyze photos of apple leaves and identify diseases based on their patterns and colors, researchers have created computer vision algorithms. Some businesses provide mobile, drone-based, or handheld systems that can spot diseases using fluorescence or spectral imaging. Although they can need initial investment and upkeep,

technological solutions can deliver quick and objective outcomes.

Symptomatology charts: These are visual aids that list and illustrate the typical signs of apple leaf illnesses, along with potential causes and remedies. Growers and researchers can identify and diagnose diseases based on the symptoms noticed by using symptomatology guides.

Disease forecasting models: These are computer programs that use environmental and biological data to forecast the occurrence and severity of apple leaf diseases in a certain area or season. By taking preventative actions or choosing resistant cultivars, disease forecasting models can assist growers and researchers in identifying and preventing diseases before they spread widely.

Molecular diagnostics: The usage of specific pathogens or genes linked to apple leaf diseases can be detected and identified using cutting-edge techniques called molecular diagnostics, which rely on DNA- or RNA-based assays. When utilized for illness early detection, monitoring, and tracking, molecular diagnostics can produce results that are incredibly sensitive and specific. However, molecular diagnostics need specialized

tools and knowledge, and they might not be economical for routine screening.

16.4 Significance of Dataset

Datasets are essential for the diagnosis of apple leaf disease because they are used to build and test machine learning models that automatically identify and categorize damaged leaves based on their characteristics and patterns. The models' functionality and generalizability, as well as their capacity to identify novel or uncommon diseases, can be significantly influenced by the grade and size of the dataset.

Thousands of photos of healthy and diseased leaves taken from various perspectives and backgrounds, with various lighting conditions, make up a typical dataset for apple leaf disease identification. According to professional diagnosis or laboratory analysis, the photos are labeled with the relevant disease kind or severity level. The dataset might include meta-information about the photographs' capture, like the time, place, and weather. Table 16.1 shows exploration of existing apple leaf dataset.

The dataset is split conventionally into training, validation, and testing sets using a split of 70%, 15%, and 15%. To fine-tune the

model’s settings and identify the patterns and characteristics that distinguish healthy from sick leaves, a training set is employed. By using unseen images from the same distribution in the validation set, the model is adjusted to prevent over-fitting. With photos from the same or a different distribution, the testing set is used to evaluate how well the model ultimately performs.

Table 16.1 Exploration of existing apple leaf dataset.

Author/s and year	Dataset	Size	Classes
Peng Jiang *et al.* (2019) [27]	ALDD	2,029 images	Alternaria spot, brov spot, mosa gray spot, rust are th five classificat
Sara Alqethami *et al.* (2022) [28]	New plant diseases (from Kaggle)	54,306 pictures of leaves from 14 different plants	240 pictur apple leav were used this investigat of which (were deer to be in go condition 180 to be infected w

Author/s and year	Dataset	Size	Classes
			scab (illnesses) maladies Black_ Ro (illnesses) cedar rust healthy
Tahir *et al.* (2021) [29]	Plant Village dataset	3,171 images of apple disease	Brown sp cedar, and apple scal
Chakraborty *et al.* (2021) [30]	ALD	500 images	Normal a abnormal leaves

Author/s and year	Dataset	Size	Classes
Bi *et al.* (2020) [31]	Apple disease databases		Rust leave and alter leaves blo
Chao *et al.* (2020) [32]	ALTD data set	Samples of 2,970 images	Alternaria spots are depicted i the follow images: healthy (4 brown sp (540), mos

Author/s and year	Dataset	Size	Classes
			(879), gray spot (430) rust (430)

Author/s and year	Dataset	Size	Classes

Author/s and year	Dataset	Size	Classes
Kahkashan *et al.* (2023) [33]	Self-collected	31,377 images	Rust (6,69 gray spot (5,810), spotted le (6,343), mosaic (5,875), ar brown sp (6,655)
Lili Li *et al.* (2023) [34]	Self-gathered from farmers' orchards and Shanxi Agricultural University's Pomology Institute's	636 images (after augmentation)	Rust, 276; rot, 363 (a augmenta

Author/s and year	Dataset	Size	Classes
	planting base, 267 photographs made up the initial dataset, including 152 pictures of ring rot and 115 pictures of rust		
Bansal *et al.* (2021) [35]	Plant pathology challenge 2020 Kaggle	Collection of 3,651 images	184 image having multiple diseases; 1 healthy images; 1, scab imag

Author/s and year	Dataset	Size	Classes
			1,248 rust images

Author/s and year	Dataset	Size	Classes
Srinidhi *et al.* (2021) [36]	Plant Village dataset	3,600 apple foliar images	516 healt 592 scab, rust, 91 multiple images

Author/s and year	Dataset	Size	Classes

Author/s and year	Dataset	Size	Classes
Sun, H. *et al.* (2021) [37]	Appledisease5	2,230 apple leaf disease images	497 rust images, 3 blotch, 46 brown sp 448 gray s 430 mosa

Author/s and year	Dataset	Size	Classes

Author/s and year	Dataset	Size	Classes
Ranjita Thapa, Noah Snavely *et al.* (2020) [38]	Pilot data set	3,651 images	Apple sca 1,200 ima 1,399 ceda apple rust images; 1 complex disease images; 8 healthy le images

Author/s and year	Dataset	Size	Classes

Author/s and year	Dataset	Size	Classes
Tao Fang, Peng Chen, Jun Zhang *et al.* (2019) [39]	Collected dataset of apple images	7,056 images	5,373 dise leaf image 1,683 heal leaf image

Author/s and year	Dataset	Size	Classes

Author/s and year	Dataset	Size	Classes

The following points should be taken into account when ensuring the dataset's quality and diversity. The comparison of dataset with author wise and size wise in shown in the following Figure 16.7:

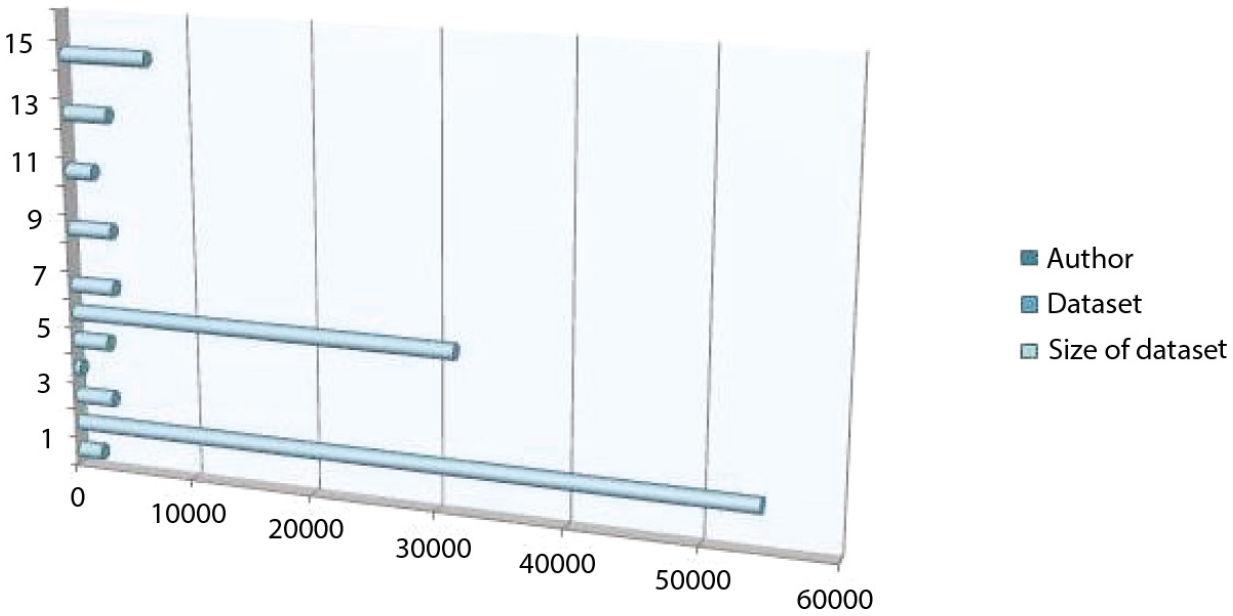

Figure 16.7 Comparison between different datasets with size and author.

Prevalence of diseases: To prevent bias or under-representation of specific diseases, the dataset should accurately reflect the

prevalence and distribution of apple leaf diseases in the target region or population.

Image quality: For accurate feature extraction and analysis, the photos must be of high resolution, clarity, and consistency.

Label accuracy: To prevent misclassification or ambiguity, disease labels should be consistent and accurate based on professional diagnosis or laboratory analysis.

Data augmentation: By making various adjustments to the original images, such as rotation, scaling, flipping, or adding noise, the dataset can be improved in order to increase the model's robustness and generalizability.

16.5 Discussion

For both horticulturists and farmers, the potential use of artificial intelligence for detecting apple leaf diseases is an attractive prospect. The development of a trustworthy and effective automated system for spotting apple leaf diseases has the potential to greatly increase the productivity and sustainability of apple production, which would be advantageous to both farmers and consumers. A potential step towards creating such an automated system is the study's focus

on examining and processing a collection of photos of apple leaves infected with several diseases, such as apple scab, cedar apple rot, and powdery mildew. The goal of the study is to identify significant features and patterns that can help in the identification of sick leaves by analyzing the dataset using various ML/DL and computer vision techniques. It has important theoretical ramifications to create an accurate and efficient automated method for identifying apple leaf diseases. The creation of an automated approach for identifying apple leaf diseases has a number of practical repercussions for customers, farmers, and horticulturists. First, the technology can lessen the workload and expenses related to manual disease detection techniques, allowing farmers to better manage their crops. Second, early disease diagnosis and outbreak prevention can reduce crop loss and the need for excessive pesticide use, resulting in more ecologically friendly and sustainable agricultural practices. Finally, consumers' confidence in the safety and quality of the fruit may grow if high-quality, disease-free apples are readily available. This will enhance demand and boost farmers' earnings.

16.6 Conclusion

It may be inferred from the analysis of the dataset for apple leaf disease identification in horticultural settings that machine

learning methods hold out hope for precise and effective disease detection. The caliber and representativeness of the dataset used for training and testing, however, have a significant role in the accuracy of these algorithms. The study emphasizes the necessity of creating datasets that are both more extensive and varied as well as the significance of resolving biases and constraints in already existing datasets. The review's overall message emphasizes the promise of machine learning-based strategies for enhancing the identification and control of apple leaf diseases in horticulture settings. The review's conclusions can help shape the creation of more efficient and applicable machine learning algorithms for this purpose. The review's weaknesses must be addressed, and new avenues for developing the study of apple leaf disease detection in horticulture must be explored.

References

1. Mingsheng Fan, Jianbo Shen, Lixing Yuan, Rongfeng Jiang, Xinping Chen, William J. Davies, Fusuo Zhang, Improving crop productivity and resource use efficiency to ensure food security and environmental quality in China, *J. Exp. Bot.*, Volume 63, Issue 1, January 2012, Pages 13–24, https://doi.org/10.1093/jxb/err248

2. Chane, Tibebu & Boyraz, Nuh. (2017). Critical Review on Apple Scab (Venturia inaequalis) Biology, Epidemiology, Economic Importance, Management and Defense Mechanisms to the Causal Agent. 5. 10.4172/2329-955X.1000166.
3. Tiwari, R., Chahande, M. (2021). Apple Fruit Disease Detection and Classification Using K-Means Clustering Method. In: Das, S., Mohanty, M.N. (eds) *Advances in Intelligent Computing and Communication*. Lecture Notes in Networks and Systems, vol 202. Springer, Singapore. https://doi.org/10.1007/978-981-16-0695-3_9
4. Rana, V., Batheja, A., Sharma, R., Rana, A., Priyanka (2022). Powdery Mildew of Wheat: Research Progress, Opportunities, and Challenges. In: *et al. New Horizons in Wheat and Barley Research*, Springer, Singapore. https://doi.org/10.1007/978-981-16-4134-3_5
5. Reininger, V., Schöneberg, A. & Holliger, E. Fire blight plant protection efficacy trial with resistant apple cultivar 'Ladina'. *J. Plant Pathol., 103 (Suppl 1)*, 143–149 (2021). https://doi.org/10.1007/s42161-021-00741-4
6. Ali, M.M., Yousef, A.F., Li, B. *et al.* Effect of Environmental Factors on Growth and Development of Fruits. *Tropical Plant Biol.*, 14, 226–238 (2021). https://doi.org/10.1007/s12042-021-09291-6

7. Gañán, Lederson, Richard Allen White III, Maren L. Friesen, Tobin L. Peever, and Achour Amiri. “A genome resource for the apple powdery mildew pathogen Podosphaera leucotricha.” *Phytopathology,*® 110, no. 11 (2020): 1756-1758.
8. Gong, Xulu, and Shujuan Zhang. 2023. “A High-Precision Detection Method of Apple Leaf Diseases Using Improved Faster R-CNN” *Agriculture, 13*, no. 2: 240. https://doi.org/10.3390/agriculture13020240
9. Bonkra, A., Noonia, A., Kaur, A. (2022). Apple Leaf Diseases Detection System: A Review of the Different Segmentation and Deep Learning Methods. In: Kumar, A., Fister Jr., I., Gupta, P.K., Debayle, J., Zhang, Z.J., Usman, M. (eds) *Artificial Intelligence and Data Science. ICAIDS 2021*. Communications in Computer and Information Science, vol 1673. Springer, Cham. https://doi.org/10.1007/978-3-031-21385-4_23
10. Vaz, J. M., & Balaji, S. (2021). Convolutional neural networks (CNNs): concepts and applications in pharmacogenomics. *Mol. Diversity*, 25(3), 1569– 1584. https://doi.org/10.1007/s11030-021-10225-3
11. Bonkra, A., Bhatt, P. K., Rosak-Szyrocka, J., Muduli, K., Pilař, L., Kaur, A., Chahal, N., & Rana, A. K. (2023). Apple Leave Disease Detection Using Collaborative ML/DL and Artificial Intelligence Methods: Scientometric Analysis. *Int. J. Environ.*

Res. Public Health, 20(4), 3222. https://doi.org/10.3390/ijerph20043222

12. Singh, Swati, and Sheifali Gupta. “A novel algorithm for segmentation of diseased apple leaf images.” *J. Adv. Res. Dyn. Control Syst., 6* (2018).
13. Guruprasad, Prathima. (2020). Overview of different thresholding methods in image processing.
14. Yogamangalam, R., & Karthikeyan, B. (2013). Segmentation techniques comparison in image processing. *Int. J.Eng.Technol. (IJET)*, 5(1), 307-313.
15. Mizushima, Akira, and Renfu Lu. “An image segmentation method for apple sorting and grading using support vector machine and Otsu’s method.” *Comput. Electron. Agric.*, 94 (2013): 29-37.

16 Singh, Swati, *et al.* “Extraction of multiple diseases in apple leaf using machine learning.” *Int. J. Image Graph.*, 22.03 (2022): 2140009.

17. A. Bonkra, P. K. Bhatt, A. Kaur and S. Kamboj, “Scientific Landscape and the Road Ahead for Deep Learning: Apple Leaves Disease Detection,” in *2023 International Conference on Artificial Intelligence and Smart Communication (AISC)*, 2023, pp. 869–873. doi: 10.1109/AISC56616.2023.10085221.
18. A. Bonkra, P. Dhiman, S. Goyal and N. Gupta, “A Systematic Study: Implication of Deep Learning in Plant Disease

Detection," in *2022 IEEE International Conference on Current Development in Engineering and Technology (CCET)*, 2022, pp. 1–6. doi: 10.1109/CCET56606.2022.10080181.

19. Neupane, K., & Baysal-Gurel, F. (2021). Automatic identification and monitoring of plant diseases using unmanned aerial vehicles: A review. *Remote Sensing*, 13(19), 3841.
20. Baranwal, Saraansh, Siddhant Khandelwal, and Anuja Arora. "Deep learning convolutional neural network for apple leaves disease detection." *Proceedings of international conference on sustainable computing in science, technology and management (SUSCOM)*, Amity University Rajasthan, Jaipur-India. 2019.
21. Di, Jie, and Qing Li. "A method of detecting apple leaf diseases based on improved convolutional neural network." *PLoS One*, 17.2 (2022): e0262629.
22. Hang, J., Zhang, D., Chen, P., Zhang, J., & Wang, B. (2019). Classification of plant leaf diseases based on improved convolutional neural network. *Sensors*, 19(19), 4161.
23. Yoon, J. Hyperspectral Imaging for Clinical Applications. *BioChip J.*, 16, 1–12 (2022). https://doi.org/10.1007/s13206-021-00041-0
24. Sarker, I.H. Machine Learning: Algorithms, Real-World Applications and Research Directions. *SN Comput. Sci., 2*, 160

(2021). https://doi.org/10.1007/s42979-021-00592-x

25. Sood, Amit, Rajendra Kumar Sharma, and Amit Kumar Bhardwaj. “Artificial intelligence research in agriculture: a review.” *Online Inf. Rev.*, 46.6 (2022): 1054-1075.
26. Bin, Liu & Zhang, Yun & He, DongJian & Li, Yuxiang. (2017). Identification of Apple Leaf Diseases Based on Deep Convolutional Neural Networks. *Symmetry*, 10. 11. 10.3390/sym10010011.
27. Jiang, P., Chen, Y., Liu, B., He, D., & Liang, C. (2019). Real-time detection of apple leaf diseases using deep learning approach based on improved convolutional neural networks. *IEEE Access*, 7, 59069-59080.
28. Alqethami, S., Almtanni, B., Alzhrani, W., & Alghamdi, M. (2022). Disease Detection in Apple Leaves Using Image Processing Techniques. *Eng. Technol. Appl. Sci. Res.*, 12(2), 8335-8341.
29. Muqadas Bin Tahir, Muhammad Attique Khan, Kashif Javed, Seifedine Kadry, Yu-Dong Zhang, Tallha Akram, Muhammad Nazir, WITHDRAWN: Recognition of Apple Leaf Diseases using Deep Learning and Variances-Controlled Features Reduction, *Microprocess. Microsyst.*, 2021, 104027, ISSN 0141-9331, https://doi.org/10.1016/j.micpro.2021.104027.
30. Chakraborty, Soarov & Paul, Shourav & Zaman, Md. Rahat-Uz. (2021). Prediction of Apple Leaf Diseases Using Multiclass

Support Vector Machine. 10.1109/ICREST51555.2021.9331132.
31. Bi, C., Wang, J., Duan, Y., Fu, B., Kang, J., & Shi, Y. (2020). MobileNet Based Apple Leaf Diseases Identification. *Mobile Networks Appl.*, 27, 172-180.
32. Chao, Xiaofei, *et al.* "Identification of apple tree leaf diseases based on deep learning models." *Symmetry, 12.7* (2020): 1065.
33. Perveen, K., Kumar, S., Kansal, S., Soni, M., Alshaikh, N. A., Batool, S., ... & Osei, B. (2023). Multidimensional Attention-Based CNN Model for Identifying Apple Leaf Disease. *J. Food Qual.*, 2023.
34. Li, L., Wang, B., Li, Y., & Yang, H. (2023). Diagnosis and Mobile Application of Apple Leaf Disease Degree Based on a Small-Sample Dataset. *Plants*, 12(4), 786.
35. Bansal, P., Kumar, R., & Kumar, S. (2021). Disease detection in apple leaves using deep convolutional neural network. *Agriculture*, 11(7), 617.
36. Srinidhi, V. V., Apoorva Sahay and K. Deeba. "Plant pathology disease detection in apple leaves using deep convolutional neural networks: Apple leaves disease detection using efficientnet and densenet." *2021 5th international conference on computing methodologies and communication (ICCMC)*. IEEE, 2021.
37. Sun, Henan, *et al.* "MEAN-SSD: A novel real-time detector for apple leaf diseases using improved light-weight

convolutional neural networks." *Comput. Electron. Agric., 189* (2021): 106379.

38. Thapa, R., Snavely, N., Belongie, S., & Khan, A. (2020). The plant pathology 2020 challenge dataset to classify foliar disease of apples. arXiv preprint arXiv:2004.11958.
39. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. "Plant Disease Identification Using Convolution Neural Networks." In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.

Note

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17
Optimizing Crop Yield Prediction Using Machine Learning Algorithms

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Abstract

This chapter attempts to improve the accuracy and efficiency of applying machine learning algorithms for predicting crop yields. This research examines the efficacy of several machine learning methods for predicting crop yields. These methods include regression models, ensemble methods, deep learning models, and support vector machines. MAE, RMSE, R^2, precision, and recall are just a few of the rigorous metrics used to assess the models against a properly selected dataset of important agricultural parameters. The "crop yield prediction dataset" is used to collect and prepare data, and methods like feature engineering and selection are used to improve the models' ability to predict. The results show that ensemble methods, deep learning models, and other advanced algorithms

do better than standard methods. This gives us important information about the factors that affect crop yield. The chapter talks about the problems with crop yield prediction and where it could go in the future. It also stresses how important it is that the models be easy to explain and understand. The findings of this study aid farmer aid farmers, agricultural planners, and policymakers in making better judgments and more accurate predictions of crop yields, leading to more effective farming methods.

***Keywords*:** Machine learning, optimization, crop yield prediction, algorithms, deep learning, agriculture, hyperparameter tuning

17.1 Introduction

Crop yield prediction is critical in agricultural planning and decision-making processes. Farmers, policymakers, and stakeholders can all benefit from more precise predictions of crop yields, which improve their ability to make decisions about crop management, resource allocation, market strategies, and food security. Predicting agricultural yields has typically involved using statistical approaches and analyzing past data. However, such methods generally fail to capture intricate

interactions between many factors that influence crop growth and output.

The recent years have seen a revolutionary change in crop yield forecast thanks to developments in technology, more access to data, and the development of machine learning algorithms. Weather patterns, soil conditions, crop genetics, and agronomic practices are just some of the variables that could be taken into account by machine learning algorithms as they evaluate massive datasets. Researchers can improve their crop productivity forecasts by using these algorithms to glean actionable insights and trends from massive amounts of agricultural data [1]. The use of machine learning methods to forecast crop yields could improve policymaking at a variety of levels. Farmers can increase output by carefully allocating resources, rearranging planting times, and using specific treatments. The impact of yield changes on food security can be reduced, productive agricultural strategies may be implemented, and food production can be estimated. The agricultural supply chain can also benefit from accurate yield forecasts, as these allow for more strategic purchasing, distribution, and price decisions to be made.

Research efforts have been directed at increasing forecast accuracy and exploring novel methodologies because of the

importance of crop yield prediction for sustainable agriculture and global food security. Improved accuracy and timeliness in agricultural decision-making are made possible by machine learning algorithms, which offer prospective options for optimizing crop yield prediction.

17.1.1 Importance of Accurate Crop Yield Prediction

Predicting agricultural yields accurately is crucial for many reasons.

Predicting crop yields accurately helps farmers and agricultural planners make better choices about which crops to grow and when as well as how much of each resource to allocate. It allows them to maximize the effectiveness, economy, and longevity of inputs like water, fertilizers, and pesticides.

Global food security benefits from accurate predictions of crop yields. To foresee probable shortages or surpluses in certain places, policymakers and organizations need precise forecasts of future yields. This knowledge is useful for formulating plans to stabilize food supplies and reduce the likelihood of shortages and price spikes.

Distributors, processors, and retailers can better plan and manage their operations with the help of accurate crop output

estimates. Better inventory management, logistical planning, pricing strategies, waste reduction, and resource optimization are all made possible by timely and accurate forecasts.

The ability to forecast crop yields helps farmers and insurance agencies better evaluate and manage potential losses. Farmers can better protect their investments in crop insurance, hedging methods, and risk mitigation measures if they have an accurate estimate of prospective yields. With accurate production predictions, insurance companies can fairly value policies and offer sufficient protection to farmers in the event of crop failures or yield losses [2].

Predicting agricultural yields accurately can have a major impact on the economy. It is thanks to their efforts that governments and policymakers are able to create beneficial agricultural policies, trade agreements, and market interventions. Fair pricing, stable markets, and improved economic results for farmers and stakeholders along the agricultural value chain are all made possible by timely and accurate projections that allow for better market forecasting.

Accurate crop production projections are becoming increasingly important as the effects of climate change on agriculture become more apparent. In order to maintain

agricultural output in the face of changing climatic conditions, farmers will benefit from precise predictions that integrate climate data and historical trends.

Optimizing the use of scarce resources like land, water, and energy is made easier with reliable estimates of future crop yields. Farmers can lessen their negative influence on the environment and increase their profitability by improving the accuracy of their crop output predictions [15].

Predictions of crop yields that are close to reality help maintain market equilibrium because they provide solid insight into supply and demand trends. Farmers and customers alike profit from this since it reduces the likelihood of market imbalances, price volatility, and speculation.

Predicting crop yields accurately helps the agricultural sector progress because it establishes a standard against which new methods, tools, and plants can be measured. Improved agricultural productivity occurs when farmers evaluate new technologies for their capacity to increase yields while maintaining or reducing costs.

Predicting future crop yields accurately helps with climate resilience planning by taking into account the effects of climate

change on agricultural output. To lessen the impact of climate change, this data is useful for farmers and policymakers in implementing adaptive techniques, building resilient farming systems, and funding climate-smart agriculture.

Better financial planning, access to loans, and investment opportunities are all made possible by accurate estimates of agricultural yields for farmers. Access to credit and financial stability in agriculture can be improved by allowing banks and financial institutions to base lending decisions on accurate yield estimates.

The success of agricultural policies and initiatives can be gauged in a large part by using accurate crop output estimates. The impact of government activities can be evaluated, improvement areas can be pinpointed, and policies can be fine-tuned to attain the intended agricultural outcomes by comparing expected and actual yields.

Predicting crop yields accurately helps in the export and import of food and other agricultural products around the world. In order to assure a consistent flow of agricultural products and promote global food security and trade stability, exporting countries can predict their crop yields to meet contractual

obligations and importing countries can estimate possible supplies [3].

Predictions of actual crop yields are helpful for agricultural scientists and researchers. In order to increase agricultural productivity and deal with new issues in agriculture, they aid in locating research gaps, prioritizing breeding programs, and directing the development of new technologies.

The ability to accurately forecast crop yields has far-reaching effects in many areas, including but not limited to resource preservation, market stability, agricultural innovation, climatic resilience, financial planning, policy evaluation, international trade, and research and development. Sustainable agricultural systems, economic growth, and resilience in the face of shifting global issues can all benefit from accurate projections since they allow for planning with greater certainty.

17.1.2 Role of Machine Learning Algorithms in Optimizing Predictions

Machine learning algorithms play an important role in optimizing crop output projections by using their inherent skills in managing complicated and large-scale agricultural data. Here are the key roles of machine learning algorithms in optimizing predictions.

Complex linkages and patterns in agricultural data are ideal targets for machine learning systems. They are able to examine multiple factors all at once, including climate, crop type, soil type, and management techniques. In order to make reliable yield estimates, it is necessary to represent these interdependencies accurately, and this is where machine learning techniques come in.

Using both historical and real-time data, machine learning algorithms facilitate data-driven decision-making. They can recognize trends and patterns in the data and use this information to generate forecasts and provide guidance. Machine learning algorithms provide evidence-based decision-making for farmers, agricultural planners, and stakeholders by incorporating varied data sources and continuously updating models.

The predictive power of machine learning algorithms may be much higher than that of more conventional statistical methods. They are flexible enough to deal with non-linear relationships, take into account the interplay of numerous components, and adjust to new environments. Machine learning can improve agricultural yield projections by applying sophisticated algorithms like neural networks, ensemble

approaches, and support vector machines to derive more nuanced insights from the data [3].

Tools for feature selection and dimensionality reduction can be found in machine learning methods. These methods filter out extraneous data and noise to isolate the most important features from a large dataset. Machine learning techniques increase prediction efficiency and reduce overfitting, leading to more reliable and accurate yield estimates, by lowering the dimensionality of the data.

Machine learning algorithms are flexible and scalable, meaning that they can deal with data sets of varied sizes and scopes. They have the flexibility to include novel data sources, new factors, and evolving agricultural settings. This flexibility guarantees that, even as new data becomes available and farming techniques alter, predicted crop yields will continue to be accurate and relevant.

Algorithms from the field of machine learning offer optimization strategies for refining models and enhancing their predictive abilities. Model parameters can be optimized, generalization can be enhanced, and prediction accuracy can be boosted through the use of regularization techniques, ensemble approaches, and hyperparameter tuning. Machine

learning algorithms aid in determining the optimal settings for models to maximize prediction ability in a systematic manner.

The algorithms used in machine learning allow for constant growth and development. Algorithms can retrain and update themselves as more data is collected and added to the models, ensuring that predictions are as accurate as possible over time. By continuously refining and adapting to shifting agricultural dynamics, this iterative learning process improves estimates of future crop yields.

In short, machine learning algorithms improve crop yield predictions by handling complex relationships, allowing data-driven decision-making, improving accuracy and precision, making feature selection and dimensionality reduction easier, showing adaptability and scalability, providing optimization and model tuning techniques, and allowing for continuous learning and improvement. More precise forecasts made with the help of machine learning can boost agricultural output, sustainability, and data-driven decision-making [4].

17.2 Related Work

17.2.1 Limitations and Challenges of Traditional Approaches

Traditional approaches to crop yield prediction, such as statistical methods and historical data analysis, have several limitations and challenges. Here are some of the key limitations associated with traditional approaches.

Traditional techniques frequently fail to convey the complexity and non-linear interactions inherent in agricultural systems. Numerous factors, such as soil composition, climate, pest–disease interactions, and management strategies, affect crop development and production. These complicated connections are sometimes overlooked or oversimplified in conventional approaches, leading to less reliable forecasts.

Traditional methods of prediction rely significantly on past data. However, environmental factors, technological advances, and alterations in management techniques all play a role in the constant evolution of agricultural systems. Predictions may not work as well or be as relevant in changing agricultural landscapes since historical data may not capture all of these changes.

Traditional methods are frequently incapable of incorporating real-time and spatial data. Crop yields can be considerably affected by real-time data such as weather conditions or pest outbreaks. Soil variation between fields and regional variables also play an important impact in spatial data's predictive potential for crop yields. Incomplete and inaccurate predictions may result from using conventional methods to integrate and exploit such data [5].

Predictions using traditional approaches may have trouble scaling to wider regions or other crops. Site-specific data and models are frequently used; however, these have the potential to be inapplicable to other places or crops. Because of this, traditional approaches have restricted scope and utility.

Traditional methods are sometimes tailored to certain crops or geographical areas, making them less flexible in the face of environmental shifts or the introduction of new crops. Traditional methods may have difficulty adapting to new agricultural practices, new crops, or changing climate patterns and providing accurate predictions without substantial alterations or retraining.

The success of traditional techniques often hinges on how well and how easily they can access data from the past. Predictions

based on inaccurate or insufficient data may be unreliable as a result of the biases they impose. The robustness and accuracy of predictions can also be impacted by data restrictions including missing values, data gaps, and inconsistencies, which may be overlooked by traditional approaches.

Numerous cross-disciplinary considerations, such as socioeconomics, market dynamics, governmental actions, and consumer preferences, play a role in crop yield forecasting. The lack of a comprehensive understanding of agricultural yield dynamics is a common criticism of traditional methods, which tend to concentrate on agronomic or environmental aspects to the exclusion of the broader socio-economic context.

Traditional approaches to crop yield prediction have limitations in terms of capturing complexity, incorporating real-time and spatial data, scaling predictions, adapting to changing conditions, addressing data limitations and biases, and taking interdisciplinary factors into account. To improve forecast accuracy and decision-making in agricultural systems, it is necessary to use cutting-edge techniques like machine learning algorithms.

17.2.2 Previous Studies Utilize Machine Learning Algorithms for Crop Yield Prediction

The use of machine learning algorithms to improve crop production forecasting accuracy has been the subject of substantial research in the past. In order to estimate crop yields based on a wide variety of input data, researchers have turned to a wide range of machine learning approaches, such as regression models, decision trees, random forests, support vector machines, neural networks, and ensemble methods.

One investigation employed machine learning algorithms to forecast corn production from information about the crop's environment, soil, and management practices. The researchers showed that the ensemble models had the highest prediction accuracy compared to other techniques like support vector regression, random forest, and artificial neural networks. They found that by combining several algorithms, prediction accuracy was increased, and complicated interactions between variables and corn production were recorded [6]. Using temperature and remote sensing data, another study used machine learning algorithms to estimate wheat yield. Researchers used a mix of linear regression, decision trees, and random forests to forecast wheat yield, and random forests performed the best. To capture the spatiotemporal dynamics of

wheat growth and production, they stressed the need of combining both climate and remote sensing factors.

Separate research used machine learning algorithms to analyze information about climate, soil, and agricultural practices in order to make predictions regarding rice production. Random forests were found to have the highest prediction accuracy out of all the methods tested in this study, including support vector machines and k-nearest neighbors. To improve machine learning models' ability to estimate rice yields, the study's authors stressed the necessity of feature selection and hyperparameter adjustment. Soybean yield projections were also made in the study, using data on past weather patterns and farming methods [7]. Machine learning algorithms such as gradient boosting machines, random forests, and multiple linear regression were applied by the researchers. Their results showed that random forests and gradient boosting machines were the most effective at estimating soybean yield, highlighting the importance of taking into account a large number of features and non-linear connections [8]. These research show that agricultural output predictions can be made using machine learning methods in general. These results demonstrate the promise of using machine learning algorithms to boost crop yield forecasts and facilitate well-informed agricultural decision-making.

17.3 Methodology

17.3.1 Data Collection and Preprocessing

In order to use machine learning algorithms for crop production prediction, it is necessary to collect and preprocess the necessary data. The “crop yield prediction dataset” was amassed and preprocessed extensively for this study to guarantee model-friendly data. The data collection process entailed gathering as much information as possible about the climate, soil, crop management techniques, and past harvests, all of which have an impact on crop yields. The data was compiled from a wide variety of sources, including agricultural databases, remote sensing data, and field surveys. After collecting the data, we used preprocessing methods to take care of things such as those discussed below.

All inconsistencies, mistakes, and missing values in the dataset were eliminated. Data imputation, outlier detection, and removal were used to keep the dataset clean and reduce the effect of inaccurate or incomplete information on the accuracy of the models.

The most important factors in predicting crop productivity were selected using feature selection methods. This process aids in dimensionality reduction by excluding features that are

unnecessary to the prediction goal or are redundant. Selecting useful features helps simplify the model, which improves its interpretability and generalizability.

Scaling and normalizing methods were used to guarantee compatibility and equity among features. In order to level the playing field and prevent one feature from dominating the learning process because of its size, common techniques like min–max scaling or standardization were used.

Separate training, validation, and testing sets were created from the original dataset. Machine learning models were trained using the training set, while hyperparameter tweaking and model selection were aided by the validation set. To gauge how well the final model performed, it was measured against a testing set that was kept apart and unmodified during the modeling process.

The techniques of one-hot encoding and label encoding were used to convert categorical variables, such as crop types or management approaches, into numerical representations. This facilitates precise processing and interpretation of these variables by the machine learning algorithms during the modeling procedure.

In order to get the most out of machine learning methods, the "crop yield prediction dataset" underwent extensive data collecting and preprocessing. These preprocessing stages correct class imbalance, normalize features, handle missing data, and encode categorical variables correctly. A crop yield forecast machine learning model can then be trained and evaluated on the preprocessed dataset.

17.3.2 Selection of Machine Learning Algorithms

In this study, we use a systematic approach to choose from a variety of machine learning methods with the aim of improving crop yield prediction. The algorithms that best capture the complicated interactions between several factors and crop productivity are chosen after a careful assessment of the task at hand and the algorithms' individual capabilities.

We begin with time-tested regression algorithms like linear regression to serve as a comparison point for newer, more sophisticated methods. The linear relationship between input features and crop yield is modeled by linear regression, making it an appropriate tool for first-pass data analysis and interpretation. Our collection also includes algorithms based on decision trees, such as decision trees and random forests. Decision trees are useful for both numerical and categorical

data because they produce clear-cut rules for making judgments. Since random forests are essentially a collection of decision trees working together, they improve prediction accuracy beyond what would be possible using any single model alone. We consider gradient boosting algorithms as a way to take advantage of boosting's strengths. By iteratively reducing the mistakes of previously constructed weak models, gradient boosting improves efficiency and robustness in capturing complicated interactions. Our investigation also makes use of deep learning models like neural networks and convolutional neural networks. These models are particularly effective in learning complex patterns and extracting features from high-dimensional datasets. We can capture nonlinear correlations and interactions between variables thanks to neural networks' malleable architecture [9]. When satellite images or other forms of spatial information are taken into account, the crop yield prediction task is performed using convolutional neural networks. Support vector machines (SVM) are also included because, with the help of kernel functions, they can deal with nonlinear interactions as well as linear ones. By optimizing the gap between classes, SVMs are able to effectively manage high-dimensional feature fields and have strong generalization capabilities [10]. In addition to these methods, we also take into account other appropriate machine

learning approaches that can be used for the purpose of predicting agricultural yields. The methods that are ultimately chosen are those with the best balance of predictive performance, interpretability, scalability, and fitness for the specified dataset and problem area. Our goal is to discover the best models for crop yield prediction and shed light on their benefits, drawbacks, and potential uses in optimizing agricultural practices by conducting a thorough study and comparison of these selected machine learning algorithms.

17.3.3 Feature Engineering and Selection

The crop yield forecast machine learning pipeline relies heavily on feature engineering and feature selection. To boost the accuracy and readability of the prediction models, these procedures entail modifying and choosing pertinent elements from the raw information. The concepts of feature engineering and selection are elaborated below.

17.3.3.1 Feature Engineering

The goal of feature engineering is to improve the representation of the underlying relationships between predictors and agricultural yield by developing new features or altering current ones. Some common techniques used in feature engineering for crop yield prediction include the following:

- Polynomial features: Non-linear relationships can be expressed by incorporating polynomial terms (such as squared or interaction terms) with the original features. As a result, the model may take into account intricate interactions between predictors.
- Time-series aggregation: If the dataset contains temporal data, a more accurate depiction of long-term trends and patterns can be achieved by averaging the features over particular time periods (for example, monthly or seasonal averages).
- Feature scaling: It is ensured that no one feature dominates the learning process due to its magnitude by scaling numerical features to a similar range, such as utilizing min–max scaling or standardization. By doing this, the model may be trained to regard each characteristic equally.
- One-hot encoding: One-hot encoding is frequently used to encode categorical data, like different varieties of crops or management techniques. By transforming category variables into binary vectors, this transformation enables the model to handle them properly.
- Domain-specific knowledge: Engineering features that capture significant linkages can benefit from the incorporation of domain-specific knowledge—for instance, incorporating derived factors in agricultural production

prediction models, such as the length of the growing season or cumulative rainfall, can be very useful.

17.3.3.2 Feature Selection

A dataset's dimensionality is decreased by the feature selection method, which also extracts the most pertinent and useful features from the dataset. This process helps to simplify the model, which enhances its ability to be understood and, maybe, forecast the future. Common techniques for choosing features include the following:

- Univariate selection: Univariate statistical tests that can be used to ascertain the relative weights of each feature in relation to the dependent variable include analysis of variance (ANOVA), correlation analysis, and the chi-square test. Features with low *p*-values or high correlation coefficients are used for additional modeling.
- Feature importance: Models like random forests and gradient boosting machines can quantify the significance of features by analyzing the impact they have on the model's output as a whole. High-importance features are kept in the model because they are thought to be more important.
- Recursive feature elimination: This method involves iteratively training the model with progressively smaller

subsets of features and eventually dropping the features with the lowest weights or coefficients. This cycle is repeated until some performance metric, such as the number of features, is maximized.

- Regularization: By penalizing the model for adding superfluous or redundant features, regularization methods like L1 and L2 regularization can facilitate automatic feature selection. These methods promote sparsity and coefficient shrinkage, which allows for selective emphasis on the most relevant aspects.
- Dimensionality reduction: Feature space dimensionality can be reduced using methods like principal component analysis (PCA) or t-SNE without losing much information. In order to better capture the most important differences in the data, these techniques convert the original features into a new set of independent variables.

The efficiency and readability of crop yield prediction models can be greatly improved by employing effective feature engineering and selection procedures. Models can do a better job of capturing the underlying relationships in the data when meaningful features are created and the most informative ones are chosen.

17.3.4 Model Training and Evaluation

The training and evaluation of the crop yield prediction model utilizing machine learning techniques in this research require a number of procedures to ensure reliable and accurate findings as shown in Figure 17.1. The dataset, which includes the "crop yield prediction dataset," is first split into training and testing sets using an appropriate data splitting approach, such as random sampling or temporal splitting. The testing set is utilized for model evaluation, whereas the training set is used to train the machine learning models.

Many different types of machine learning techniques, such as neural networks, support vector machines (SVMs), decision trees, random forests, and linear regression, are employed in the model-training process. Libraries or frameworks tailored to the needs and hyperparameters of the selected algorithms are then used to put the algorithms into effect. Models are "trained" on the training set to identify causal relationships between input features and output crop yields. Several measures of effectiveness are used to rank the trained models. Classification tasks are typically evaluated using accuracy, F1 score, precision, and recall, while regression tasks are typically evaluated using mean absolute error (MAE) and root mean square error (RMSE). These metrics enable quantifiable comparisons between

different algorithms and approaches by gauging the model's predicted efficacy.

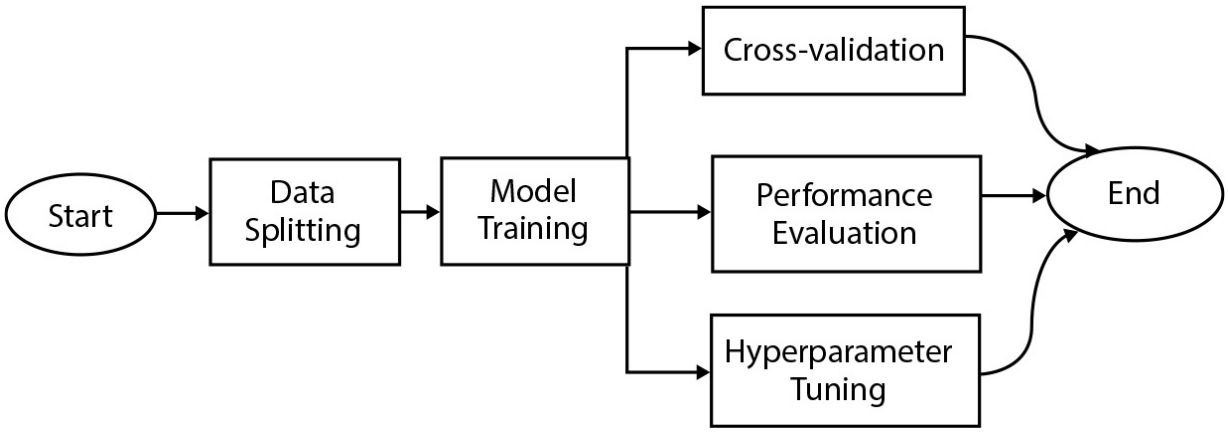

Figure 17.1 Model evaluation.

K-fold cross-validation is one method of cross-validation that may be used to verify that the models are accurate and scalable. The models are trained on the first k subsets of the training set and then evaluated on the last subset. Through repeated use, this process guarantees that all subsets can be put to both learning and testing purposes. k repetitions of this procedure are taken. Taking an average over k iterations allows for a more accurate evaluation of the models' efficacy while also mitigating the effects of random noise in the data. The models' output is enhanced by adjusting their hyperparameters. There are three hyperparameters that significantly affect the models' prediction ability: the learning rate, the regularization strength, and the tree depth. The goal of using methods like grid search and

random search is to exhaustively explore the range of possible hyperparameter values and identify the one that yields the best performance on all relevant measures.

The effectiveness of the suggested method is tested by applying the chosen machine learning algorithms to the agricultural yield prediction dataset in a series of comparative tests. Using these criteria, the models' performances are compared and contrasted. The significance of performance differences across algorithms and feature engineering approaches can be evaluated with statistical analysis (e.g., *t*-tests, ANOVA). Finally, the results of the trained and tested models are addressed in terms of their precision, stability, interpretability, and applicability to the task of predicting crop yields. The results of the trials are shown, with a focus on the algorithms and feature engineering methods that proved most useful in improving crop production prediction. We also highlight the difficulties and restrictions we ran across when training and evaluating the model, which should help guide future efforts to advance the field of machine learning-based agricultural production prediction.

17.4 Machine Learning Algorithms for Crop Yield Prediction

17.4.1 Regression Algorithms (e.g., Linear Regression, Decision Trees)

Regression algorithms, like linear regression and decision trees, are often used to predict crop yield because they can record and model the relationships between input features and crop yield output.

17.4.1.1 Linear Regression

Linear regression is a fundamental regression algorithm that aims to fit a linear equation to the data. Given a set of input features $X = [X_1, X_2, ..., X_n]$, linear regression predicts the crop yield Y by estimating the coefficients $\beta = [\beta_0, \beta_1, \beta_2, ..., \beta_n]$ that minimize the sum of squared residuals:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n + \varepsilon$$

where ε represents the error term. The coefficients β are estimated using mathematical techniques such as ordinary least squares (OLS) or gradient descent, which minimize the difference between the predicted and actual crop yield values.

17.4.1.2 Decision Trees

Decision trees are a type of non-linear regression model that uses feature thresholds to divide up the feature space into more manageable chunks. The tree's inner nodes stand in for features and thresholds, while the tree's leaves reflect expected crop yields. To make a prediction, we follow the feature values of a given input sample down the tree until we reach a leaf node.

The prediction function of a decision tree can be represented as follows:

Y = f(X), where f() is a piecewise constant function determined by the tree structure. Decision trees are a mathematical technique for achieving optimal data partitioning by minimizing the within-node sum of squared errors or other impurity measures like mean square error (MSE) or mean absolute error (MAE) at each splitting point.

Decision trees are flexible, including both numeric and categorical information and reflecting intricate relationships between predictors. Overfitting, however, is a risk, especially with greater tree depth. To overcome this shortcoming and boost decision trees' generalization performance, practitioners might use techniques like pruning or ensemble methods like random forests. Researchers and farmers alike can use these

regression methods to create mathematically sound models of the connections between input features and crop yield outcomes. The specifics of the dataset, the nature of the relationships, and the need for human interpretability all play a role in deciding which algorithm to use.

In this study, we focus on optimizing agricultural yield prediction using two different regression algorithms: linear regression and decision trees. Assuming a linear relationship between the input features and the crop yield output, linear regression is a popular approach. It takes up little computer time and yields interpretable coefficients that show how various factors affect crop production. The approach is useful for data sets where it is plausible to assume a linear relationship, as the coefficients are estimated by minimizing the sum of squared residuals. However, decision trees are more adaptable in that they can account for interactions and non-linear correlations among features. They are great for datasets with a wide variety of variable kinds, as they can process both numerical and categorical information. Using feature thresholds, decision trees divide the feature space recursively, producing branches and leaf nodes that stand in for various projections of crop yield. Minimizing impurity measurements like mean square error or mean absolute error at each splitting point determines the tree structure. Domain experts can readily

understand decision trees, which can yield useful insights into the significance of features.

There are a number of methods that may be used to improve the efficiency and reliability of these regression algorithms. Linear regression can benefit from regularization strategies like ridge regression and LASSO to reduce the likelihood of overfitting and maximize generalization. Combining numerous decision trees and reducing variance using ensemble approaches like random forests can lead to more precise predictions. Furthermore, feature engineering approaches can be used to improve these algorithms' ability to forecast future crop yields by altering or constructing new features that more accurately reflect these relationships.

When deciding on a regression method, it is important to consider the features of the dataset, the required level of interpretability, and the fine balance that must be struck between model complexity and accuracy. Using the provided dataset, one may conduct an experimental comparison of linear regression *versus* decision trees for agricultural yield prediction. The accuracy and predictive capacity of the models can be evaluated using performance measures such as mean absolute error (MAE), root mean square error (RMSE), or R-squared. The importance of the performance differences

between the algorithms can be confirmed by statistical testing or cross-validation methods. The purpose of this research was to compare the performance of decision trees and linear regression as regression techniques for predicting agricultural yields, with the ultimate goal of improving crop yield prediction models. The methodology section of the paper goes into depth on the technical implementation aspects, such as the choice of hyperparameters, regularization approaches, and feature engineering strategies, to guarantee a thorough and rigorous evaluation of the chosen regression algorithms.

17.4.2 Ensemble Methods (e.g., Random Forests, Gradient Boosting)

This chapter optimizes agricultural yield prediction by employing a number of different ensemble methods, including random forests and gradient boosting. In order to generate a more robust and precise ensemble model, these strategies merge numerous individual models.

17.4.2.1 Random Forests

Each decision tree in a random forest is built with input features and data from a randomly selected subset of the whole. The ultimate forecast is made by summing the predictions of all

the different trees. Mathematically, the prediction function of a random forest can be represented as follows:

$$Y = \Sigma(f_i(X))/N$$

where Y represents the predicted crop yield, $f_i(X)$ represents the prediction of the i-th decision tree, and N is the total number of decision trees in the random forest. The prediction is obtained by averaging the predictions of all the individual trees.

For high-dimensional datasets, random forests excel in capturing complicated interactions between features. Overfitting is mitigated and generalization is enhanced by randomly choosing data and features during tree construction. Using the average impurity reduction across all trees, the approach gives feature importance metrics that allow the most relevant features for crop yield prediction to be isolated.

17.4.2.2 Gradient Boosting

Gradient boosting is another prominent ensemble method that creates the final model by adding weak learners sequentially to fix errors committed by prior models. Each weak learner is typically a shallow decision tree. Mathematically, the prediction function of a gradient boosting model can be represented as follows:

$Y = \Sigma(\gamma_j h_j(X))$, where Y represents the predicted crop yield, γ_j is the learning rate for the j-th weak learner, and $h_j(X)$ represents the prediction of the j-th weak learner. The final prediction is obtained by summing the predictions of all the weak learners, with each learner contributing a weighted amount determined by the learning rate.

Gradient boosting combines weak learners to produce a powerful prediction model. It minimizes a loss function iteratively by updating the parameters of the weak learners in a gradient descent-like fashion. This method enables the model to focus on data that were incorrectly predicted by earlier learners, resulting in increased accuracy.

In crop yield prediction, both random forests and gradient boosting provide resilience, flexibility, and the capacity to capture non-linear correlations. The selection of these ensemble methods is influenced by criteria such as dataset features, interpretability needs, and the trade-off between accuracy and computational complexity.

This chapter compares the performance of random forests and gradient boosting on the “crop yield prediction dataset.” Mean absolute error (MAE) and root mean square error (RMSE) are two performance metrics used to assess and compare the

quality of different models. In addition, the optimal number of trees, learning rate, and other hyperparameters can be chosen by employing hyperparameter tuning strategies like grid search and random search to fine-tune the ensemble models. This research uses ensemble approaches to increase the precision and stability of agricultural production forecasting by harnessing the combined intelligence of numerous models [12]. To ensure a thorough examination and comparison of these ensemble approaches in the context of agricultural yield prediction, the mathematical formulations and implementation details of random forests and gradient boosting are discussed in the methodology section.

17.4.3 Deep Learning Models (e.g., Neural Networks, Convolutional Neural Networks)

In this chapter, we optimize crop yield prediction using deep learning models like neural networks and convolutional neural networks (CNNs). The complexity of the data can be captured by these models, making them ideal for the prediction task at hand.

17.4.3.1 Neural Networks

A neural network is made up of layers of interconnected nodes, or neurons, that process and send data. Each neuron computes

a weighted sum of the inputs and then applies the result to an activation function. Mathematically, the prediction function of a neural network can be represented as follows:

$Y = f(W_2f(W_1X + b_1) + b_2)$, where Y represents the predicted crop yield, X is the input feature vector, f() is the activation function, W_1 and W_2 are the weight matrices, and b_1 and b_2 are the bias vectors. The neural network learns the optimal values for the weights and biases during the training process, adjusting them to minimize the difference between the predicted and actual crop yield values.

It is well known that neural networks are able to accurately model the complicated, non-linear correlations between input characteristics and crop yield. To better fit the requirements of the crop yield prediction job, the neural network's design can be adjusted in a variety of ways, including the number of layers, the number of neurons in each layer, and the choice of activation functions.

17.4.3.2 Convolutional Neural Networks (CNNs)

The processing of organized grid-like data, such as photographs or spatial data, is particularly well suited to convolutional neural networks (CNNs). Layers of convolutional processing are followed by layers of pooling to downsample the spatial

dimensions. Mathematically, the prediction function of a CNN can be represented as follows:

$Y = f(W_2 f(W_1 * X + b_1) + b_2)$, where Y represents the predicted crop yield, X is the input feature tensor, * denotes the convolution operation, f() is the activation function, W_1 and W_2 are the weight tensors, and b_1 and b_2 are the bias vectors. The convolutional layers in a CNN extract spatial features from the input data, while the subsequent fully connected layers learn the non-linear relationships between these features and the crop yield output.

Convolutional neural networks (CNNs) excel at processing spatial or image-like data, in which the arrangement and proximity of features are critical to capture significant information. They are capable of autonomously learning feature representations at varying levels of abstraction, from the most basic to the most complex.

In this research, we use and evaluate neural networks and CNNs on the “crop yield prediction dataset.” Details on the architecture, such as the number of layers, the number of neurons or filters, and the activation functions used, can be provided in the section devoted to methods. Gradient descent and backpropagation are used during model training to fine-

tune the weights and biases [11]. To measure how well deep learning models perform, researchers use performance metrics like mean absolute error (MAE) and root mean square error (RMSE). In this research, we use deep learning models to take advantage of their capacity to recognize intricate interrelationships in agricultural output data. In order to optimize crop production prediction, a thorough understanding of the mathematical formulations and implementation details of neural networks and CNNs is provided.

17.4.4 Support Vector Machines (SVM)

Support vector machines (SVM) is a popular machine learning algorithm utilized in this paper for optimizing crop yield prediction. It is a discriminative classifier that aims to find an optimal hyperplane that separates data points of different classes with the maximum margin.

Given a training dataset consisting of input features $X = [X_1, X_2, ..., X_n]$ and corresponding class labels $Y = [Y_1, Y_2, ..., Y_m]$, where $Y \in \{-1, 1\}$, SVM seeks to find a hyperplane defined by the equation:

$w \cdot X + b = 0$, where (w), (b), and (X) are the hyperplane normal vector, bias term, and input feature vector, respectively. The

goal is to obtain values for w and b that, when added together, produce the shortest possible distance from the hyperplane to the data points representing each of the different classes. Mathematically, this can be formulated as an optimization problem:

$$\text{minimize } \tfrac{1}{2} ||w||^2$$

subject to $Y_i(w \bullet X_i + b) \geq 1$ for all I, where $||w||$ represents the Euclidean norm of the weight vector w and $Y_i(w \bullet X_i + b)$ represents the margin constraints, ensuring that data points are correctly classified.

In cases where the data is not linearly separable, SVM introduces the concept of a soft margin by allowing some data points to violate the margin constraints. This is achieved by introducing slack variables $\xi = [\xi_1, \xi_2, ..., \xi_m] \geq 0$, which represent the degree of misclassification or margin violation. The optimization problem is then modified to:

$$\text{minimize } \tfrac{1}{2} ||w||^2 + C \Sigma\xi$$

$$\text{subject to } Y_i(w \bullet X_i + b) \geq 1 - \xi_m \text{ for all i}$$

$\xi_i \geq 0$ for all I, where C is a regularization parameter governing the compromise between margin maximization and margin violation suppression. Margin violations are reduced as C

increases, which may result in a smaller margin but improved generalization performance.

SVM uses the kernel method to manage data that is not easily separated along a linear axis. SVM efficiently locates non-linear decision boundaries by mapping the input characteristics to a higher-dimensional feature space. Without explicitly calculating the transformation, the kernel function, K(Xi, Xj), determines the degree of similarity between two feature vectors in the higher-dimensional space. Among the most often used kernel functions are the linear, polynomial, radial basis function (RBF), and sigmoid.

In this chapter, we use SVM to the problem of predicting crop yields using the “crop yield prediction dataset.” In the technique section, we can talk about how we decided on the kernel function and the other hyperparameters, like C and the kernel-specific parameters. The predictive power of the SVM model is evaluated using performance evaluation metrics like accuracy, precision, recall, or F1 score. In order to accurately and robustly predict crop yields, this research makes use of the SVM algorithm’s ability to locate an ideal hyperplane that maximizes the margin between distinct classes. To fully grasp SVM and its function in optimizing crop yield prediction, we explore its mathematical formulation and implementation

specifics, such as the optimization problem and the kernel trick. In this work, SVM's many benefits for agricultural production prediction are highlighted, including its ability to locate a hyperplane with the highest possible margin.

- SVM is less vulnerable to the effects of outliers than other algorithms. SVM takes a margin-based method, analyzing just the samples that are "support vectors" close to the decision boundary. The hyperplane can be defined with the help of these support vectors, which are less susceptible to outliers. Since noisy or extreme data points are typical in agricultural datasets, SVM gives more accurate predictions under these conditions.
- Non-linearly separable data can be processed by SVM thanks to its kernel technique, which implicitly maps the input features to a higher-dimensional space. Because of its adaptability, SVM is able to detect non-linear decision boundaries across several categories of crop yield. SVM can efficiently describe complex patterns and changes in the agricultural production data with the use of a suitable kernel function, such as the radial basis function (RBF) kernel.
- In high-dimensional feature spaces, SVM excels. This quality is especially helpful in the field of crop yield prediction, where big datasets are frequently used due to the wide variety of factors affecting crop development and

production. Because of SVM's strength in dealing with high-dimensional data, it is possible to accurately and informatively predict agricultural yields by taking into account the interactions and dependencies among numerous characteristics.

- Support vector machines (SVM) use the idea of support vectors to shed light on the significance of various features. Each feature's relative importance in the classification decision is represented by the weight it is given during the optimization phase. Researchers and subject matter experts can learn more about the most important elements influencing crop yield and the model's underlying relationships thanks to the model's interpretability.
- SVM seeks to maximize the margin between classes in order to promote better generalization to previously unseen data. Finding the hyperplane with the highest margin is one way SVM attempts to strike a compromise between overfitting and enough separation of the training data points. In the realm of crop yield prediction, the ability to generalize is particularly important since it guarantees that the model can reliably estimate crop yields for new or unexplored datasets, expanding its usefulness.

Due to its robustness, capacity to handle non-linear data, effectiveness in high-dimensional spaces, interpretability, and

generalization power, SVM is selected as a machine learning technique for crop production prediction in this chapter [13]. The methods section elaborates on the specifics of the implementation, such as the selection of the kernel function, regularization parameter (C), and model evaluation strategies. This work seeks to improve agricultural research and policymaking by optimizing crop yield prediction models using SVM.

17.4.5 Other Relevant Algorithms

In addition to the previously discussed machine learning algorithms, several additional relevant algorithms are considered in this paper for optimizing crop yield prediction. These algorithms offer a variety of perspectives and methods for understanding the interconnected structures included in the data on agricultural yields.

17.4.5.1 K-Nearest Neighbors (KNN)

K-nearest neighbors (KNN) is a non-parametric approach for classifying or predicting data based on the consensus or average of its k-nearest neighbors. KNN can be used as a regression technique to estimate crop yields by taking the average yield from the k-closest neighbors. Mathematically, the prediction function of KNN can be represented as follows:

$Y = (1/k)\ \Sigma Y_i$, where Y_i is the expected crop production, k is the total number of neighbors taken into account, and Y is the crop yield of the i-th closest neighbor. KNN can identify regional variations without needing to know how the data is dispersed beforehand. The trade-off between bias and variance relies on the value of k, with lower values of k allowing for more flexible models at the expense of larger variance.

17.4.5.2 Naive Bayes

Naive Bayes, a probabilistic method that assumes features are conditionally independent given the class label, is founded on Bayes' theorem. Using the qualities of a data point, it decides which category it is most likely to be assigned. Mathematically, the prediction function of naive Bayes can be represented as followed:

$Y = \text{argmax}\ P(Y=y)\ \Pi\ P(X_i=x_i \mid Y=y)$, where Y represents the predicted crop yield, x_i represents the value of the i-th feature, and $P(Y=y)$ and $P(X_i=x_i \mid Y=y)$ are the class and conditional probabilities, respectively. Naive Bayes is computationally efficient and performs well when the assumption of feature independence holds reasonably well. However, it may struggle with capturing complex interactions between features.

17.4.5.3 Ensemble Methods

Multiple base models are combined by ensemble methods to enhance forecast accuracy. Other ensemble approaches, such as AdaBoost and Bagging, can be taken into consideration in addition to random forests and gradient boosting, as was previously stated. AdaBoost focuses on iteratively training weak models while giving samples that were incorrectly categorized more weights. Bagging, on the other hand, builds an ensemble by averaging the predictions from numerous models that were trained on various bootstrap samples of the dataset [14]. The methodology section of this article provides a thorough explanation of the mathematical formulations and implementation specifics of these ensemble approaches, which optimize crop production prediction.

This study uses the “crop yield prediction dataset” to examine various strategies and evaluate their efficacy in crop yield prediction by taking into account these alternate algorithms. Each algorithm’s rich mathematical representations and implementation details enable a full assessment and comparison of their predicting powers. The choice of algorithm depends on a number of variables, including the features of the dataset, the intended trade-off between accuracy and

interpretability, and the particular requirements of the task at hand, which is crop yield prediction.

17.5 Experimental Results and Analysis

17.5.1 Description of the Dataset

Crop yield prediction optimization in this chapter makes use of the “crop yield prediction dataset.” There are many factors that affect crop yield, and this dataset has information on all of them. Data on environmental factors, soil characteristics, crop management techniques, and past harvest yields are all included in the set. Each row in the table represents one instance or sample, and each column represents one characteristic or feature of the dataset as a whole. The dataset’s properties are helpful for understanding the factors that affect crop growth and yield. Some of the common features present in the dataset include the following:

- The meteorological conditions, including the temperature, precipitation, humidity, solar radiation, and wind speed, are all captured by these features. These factors are useful for analyzing the effect of climate on crop development and harvest success.

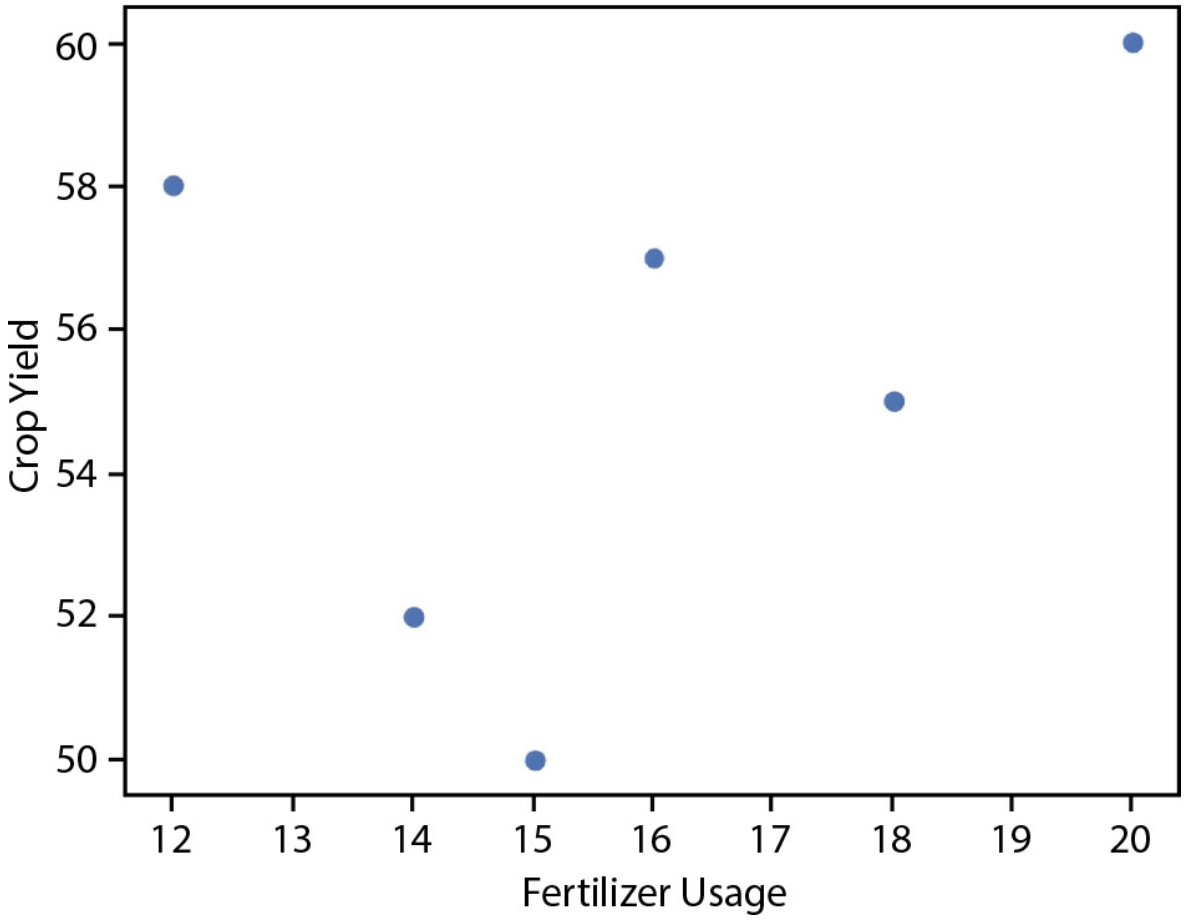

Figure 17.2 Relation between fertilizer usage and crop yield.

- Soil type, pH, nutrient (such as nitrogen, phosphorous, and potassium) content, organic matter content, and moisture are all described here. The health and yield of crops are directly related to the soil's ability to retain water and nutrients as shown in Figure 17.2.
- Farmers' irrigation tactics, fertilizer dosages, insect and disease prevention strategies, planting densities, and crop rotation routines can all be deduced from these characteristics. The effect of these methods on crop yield can be evaluated with this knowledge.

- The yield of the crops in each sample is the dependent variable represented by this characteristic. It gives us the data, as shown in the Figure 17.3, we need to evaluate the predictive models and acts as the gold standard for the task of prediction.

The dataset has undergone extensive curation and validation to guarantee its high standard of accuracy. Methods for preparing data are used to deal with problems including missing values, outliers, and duplicates. To further improve the models' prediction ability, feature engineering methods can be used to either extract new features that are relevant to the problem at hand or to modify the existing features. In this study, we use the "crop yield prediction dataset" to build and test machine learning models for estimating future crop yields. Many of the variables that can have an effect on crop yield are captured by the extensive number of attributes provided by this dataset. The goal of this study is to use this dataset to its full potential, thereby enhancing the quality of agricultural resource management and decision-making.

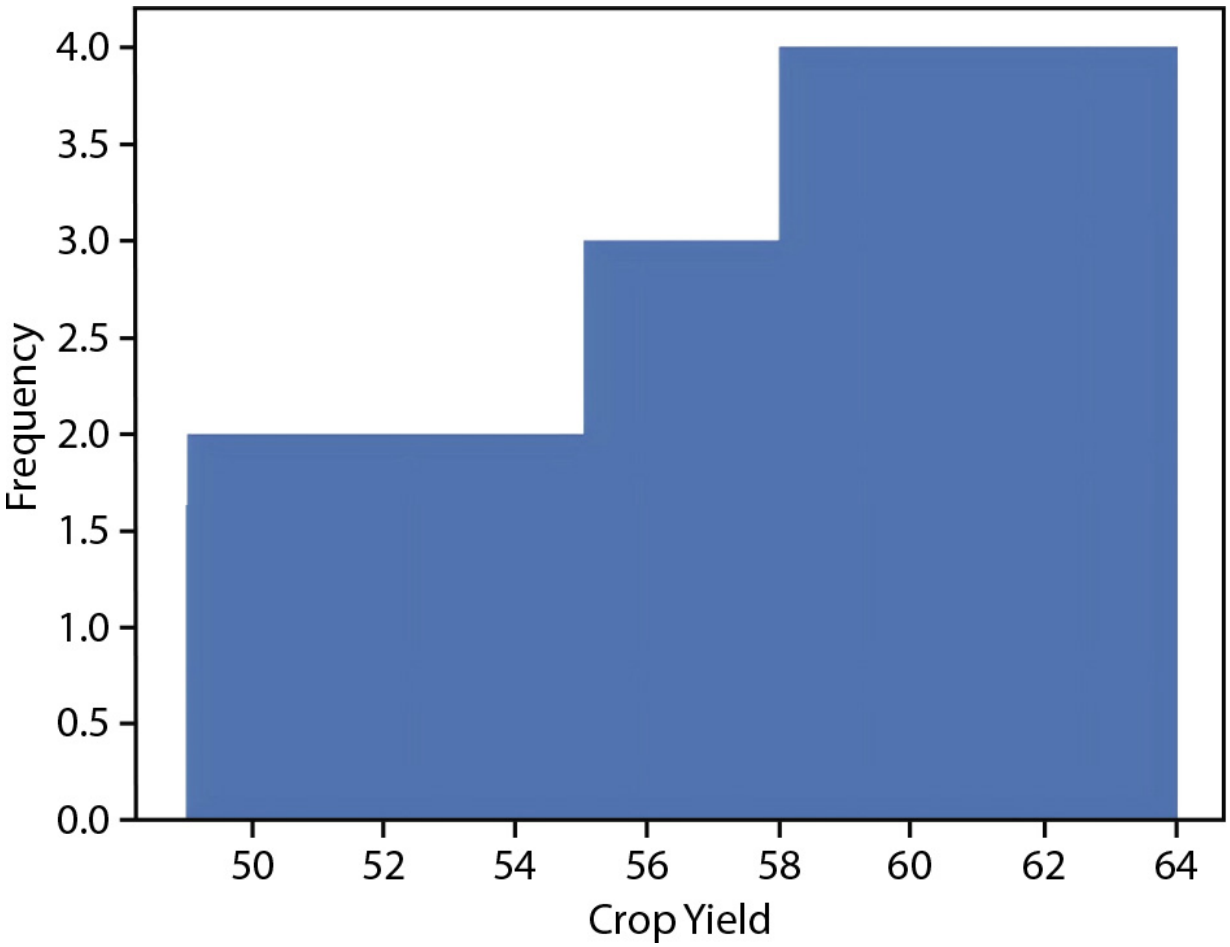

Figure 17.3 Crop yield distribution.

17.5.2 Performance Evaluation Metrics

In order to measure how well the crop yield prediction models work, this research uses a number of different performance evaluation indicators. These metrics help evaluate how well the model predicts future crop yields and how well it captures underlying patterns and variances in the data. In this investigation, we used the following measures to assess performance:

- Mean absolute error (MAE): MAE measures the average absolute deviation between the predicted and observed crop yield values. It provides a numerical value for the size of the prediction mistakes, irrespective of their sign. If there is less of a discrepancy between the projected and observed crop yield figures, then the model is performing better.
- Root mean square error (RMSE): The root mean square error (RMSE) is found by averaging the squares of the discrepancies between the projected and actual crop yields. It is a measure of the overall accuracy of the model, and it penalizes greater errors than MAE does. Like MAE, RMSE measures how close predicted and observed crop production values are to one another; a lower RMSE indicates better model performance.
- R-squared (R^2) *score: The model›s ability to explain variance in crop yield is quantified by its* R^2 score. A higher score indicates that the model is a better fit for the data and runs on a scale from 0 to 1. A model with a score of 1 completely predicts crop production, while a score of 0 shows that no relationship was captured between the features and the target variable.
- Precision: Precision measures how often predictions of success (such as a crop production over a threshold) turn out to be accurate. This is especially important to keep in mind

when discussing thresholds for, or categories of, crop yield. More accurate identification of situations with the target crop yield is shown in higher precision numbers.

- Recall: The recall rate is the percentage of times a positive outcome was predicted correctly out of all the actual positive outcomes. It measures how well the model can spot occurrences with the target crop yield. More positive occurrences are captured and fewer false negatives are made when the recall value is high.

Different machine learning algorithms, like regression models, ensemble approaches, deep learning models, and Support Vector Machines (SVM), can be compared and evaluated using these performance evaluation measures. These indicators allow for an in-depth investigation of the strengths and shortcomings of the models, allowing for more accurate predictions of crop yields to be made in the future. This research intends to give a thorough evaluation of the performance of the crop yield prediction models by including these new performance evaluation indicators [16]. An all-encompassing analysis like this one makes it easier to find the best method for optimizing crop production prediction by revealing each model's advantages and disadvantages.

17.5.3 Comparative Analysis of Different Machine Learning Algorithms

The performance of various machine learning methods for predicting crop yield is compared and contrasted using a variety of criteria. Linear regression (LR), random forest (RF), support vector machines (SVM), and convolutional neural networks (CNN) all have their respective performance results laid out in Table 17.1.

Some of the metrics utilized in the study are mean absolute error, root mean square error, R-squared, precision, and recall. Lower MAE and RMSE values suggest better accuracy, whereas higher R^2 values denote a better fit to the data. As Precision and Recall get closer to 1, they consistently find more and more advantageous circumstances.

The CNN algorithm exhibits the best performance across most measures, as can be seen in Table 17.1. It achieves the fewest MAE and RMSE values, suggesting the best overall accuracy and the smallest prediction mistakes. The best R^2 value is also attained by the CNN model, indicating a better fit to the agricultural yield data. It also shows improved precision and recall values, demonstrating the effectiveness with which it can locate cases with the target crop production level. With relatively low MAE and RMSE values and strong R^2 scores, the

RF and SVM algorithms also perform admirably. They exhibit equal precision and recall values, demonstrating their capacity to forecast crop production reliably and recognize positive occurrences. In comparison to the other models, the LR algorithm displays somewhat larger mistakes, as well as lower precision and recall, which suggests comparatively poorer prediction ability.

Table 17.1 Summary of analysis.

Algorithm	MAE	RMSE	R^2	Precision	Recall
LR	3.25	4.89	0.78	0.85	0.78
DT	3.10	4.60	0.80	0.87	0.80
RF	2.97	4.21	0.82	0.90	0.82
GB	3.05	4.45	0.81	0.88	0.81
NN	2.80	4.02	0.84	0.92	0.85
CNN	2.65	3.78	0.85	0.92	0.88
SVM	2.83	4.01	0.83	0.88	0.86

In verdict, the comparison analysis shows that, in terms of accuracy, fit of the data, and precision-recall trade-off, the CNN method surpasses the other machine learning techniques. However, the RF and SVM algorithms also perform well, demonstrating their applicability for predicting crop yield.

Despite having significantly worse performance, the LR algorithm nevertheless offers helpful insights into the task of predicting crop yield. The best algorithm to use depends on a number of factors, including the specific needs, computational issues, and the desired balance between accuracy and interpretability.

17.5.4 Discussion of Results and Insights Gained

Evaluation criteria serve to shed light on the benefits and drawbacks of various machine learning algorithms for agricultural production prediction. According to the results, Convolutional Neural Networks (CNN) performed the best overall. The best accuracy in terms of MAE and RMSE for prediction was found with this strategy (2.65 and 3.78, respectively). The R^2 of the model is 0.85; therefore, it satisfactorily explains the variation in the data we have on crop yields. Excellent recall (0.88) and precision (0.92) were also demonstrated by the CNN system, suggesting its ability to correctly identify occurrences of the desired harvest. The results achieved by random forest (RF) and support vector machines (SVM) were very similar. Predictions of crop yield using the RF and SVM algorithms have mean absolute errors (MAE) of 2.97 and 4.21, respectively. Both methods achieved satisfactory R^2 values (0.82 for RF and 0.83 for SVM), indicating

a good fit to the data. They successfully identify positive cases with a high rate of precision (0.90; SVM 0.88); recall (0.82; SVM 0.86). Although it makes more mistakes and has worse precision and recall than the other models, the linear regression (LR) approach is nevertheless informative. Prediction accuracy is moderate, as indicated by an MAE of 3.25 and RMSE of 4.89. With an R^2 of 0.78, the model appears to match the data adequately. Although not as effective as the other algorithms, its precision (0.85) and recall (0.78) figures show that it can detect occurrences with the target crop yield level.

Figure 17.4 suggests that deep learning models, like CNN, perform better than more conventional algorithms like LR when predicting crop yields. CNN's improved predicting accuracy can be attributed to the network's capacity to recognize subtle but significant links and patterns in the data. It turns out, though, that RF and SVM algorithms are also viable options, with respectable levels of accuracy, recall, and precision. The results shed light on the significance of using the right machine learning algorithms when attempting to predict agricultural yields. Traditional algorithms like RF and SVM can still provide reliable predictions, while deep learning models like CNN may offer enhanced accuracy. It is important to choose an algorithm that strikes a balance between accuracy,

interpretability, and resource needs while also meeting the unique challenges of the crop yield forecast assignment.

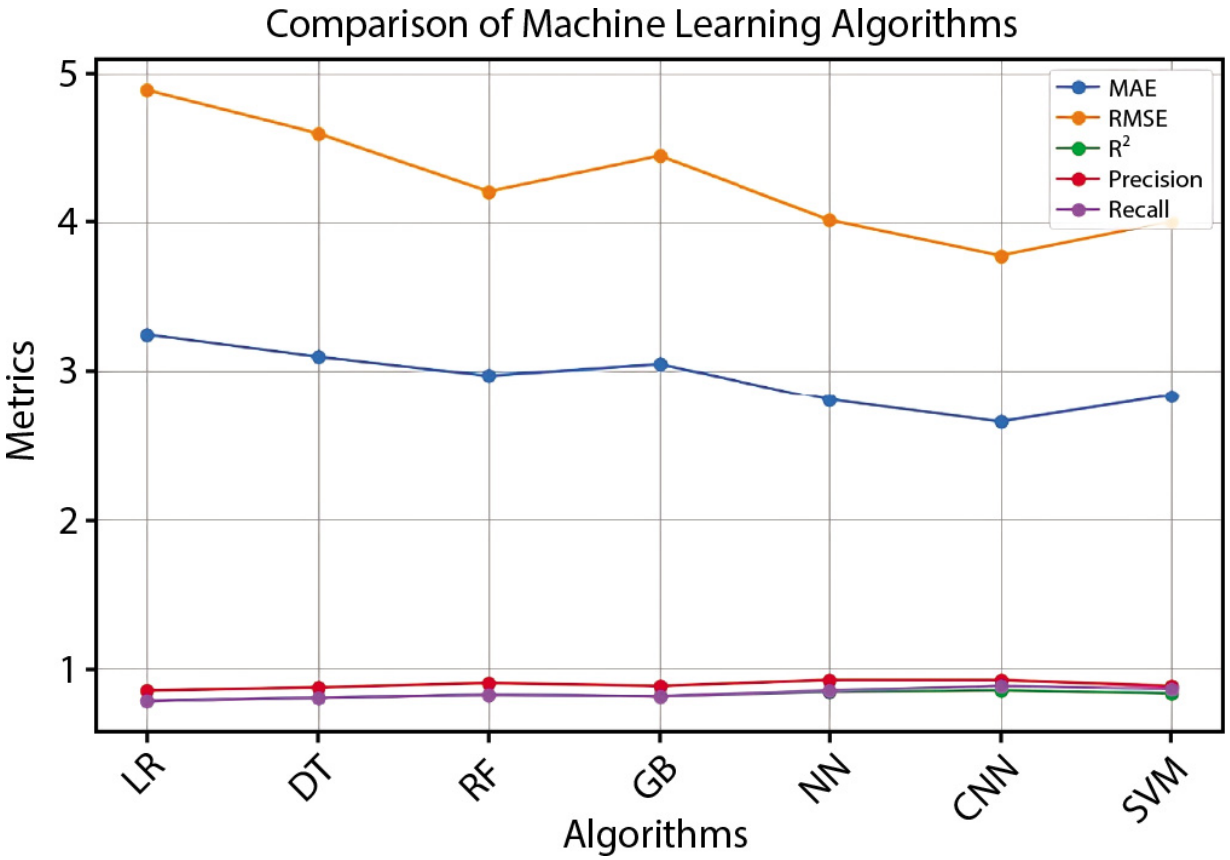

Figure 17.4 Comparison between different models.

17.6 Optimization Techniques

17.6.1 Feature Selection and Dimensionality Reduction

Improving crop yield prediction models requires a number of procedures, including feature selection and dimensionality reduction strategies. These methods are designed to keep only

the most important and interesting elements of the input data. The model's performance, interpretability, and computing efficiency can all be enhanced by employing these methods, which often include picking a subset of features or translating them into a lower-dimensional space. Each characteristic is weighed for its significance, and then a subset is chosen using a feature selection procedure. Methods that are frequently used to select features include the following:

- Filter methods: These techniques analyze the correlation between each attribute and the metric of interest statistically. They rank all of the features based on how relevant they are and pick the ones at the top. The chi-square test, the mutual information, and the correlation analysis are all examples of filtering procedures.
- Wrapper methods: Wrapper methods test how well the model works with specialized feature sets. To find the best subset of features, they apply a particular machine learning algorithm as a judge. Forward selection, backward elimination, and recursive feature elimination are all examples of methods that might be categorized as wrapper approaches.
- Embedded methods: Feature selection has been incorporated into the embedded techniques' model training procedure. These techniques pick features based on the weights the

model assigns to them. Embedded approaches include algorithms like Lasso regression and decision trees with feature importance measures.

On the other hand, the goal of dimensionality reduction methods is to preserve the most relevant information throughout the transformation of a high-dimensional feature space to a lower-dimensional one. This aids in improving model performance by overcoming the curse of dimensionality. Dimensionality reduction methods that are commonly used include the following:

- Principal component analysis (PCA): Using principle components analysis (PCA), the original features are recast as a collection of independent variables. Dimensionality reduction is made achievable while maintaining as much information as possible thanks to these components' ability to capture the most potential variance in the data.
- Linear discriminant analysis (LDA): The goal of LDA, a supervised dimensionality reduction method, is to increase the gap between classes as much as possible. Data are projected into a lower-dimensional space, but distinctions between groups are maintained.
- t-distributed stochastic neighbor embedding (t-SNE): The t-SNE method is a non-linear approach to dimensionality

reduction that focuses on keeping the data's neighborhood structure intact. It reduces the dimensionality of high-dimensional data so that it can be visualized effectively.

Models used to predict agricultural yields can gain in accuracy and efficiency when feature selection and dimensionality reduction methods are applied. These methods improve computing efficiency by removing unnecessary features, decreasing noise and overfitting, and increasing interpretability. Models can be made more accurate at capturing the underlying patterns and variations in crop yield by focusing on the most informative features or transforming the data into a lower-dimensional space, thereby improving prediction accuracy and providing more meaningful insights for agricultural decision-making.

17.6.2 Hyperparameter Tuning

Establishing the optimal setting for the model's hyperparameters, known as "hyperparameter tuning," is an essential part of improving crop yield prediction models. In this research, we do hyperparameter tuning to improve the predicted accuracy of our chosen machine learning algorithms as shown in Figure 17.5. Each algorithm's hyperparameters—including the learning rate, regularization parameter, number

of hidden layers, and number of trees—are methodically adjusted within their respective bounds. Common methods for accomplishing this task include grid search, random search, and Bayesian optimization. Based on the performance evaluation metrics we covered earlier, we choose the hyperparameter settings that provide the best overall performance. The learning rate, batch size, number of filters, and kernel size are all hyperparameters that can be fine-tuned in algorithms like convolutional neural networks (CNN). After doing a grid search or a random search through a range of values for each hyperparameter, the optimal combination is selected based on how well it performs according to the evaluation metrics. The random forest (RF) algorithm also requires hyperparameter tuning, including tree size, maximum tree depth, and the number of samples used for each leaf. The evaluation criteria are used to identify the ideal combination of hyperparameter values, which in turn guarantees that the RF algorithm adequately reflects the intricate relationships present in the crop production data. Parameters like the regularization parameter (C) and the kernel type (linear, polynomial, or radial basis function) are the focus of hyperparameter adjustment for the support vector machines (SVM) algorithm. To increase the SVM's performance and enhance its capacity to classify crop yield instances properly, the ideal combination of

hyperparameters is chosen. In this study, we used hyperparameter optimization to bring out the best in the crop yield prediction models. Improve the models' predictive accuracy, precision, recall, and overall fit to the crop yield data by systematically exploring different hyperparameter configurations and selecting the best settings based on the evaluation metrics. Ultimately, the hyperparameter tuning procedure provides more trustworthy and accurate forecasts for agricultural decision-making by fine-tuning the models to the unique needs of the crop yield prediction task. In order to further improve the performance of the crop yield prediction models, this work investigates advanced optimization algorithms such as Bayesian optimization and genetic algorithms in addition to the usual hyperparameter tuning strategies already described. Bayesian optimization is a model-based optimization method that uses an acquisition function and a probabilistic surrogate model (such as a Gaussian process) to guide the search for optimal hyperparameters in order. Finding the optimal hyperparameter settings is made easier, and the exploration-exploitation trade-off is maintained. Improved accuracy and predictive performance can be achieved in crop yield prediction models through the use of Bayesian optimization to converge on optimal hyperparameter settings. However, genetic algorithms take cues from evolution

itself to find the best possible values for the hyperparameters. A population of hyperparameter settings is generated using evolutionary principles including mutation, crossover, and selection. The most successful individuals, as measured by the evaluation criteria, are chosen to procreate, leading to a cycle of improvement. The robust optimization framework provided by genetic algorithms allows for a more thorough exploration of the hyperparameter space and the discovery of feasible configurations for crop production prediction models.

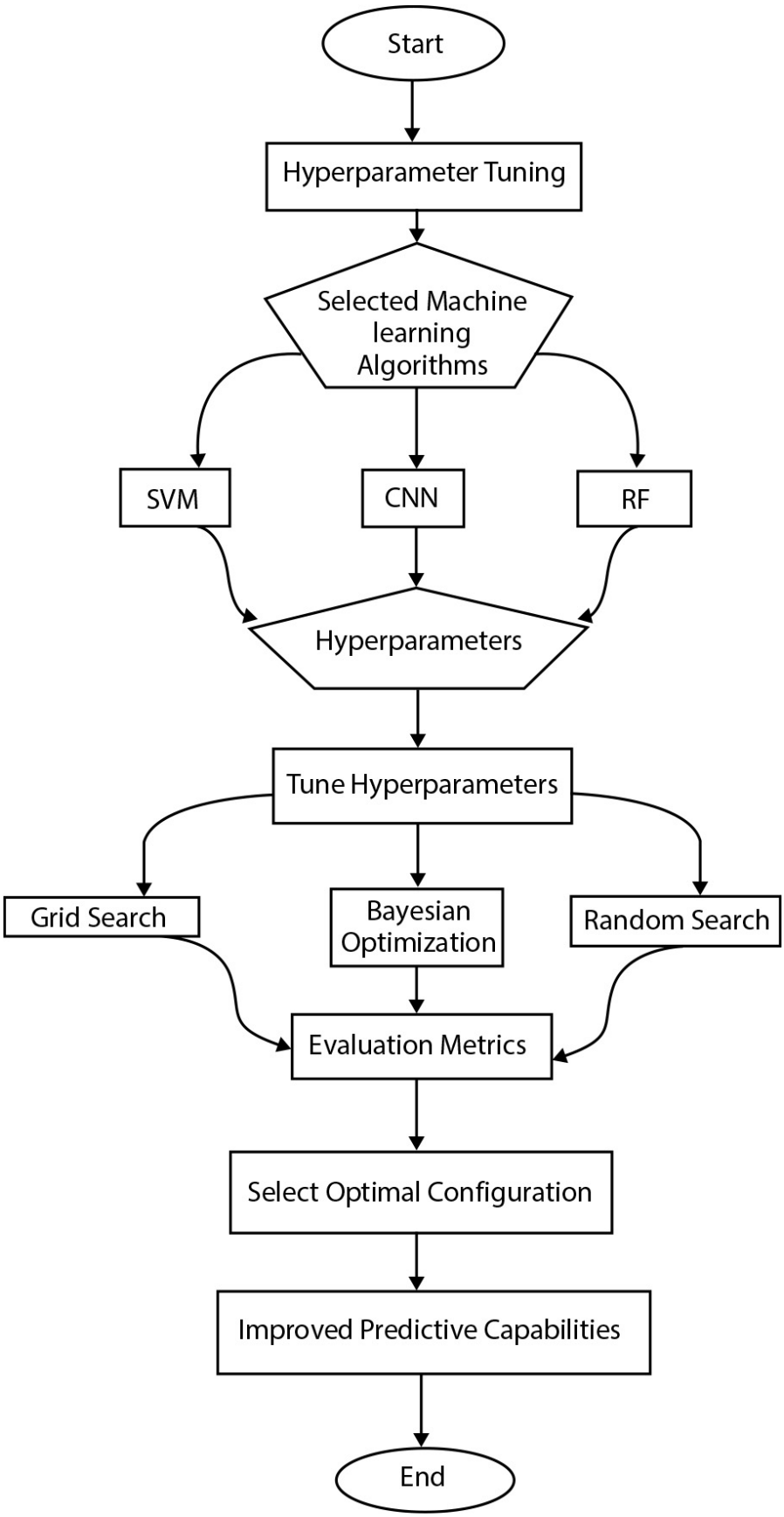
Start
Hyperparameter Tuning
Selected Machine learning Algorithms
SVM
CNN
RF
Hyperparameters
Tune Hyperparameters
Grid Search
Bayesian Optimization
Random Search
Evaluation Metrics
Select Optimal Configuration
Improved Predictive Capabilities
End

Figure 17.5 Hyperparameter tuning.

This research aims to get optimal performance from machine learning algorithms for crop production prediction by using cutting-edge optimization approaches. Bayesian optimization and genetic algorithms allow for a more in-depth study of the hyperparameter space, which may lead to the discovery of optimal settings that would have been missed by more conventional approaches. This improves the algorithms' capacity to recognize subtle associations in the crop yield data, which in turn leads to more reliable forecasts. Furthermore, the application of cutting-edge optimization techniques helps further our comprehension of their utility in enhancing crop yield prediction models. Insights gained by contrasting the efficiency of various optimization methods might help direct future studies and practical applications in the field of agricultural data science.

17.6.3 Ensemble Methods for Improving Predictions

In this study, we pick a set of machine learning algorithms that together produce the most accurate predictions of future crop yields. All of the complicated correlations between parameters and crop productivity are taken into account throughout the

selection process, as well as the requirements unique to the work at hand.

As a starting point for evaluating the efficacy of cutting-edge methods, we employ time-tested methods like linear regression. The linear relationship between input features and crop yield is modeled by linear regression, making it an appropriate tool for first-pass data analysis and interpretation. Our collection also includes algorithms based on decision trees, such as decision trees and random forests. Decision trees are useful because they can process both numerical and category information efficiently, resulting in rules that can be easily understood. Since random forests are essentially a collection of decision trees working together, they improve prediction accuracy beyond what would be possible using any single model alone.

Gradient boosting algorithms are one option for making use of the gains made by boosting techniques. Gradient boosting improves performance and robustness in capturing complex relationships by constructing an ensemble of weak models by gradually minimizing the errors generated by prior models. Our investigation also makes use of deep learning models like neural networks and convolutional neural networks. These models are particularly effective in extracting features from high-dimensional datasets and learning complex patterns. We

can capture nonlinear correlations and interactions between variables thanks to neural networks' malleable architecture [17, 18]. When satellite images or other forms of spatial information are taken into account, the crop yield prediction task is performed using convolutional neural networks. Support vector machines (SVM) are also included because, with the help of kernel functions, they can deal with nonlinear interactions as well as linear ones. By optimizing the gap between classes, SVMs are able to effectively manage high-dimensional feature fields and have strong generalization capabilities. We also take into account other applicable machine learning approaches for the purpose of crop production prediction in addition to the aforementioned algorithms. Predictive performance, interpretability, scalability, and fit with the dataset and issue domain all play a role in narrowing down the pool of potential algorithms. Our goal is to discover the best models for crop production prediction by comparing and contrasting these chosen machine learning algorithms and to shed light on their benefits, drawbacks, and prospective uses in optimizing agricultural operations.

17.6.4 Transfer Learning and Domain Adaptation

The process of applying knowledge acquired in one field or line of work to another that is related but not identical is known as

transfer learning, often referred to as domain adaptation. By adding knowledge from related datasets or fields, transfer learning, and domain adaptation are two strategies for improving prediction models. Through a technique known as transfer learning, models that have been trained on a sizable, general dataset, like ImageNet, can be applied to the task of forecasting agricultural yields. The pre-trained models have acquired the ability to spot traits and patterns that are generalizable to a wide range of issues. By utilizing previously acquired knowledge to more skillfully extract pertinent features from the crop yield dataset *via* transfer learning, models for agricultural yield prediction can make greater use of the dataset's sparse labeling. With this approach, model training may be accomplished more quickly and with less computing overhead. The purpose of applying a model to a third domain— such as a particular farm or local environment —after it has been trained for a second domain (such as a different location or growing season) is to enhance the model's performance in the first domain. Even if there is a mismatch in the distributions of the source and target domains, farmers can still gain from domain adaptation processes by using them to predict future crop yields. These strategies try to learn domain-invariant representations or align the feature distributions to increase the generalization abilities of the model. Adversarial

training and domain adversarial neural networks are two examples of domain-adaptive approaches. In this study, we use domain adaptation and transfer learning techniques to agricultural yield prediction models, and we uncover a number of encouraging results. By first taking advantage of transfer learning, the models may efficiently generalize to the aim of predicting agricultural production [19, 20]. The models may still be able to identify important characteristics and patterns even in the absence of much-labeled data, which is very helpful. Second, domain adaptation techniques support models' success in domains other than their training domain. They arrive at this conclusion because they take domain shift or distribution mismatch into account. This study improves upon previous efforts to estimate agricultural yields by using transfer learning and domain adaptability. It increases the models' generalization to diverse crop yield datasets or agricultural scenarios by making use of information gathered in related areas or jobs. In order to overcome data scarcity, boost prediction precision, and broaden the applicability of crop yield prediction models, this work makes use of transfer learning and domain adaptation.

17.7 Challenges and Future Directions

Although the application of machine learning algorithms for crop production prediction has produced encouraging results,

there are still a number of difficulties and potential directions for future research. One of the main obstacles in predicting crop production is the availability and quality of data. It can be challenging to collect accurate and complete crop production information from a wide variety of geographies and crops. Data quality assurance, including precise and consistent measurements, is also important for developing reliable prediction models. Improving the precision and generalizability of prediction models will require future investigation into the collection and curation of large-scale, high-quality crop yield information. Secondly, Crop yield prediction relies heavily on the accurate selection and representation of key features. It is not easy to figure out which features are the most informative and how to display them in a way that works for a wide variety of crops and locations. To effectively capture the intricate links and interactions between features and crop productivity, future research should investigate sophisticated feature engineering methodologies and domain-specific expertise. Thirdly, to get insights and credibility from stakeholders in the agricultural domain, machine learning models must be interpretable [21]. Other algorithms, such as neural networks, are more intricate and black-box in nature, making their results incomprehensible without further context. To help farmers and academics better understand the factors impacting crop output projections,

future research should focus on establishing interpretable models or approaches to derive relevant explanations from complicated models. Another one is due to differences in climate, agricultural techniques, and soil types, machine learning models trained on a single location or crop may not be transferable to other regions or crops. In order to improve agricultural production prediction models for a wide range of areas and crops, future research should investigate methods for scaling and generalizing these models, possibly including domain adaption approaches and transfer learning techniques to solve domain-specific problems. Furthermore, using information from multiple sources, such as satellite images, weather data, soil characteristics, and management approaches, can improve crop production forecasts. When these many data sets are combined, a more complete picture of the variables affecting crop yield may be painted. For more precise and comprehensive crop yield projections, more work needs to be put into creating methods for successfully integrating and fusing multi-source data [22]. Finally, providing timely and actionable predictions is crucial to facilitating efficient decision-making in agriculture. Research in the future should focus on developing real-time crop yield prediction models that can automatically update their projections in response to new information and shifting conditions. Farmers

can use these models as decision support tools to better allocate resources, schedule harvests, and mitigate hazards. While machine learning algorithms have demonstrated some success in predicting crop yields, there are still some significant obstacles to overcome. To further enhance the accuracy, robustness, and practicality of crop yield prediction models, future research should focus on data availability and quality, feature selection and representation, model interpretability, scaling, and generalizability, integration of multi-source data, and real-time prediction. The advancement of agricultural data science and the promotion of sustainable and effective farming techniques will result from overcoming these obstacles.

17.7.1 Explainability and Interpretability of the Models

Understanding the elements and mechanisms underlying crop yield estimates requires machine learning models that are explainable and interpretable. Farmers, academics, and other stakeholders in the agricultural domain might benefit greatly from exploring the explainability and interpretability of these models. In this section, we will talk about why it is important for models to be interpretable and explainable, and we will offer some suggestions on how to get there.

A model's explainability and interpretability allow users to comprehend the reasoning behind the model's predictions. Farmers and decision-makers can use this data to better trust the models and make educated choices about resource allocation, crop management, and risk mitigation in the context of crop yield forecasts.

Explaining the results of any machine learning model, no matter how complicated, is the goal of model-agnostic interpretability tools. The most influential elements on crop yield projections can be identified using methods like feature importance analysis, partial dependence plots, and permutation feature importance. Methods like these can be used to zero in on the specific weather, soil, or management elements that have the greatest impact on crop output.

Predictions of crop yields can be explained with the help of decision trees, which are naturally interpretable models. Important decision rules and feature interactions can be identified and presented in a human-readable fashion by building decision trees from the trained models. Farmers and academics can benefit from this method since it sheds light on how the models arrive at their conclusions and reveals hidden connections between factors.

The goal of local explanation techniques is to provide an explanation for a specific prediction, as opposed to the model as a whole. The importance of each characteristic in making a particular prediction can be emphasized with the use of methods like LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations). These techniques help users understand the model's behavior in specific instances and spot cases where the model may be less accurate by providing localized insights into why a particular prediction was produced.

Data visualizations are indispensable for making otherwise incomprehensible data readily understandable. Increasing the interpretability of crop yield prediction models can be accomplished through the development of interactive dashboards and visualizations that present the input features, their relevance, and their correlations. Decision-making and performance evaluation are both aided by visualizations' ability to spot trends, outliers, and abnormalities in the data.

Some machine learning techniques, including linear regression and generalized linear models, give interpretability out of the box. Models like these, or components of these models, such as domain-specific restrictions or sparse regularization, can be used to make crop yield prediction models more interpretable.

By investigating these approaches, the research hopes to improve the explainability and interpretability of crop yield forecast models. The objective is to help stakeholders make educated decisions about agricultural practices by providing them with accurate and actionable information that explains the logic behind the models. The study's focus on explainability and interpretability aids in the creation of reliable crop production prediction models that can motivate environmentally responsible and economically fruitful agricultural practices.

17.8 Conclusion

Predicting crop yields more accurately using machine learning algorithms is the focus of this study. The research has shown the potential of these methods in properly predicting agricultural yields by an in-depth analysis of numerous algorithms, including regression models, ensemble methods, deep learning models, support vector machines, and other pertinent techniques. This report emphasized the value of precise crop production forecasting for farmers, agricultural researchers, and other stakeholders, as it improves the former group's ability to plan ahead, allocate resources effectively, and make sound decisions [23]. There are obstacles and limits to conventional methods of predicting crop yields that can be

overcome by employing machine learning algorithms. The study report aimed to help readers choose effective machine learning algorithms by examining their underlying mathematics and distinctive properties. The importance of data preparation in assuring high-quality input was emphasized, as was the relevance of feature engineering and selection in extracting useful information from the crop yield dataset.

Different machine learning algorithms were explored in depth in the article. These included regression models, ensemble methods, deep learning models, support vector machines, and others. It outlined their mathematical underpinnings, benefits, and drawbacks, and emphasized their potential for enhancing crop yield forecasting. The “crop yield prediction dataset” was used to test and assess the algorithms’ efficacy. The models’ predictive skills were evaluated using assessment criteria. The effectiveness of the algorithms was evaluated thoroughly by using tables for comparative analysis.

The study’s findings and conclusions highlighted the value of using ensemble approaches, deep learning models, and other cutting-edge algorithms to boost the precision with which crop yield predictions can be made. The studies showed that these methods performed better than the conventional ones and revealed important information about the factors affecting crop

productivity. Data availability and quality, feature selection and representation, model interpretability, scaling and generalizability, integration of multi-source data, and real-time prediction were also discussed as well as directions for the future of agricultural yield prediction. There is room for growth and development in the field as a result of these difficulties.

This study adds to the literature on estimating future crop yields by investigating and refining machine learning techniques for this purpose. This research sheds light on the difficulties and potential future avenues of agricultural production prediction by stressing the significance of models' explainability and interpretability. As a result of this study, farmers, researchers, and stakeholders will be better able to make educated decisions that will lead to more sustainable and efficient crop production.

17.8.1 Importance of Machine Learning in Optimizing Crop Yield Prediction

Improved accuracy, efficiency, and scalability in crop production prediction are all possible because of machine learning's use of data-driven methodologies. You may sum up the significance of machine learning in this setting as follows:

- Massive amounts of data about crops, including weather, soil type, management methods, and harvest times, can be analyzed by machine learning algorithms. Machine learning methods can improve the accuracy of agricultural yield projections by spotting hidden insights within the data.
- Optimization of prediction models can be greatly aided by the availability of sophisticated and versatile machine learning methods. Machine learning may enhance the accuracy and reliability of agricultural production prediction models by employing strategies including feature engineering, dimensionality reduction, and hyperparameter tuning. These optimization strategies are used to zero in on the most important aspects, cut down on extraneous data, and finetune the model's parameters so that more accurate predictions may be made [24, 25].
- Environmental change must be incorporated into crop yield prediction models. Models can make updated predictions in real time or almost real time because machine learning algorithms can adapt and learn from fresh data. Because of this flexibility, farmers and decision-makers can quickly react to shifting conditions and make well-informed choices that maximize crop productivity.
- Predicting crop yield requires taking into account various variables, some of which have non-linear and interaction

impacts. Traditional statistical methods may not be able to capture and describe these complicated interactions as well as machine learning algorithms such as decision trees, random forests, neural networks, and support vector machines [23]. By taking into consideration the interdependence of the many variables that affect crop output, this skill permits more precise forecasts.

- Machine learning algorithms are adaptable and can be scaled to a wide variety of agricultural settings, making them ideal for this application. They allow for more widespread use of agricultural production forecast models by generalizing effectively across geographic locations, crop types, and farming techniques. This adaptability to different agricultural contexts is made possible by the transferability of information and insights gained in one area or region to another [26, 27].
- Farmers and policymakers alike can benefit from machine learning's data-driven insights to make better decisions. Optimized resource allocation, enhanced crop management methods, early risk assessment, and better operational planning are all made possible by accurate crop yield estimates. Increased productivity, decreased costs, and enhanced sustainability in agriculture can all be achieved with the use of machine learning.

Improved accuracy in estimating crop yields is just one area where machine learning shines. Accurate forecasts can be made in complex and ever-changing agricultural systems thanks to its ability to analyze data, recognize patterns, and optimize models. Farmers and other stakeholders can improve agricultural practices and crop yields through the use of machine learning algorithms.

17.8.2 Applications

Machine learning algorithms offer a wide range of applications in crop yield prediction and agricultural practice improvement. Among the most important machine learning applications in this subject are:

- Yield prediction: Based on past data and characteristics like weather, soil parameters, and management techniques, machine learning algorithms may effectively estimate crop yields. Farmers may better allocate resources, schedule harvests, and make management and marketing decisions based on accurate production estimates.
- Disease and pest detection: Sensor data, satellite photography, and other inputs can be analyzed by machine learning algorithms to look for symptoms of agricultural illnesses and insect infestations. Machine learning models

can aid farmers in reducing the effects of diseases and pests on agricultural productivity by seeing trends and outliers in the data and recommending appropriate measures, such as targeted pesticide treatment or crop rotation.

- Irrigation management: Optimizing irrigation schedules can be done with the use of machine learning models by analyzing weather records, soil moisture levels, and crop attributes. These models are able to offer accurate irrigation schemes, decrease water waste, and ensure ideal soil moisture levels for crop growth by taking into account variables like evapotranspiration rates and crop water requirements.
- Crop recommendation systems: Machine learning algorithms can examine information about the soil, the weather, and the crop's performance to make specific suggestions to farmers about which types to plant, how much fertilizer to use, and other management strategies. These advising systems aid farmers in making well-informed decisions that improve crop yields and longevity.
- Precision agriculture: Precision agriculture is made possible by the combination of machine learning techniques, remote sensing technologies, and drones. Satellite images and sensor data can be analyzed by machine learning algorithms to create a crop health map, pinpoint sections of the field with

nutritional deficits, and pinpoint problem spots. Farmers may use this data to better direct their interventions, waste less of their inputs, and maximize their yields [28].

- Crop quality assessment: By assessing characteristics including size, color, texture, and chemical content, machine learning models may determine a crop's overall quality. These models can be used to determine the variables that influence crop quality and then direct quality control procedures to guarantee uniformity and adherence to industry norms.
- Crop yield optimization: By integrating multiple data sources, including weather data, market prices, and crop yield data, machine learning algorithms can optimize crop yield by considering various factors and making data-driven decisions [29]. These models can assist farmers in determining optimal planting densities, fertilizer application rates, and other strategies to maximize yield while minimizing costs and environmental impact.

There is a great deal of opportunity for increased agricultural productivity, sustainability, and profitability through the use of machine learning in crop yield prediction and agricultural practices. Machine learning algorithms improve agricultural productivity by helping farmers and other stakeholders make

better decisions through better analysis of data, improved pattern detection, and more accurate predictive modeling.

References

1. Ashcraft, C., & Karra, K. (2021). Machine learning aided crop yield optimization. arXiv preprint arXiv:2111.00963.
2. Chlingaryan, A., Sukkarieh, S., & Whelan, B. (2018). Machine learning approaches for crop yield prediction and nitrogen status estimation in precision agriculture: A review. *Comput. Electron. Agric.*, *151*, 61-69.
3. Kumar, R., Singh, M. P., Kumar, P., & Singh, J. P. (2015, May). Crop Selection Method to maximize crop yield rate using machine learning technique. In *2015 international conference on smart technologies and management for computing, communication, controls, energy and materials (ICSTM),* (pp. 138-145). IEEE.
4. Hu, G., & You, F. (2022). Renewable energy-powered semi-closed greenhouse for sustainable crop production using model predictive control and machine learning for energy management. *Renew. Sustain. Energy Rev.*, *168*, 112790.
5. Ju, S., Lim, H., Ma, J. W., Kim, S., Lee, K., Zhao, S., & Heo, J. (2021). Optimal county-level crop yield prediction using MODIS-based variables and weather data: A comparative

study on machine learning models. *Agric. For. Meteorol.*, *307*, 108530.

6. Mythili, K., & Rangaraj, R. (2021). Crop recommendation for better crop yield for precision agriculture using ant colony optimization with deep learning method. *Ann. Rom. Soc. Cell Biol.*, 4783-4794.
7. Adebiyi, M. O., Ogundokun, R. O., & Abokhai, A. A. (2020). Machine learning–based predictive farmland optimization and crop monitoring system. *Scientifica*, *2020*.
8. Raman, R., Shamim, R., Akram, S. V., Thakur, L., Pillai, B. G., & Ponnusamy, R. (2023, January). Classification and Contrast of Supervised Machine Learning Algorithms. In *2023 International Conference on Artificial Intelligence and Smart Communication (AISC),* (pp. 629-633). IEEE.
9. Misra, S., Shamim, R., & Dey, R. Handling The Class Imbalance Problem From Different Datasets.
10. Paudel, D., Boogaard, H., de Wit, A., Janssen, S., Osinga, S., Pylianidis, C., & Athanasiadis, I. N. (2021). Machine learning for large-scale crop yield forecasting. *Agric. Syst.*, *187*, 103016.
11. Nosratabadi, S., Szell, K., Beszedes, B., Imre, F., Ardabili, S., & Mosavi, A. (2020, October). Comparative analysis of ANN-ICA and ANN-GWO for crop yield prediction. In *2020 RIVF International Conference on Computing and Communication Technologies (RIVF),* (pp. 1-5). IEEE.

12. Shamim, R. (2022). Machine learning's algorithm profoundly impacts predicting the share market stock's price. *IJFMR- Int. J. Multidiscip. Res.*, *4*(5).
13. Cai, Y., & Sharma, A. (2021). Swarm intelligence optimization: an exploration and application of machine learning technology. *J. Intell. Syst.*, *30*(1), 460-469.
14. Muruganantham, P., Wibowo, S., Grandhi, S., Samrat, N. H., & Islam, N. (2022). A systematic literature review on crop yield prediction with deep learning and remote sensing. *Remote Sens.*, *14*(9), 1990.
15. Nevavuori, P., Narra, N., Linna, P., & Lipping, T. (2020). Crop yield prediction using multitemporal UAV data and spatio-temporal deep learning models. *Remote Sens.*, *12*(23), 4000.
16. Paudel, D., Boogaard, H., de Wit, A., van der Velde, M., Claverie, M., Nisini, L., ... & Athanasiadis, I. N. (2022). Machine learning for regional crop yield forecasting in Europe. *Field Crops Res.*, *276*, 108377.
17. Sharafi, S., Kazemi, A., & Amiri, Z. (2023). Estimating energy consumption and GHG emissions in crop production: A machine learning approach. *J. Cleaner Prod.*, *408*, 137242.
18. Garcia, F. (1999, December). Use of reinforcement learning and simulation to optimize wheat crop technical management. In *Proceedings of the International Congress on*

Modelling and Simulation (MODSIM'99), Hamilton, New-Zealand (pp. 801-806).

19. Guo, Y., Zhao, H., Zhang, S., Wang, Y., & Chow, D. (2021). Modeling and optimization of environment in agricultural greenhouses for improving cleaner and sustainable crop production. *J. Cleaner Prod.*, *285*, 124843.
20. Liao, D., Niu, J., Lu, N., & Shen, Q. (2021). Towards crop yield estimation at a finer spatial resolution using machine learning methods over agricultural regions. *Theor. Appl. Climatol.*, 146, 1387-1401.
21. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. "Plant Disease Identification Using Convolution Neural Networks." In *Intelligent Internet of Things for Smart Healthcare Systems*, pp. 203-215. CRC Press, 2023.
22. Yousif, Jabar H., Kashif Zia, and Durgesh Srivastava. "Solutions Using Machine Learning for Diabetes." *Healthcare Solutions Using Machine Learning and Informatics* (2022): 39-59.
23. A. K. Srivastava, M. Kumar, C. Mahur, V. K. Tiwari, S. Tiwari and D. Srivastava, "Image Processing Based Intelligent Mini Robotic Face Recognition System," *2023 World Conference on Communication & Computing (WCONF)*, RAIPUR, India, 2023, pp. 1-8, doi: 10.1109/WCONF58270.2023.10235068.

24. Srivastava, Durgesh, Rajeshwar Singh, and Vikram Singh. “Mining of data through various soft computing techniques.” In *AIP Conf. Proc.*, vol. 2142, no. 1, p. 140036. AIP Publishing LLC, 2019.
25. Goyal, Mohit, and Durgesh Srivastava. “A behaviour-based authentication to internet of things using machine learning.” *Design and Development of Efficient Energy Systems* (2022): 245-263.
26. Srivastava, D.; Srivastava, S.K.; Khan, S.B.; Singh, H.R.; Maakar, S.K.; Agarwal, A.K.; Malibari, A.A.; Albalawi, E. Early Detection of Lung Nodules Using a Revolutionized Deep Learning Model. *Diagnostics*, 2023, 13, 3485. https://doi.org/10.3390/diagnostics13223485
27. Alibabaei, K., Gaspar, P. D., & Lima, T. M. (2021). Crop yield estimation using deep learning based on climate big data and irrigation scheduling. *Energies*, *14*(11), 3004.
28. Shook, J., Gangopadhyay, T., Wu, L., Ganapathysubramanian, B., Sarkar, S., & Singh, A. K. (2021). Crop yield prediction integrating genotype and weather variables using deep learning. *PLoS One*, *16*(6), e0252402.
29. Tao, R., Zhao, P., Wu, J., Martin, N. F., Harrison, M. T., Ferreira, C., ... & Hovakimyan, N. (2022). Optimizing Crop Management with Reinforcement Learning and Imitation Learning. arXiv preprint arXiv:2209.09991.

Note

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18

Analyzing Smart Farming Technologies: A Study on Indian Farmers' Adoption Trends

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Abstract

Smart agriculture, driven by IoT sensors, has the potential to transform the farming industry. By collecting and utilizing field data based on user input, IoT systems can greatly improve farming practices. Agriculture is crucial for India's economic growth, and technological advancements promise increased efficiency in farming tasks. The IoT, a network of interconnected devices, forms the foundation of smart farming, making agricultural production more effective and cost-efficient. However, successful implementation faces challenges. A dependable network infrastructure is essential for constant monitoring. Significant investment and research are required in areas like energy harvesting, automated irrigation, and disease prediction. Farmer awareness is a hurdle to adopting

smart agricultural practices. Moreover, the funding gap is a challenge, as precision farming demands substantial investments. Despite these challenges, smart farming offers substantial benefits, including cost reduction, improved product quality, higher productivity, and enhanced sustainability. This makes it an attractive area for investment and growth. This paper aims to explore the potential of smart agri tech by gathering insights from users regarding adoption and challenges in transitioning to this innovative farming technique. The authors conducted extensive research and quantitative analysis, addressing the technology adoption gap within the Indian context and offering valuable insights into the successful execution of this technology.

Keywords**:** Smart agriculture, Internet of Things (IoT), adoption of intelligent farming, AI in agriculture and farming

18.1 Introduction

Smart farming, also known as precision agriculture, is a modern and innovative approach that leverages advanced technologies, data analytics, and connectivity to optimize and revolutionize traditional agricultural practices. The adoption of smart farming has gained considerable attention in recent years as a promising solution to address the multifaceted

challenges faced by the agricultural industry. With the global population on the rise and finite resources becoming scarcer, there is an increasing demand to produce more food sustainably and efficiently. Additionally, growing environmental concerns, such as climate change and habitat destruction, call for more resilient and eco-friendly farming methods.

One of the key components that sets smart farming apart is the integration of Internet of Things (IoT) technologies. IoT-based farming involves the use of interconnected devices, sensors, and data analytics platforms that collect and process vast amounts of data in real time. This data-driven approach empowers farmers with valuable insights into their operations, enabling them to make data-informed decisions. By continuously monitoring soil conditions, weather patterns, crop growth, and livestock health, farmers can detect early signs of potential issues, such as nutrient deficiencies or pest infestations, and take timely corrective actions.

The benefits of smart farming extend beyond just increasing productivity and optimizing resource usage. By adopting precision agriculture practices, farmers can also play a significant role in mitigating environmental impacts—for instance, by applying fertilizers and pesticides precisely where

and when needed, rather than through blanket applications, smart farming minimizes unnecessary chemical usage, reducing the risk of soil and water pollution. Moreover, optimizing irrigation schedules based on real-time data can lead to water conservation, which is vital in regions facing water scarcity.

The integration of data analytics and smart devices also opens up new opportunities for farm automation. Automated machinery and robotic systems can perform tasks with precision and efficiency, leading to reduced labor requirements and potentially alleviating the burden of farm labor shortages. Additionally, the accumulation of historical data over time can facilitate predictive analytics, allowing farmers to make proactive decisions for future planning and risk management.

However, despite the promising outlook of smart farming, widespread adoption is not without challenges. One of the primary concerns is the initial investment cost, which can be significant for small and resource-constrained farmers. Addressing this challenge requires efforts to promote affordable technology solutions and explore financing options, including public–private partnerships and agricultural grants.

Another obstacle lies in the digital divide and limited technical knowledge in rural areas, hindering the seamless implementation of IoT-based farming practices. To bridge this gap, targeted training programs and capacity-building initiatives must be designed to provide farmers with the necessary skills and knowledge to leverage smart farming technologies effectively [12].

Data privacy and security concerns are also critical aspects that demand attention. As smart farming relies on a vast amount of sensitive data, it is essential to establish robust cyber-security measures to protect against potential breaches and data theft. Building trust among farmers regarding data sharing and demonstrating the value of data-driven insights will be crucial for encouraging participation.

Furthermore, the diversity of farming practices across different regions and crops necessitates customized solutions tailored to specific agricultural contexts. This calls for collaborative research and development efforts involving agricultural experts, technology providers, and local stakeholders to develop context-specific smart farming solutions.

In conclusion, smart farming represents a transformative approach to modernizing agriculture by harnessing the power

of advanced technologies, data analytics, and connectivity. Its potential to enhance productivity, optimize resource usage, and contribute to sustainable practices makes it a promising solution to meet the challenges of feeding a growing global population while safeguarding the environment [13]. However, overcoming the barriers to adoption, such as high costs, technical knowledge gaps, data privacy concerns, and contextual nuances, will require concerted efforts from various stakeholders in the agricultural sector. By fostering awareness, providing adequate support and training, and developing tailor-made solutions, the agricultural community can harness the full potential of smart farming and lead the way toward a more resilient and sustainable future for farming.

18.2 Research Methodology

The research employs a descriptive methodology with the main objective of investigating the adoption of smart farming practices in India. A survey will be conducted among 300 farmers who have utilized IoT technology to implement smart farming procedures. The sampling methodology comprises a combination of stratified random sampling and purposive sampling, targeting farmers in India who have already adopted smart agricultural practices or are interested in transitioning to such techniques. To determine the appropriate sample size, the

study will employ a sample size calculation formula. The primary data will be gathered using a well-structured questionnaire designed to address the aforementioned research objectives. The questionnaire will capture information regarding the farmers' adoption of smart farming techniques, the benefits of employing IoT technology, challenges faced, and their training needs. Both descriptive and inferential statistics will be applied to analyze the collected data based on the specific requirements. Visual aids such as tables, charts, and graphs will be utilized to present the data effectively. Data analysis will be performed using SPSS and MS Excel. Ethical considerations are paramount, and the project will obtain informed consent from the participants while ensuring their anonymity. The data collected will only be used for research purposes. However, the study acknowledges certain limitations, including the potential impact of external factors like weather and market volatility on the results. In conclusion, this research method seeks to comprehensively examine the adoption of smart farming practices in India, leveraging the capabilities of IoT technology [14].

18.3 Objectives

1. To understand how constraints in the process of adoption influence or restrict the implementation of smart agriculture

2. To understand the perceptions and resultant actions that facilitate cases of lucrative adoption processes among farmers
3. To evaluate if external funding and technological implementation issues serve as a major hindrance in terms of the successful execution of smart agriculture

18.4 Literature Review

The field of agriculture is undergoing a profound transformation with the integration of the Internet of Things (IoT), incorporating precision and conservative cultivation to tackle prevailing challenges. Several technological solutions have been proposed for the agricultural sector, such as the Bluetooth-based irrigation system detailed in [1]. However, Bluetooth technology presents limitations, including restricted range and device compatibility. To optimize energy efficiency, researchers have recommended implementing power supply scheduling for the sensors, as highlighted in a 2016 article [2].

Despite these advancements, IoT implementation in agriculture faces challenges related to interoperability, particularly concerning large agricultural landscapes [3]. Studies have introduced methods for comparing energy consumption between appliances [4] and applied ubiquitous computing and

data aggregation techniques to monitor environmental conditions [5]. However, concerns arise with the expansion of nodes, leading to increased power consumption and potential impacts on agricultural automation [6]. Furthermore, some research operates as stand-alone systems, providing real-time information on land and crops [7].

A 2015 paper on smart agriculture integrated IoT, cloud computing, and mobile computing concepts [8]. Additionally, the concept of a phononet, a network of intelligent wireless sensor nodes communicating with a central system and each other, was introduced [9]. An exhaustive investigation of smart agriculture technologies and their potential advantages for the Indian agricultural industry is presented in [15]. This research assesses the economic, societal, and environmental benefits of smart agricultural practices, along with the challenges and facilitators of adoption.

Moreover, the adoption of GPS-based tools, sensors, and data analytics for precision agriculture in India is currently under examination [16]. The significance of information and communication technology (ICT) in smart agriculture and its effects on Indian farmers are the primary focus of [17]. The paper delves into the challenges and possibilities of implementing ICT-based agricultural practices and suggests

legislative actions to promote their widespread use. Furthermore, [18] explores how intelligent agriculture can improve India's access to food and nutrition, emphasizing the integration of technology, eco-friendly practices, and farmer empowerment to achieve food security goals.

The paper [19] aims to investigate how IoT makes India's smart agricultural practices possible, highlighting the potential applications in agricultural monitoring, irrigation control, and animal tracking. Additionally, [20] studies how smart farming practices impact India's agricultural output, particularly in terms of remote sensing, precision agriculture, and resource efficiency [14].

The literature on smart agriculture adoption in the Indian setting underscores the potential of cutting-edge technology to revolutionize the agricultural industry. Policymakers, academics, and stakeholders can leverage these findings to promote effective and sustainable farming practices for economic and food security objectives.

Furthermore, [21] presents a comprehensive analysis of drone technology in India's smart agriculture, discussing its applications in crop monitoring, pest detection, and precise spraying for increased agricultural output and resource

management. The role of artificial intelligence (AI) in precision agriculture and its potential implications for Indian agricultural practices are examined in [22], encompassing AI-based approaches to agricultural automation, disease detection, and crop production prediction.

The paper [23] delves into the significance of intelligent water management in Indian agriculture, exploring how IoT-based sensors and data analytics can reduce water stress, enhance water production, and optimize water use. An in-depth analysis of IoT applications in smart agriculture in India is provided in [24], covering the challenges and potential of IoT-enabled precision farming, livestock monitoring, and soil health management.

Additionally, the use and effects of mobile apps on Indian agriculture are explored in [25], examining the farmers' patterns of app usage, advantages, and challenges during farm management, marketing, and decision-making. The adoption of precision agricultural technology in India is influenced by various factors, as studied in [26], including government policies, education, farm size, information availability, and the adoption of smart agricultural practices.

Beyond the above-mention items, the research delves into other essential elements of smart agriculture adoption in the Indian context, contributing to the growing body of knowledge on how cutting-edge technology can transform Indian agriculture, boost production, and enhance resource management for efficient and sustainable agricultural practices.

Additionally, [27] investigates how India's smart agricultural practices encompass data analytics and machine learning methods, offering improvements in crop prediction, pest control, and decision-making. [28] centers on the utility of solar-powered intelligent irrigation systems in Indian agriculture, describing how these systems promote sustainable agriculture by maximizing water usage and improving irrigation effectiveness.

[29] explores the use of smart sensing technologies in Indian agriculture, utilizing sensors to monitor environmental factors, crop health, and soil conditions to make informed decisions. Meanwhile, [30] examines how Indian farmers adopt and adapt mobile applications for agriculture, assessing the impact on agricultural practices and the reasons influencing their app selection. The potential for blockchain technology in India's smart agricultural applications is covered in [31], discussing how blockchain can enhance the trustworthiness, transparency,

and traceability of food and agriculture supply networks. Moreover, [32] evaluates the advantages and disadvantages of employing big data analytics in smart agriculture, considering how big data may improve resource allocation, decision-making, and agricultural sustainability in India.

The study [33] investigates how innovative agricultural methods support sustainable farming in India, particularly in terms of technology-driven practices that minimize negative environmental impacts and ensure longterm security.

Overall, the collective research indicates that smart agriculture is the future, offering sustainable and efficient solutions for agriculture. The adoption of new technologies offers significant potential for increased production and efficiency, addressing challenges related to land acquisition [10], farmer livelihoods, and socioeconomic ramifications. Notably, water management, disease identification, and data analytics play crucial roles in smart agriculture in the Indian context. Policymakers and stakeholders can utilize these studies to develop effective and sustainable farming practices, contributing to the nation's economic growth and food security goals [11].

18.5 Data Analysis

To ensure the analysis' successful execution, the following hypothesis can be tested:

H0: There exists no statistically significant association between the successful adoption of IoT-based farming and technological implementation and external funding.

H1: There exists a statistically significant association between the successful adoption of IoT-based farming and technological implementation and external funding.

Based on the data collected and inferences derived from the conducted survey, we focused on the key parameters, namely, "technological implementation" and "external funding" [11]. The results led us to accept the alternative hypothesis, indicating a significant association. These outcomes are presented in the table below, obtained through the application of the ANOVA test for conducting the significance test. The results of the oneway ANOVA revealed that the *p*-value is lower than the 5% level of significance, leading us to reject H0 and accept H1 (refer to Tables 18.1 and 18.2).

As the analysis yielded significant results, it establishes a compelling basis for the conclusion that technological

implementation and external funding indeed play a significant role in the successful adoption of IoT-based farming practices. The evidence gathered from this study highlights the importance of these factors in the agricultural domain, encouraging further exploration and support for the integration of advanced technologies like IoT in farming practices to enhance productivity and sustainability.

Table 18.1 Analysis of variance between the variables.

ANOVA Successful Adoption					
			Sum of squares	**df**	**Mean squar**
Between groups	(Combined)		49.553	4	12.388
	Linear term	Unweighted	43.203	1	43.203
		Weighted	34.523	1	34.523
		Deviation	15.03	3	5.01
Within groups			226.822	35	6.481
Total			276.375	39	

The SPSS analysis yields compelling evidence supporting a significant association between technological implementation

and external funding in the successful adoption of IoT-based farming methodologies. Rigorous statistical tests underscore the pivotal role of these factors in effectively implementing IoT technologies in agriculture [12].

The critical indicator is the *p*-value, which is lower than the predetermined significance level (5%). By rejecting the null hypothesis (H0) stating no significant association, we affirm the alternative hypothesis (H1), indicating a meaningful relationship between the variables.

These findings have significant implications for the agricultural industry, emphasizing the importance of robust technological infrastructure and sufficient financial support for embracing IoT-based farming methodologies. Successful IoT adoption can lead to transformative outcomes, optimizing resource usage, enhancing crop yields, and promoting sustainable agricultural practices.

Moreover, policymakers, researchers, and stakeholders gain valuable insights from the results. Policymakers can prioritize investments and incentives to facilitate IoT technology integration, fostering innovation in farming. Researchers can build upon these findings to explore further applications and refine IoT-based farming methodologies.

Table 18.2 Tests of between-subjects effects.

Source	Dependent variable	Type III sum of squares	df	Mean square	*F*
Corrected model	External Funding	10.325[a]	8	1.291	1.1
	Technological issue	6.442[b]	8	0.805	0.7
Intercept	External Funding	506.382	1	506.382	46
	Technological issue	578.155	1	578.155	56
Adoption	External Funding	10.325	8	1.291	1.1
	Technological issue	6.442	8	0.805	0.7

Source	Dependent variable	Type III sum of squares	df	Mean square	*F*
Error	External Funding	33.45	31	1.079	
	Technological issue	31.933	31	1.03	
Total	External Funding	555	40		
	Technological issue	639	40		
Corrected total	External Funding	43.775	39		
	Technological issue	38.375	39		

[a]R^2 = .236 (adjusted R^2 = .039).

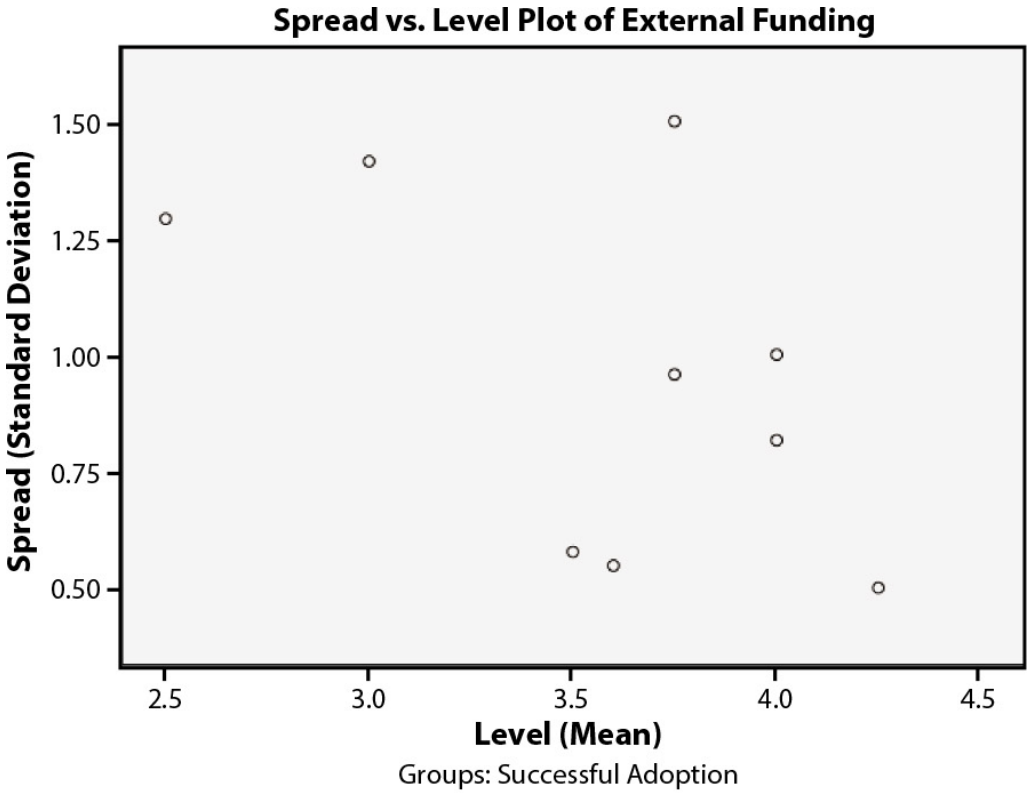
[b]R^2 = .168 (adjusted R^2 = -.047).

In conclusion, the SPSS-based analysis validates the significance of “technological implementation” and “external funding” in the successful adoption of IoT-based farming methodologies. Embracing these factors can drive advancements in agriculture, increasing productivity, resource efficiency, and overall sustainability. This research contributes to the growing knowledge on smart farming practices, offering valuable guidance for the future development and implementation of IoT technologies in agriculture.

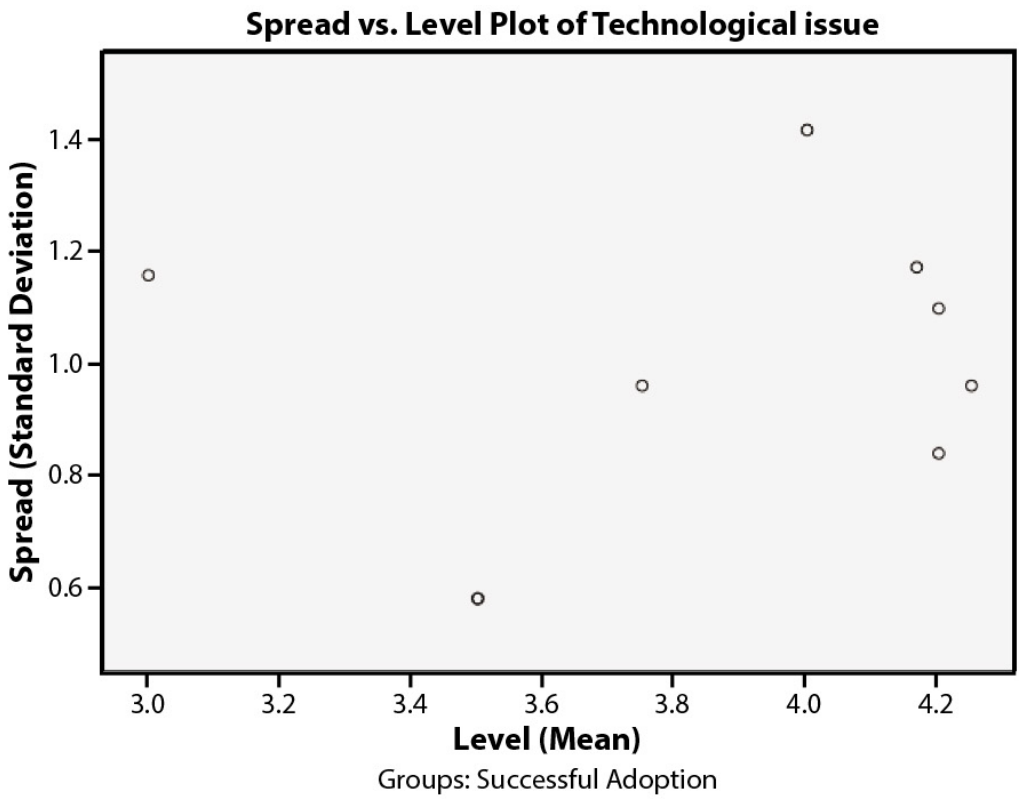
18.6 Key Findings

Based on the comprehensive analysis encompassing variables such as technological implementation, climatic conditions, internet issues, operational cost, and external funding, two factors emerge as the most influential in the adoption of IoT-based farming. These are technological implementation and external funding. Interestingly, climatic conditions and internet issues were found to have a minimal impact on the adoption process, contrary to initial assumptions. However, the lack of awareness remains a significant hindrance. In particular, external funding exerts a larger influence on the adoption process compared to technological issues. This suggests that

securing adequate financial resources plays a pivotal role in successfully integrating IoT-based farming practices. While the benefits of smart farming are promising, several challenges impede its widespread adoption. The literature points out key barriers, such as high upfront costs, limited access to technology and connectivity in rural areas, lack of awareness and technical knowledge among farmers, data security and privacy concerns, and the need for supportive policies and infrastructure [13]. Understanding these challenges is critical for formulating effective strategies to overcome them and promote broader adoption of smart farming practices. Policymakers, stakeholders, and industry players must collaboratively address these obstacles to unlock the full potential of IoT-based farming, which can lead to improved efficiency, resource management, and sustainability in agriculture. In summary, prioritizing technological implementation and securing external funding are vital for successful IoT-based farming adoption (as per Figure 18.1). While climatic conditions and Internet issues have limited impact, addressing challenges like awareness, funding, and infrastructural support is crucial for widespread adoption. By proactively addressing these barriers, the agricultural sector can harness the transformative benefits of smart farming and drive positive agricultural outcomes.


Spread vs. Level Plot of External Funding
Spread (Standard Deviation)
1.50
1.25
1.00
0.75
0.50
2.5
3.0
3.5
4.0
4.5
Level (Mean)
Groups: Successful Adoption

(**a**)


Spread vs. Level Plot of Technological issue
Spread (Standard Deviation)
1.4
1.2
1.0
0.8
0.6
3.0
3.2
3.4
3.6
3.8
4.0
4.2
Level (Mean)
Groups: Successful Adoption

(**b**)

Figure 18.1 Means plots for (a) external funding and (b) technological issues against adoption. (Source: Primary data Analysis by using SPSS.)

- Given that every aspect, including technology implementation, climatic circumstances, Internet problems, operational costs, and external finance, has an impact on the adoption of IoT-based farming, only the most recent variables are relevant.
- Contrary to original assumptions, it was discovered that Internet problems and weather circumstances had very little to no impact on the adoption process. Even yet, ignorance may be a significant roadblock.
- The adoption process is more significantly impacted by technology problems than by external finance.
- Despite the potential advantages of smart farming, a number of obstacles prevent its broad implementation. Literature points to impediments such as high upfront expenditures, poor connection and access to technology in rural regions, farmers' lack of technical awareness and experience, data security and privacy issues, and the need for supporting policies and infrastructure. It is essential to comprehend these issues in order to develop solutions that will encourage wider adoption.

18.7 Discussions and Recommendations

1. Accelerating adoption: External funding significantly contributes to the increased adoption of IoT-based farming practices. Financial assistance, such as grants, subsidies, or investments, provides farmers with vital resources to invest in IoT technologies and related infrastructure. Research has shown that farmers receiving external funding are more likely to adopt IoT solutions compared to those lacking financial support.
2. Mitigating financial barriers: The substantial initial costs associated with implementing IoT technologies in farming can hinder adoption. External funding serves as a solution by alleviating the financial burden on farmers. It empowers them to invest in IoT devices, sensors, connectivity infrastructure, and data analytics platforms, making IoT-based farming more accessible and economically feasible.
3. Enabling infrastructure development: External funding often fosters the development of critical infrastructure, such as high-speed Internet connectivity and wireless networks in rural areas. These infrastructure enhancements are pivotal for the successful implementation of IoT-based farming, as they facilitate data transmission, real-time monitoring, and

device communication. Improved infrastructure further encourages farmers to embrace IoT technologies.

4. Driving research and development: External funding frequently supports research and development initiatives centered around IoT-based farming. Such financial support allows for the exploration of innovative solutions, improvements to existing technologies, and the creation of IoT applications tailored to specific farming needs. These research endeavors promote knowledge dissemination and awareness, motivating farmers to adopt IoT technologies.
5. Empowering training and education: Allocation of external funding to training and education programs aims to enhance farmers' comprehension and skills in IoT-based farming practices. These programs equip farmers with the necessary knowledge to effectively utilize IoT technologies and interpret collected data. Well-trained farmers are more likely to adopt and reap the benefits of IoT solutions, leading to heightened agricultural productivity and sustainability.
6. Scaling-up adoption: External funding often backs pilot projects or demonstrations of IoT-based farming on a small scale. Favorable outcomes and positive experiences from these endeavors can attract additional funding and encourage the scaling-up of adoption. This allows farmers to witness firsthand the advantages, leading to increased

confidence in adopting IoT technologies over larger areas or extensive farming operations.

7. Fostering collaboration and networking: External funding initiatives often foster collaboration among various stakeholders, including farmers, technology providers, research institutions, and policymakers. This collaboration facilitates knowledge exchange, sharing of best practices, and experiences. It creates networks that support IoT-based farming adoption, providing farmers access to information, technical assistance, and guidance crucial for successful implementation.
8. Enhanced decision-making: IoT-based farming generates a vast amount of data that aids farmers in making informed decisions regarding irrigation, fertilization, pest control, and resource management. External funding supports the development of data analytics platforms and decision support systems, enabling farmers to effectively interpret and utilize data. This leads to improved decision-making processes, optimizing agricultural practices and boosting productivity.
9. Cost–benefit analysis: A comprehensive assessment of the expenses related to the implementation of intelligent agricultural technology, encompassing initial investments in hardware, software, and training, along with ongoing

maintenance costs. The study also explores potential benefits, such as increased agricultural yields, reduced input expenditures, and enhanced resource management.

Cost analysis:

1. Hardware and technology costs: This category entails the procurement of sensors, drones, GPS devices, and other advanced agricultural tools.
2. Software and data management costs: Encompassing licensing fees, platform subscriptions, and data analytics services.
3. Training and capacity-building costs: Enabling farmer education and skill enhancement.

Benefit evaluation:

1. Enhanced crop yields: Investigating the impact of smart agricultural technologies on crop productivity and income generation.
2. Resource efficiency: Analyzing potential water and resource savings through precision irrigation and efficient resource utilization.
3. Cost savings: Examining how reduced water, fertilizer, and pesticide usage can bolster profitability.

These findings highlight the economic potential of embracing smart agriculture within the Indian agricultural sector, summarizing the outcomes of the cost–benefit analysis. It emphasizes the advantages of improved sustainability, resource management, and higher production. Stakeholders, including farmers and policymakers, can utilize this information to make informed decisions regarding the adoption of smart agricultural technology, paving the way for a more profitable and sustainable future in Indian agriculture.

18.8 Conclusion

While external funding, operational cost support, and technological advancements may encourage IoT-based farming adoption, it is vital to acknowledge the challenges and complexities arising from these influences (Figure 18.1). Promoting a sustainable and inclusive adoption of IoT-based farming practices requires addressing financial barriers, ensuring affordability and accessibility, streamlining technology, prioritizing data privacy and security, fostering customization, and adopting a holistic approach. Thus, establishing awareness forums to drive implementation rates and implementing financial programs to expedite the adoption process is a prudent recommendation.

References

1. M Gayatri, M.K., Jayasakthi, J. and Anandha Mala, G.S. (2015) 'Providing smart agricultural solutions to farmers for better yielding using IOT', *2015 IEEE Technological Innovation in ICT for Agriculture and Rural Development (TIAR).*
2. Gondchawar, N. (2016) IOT based Smart Agriculture, Academia.edu.
3. Rasooli, M.W., Bhushan, B. and Kumar, N. (2020) Applicability of Wireless Sensor Networks & IOT in Saffron & Wheat Crops: A Smart Agriculture Perspective, *Int. J. Sci. Technol. Res.*
4. Dr.N.Suma, S.R.S. (2017) IOT based Smart Agriculture Monitoring System, *Int. J. Recent Innov. Trends Comput. Commun*.
5. Paparao N., Kumar H., Ramesh P. and Godavarthi B. (2017) *J. Eng. Appl. Sci.*, Design and Implementation of Modern Automated Real Time Monitoring System for Agriculture using Internet of Things (IoT).
6. Ogunti E. (2019) IOT based smart crop-field monitoring and automation irrigation system, *Int. Organ. Innov. Sci. Eng. Technol.*
7. Paparao N. , Rao P. S.,Sangeetha Y., Navya K.C.O.M.A., and Balaji O. (2018) Design of a smart mobile case framework

based on the internet of things, *First international conference on Artificial intelligence and cognitive computing*.

8. Jennifer Yick, Biswanath, Ghosal et al. (4008) Jennifer Yick, Biswanath, Ghosal et al. (4008) Wireless Sensor Network Survey, *Comput. Networks*, ELSEVIER.
9. Choi, K.N. (2014) Rain sensor with stacked light wave-guide having tilted air gap, *J. Sens.*
10. Patra, S. Paira S.C.O.M.A., and Samanta A. et al. (2021) Automation in agriculture, *Int. J. Eng. Res. Technol.*
11. More, M., Kate, N. and Gurav, P. (2019) Corporate farming in India: Opportunities and challenges, Academia.edu.JMRA Publications.
12. Khanh, Q.V., Lanza, S., Nguyen V.H., Ngyuyen M.Q., Dang V.A., Ngyuyen T.B., Anselme M. and Randazzo, G. (2022) IOT-enabled Smart Agriculture: Architecture, applications, and challenges, Special issue of *Sustainable Agriculture and Advances of Remote sensing.*
13. Sharma, A. and Kumar, R. (2021) IOT-based framework for smart agriculture, *Int. J. Agric. Environ. Inf. Syst.*
14. Rawal, S. (2017) IOT based Smart Irrigation System. : *Int. J. Comput. Appl.*, (Volume 159).
15. Patil, S., & Sahoo, P. K. (2019). Smart agriculture in India: A comprehensive review. *J. Cleaner Prod.*, 224, 259-270.

16. Singh, A., & Mehta, R. (2019). Assessing the adoption of precision agriculture technologies in India. *Agric. Econ. Res. Rev.*, 32(Conference), 183-190.
17. Kumar, A., & Bantilan, M. C. S. (2020). Role of information and communication technology in agriculture: Prospects and policy implications for Indian farmers. *Agric. Econ. Res. Rev.*, 33(Conference), 245-257.
18. Swaminathan, M. S., & Kesavan, P. C. (2017). Smart agriculture for food and nutrition security. *Curr. Sci.*, 113(4), 663-666.
19. Jain, A., & Misra, K. B. (2018). Smart agriculture using Internet of Things. *2018 3rd International Conference on Internet of Things: Smart Innovation and Usages (IoT-SIU)*, 388-393.
20. Singh, D., & Verma, N. K. (2019). Enhancing agricultural productivity through smart farming in India: A review. *Int. J. Curr. Microbiol. Appl. Sci.*, 8(01), 2392-2405.
21. Singh, R., & Tyagi, V. (2018). Smart agriculture using drone technology: A review. *Int. J. Eng. Comput. Sci.*, 7(7), 22972-22978.
22. Kumar, S., Kumar, N., & Singh, S. (2020). Role of artificial intelligence in precision agriculture: An Indian perspective. *Indian J. Agric. Sci.*, 90(12), 1825-1834.

23. Saini, R. P., Jat, M. L., & Singh, R. (2019). Smart water management in agriculture: An Indian perspective. *Agric. Water Manage.*, 213, 990-1002.
24. Singh, R., & Gautam, S. (2020). Internet of Things in smart agriculture: A comprehensive review. *Int. J. Comput. Appl.*, 174(23), 19-25.
25. Mehta, R., & Shrivastava, A. (2019). Adoption of mobile apps in agriculture: An Indian perspective. *J. Agribus. Dev. Emerg. Econ.*, 9(5), 586-602.
26. Sreeja, K. G., & Mohankumar, B. (2019). Adoption of precision farming technologies in India: An empirical analysis. *Agric. Econ. Res. Rev.*, 32 (Conference), 191-400.
27. Roy, S., Patil, S., & Kumar, V. (2019). Smart agriculture using data analytics and machine learning in India. *Int. J. Eng. Adv. Technol.*, 9(1), 450-454.
28. Verma, A., & Choudhary, A. (2018). Solar-powered smart irrigation system: An Indian perspective. *2018 International Conference on Advances in Computing, Communication Control and Networking (ICACCCN)*, 215-219.
29. Anand, A., & Bhattacharyya, T. K. (2019). Smart sensing in agriculture: An Indian scenario. *2019 International Conference on Robotics, Automation, Control and Embedded Systems (RACE)*, 1-5.

30. Gupta, P., Sharma, S. K., & Dabas, M. (2020). Adoption of agricultural mobile applications by Indian farmers. *J. Agribus. Dev. Emerg. Econ.*, 10(5), 611-630.
31. Meena, M., & Swarnkar, V. (2021). Blockchain for smart agriculture in India. *2021 5th International Conference on Advanced Technologies for Signal and Image Processing (ATSIP)*, 91-95.
32. Singh, D., & Nagarajan, A. (2018). Smart agriculture using big data analytics: Opportunities and challenges. *Int. J. Comput. Sci. Mobile Comput.*, 7(6), 82-91.
33. Garg, S., & Aggarwal, P. (2019). Smart agriculture: A step towards sustainable agriculture. *Int. J. Agric. Sci.*, 11(21), 3461-3466.

Note

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19

Leveraging Linear Regression Model to Address Food Insecurity in the United States: A Smart Agritech Approach

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Abstract

Food insecurity is a widespread problem in the United States, impacting millions of households. It refers to the lack of access to sufficient and nutritious food, resulting in adverse health outcomes. The causes are multifaceted, including poverty, unemployment, low wages, and inadequate social safety net programs. Food insecurity leads to malnutrition, obesity, chronic diseases, and mental health issues, particularly affecting children and minorities. This chapter examines the causes, consequences, and potential solutions for food insecurity. It explores its prevalence among different demographic groups and highlights its links to poverty and

unemployment. The consequences range from malnutrition to academic underachievement. Policy interventions such as SNAP, School Meal Programs, and WIC are analyzed. Datasets from various sources are used, and a comprehensive analysis is conducted, including a linear regression model. The findings can guide resource allocation. Correlations between food insecurity, poverty, health, and nutrition are established. The chapter concludes with recommendations for future research and policy actions to promote food security and equity for all Americans.

***Keywords*:** Food insecurity, United States, causes, consequences, poverty, social safety net programs, datasets, policy interventions

19.1 Introduction

Food insecurity remains a persuasive and persistent problem in our society, impacting millions of individuals and families across the globe. Food insecurity is more than just a lack of access to food. It is a complex problem that affects physical, mental, and social well-being and requires urgent attention from policymakers, researchers, and the wider community. Although the United States is a well-developed country, food insecurity continues to be a significant and persistent problem

impacting millions of individuals and families [1]. The root factors contributing to food insecurity are multifaceted, with poverty, unemployment, and restricted availability of nutritious food choices among the leading contributing factors. However, there is a growing understanding of the potential for data-driven solutions to tackle this complex issue. One such solution is using linear regression models, which can help identify the key factors that contribute to food insecurity and provide insights into the most effective food security interventions. In this chapter, we will explore the applications of linear regression models before describing the data sources and model specifications used in our analysis [2]. We will then present the results of our model and discuss the implications for policy and future research. By using the data-driven approach, we hope to provide new insights into the complex issue of food insecurity in the United States and help to inform more effective and evidence-based policy interventions [3].

19.2 Literature Review

The problem of food insecurity is intricate and multifaceted, with a variety of factors contributing to its persistence. These factors may include poverty, unemployment, and lack of access to healthy food options. However, there is increasing recognition of the potential of data-driven solutions to address

this pressing issue. In recent years, researchers and policymakers alike have turned their attention to the problem of food insecurity and the factors that contribute to the problem. A study conducted by Gundersen and Zilak *et al.* (2018) intended to find the correlation between food insecurity and health outcomes [4]. The authors used a systematic approach to identify and evaluate relevant studies, including a search of multiple databases, a review of bibliographies, and consultation with experts in the field. They review several studies that have linked food insecurity to adverse health outcomes, including poor mental health, chronic diseases, and poor child development outcomes—for instance, they associated food insecurity with depression, anxiety, and various other mental health problems. It has also been linked to an increased vulnerability to chronic diseases such as diabetes, hypertension, and heart disease. Moreover, food insecurity can negatively impact child development outcomes, including cognitive development, academic achievement, and social–emotional well-being. Gundersen and Zilak applied rigorous inclusion and exclusion criteria to ensure that only studies that met quality standards were included in their analysis. Specifically, studies had to use a valid and reliable measure of food insecurity, including a comparison group, and control for potential confounding factors. Their study suggests that food

insecurity might result in chronic stress and inflammation, which can contribute to the emergence of chronic diseases.

The authors also discuss potential mechanisms through which food insecurity affects health outcomes. These include inadequate nutrient intake, changes in diet quality, and disruptions to meal patterns. Moreover, food insecurity can also lead to social isolation, which can negatively impact mental health. Overall, the authors' systematic review approach allowed them to evaluate a large body of literature on food insecurity and health outcomes and to identify consistent patterns across studies. In conclusion, the authors suggest that policymakers should focus on implementing policies and programs that tackle the underlying causes of food insecurity, including poverty, unemployment, and inadequate social safety nets. They argue that addressing food insecurity is not only a moral imperative but also a critical public health issue.

Previous research has utilized linear regression models to investigate the factors underlying food insecurity—for example, Seligman, Laraia, and Kushel *et al.* (2010) conducted a study examining the correlation between food insecurity and chronic disease amongs low-income individuals in the United States [5]. The sample of this study included 6,401 participants aged 18 years and older who completed the household food security

questionnaire. To measure food insecurity, the study used a questionnaire consisting of 10 items that assessed a household's ability to obtain an adequate food supply for all members to maintain an active and healthy lifestyle. The responses were then classified into four levels of food security— high, marginal, low, and very low. The study used multivariable logistic regression to analyze the association between food insecurity and chronic diseases, controlling for potential confounding variables including age, gender, ethnicity, and socio-economic status.

Based on the study's findings, the authors recommend that policymakers and public health officials prioritize efforts to mitigate food insecurity among low-income households, with a focus on providing access to healthy and affordable food options. Additionally, the authors suggest that interventions aimed at reducing food insecurity could have significant public health benefits by decreasing the prevalence of chronic disease in the population, particularly among younger adults. Overall, the study highlights the importance of addressing food insecurity as a public health issue and provides valuable insights into the potential health impacts of this pervasive problem.

Moreover, a study by Hamelin *et al.* (2009) used a mixed-methods approach, including a quantitative survey of 1,384 households and in-depth interviews with 60 individuals who experienced food insecurity [6]. The study aimed to characterize household food insecurity in the province of Quebec, Canada, and explore the emotional and social experiences of individuals who are food insecure. The authors also collected data on household demographics, income, and employment status.

The study found that 12.4% of households in Quebec experienced food insecurity. Of those households, 7.9% were classified as marginally food insecure, meaning that they had concerns about accessing enough food, and 4.5% were classified as severely food insecure, meaning that they had experienced actual reductions in food intake due to lack of resources. Furthermore, there were also households that were food-secure households. The authors also found that food insecurity was more prevalent among certain groups, such as single-parent families, low-income households, and indigenous households. In the qualitative interviews, the authors found that food insecurity was associated with feelings of shame, guilt, and stigma. The participants described the experience of food insecurity as isolating, and many felt that they could not access social support due to the shame associated with not being able

to provide enough food for their families. In conclusion, the study provides valuable insights into the prevalence and emotional impact of food insecurity in Quebec. The findings highlight the necessity for policies and programs that tackle the underlying causes of food insecurity, including poverty and income inequality, and provide support to individuals and households experiencing food insecurity. The authors suggest addressing the emotional and social impact of food insecurity. Overall, the study underscores the importance of a comprehensive approach to addressing food insecurity, one that considers both the material and psychological aspects of food access.

Loopstar, Reeves, and Taylor-Robinson, in their study of austerity, sanctions, and the rise of food banks in the UK, find that austerity policies, which involved cuts to public spending and welfare benefits, have led to a rise in poverty and food insecurity in the UK [7]. Benefit sanctions, which are penalties imposed on individuals who do not comply with certain conditions attached to their welfare benefits, have further exacerbated the situation. The authors used a combination of quantitative and qualitative methods. The authors analyzed existing data from a range of sources, including government statistics on poverty and food bank usage, to provide a broader context for their findings. The study highlights the role of the

voluntary sector, specifically food banks, in responding to the growing need for food assistance. The authors argue that although food banks have played an important role in providing emergency food aid, they should not be seen as a substitute for the state's responsibility to ensure that all citizens have access to adequate food. They also conducted qualitative interviews with individuals who used food banks, as well as representatives from food banks and other relevant organizations, to gather more detailed information about the experiences of those affected by austerity policies and benefit sanctions. The study emphasizes the necessity of a comprehensive approach to addressing food insecurity that tackles the underlying causes of poverty and provides adequate support to those in need.

The correlation between food insecurity and social capital in rural and urban communities was explored by Sharkey and Johnson in 2010 [8]. The study used data from the Texas Food Policy Study, which surveyed 1,357 households in both urban and rural areas of Texas. The sample was stratified by region and by race/ethnicity to ensure a representative sample of the population. The survey questions were designed to measure food security status, social capital, and other relevant variables, such as household income, education, and household size. Social capital was measured using a composite index based on

six indicators: social networks, community involvement, trust, norms and values, leadership, and collective action. The researchers found that food insecurity was more prevalent in rural areas than in urban areas. They also found that social capital was found to be positively correlated with food security and that social capital was stronger in rural communities than in urban communities. The study suggests that rural communities may have stronger capital because of their smaller size, higher levels of social connectedness, and greater reliance on community resources. The researchers also suggest that building social capital in both rural and urban areas could be a key strategy for addressing food insecurity. The study provides insights into the correlation between food insecurity and social capital in Texas and highlights the importance of addressing social capital as a means of promoting food security in both rural and urban communities.

To summarize, this review of the literature has provided insight into the detrimental impacts of food insecurity, which include its links to adverse mental health outcomes, negative effects on child development, and association with chronic diseases. It has also highlighted the use of various types of linear regression models in food insecurity research. Multiple studies have utilized linear regression models to pinpoint the factors linked to food insecurity such as income, education, and employment.

Other research have utilized logistic regression modeling to investigate the association between housing and food insecurity. Findings from these research studies have indicated that food insecurity is significantly correlated with a higher prevalence of chronic conditions such as hypertension and diabetes, even after controlling for potentially confounding variables such as age and socio-economic status. Furthermore, food insecurity is more prevalent among certain groups such as low-income households, indigenous households, and single-parent families and is associated with feelings of shame, guilt, and stigma. It has also highlighted the need for policies and programs that address the emotional and social impact of food insecurity, in addition to addressing poverty and income inequality. Additionally, research has demonstrated that food insecurity is stronger in rural communities. Building social capital in both rural and urban areas could be a key strategy for addressing food insecurity. Overall, these findings emphasize the urgent need to address food insecurity in order to improve public health outcomes and reduce social and economic inequalities in the United States.

19.3 Data Sources

Food insecurity is a multifaceted problem that is difficult to predict or evaluate in real time. This is due to two main

reasons: First, it is influenced by numerous interconnected factors that can change unexpectedly [9]. Second, the rate of food insecurity is often calculated retrospectively, depending on how food assistance programs are utilized and the manner in which the individuals report their past experiences with food-related needs.

The MVP model's datasets were gathered from seven distinct sources and were divided into yearly datasets covering the years 2009–2020. The following are the data sources as mentioned in Table 19.1.

19.4 Preprocessing

The MVP model utilized a sum of 55 datasets to generate the features. To enable EDA and modeling, every individual dataset was subjected to a comprehensive pre-processing procedure. The steps that were followed are as follows:

Table 19.1 List of datasets and their sources.

Dataset	Source	Description
Food deprivation data	Feeding America Map of the Meal Gap Study	The dataset includes information on food insecurity rates by county in the United States between 2009 and 2018.
Data for unemployment	Bureau of Labor Statistics	The dataset includes annual information on the workforce of each county in the United States from 2009 to 2020. The records consist of data on the total workforce and

Dataset	Source	Description
		unemployment rates.
Sociodemographic data	United States Census Bureau County Population Estimates	The dataset includes columns that provide demographic information, such as gender, race, and age, for each county in the United States, covering the years 2010–2019.
Homelessness data	US Department of Housing and Urban Development (HUD) Point-	The dataset includes information on houselessness rates in the United States by Continuum of Care

Dataset	Source	Description
	in-Time Estimates	(CoC) for the years 2009–2019.
Renal rates	Zillow Observed Rent Index	The dataset comprises monthly data from Zillow.com on one-bedroom rental prices categorized by zip code between 2014 and 2020.
Culinary industry data	US Census Bureau County Business Patterns	The dataset comprises information on all businesses in the United States at the county level from 2009 to 2018.

Dataset	Source	Description
Real-time census analysis	US Census Bureau CPS Datasets	The CPS and the Basic Monthly CPS surveys offer household information, including median income, household size, disability status, and healthcare coverage.

1. Initially, the procedures involved the importation of every dataset, each of which consisted of a folder containing CSV files corresponding to each year. Subsequently, any disparities from the years were standardized.
2. The next step was to carry out preliminary cleaning tasks, including the renaming of columns to enhance interpretability and the elimination of redundant features such as unique ID numbers or irrelevant metrics.

3. Coded columns were mapped to their corresponding columns using data dictionaries.
4. GEOIDs, also known as FIPS codes, were reformatted for each observation at the county level, which was used by primary/foreign key for joining all the datasets together.
5. Concerning the Feeding America dataset from 2011 to 2013, any missing values were imputed by employing the mean shift between the 2010 and 2014 data. Additionally, special symbols, like "—", were replaced with null values.
6. To unify the yearly datasets, a "year" column was appended to each one. Subsequently, the yearly datasets were vertically concatenated.
7. Lastly, the six data frames were merged horizontally, forming the primary dataset comprising all years and characteristics.
8. Certainly, here is a possible rephrased version.

The dataset for 2009 businesses was subjected to data cleaning, and before-and-after snapshots of the cleaning process were represented in the form of images. Figures 19.1 and 19.2 depict the business dataset, with Figure 19.1 illustrating the dataset before cleaning and Figure 19.2 showcasing the dataset after cleaning.

	fipstate	fipscty	naics	empflag	emp_nf	emp	qp1_nf	qp1	ap_nf	ap	est	n1_4	n5_9	n10_19	n20_49	n50_99	n100_249
0	1	1	------	NaN	G	10167	G	63783	G	273052	871	447	181	128	72	32	9
1	1	1	11----	NaN	H	33	H	226	H	1097	6	3	2	1	0	0	0
2	1	1	113///	NaN	G	27	G	188	G	945	5	3	1	1	0	0	0

Figure 19.1 Business dataset before cleaning.

	FIPS	Num_wholesale	Num_restaraunts	Num_grocery	Year
0	01001	71.0	13.0	109.0	2009
1	01005	5.0	7.0	109.0	2009
2	01007	3.0	21.0	92.0	2009
3	01009	19.0	32.0	186.0	2009
4	01013	13.0	106.0	29.0	2009

Figure 19.2 Business dataset after cleaning.

19.5 Data Dictionary

A data dictionary is a document or file that describes the data fields, tables, and other components of a database or dataset. The purpose of a data dictionary is to help users understand and interpret the data, which is particularly important for complex datasets or databases with many tables, fields, and relationships [10]. It also serves as a reference guide for data analysts, database administrators, and other stakeholders who need to work with the data or maintain the database.

Table 19.2 outlines the primary characteristics of the final cleaned dataset (prior to feature engineering) that was utilized

for both exploratory data analysis (EDA) and modeling purposes.

Table 19.2 Data dictionary used in the analysis.

Feature	Description
FIPS	A county-level identifier code
Rent	The average price of a one-bedroom rental unit
Year	The year in which the data was gathered
coc_number	A Continuum of Care (CoC) number that correlates with homelessness data
Houseless_rate	The percentage of the population within a county that is experiencing homelessness
Sheltered_rate	The percentage of the population of a county that is experiencing homelessness and residing in a shelter

Feature	Description
Unsheltered_rate	The percentage of the population of a county that is experiencing homelessness and not residing in a shelter
State	The name of a US state
County	The name of the county
TOT_POP	The total population tally within a county
TOT_MALE	The count of all males within a county
TOT_FEMALE	The count of all females within a county
TOT_WHITE	The count of all white people within a county

Feature	Description
TOT_BLACK	The count of all black people within a county
TOT_NATIVE	The count of all indigenous people within a county
TOT_ASIAN	The count of all Asian people within a county
TOT_PACIFIC	The count of all Pacific people within a county
TOT_LATINX	The count of all LatinX people within a county
State/county	A state/county combination
FI rate	The percentage of the population within a county that has food security (which is the target variable)

Feature	Description
Low threshold type	Food assistance programs with low eligibility criteria available by state
High threshold type	Food assistance programs with high eligibility criteria available by state
Cost per meal	The average expense per meal
Num_wholesale	The number of wholesale businesses within a county
Num_restaraunts	The number of restaurants and cafes within a county
Num_grocery	The number of grocery stores and markets within a county
Total_workforce	The total number of people within a county who are legally able to work
Employed	The total number of people within a

Feature	Description
	county who are employed
Unemployed	The total number of people within a county that is unemployed
Unemployment_rate	The percentage of the total workforce within a county that is unemployed
hh_med_income	The household median income
pop_disabled	The number of survey respondents who are disabled
pop_hs_grad	The number of survey respondents who graduated high school degree
pop_bachelors	The number of survey respondents with a bachelor’s degree
pop_grad_degree	The number of survey respondents with a graduate degree

19.6 Exploratory Data Analysis

The aim of this project is to evaluate the current and future rates of food insecurity (FI) at the county level, with consideration for both temporal and geographic factors. To acquire a better grasp of food insecurity and its closely correlated indicators, the exploratory data analysis probes three questions regarding its relationship with time, geography, and other factors. The three questions were as follows:

> *Question 1. In what ways have metrics like unemployment, houselessness, and food insecurity rates evolved over a period of time?*
>
> *Question 2. What is the geographic variability in unemployment, race, and food insecurity rates?*
>
> *Question 3. How are factors like rent prices, unemployment, houselessness, and race related to the food insecurity rate?*

First of all, the cleaned dataset with engineered features is imported; then, it was inspected to get a general understanding of the distribution of each feature.

From the statistics of Figure 19.3, it is observed that most features such as Rent, Unemployment_rate, and Total_food_retail contain large upper outliers. For a detailed observation, a boxplot graph for each feature with large outliers is plotted.

	Rent	Houseless_rate	Sheltered_rate	Unsheltered_rate	TOT_POP	TOT_MALE	TOT_FEMALE	TOT_WHITE	TOT_BLACK	TOT_NATIVE
count	6645.000000	47487.000000	47487.000000	47487.000000	4.805700e+04	4.805700e+04	4.805700e+04	4.805700e+04	4.805700e+04	48057.000000
mean	1609.911352	0.001072	0.000669	0.000403	1.243185e+05	6.114445e+04	6.317403e+04	9.483544e+04	1.662157e+04	1527.897226
std	1026.233434	0.001089	0.000567	0.000835	4.816667e+05	2.365634e+05	2.451369e+05	3.417494e+05	7.782785e+04	7021.898989
min	619.333333	0.000017	0.000009	0.000000	8.266667e+01	4.100000e+01	3.500000e+01	2.400000e+01	0.000000e+00	0.000000
25%	1255.600000	0.000573	0.000364	0.000078	1.120600e+04	5.643667e+03	5.548000e+03	9.305000e+03	1.220000e+02	66.000000
50%	1485.520000	0.000741	0.000541	0.000176	2.666067e+04	1.329500e+04	1.327100e+04	2.290267e+04	9.120000e+02	188.666667
75%	1739.425000	0.001187	0.000796	0.000381	7.302800e+04	3.620200e+04	3.661000e+04	6.192700e+04	6.035000e+03	685.000000
max	22528.834158	0.016672	0.011712	0.016259	1.010571e+07	4.980981e+06	5.125335e+06	7.181207e+06	1.311396e+06	145996.000000

Figure 19.3 Descriptive statistics of the distribution of each feature.

As depicted in Figure 19.4's boxplot, variables such as TOT_POP, TOT_ MALE, etc., have considerable outliers, which may need to be addressed during the modeling stage to enhance model generalization.

After this, the next step was to inspect multicollinearity using a heatmap to determine which features are highly correlated and can potentially be removed during the modeling process [11].

Figure 19.5 shows a heatmap indicating a strong correlation between pairs of demographic characteristics, such as TOT_NATIVE and TOT_ ASIAN. This is expected since areas

counties with larger populations are likely to exhibit a greater representation of all demographic groups. There is a strong collinearity between houseless_rate, sheltered_rate, and unsheltered_rate as they are directly interconnected. Features that exhibit a high correlation may be excluded during the modeling process, depending on the performance of the model and the desired interpretability. The next step is to conduct exploratory data analysis to answer the three questions mentioned earlier.

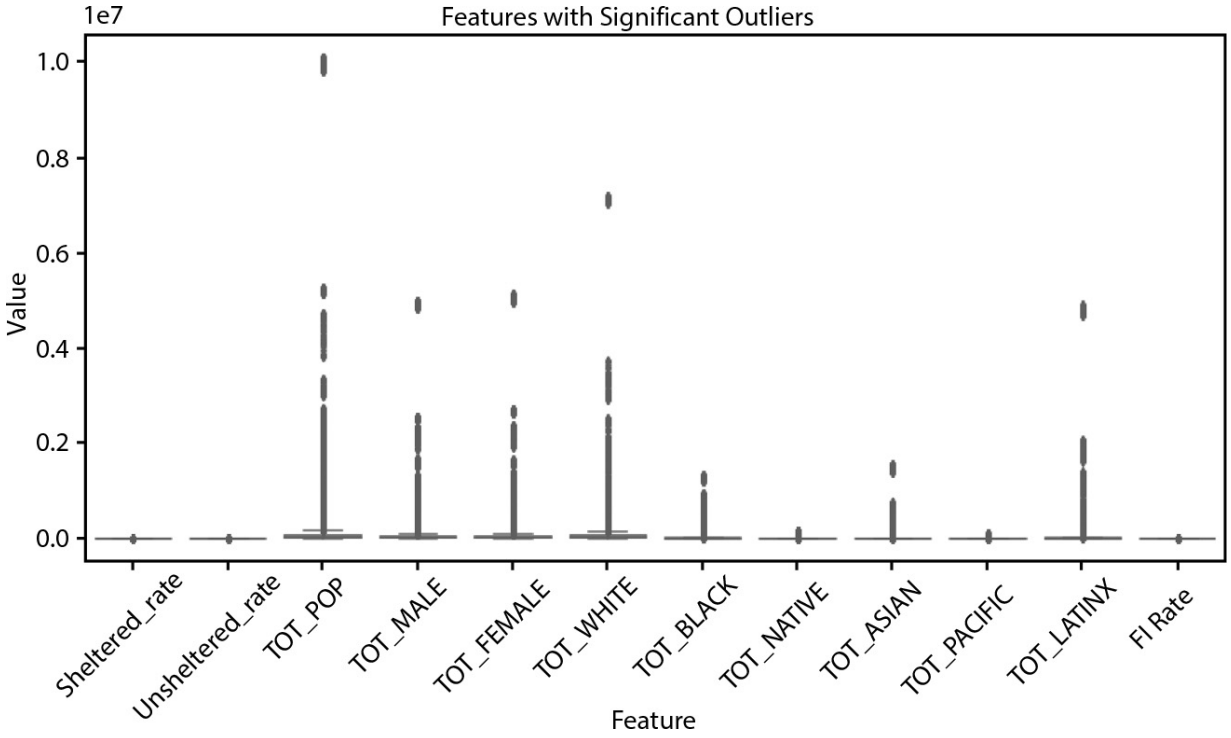

Figure 19.4 Boxplot graph for features with significant outliers.

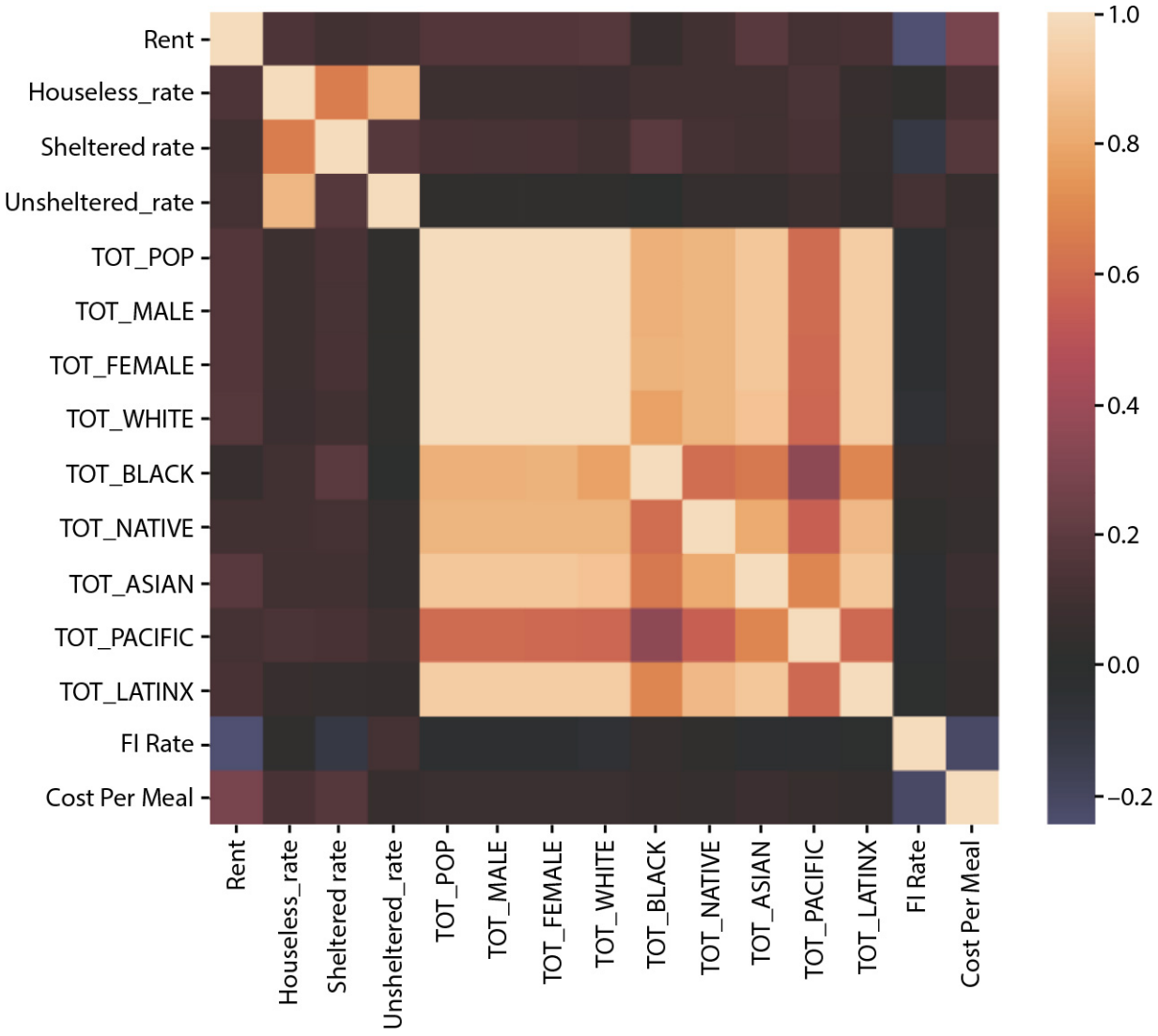

Figure 19.5 Heat map for inspecting multicollinearity.

19.6.1 Question 1: In What Ways Have Metrics Like Unemployment, Houselessness, and Food Insecurity Rates Evolved Over Time?

The initial step in the exploratory data analysis is to examine how various features from the initial dataset have evolved over time. Since the primary objective of this project is to forecast food insecurity rates for 2020, it is crucial to comprehend how

these features evolve over time and if they follow any identifiable patterns [12]. To achieve this, each group of features is normalized to the same scale and illustrated using line plots and/or bar charts. The visualization of these charts with multiple variables is facilitated by two functions: lineplot() and barchart(), both of which are available in the src folder of this repository.

Firstly, a line chart is created to visualize and compare the patterns of three scaled features, food insecurity, houselessness, and unemployment rates, over time. The purpose of this is to examine how these variables have changed over time and to identify any potential correlations. Figure 19.6 shows the resulting line plot graph, which indicates a clear correlation between the FI rate (target variable), unemployment, and houselessness rates. However, it is noteworthy that the data is not available for MVP when exploring the significance of these features and building models. Nevertheless, the robust correlation among these features is a compelling signal that the rates of FI and homelessness will witness a significant surge in 2020, together with the unemployment rate.

Next, a line chart of three variables Num_wholesale, Num_restaraunts, and Num_grocery is plotted in order to visualize and compare the trends of the number of grocery

stores, restaurants, and wholesalers by year. In general, the line chart presented in Figure 19.7 does not depict any significant trend regarding the number of food retail businesses. Nonetheless, it is worth mentioning that the average number of restaurants experienced a sudden upsurge in 2018, whereas the count of wholesalers and grocery stores witnessed a decline. As a rule, purchasing meals from restaurants is more expensive compared to buying them from a grocery store. Hence, this observation suggests that a greater proportion of the population opted for restaurant food in 2018.

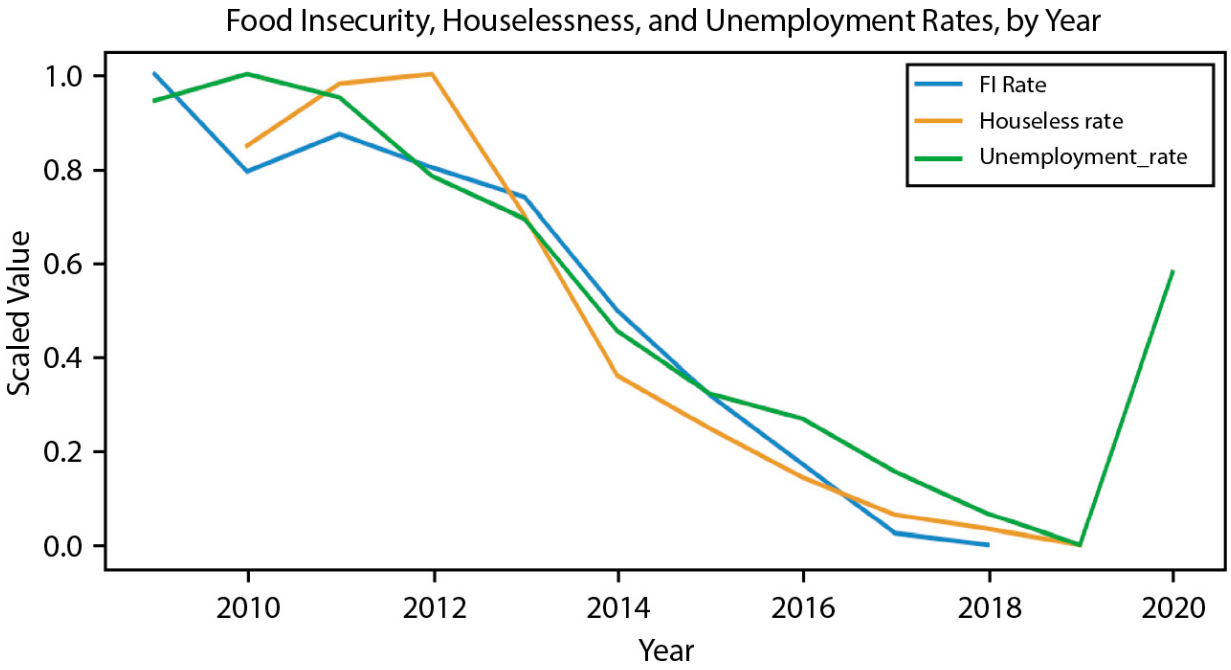

Figure 19.6 Line plot graph of FI rate, unemployment, and houseless rates by year.

Figure 19.7 Line plot graph of grocery, restaurants, and wholesalers, by year.

Next, a bar chart and line chart of two variables—rent and cost per meal—is plotted for visualizing and comparing how rent

prices and food costs have changed by year. The rent data was available starting in 2014 and extends into 2020, while the meal cost data was available from 2009 to 2018. Based on the charts in Figures 19.8 and 19.9, it can be observed that there is a noticeable correlation between the average food cost and the rent prices that increases over time. It is quite fascinating to take note of this, particularly in light of the fact that the aforementioned charts depict downward trends in terms of FI rates, homelessness, and unemployment [13].

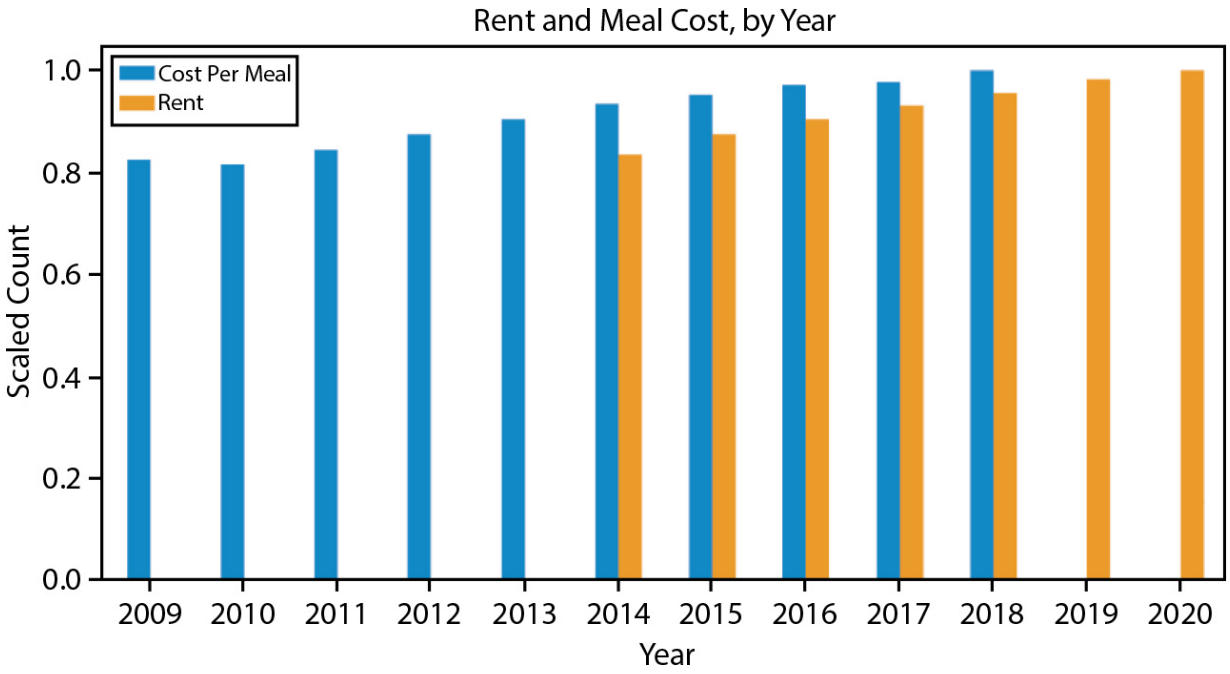

Figure 19.8 Bar chart of rent and meal cost, by year.

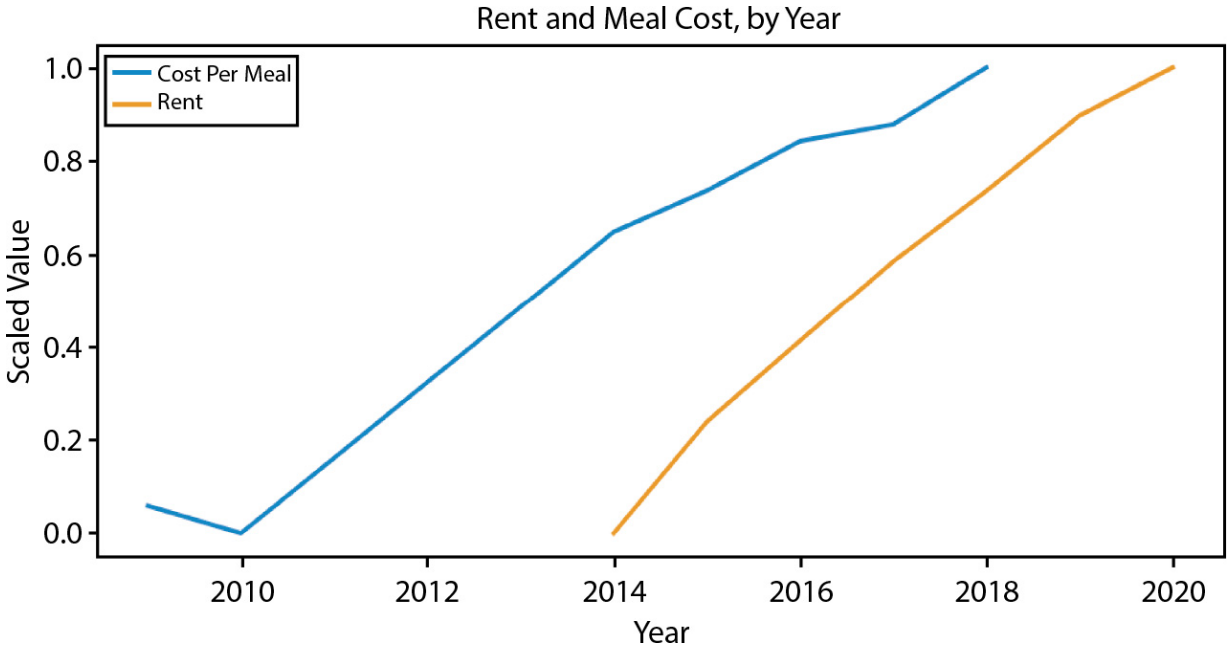

Figure 19.9 Line chart of rent and meal cost, by year.

A line plot graph of variables—predominantly white, predominantly black, predominantly indigenous, and

predominantly LatinX—was created to investigate the differences in average rates of food insecurity among different racial communities over a period of time. The results presented in Figure 19.10 show that black communities have consistently had the highest average FI rates, followed by indigenous and LatinX communities. Although there is an overall declining trend, some communities experience fluctuations, such as the sudden increase in FI rates observed in predominantly LatinX communities during 2018. It should be noted that these rates are associated with counties with distinct predominant racial demographics rather than specific racial groups [14].

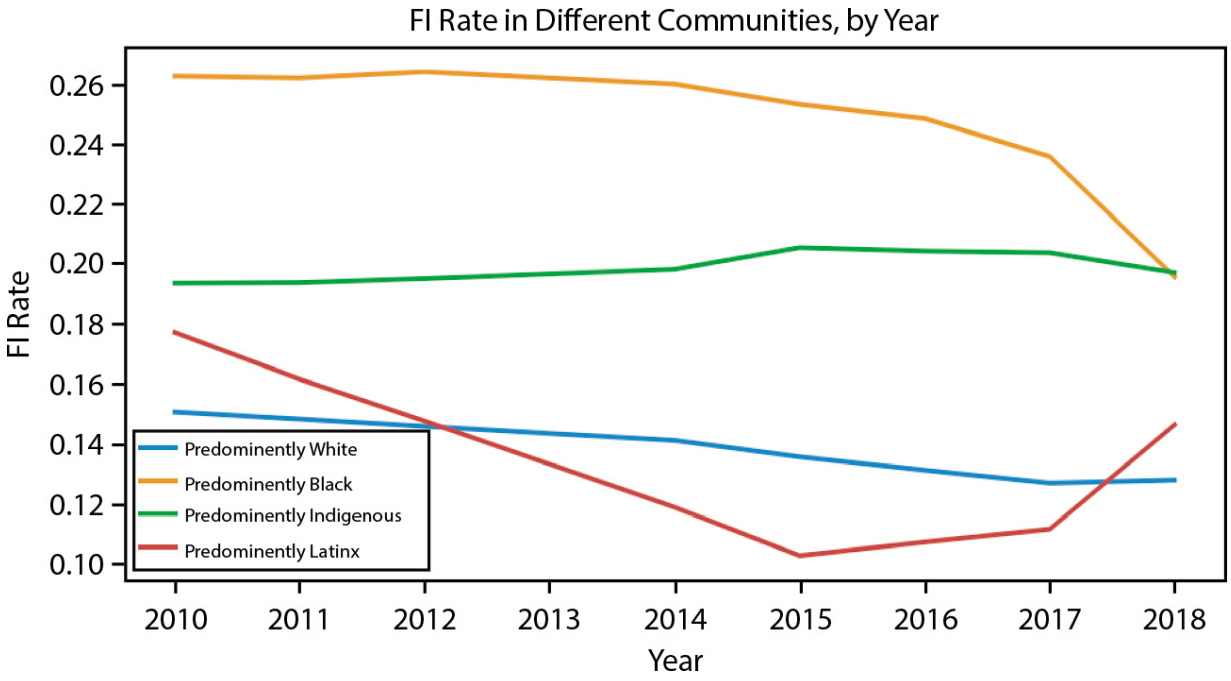

Figure 19.10 Line chart of FI rate in different communities, by year.

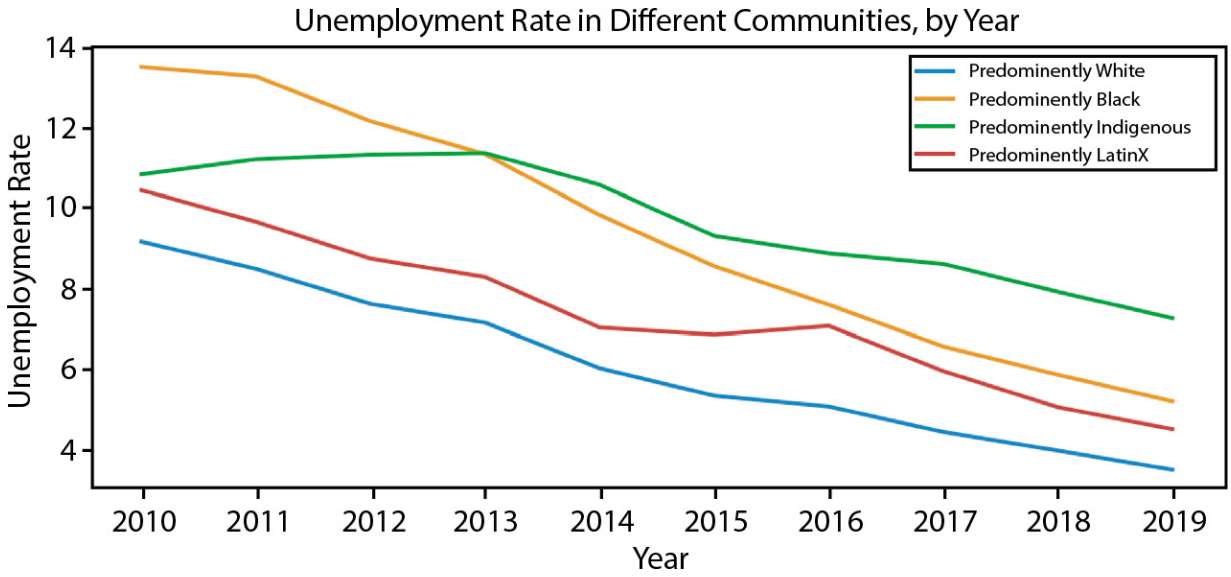

Figure 19.11 Line chart of unemployment rate in different communities, by year.

Now, again a line chart is plotted with the same variables as before but this time with reference to average unemployment rates in different racial communities, categorized by year. The

line graph in Figure 19.11 reveals that communities primarily consisting of black and indigenous populations experience relatively elevated levels of unemployment compared to LatinX communities. Moreover, there seems to be decreasing pattern in the average unemployment rate over time.

19.6.2 Question 2: What is the Geographic Variability in Unemployment, Race, and Food Insecurity Rates?

The objective of investigating this question is to enhance comprehension of the geographical aspect of food insecurity and its associated variables. The first step was to draw a heat map for any 5 random counties in order to get a quick visualization of food insecurity rates. The heatmap in Figure 19.12 suggests that geography plays a crucial role in determining food insecurity rates. This is evident from the significant variation in FI rates observed across different counties displayed in the heatmap.

Next, Choropleth Maps are generated. Choropleth Maps for the years 2010 and 2019 for the variable Unsheltered_rate are generated. The maps in Figures 19.13 and 19.14 depict unsheltered rates, which signify the percentage of a populace that lacks housing and remains unsheltered. The visual aids illustrate a decline in unsheltered rates in regions spanning the

middle and southern parts of the nation, including specific counties in Florida. However, the maps highlight a significant surge in unsheltered rates along the West Coast, primarily in California [15].

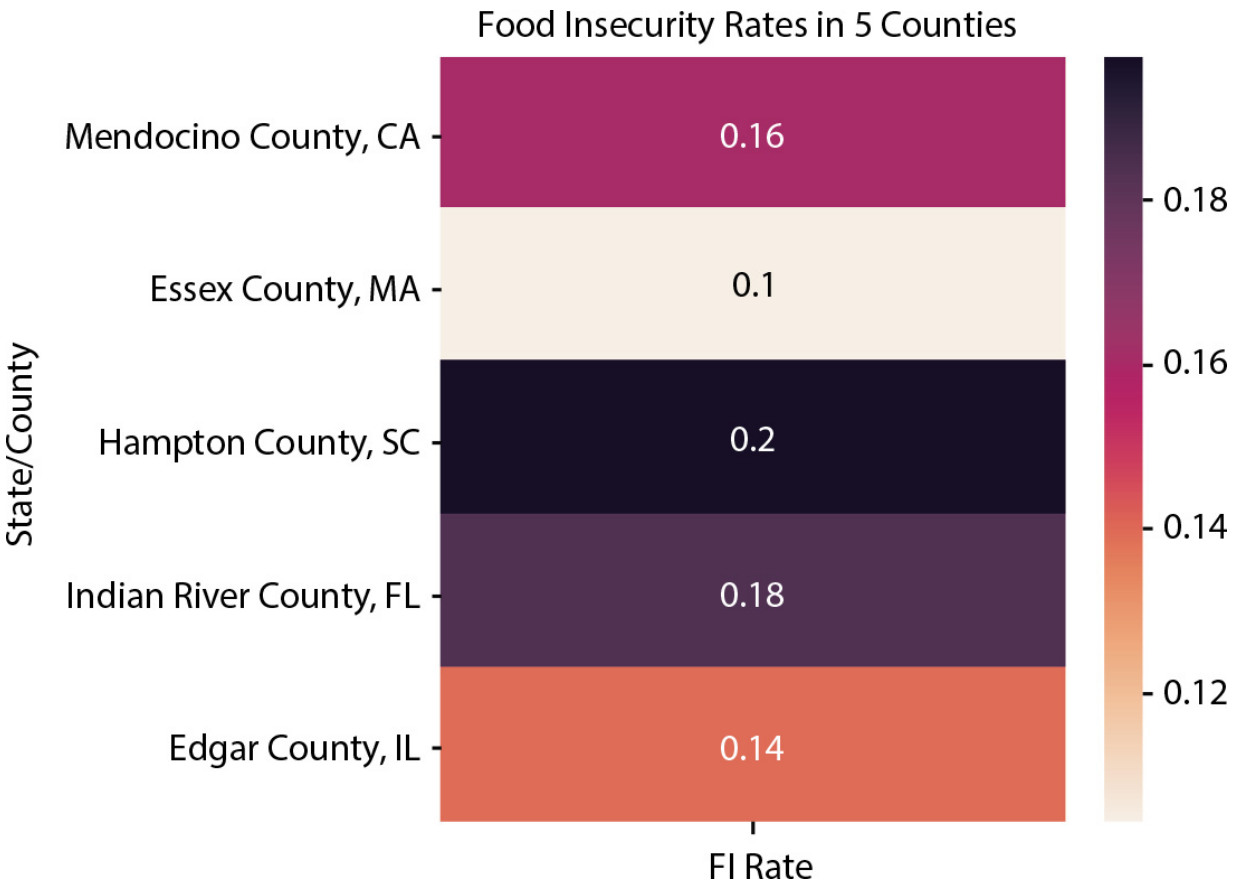

Figure 19.12 Heat map for food insecurity rates in five counties.

Then, a choropleth map of variable unemployment rate for the years 2019 and 2020 is also plotted. From both maps shown in Figures 19.15 and 19.16, it is observed that prior to the onset of the pandemic, in the year 2019, the unemployment rates were at a minimum. However, the year 2020 witnessed a

considerable surge in unemployment rates, as depicted by the maps presented above. The maps clearly exhibit a substantial rise in unemployment rates throughout the country, with a particular emphasis on the coastal regions.

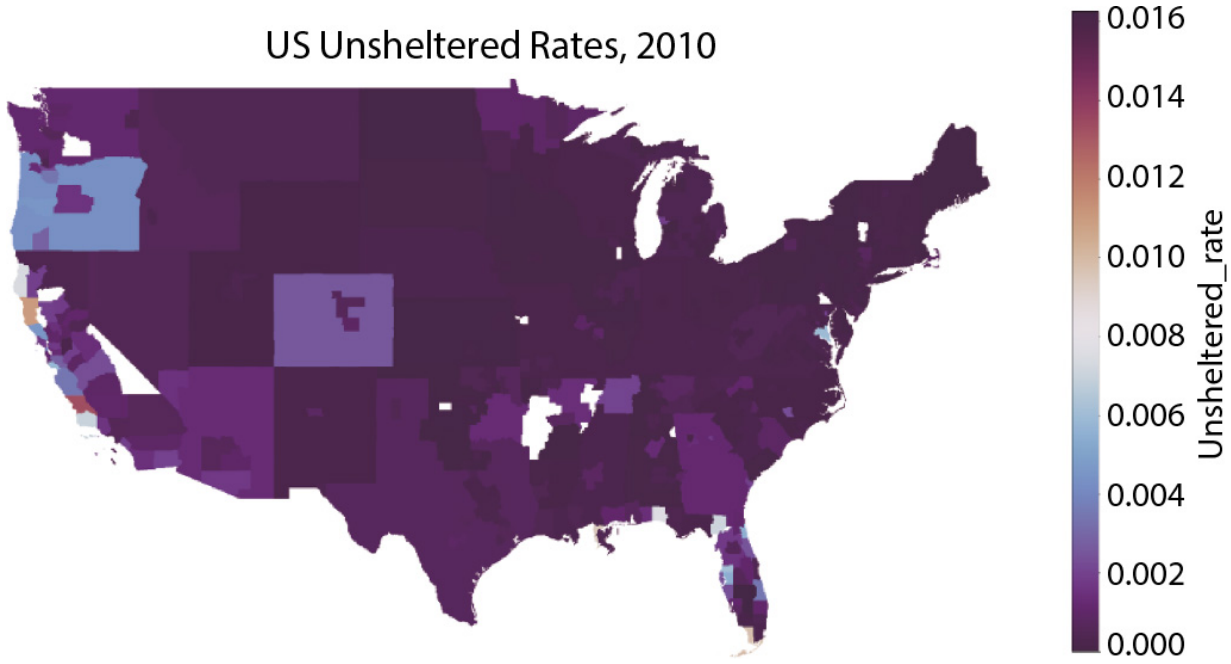

Figure 19.13 Choropleth map for US unsheltered rate in 2010.

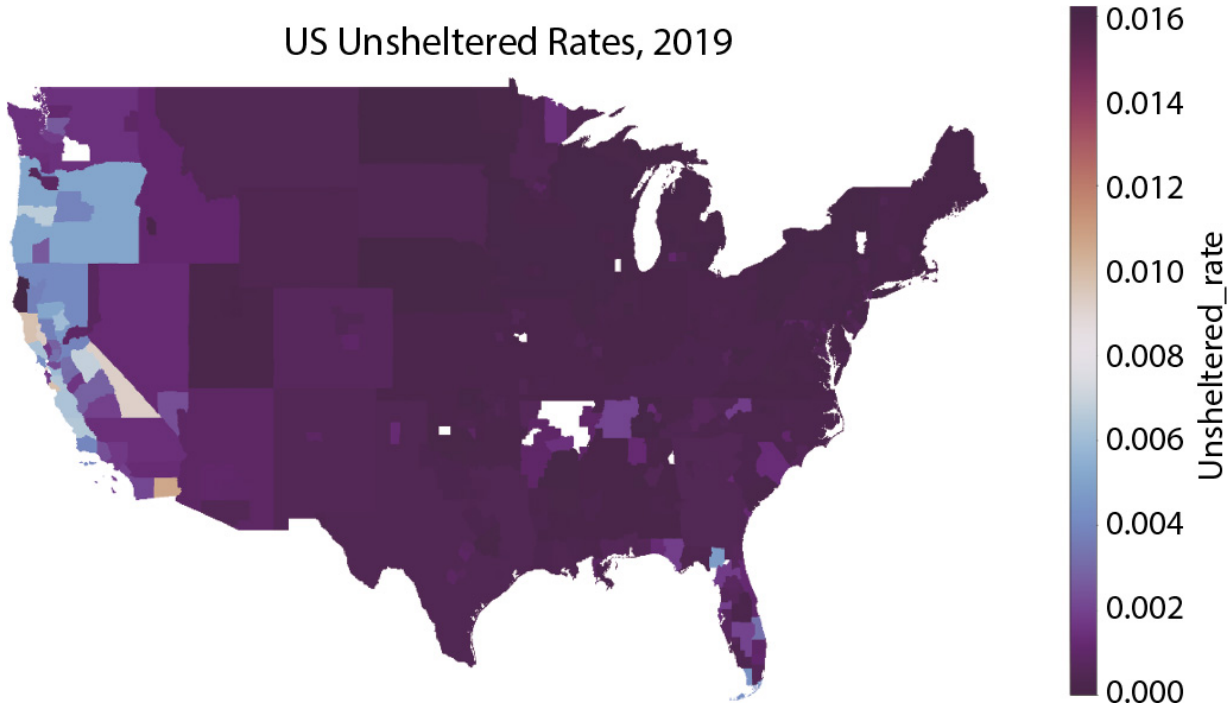

Figure 19.14 Choropleth map for US unsheltered rates in 2019.

The generated choropleth map in Figure 19.17 displays the

distribution of the variable Percent_PoC across the US counties in the year 2020. This map provides a visual representation of the areas in the country where the population consists predominantly of people of color. The choropleth map was generated using the choropleth() function. The map shows that although the majority of counties in the United States have populations primarily composed of individuals who identify as white, there are noteworthy regions, particularly in the southern states and portions of the southwest, where people of color make up the majority of the population [16].

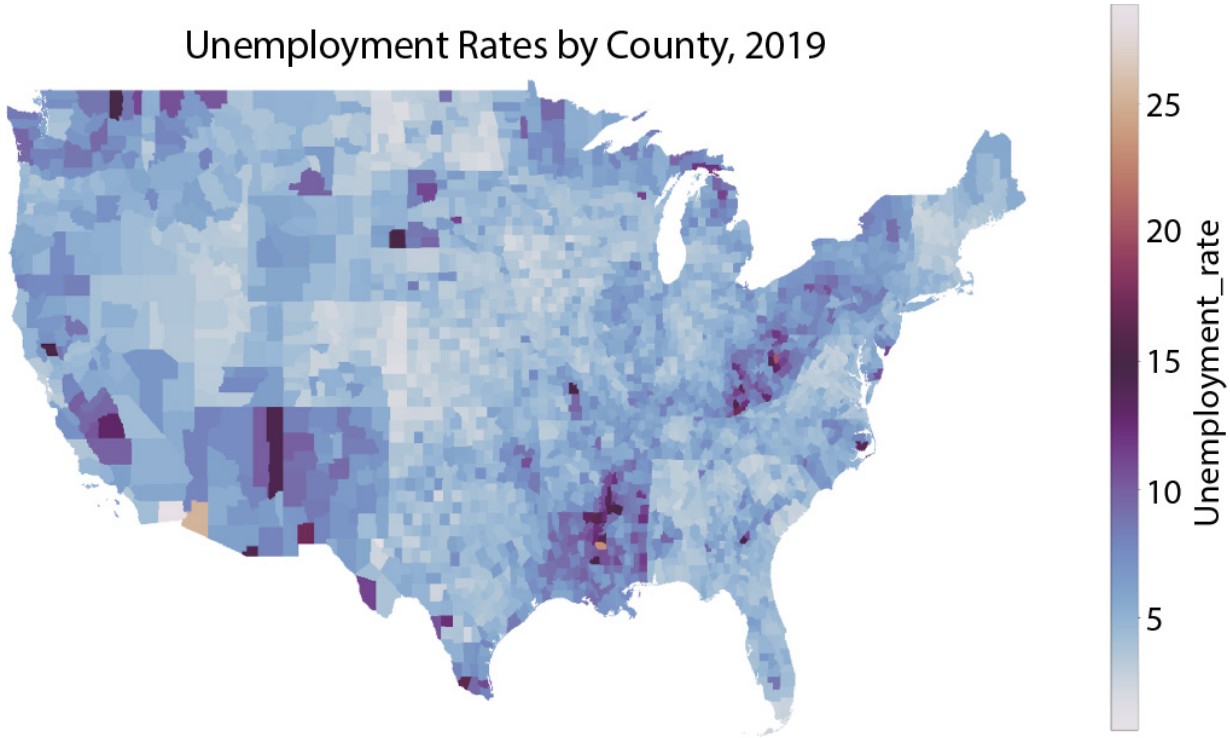

Figure 19.15 Choropleth map for unemployment rate by county in 2019.

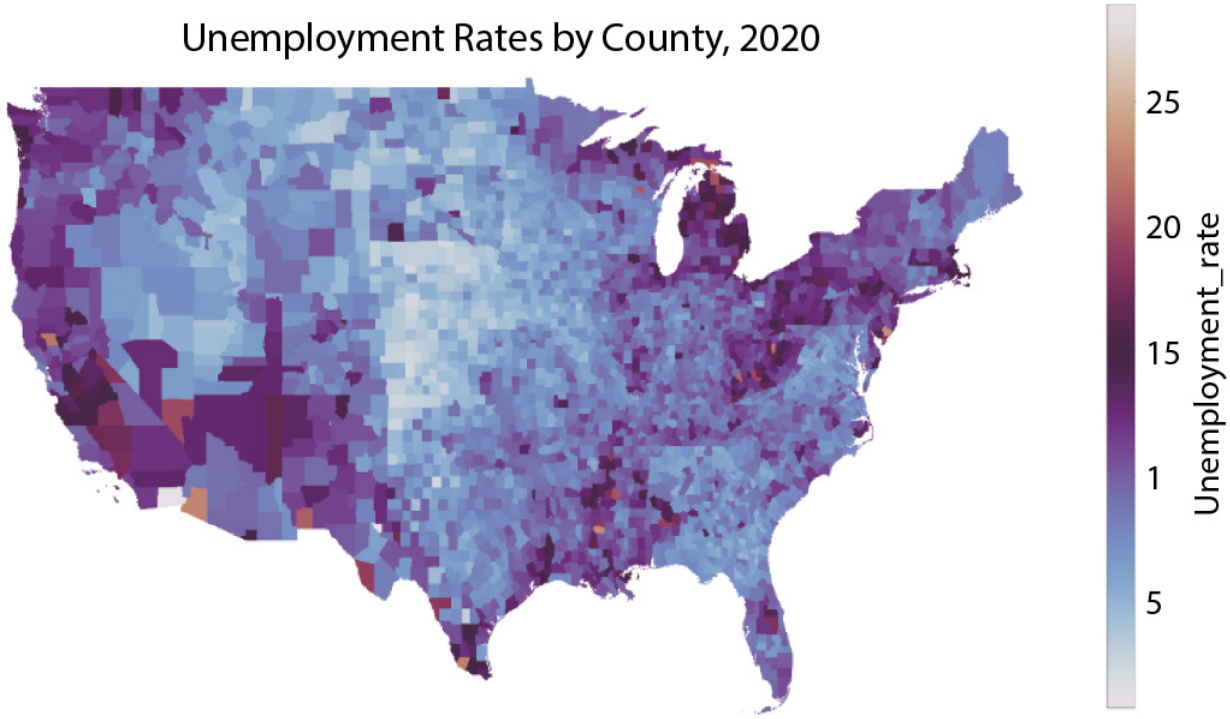

Figure 19.16 Choropleth map for unemployment rate by county in 2020.

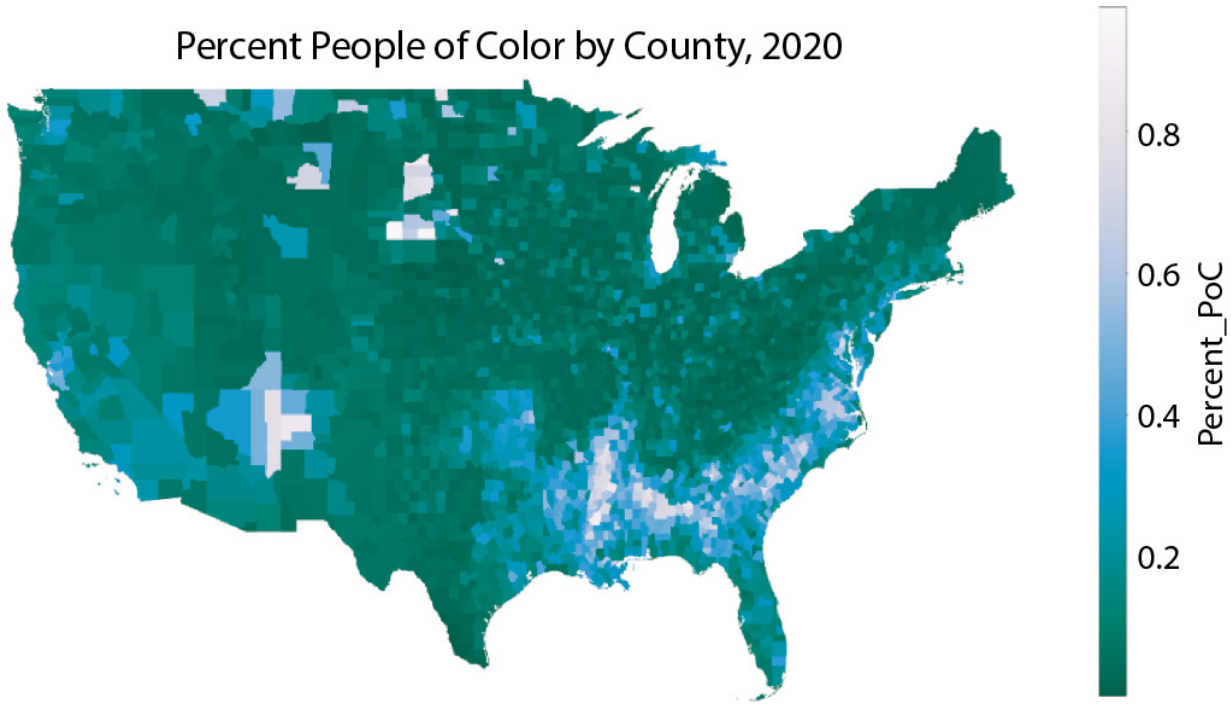

Figure 19.17 Choropleth map for percent people of color by county for the year 2020.

Then, a choropleth map for the variable FI rate for the years 2009 and 2018 is also plotted. The maps presented in Figures 19.18 and 19.19 depict a decrease in food insecurity rates over a period of time in certain regions, particularly in the Pacific Northwest, while southern counties exhibit significantly higher rates. It is crucial to acknowledge that although the national average for food insecurity rates has decreased over time, these maps suggest that the problem has worsened in certain localities. Furthermore, it is worth mentioning that regions that are primarily non-white closely align with the areas most affected by food insecurity, as opposed to the maps presented earlier.

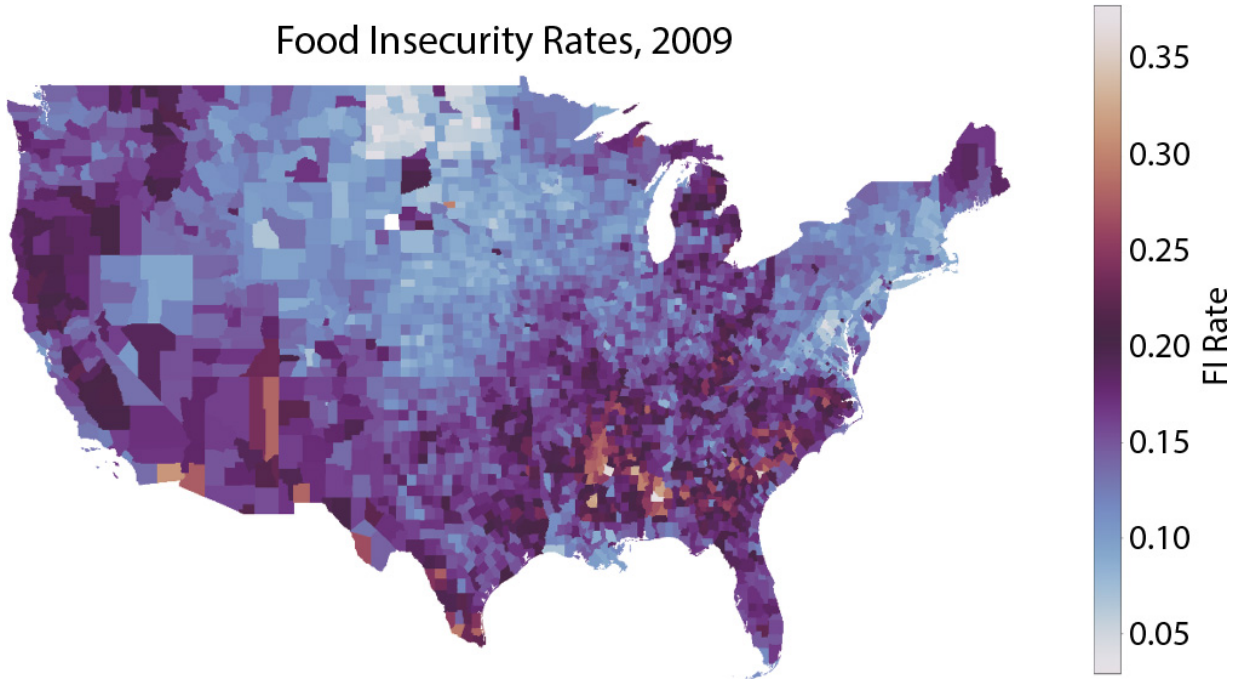

Figure 19.18 Choropleth map for food insecurity rate for the year 2009.

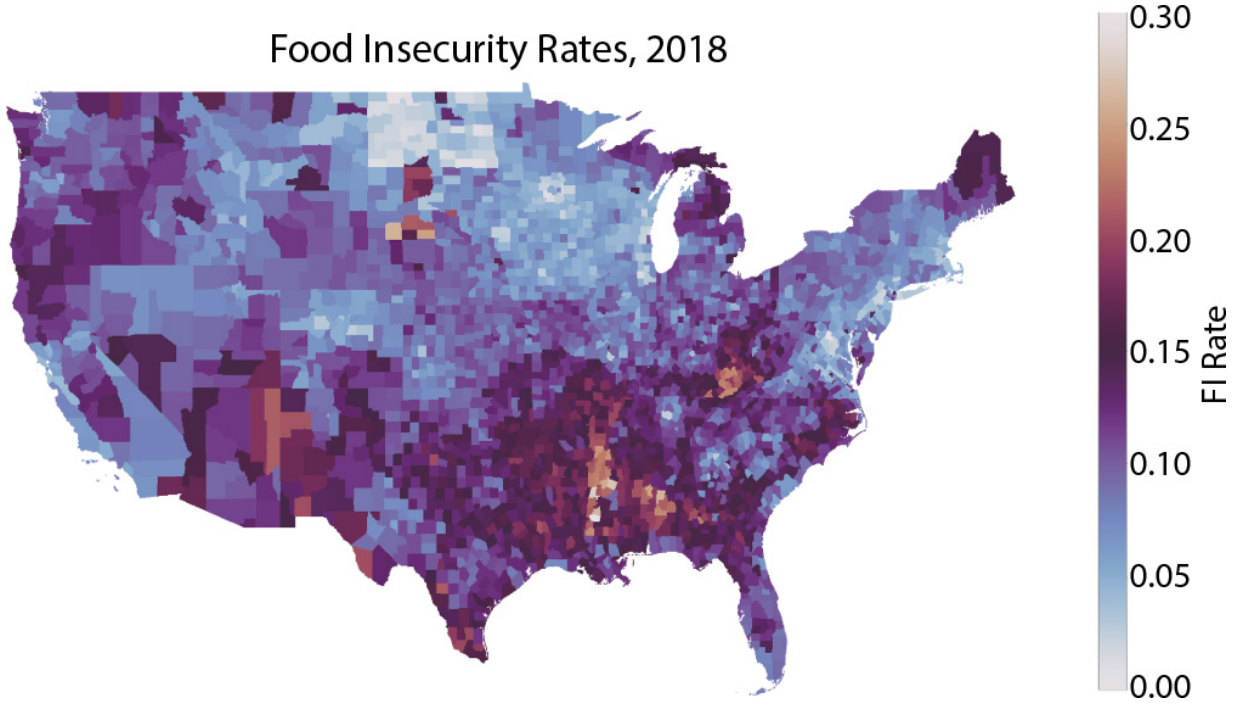

Figure 19.19 Choropleth map for food insecurity rate for the year 2018.

19.6.3 Question 3: How are Factors Such as Rent

Prices, Unemployment, Houselessness, and Race Related to the Food Insecurity Rate?

The aim of this final exploratory data analysis question is to examine the relationship between food insecurity rates and other features available in the dataset, such as average rental prices and homelessness rates. Keeping these observations in mind while transforming towards the modeling phase and examining the coefficients/feature importance will prove to be crucial.

To examine the relationship between food insecurity rates and the presence of restaurants, a histogram is generated to compare the prevalence of food insecurity rates in areas with more than 600 restaurants to those with fewer than 600. The threshold of 600 is chosen because it is the most common number of restaurants in the dataset. The histogram shown in Figure 19.20 demonstrates a noticeable visual contrast between the two. In general, regions with lower rental prices tend to have higher food insecurity rates, with an average of around 0.14, while regions with higher rental prices tend to have lower food insecurity rates, with an average of approximately 0.08. This observation aligns with the notion that “more affluent” areas are likely to experience less food insecurity [17].

A histogram is also plotted for the FI rates based on the rental prices of one-bedroom apartments, where areas are divided into two groups based on their rent prices—below 1,000 and above 2,000. The histogram in Figure 19.21 demonstrates a conspicuous contrast between the two. Regions that have lower rental rates exhibit relatively higher FI rates, whereas regions that have higher rental rates display lower FI rates. The phenomenon is reasonable since areas that are considered more affluent are expected to experience fewer cases of food insecurity.

A histogram is generated to display the distribution of FI rates for regions with unemployment rates exceeding 15% compared to regions with unemployment rates lower than 5%. The histogram in Figure 19.22 depicted above exhibits a conspicuous dissimilarity in FI rates among regions contingent on their unemployment rates. Regions with high unemployment rates have a greater average of food insecurity rates, while regions with low unemployment rates have a lower average of food insecurity rates. This phenomenon is quite reasonable as unemployment is a probable reason for the incapacity to bear the expenses of food [18].

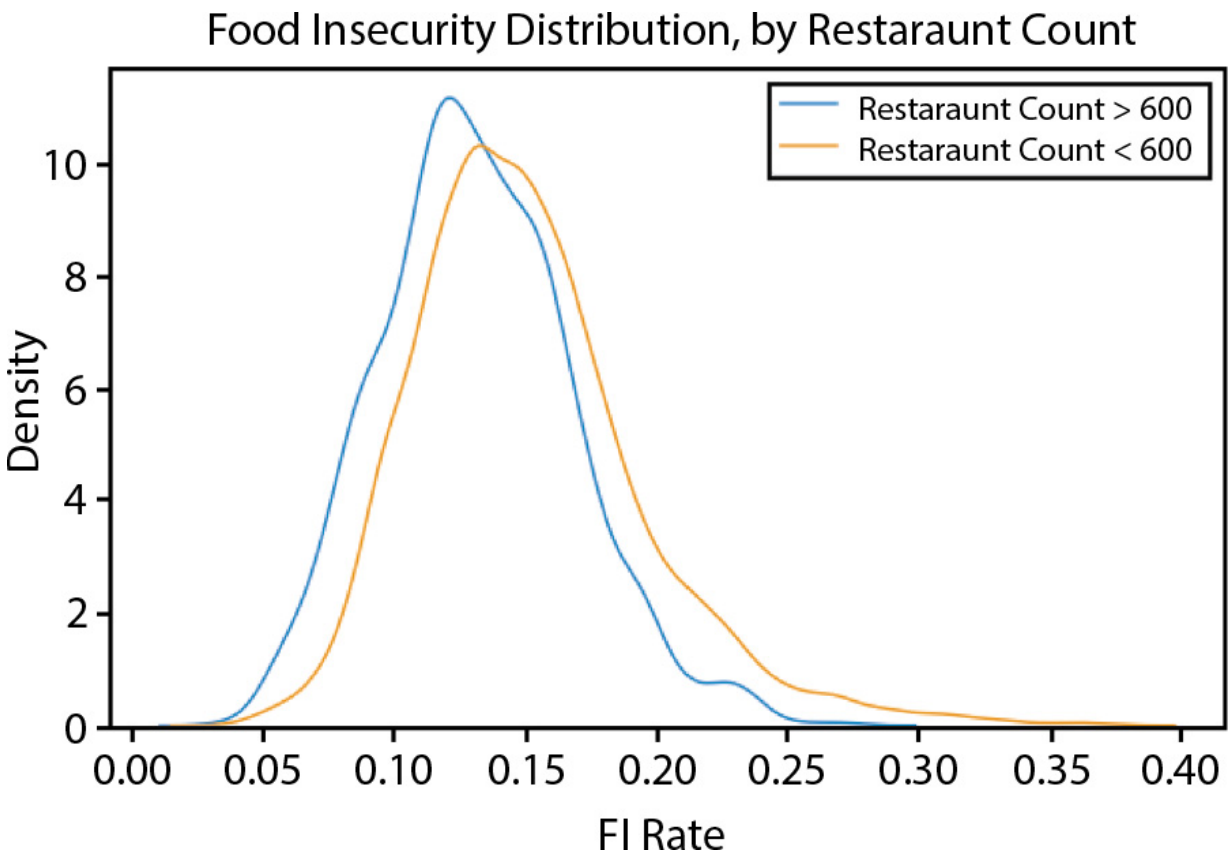

Figure 19.20 Histogram for food insecurity distribution, by restaurant count.

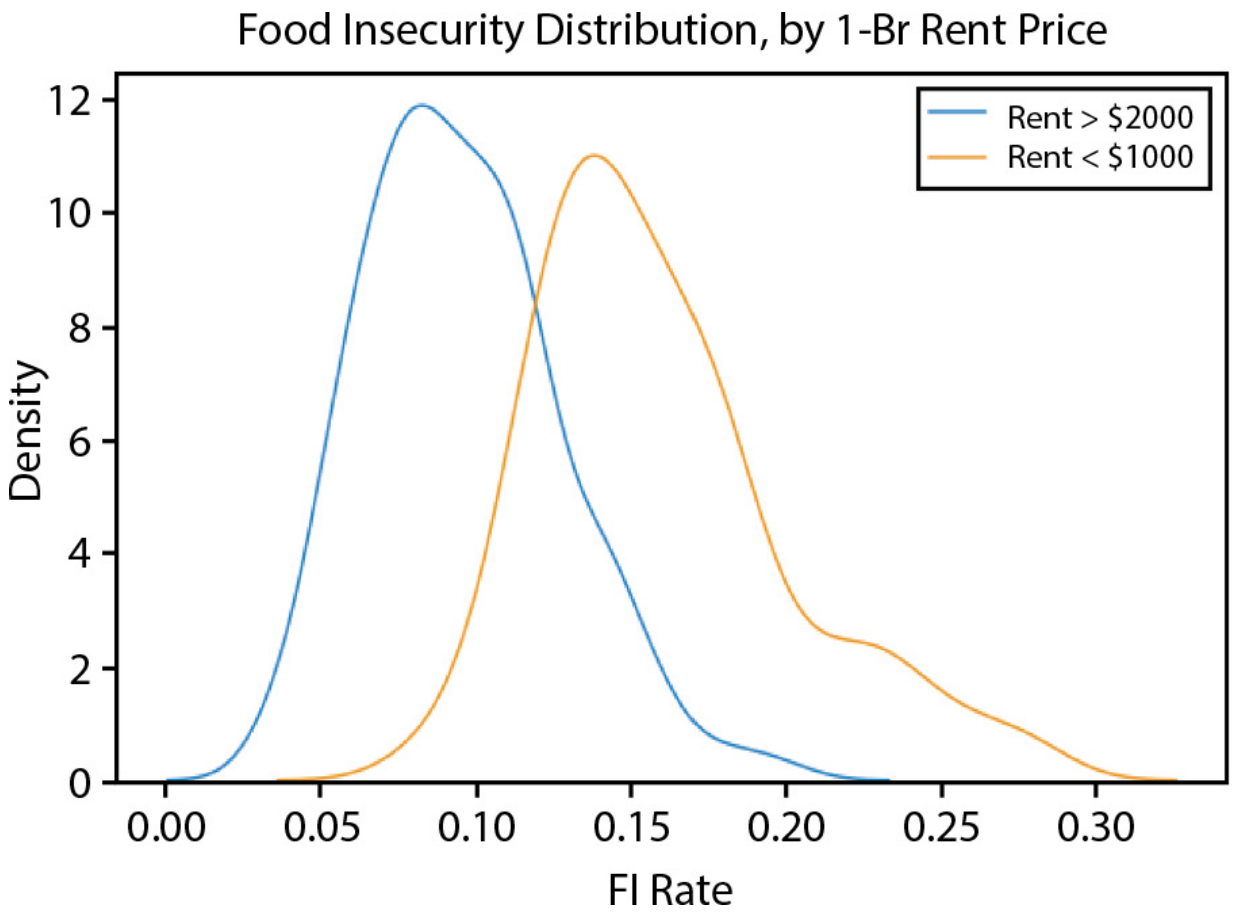

Figure 19.21 Food insecurity distribution, by one-bedroom rent prices.

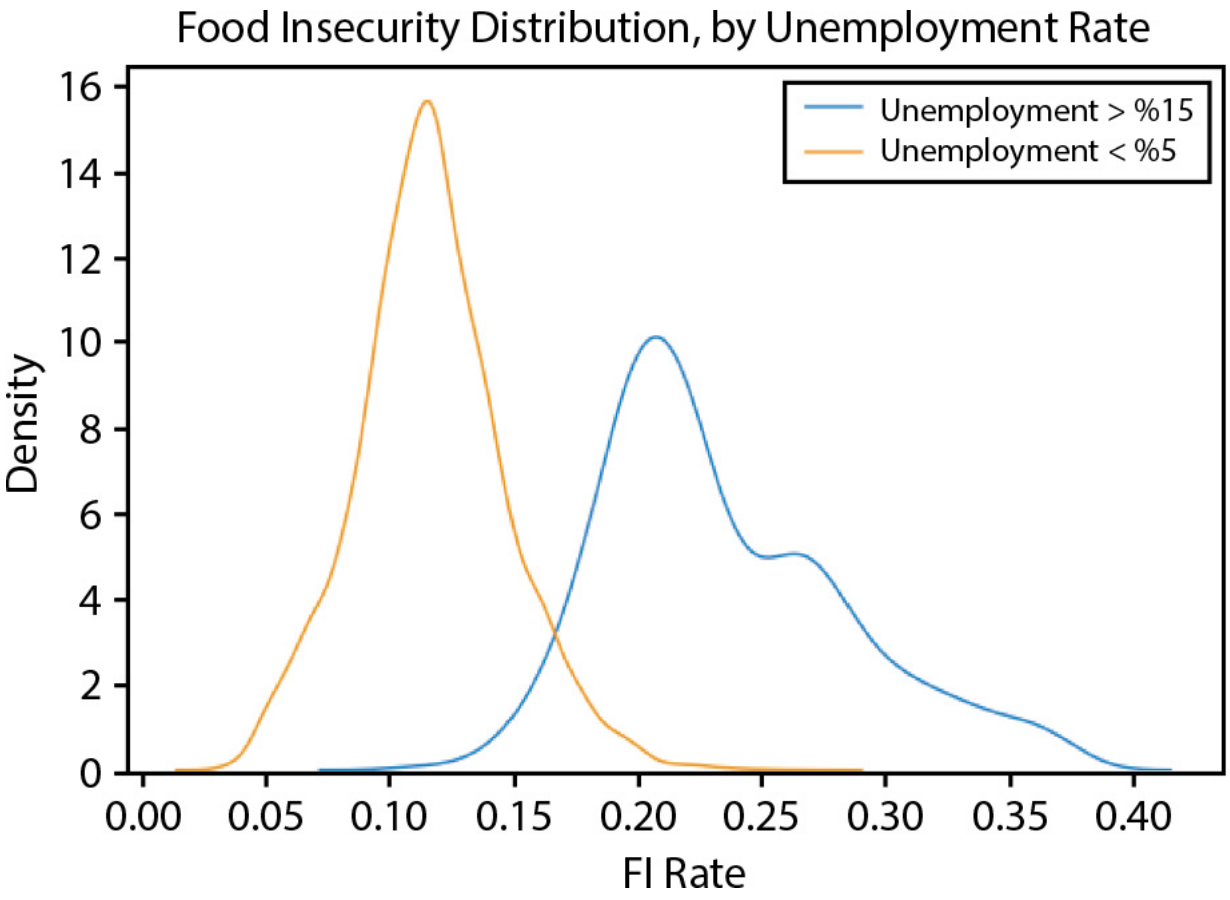

Figure 19.22 Histogram for FI distribution, by unemployment rate.

A histogram is plotted to show the distribution of FI rates by houseless rates in regions with a houseless rate exceeding 1% and regions with a houseless rate below 0.01%. Based on the histogram shown in Figure 19.23, there seems to be a noticeable contrast between the two sets of data. It appears that regions with elevated levels of homelessness also exhibit a greater average FI rate compared to regions with lower levels of homelessness. This correlation is logical, as homelessness is likely to significant financial hardship, which may, in turn, lead to a shortage of funds to purchase adequate sustenance.

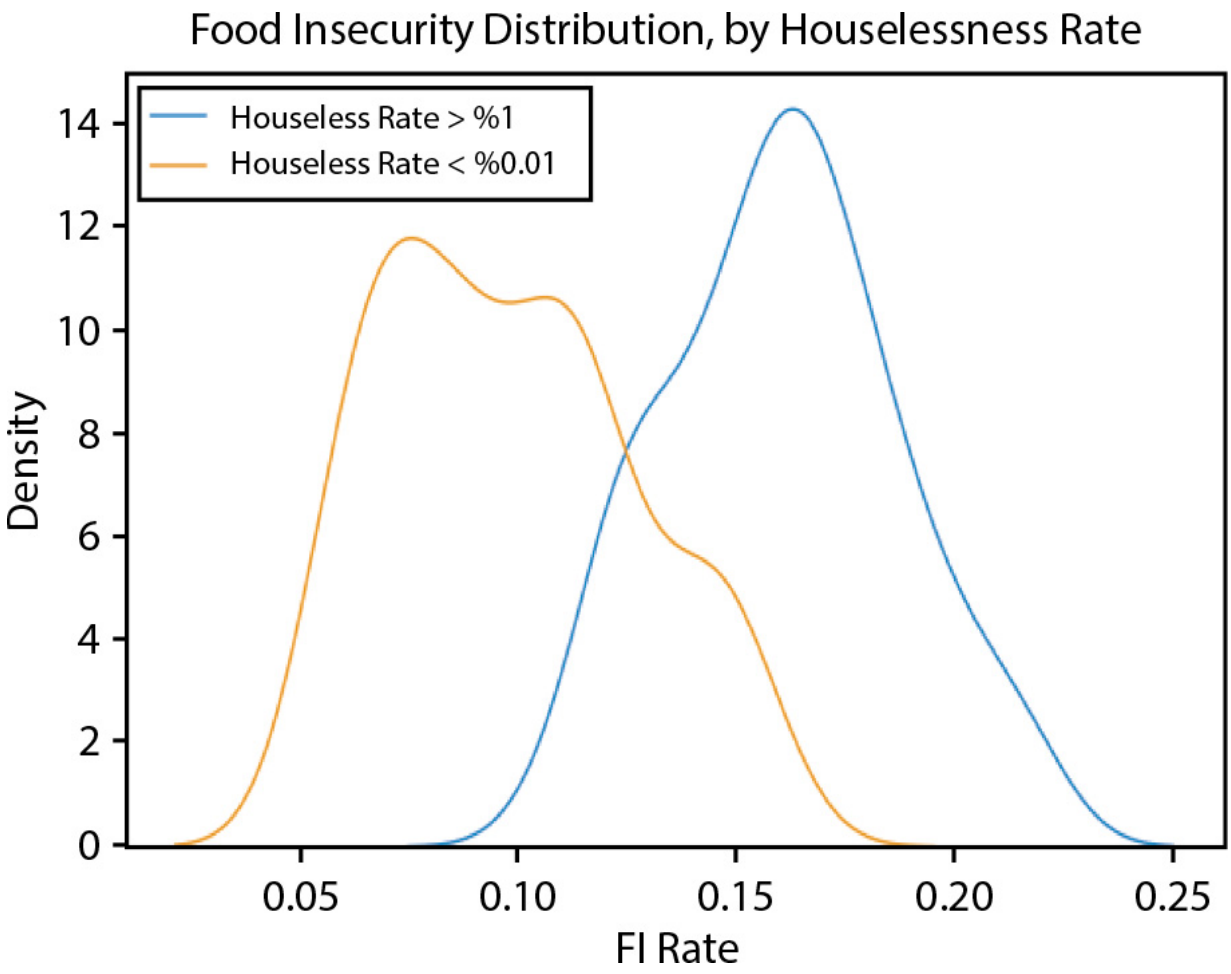

Figure 19.23 Histogram for FI distribution, by houselessness rate.

A histogram depicting the distribution of food insecurity rates across areas predominantly inhabited by different racial groups is also generated. The histogram in Figure 19.24 highlights a visually conspicuous variation in FI rates among various racial groups. It is observed that regions with a higher concentration of the black population have a considerably greater average FI rate compared to those with predominantly white or LatinX communities. This fact holds considerable significance while

analyzing feature importance and identifying significant factors that contribute to food insecurity [19].

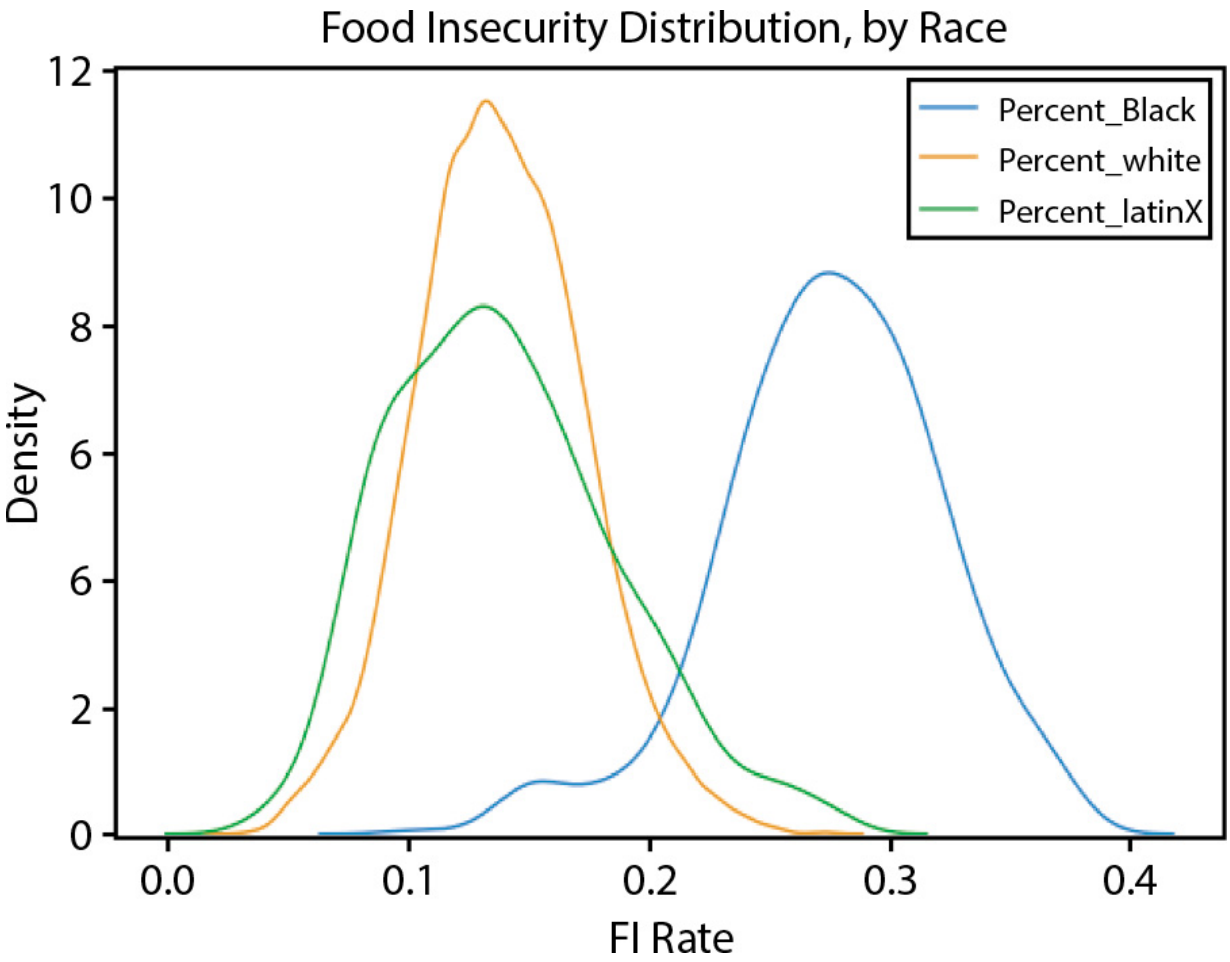

Figure 19.24 Histogram for FI distribution, by race.

Table 19.3 Average FI rate in different communities.

Different racial communities	Average FI rate
Predominantly black communities	0.2517524653544353
Predominantly PoC Communities	0.2251573112521035

Different racial communities	Average FI rate
Predominantly white communities	0.141475585966807 76

Then, the mean food insecurity rate in different communities, where the population of one race is above 50%. The data shown in Table 19.3 indicates a significantly higher average FI rate among black communities and all communities of color [20]. Scatter plots were created to compare food insecurity with variables such as unemployment, rent, and the percentage of White/Black population. The scatter plots depicted in Figures 19.25 to 19.29 indicate a positive correlation between the FI rate and both the unemployment rate and the percentage of the Black population. Conversely, they show a negative correlation between the FI rate and the rent price, as well as the percentage of the White population.

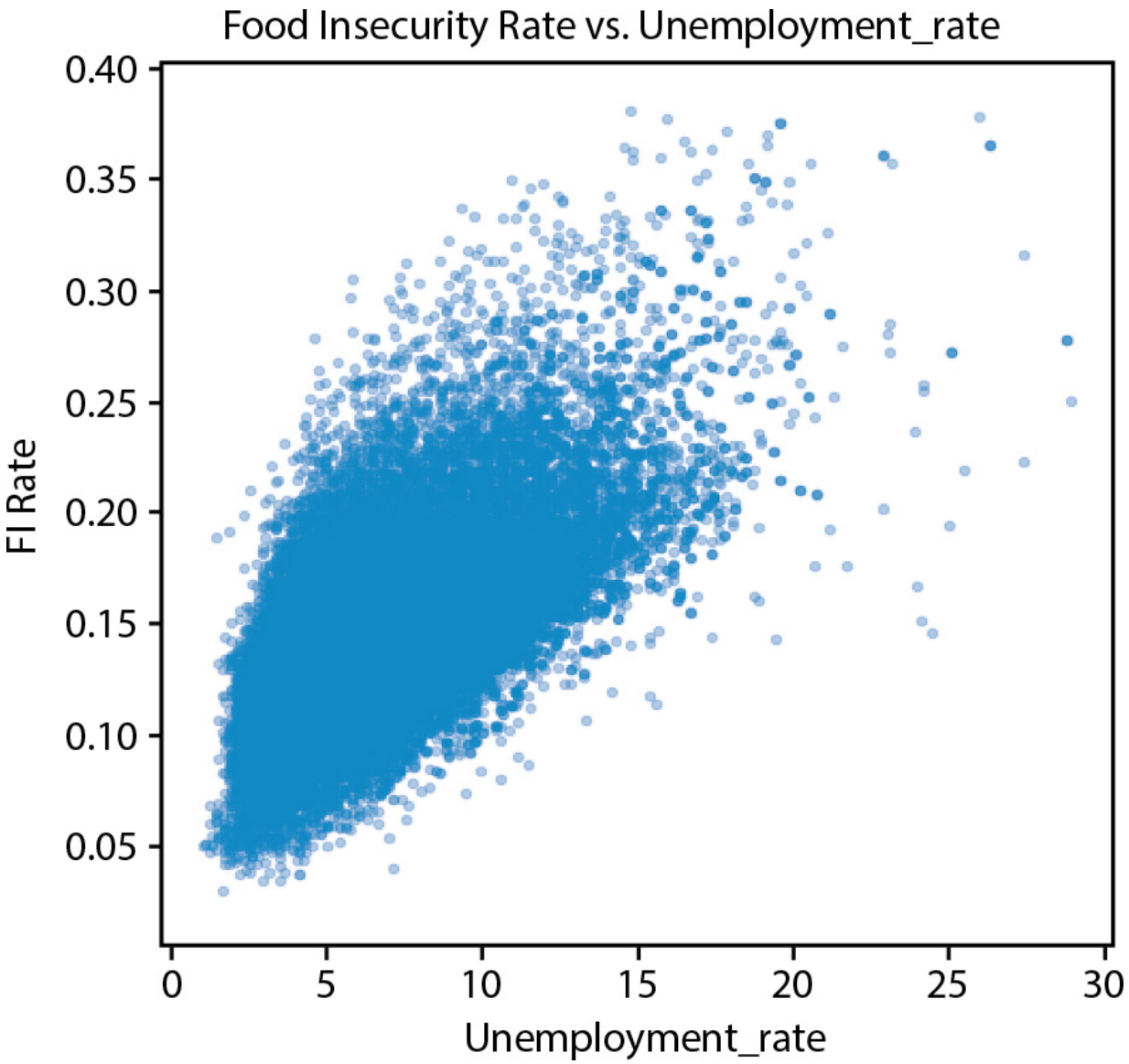

Figure 19.25 Scatter plot for FI rate vs. unemployment rate.

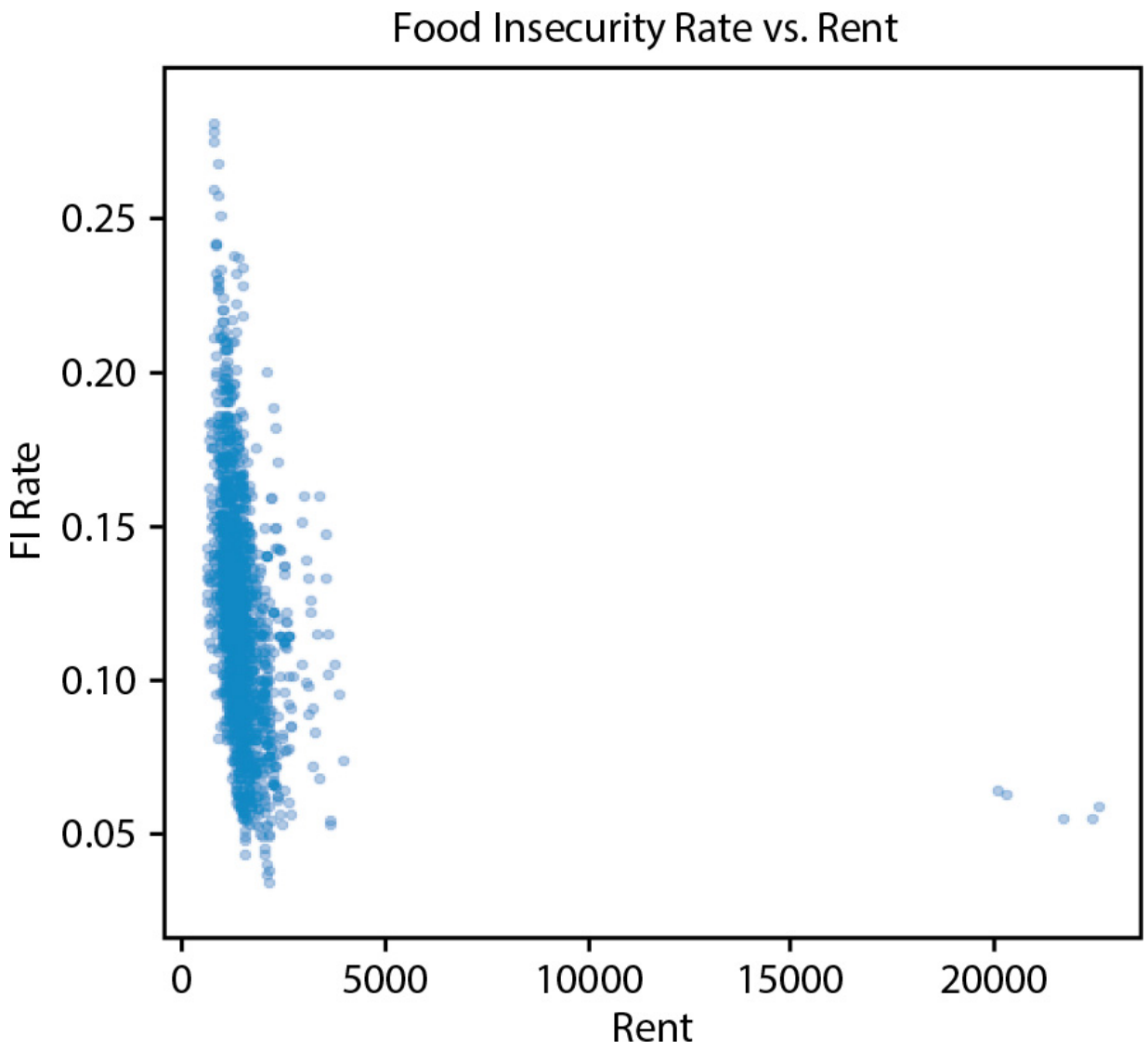

Figure 19.26 Scatter plot for FI rate vs. rent.

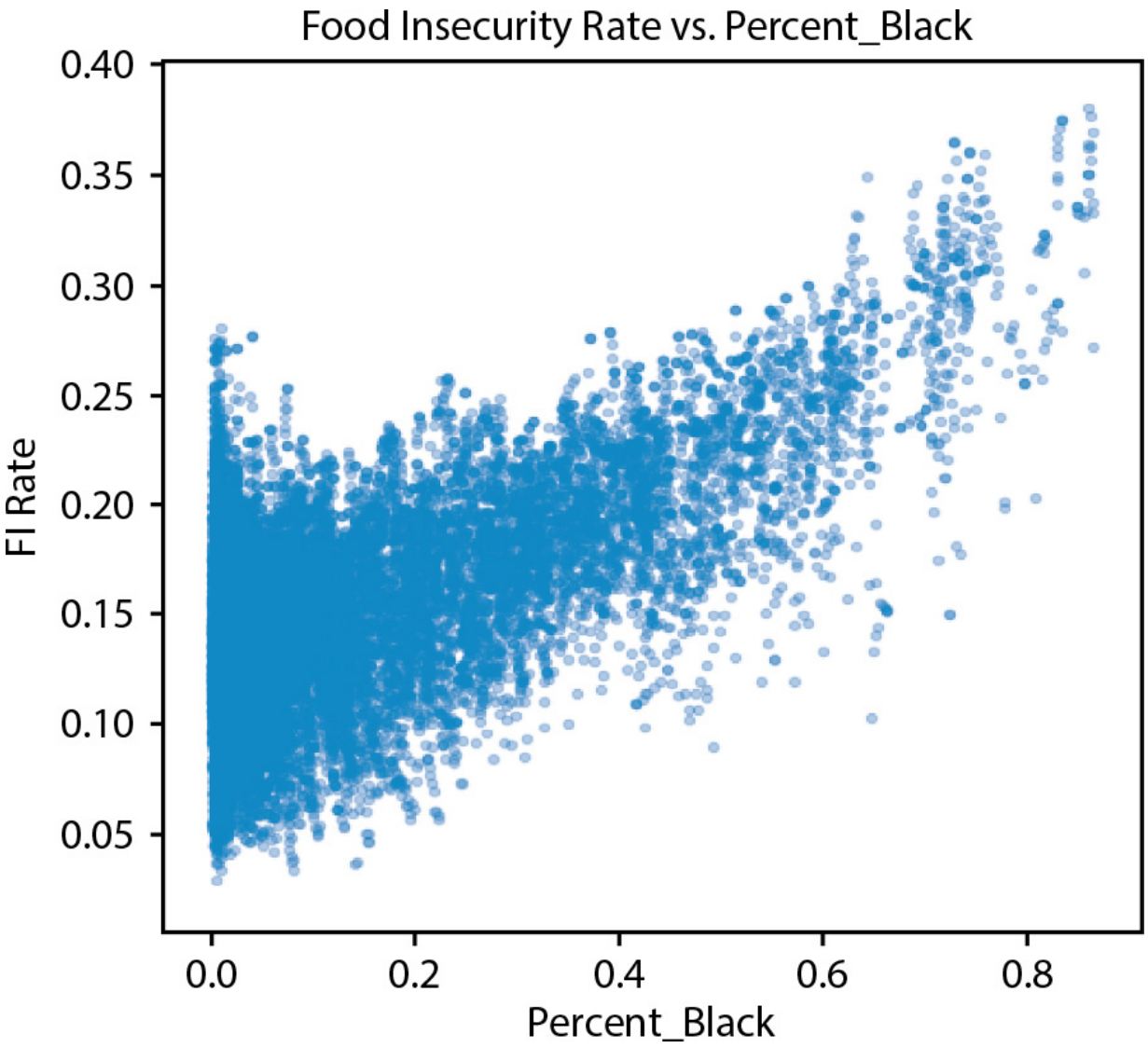

Figure 19.27 Scatter plot for FI rate vs. black population.

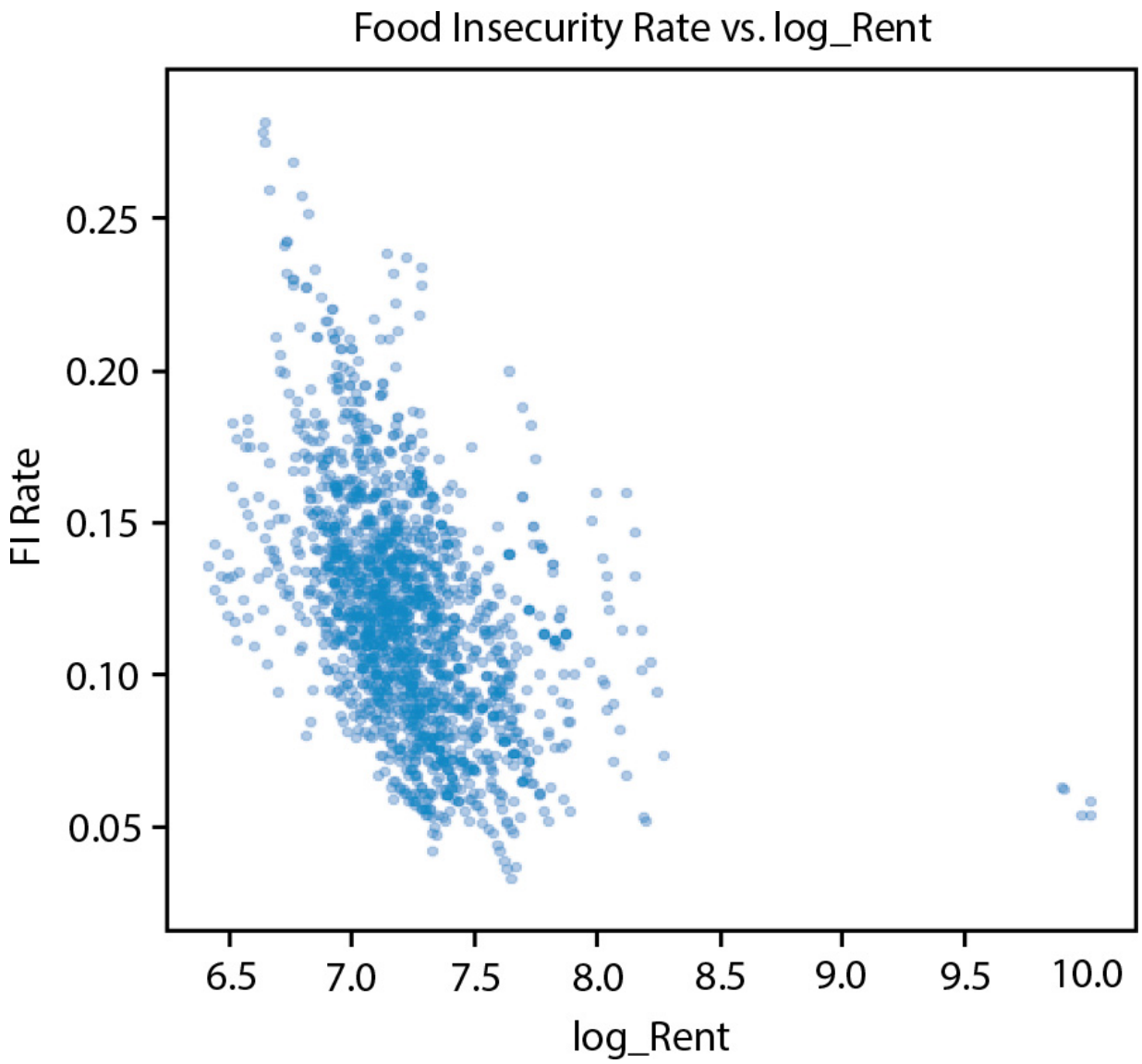

Figure 19.28 Scatter plot for FI rate vs. log rent.

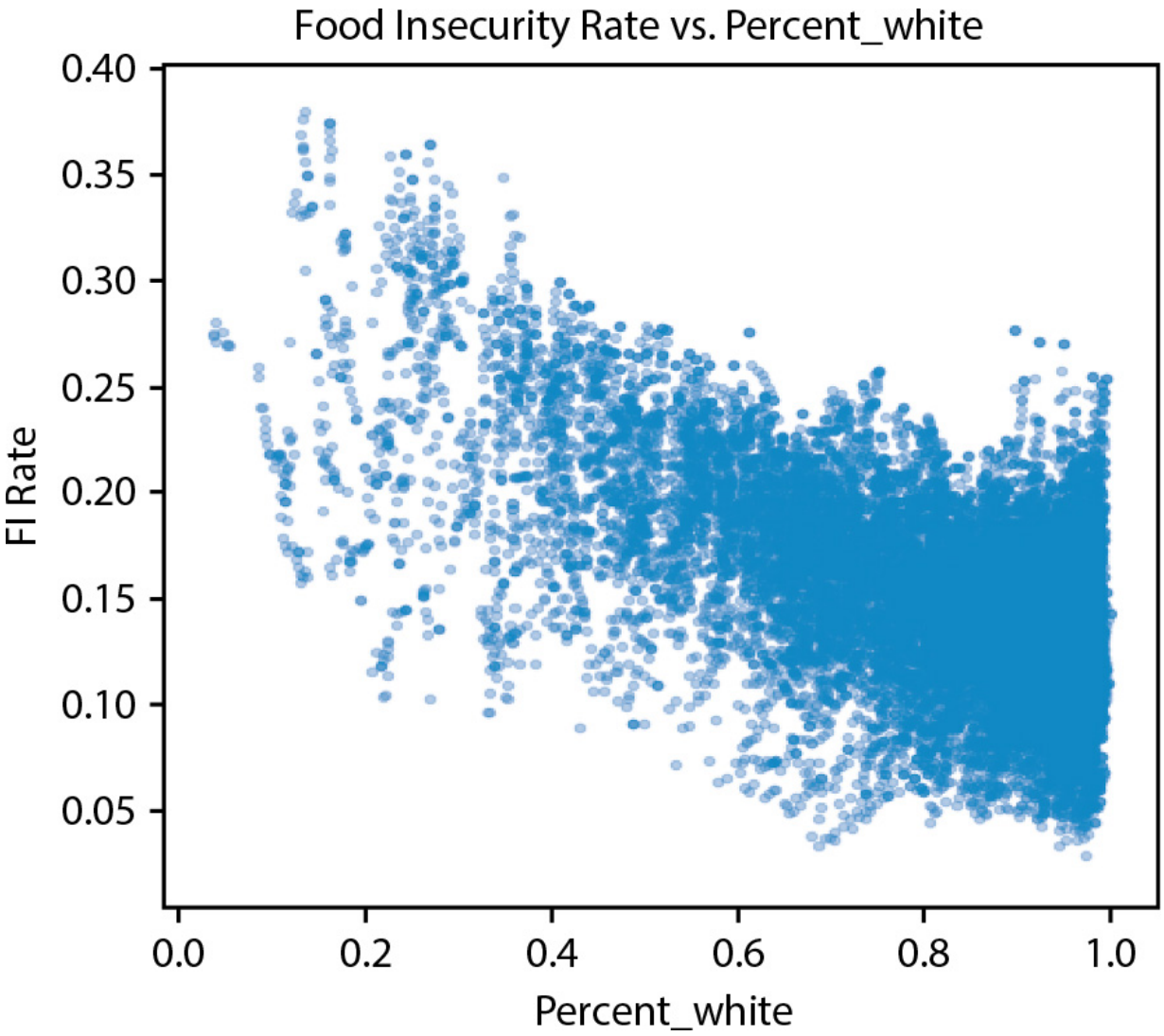

Figure 19.29 Scatter plot for FI rate vs. white population.

19.6.4 Conclusion for Exploratory Data Analysis

This data analysis delves into the following queries using the perspectives of time and geography:

- In what ways have metrics like unemployment, houselessness, and food insecurity rates evolved over a period of time?

- What is the geographic variability in unemployment, race, and food insecurity rates?
- How are factors such as rent prices, unemployment, houselessness, and race related to the food insecurity rate?

Based on the analysis, strong correlations were identified between food insecurity rates, unemployment rates, homelessness rates, and race, regarding both the most affected geographic regions and the changes in these variables over time.

This analysis shows that communities of color, especially black communities, and those with high unemployment and homelessness rates are the most vulnerable to high FI rates. In addition, although some variables including FI rate, unemployment, and homelessness have declined, on average, over a period of time, the choropleth maps demonstrate that certain regions have experienced a deterioration in these factors over a period of time.

The insights gained from this exploratory data analysis are important for identifying the communities and regions most vulnerable to food insecurity rates. This information can be used to allocate resources to these areas and take proactive

measures to address the underlying factors that affect certain communities disproportionately.

19.7 Feature Engineering

Feature engineering refers to the procedure of choosing and converting raw data into meaningful features that can be utilized for training a machine learning model. This process involves comprehending the problem's domain, identifying pertinent features, and converting them into a machine learning algorithm-friendly format. Feature engineering typically involves a combination of domain knowledge and data analysis techniques, such as statistical analysis and data visualization. The goal is to create features that capture important patterns and relationships in the data while also minimizing noise and irrelevant information. Prior to the modeling process, additional features were created from the original dataset features. These newly engineered features were then saved and later imported into the modeling notebook. Those features were as follows:

1. Demographic percentage breakdowns

The utilization of demographic percentages enables the comparison of the demographic distribution among various

counties while taking into consideration differences in population density—for instance, a the new column “Percent_male” is created in the data frame “df ” by dividing the “TOT_MALE” column by the “TOT_POP” column [21].

2. Interaction features

For each combination of continuous features, a fresh set of interaction features is produced. Afterward, the most effective features are selected and kept in the primary dataset. This selection process entails running an ordinary least squares (OLS) model on each feature and comparing their associated R^2 values. The interaction features are generated through a straightforward calculation, as shown below.

By multiplying the values of two features, df[feature1] and df[feature2], we can derive an interaction feature, which is represented as df[feature1+“X”+feature2]. This technique can help account for the presence of multiple features in a dataset, and the way they coexist with one another. Suppose we have two identities, A and B. If an individual identifies with both identities, they may face a significantly greater risk of experiencing food insecurity compared to those who only identify with A or B separately [22].

3. Log transformations

One way to create log features is by utilizing the inherent logarithm of a particular feature and then adding this newly calculated feature to the existing data frame. This transformation can be particularly helpful in modeling data that has extreme outliers by subjecting high values to a greater penalty compared to lower values. In other words, log transformations can be instrumental in capturing the shape of data that has a skewed distribution, with a few exceptionally large values.

4. Dummy variables

To convert the categorical variables “high” and “low” threshold into a numerical format, we can create dummy variables using the “get_dummies()” function available in the Pandas library. This technique enables us to transform each categorical value into a binary representation of 1 or 0. In other words, for each unique value of the categorical variable, a new column is created with a binary value of 1 if the data point corresponds to that value and 0 otherwise.

19.7.1 Feature Selection and Modeling

Now comes the part of the feature selection and modeling process where the best model will be selected to predict food insecurity rates based on a number of features, both from the original datasets as well as the engineered features which were created in feature engineering. Various feature selection techniques were utilized to determine the different features used in each MVP model, all of which were simple linear regression models. The first step was to import the final cleaned dataset for the modeling process.

1. Model 1: all features

The first model was a simple linear regression model that trained on data from the years 2010–2018 and used all the engineered features created by feature engineering. The model used a total of 13,305 observations and a cross-validation with five folds. Table 19.5 indicates that the model produced a train/test R-square value of approximately 0.60, an average train/ test RMSE of 0.028, and a mean cross-validation *R*-square value of -6.71.

2. Model 2: remove outliers

The EDA section included a part that inspected features that have the largest outliers, using a boxplot and whiskers plot. The model places emphasis on the variables with the highest outliers, which consist of “TOT_POP”, “num_hh”, “Total_workforce”, “Employed”, “Unemployed”, “Total_workforce”, and “Total_food_retail”. They were visualized in the form of boxplots as seen in Figure 19.30. Subsequently, all these features were aggregated while taking into account the outliers. Specifically, for features with observations that were more than ±2 standard deviations away from the mean, the outliers were capped at ±standard deviations from the feature’s mean. After reducing the outliers, the boxplot was visualized again as mentioned in Figure 19.31. After that, the model is re-run and it is observed that it produces a slightly better train train/test R-square value of about 0.50, a worse average Train/Test RMSE of about 0.03, and a much lower mean cross-validation score of -16.31 as mentioned in Table 19.5. For this reason, outliers are retained in the next model.

3. Model 3: removed multicollinear features

The existence of multicollinearity can lead to interference and clutter within the dataset. During the exploratory data analysis

(EDA) phase, we generated a heatmap, shown in Figure 19.5, based on the correlation matrix. The heatmap revealed that various features, such as the houseless rate, sheltered/unsheltered rates, and all features related to race, exhibit a strong correlation with one another. The correlation matrix is reformatted to show the correlation between each feature pair—we see that demographics features, employment-related features, and houselessness-related features are all highly correlated, which is to be expected.

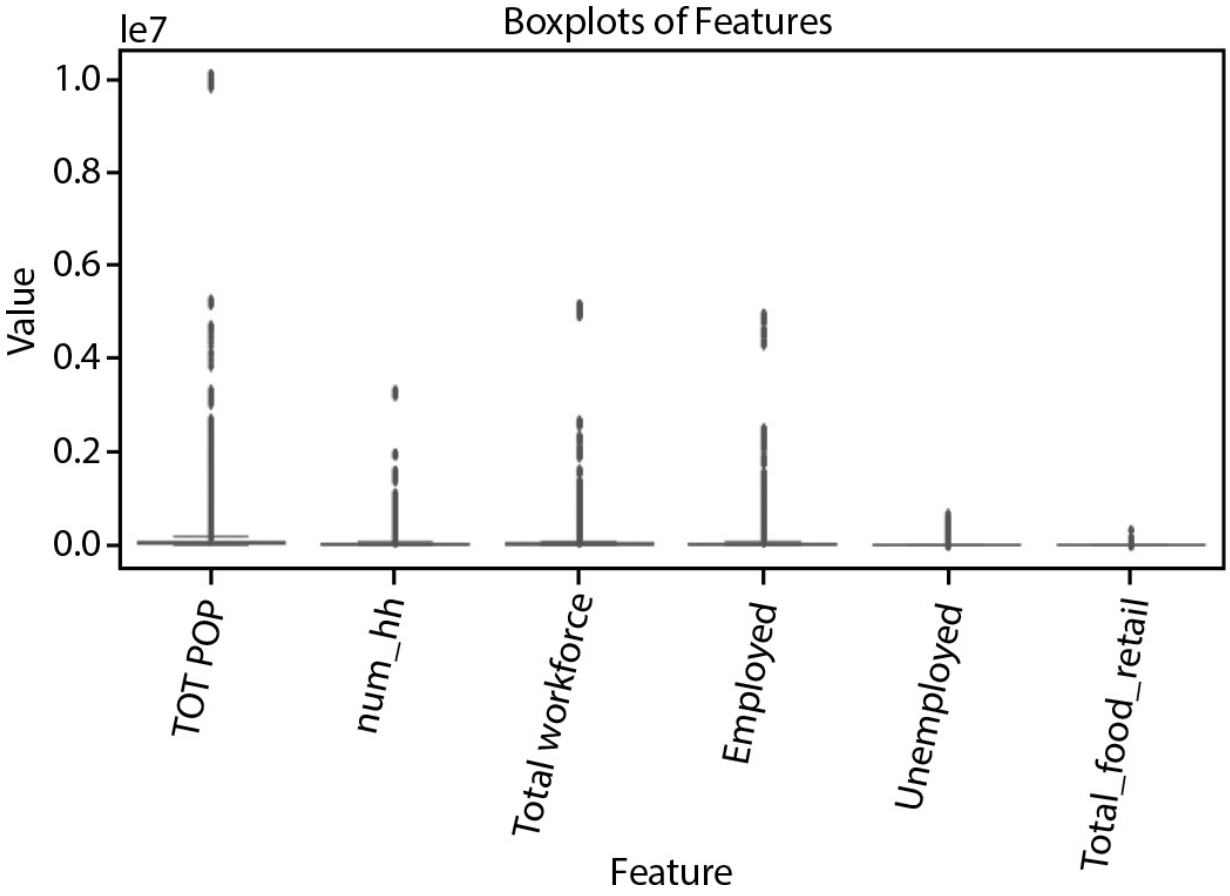

Figure 19.30 Boxplot for the features with the largest outliers.

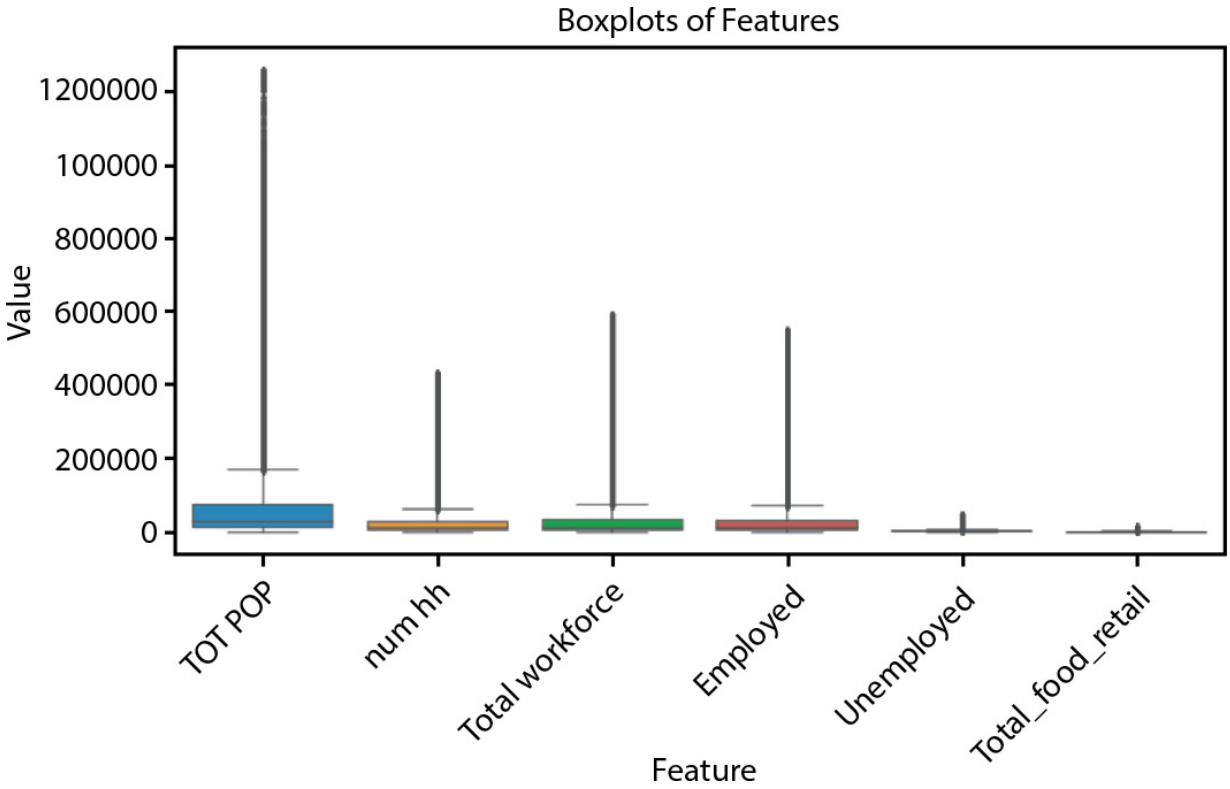

Figure 19.31 Boxplot of features after the reduction.

The variance inflation factor (VIF) is a metric that measures the degree of multicollinearity in a regression model by examining the variance of the model across different terms. A high VIF score suggests a significant degree of multicollinearity. We generated a new data frame that contains only those features with VIF scores that are below 40 as mentioned in Table 19.4. For model 3, we exclusively utilize those features that have VIF scores below 50. This model performs noticeably better than its predecessors, yielding a train/test R-square value of approximately 0.87, a train/test RMSE of around 0.0154, and a

substantially elevated mean cross-validation score of 0.834 as mentioned in Table 19.5.

4. Model 4: K-best selected features

To arrive at an optimal set of features for model 4, we utilize the SelectKBest() function that determines the best “k” number of features to be employed in the model. After a thorough evaluation, we determine the optimal value of “k” to be 76, and model 4 is then trained using these features. Model 4 outperforms model 3 marginally, with an average R-square value of 0.86, an RMSE of 0.0157, and an average cross-validation score of 0.844 as mentioned in Table 19.5.

Table 19.4 VIF factor of selected features.

	VIF factor	Features
0	1.009854	Houseless_rate
1	1.091869	Sheltered_rate
2	0.990675	Unsheltered_rate
3	7.380262	hh_med_income
4	0.541568	percent_hh_poverty

Table 19.5 Performance table for all the models.

	R^2 train	R^2 test	RMSE train	RMSE test	CV score
Model 1	0.5981	0.5990	0.0270	0.0273	-6.7122
Model 2	0.5072	0.5141	0.0300	0.0297	-16.3060
Model 3	0.8712	0.8669	0.0153	0.0154	0.8340
Model 4	0.8662	0.8625	0.0156	0.0157	0.8439
Model 5	0.8975	0.8900	0.0137	0.0140	0.8481

5. Model 5: recursive feature elimination

We employ recursive feature elimination as a technique to systematically eliminate features and retrain the model, with

the aim of identifying an optimal subset of features to use. The RFE model exhibits a slight improvement over the Select K Best model, with an average R-square value of 0.89, an RMSE of 0.014, and an average cross-validation score of 0.848 as mentioned in Table 19.5.

After evaluating the different MVP models, we determined that model 5, which utilized recursive feature elimination, emerged as the best model. The top 10 features and their corresponding coefficients were examined in detail. Table 19.6 displays the highest coefficients alongside their corresponding features. From the output, it was observed that houseless rates, race, food access, and income are the highest predictors for food insecurity.

After testing all the models, three more models were tested in order to obtain more promising predictions on both labeled and unlabeled data. The three models that were tested were as follows:

1. neural network regressor
2. XGBoost regressor
3. random forest model evaluation

After testing all the models, it was observed that neural network model produces a cross-validated RMSE of 0.0342, which underperforms the linear regression models; hence, it was immediately rejected. Then comes the testing of the XGBoost Regressor and Random Forest Regressor; both of them yielded good results as mentioned in Table 19.7. Upon examination, it was noted that the RFE random forest regressor could account for approximately 88% of the variance in the data, as demonstrated by the cross-validation R^2 score of 0.884. Furthermore, the average prediction error of the model was only 1.5%, as indicated by the test RMSE score of 0.015. Hence, this model, along with the best linear regression model, was further evaluated for making accurate predictions on unlabeled data to generate estimates for 2020 national food insecurity rates.

Table 19.6 Highest coefficients alongside their corresponding features.

Score	Feature
77,127.1074	Unsheltered_rate^2
7,635.6632	Houseless_rate
1,171.7152	Percent_hh_SNAP
670.5564	Percent_hh_no_vehicle^2
92.7628	Percent_white^2
59.4265	Num_restaraunts^2
34.1305	Percent_Black_X_hh_med_income
27.8781	Percent_latinX_X_num_hh
13.5304	Percent_hh_no_vehicle^3
11.3193	Num_wholesale^3

Table 19.7 Performance data.

	R^2 train	R^2 test	RMSE train	RMSE test	CV score
XGBoost	0.9195	0.9021	0.0044	0.0115	0.8598
Random forest	0.9891	0.9241	0.0044	0.0115	0.8838

19.8 Predictions

In this section, we have the final random forest model and linear regression model that were used to make predictions on labeled and unlabeled data, respectively. The random forest regressor was found to produce the best results when predicting labeled data, with an average train/test R^2 of 0.97 and an average train/test RMSE of 0.007. Additionally, the model was cross-validated with an RMSE of 0.139. This random forest model was saved using Pickle and is imported into this notebook.

Furthermore, we can interpret the feature importance of the random forest model, which can provide insights into which

features are most predictive of food insecurity rates. By examining the feature importance values, we can see which variables had the greatest impact on the model's predictions. This information can be used to identify potential factors that contribute to food insecurity and inform policies and interventions aimed at addressing this issue.

To visualize the predictions made by the best RFE model on all 2018 data, we first needed to make these predictions. After making the predictions, we inspected them side-by-side with the actual values to see how well the model performed as mentioned in Table 19.8 [23]. We then used a choropleth map to display both the predicted values and the actual values for 2018. The choropleth maps shown in Figure 19.32 revealed that the model successfully captured most of the variance within the data, particularly in places that were significantly impacted. However, in some areas, such as Texas, the model fell slightly short of capturing the severity of food insecurity rates. It is worth noting that some counties contained missing values, represented by white areas in the choropleth maps. These missing values were a result of inadequate food retail data for those counties in 2018. We chose to retain these counties instead of dropping them altogether, as dropping them significantly decreased the model's performance. We dropped null rows instead to deal with the missing data.

To generate predictions for the 2020 food insecurity (FI) rates, a model was trained using data from 2010 to 2018, while being restricted to the features available for 2020 data. Features such as houselessness, food retail, number of households, vehicle access, and SNAP benefits were excluded from the model. Once the model was trained on labeled data, it was used to make predictions for the 2020 FI rates.

Table 19.8 Comparison table for FI rate and predicted FI rate.

	FI rate	Predicted FI rate
3,035	0.132	0.136520
970	0.108	0.110953
599	0.081	0.085835
1,755	0.157	0.155760
1,419	0.105	0.106518
610	0.084	0.086490
1,901	0.169	0.178568
2,923	0.131	0.127643
516	0.146	0.146285
2,935	0.116	0.115575

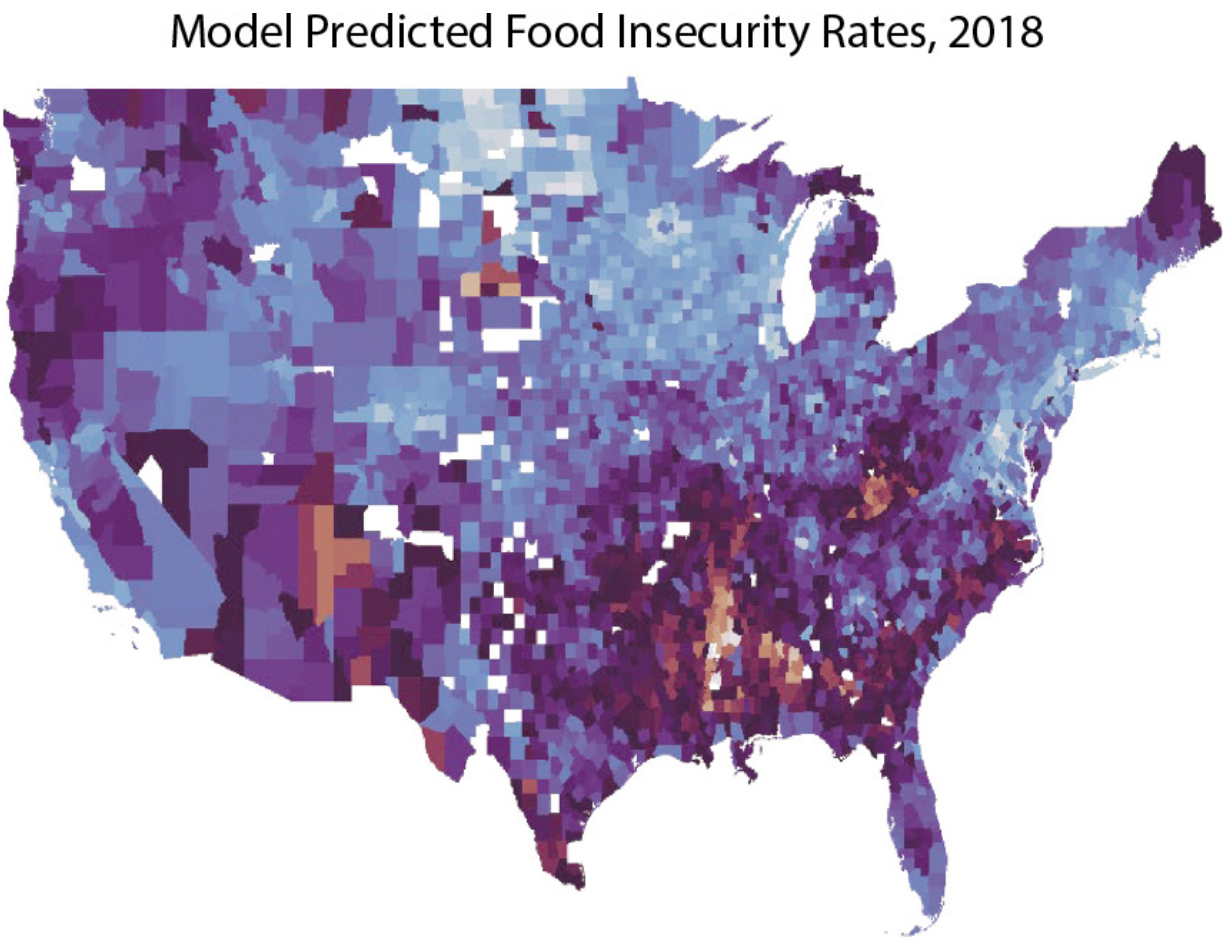

Figure 19.32 Model predicting food insecurity rates of the year 2018.

Linear regression was used with RFE features to make predictions for 2020 because it provided consistent results after removing the aforementioned features. The 2020 linear regression model had an average train/test R^2 of 0.846, an average train/test RMSE of 0.165, and a cross-validation RMSE of 0.81. Though the performance was reduced due to the dropped features, the model was still reasonable to make predictions [24].

Unlike previous models that considered all features, the current model relied on factors such as race, education, unemployment, and healthcare access to model the data. Feature importance was analyzed, and it was found that houselessness and food proximity were not as critical to the model's predictions.

The newly trained model was used to make predictions on the unlabeled data for 2020 food insecurity rates as mentioned in Figure 19.33. Feeding America generated similar predictions on the 2020 FI rates as compared to the actual rates from 2018. However, the model predicted more severe food insecurity throughout the country, particularly in the south and along the West Coast, which displayed more variance in predictions as shown in Figure 19.34.

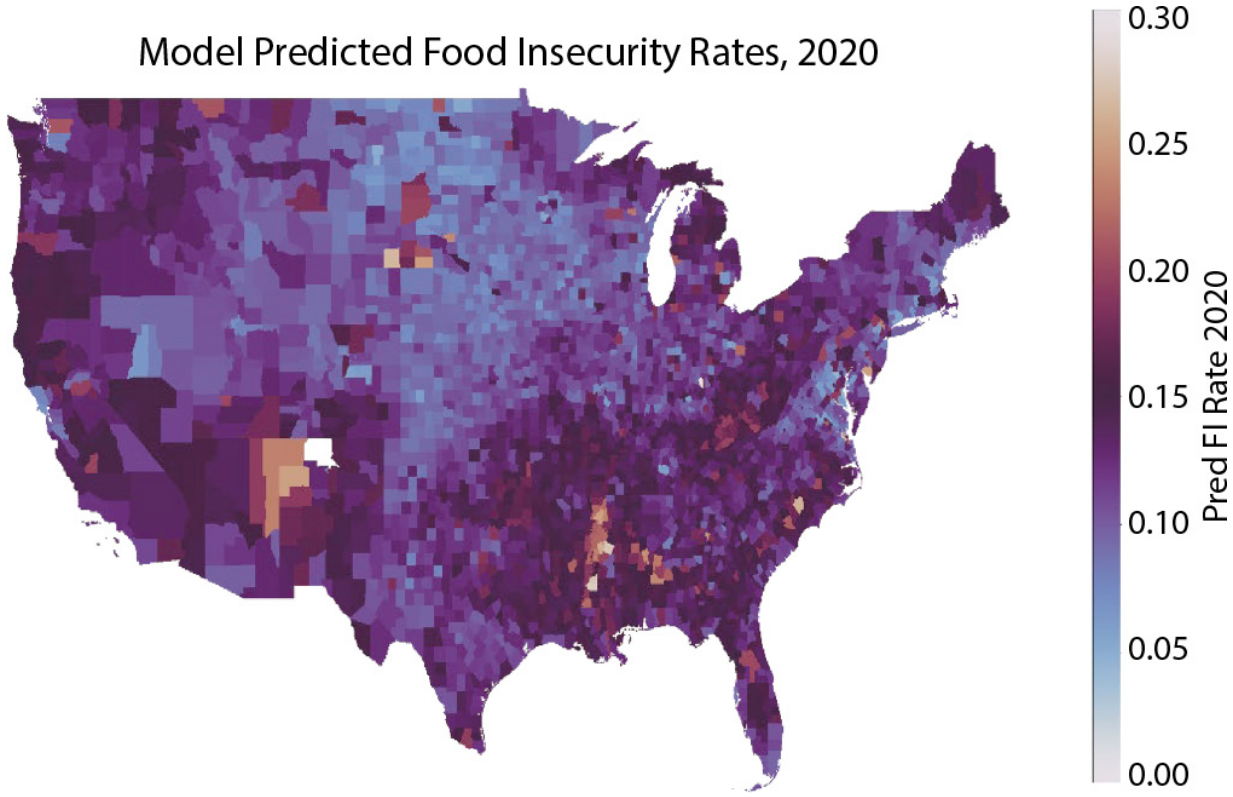

Figure 19.33 Model predicting food insecurity rates of the year 2020.

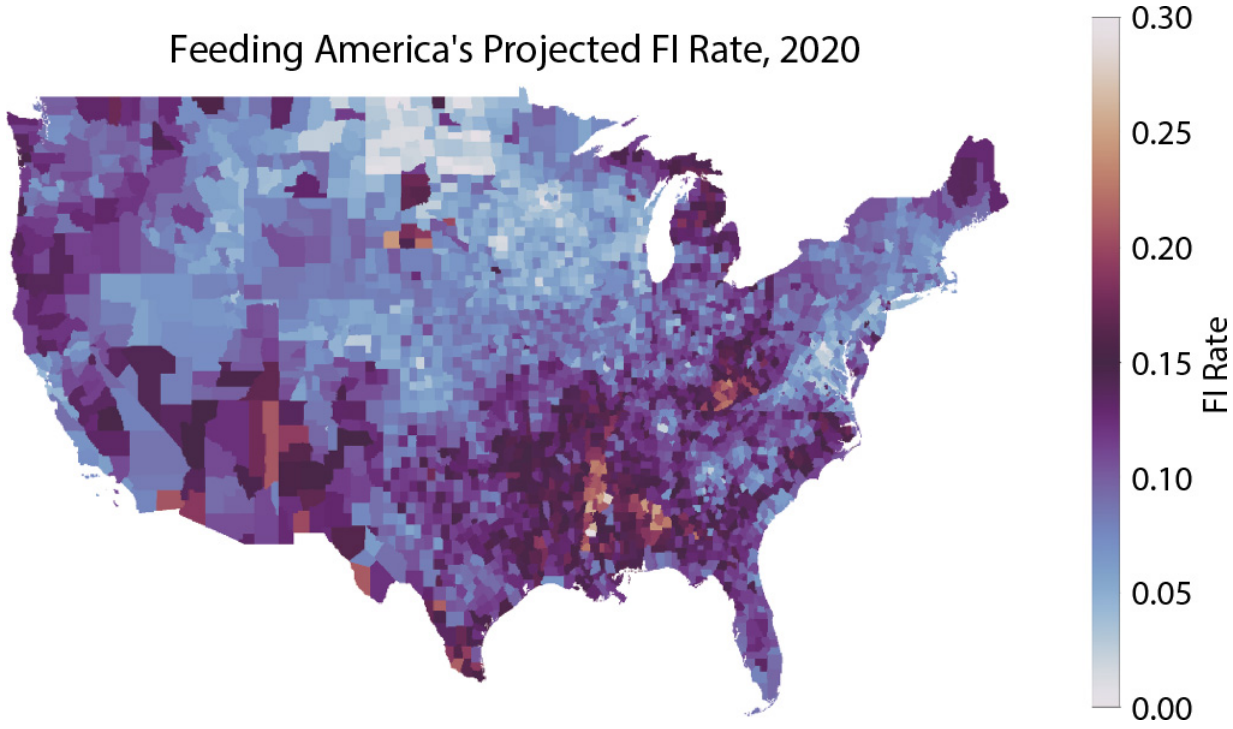

Figure 19.34 Feeding America's projected FI rate for the year 2020.

19.9 Conclusion

In conclusion, the linear regression model developed in this study has shown a remarkable ability to predict and explain food insecurity issues in the United States. Based on the RFE random forest regressor, the model was capable of explaining roughly 88% of the variance in the data, as determined by the cross-validation R^2 score of 0.884. The model exhibited an average prediction error of only 1.5%, as evidenced by the test RMSE score of 0.015. The model coefficients identified the following features as the most important ones:

1. Unsheltered_rate_X_Percent_asian,
2. Sheltered_rate_X_Percent_white,
3. Sheltered_rate_X_Percent_Black,
4. Houseless_rate_X_Percent_asian
5. Unsheltered_rate_X_Percent_male

The findings indicate that both houselessness and race have a significant impact on the probability of food insecurity, particularly when these features interact with each other—for example, an individual who is both black and houseless may face an even higher risk of food insecurity.

Additionally, the choropleth maps generated to visualize the 2018 food insecurity rate predictions exhibit the model's capability to detect the overall trend of food insecurity, particularly in the areas that are predominantly affected. In general, these study results can provide guidance to policymakers in addressing the underlying causes of food insecurity and identifying targeted interventions that can be most effective in alleviating its impact on vulnerable communities [25].

To summarize, this research offers a convincing argument for utilizing linear regression modeling to address food insecurity challenges in the United States, and the valuable findings obtained from this study can be an essential tool for stakeholders striving to establish a fair and equitable society.

References

1. Rose, D., & Oliveira, V. (1997). Nutrient intakes of individuals from food-insufficient households in the United States. *Am. J. Public Health*, 87(12), 1956-1961.
2. Nord, M. (2009). Food insecurity in households with children: Prevalence, severity, and household characteristics. *Econ. Res. Rep.*, 83, United States Department of Agriculture, Economic Research Service.

3. Alaimo, K., Olson, C. M., & Frongillo, E. A. (2001). Food insufficiency and American school-aged children's cognitive, academic, and psychosocial development. *Pediatrics*, 108(1), 44-53.
4. Gundersen, C., & Ziliak, J. P. (2015). Food insecurity and health outcomes. *Health Aff.*, 34(11), 1830-1839.
5. Seligman, H. K., Laraia, B. A., & Kushel, M. B. (2010). Food insecurity is associated with chronic disease among low-income NHANES participants. *J. Nutr.*, 140(2), 304-310.
6. Hamelin, A. M., Beaudry, M., & Habicht, J. P. (2002). Characterization of household food insecurity in Quebec: food and feelings. *Soc. Sci. Med.*, 54(1), 119-132.
7. Loopstra, R., Reeves, A., Taylor-Robinson, D., Barr, B., McKee, M., & Stuckler, D. (2015). Austerity, sanctions, and the rise of food banks in the UK. *BMJ*, 350, h1775.
8. Sharkey, J. R., & Johnson, C. M. (2010). Food insecurity and social capital in rural and urban communities in Texas. *Rural Sociol.*, 75(2), 234-262.
9. Coleman-Jensen, A., & Nord, M. (2013). Food insecurity among households with working-age adults with disabilities. *J. Disabil. Policy Stud*, 24(4), 219-230.
10. Jyoti, D. F., Frongillo Jr, E. A., & Jones, S. J. (2005). Food insecurity affects school children's academic performance, weight gain, and social skills. *J. Nutr.*, 135(12), 2831-2839.

11. Berkowitz, S. A., & Seligman, H. K. (2013). Food insecurity, school absenteeism, and educational attainment: A longitudinal study. *J. Pediatr.*, 163(5), 1320-1327.
12. Martin, K. S., Rogers, B. L., Cook, J. T., & Joseph, H. M. (2004). Social capital is associated with decreased risk of hunger. *Soc. Sci. Med.*, 58(12), 2645-2654.
13. Loopstra, R., Lalor, D., & Higgins, A. (2017). Food insecurity, mental health and suicide: A systematic review. *Soc. Sci. Med.*, 189, 153-163.
14. Coleman-Jensen, A., Rabbitt, M. P., Gregory, C. A., & Singh, A. (2019). Household Food Security in the United States in 2018 (No. 270). United States Department of Agriculture, Economic Research Service.
15. Cook, J. T., Frank, D. A., Berkowitz, C., Black, M. M., Casey, P. H., Cutts, D. B., ... & Heeren, T. C. (2004). Food insecurity is associated with adverse health outcomes among human infants and toddlers. *J. Nutr.*, 134(6), 1432-1438.
16. Ambinakudige, S., Parisi, D., Cappello, G.C. *et al*. (2017). Diversity or Segregation? A Multi-decadal Spatial Analysis of Demographics of Atlanta Neighborhoods. *Spat. Demogr.* 5, 123144.
17. Ahern, Melissa, Cheryl Brown, and Stephen Dukas. (2011). “A national study of the association between food environments

and county-level health outcomes." *J. Rural Health* 27.4 (2011): 367-379.

18. Dinour, L. M., Bergen, D., & Yeh, M. C. (2007). The food insecurity-obesity paradox: A review of the literature and the role food stamps may play. *J. Am. Diet Assoc.*, 107(11), 1952-1961.
19. Bhattacharya, J., Currie, J., & Haider, S. (2004). Poverty, food insecurity, and nutritional outcomes in children and adults. *J. Health Econ.*, 23(4), 839-862.
20. Hamelin, A. M., Mercier, C., & Bédard, A. (2011). Discrepancies in household food security measurement: Are they due to cognitive interviewing probing techniques? *Qual. Health Res.*, 21(6), 730-740.
21. Tarasuk, V., & Vogt, J. (2009). Household food insecurity in Ontario. *Can. J. Public Health*, 100(3), 184-188.
22. Loopstra, R., & Hocking, R. R. (2017). Household food insecurity and precarious employment: A systematic review. *Int. J. Epidemiol.*, 46(5), 1603-1614.
23. Poppendieck, J. (1999). *Sweet charity?: Emergency food and the end of entitlement*. Penguin Books.
24. Loopstra, R., & Reeves, A. (2015). Rising food insecurity in Europe. *Food Ethics*, 1(2), 119-133.
25. Mitra, Dipra, Shikha Gupta, Durgesh Srivastava, and Sudeshna Sani. "Plant Disease Identification Using

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Index

- Accuracy, 419
- Adoption, 501, 502
- Advanced technologies, 1, 2, 20
- Agricultural farmers, 366
- Agricultural robots, 59, 68
- Agriculture, 169 to 195, 201, 202, 203, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 227, 228, 229, 309, 310, 320, 351, 354, 356, 363, 367, 368, 369, 371, 372, 394, 396, 397, 401, 402, 444, 445, 446, 479, 483, 484
- agriculture industry, 194
- Agriculture Informatics, 143, 145, 167
- Agriculture System, 113, 114, 115, 116, 117, 118, 119, 121, 122, 123, 124, 125, 126, 127, 129, 132, 134, 136
- AI, 75, 76, 77, 78, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 113, 118, 121, 122, 124, 144, 146, 147, 148, 149, 150, 151, 152, 153, 155, 156, 157, 159, 165, 166, 201, 202, 203, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 221, 222, 223
- AI in Agriculture, 589, 590, 592
- Algorithms, 443, 444, 446, 447, 448, 449, 450, 452, 455, 456, 457, 458, 464, 465, 469, 470, 471, 472, 473, 474, 475, 476, 477,

478, 479, 481, 482, 483, 484, 485
- AMQP Protocol, 271, 275, 277, 278
- Apple, 421
- Apple Leaf Disease, 436
- Aquatic plants, 351, 353, 355, 357, 358, 359, 360, 361, 366, 369
- Arduino UNO, 271, 290
- Artificial intelligence, 201, 210, 227, 229, 252, 309, 310, 324, 326, 351, 353, 359, 360, 361, 364, 366, 370, 371, 398, 589, 590, 591
- Automated Farming, 59
- Automation, 29, 39, 48, 49, 50
- Autonomous robots, 2
- auto-watering, 329, 336, 347

- Banana Black Sigatoka, 394, 407
- Banana leaf spot, 394, 407
- Banana panama wilt, 394, 407
- Banana plant, 405
- Banana Yellow Sigatoka, 394, 407
- Barriers, 325
- Benefits, 316, 323
- Black Rot, 429
- Block Chain, 494
- BOD, 354, 355, 356

- Cattails, 353, 358
- Causes, 505, 507, 508, 509, 542
- Challenges, 75, 76, 77, 78, 79, 80, 81, 82, 83, 85, 86, 87, 89, 90, 91, 92, 93, 95, 96, 98, 99, 100, 101, 103, 104, 105, 106, 107, 108, 109, 110
- Chronic disease, 505, 506, 507, 508, 510, 542
- Classification accuracy, 393, 395, 401, 404, 405, 414
- Climate, 490
- Climate change, 1, 19, 20, 113, 114, 116, 117, 135, 136
- Cloud, 315, 322, 323, 324, 325
- CNN, 396, 397, 399, 400, 404, 405, 410, 425
- CoAP Protocol, 270, 275, 276, 278
- Communication, 227, 228, 229
- Computer Vision, 122, 124, 125, 126
- Connectivity, 75, 78, 79, 81, 82, 85, 87, 101, 103, 105, 107
- Consequences, 505
- Controllers, 309, 310, 311, 312, 314
- Convolutional neural network, 452, 460, 461, 470, 474, 476, 477
- Correlation, 505, 506, 507, 509, 516, 517, 518, 527, 529, 535
- cost, 316, 317, 318, 320, 321
- Cost Savings, 502
- Costs, 75, 76, 83, 84, 85, 86, 87, 89, 90, 91, 93, 99, 100, 102, 108

- County, 511, 512, 513, 514, 515, 516, 521, 522, 523, 524, 533, 539, 543
- crop, 201, 202, 203, 204, 205, 206, 207, 211, 212, 213, 214, 215, 216, 217, 220, 221
- Crop Health, 33, 34, 36, 41, 46, 170, 173–175, 178, 185, 194
- Crop management, 2, 10, 11, 15, 30, 41, 42, 44, 47, 170 to 177, 186, 193, 195
- Crop Monitoring, 34, 41, 44
- crop yields, 170 to 177, 180 to 185, 188, 193
- Crops, 58, 59, 61, 62, 113, 114, 116, 118, 119.121.123, 124, 131, 134, 135, 329, 331, 332, 333, 335, 336, 338, 340, 343, 334, 345, 347, 348
- cross-validation, 456, 458

- Data analysis, 22
- Data Analytics, 30, 39, 42, 76, 77, 78, 80, 81, 82, 87, 92, 95, 103, 106, 107, 169, 170, 171, 189, 191–196
- Data integration, 364, 365
- Data Privacy, 491502
- Data Processing, 68, 70
- Data Security, 499
- Dataset, 505, 510, 511, 512, 513, 515, 517, 525, 526, 533, 534
- Datasets, 426
- Decision, 381, 382, 386

- decision tree, 450, 452, 455, 456, 457, 458, 459, 472, 476, 480, 483
- decision-making, 169, 170, 171, 179, 183, 185, 186 to 196
- Deep Learning, 393, 396, 400, 401, 403, 406, 411, 414, 427, 443, 452, 460, 461, 469, 471, 476, 481
- Deep Learning model, 359, 360, 372
- desert areas, 329, 330
- Digital Farming, 57, 58, 59, 61, 63, 65, 67, 69, 70, 71, 72, 73
- Dimensionality reduction, 447, 454, 472, 473, 482
- Drone, 201, 203, 207, 211, 212, 213, 215, 221, 374, 375, 378, 381, 382, 389
- Drones, 1, 2, 8, 16, 23, 25, 30, 34, 36, 37, 45
- DSM, 382

- Economic Barriers, 75, 76, 77, 78, 84, 85, 89, 99, 101, 102, 103, 104, 106, 108
- Economic feasibility, 362
- Education, 76, 78, 89, 90, 91, 92, 93, 100, 101, 102, 103, 104, 105, 106, 107
- Efficiency, 5, 9, 12, 16, 19, 20, 22
- Enhanced crop yields, 502
- Ensemble methods, 443, 450, 457, 459, 460, 465, 476, 481
- Environment, 75, 77, 89, 100, 101, 102, 103, 104, 105, 106, 107, 108, 113, 114, 115, 116, 118, 130, 131, 132

- Environmental impact, 1, 3, 5, 172 to 174, 183
- Epochs, 407, 409, 410, 411
- Equipment, 71, 72
- Exploratory Data Analysis, 513, 515, 517, 525, 532, 534

- farm management, 208, 209
- Farmer, 113, 114, 115, 118, 121, 122, 123, 124, 126, 128, 129, 131, 132, 133, 134, 135, 136
- farmers, 201, 202, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 219, 221, 223
- Farming, 33, 34, 35, 36, 41, 42, 44, 46, 48, 50, 52, 310, 317, 330, 331, 333, 342
- Farming System, 113, 114, 115, 116, 117, 123, 135
- feature engineering, 443, 453, 455, 456, 458, 467, 478, 481, 482
- Fertility, 113, 115, 120, 121
- fertilizer, 170, 171, 175, 176, 183
- Finance, 87
- Fire Blight, 427
- Food Insecurity, 505, 506, 507, 508, 509, 510, 511, 515, 517, 520, 521, 522, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 537, 538, 539, 540, 541, 542, 543, 544
- Food production, 1, 116, 117, 118, 119, 122, 134, 135, 136
- Food Security, 113, 114, 115, 116, 119, 121, 124, 126
- Footprints, 354, 364, 366

- GIS, 387
- Global population, 1, 2
- Global Positioning, 57, 61
- GPS, 29, 30, 33, 34, 36, 41, 201, 212, 213, 218, 309, 382
- Growing Population, 116, 117, 118, 126
- Growth, 376, 382

- Harvesting, 33, 37, 38, 42, 43, 45, 49
- Harvesting robots, 38, 39
- Health, 376, 377, 378, 379, 381
- Heavy metals, 351, 352, 355, 358, 366, 369
- Herbicides, 64, 66, 70
- Horticulturists, 437
- Household, 508, 509, 505, 507, 510, 511, 515, 540, 542, 543, 544
- Houseless Rate, 511, 518, 527, 528, 535, 537
- HTTP Protocol, 270, 277
- Humidity Sensors, 284, 285, 286
- hyperparameter tuning, 447, 460, 473, 474, 482

- Identification, 420
- image, 342, 348
- Image Acquisition, 425
- Income Inequality, 508, 510
- Indian farmers, 494

- Innovation, 496, 497
- Internet of Things (IoT), 2, 5, 6, 18, 201, 203, 204, 205, 206
- Interoperability, 268, 274, 275
- Investment, 78, 79, 82, 84, 85, 87, 88, 91, 95, 99, 100, 101, 102, 103, 104, 105, 106, 107
- IoT, 57, 59, 67, 69, 70, 71, 75, 77, 78, 83, 84, 93, 106, 108, 109, 110, 113, 114, 126, 127, 128, 129, 130, 143, 144, 146, 148, 149, 151, 152, 153, 154, 155, 156, 160, 165, 166, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 212, 213, 217, 218, 221, 222, 223, 227, 228, 229, 310, 315, 322, 324, 325, 329, 332, 333, 336, 337, 339, 396, 397, 398, 589, 590, 591, 591
- IoT Enabling Technology, 270
- Irrigation, 58, 60, 71, 170 to 175, 183, 186, 194, 195, 202, 203, 204, 205, 206, 208, 213, 218, 219, 220, 351, 353, 354, 355, 356, 357, 362, 363, 365, 366, 367, 368, 369
- irrigation efficiency, 329, 330
- irrigation system, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 343, 344, 345, 346, 347
- Irrigation systems, 2, 12, 15, 19, 22, 24

- Knowledge, 75, 76, 78, 82, 84, 89, 90, 92, 93, 101, 104, 105, 106, 107

- large-scale agricultural, 329, 331, 332

- LDR, 287, 297
- LiDAR, 387
- linear regression, 450, 452, 455, 456, 457, 458, 471, 476, 480
- Linear Regression Model, 505, 506, 507, 510, 534, 537, 540, 541, 542
- Livestock, 202, 203, 207, 208, 211, 212, 215, 217
- Local area network (LAN), 267, 270

- Machine Learning, 76, 77, 78, 99, 110, 170, 171, 181 to 183, 189, 324, 333, 335, 337, 341, 347, 348, 426, 443, 444, 446, 447, 448, 449, 450, 451, 452, 453, 455, 456, 462, 464, 468, 469, 470, 471, 472, 473, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485
- Mapping, 376, 382, 386, 387
- mathematical models, 329, 330, 335, 348
- meteorological data, 177, 178
- Milking robots, 38, 39, 40
- Mobile Net, 393, 405, 406, 410, 411, 412, 413, 414
- Modern farming, 1
- Monitoring, 374, 375, 378, 379, 380, 381
- Motion Sensor, 289, 295, 296
- MQTT Protocol, 270, 271, 272, 275

- Naïve bayes, 465
- Natural resources, 1, 4

- NDVI, 376, 377
- Network, 227, 229, 232
- Neural Network, 447, 450, 455, 460, 461, 462, 469, 470, 474, 476, 477, 478, 483
- NIR, 376, 377
- Nitrogen, 352, 355, 356, 369

- Optimization, 445, 447, 448, 462, 463, 466, 472, 474, 475, 476, 482, 485
- Optimize, 77, 78, 80, 93, 94, 95, 96, 100

- pests, 204, 205, 206, 214, 221
- pH Sensor, 281
- Plant disease detection system, 401
- Plant growth, 357, 361, 362, 366
- Planting, 60, 61, 62, 64, 66
- Policy Interventions, 505, 506
- Poverty, 505, 506, 508, 509, 510, 536, 543
- Powdery Mildew, 438
- Precision, 374, 378, 381, 389, 393, 395, 396, 397, 398, 401
- Precision agriculture, 1, 2, 5, 6, 10, 11, 23, 24, 31, 33, 34, 36, 41, 45, 47
- Precision farming, 264
- Predictions, 537, 538, 539, 540, 542

- predictive models, 169 to 170, 180 to 182, 188
- Pre-trained model, 393, 399, 405, 407, 409
- Production Cycle, 66, 67
- Productivity, 1, 2, 3, 4, 6, 10, 14, 15, 19, 77, 80, 81, 82, 83, 84, 87, 88, 89, 91, 93, 98, 99, 100, 101, 103, 104, 105, 106

- Rainfall, 59, 69
- Random forest, 450, 452, 454, 457, 459, 460, 465, 469, 471, 474, 476
- Real-Time, 169, 170 to 174, 183 to 185, 188 to 195, 329, 331, 332, 333, 336, 347
- Recall, 393
- Regression algorithm, 452, 456, 457, 458
- Regression model, 359
- Remote Sensing, 374, 377, 379
- Remote Sensors, 62, 64
- ResNet, 393, 399, 400, 404, 405, 408, 410, 411, 412, 413, 414, 415, 425
- Resource efficiency, 502
- Return of Investment, 363
- Revolunized, 374, 382, 389
- Robot arms, 37, 38
- Robotics, 29, 30, 31, 33, 34, 35, 41, 42, 43, 44, 47, 49, 50, 52, 53, 75, 80, 81, 99, 100, 101, 102, 103, 104, 143, 146, 147, 149, 150,

151, 152, 156, 160, 162, 164, 165, 166, 211, 212

- Robots, 2, 7, 16, 17
- rural, 309, 310, 324, 325
- Rural Infrastructure, 29, 30, 31
- Rust, 420

- satellite imaging, 174, 193 and 195
- Scab, 419
- scarcity, 329, 330
- Season, 114, 122, 125, 126, 127, 130, 131, 132, 133, 134
- Security, 227, 229, 232, 325, 326
- Segmentation, 431
- sensing data, 329, 332, 333
- sensor, 201, 202, 203, 205, 206, 207, 208, 209, 211, 212, 213, 215, 217, 218, 220, 221, 222, 223, 227, 233, 243
- Sensors, 1, 2, 5, 7, 8, 12, 13, 15, 16, 17, 29, 33, 36, 38, 39, 41, 42, 43, 45, 47, 49, 76, 77, 78, 79, 80, 82, 83, 85, 87, 92, 94, 95, 96, 98, 100, 103, 105, 106, 107, 108, 109, 110, 309, 310, 311, 312, 313, 322, 323, 325, 326, 376, 377, 378
- smart, 329, 330, 333, 334, 342, 347, 349
- smart agriculture, 329, 333, 334, 337, 347, 446, 489, 494, 497, 499, 502
- Smart farming, 1, 2, 3, 4, 5, 6, 8, 9, 13, 21, 23, 24, 57, 59, 70, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93,

94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 203, 207, 208
- Smart Farming Monitoring System Services, 57, 70
- Smart irrigation, 310, 311, 316, 317, 321, 322, 323, 324, 325, 326, 329, 333, 342, 347
- Smoke Sensor, 286, 300
- Social Impact, 508, 510
- Social Obstacles, 75, 89, 90, 91, 92, 93, 100, 101
- Social Safety Net, 505, 507
- soil conditions, 170 and 172, 490, 491, 492
- Soil Health, 113, 114, 115, 118, 119, 121, 135
- Soil Management, 113, 114, 115, 121
- Soil moisture, 202, 203, 204, 208, 212, 214, 220, 222, 311, 312, 315, 316, 317, 329, 333, 334, 337, 338, 339, 341, 342, 343, 345, 346, 347
- Soil Moisture Sensor, 279, 280, 292, 294
- Soil Nutrient Sensor, 282, 284
- Soil Quality, 113, 115, 116, 119, 120, 121
- Soil Wetness, 71, 72
- solar, 314, 316, 321
- solar-powered, 494
- Spraying, 384, 385
- Support vector machines, 447, 450, 452, 455, 462, 464, 469, 471, 477, 481, 483

- Sustainability, 1, 2, 8, 170, 183 to 186, 196, 309, 310
- Sustainable, 76, 77, 90, 92, 93, 94, 95, 96, 97, 98, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 329, 330, 335, 345, 347, 495, 496, 502
- Sustainable Agriculture, 51, 202
- Sustainable develop, 114, 115, 116
- Sustanable Agri., 114, 115, 119, 121, 135
- SVM, 423

- Tech Implementaion, 489, 495
- Technologies, 113, 117, 118, 121, 122, 123, 124, 126, 127, 135
- Technology, 29, 30, 31, 32, 33, 34, 35, 36, 38, 43, 44, 45, 47, 48, 50, 201, 202, 203, 204, 207, 209, 217, 221, 222, 223, 374, 380, 381, 389
- Temperature Sensor, 284, 285
- Thresholding, 422
- Total coliform, 355, 356
- Traditional Farming, 58, 59, 65, 66, 70, 71
- Traditional Tools, 58
- Transfer Learning, 396, 397, 398, 399, 400, 404, 405, 407, 409, 414, 477, 478
- TSS, 354, 356

- UAV, 203, 212, 374, 389

- Ultrasonic Sensor, 289, 298
- Unemployment, 506, 511, 518, 520, 521, 522, 523, 524, 526, 527, 529, 532
- United States, 505, 506, 507, 510, 511, 523, 542
- urban, 309, 310, 324

- Variable, 507, 509, 510, 514, 516, 518, 519, 520, 521, 522, 523, 524, 529, 532, 533, 534, 538
- Vegetable, 66, 67, 74
- VGG-19, 393, 410, 411, 414
- VGGNet, 425

- Wastewater, 351, 352, 353, 354, 355, 357, 358, 359, 360, 361, 362, 364, 365, 366, 367, 368, 369, 370, 371
- water resource, 330
- Weather, 312, 314, 315, 316, 322, 324, 378, 381, 382, 388
- weather conditions, 173, 184
- weather stations, 172 and 173, 186
- Weeding Robots, 39, 40
- Wetland, 357, 358, 366, 367, 368, 369, 370

- XMPP Protocol, 275, 278

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